



**TENGIZCHEVROIL
ТЕНГИЗШЕВРОЙЛ**

**SPECIFICATION
ТЕХНИЧЕСКИЕ ТРЕБОВАНИЯ**

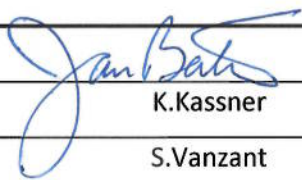
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ДАННЫЙ ДОКУМЕНТ ЯВЛЯЕТСЯ КОНТРОЛИРУЕМЫМ. НЕ ВНОСИТЬ НЕУТВЕРЖДЕННЫХ ИЗМЕНЕНИЙ**

**DETAILED PIPING LINE CLASS
SPECIFICATIONS
ДЕТАЛЬНАЯ СПЕЦИФИКАЦИЯ
ТРУБОПРОВОДОВ ПО КЛАССАМ**

**This document is dual language. Ensure both versions are modified.
Данный документ выполнен на двух языках. Убедитесь, что изменения внесены в обе версии.**

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REVISIONS РЕДАКЦИИ			PRELIM APPROVALS ПРЕДВАРИТЕЛЬНЫЕ УТВЕРЖДЕНИЯ			FINAL APPROVALS ОКОНЧАТЕЛЬНЫЕ УТВЕРЖДЕНИЯ			

REVISION DESCRIPTION SHEET
ПЕРЕЧЕНЬ РЕДАКЦИЙ

<u>REV.</u> <u>РЕД.</u>	<u>PARA.</u> <u>ПУНКТ</u>	<u>REVISION DESCRIPTION</u> <u>ОПИСАНИЕ РЕДАКЦИИ</u>
A	All / Все	Issued for internal review Выпущен для внутреннего рассмотрения
B	None / Нет	Issued for Client review Выпущен для рассмотрения Заказчиком
0	All / Все	Line class revision history and index summary added. Non-applicable classes removed. Small size butt weld fittings added. Bolting, thread sizes, temperatures revised. Добавлено описание редакций и сводная таблица классов трубопроводов. Удалены неприменимые классы трубопроводов. Добавлены сварные встык фитинги малого диаметра. Пересмотрены размеры болтовых соединений и резьбы и значения температуры.
1	Table 1 / Таблица 1	Rev 1 added Добавлена Ред. 1
1	3	Section 3 reworded to reflect the incorporation of required modifications to line classes. Изменена формулировка раздела 3 для отражения включения необходимых изменений в классах линий.
1	Appdx. A / Прил. А	Revised to reflect line class changes requested by TCO Документ пересмотрен для отражения изменений в классах линий, затребованных ТШО
1A	Appdx. A and Table 1/ Прил. А и Таблица 1	Revised to reflect line class changes requested by TCO and additions of new classes. Документ пересмотрен для отражения изменений в классах линий, затребованных ТШО и добавление новых классов.
2	Appdx. A and Table 1/ Прил. А и Таблица 1	Text modified as per TCO comments Текст заменен в соответствии с комментариями ТШО
3	Appdx. A and Table 1&2 Приложение А и Таблица 1&2	Revised to reflect line class changes requested by TCO Документ пересмотрен для отражения изменений в классах линий, затребованных ТШО.

3	Appdx. A and Table 1&2 Приложение А и Таблица 1&2	Revised to reflect the incorporation of new line classes: 150H23, 150K5F, 150PE6, 150PE4, 150VS01 600C80, 900K23, 900K5D, 10000H5A, 10000K5A Документ пересмотрен для отражения добавленных новых классов линий: 150H23, 150K5F, 150PE6, 150PE4, 150VS01 600C80, 900K23, 900K5D, 10000H5A, 10000K5A
3	Appdx. A and Table 1&2 Приложение А и Таблица 1&2	Revision 2 was accepted/adopted by TCO. Rev. 3 was received from FGP and it was decided that Base Business would take care of all future revisions specific to Base Business needs. FGP will carry on with project specific revisions until handover. Revision 3 reflects changes made by FGP and base business due to Base Business specific needs. Редакция 2 была принята ТШО. Ред. 3 была получена от ПБР, и было решено, что Основное производство будет заниматься всеми последующими редакциями, относящимися к Основному производству. ПБР будет продолжать заниматься редакциями, относящимися к проекту до момента передачи. Ред. 3 отражает изменения, выполненные ПБР и Основным производством в связи с потребностями Основного производства.
3	Appdx. A Приложение А	Non-applicable/ repeated legacy piping items removed and not available items from legacy section moved to line specification : Не применимые / повторные устаревшие номера классов удалены, и недоступные классы из раздела по устаревшим данным перенесены в спецификацию линий: 150A20, 150CP2, 150H21, 150H25, 150H26, 150H2G, 150K21, 150K26, 150M06, 150P21, 150P23, 150P24, 150PE2; 300H21, 300H25, 300H2G, 300K21, 300K5D, 300M06, 300P21; 600H21, 600H2G, 600K21, 600P21; 900H21, 900K21, 900P21; 1500H1, 1500K1, 1500P1, 2500H1, 2500P1

3	Appdx. A Приложение А	Legacy Specification removed from specification: Устаревшие спецификации удалены из спецификации: 150H22, 150H5C, 150H5H, 150J20, 150JK0, 150K5D, 150P26, 150PF2, 150PJ1, 150PV2 300H22, 600CP2, 600H22, 600H23, 600H5G, 600H5I, 600K22, 600K5B, 600K5C, 600PL; 900H22, 900H2G, 900H5F, 900K5A, 10000A, 10000K, 10000P, PVC
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TABLE 1 - LINE CLASS REVISION RECORD

LINE CLASS	REVISION							
	0	1	(1A)	2	3	4	5	6
150A20	08/04/13	02/08/13	-	-	01/18			
150C80	-	-	26/06/14	02/10/14	01/18			
150CP2	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150D01	-	-	26/06/14	02/10/14	01/18			
150H21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150H23	-	-	-	-	01/18			
150H25	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150H26	08/04/13	02/08/13	-	-	01/18			
150H2G	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150H5E	-	-	26/06/14	02/10/14	01/18			
150K21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150K23	-	-	-	-	01/18			
150K26	-	02/08/13	-	-	01/18			
150K5E	-	-	26/06/14	02/10/14	01/18			
150K5F	-	-	-	-	01/18			
150M40	08/04/13	02/08/13	-	-	01/18			
150MO6	08/04/13	02/08/13	-	-	01/18			
150P21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150P22	-	02/08/13	26/06/14	02/10/14	01/18			
150P23	08/04/13	02/08/13	-	-	01/18			
150P24	08/04/13	02/08/13	-	-	01/18			
150PE2	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150PE3	-	-	26/06/14	02/10/14	01/18			
150PE4	-	-	-	-	01/18			
150PE6	-	-	-	-	01/18			
150PE7	-	-	26/06/14	02/10/14	01/18			
150PE8	-	-	26/06/14	02/10/14	01/18			
150PE9	-	-	26/06/14	02/10/14	01/18			
150VC01	-	-	26/06/14	02/10/14	01/18			
150VG01	-	-	26/06/14	02/10/14	01/18			
150VS01	-	-	-	-	01/18			
300C80	-	-	26/06/14	02/10/14	01/18			
300H21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
300H25	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
300H2G	08/04/13	02/08/13	-	-	01/18			
300H5F	-	-	26/06/14	02/10/14	01/18			
300H5P	-	-	26/06/14	02/10/14	01/18			
300K21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
300K5A	-	-	26/06/14	02/10/14	01/18			
300K5D	08/04/13	02/08/13	-	-	01/18			
300K5E	-	-	26/06/14	02/10/14	01/18			
300K5F	-	-	26/06/14	02/10/14	01/18			
300K5G	-	-	26/06/14	02/10/14	01/18			
300MO6	08/04/13	02/08/13	-	-	01/18			
300P21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
300P22	-	02/08/13	26/06/14	02/10/14	01/18			
300PF2	08/04/13	02/08/13	-	-	01/18			
600A80	-	-	26/06/14	02/10/14	01/18			
600C80	-	-	-	-	01/18			
600H21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
600H25	-	-	26/06/14	02/10/14	01/18			
600H2G	08/04/13	02/08/13	-	-	01/18			
600H5M	-	-	26/06/14	02/10/14	01/18			
600H5P	-	-	26/06/14	02/10/14	01/18			

LINE CLASS	REVISION							
	0	1	(1A)	2	3	4	5	6
600K21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
600K5D	-	-	26/06/14	02/10/14	01/18			
600K5E	-	-	26/06/14	02/10/14	01/18			
600K5F	-	-	26/06/14	02/10/14	01/18			
600K5G	-	-	26/06/14	02/10/14	01/18			
600K5H	-	-	26/06/14	02/10/14	01/18			
600P21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
600P22	-	02/08/13	26/06/14	02/10/14	01/18			
900A80	-	02/08/13	26/06/14	02/10/14	01/18			
900C80	-	-	26/06/14	02/10/14	01/18			
900D01	-	-	26/06/14	02/10/14	01/18			
900H21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
900H25	-	-	26/06/14	02/10/14	01/18			
900H5H	-	-	26/06/14	02/10/14	01/18			
900K21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
900K23	-	-	-	-	01/18			
900K5B	-	-	26/06/14	02/10/14	01/18			
900K5C	-	-	26/06/14	02/10/14	01/18			
900K5D	-	-	-	-	01/18			
900P21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
900P22	-	02/08/13	26/06/14	02/10/14	01/18			
1500A60	-	-	26/06/14	02/10/14	01/18			
1500F22	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
1500H1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
1500K1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
1500P1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
2500A60	-	-	26/06/14	02/10/14	01/18			
2500F22	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
2500H1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
2500K1	-	-	26/06/14	02/10/14	01/18			
2500P1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
2500P2	-	-	26/06/14	02/10/14	01/18			
10000A60	-	-	26/06/14	02/10/14	01/18			
10000F22	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
10000H5A	-	-	-	-	01/18			
10000K5A	-	-	-	-	01/18			
10000X1	-	-	26/06/14	02/10/14	01/18			
10000X22	-	-	26/06/14	02/10/14	01/18			

NOTE:-

Piping Line Classes marked with an * * are not used on **FGP/WPMP**.

1. SCOPE

This line class specifications contained in Appendix A of this document define the requirements for bulk pipe, fittings, flanges and other items to be used in the relevant line classes for Tengizchevroil's facility at Tengiz in the Republic of Kazakhstan. This document is to be read in conjunction with document PIM-SU-5112-TCO, Piping Material Line Classes, which includes the full Line Class Index and all Branch Tables.

2. SPECIFICATIONS

Reference is also made to the following TCO Engineering Specification
PIM-SU-5112-TCO Piping Material Line Classes

3. INTRODUCTION

The detailed line class specifications contained in Appendix A are to be used on the Future Growth Project / Well Pressure Management Project (FGP/WPMP) and all further developments at Tengizchevroil's facility at Tengiz. They are developments of those used on the TCO Asset Development Project (SGI/SGP) with significant amendments as required by TCO for present and future work.

Components used on the SGI/SGP Project that will not be permitted on FGP/WPMP or future projects can be found in the archived legacy specification. These shall only be used for in kind or brown field tie-in purposes.

TABLE 2 - LINE CLASS INDEX SUMMARY

LINE CLASS	RATING #	MAT'L	PIPE ASTM	CORR ALL mm	DESIGN CONDITIONS				SIZE NPS in
					MIN TEMP		MAX TEMP		
					barg	°C	barg	°C	
150A20	150	ALLOY 20	B464 N08020	0	20	-46	13.8	200	1/2 - 4
150C80	150	INC 825+ LTCS	A333 GR 6	1	19.6	-46	13.8	200	3/4-24&48
150CP2	150	PTFE + LTCS	A333 GR 6	0	16	-46	16	150	1/2 - 24
150D01	150	Super Duplex	S32760	1	20	-46	17.7	100	3/4 - 24
150H21	150	LTCS	A333 GR 6	1.5	19.6	-46	8.6	343	1/2 - 48
150H23	150	LTCS	A333 GR 6	6	19.6	-46	13.8	200	3/4 - 20
150H25	150	LTCS	A333 GR 6	3	19.6	-46	8.6	343	1/2 - 40
150H26	150	LTCS	A333 GR 6	3	5	-36	5	343	1/2 - 92
150H2G	150	GLTCS	A333/6 GV	3	15.8	-46	15.8	150	3/4 - 36
150H5E	150	LTCS	API 5L X60	1.5	10	-40	10	75	3/4 - 28
150K21	150	LTCS	A333 GR 6	3	19.6	-46	12.7	232	3/4 - 56
150K23	150	LTCS	A333 GR 6	6	19.6	-46	13.8	200	3/4 - 20
150K26	150	LTCS	A333 GR 6	3	5.0	-36	5.0	343	3/4 - 92
150K5E	150	HOLD + LTCS	A333 GR 6	6	10	-40	10	85	16 - 18
150K5F	150	LTCS	A333 GR 6	3	17.7	-40	17.7	100	3/4 - 24
150M40	150	MON 400	B165 N04400	0	15.9	-36	12.3	215	3/4 - 2
150MO6	150	6% Mo	A312 S31254	1	20.0	-36	13.8	200	1/2 - 10
150P21	150	SS 316/L	A312 316/L	1	19	-120*	14.1	180	1/2 - 42
150P22	150	SS 316/L	A312 316/L	1	19	-46	1.5	535	1/2 - 42
150P23	150	SS 316/L	A312 316/L	0	10	-40	10	145	1/2 – 2
150P24	150	SS 316/L	A312 316/L	0	19	-40	13.7	200	1/2 – 1
150PE2	150	HPPE	SDR 11 HD	0	16 HOLD	-46	16 HOLD	40	2 – 40
150PE3	150	HDPE	SDR 11 HD	0	4.7	-40	4.7	40	2 - 24
150PE4	150	HDPE	SDR 11 HD	0	16 HOLD	-46	16 HOLD	40	2 - 24
150PE6	150	HDPE	SDR 11 HD	0	14.5	-40	14.5	25	6 - 8
150PE7	150	HDPE	PE-RT	0	10	-40	7.1	80	18 only
150PE8	150	HDPE	SDR 11 HD	0	9.3	-40	9.3	25	2 - 24
150PE9	150	HDPE	SDR 11 HD	0	9.3	-40	9.3	40	2 - 24
150VC01	150	LTCS	A333 GR 6	3	15.8	-46	15.8	110	3/4 - 24
150VG01	150	GLTCS	A333 GR 6	3	15.8	-46	15.8	110	3/4 - 36
150VS01	150	SS 316/L	A312 316/L	1	15.8	-46	15.8	110	1 - 16
300C80	300	INC 825 +LTCS	A333 GR 6	1	51.1	-46	42.7	230	3/4 - 42
300H21	300	LTCS	A333 GR 6	1.5	51.1	-46	37.9	343	3/4 - 48
300H25	300	LTCS	A333 GR 6	3	51.1	-46	37.9	343	3/4 - 24
300H2G	300	GLTCS	A333/6 GV	1.5	32	-29	32	150	3/4 - 3
300H5F	300	LTCS	API 5L X60	1.5	47	-40	47	90	3/4 - 20
300H5P	300	LTCS	API 5L X60	1.5	40	-40	40	80	3/4 - 28
300K21	300	LTCS	A333 GR 6	3	51.1	-46	41.5	260	3/4 60
300K5A	300	LTCS	API 5L X60	3	40	-40	40	88	3/4 - 28
300K5D	300	LTCS	A333 GR 6	3	51.1	-40	37.9	343	3/4 - 6
300K5E	300	LTCS	API 5L X60	3	47	-40	40	95	3/4 - 28
300K5F	300	LTCS	API 5L X60	3	47.5	-40	45.8	85	3/4 - 28
300K5G	300	INC 825+LTCS	API 5L X60	0	B31.8	-40	B31.8	95	1/2 - 16
300MO6	300	6% MO	A312 S31254	1	51.7	-36	47.8	130	1/2 - 3
300P21	300	SS 316/L	A312 316/L	1	49.6	-120*	36.8	180	1/2 - 24
300P22	300	SS 316/L	A312 316/L	1	49.6	-46	30.3	350	1/2 - 24
300PF2	300	GRE	D2996	0	40	-46	40	130	2 - 3
600A80	600	INC 825	B423 N08825	1	103.4	-46	94.3	230	3/4 - 30
600C80	600	INC 825+LTCS	A333 GR 6	1	102.1	-46	87.6	200	3/4 - 16
600H21	600	LTCS	A333 GR 6	1.5	102.1	-46	75.7	343	3/4 - 30
600H25	600	LTCS	A333 GR 6	3	102.1	-46	75.7	343	3/4 - 24
600H2G	600	GLTCS	A333/6 GV	1.5	102.1	-40	102.1	38	3/4 – 3
600H5M	600	LTCS	API 5L X60	3	67.5	-46	67.5	75	3/4 - 4

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LINE CLASS	RATING #	MAT'L	PIPE ASTM	CORR ALL mm	DESIGN CONDITIONS				SIZE NPS in
					MIN		MAX		
					barg	°C	barg	°C	
600H5P	600	LTCS	API 5L X60	1.5	93	-40	75	100	3/4 - 28
600K21	600	LTCS	A333 GR 6	3	102.1	-46	83.0	260	3/4 - 42
600K5D	600	LTCS	API 5L X60	3	B31.8	-40	B31.8	90	3/4 - 28
600K5E	600	LTCS	API 5L X60	3	93	-40	93	88	3/4 - 28
600K5F	600	LTCS	API 5L X60	3	B31.8	-40	B31.8	100	3/4 - 28
600K5G	600	INC 825+LTCS	API 5L X60	0	B31.8	-40	B31.8	95	3/4 - 16
600K5H	600	LTCS	API 5L X60	3	93	-40	93	95	3/4 - 28
600P21	600	SS 316/L	A312 316/L	1	99.3	-120*	73.6	180	3/4 – 48
600P22	600	SS 316/L	A312 316/L	1	99.3	-46	60.7	350	3/4 – 48
900A80	900	INC 825	B423 N08825	1	103.4	-46	103.4	300	3/4 - 24
900C80	900	INC 825+ LTCS	A333 GR 6	1	153.2	-46	131.4	200	3/4 - 36
900D01	900	Super Duplex	S32760	1	155.1	-46	152	100	3/4 - 16
900H21	900	LTCS	A333 GR 6	1.5	153.2	-46	97	410	3/4 - 30
900H25	900	LTCS	A333 GR 6	3	153.2	-46	97	410	3/4 - 24
900H5H	900	LTCS	API 5L X60	1.5	132	-40	132	80	3/4 - 6
900K21	900	LTCS	A333 GR 6	3	153.2	-46	124.5	260	3/4 - 36
900K23	900	LTCS	A333 GR 6	6	153.2	-46	131.4	200	1 - 16
900K5B	900	LTCS	API 5L X60	3	132	-40	132	90	4 - 20
900K5C	900	LTCS	API 5L X60	3 HOLD	140	-40	140	90	6 - 8
900K5D	900	LTCS	API 5L X60	3	132	-40	132	120	4 - 20
900P21	900	SS 316/L	A312 316/L	1	148.9	-140*	91	350	3/4 - 36
900P22	900	SS 316/L	A312 316/L	1	148.9	-46	91	350	3/4 - 36
1500A60	1500	INC 625	B444 N06625	1	184	-100	184	110	3/4 - 3
1500F22	1500	LACS	A182 F22M	3	258.6	-40	250.1	155	3/4 - 30
1500H1	1500	LTCS	A333 GR 6	1.5	255.3	-46	189.4	343	3/4 - 18
1500K1	1500	LTCS	A333 GR 6	3	255.3	-46	228.9	126	3/4 - 20
1500P1	1500	SS 316/L	A312 316/L	1	248.2	-50	211	100	3/4 - 4
2500A60	2500	INC 625	B444 N06625	1	324	-100	324	110	3/4 - 4
2500F22	2500	LACS	A182 F22M	3	430.9	-40	410.5	180	3/4 - 26
2500H1	2500	LTCS	A333 GR 6	1.5	425.5	-46	365	200	3/4 - 16
2500K1	2500	LTCS	A333 GR 6	3	365	-46	365	200	3/4 - 4
2500P1	2500	SS 316/L	A312 316/L	1	413.7	-46	339.2	120	3/4 - 4
2500P2	2500	SS 316/L	A312 316/L	1	350	-46	350	120	3/4 - 4
10000A60	10000	INC 625	B444 N06625	1	600	-100	600	110	3/4 - 8
10000F22	10000	LACS	A182 F22M	3	600	-40	600	130	3/4 - 22
10000H5A	10000	LTCS	API 5L X60	1.5	600	-40	600	110	2 - 4
10000K5A	10000	API 5L X60	API 5L X60	1.5	600	-36	600	90	3 - 3
10000X1	10000	LTCS	API 5L X60	3	600	0	600	110	2 & 14
10000X22	10000	F22M	A182 F22M	3	600	-40	600	110	14 ONLY

NOTE:-

Piping Line Classes marked with an * * are not used on **FGP/WPMP**.

Temperatures marked with *. No manual valves are to be operated at temperatures lower than -100°C. Valves required to be operable at temperatures below this are to be reviewed by Piping Materials and considered SP items.

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ТАБЛИЦА 1 – РЕГИСТРАЦИЯ ИЗМЕНЕНИЙ КЛАССОВ ЛИНИЙ

КЛАСС ЛИНИИ	РЕДАКЦИЯ							
	0	1	(1A)	2	3	4	5	6
150A20	08/04/13	02/08/13	-	-	01/18			
150C80	-	-	26/06/14	02/10/14	01/18			
150CP2	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150D01	-	-	26/06/14	02/10/14	01/18			
150H21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150H23	-	-	-	-	01/18			
150H25	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150H26	08/04/13	02/08/13	-	-	01/18			
150H2G	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150H5E	-	-	26/06/14	02/10/14	01/18			
150K21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150K23	-	-	-	-	01/18			
150K26	-	02/08/13	-	-	01/18			
150K5E	-	-	26/06/14	02/10/14	01/18			
150K5F	-	-	-	-	01/18			
150M40	08/04/13	02/08/13	-	-	01/18			
150MO6	08/04/13	02/08/13	-	-	01/18			
150P21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150P22	-	02/08/13	26/06/14	02/10/14	01/18			
150P23	08/04/13	02/08/13	-	-	01/18			
150P24	08/04/13	02/08/13	-	-	01/18			
150PE2	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
150PE3	-	-	26/06/14	02/10/14	01/18			
150PE4	-	-	-	-	01/18			
150PE6	-	-	-	-	01/18			
150PE7	-	-	26/06/14	02/10/14	01/18			
150PE8	-	-	26/06/14	02/10/14	01/18			
150PE9	-	-	26/06/14	02/10/14	01/18			
150VC01	-	-	26/06/14	02/10/14	01/18			
150VG01	-	-	26/06/14	02/10/14	01/18			
150VS01	-	-	-	-	01/18			
300C80	-	-	26/06/14	02/10/14	01/18			
300H21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
300H25	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
300H2G	08/04/13	02/08/13	-	-	01/18			
300H5F	-	-	26/06/14	02/10/14	01/18			
300H5P	-	-	26/06/14	02/10/14	01/18			
300K21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
300K5A	-	-	26/06/14	02/10/14	01/18			
300K5D	08/04/13	02/08/13	-	-	01/18			
300K5E	-	-	26/06/14	02/10/14	01/18			
300K5F	-	-	26/06/14	02/10/14	01/18			
300K5G	-	-	26/06/14	02/10/14	01/18			
300MO6	08/04/13	02/08/13	-	-	01/18			
300P21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
300P22	-	02/08/13	26/06/14	02/10/14	01/18			
300PF2	08/04/13	02/08/13	-	-	01/18			
600A80	-	-	26/06/14	02/10/14	01/18			
600C80	-	-	-	-	01/18			
600H21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
600H25	-	-	26/06/14	02/10/14	01/18			
600H2G	08/04/13	02/08/13	-	-	01/18			
600H5M	-	-	26/06/14	02/10/14	01/18			
600H5P	-	-	26/06/14	02/10/14	01/18			

КЛАСС ЛИНИИ	РЕДАКЦИЯ							
	0	1	(1A)	2	3	4	5	6
600K21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
600K5D	-	-	26/06/14	02/10/14	01/18			
600K5E	-	-	26/06/14	02/10/14	01/18			
600K5F	-	-	26/06/14	02/10/14	01/18			
600K5G	-	-	26/06/14	02/10/14	01/18			
600K5H	-	-	26/06/14	02/10/14	01/18			
600P21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
600P22	-	02/08/13	26/06/14	02/10/14	01/18			
900A80	-	02/08/13	26/06/14	02/10/14	01/18			
900C80	-	-	26/06/14	02/10/14	01/18			
900D01	-	-	26/06/14	02/10/14	01/18			
900H21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
900H25	-	-	26/06/14	02/10/14	01/18			
900H5H	-	-	26/06/14	02/10/14	01/18			
900K21	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
900K23	-	-	-	-	01/18			
900K5B	-	-	26/06/14	02/10/14	01/18			
900K5C	-	-	26/06/14	02/10/14	01/18			
900K5D	-	-	-	-	01/18			
900P22	-	02/08/13	26/06/14	02/10/14	01/18			
1500A60	-	-	26/06/14	02/10/14	01/18			
1500F22	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
1500H1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
1500K1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
1500P1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
2500A60	-	-	26/06/14	02/10/14	01/18			
2500F22	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
2500H1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
2500K1	-	-	26/06/14	02/10/14	01/18			
2500P1	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
2500P2	-	-	26/06/14	02/10/14	01/18			
10000A60	-	-	26/06/14	02/10/14	01/18			
10000F22	08/04/13	02/08/13	26/06/14	02/10/14	01/18			
10000H5A	-	-	-	-	01/18			
10000K5A	-	-	-	-	01/18			
10000X22	-	-	26/06/14	02/10/14	01/18			

ПРИМЕЧАНИЕ :

Классы трубопроводов, отмеченные звездочкой * *, не используются на проекте ПБР/ПУУД.

1 ОБЛАСТЬ ПРИМЕНЕНИЯ

Спецификация трубопроводов по классам, включенная в Приложение А настоящего документа, определяет требования к партиям труб, фитингам, фланцам и другим соответствующим трубопроводным изделиям, используемым в линиях соответствующих классов на производственных объектах компании «Тенгизшевройл» на месторождении Тенгиз в Республике Казахстан. Настоящий документ следует рассматривать вместе с документом № PIM-SU-5112-ТСО «Классы материалов трубопроводов», который содержит полный Указатель классов трубопроводов и Таблицы с перечислением всех отводов.

2 ТЕХНИЧЕСКИЕ УСЛОВИЯ

Необходимо включить также ссылку на следующие Технические условия ТШО:

PIM-SU-5112-ТСО Классы материалов трубопроводов

3 ВВЕДЕНИЕ

Детальная спецификация трубопроводов по классам, включенная в Приложение А, предназначена для использования на Проекте будущего расширения и Проекте управления устьевым давлением (ПБР/ПУУД) и на всех будущих проектах развития производственных объектов компании «Тенгизшевройл» на Тенгизе. Данная спецификация представляет собой модификацию документа, использовавшегося на Проекте развития основных средств ТШО (ЗСГ/ПВП), со значительными дополнениями, требующимися ТШО для текущих и будущих работ.

Компоненты, использовавшиеся на Проекте ЗСГ/ПВП, применение которых на ПБР/ПУУД или на будущих проектах не разрешено, могут быть найдены в архивных Устаревших спецификациях. Они используются исключительно для замены или выполнения врезок в существующие объекты.

ТАБЛИЦА 2 - СВОДНЫЙ УКАЗАТЕЛЬ ТРУБОПРОВОДОВ

КЛАСС ЛИНИИ	НОМИН АЛ #	МАТЕРИАЛ	ТРУБА ПО ASTM	ДОПУСК НА КОРРОЗИЮ, мм	РАСЧЕТНЫЕ УСЛОВИЯ				НОМ. ДИАМЕТР, дюймы
					МИН.ТЕМП.		МАКС. ТЕМП.		
					бар изб.	°C	бар изб.	°C	
150A20	150	СПЛАВ 20	B464 N08020	0	20	-46	13.8	200	1/2 - 4
150C80	150	ИНК 825+ НТУС	A333 GR 6	1	19.6	-46	13.8	200	3/4-24&48
150CP2	150	PTFE + НТУС	A333 GR 6	0	16	-46	16	150	1/2 - 24
150D01	150	Super Duplex	S32760	1	20	-46	17.7	100	3/4 - 24
150H21	150	НТУС	A333 GR 6	1.5	19.6	-46	8.6	343	1/2 - 48
150H23	150	НТУС	A333 GR 6	6	19.6	-46	13.8	200	3/4 - 20
150H25	150	НТУС	A333 GR 6	3	19.6	-46	8.6	343	1/2 - 40
150H26	150	НТУС	A333 GR 6	3	5	-36	5	343	1/2 - 92
150H2G	150	GLTCS	A333/6 GV	3	15.8	-46	15.8	150	3/4 - 36
150H5E	150	НТУС	API 5L X60	1.5	10	-40	10	75	3/4 - 28
150K21	150	НТУС	A333 GR 6	3	19.6	-46	12.7	232	3/4 - 56
150K23	150	НТУС	A333 GR 6	6	19.6	-46	13.8	200	3/4 - 20
150K26	150	НТУС	A333 GR 6	3	5.0	-36	5.0	343	3/4 - 92
150K5E	150	УТОЧН. + НТУС	A333 GR 6	6	10	-40	10	85	16 - 18
150K5F	150	НТУС	A333 GR 6	3	17.7	-40	17.7	100	3/4 - 24
150M40	150	MON 400	B165 N04400	0	15.9	-36	12.3	215	3/4 - 2
150MO6	150	6% Мо	A312 S31254	1	20.0	-36	13.8	200	1/2 - 10
150P21	150	SS 316/L	A312 316/L	1	19	-120*	14.1	180	1/2 - 42
150P22	150	SS 316/L	A312 316/L	1	19	-46	1.5	535	1/2 - 42
150P23	150	SS 316/L	A312 316/L	0	10	-40	10	145	1/2 – 2
150P24	150	SS 316/L	A312 316/L	0	19	-40	13.7	200	1/2 – 1
150PE2	150	HPPE	SDR 11 HD	0	16 УТОЧН.	-46	16 УТОЧН.	40	2 – 40
150PE3	150	HDPE	SDR 11 HD	0	4.7	-40	4.7	40	2 - 24
150PE4	150	HDPE	SDR 11 HD	0	16 УТОЧН.	-46	16 УТОЧН.	40	2 - 24
150PE6	150	HDPE	SDR 11 HD	0	14.5	-40	14.5	25	6 - 8
150PE7	150	HDPE	PE-RT	0	10	-40	7.1	80	18 only
150PE8	150	HDPE	SDR 11 HD	0	9.3	-40	9.3	25	2 - 24
150PE9	150	HDPE	SDR 11 HD	0	9.3	-40	9.3	40	2 - 24
150VC01	150	НТУС	A333 GR 6	3	15.8	-46	15.8	110	3/4 - 24
150VG01	150	GLTCS	A333 GR 6	3	15.8	-46	15.8	110	3/4 - 36
150VS01	150	SS 316/L	A312 316/L	1	15.8	-46	15.8	110	1 - 16
300C80	300	INC 825 +LTCS	A333 GR 6	1	51.1	-46	42.7	230	3/4 - 42
300H21	300	НТУС	A333 GR 6	1.5	51.1	-46	37.9	343	3/4 - 48
300H25	300	НТУС	A333 GR 6	3	51.1	-46	37.9	343	3/4 - 24
300H2G	300	GLTCS	A333/6 GV	1.5	32	-29	32	150	3/4 - 3
300H5F	300	НТУС	API 5L X60	1.5	47	-40	47	90	3/4 - 20
300H5P	300	НТУС	API 5L X60	1.5	40	-40	40	80	3/4 - 28
300K21	300	НТУС	A333 GR 6	3	51.1	-46	41.5	260	3/4 60
300K5A	300	НТУС	API 5L X60	3	40	-40	40	88	3/4 - 28
300K5D	300	LTCS	A333 GR 6	3	51.1	-40	37.9	343	3/4 - 6
300K5E	300	LTCS	API 5L X60	3	47	-40	40	95	3/4 - 28
300K5F	300	НТУС	API 5L X60	3	45.8	-40	45.8	85	3/4 - 28
300K5G	300	ИНК 825+НТУС	API 5L X60	0	B31.8	-40	B31.8	95	1/2 - 16
300MO6	300	6% MO	A312 S31254	1	51.7	-36	47.8	130	1/2 - 3
300P21	300	SS 316/L	A312 316/L	1	49.6	-120*	36.8	180	1/2 - 24
300P22	300	SS 316/L	A312 316/L	1	49.6	-46	30.3	350	1/2 - 24
300PF2	300	GRE	D2996	0	40	-46	40	130	2 - 3
600A80	600	ИНК 825	B423 N08825	1	103.4	-46	94.3	230	3/4 - 30
600C80	600	INC 825+LTCS	A333 GR 6	1	102.1	-46	87.6	200	3/4 - 16
600H21	600	НТУС	A333 GR 6	1.5	102.1	-46	75.7	343	3/4 - 30
600H25	600	НТУС	A333 GR 6	3	102.1	-46	75.7	343	3/4 - 24
600H2G	600	GLTCS	A333/6 GV	1.5	102.1	-40	102.1	38	3/4 – 3
600H5M	600	НТУС	API 5L X60	3	67.5	-46	67.5	75	3/4 - 4

КЛАСС ЛИНИИ	НОМИН АЛ #	МАТЕРИАЛ	ТРУБА ПО ASTM	ДОПУСК НА КОРРОЗИЮ, мм	РАСЧЕТНЫЕ УСЛОВИЯ				НОМ. ДИАМЕТР, дюймы
					МИН.ТЕМП.		МАКС. ТЕМП.		
					бар изб.	°C	бар изб.	°C	
600H5P	600	НТУС	API 5L X60	1.5	93	-40	75	100	3/4 - 28
600K21	600	НТУС	A333 GR 6	3	102.1	-46	83.0	260	3/4 - 42
600K5D	600	НТУС	API 5L X60	3	B31.8	-40	B31.8	90	3/4 - 28
600K5E	600	НТУС	API 5L X60	3	93	-40	93	88	3/4 - 28
600K5F	600	НТУС	API 5L X60	3	B31.8	-40	B31.8	100	3/4 - 28
600K5G	600	ИНК825+ НТУС	API 5L X60	0	B31.8	-40	B31.8	120	3/4 - 16
600K5H	600	НТУС	API 5L X60	3	93	-40	93	95	3/4 - 28
600P21	600	SS 316/L	A312 316/L	1	99.3	-120*	73.6	180	3/4 – 48
600P22	600	SS 316/L	A312 316/L	1	99.3	-46	60.7	350	3/4 – 48
900A80	900	ИНК 825	B423 N08825	1	103.4	-46	103.4	300	3/4 - 24
900C80	900	ИНК825+ НТУС	A333 GR 6	1	153.2	-46	131.4	200	3/4 - 36
900D01	900	Super Duplex	S32760	1	155.1	-46	152	100	3/4 - 16
900H21	900	НТУС	A333 GR 6	1.5	153.2	-46	97	410	3/4 - 30
900H25	900	НТУС	A333 GR 6	3	153.2	-46	97	410	3/4 - 24
900H5H	900	НТУС	API 5L X60	1.5	132	-40	132	80	3/4 - 6
900K21	900	НТУС	A333 GR 6	3	153.2	-46	124.5	260	3/4 - 36
900K23	900	НТУС	A333 GR 6	6	153.2	-46	131.4	200	1 - 16
900K5B	900	НТУС	API 5L X60	3	132	-40	132	90	4 - 20
900K5C	900	НТУС	API 5L X60	3 УТОЧН.	140	-40	140	90	6 - 8
900K5D	900	LTCS	API 5L X60	3	132	-40	132	120	4 - 20
900P21	900	SS 316/L	A312 316/L	1	148.9	-140*	91	350	3/4 - 36
900P22	900	SS 316/L	A312 316/L	1	148.9	-46	91	350	3/4 - 36
1500A60	1500	ИНК 625	B444 N06625	1	184	-100	184	110	3/4 - 3
1500F22	1500	LACS	A182 F22M	3	258.6	-40	250.1	155	3/4 - 30
1500H1	1500	НТУС	A333 GR 6	1.5	255.3	-46	189.4	343	3/4 - 18
1500K1	1500	НТУС	A333 GR 6	3	255.3	-46	228.9	126	3/4 - 20
1500P1	1500	SS 316/L	A312 316/L	1	248.2	-50	211	100	3/4 - 4
2500A60	2500	ИНК 625	B444 N06625	1	324	-100	324	110	3/4 - 4
2500F22	2500	LACS	A182 F22M	3	430.9	-40	410.5	180	3/4 - 26
2500H1	2500	НТУС	A333 GR 6	1.5	425.5	-46	365	200	3/4 - 16
2500K1	2500	НТУС	A333 GR 6	3	365	-46	365	200	3/4 - 4
2500P1	2500	SS 316/L	A312 316/L	1	413.7	-46	339.2	120	3/4 - 4
2500P2	2500	SS 316/L	A312 316/L	1	350	-46	350	120	3/4 - 4
10000A60	10000	ИНК 625	B444 N06625	1	600	-100	600	110	3/4 - 8
10000F22	10000	LACS	A182 F22M	3	600	-40	600	130	3/4 - 22
10000H5A	10000	LTCS	API 5L X60	1.5	600	-40	600	110	2 - 4
10000K5A	10000	API 5L X60	API 5L X60	1.5	600	-36	600	90	3 - 3
10000X1	10000	LTCS	API 5L X60	3	600	0	600	110	2 & 14
10000X22	10000	F22M	A182 F22M	3	600	-40	600	110	14 ONLY

ПРИМЕЧАНИЕ :

Классы трубопроводов, отмеченные звездочкой * *, не используются на проекте ПБР/ПУУД.

Значения температуры отмечены *. Ручные клапаны, используемые при температурах ниже -100 °C, отсутствуют. Клапаны, которые предполагается использовать при температурах ниже данного значения, будут рассмотрены отделом трубных материалов и учтены в качестве изделий особого назначения.

APPENDIX A.	DETAILED PIPING LINE CLASS SPECIFICATIONS
ПРИЛОЖЕНИЕ А.	ДЕТАЛЬНАЯ СПЕЦИФИКАЦИЯ ТРУБОПРОВОДОВ ПО КЛАССАМ

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
10						***** BALL VALVES *****		032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE FLAT FACE FLANGES WHEN MATING TO FRP FLANGES. ALL FLANGE CONNECTIONS TO BE SUPPLIED WITH SPLASH GUARDS AROUND THE FLANGES. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. ONLY FOR MATCHING 300# FLANGED CONTROL VALVES. USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING. A STRAIGHT TEE WITH A CONCENTRIC REDUCER ON THE TEE BRANCH MAY BE USED AS A SUSTITUTE FOR A REDUCING TEE. USE INSERT FOR REDUCING ON STRAIGHT SOCKET WELD TEE.							
20																
30																
40	C		Ball V	BA1D1	3/4 - 2	BALL RB 150# RF ALLOY 20 BODY & TRIM 3 PIECE BOLTED BODY STANDARD PORT		B1								
50								FF3								
60	C		Ball V	BA1D2	3/4 - 2	BALL RB 2000# WOG SW ALLOY 20 BODY & TRIM 3 PIECE BOLTED BODY STANDARD		FR6								
70								FS1								
80						***** PIPE *****		FS9								
90								SC1								
100																
110	C		Piping	5731001	1/2 - 2	PIPE SCH 40S SMLS ALLOY 20 B464 UNS N08020 BE										
120			Piping	5731002	3 - 4	PIPE SCH 10S SMLS ALLOY 20 B464 UNS N08020 BE										
130						***** FITTINGS *****		ZA6								
140																
150								ZG3								
160	C		90 Deg	5731011	1/2 - 2	ELL BW 90 DEG LR SCH 40S ALLOY 20 B366 UNS N08020										
170			90 Deg	5731013	3 - 4	ELL 90 DEG LR SCH 10S ALLOY 20 B366 UNS N08020		ZG4								
180																
190	C		Tee	5464271	3/4 - 2	TEE BW SCH 40S ALLOY 20 B366 UNS N08020										
200			Tee	5731021	3 - 4	TEE BW SCH 10S ALLOY 20 B366 UNS N08020										
210																
220		ZG3	Reduci	5731022	3/4 - 1 1/2	TEE BW RED SCH 40S ALLOY 20 B366 UNS N08020	1 - 2									
230			Reduci	5731023	1 1/2 - 2	TEE BW RED SCH 10S/SCH 40S ALLOY 20 B366 UNS N08020	3 - 4									
240			Reduci	5731024	3 - 3	TEE BW RED SCH 10S ALLOY 20 B366 UNS N08020	4 - 4									
250																
260	C		Concen	5731030	3/4 - 1 1/2	REDUCER CONC SCH 40S ALLOY 20 B366 UNS N08020	1 - 2									
270			Concen	5731031	1 1/2 - 2	REDUCER CONC SCH 10S/SCH 40S ALLOY 20 B366 UNS N08020	3 - 4									
280																
290																
300	C		Eccent	5464274	3/4 - 1 1/2	REDUCER ECC SCH 40S ALLOY 20 B366 UNS N08020	1 - 2									
310			Eccent	5731040	1 1/2 - 2	REDUCER ECC SCH 10S/SCH 40S ALLOY 20 B366 UNS N08020	3 - 4									
320																
330			Weldol	5731050	1/2 - 1	WELDOLET SCH 40S ALLOY 20 B462 UNS N08020	2 - 2									
340			Weldol	5731051	1/2 - 1 1/2	WELDOLET SCH 10S/SCH 40S ALLOY 20 B462 UNS N08020	3 - 4									
350																
360			90 Deg	5731012	1/2 - 2	ELL 90 DEG 3000# SW ALLOY 20 B462 UNS N08020										
370		ZG4	Tee	5731020	1/2 - 2	TEE 3000# SW ALLOY 20 B462 UNS N08020										
380																
390						***** INSERTS *****										
400																
410		ZG4	Bushin	5731080	3/4 - 1	INSERT 3000# SW ALLOY 20 B462 UNS N08020	1 - 1 1/2									
420																
430		FR6														
440		FS1				***** PIPE FLANGES *****										
450																
460		SC1	Flange	5731067	3/4 - 3/4	FLG SCRD (FNPT) 150# RF ALLOY 20 B462 UNS N08020										
470																
480			Flange	5731060	1/2 - 2	FLG SW 150# RF ALLOY 20 B462 UNS N08020 SCH 40S										
490		FF3	Flange	5731063	1/2 - 2	FLG SW 150# FF ALLOY 20 B462 UNS N08020 SCH 40S										
500																
510	C		Flange	5731066	1/2 - 2	FLG WN 150# RF ALLOY 20 B462 UNS N08020 SCH 40S										
520			Flange	5731069	3 - 4	FLG WN 150# RF ALLOY 20 B462 UNS N08020 SCH 10S										
530																
540																
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP								
						ALL SIZES- BUTTWELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS			
							0 MM	17NOV16	SMS	C	1 OF 2	5	150A20			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
550	C		Blind	5731068	1/2 - 4	FLG BLIND 150# RF ALLOY 20 B462 UNS N08020	3/4 - 1 1/2 2 - 2							
560														
570	C	FS9	Flange	5731061	1/2 - 2	FLG WN 300# RF ALLOY 20 B462 UNS N08020 SCH 40S								
580			Flange	5731065	3 - 4	FLG WN 300# RF ALLOY 20 B462 UNS N08020 SCH 10S								
590						***** SWAGE NIPPLES *****								
600														
610														
620			Concen	5721931	1/2 - 1/2	SWAGE CONC SCH 40S/80S B366 UNS N08020 PLE/TSE								
630			Concen	5721932	1/2 - 1/2	SWAGE CONC SCH 40S/80S B366 UNS N08020 BLE/TSE								
640														
650														
660		FS1				***** GASKETS *****								
670														
680	C		GASFRS	5673043	3/4 - 4	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR								
690														
700			GASFR	5731070	1/2 - 4	GASKET 150# LOW STRESS TEFLON ENVELOPE NEOPRENE FILLER, GARLOCK MODEL 370								
710														
720		FS9	GASFR	5731071	1/2 - 4	GASKET 300# LOW STRESS TEFLON ENVELOPE NEOPRENE FILLER, GARLOCK MODEL 370								
730														
740														
750														
760						***** BOLTING *****								
770														
780														
790	C	B1	BOLS	5675770	1/2 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
800														
810						***** BRANCHES *****								
820														
830														
840						USE BRANCH CHART 150A20								
850						***** CONSTRUCTION DETAILS *****								
860														
870														
880						SEAL WELDING -----FOR THREADED CONNECTIONS ONLY								
890						CARPENTER 20 IS RECOMMENDED FOR								
900						THE WELDING OF THIS MATERIAL								
910						PIPE BENDS ----- NONE								
920						THREAD SEALANT ----- PTFE PASTE								
							SERVICE							
							DESIGN ANSI B16.5 MG 3.1 CODE ASME B31.3 AS, AP NOTE - PIPING 1/2" NB & BELOW IN THIS PIPING CLASS SHALL ONLY BE USED WITHIN THE IRCC SCOPE OF THE EVAPORATOR BRINE CONCENTRATOR SYSTEM. PIPING 3/4" NB IS PERMITTED TO BE USED OUTSIDE THE LIMITS OF THE IRCC SCOPE OF THE PROJECT.							
							MATERIAL: ALLOY 20 FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 200 °C COMMENT: 1/2" & BELOW - FOR VENDOR PACKAGES ONLY							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						ALL SIZES- BUTTWELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
						0 MM		17NOV16	SMS	C	2 OF 2	5	150A20	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10		PP9				*****VALVES*****		810	NIPPLE TO BE CUT FROM SCH 160 PIPE ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY. ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
20		VA5				*****MODULAR VALVES*****		B1						
30								FA8						
40								FS1						
50														
60	B		2BV	K01AM10	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV		FF5						
70														
80	B	VAV	2BV	K03AM10	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV		PP9						
90														
100	B	FS1	2BVR	M01AM10	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR		SB6						
110														
120	B	FS1	2BVR	M03AM10	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR		UJ2						
130														
140	B	VAW	R2BVR	M03EM10	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	VA5						
150								VAV						
160	B	VAW	R2BVR	M03EM10	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	VAV						
170								VB2						
180	B	VB2				*****BALL VALVES*****		VC1						
190														
200	B		Ball V	A01AM09	3 - 3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO		VC2						
210	B		Ball V	A01AR09	4 - 6	BALL 150# RF LTCS + CLAD INC 825 INC 625 TRIM FLOATING SOFT RB LO								
220	B		Ball V	A01AR15	8 - 8	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB GO		VQ2						
230														
240	B		Ball V	B01AM09	3/4 - 2	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO		ZA6						
250		VA5	Ball V	B01AM09	3 - 3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO								
260	B	VA5	Ball V	B01AR11	4 - 6	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT FB LO								
270	B	VA5	Ball V	B01AR15	8 - 20	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT FB GO								
280														
290	B	FS1	Ball V	B03AM09	3/4 - 2	BALL 300# RF INC 825 INC 625 TRIM FLOATING SOFT FB LO								
300	B	FS1	Ball V	A03AM09	3 - 3	BALL 300# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO								
310														
320						*****BUTTERFLY VALVES*****								
330														
340			Butter	T01AR63	10 - 24	BUTTERFLY 150# RF LTCS + CLAD INC 825 INC TRIM 625 TOV METAL DBL FLG SHORT PAT GO								
350														
360						*****GATE VALVES*****								
370														
380			Gate V	G01AM49	3/4 - 3	GATE 150# RF INC 825 INC 625 TRIM WEDGE HW								
390			Gate V	G01AR43	4 - 8	GATE 150# RF LTCS + CLAD INC 825 INC TRIM 625 WEDGE HW								
400	B	VA5	Gate V	G01AR43	10 - 10	GATE 150# RF LTCS + CLAD INC 825 INC TRIM 625 WEDGE HW								
410	B	VA5	Gate V	G01AR44	12 - 24	GATE 150# RF LTCS + CLAD INC 825 INC TRIM 625 WEDGE GO								
420														
430						*****GLOBE VALVES*****								
440														
450			Globe	L01AM49	3/4 - 3	GLOBE 150# RF INC 825 INC TRIM 625 OS&Y HW								
460			Globe	L01AR49	4 - 6	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y HW								
470			Globe	L01AR44	8 - 10	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y GO								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						ALL SIZES- BUTTWELD PWHT FOR SIZES 4" AND LARGER		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								1 MM	24MAR17	SMS	B	1 OF 6	5	150C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
480	B	VC1	Check	C01AM51	3/4 - 1 1/2	*****CHECK VALVES***** CHECK PISTON 150# RF INC 825 INC TRIM 625						
490			Check	C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED						
500			Check	C01AR46	4 - 6	CHECK DUAL WAFER LUG 150# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED						
510		VC2	Check	C01AR53	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS + CLAD INC 825 INC TRIM 625						
520					*****PIPES*****							
530		VC2	Piping	5732001	3/4 - 1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE						
540			Piping	5732002	1 1/2 - 1 1/2	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 PE						
550			Piping	5361603	2 - 3	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 BE						
560			Piping	5358505_C80	4 - 6	PIPE SCH STD SMLS LTCS A333 GR.6 WITH 3MM INC 825 INTERNAL CLADDING						
570			Piping	5358501_C80	8 - 16	PIPE SCH 20 SMLS LTCS A333 GR.6 WITH 3MM INC 825 INTERNAL CLADDING						
580			Piping	5358597_C80	18 - 18	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 WITH 3MM INC 825 INTERNAL CLADDING						
590	B	810	Piping	5358576_C80	20 - 24	PIPE SCH STD EFW LTCS A671 CC60 CL.22 WITH 3MM INC 825 INTERNAL CLADDING						
600			Piping	5358577_C80	48 - 48	PIPE SCH XS EFW LTCS A671 CC60 CL.22 WITH 3MM INC 825 INTERNAL CLADDING						
610			Piping	5732003	1/2 - 1/2	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE						
620					*****FITTINGS*****							
630			90 Deg	5425111	3/4 - 1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825						
640		810	90 Deg	5424722	1 1/2 - 3	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPS UNS N08825						
650			90 Deg	5392438_C80	4 - 6	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING						
660			90 Deg	5392276_C80	8 - 16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING						
670			90 Deg	5392280_C80	18 - 18	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING						
680			90 Deg	5392496_C80	20 - 24	ELL 90 DEG LR SCH STD LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING						
690	B	810	45 Deg	5425113	3/4 - 1	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825						
700			45 Deg	5424727	1 1/2 - 3	ELL 45 DEG SCH 40S ALLOY 825 B366-WPS UNS N08825						
710			45 Deg	5392444_C80	4 - 6	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING						
720			45 Deg	5392282_C80	8 - 16	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING						
730			45 Deg	5392281_C80	18 - 18	ELL BW 45 DEG LR SCH20 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING						
740			45 Deg	5392607_C80	20 - 24	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING						
750		810	Tee	5425119	3/4 - 1	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825						
760			Tee	5424725	1 1/2 - 3	TEE SCH 40S ALLOY 825 B366-WPS UNS N08825						
770			Tee	5392447_C80	4 - 6	TEE BW SCH STD LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING						
780			Tee	5392285_C80	8 - 16	TEE BW SCH 20 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING						
790			Tee	5392295_C80	18 - 18	TEE BW SCH 20 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING						
800	B	810	Tee	5392405_C80	20 - 24	TEE SCH STD LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING						
810												
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SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
890			Reduci	5425126	3/4 - 3/4	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1 - 1										
900			Reduci	5425124	3/4 - 1	TEE RED SCH 40S/80S ALLOY 825 B366-WPS UNS N08825	1 1/2 - 2										
910			Reduci	5424724	1 1/2 - 2	TEE RED SCH 40S ALLOY 825 B366-WPS UNS N08825	2 - 3										
920			Reduci	5425128	1 1/2 - 3	TEE RED SCH STD/SCH 40S ALLOY 825 B366-WPS UNS N08825	4 - 6										
930			Reduci	5392445_C80	4 - 4	TEE RED BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 6										
940			Reduci	5392289_C80	4 - 6	TEE RED BW SCH 20/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	8 - 16										
950			Reduci	5392209_C80	8 - 14	TEE RED BW SCH 20 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10 - 16										
960			Reduci	5392292_C80	8 - 16	TEE RED BW SCH 20 STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18 - 18										
970			Reduci	5556912_C80	8 - 18	TEE RED BW SCH STD/ SCH 20 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	20 - 24										
980			Reduci	5392428_C80	20 - 20	TEE RED BW SCH STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24 - 24										
990			Cap	5424270	3/4 - 1	CAP SCH 80S ALLOY 825 B366 UNS N08825											
1000			Cap	5424219	1 1/2 - 3	CAP SCH 40S ALLOY 825 B366 UNS N08825											
1010			Cap	5392448_C80	4 - 6	CAP BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING											
1020			Cap	5392286_C80	8 - 18	CAP BW SCH 20 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING											
1030			Cap	5392448_C80	20 - 24	CAP BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING											
1040			Concen	5425122	3/4 - 3/4	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1										
1050				5425143	3/4 - 1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2										
1060				5424733	1 1/2 - 2	REDUCER CONC SCH 40S ALLOY 825 B366-WPS N08825	2 - 3										
1070				5425146	1 1/2 - 3	REDUCER CONC SCH STD/40S ALLOY 825 B366-WPS N08825	4 - 6										
1080				5392449_C80	4 - 4	REDUCER CONC SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 6										
1090			Concen	5392290_C80	4 - 6	REDUCER CONC SCH 20/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	8 - 14										
1100			Concen	5392211_C80	8 - 14	REDUCER CONC SCH 20 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10 - 16										
1110			Concen	5392211_C80	8 - 14	REDUCER CONC SCH 20 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10 - 16										
1120			Concen	5392287_C80	10 - 16	REDUCER CONC SCH 20 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18 - 18										
1130			Concen	5392500_C80	12 - 18	REDUCER CONC SCH STD/20 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	20 - 24										
1140			Concen	5392505_C80	20 - 20	REDUCER CONC SCH STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24 - 24										
1150			Eccent	5425123	3/4 - 3/4	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1										
1160				5425145	3/4 - 1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2										
1170				5424734	1 1/2 - 2	REDUCER ECC SCH 40S ALLOY 825 B366-WPS N08825	2 - 3										
1180				5425148	1 1/2 - 3	REDUCER ECC SCH STD/40S ALLOY 825 B366-WPS N08825	4 - 6										
1190				5392450_C80	4 - 4	REDUCER ECC SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 6										
1200			Eccent	5392450_C80	4 - 4	REDUCER ECC SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 6						MATERIAL: CARBON STEEL / CRA FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 200 °C				
1210			Eccent	5392291_C80	4 - 6	REDUCER ECC SCH 20/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	8 - 16										
1220			Eccent	5392212_C80	8 - 14	REDUCER ECC SCH 20 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	10 - 16										
1230			Eccent	5392288_C80	10 - 16	REDUCER ECC SCH 20 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18 - 18										
1240			Eccent	5392508_C80	12 - 18	REDUCERECC SCH STD/20 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	20 - 24										
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES- BUTTWELD PWHT FOR SIZES 4" AND LARGER	CORR. ALLOW. 1 MM	DATE 24MAR17	BY SMS	REV B	FGP/WPMP SHEET 3 OF 6		PA 5	CLASS 150C80			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1250			Eccent	5392509_C80	20 - 20	REDUCERECC SCH STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24 - 24						
1260													
1270			Nipofl	5504121	3/4 - 1	NIPOFLANGE 150# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2 - 3						
1280			Nipofl	5504123	3/4 - 1	NIPOFLANGE 150# RF SCH STD/80S ALLOY 825 B564 UNS N08825	4 - 6						
1290			Nipofl	5504150	3/4 - 1	NIPOFLANGE 150# RF SCH 20/80S ALLOY 825 B564 UNS N08825	8 - 18						
1300			Nipofl	5504123	3/4 - 1	NIPOFLANGE 150# RF SCH STD/80S ALLOY 825 B564 UNS N08825	20 - 24						
1310			Nipofl	5504122	1 1/2 - 2	NIPOFLANGE 150# RF SCH 40S ALLOY 825 B564 UNS N08825	3 - 3						
1320			Nipofl	5504124	1 1/2 - 2	NIPOFLANGE 150# RF SCH STD/40S ALLOY 825 B564 UNS N08825	4 - 6						
1330			Nipofl	5504155	1 1/2 - 2	NIPOFLANGE 150# RF SCH 20/40S ALLOY 825 B564 UNS N08825	8 - 18						
1340			Nipofl	5504124	1 1/2 - 2	NIPOFLANGE 150# RF SCH STD/40S ALLOY 825 B564 UNS N08825	20 - 24						
1350													
1360		FS1	Nipofl	5513491	1 - 1	NIPOFLANGE 300# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	1 1/2 - 2						
1370		FS1	Nipofl	5513495	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40S ALLOY 825 B564 UNS N08825	2 - 3						
1380													
1390		UJ2	Nipofl	5513479	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 ALLOY 825 B564 UNS N08825	4 - 6						
1400		UJ2	Nipofl	5513420	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 ALLOY 825 B564 UNS N08825	8 - 18						
1410		UJ2	Nipofl	5513479	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 ALLOY 825 B564 UNS N08825	20 - 24						
1420													
1430		FF5	Endole	5504198	3/4 - 1	NIPOFLANGE FLAT BOTTOM 150# RF SCH 80S ALLOY 825 B564 UNS N08825							
1440		FF5	Endole	5504199	1 1/2 - 1 1/2	NIPOFLANGE FLAT BOTTOM 150# RF SCH 40S ALLOY 825 B564 UNS N08825							
1450		FF5	Endole	5504518	3/4 - 1	NIPOFLANGE FLAT BOTTOM 300# RF SCH 80S ALLOY 825 B564 UNS N08825							
1460		FF5	Endole	5504519	1 1/2 - 1 1/2	NIPOFLANGE FLAT BOTTOM 300# RF SCH 40S ALLOY 825 B564 UNS N08825							
1470													
1480			Weldol	5513437	3/4 - 1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	2 - 3						
1490			Weldol	5513417	3/4 - 1	WELDOLET SCH STD/80S ALLOY 825 B564 UNS N08825	4 - 6						
1500			Weldol	5513442	3/4 - 1	WELDOLET SCH 20/80S ALLOY 825 B564 UNS N08825	8 - 18						
1510			Weldol	5513417	3/4 - 1	WELDOLET SCH STD/80S ALLOY 825 B564 UNS N08825	20 - 24						
1520			Weldol	5513418	1 1/2 - 2	WELDOLET SCH STD/ 40S ALLOY 825 B564 UNS N08825	4 - 6						
1530			Weldol	5513441	1 1/2 - 3	WELDOLET SCH 20/ 40S ALLOY 825 B564 UNS N08825	8 - 18						
1540			Weldol	5513418	1 1/2 - 3	WELDOLET SCH STD/ 40S ALLOY 825 B564 UNS N08825	20 - 24						
1550			Weldol	5513440	4 - 6	WELDOLET SCH 20/ STD ALLOY 825 B564 UNS N08825	12 - 18						
1560			Weldol	5513439	4 - 6	WELDOLET SCH STD ALLOY 825 B564 UNS N08825	20 - 24						
1570			Weldol	5513444	8 - 8	WELDOLET SCH STD/ 20 ALLOY 825 B564 UNS N08825	24 - 24						
1580													
1590		UJ2	Elbole	5516626	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 ALLOY 825 B564 UNS N08825	4 - 6						
1600		UJ2	Elbole	5516629	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 ALLOY 825 B564 UNS N08825	8 - 18						
1610		UJ2	Elbole	5516626	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 ALLOY 825 B564 UNS N08825	20 - 24						
1620													
1630		FS1				*****PIPE FLANGES*****							
1640													
1650		FA8	Reduci	5614311	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 150# RF ALLOY 825 B564 UNS N08825	3/4 - 1 1/2						
1660													
1670		FA8	Reduci	5614310	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF ALLOY 825 B564 UNS N08825	3/4 - 1 1/2						
1680													
1690			Flange	5851200	1/2 - 3/4	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
1700			Flange	5851200	2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
1710			Flange	5851203	3/4 - 3/4	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
1720			Flange	5851209	1 - 1	FLANGE ADAPTER 150# RF 316 SS X 25 MM TUBE							
1730													
1740			Flange	5569030	1/2 - 1	FLG WN 150# RF ALLOY 825 SCH 80S B564 UNS N08825							
1750			Flange	5569026	1 1/2 - 3	FLG WN 150# RF ALLOY 825 SCH 40S B564 UNS N08825							
1760			Flange	5556261_C80	4 - 6	FLG WN 150# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE							
			Flange	5556265_C80	8 - 18	FLG WN 150# RF LTCS A350 LF2 SCH 20 WITH 3MM INC 825 CLAD FACE & BORE							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTTWELD PWHT FOR SIZES 4" AND LARGER	CORR. ALLOW. 1 MM						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1770	B		Flange	5556261_C80	20 - 24	FLG WN 150# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE									
1780			Flange	5556266_C80	48 - 48	FLG WN 150# RF LTCS A350 LF2 SCH XS B16.47 SERIES A WITH 3MM INC 825 CLAD FACE & BORE									
1790			Flange	5569290	1/2 - 1	FLG WN 300# RF ALLOY 825 SCH 80S B564 UNS N08825									
1800				5569276	1 1/2 - 3	FLG WN 300# RF ALLOY 825 SCH 40S B564 UNS N08825									
1810				Flange	5556473_C80	4 - 6							FLG WN 300# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE		
1820				Flange	5556474_C80	8 - 18							FLG WN 300# RF LTCS A350 LF2 SCH 20 WITH 3MM INC 825 CLAD FACE & BORE		
1830			Flange	5556473_C80	20 - 24	FLG WN 300# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE									
1840			B	Flange	5556477_C80	48 - 48							FLG WN 300# RF LTCS A350 LF2 SCH XS B16.47 SERIES A WITH 3MM INC 825 CLAD FACE & BORE		
1850				Flange	5570276	2 - 2							FLG WN 150Q# RF ALLOY 825 SCH 40S B564 UNS N08825		
1860					5557400	3 - 3							FLG WN 900# RF ALLOY 825 SCH 40S B564 UNS N08825		
1870	Blind	5617150			1/2 - 3	FLG BLIND 150# RF ALLOY 825 B564 UNS N08825									
1880		5615566_C80			4 - 24	FLG BLIND 150# RF LTCS A350 LF2 WITH 3MM INC 825 CLAD FACE									
1890	Blind	5617159		1/2 - 3	FLG BLIND 300# RF ALLOY 825 B564 UNS N08825										
1900		5615576_C80		4 - 24	FLG BLIND 300# RF LTCS A350 LF2 WITH 3 MM INC 825 CLAD FACE										
1910	810	FS1		Orific	5650830	2 - 3	WN ORIF SET 300# RF ALLOY 825 B564 UNS N08825 SCH 40S								
1920					5623944_C80	4 - 6	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE								
1930					Orific	5623941_C80	8 - 18	WN ORIF SET 300# RF LTCS A350 LF2 SCH 20 WITH 3MM INC 825 CLAD FACE & BORE							
1940			5623944_C80			20 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE								
1950			***** PIPE NIPPLES *****												
1960			NIPPLE PBE CUT FROM PIPE												
1970			***** GASKETS *****												
1980			B		GASFRS	5673653	1/2 - 24	GASKET 150# ALLOY 825 SPIRAL WOUND GRAPHITE IR							
1990						5674016	48 - 48	GASKET 150# ALLOY 825 KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2000			B		GASFRS	5673657	1/2 - 24	GASKET 300# ALLOY 825 SPIRAL WOUND GRAPHITE IR							
2010	5674034	48 - 48		GASKET 300# ALLOY 825 KAMMPROFILE GRAPHITE ASME B16.47 SERIES A											
2020	B	GASFRS	5673800	2 - 2	GASKET 1500# ALLOY 825 SPIRAL WOUND GRAPHITE IR										
2030			5673700	3 - 3	GASKET 900# ALLOY 825 SPIRAL WOUND GRAPHITE IR										
2040	B1			***** BOLTING *****											
2050				SB6	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
2060															
2070															
2080															
2090															
2100	CPSBOL	5675675	12 - 48	JACK SCREW HEX HEAD A320 GR L7M											
2110	FGP / WPMP					CONSTRUCTION					REFERENCE / LARGE END SIZES				
2120	TENGIZCHEVROIIL - TENGIZCHEVROIIL					ALL SIZES- BUTTWELD					CORR. ALLOW.				
2130	TCO					PWHT FOR SIZES 4" AND LARGER					DATE				
2140	TENGIZ, KAZAKHSTAN										BY				
2150											REV				
2160											SHEET				
2170											PA				
2180											CLASS				
2190											1 MM				
2200											24MAR17				
2210											SMS				
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											5 OF 6				
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											150C80				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
2220	B	FS1	Specta	5749681 5749580_C80	3/4 - 3 4 - 14	**** BLINDS & SPACERS **** SPECTACLE BLIND 150# RF ALLOY 825 B424 UNS N08825 SPECTACLE BLIND 150# RF LTCS A 516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE SPECTACLE BLIND 300# RF ALLOY 825 B424 UNS N08825 SPECTACLE BLIND 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE PADDLE BLANK 150# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE PADDLE BLANK 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A WITH 3MM INC 825 CLAD FACE PADDLE SPACER 150# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE PADDLE SPACER 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A WITH 3MM INC 825 CLAD FACE & BORE ***** BRANCHES ***** USE BRANCH CHART 150C80 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE FOR FABRICATION AND WELDING DETAILS SEE DOC. № L-ST-2024	3 - 48 3 - 48	SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 GS, PHC, WSR 				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10						***** VALVES FOR CHEMICAL LINES *****		032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.						
20								814	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.						
30	C	VC3	Check	C01AP76	3/4 - 8	CHECK 150# RF LTCS VERTICAL BALL TYPE PTFE LINED		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
40	C	VC5	Check	C01AP77	3/4 - 4	CHECK 150# RF LTCS SPRUNG HORIZONTAL BALL TYPE PTFE LINED		GP1	USE GASKET ONLY WHEN MATING TO A METAL FLANGE SURFACE.						
50								GP2	FOR USE BETWEEN PTFE (RF) AND RUBBER LINED (RF) CONNECTIONS						
60	C		Ball V	B01AP39	3/4 - 2	BALL 150# RF LTCS PTFE LINED FLOATING SOFT FB LO									
70	E		Ball V	B01AP49	3 - 6	BALL 150# RF LTCS PTFE LINED FLOATING SOFT RB LO									
80	E		Ball V	B01AP50	8 - 8	BALL 150# RF LTCS PTFE LINED FLOATING SOFT RB GO		GP3	FOR USE BETWEEN PTFE (RF) AND RUBBER LINED (FF) CONNECTIONS.						
90								PM6	THIS PIPE IS SUPPLIED CUT TO LENGTH WITH INTEGRAL FLANGES AND LINING. SPECIFY LENGTH OF EACH PIECE.						
100						***** VALVES FOR WATER LINES *****		SB7	SPECTACLE BLIND MUST BE INSTALLED AGAINST FIXED END OF PIPE SPOOL PIECE.						
110								VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.						
120	C	ZA4	Gate V	G01AB47	3/4 - 8	GATE 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2 HW		VC3	INSTALL THESE CHECK VALVES IN THE VERTICAL POSITION WITH FLOW UPWARD ONLY.						
130								VC5	SPRUNG CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.						
140	C	ZA4	Butter	UA1B2	10 - 24	BUTTERFLY 150# RF DOUBLE FLANGED TYPE TRIPLE OFFSET DESIGN FIRE SAFE GEAR OP BS 1400 GR.AB2		VU9	USE 1/2" NB DUPLEX SS FLANGE, NIPPLES AND BALL VALVE FOR ORIFICE TAP BLOCK VALVES ONLY.						
150								ZA4	COMPONENT SHALL ONLY BE USED ON WATER SERVICES (WRA, WP, WE)						
160	C	ZA4	Globe	L01AB47	3/4 - 6	GLOBE 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2 HW		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
170	C	ZA4	Globe	L01AB58	8 - 8	GLOBE 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2 GO									
180															
190		VC2	Check	C01AB78	2 - 6	CHECK DUAL WAFER LUG 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2 THROU BOLTED									
200		VC2	Check	C01AB59	8 - 24	CHECK DUAL DBL FLG 150# RF ALUMINIUM BRONZE BS 1400 GR. AB2									
210		ZA4				***** PIPE *****									
220															
230															
240	D	PM6	Piping	5717206	3/4 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 PTFE LINED 150# FLANGED ENDS									
250															
260		PM6	Piping	5717205	3 - 10	PIPE SCH 40 SMLS LTCS A333 GR.6 PTFE LINED 150# FLANGED ENDS									
270															
280	C		Piping	5717204	12 - 24	PIPE SCH 20 SMLS LTCS A333 GR.6 PTFE LINED 150# FLANGED ENDS									
290															
300	D	814	Piping	5366119	1/2 - 1/2	PIPE SCH 160 SMLS DUPLEX STL S31803									
310		VU9				***** FITTINGS *****									
320															
330															
340	C		90 Deg	5717620	3/4 - 10	ELL 90 DEG 150# FLANGED LTCS PTFE LINED									
350	C		90 Deg	5717630	12 - 24	ELL 90 DEG 150# FLANGED SCH 20 LTCS PTFE LINED									
360															
370	C		45 Deg	5717621	3/4 - 10	ELL 45 DEG 150# FLANGED LTCS PTFE LINED									
380	C		45 Deg	5717631	12 - 24	ELL 45 DEG 150# FLANGED SCH 20 LTCS PTFE LINED									
390															
400	C		Tee	5717622	3/4 - 10	TEE EQUAL 150# FLANGED LTCS PTFE LINED									
410	C		Tee	5717632	12 - 24	TEE EQUAL 150# FLANGED SCH 20 LTCS PTFE LINED									
420															
430	C		Reduci	5717623	3/4 - 8	TEE REDUCING 150# FLANGED LTCS PTFE LINED	1 - 10								
440	C		Reduci	5717636	4 - 10	TEE REDUCING 150# FLANGED SCH 20 X SCH 40 LTCS PTFE LINED	12 - 24								
450															
460	C		Reduci	5717633	12 - 20	TEE REDUCING 150# FLANGED SCH 20 LTCS PTFE LINED	14 - 24								
470															
480	C		Tee	5717627	3/4 - 2	INSTRUMENT TEE 150# FLANGED LTCS PTFE LINED	1 - 10								
490	C		Tee	5717637	3/4 - 2	INSTRUMENT TEE 150# FLANGED SCH 20 X SCH XS LTCS PTFE LINED	12 - 24								
500															
510	C		Concen	5717624	3/4 - 8	REDUCER CONCENTRIC 150# FLANGED LTCS PTFE LINED	1 - 10								
520	C		Concen	5717638	6 - 10	REDUCER CONCENTRIC 150# SCH 20 X SCH 40 FLANGED LTCS PTFE LINED	12 - 12								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION LTCS PIPING + PTFE LINED. ALL SIZES-FLANGED.		CORR. ALLOW. 0 MM	DATE 17MAR17	BY KD	REV D	SHEET 1 OF 2		PA 5	CLASS 150CP2

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
530	C		Concen	5717634	12 - 20	REDUCER CONCENTRIC 150# SCH 20 FLANGED LTCS PTFE LINED	14 - 24						
540													
550	C		Ecccent	5717625	3/4 - 8	REDUCER ECCENTRIC 150# FLANGED LTCS PTFE LINED	1 - 10						
560	C		Ecccent	5717639	8 - 10	REDUCER ECCENTRIC 150# SCH 20 X SCH 40 FLANGED LTCS PTFE LINED	12 - 12						
570													
580	C		Ecccent	5717635	12 - 20	REDUCER ECCENTRIC 150# SCH 20 FLANGED LTCS PTFE LINED	14 - 24						
590													
600						***** PIPE FLANGES *****							
610													
620	C		Reduci	5717628	1/2 - 20	REDUCING FLG 150# PTFE LINED LTCS A350 LF2	3/4 - 24						
630	C		Blind	5715115	1/2 - 24	FLG BLIND 150# PTFE LINED LTCS A350 LF2							
640	C		Blind	5715130	1/2 - 1/2	FLG BLIND 300# PTFE LINED LTCS A350 LF2							
650													
660						***** SPACERS *****							
670													
680	C	GP2	Paddle	5717618	3/4 - 12	STANDARD RING SPACER 150# PTFE							
690	C	GP3	Paddle	5717619	3/4 - 12	STANDARD FULL FACE SPACER 150# PTFE							
700													
710						***** GASKETS *****							
720													
730	C	GP1	GASFR	5664201	1/2 - 24	GASKET 150# RF FLAT RING PTFE							
740	C	GP1	GASFR	5664204	1/2 - 1/2	GASKET 300# RF FLAT RING PTFE							
750													
760		B1				***** BOLTING *****							
770													
780	B		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
790													
800						***** SPECTACLE BLINDS *****							
810													
820		SB7	Specta	5749585	1 - 12	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 PTFE LINED AND FACED							
830													
840			Paddle	5750024	16 - 16	PADDLE BLANK 150# RF LTCS A516 GR.70 PTFE LINED AND FACED							
850													
860			Paddle	5750243	16 - 16	PADDLE SPACER 150# RF LTCS A516 GR.70 PTFE LINED AND FACED							
870													
880						*****PIPE NIPPLES*****							
890													
900	D	814 VU9				NIPPLES PBE TO BE MADE FROM PIPE							
910													
920													
930	D	VU9	Nipple	5542170	1/2 - 1/2	NIPPLE SCH 80S A790 UNS S31803 POE/TOE 3" LONG							
940	D	VU9	Nipple	5542172	1/2 - 1/2	NIPPLE SCH 80S A790 UNS S31803 TOE/POE 7 IN LONG							
950													
960													
970						***** BRANCHES *****							
980													
990						USE BRANCH CHART 150CP2							
1000													
1010						*****CONSTRUCTION DETAILS*****							
1020													
1030		ZA6				SEAL WELDING-----FOR THREADED CONNECTIONS ONLY							
1040						PIPE BENDS-----NONE							
1050						THREAD SEALANT-----NONE							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP						
						LTCs PIPING + PTFE LINED. ALL SIZES-FLANGED.	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
						0 MM	17MAR17	KD	D	2 OF 2	5	150CP2	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10		PP9						810	NIPPLE TO BE CUT FROM SCH 160 PIPE						
20								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
30								FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.						
40		VB2				*****BALL VALVES*****		FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.						
50	B		Ball V	A01AX17	3 - 4	BALL 150# RF SUPER DUPLEX F55 FLOATING RB SOFT LO		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
50	B		Ball V	A01AX34	6 - 6	BALL 150# RF SUPER DUPLEX F55 FLOATING RB SOFT LO		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
60	B		Ball V	A01AX37	8 - 8	BALL 150# RF SUPER DUPLEX F55 TRUNNION RB SOFT GO		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.						
70								VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.						
80		VA5	Ball V	B01AX17	1/2 - 2	BALL 150# RF SUPER DUPLEX F55 FLOATING FB SOFT LO		VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.						
90		VA5	Ball V	B01AX17	3 - 4	BALL 150# RF SUPER DUPLEX F55 FLOATING FB SOFT LO		VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.						
100	B	VA5	Ball V	B01AX34	6 - 6	BALL 150# RF SUPER DUPLEX F55 FLOATING FB SOFT LO		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.						
110	B	VA5	Ball V	B01AX37	8 - 8	BALL 150# RF SUPER DUPLEX F55 TRUNNION FB SOFT GO									
120															
130		FS1	Ball V	B03AX17	1/2 - 2	BALL 300# RF SUPER DUPLEX F55 FLOATING FB SOFT LO									
140		FS1	Ball V	A03AX17	3 - 3	BALL 300# RF SUPER DUPLEX F55 FLOATING RB SOFT LO									
150						*****BUTTERFLY VALVE*****									
160	B														
170			Butter	T01AX67	10 - 24	BUTTERFLY 150# RF SUPER DUPLEX A6 TOV METAL DBL FLG SHORT PAT GO									
180						*****GLOBE VALVES*****									
190															
200															
210															
220			Globe	L01AX84	3/4 - 1 1/2	GLOBE 150# RF SUPER DUPLEX F55 OS&Y HW									
230			Globe	L01AX85	2 - 6	GLOBE 150# RF SUPER DUPLEX A6 OS&Y HW									
240			Globe	L01AX83	8 - 10	GLOBE 150# RF SUPER DUPLEX A6 OS&Y GO									
250															
260						*****CHECK VALVES*****									
270															
280															
290		VC1	Check	C01AX86	3/4 - 1 1/2	CHECK PISTON 150# RF SUPER DUPLEX F55									
300		VC2	Check	C01AX87	2 - 6	CHECK DUAL WAFER LUG 150# RF SUPER DUPLEX A6 THROU BOLTED									
310		VC2	Check	C01AX88	8 - 24	CHECK DUAL DBL FLG 150# RF SUPER DUPLEX A6									
320															
330						*****PIPES*****									
340															
350															
360			Piping	5366231	3/4 - 1	PIPE SCH 80S SMLS SUPER DUPLEX S32760 PE									
370			Piping	5366232	1 1/2 - 1 1/2	PIPE SCH 40S SMLS SUPER DUPLEX S32760 PE		WP							
380			Piping	5366222	2 - 6	PIPE SCH 40S SMLS SUPER DUPLEX S32760 BE		DESIGN: ASME B16.5 MG 2.8							
390	B		Piping	5366223	8 - 8	PIPE SCH 10S SMLS SUPER DUPLEX S32760 BE		CODE: ASME B31.3							
400	B		Piping	5366224	10 - 24	PIPE SCH 10S EFW SUPER DUPLEX S32760-WX BE									
410	B														
420		810	Piping	5366220	1/2 - 1/2	PIPE SCH 160 SMLS SUPER DUPLEX S32760 BE									
430						***** FITTINGS *****									
440															
450															
460			90 Deg	5461101	3/4 - 1	ELL 90 DEG LR SCH 80S SUPER DUPLEX S32760		MATERIAL:	SUPER DUPLEX FLANGE						
470			90 Deg	5461102	1 1/2 - 6	ELL 90 DEG LR SCH 40S SUPER DUPLEX S32760		FLANGE RATING:	150 RF						
480			90 Deg	5461103	8 - 8	ELL 90 DEG LR SCH 10S SUPER DUPLEX S32760		TEMPERATURE RANGE:	-46 / 100 °C						
490			90 Deg	5461104	10 - 24	ELL 90 DEG LR SCH 10S EFW SUPER DUPLEX S32760-W									
500															
510			45 Deg	5461111	3/4 - 1	ELL 45 DEG LR SCH 80S SUPER DUPLEX S32760									
520			45 Deg	5461112	1 1/2 - 6	ELL 45 DEG LR SCH 40S SUPER DUPLEX S32760									
530			45 Deg	5461113	8 - 8	ELL 45 DEG LR SCH 10S SUPER DUPLEX S32760									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP							
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS		
							1 MM	17MAR17	JL	B	1 OF 4	5	150D01		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
560	B		45 Deg	5461114	10 - 24	ELL 45 DEG LR SCH 10S EFW SUPER DUPLEX S32760-W		SERVICE WP DESIGN: ASME B16.5 MG 2.8 CODE: ASME B31.3 MATERIAL: SUPER DUPLEX FLANGE FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 100 °C						
570			Tee	5461121	3/4 - 1	TEE BW SCH 80S SUPER DUPLEX S32760								
580			Tee	5461122	1 1/2 - 6	TEE BW SCH 40S SUPER DUPLEX S32760								
590			Tee	5461123	8 - 8	TEE BW SCH 10S SUPER DUPLEX S32760								
600			Tee	5461124	10 - 24	TEE BW SCH 10S EFW SUPER DUPLEX S32760-W								
610														
620														
630			Reduci	5461131	3/4 - 3/4	TEE RED SCH 80S SUPER DUPLEX S32760	1 - 1							
640			Reduci	5461132	3/4 - 1	TEE RED SCH 40S/80S SUPER DUPLEX S32760	1 1/2 - 2							
650			Reduci	5461133	1 1/2 - 4	TEE RED SCH 40S SUPER DUPLEX S32760	2 - 6							
660			Reduci	5461130	4 - 6	TEE RED SCH 10S/40S SUPER DUPLEX S32760	8 - 8							
670			Reduci	5461134	4 - 6	TEE RED SCH 10S/40S EFW SUPER DUPLEX S32760-W	10 - 16							
680			Reduci	5461135	8 - 20	TEE RED SCH 10S EFW SUPER DUPLEX S32760-W	10 - 24							
690														
700			Cap	5461343	3/4 - 1	CAP SCH 80S SUPER DUPLEX S32760								
710			Cap	5461283	1 1/2 - 6	CAP SCH 40S SUPER DUPLEX S32760								
720			Cap	5461233	8 - 24	CAP SCH 10S SUPER DUPLEX S32760								
730														
740			Concen	5461151	3/4 - 3/4	REDUCER CONC SCH 80S SUPER DUPLEX S32760	1 - 1							
750			Concen	5461152	3/4 - 1	REDUCER CONC SCH 40S/80S SUPER DUPLEX S32760	1 1/2 - 2							
760			Concen	5461153	1 1/2 - 4	REDUCER CONC SCH 40S SUPER DUPLEX S32760	2 - 6							
770			Concen	5461150	4 - 6	REDUCER CONC SCH 10S/40S SUPER DUPLEX S32760	8 - 8							
780			Concen	5461154	4 - 6	REDUCER CONC SCH 10S/40S EFW SUPER DUPLEX S32760-W	10 - 14							
790			Concen	5461155	8 - 20	REDUCER CONC SCH 10S EFW SUPER DUPLEX S32760-W	10 - 24							
800														
810			Ecccent	5461161	3/4 - 3/4	REDUCER ECC SCH 80S SUPER DUPLEX S32760	1 - 1							
820			Ecccent	5461162	3/4 - 1	REDUCER ECC SCH 40S/80S SUPER DUPLEX S32760	1 1/2 - 2							
830			Ecccent	5461163	1 1/2 - 4	REDUCER ECC SCH 40S SUPER DUPLEX S32760	2 - 6							
840			Ecccent	5461160	4 - 6	REDUCER ECC SCH 10S/40S SUPER DUPLEX S32760	8 - 8							
850			Ecccent	5461164	4 - 6	REDUCER ECC SCH 10S/40S EFW SUPER DUPLEX S32760-W	10 - 14							
860			Ecccent	5461165	8 - 20	REDUCER ECC SCH 10S EFW SUPER DUPLEX S32760-W	10 - 24							
870														
880							*****BRANCH FITTINGS*****							
890														
900			Nipofl	5512135	3/4 - 1	NIPOFLANGE 150# RF SCH 40S/80S SUPER DUPLEX F55	2 - 6							
910			Nipofl	5512136	3/4 - 1	NIPOFLANGE 150# RF SCH 10S/80S SUPER DUPLEX F55	8 - 24							
920			Nipofl	5512137	1 1/2 - 1 1/2	NIPOFLANGE 150# RF SCH 40S SUPER DUPLEX F55	2 - 6							
930			Nipofl	5512138	1 1/2 - 1 1/2	NIPOFLANGE 150# RF SCH 10S/40S SUPER DUPLEX F55	8 - 24							
940														
950			FS1	Flange	5512145	1 - 1	NIPOFLANGE 300# RF SCH 40S/80S SUPER DUPLEX F55							1 1/2 - 2
960			FS1	Flange	5512146	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40S SUPER DUPLEX F55							2 - 3
970														
980			UJ2	Nipofl	5512147	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/160 SUPER DUPLEX F55							4 - 6
990			UJ2	Nipofl	5512148	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 10S/160 SUPER DUPLEX F55							8 - 24
1000														
1010			Nipofl	5512139	3/4 - 3/4	NIPOFLANGE 300# RF SCH 10S/80S SUPER DUPLEX F55	16 - 16							
1020														
1030			Weldol	5512124	3/4 - 1	WELDOLET SCH 40S/80S SUPER DUPLEX F55	3 - 6							
1040			Weldol	5512123	3/4 - 1	WELDOLET SCH 10S/80S SUPER DUPLEX F55	8 - 24							
1050			Weldol	5512126	1 1/2 - 2	WELDOLET SCH 40S SUPER DUPLEX F55	4 - 6							
1060			Weldol	5512125	1 1/2 - 2	WELDOLET SCH 10S/40S SUPER DUPLEX F55	12 - 12							
1070			Weldol	5512125	2 - 2	WELDOLET SCH 10S/40S SUPER DUPLEX F55	16 - 16							
1080			Weldol	5512127	8 - 8	WELDOLET SCH 10S SUPER DUPLEX F55	24 - 24							
1090														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY JL	REV B	FGP/WPMP SHEET 2 OF 4		PA 5	CLASS 150D01

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1100	B	UJ2	Elbole	5516164	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40S/160 SUPER DUPLEX F55	4 - 6						
1110		UJ2	Elbole	5516165	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 10S/160 SUPER DUPLEX F55	8 - 24						
1120		FS1				*****PIPE FLANGES*****							
1130													
1140		FA8	Reduci	5600671	1/2 - 1/2	FLG SCRCD (FNPT) REDUCING 150# RF SUPER DUPLEX F55	3/4 - 1 1/2						
1150													
1160		B	Flange	5851220	1/2 - 3/4	FLANGE ADAPTER 150# RF SUPER DUPLEX F55 X 12 MM TUBE							
1170			Flange	5851220	2 - 2	FLANGE ADAPTER 150# RF SUPER DUPLEX F55 X 12 MM TUBE							
1180		B	Flange	5590416	3/4 - 1	FLG WN 150# RF SUPER DUPLEX 80S F55							
1190			Flange	5590412	1 1/2 - 6	FLG WN 150# RF SUPER DUPLEX 40S F55							
1200		B	Flange	5590408	8 - 24	FLG WN 150# RF SUPER DUPLEX 10S F55							
1210													
1220		B	Flange	5590441	3/4 - 1	FLG WN 300# RF SUPER DUPLEX 80S F55							
1230			Flange	5590437	1 1/2 - 6	FLG WN 300# RF SUPER DUPLEX 40S F55							
1240		B	Flange	5590433	8 - 24	FLG WN 300# RF SUPER DUPLEX 10S F55							
1250													
1260		B	Flange	5590476	8 - 8	FLG WN 900# RF SUPER DUPLEX 10S F55							
1270													
1280		810	Blind	5616856	1/2 - 24	FLG BLIND 150# RF SUPER DUPLEX F55							
1290			Blind	5616863	1/2 - 1/2	FLG BLIND 300# RF SUPER DUPLEX F55							
1300		B1				*****ORIFICE FLANGES*****							
1310													
1320		FS1	Orific	5650851	2 - 6	WN ORIF SET 300# RF SUPER DUPLEX 40S F55							
1330			Orific	5650852	8 - 24	WN ORIF SET 300# RF SUPER DUPLEX 10S F55							
1340		B				***** PIPE NIPPLES *****							
1350						NIPPLE PBE CUT FROM PIPE							
1360		SB6				***** GASKETS *****							
1370													
1380		FS1	GASFRS	5673377	1/2 - 24	GASKET 150# DUPLEX 220S SP WND GRAPHITE IR							
1390			GASFRS	5673381	1/2 - 24	GASKET 300# DUPLEX 220S SP WND GRAPHITE IR							
1400		B	GASFRS	5673389	8 - 8	GASKET 900# DUPLEX 220S SP WND GRAPHITE IR							
1410													
1420		B1				***** BOLTING *****							
1430													
1440		SB6	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1450			CPSBOL	5675675	12 - 24	JACK SCREW HEX HEAD A320 GR L7M							
1460		FS1				**** BLINDS & SPACERS ****							
1470													
1480		B	Specta	5749836	3/4 - 14	SPECTACLE BLIND 150# RF SUPER DUPLEX STL S32760							
1490			Specta	5749837	1 - 10	SPECTACLE BLIND 300# RF SUPER DUPLEX STL S32760							
1500		B1	Paddle	5750106	16 - 24	PADDLE BLANK 150# RF SUPER DUPLEX STL S32760							
1510			Paddle	5750107	12 - 24	PADDLE BLANK 300# RF SUPER DUPLEX STL S32760							
1520		SB6											
1530													
1540		FS1											
1550													
1560		B											
1570													
1580		B1											
1590													
1600		SB6											
1610													
1620		FS1											
1630													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1640 1650 1660 1670 1680 1690 1700 1710 1720 1730 1740 1750 1760 1770 1780	B B B		Reinfo Reinfo	RWELD RWELD	2 - 16 2 - 16	***** BRANCHES ***** USE BRANCH CHART 150H21 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE FOR FABRICATION AND WELDING DETAILS SEE DOC. № L-ST-2024	3 - 24 3 - 24						
								SERVICE					
								WP DESIGN: ASME B16.5 MG 2.8 CODE: ASME B31.3					
								MATERIAL: SUPER DUPLEX FLANGE FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 100 °C					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP					
						ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY JL	REV B	SHEET 4 OF 4	PA 5	CLASS 150D01

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10		PP9				*****VALVES*****		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. NIPPLE TO BE CUT FROM SCH 160 PIPE B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FA4 SLIP ON FLANGE SHALL ONLY BE USED WHERE INSTRUMENT ENCLOSURES ARE INSTALLED. FA8 USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. FF5 FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY. FS1 USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. NC1 USE SW FULL COUPLING FOR LINE COUPLING ONLY. USE SOCKOLET OR SOCKETWELD TEE FOR SOCKETWELD BRANCH CONNECTIONS. SEE BRANCH CHART. PG1 USE FOR PRESSURE GAUGES WITH CHEMICAL SEALS. PJ9 FOR USE IN NON-JACKETED SULPHUR LINES ONLY. PP9 ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 SB6 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.				
20													
30		VB2				*****BALL VALVES*****							
40													
50			Ball V	A01AC01	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO							
60													
70			Ball V	A01AC02	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO							
80													
90			Ball V	A01AC03	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO							
100													
110			Ball V	A01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO							
120													
130			Ball V	A01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO							
140													
150			Ball V	A01AC08	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO							
160													
170													
180													
190	D		Ball V	B01AC01	1/2 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
200													
210	D		Ball V	B01AC02	1/2 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
220													
230		VA5	Ball V	B01AC01	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
240													
250		VA5	Ball V	B01AC02	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
260													
270	C	VA5	Ball V	B01AC03	4 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
280													
290	D	VA5	Ball V	B01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
300													
310	D	VA5	Ball V	B01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO							
320													
330	C	VA5	Ball V	B01AC07	10 - 24	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO							
340													
350	D	VA5	Ball V	B01AC08	8 - 24	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO							
360													
370		VA5	Ball V	B01JC07	26 - 48	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A							
380													
390		VA5	Ball V	B01JC08	26 - 48	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A							
400													
410													
420	C	VA5	Ball V	B01AC96	3/4 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO 2800MM EXT							
430													
440	C	VA5	Ball V	A01AC96	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO 2800MM EXT							
450													
460													
470	D	VA5	Ball V	B03AC01	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
480													
490	D	VA5	Ball V	B03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
500													
510	D	VA5	Ball V	B03AC02	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
520													
530	C	VA5	Ball V	A03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 1 OF 6 PA 5 CLASS 150H21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
540	C	VA5	Ball V	A03AC02	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY. FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. FOR USE IN SULPHUR SERVICE ONLY. USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
550						*****BUTTERFLY VALVE*****		VA5							
560								VAB							
570			Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO		VAF							
580			Butter	T01JC61	26 - 48	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A		VAG							
590						*****GATE VALVES*****		VB2							
600								VC1							
610			Gate V	G01AC48	3/4 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW									
620			Gate V	G01AC41	2 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW									
630	D	VA5	Gate V	G01AC41	10 - 10	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW									
640	D	VA5	Gate V	G01AC42	12 - 24	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE GO		VC2							
650															
660															
670															
680	D		Gate V	G03AC41	3 - 4	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW		VJ2							
690						*****THROUGH CONDUIT GATE VALVES *****		VU2							
700								ZA6							
710															
720	D	VA5	Condui	W01AC70	10 - 10	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW									
730															
740	D	VA5	Condui	W01AC71	12 - 24	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO									
750						*****GLOBE VALVES*****									
760															
770															
780			Globe	L01AC48	3/4 - 1 1/2	GLOBE 150# RF LTCS A350 LF2 TRIM 16 OS&Y HW									
790			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW									
800			Globe	L01AC42	8 - 10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO									
810						*****CHECK VALVES*****									
820															
830															
840	D	VC1	Check	C01AC50	3/4 - 1 1/2	CHECK PISTON 150# RF LTCS A350 LF2 TRIM 16									
850		VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED									
860		VC2	Check	C01AC52	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16									
870		VC2	Check	C01JC52	26 - 48	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A									
880						*****PIPES*****									
890															
900															
910	D		Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE									
920			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE									
930			Piping	5358505	3 - 14	PIPE SCH STD SMLS LTCS A333 GR.6 BE									
940			Piping	5358501	16 - 16	PIPE SCH 20 SMLS LTCS A333 GR.6 BE									
950			Piping	5358597	18 - 24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE									
960	C		Piping	5358576	26 - 38	PIPE SCH STD EFW LTCS A671 CC60 CL.22 BE									
970	C		Piping	5358577	40 - 48	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE									
980															
990		810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE									
1000															
1010						***** FITTINGS *****									
1020															
1030	D		90 Deg	5392600	1/2 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 1.5 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 SMS D 2 OF 6 5 150H21							

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040			90 Deg	5392438	3 - 14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6							
1050			90 Deg	5392276	16 - 16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6							
1060			90 Deg	5392280	18 - 24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W							
1070			90 DEG	5392241	24 - 24	ELL BW 90 DEG SR SCH 20 LTCS A420 WPL6-W							
1080	C		90 Deg	5392496	26 - 38	ELL 90 DEG LR SCH STD LTCS A420 WPL6-W							
1090	C		90 Deg	5392497	40 - 48	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W							
1100													
1110	C		45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1120			45 Deg	5392444	3 - 14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6							
1130			45 Deg	5392282	16 - 16	ELL BW 45 DEG SCH 20 LTCS A420 WPL6							
1140			45 Deg	5392281	18 - 24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W							
1150	C		45 Deg	5392607	26 - 38	ELL BW 45 DEG SCH STD LTCS A420 WPL6-W							
1160	C		45 Deg	5392608	40 - 48	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W							
1170													
1180	D		Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6							
1190			Tee	5392447	3 - 14	TEE BW SCH STD LTCS A420 WPL6							
1200			Tee	5392285	16 - 16	TEE BW SCH 20 LTCS A420 WPL6							
1210			Tee	5392295	18 - 24	TEE BW SCH 20 LTCS A420 WPL6-W							
1220	C		Tee	5392405	26 - 38	TEE SCH STD LTCS A420 WPL6-W							
1230	C		Tee	5392406	40 - 48	TEE BW SCH XS LTCS A420 WPL6-W							
1240													
1250	D		Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2						
1260			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 4						
1270			Reduci	5392445	3 - 12	TEE RED BW SCH STD LTCS A420 WPL6	4 - 14						
1280			Reduci	5392289	6 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6	16 - 16						
1290			Reduci	5392296	8 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6-W	18 - 24						
1300			Reduci	5392428	14 - 14	TEE RED BW SCH STD LTCS A420 WPL6-W	26 - 32						
1310			Reduci	5392292	16 - 20	TEE RED BW SCH 20 LTCS A420 WPL6-W	18 - 24						
1320	C		Reduci	5392320	16 - 24	TEE RED SCH STD/20 LTCS A420 WPL6-W	26 - 38						
1330	C		Reduci	5392475	18 - 24	TEE RED BW SCH XS/20 LTCS A420 WPL6-W	40 - 40						
1340	C		Reduci	5392428	26 - 36	TEE RED BW SCH STD LTCS A420 WPL6-W	28 - 38						
1350	C		Reduci	5392476	36 - 38	TEE RED SCH XS/STD LTCS A420 WPL6-W	40 - 42						
1360	C		Reduci	5392413	40 - 46	TEE RED SCH XS LTCS A420 WPL6-W	44 - 48						
1370													
1380	C		Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6							
1390			Cap	5392448	3 - 14	CAP BW SCH STD LTCS A420 WPL6							
1400			Cap	5392286	16 - 24	CAP BW SCH 20 LTCS A420 WPL6							
1410	C		Cap	5392448	26 - 38	CAP BW SCH STD LTCS A420 WPL6							
1420	C		Cap	5392610	40 - 48	CAP BW SCH XS LTCS A420 WPL6							
1430													
1440	C		Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2						
1450	C		Concen	5392452	1 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4						
1460			Concen	5392449	3 - 12	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 14						
1470			Concen	5392290	8 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6	16 - 16						
1480			Concen	5392297	10 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18 - 20						
1490			Concen	5392287	16 - 20	REDUCER CONC SCH 20 LTCS A420 WPL6-W	18 - 24						
1500			Concen	5392500	18 - 24	REDUCER CONC SCH STD/20 LTCS A420 WPL6-W	26 - 36						
1510	C		Concen	5392505	26 - 36	REDUCER CONC SCH STD LTCS A420 WPL6-W	28 - 38						
1520	C		Concen	5392506	30 - 38	REDUCER CONC SCH XS/STD LTCS A420 WPL6-W	40 - 48						
1530	C		Concen	5392507	40 - 46	REDUCER CONC SCH XS LTCS A420 WPL6-W	42 - 48						
1540													
1550	C		Eccent	5392461	1/2 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	3/4 - 2						
1560			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4						
1570			Eccent	5392450	3 - 12	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 14						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 1.5 MM						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1580			Eccent	5392291	8 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6	16 - 16						
1590			Eccent	5392298	10 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18 - 20						
1600			Eccent	5392288	16 - 20	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18 - 24						
1610			Eccent	5392508	18 - 24	REDUCER ECC SCH STD/20 LTCS A420 WPL6-W	26 - 36						
1620	C		Eccent	5392509	26 - 36	REDUCER ECC SCH STD LTCS A420 WPL6-W	28 - 38						
1630	C		Eccent	5392510	30 - 38	REDUCER ECC SCH XS/STD LTCS A420 WPL6-W	40 - 48						
1640	C		Eccent	5392511	40 - 46	REDUCER ECC SCH XS LTCS A420 WPL6-W	42 - 48						
1650													
1660			Nipofl	5504001	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	2 - 2						
1670			Nipofl	5504002	3/4 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	3 - 14						
1680			Nipofl	5504016	3/4 - 2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2	16 - 24						
1690	C		Nipofl	5504002	3/4 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	26 - 38						
1700	C		Nipofl	5504001	3/4 - 2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	40 - 48						
1710													
1720	C		Nipofl	5504020	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	1 1/2 - 2						
1730	C		Nipofl	5504021	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 6						
1740	D		Nipofl	5504021	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	8 - 14						
1750	D		Nipofl	5504017	3/4 - 3/4	NIPOFLANGE 300# RF SCH 20/XS LTCS A350 LF2	16 - 24						
1760	D		Nipofl	5504021	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	36 - 38						
1770													
1780		UJ2	Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14						
1790		UJ2	Nipofl	5504019	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24						
1800	C	UJ2	Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	26 - 38						
1810	C	UJ2	Nipofl	5504023	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	40 - 48						
1820													
1830	C	FF5	Endole	5504500	3/4 - 1 1/2	NIPOFLANGE FLAT BOTTOM 150# RF SCH XS LTCS A350 LF2							
1840													
1850	C		Weldol	5505335	3/4 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2						
1860	C		Weldol	5505179	3/4 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 14						
1870	C		Weldol	5505572	3/4 - 2	WELDOLET SCH 20/XS LTCS A350 LF2	16 - 24						
1880	C		Weldol	5505179	3/4 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	26 - 38						
1890	C		Weldol	5505333	3/4 - 2	WELDOLET SCH XS LTCS A350 LF2	40 - 48						
1900			Weldol	5505164	3 - 4	WELDOLET SCH STD LTCS A350 LF2	8 - 14						
1910			Weldol	5505571	3 - 8	WELDOLET SCH 20/STD LTCS A350 LF2	16 - 24						
1920	C		Weldol	5505164	3 - 12	WELDOLET SCH STD LTCS A350 LF2	26 - 38						
1930	C		Weldol	5505334	3 - 12	WELDOLET SCH XS/STD LTCS A350 LF2	40 - 48						
1940													
1950		UJ2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14						
1960		UJ2	Elbole	5516619	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24						
1970	C	UJ2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	26 - 38						
1980	C	UJ2	Elbole	5516623	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	40 - 48						
1990													
2000		FS1				*****PIPE FLANGES*****							
2010													
2020		FA8	Reduci	5599692	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 150# RF LTCS A350 LF2	3/4 - 1 1/2						
2030													
2040	D	FA8	Reduci	5599703	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 2						
2050													
2060	C		Flange	5851200	1/2 - 3/4	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
2070	C		Flange	5851200	1 1/2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
2080	C		Flange	5851209	1 - 1	FLANGE ADAPTER 150# RF 316 SS X 25 MM TUBE							
2090	D		Flange	5851203	3/4 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
2100													
2110	D		Flange	5556216	5 - 5	FLG WN 150# FF LTCS A350 LF2 SCH STD							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY SMS	REV D	SHEET 4 OF 6	PA 5	CLASS 150H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2120	C		Flange	5556264	1/2 - 2	FLG WN 150# RF LTCS A350 LF2 SCH XS		SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 AS, BB, BD, BF, BFC, BFH, BFL, BFU, BV, CC, CH, CS CT, DCS, DHC, DL, DMD, DLC, DS, DSL, DW, EG, FG, FGH, FM, GCL, GE, GHC, GRC, GRI, GRK, GSL, GST, GW, HOR, MEA, NH, NL, PBT, PGC, PHC, PL, PP, SL, SLL, SM, SVL, T, TH, TL, TLL, TM, TVL, TX, VA, VCB, VE, VI, VP, VU, WBL, WC, WCS, WCR, WD, WDM, WE, WHS, WM, WP, WPS, WR, WRA, WSC, WSR, WT WU MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 343 °C					
2130			Flange	5556261	3 - 14	FLG WN 150# RF LTCS A350 LF2 SCH STD							
2140			Flange	5556265	16 - 24	FLG WN 150# RF LTCS A350 LF2 SCH 20							
2150	C		Flange	5556262	26 - 38	FLG WN 150# RF LTCS A350 LF2 SCH STD B16.47 SERIES A							
2160	C		Flange	5556266	40 - 48	FLG WN 150# RF LTCS A350 LF2 SCH XS B16.47 SERIES A							
2170													
2180			Flange	5556476	1/2 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS							
2190			Flange	5556473	3 - 14	FLG WN 300# RF LTCS A350 LF2 SCH STD							
2200			Flange	5556474	16 - 24	FLG WN 300# RF LTCS A350 LF2 SCH 20							
2210	C		Flange	5556475	26 - 38	FLG WN 300# RF LTCS A350 LF2 SCH STD B16.47 SERIES A							
2220	C		Flange	5556477	40 - 48	FLG WN 300# RF LTCS A350 LF2 SCH XS B16.47 SERIES A							
2230													
2240	C		Flange	5557036	3/4 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS							
2250	C		Flange	5557033	3 - 12	FLG WN 600# RF LTCS A350 LF2 SCH STD							
2260													
2270	C		Flange	5555935	3 - 6	FLG WN 900# RF LTCS A350 LF2 SCH STD							
2280													
2290	C		Flange	5557393	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH XS							
2300													
2310	D		Flange	5557704	3/4 - 2	FLG WN 2500# RTJ LTCS A350 LF2 SCH XS							
2320													
2330	C		Blind	5615566	1/2 - 24	FLG BLIND 150# RF LTCS A350 LF2							
2340			Blind	5615570	26 - 48	FLG BLIND 150# RF LTCS A350 LF2 B16.47 SERIES A							
2350													
2360	C		Blind	5615576	1/2 - 24	FLG BLIND 300# RF LTCS A350 LF2							
2370			Blind	5615579	26 - 48	FLG BLIND 300# RF LTCS A350 LF2 B16.47 SERIES A							
2380													
2390	C		Blind	5615598	1 - 6	FLG BLIND 600# RF LTCS A350 LF2							
2400													
2410	C		Blind	5615605	3 - 6	FLG BLIND 900# RF LTCS A350 LF2							
2420													
2430	C		Blind	5615616	3/4 - 2	FLG BLIND 1500# RF LTCS A350 LF2							
2440													
2450						*****ORIFICE FLANGES*****							
2460													
2470			Orific	5623947	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS							
2480			Orific	5623944	3 - 14	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD							
2490			Orific	5623941	16 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 20							
2500	C		Orific	5624801	26 - 38	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD ASME B16.47 SERIES A							
2510													
2520	C		Orific	5624800	40 - 48	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS ASME B16.47 SERIES A							
2530													
2540													
2550						***** PIPE NIPPLES							
2560													
2570		810				***** NIPPLE PBE CUT FROM PIPE							
2580													
2590		FS1				***** GASKETS *****							
2600													
2610			GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
2620			GASFRS	5674011	26 - 48	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2630													
2640			GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR							
2650	D		GASFRS	5674032	26 - 48	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY SMS	REV D	SHEET 5 OF 6	PA 5	CLASS 150H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2660	D		GASFRS	5673103	3/4 - 12	GASKET 600# 316SS SP WND GRAPHITE IR	GARLOCK 3000						
2670													
2680	C		GASFRS	5673117	3 - 4	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
2690													
2700	C		GASFRS	5672251	2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
2710													
2720	D		GASFF	5669711	5 - 5	GASKET 150# FF NON-ASB PART AGT 1/16"THK							
2730													
2740	D		GASFRS	5672271	2 - 2	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR							
2750													
2760		B1				***** BOLTING *****							
2770													
2780	B		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
2790		SB6	CPSBOL	5675675	12 - 48	JACK SCREW HEX HEAD A320 GR L7M							
2800		B1											
2810													
2820		FS1				**** BLINDS & SPACERS ****							
2830													
2840	C		Specta	5749580	3/4 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70							
2850			Specta	5749581	1 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
2860													
2870			Paddle	5750023	16 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70							
2880			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70							
2890													
2900			Paddle	5750240	16 - 24	PADDLE SPACER 150# RF LTCS A516 GR.70							
2910			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70							
2920													
2930			Paddle	5750026	26 - 48	PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A							
2940			Paddle	5750028	26 - 48	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A							
2950			Paddle	5750196	26 - 48	PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A							
2960			Paddle	5750198	26 - 48	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A							
2970													
2980						***** BRANCHES *****							
2990													
3000						USE BRANCH CHART 150H21							
3010													
3020	C		Reinfo	5358576_R	14 - 16	REINFORCING PAD FROM PIPE A671 CC60 STD WT	34 - 38	SERVICE					
3030	C		Reinfo	5358576_R	14 - 16	REINFORCING PAD FROM PIPE A671 CC60 STD WT	34 - 38	DESIGN: ASME B16.5 MG 1.1					
3040							40 - 48	CODE: ASME B31.3					
3050	C		Reinfo	5358577_R	14 - 20	REINFORCING PAD FROM PIPE A671 CC60 SCH XS		AS, BB, BD, BF, BFC, BFH, BFL, BFU, BV, CC, CH, CS CT, DCS, DHC, DL, DMD, DLC, DS,					
3060	C		Reinfo	5358577_R	14 - 20	REINFORCING PAD FROM PIPE A671 CC60 SCH XS	40 - 48	DSL, DW, EG, FG, FGH, FM, GCL, GE, GHC, GRC, GRI, GRK, GSL, GST, GW, HOR, MEA,					
3070							3 - 48	NH, NL, PBT, PGC, PHC, PL, PP, SL, SLL, SM, SVL, T, TH, TL, TLL, TM, TVL, TX, VA, VCB,					
3080	C		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN		VE, VI, VP, VU, WBL, WC, WCS, WCR, WD, WDM,					
3090	C		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 48	WE, WHS, WM, WP, WPS, WR, WRA, WSC, WSR, WT WU					
3100								MATERIAL: LOW TEMP CARBON STEEL					
3110						***** CONSTRUCTION DETAILS *****							
3120													
3130	D	ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
3140						PIPE BENDS ----- NONE							
3150						THREAD SEALANT ----- PTFE PASTE							
3160													
3170													
3180						FOR FABRICATION AND WELDING DETAILS							
3190						SEE DOC. N° L-ST-2024							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY SMS	REV D	SHEET 6 OF 6	PA 5	CLASS 150H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
10			2BV	K01AM10	1/2 - 2	***** VALVES *****	2 - 2	812	NIPPLE TO BE CUT FROM SCH XXS PIPE								
20						B1		ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.									
30						FA8		USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.									
40						FS1		USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.									
50						SB6		USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.									
60								UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.								
70						VA5		USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.									
80								VA5	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT								
90						FS1		L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.									
100								VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.								
110						VAW		VC1					SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.				
120								VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.								
130						VAW											
140																	
150						VB2											
160																	
170						VA5											
180																	
190						VA5											
200																	
210						VA5											
220																	
230						FS1											
240																	
250						FS1											
260																	
270	Butter																
280																	
290	Gate V																
300																	
310	Gate V																
320																	
330	Globe																
340																	
350	Globe																
360																	
370	Globe																
380																	
390	Globe																
400																	
410	Globe																
420																	
430	Globe																
440																	
450	Globe																
460																	
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 6 MM	FGP/WPMP DATE 17MAR17 BY KD REV A SHEET 1 OF 4 PA 5 CLASS 150H23								

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
470						***** CHECK VALVES *****							
480													
490		VC1	Check	C01AM51	3/4 - 1 1/2	CHECK PISTON 150# RF INC 825 INC TRIM 625							
500		VC2	Check	C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED							
510		VC2	Check	C01AR46	4 - 6	CHECK DUAL WAFER LUG 150# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED							
520		VC2	Check	C01AR53	8 - 16	CHECK DUAL DBL FLG 150# RF LTCS + CLAD INC 825 INC TRIM 625							
530						***** PIPE *****							
540													
550		812	Piping	5358511	3/4 - 1 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE							
560			Piping	5358513	2 - 2	PIPE SCH XXS SMLS LTCS A333 GR.6 BE							
570			Piping	5358514	3 - 16	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
580			Piping	5358577	18 - 24	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE							
590						***** FITTINGS *****							
600													
610													
620			90 Deg	5392870	3/4 - 2	ELL 90 DEG LR XXS LTCS A420 WPL6							
630			90 Deg	5392600	3 - 16	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
640			90 Deg	5392497	18 - 24	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W							
650													
660			45 Deg	5392876	3/4 - 2	ELL 45 DEG LR XXS LTCS A420 WPL6							
670			45 Deg	5392606	3 - 16	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
680			45 Deg	5392608	18 - 24	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W							
690													
700			Tee	5392879	3/4 - 2	TEE SCH XXS LTCS A420 WPL6							
710			Tee	5392609	3 - 16	TEE BW SCH XS LTCS A420 WPL6							
720			Tee	5392406	18 - 24	TEE BW SCH XS LTCS A420 WPL6-W							
730													
740			Reduci	5392877	3/4 - 1 1/2	TEE RED SCH XXS LTCS A420 WPL6	1 - 2						
750			Reduci	5392897	1 1/2 - 2	TEE RED BW SCH XS/XXS LTCS A420 WPL6	3 - 4						
760			Reduci	5392661	3 - 14	TEE RED BW SCH XS LTCS A420 WPL6	4 - 16						
770			Reduci	5392413	8 - 20	TEE RED SCH XS LTCS A420 WPL6-W	18 - 24						
780													
790			Cap	5392880	3/4 - 2	CAP SCH XXS LTCS A420 WPL6							
800			Cap	5392610	3 - 24	CAP BW SCH XS LTCS A420 WPL6							
810													
820			Concen	5392881	3/4 - 1 1/2	REDUCER CONC SCH XXS LTCS A420 WPL6	1 - 2						
830			Concen	5392898	1 1/2 - 2	REDUCER CONC SCH XS/XXS LTCS A420 WPL6	3 - 4						
840			Concen	5392458	3 - 14	REDUCER CONC SCH XS LTCS A420 WPL6	4 - 16						
850			Concen	5392507	10 - 20	REDUCER CONC SCH XS LTCS A420 WPL6-W	18 - 24						
860													
870			Eccent	5392882	3/4 - 1 1/2	REDUCER ECC SCH XXS LTCS A420 WPL6	1 - 2						
880			Eccent	5392899	1 1/2 - 2	REDUCER ECC SCH XS/XXS LTCS A420 WPL6	3 - 4						
890			Eccent	5392461	3 - 14	REDUCER ECC SCH XS LTCS A420 WPL6	4 - 16						
900			Eccent	5392514	10 - 20	REDUCER ECC SCH XS LTCS A420 WPL6-W	18 - 24						
910													
920			Nipofl	5504211	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XXS LTCS A350 LF2	2 - 2						
930			Nipofl	5504212	3/4 - 2	NIPOFLANGE 150# RF SCH XS/XXS LTCS A350 LF2	3 - 24						
940													
950			Nipofl	5504215	1 - 1 1/2	NIPOFLANGE 300# RF SCH XXS LTCS A350 LF2	1 1/2 - 2						
960			Nipofl	5504216	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2	3 - 3						
970													
980		UJ2	Nipofl	5504213	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2	4 - 16						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 6 MM	DATE 17MAR17	BY KD	REV A	FGP/WPMP SHEET 2 OF 4	PA 5	CLASS 150H23

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
990			Weldol	5505197	3/4 - 1	WELDOLET SCH XXS LTCS A350 LF2	2 - 2						
1000			Weldol	5505180	3/4 - 2	WELDOLET SCH XS/XXS LTCS A350 LF2	3 - 24						
1010			Weldol	5505335	3 - 4	WELDOLET SCH XS LTCS A350 LF2	8 - 24						
1020													
1030		UJ2	Elbole	5516762	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2	4 - 16						
1040		FS1				***** PIPE FLANGES *****							
1050		FA8	Reduci	5599692	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 150# RF LTCS A350 LF2	3/4 - 1 1/2						
1060		FA8	Reduci	5599703	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 3/4						
1070													
1080			Flange	5851200	1/2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
1090			Flange	5851203	1/2 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
1100													
1110			Flange	5556221	1/2 - 2	FLG WN 150# RF LTCS A350 LF2 SCH XXS							
1120			Flange	5556264	3 - 24	FLG WN 150# RF LTCS A350 LF2 SCH XS							
1130													
1140			Flange	5556481	3/4 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XXS							
1150			Flange	5556476	3 - 16	FLG WN 300# RF LTCS A350 LF2 SCH XS							
1160													
1170			Blind	5615566	3/4 - 24	FLG BLIND 150# RF LTCS A350 LF2							
1180			Blind	5615576	3/4 - 16	FLG BLIND 300# RF LTCS A350 LF2							
1190													
1200						***** ORIFICE FLANGES *****							
1210													
1220			Orific	5623950	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XXS							
1230			Orific	5623947	3 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS							
1240						***** PIPE NIPPLES *****							
1250													
1260													
1270		812				NIPPLE PBE - CUT FROM PIPE							
1280													
1290													
1300													
1310													
1320													
1330													
1340		FS1				***** GASKETS *****							
1350													
1360			GASFRS	5673043	3/4 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
1370			GASFRS	5673033	3/4 - 16	GASKET 300# 316SS SP WND GRAPHITE IR							
1380													
1390													
1400		B1				***** BOLTING *****							
1410													
1420													
1430			BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1440		SB6	CPSBOL	5675675	12 - 16	JACK SCREW HEX HEAD A320 GR L7M							
1450													
1460						***** BLINDS AND SPACERS *****							
1470													
1480													
1490			Specta	5749580	3/4 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70							
1500			Specta	5749581	1 1/2 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
1510													
1520													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 6 MM		FGP/WPMP DATE 17MAR17 BY KD REV A SHEET 3 OF 4 PA 5 CLASS 150H23			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1530			Paddle	5750023	16 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70							
1540			Paddle	5750100	12 - 16	PADDLE BLANK 300# RF LTCS A516 GR.70							
1550													
1560			Paddle	5750240	16 - 24	PADDLE SPACER 150# RF LTCS A516 GR.70							
1570			Paddle	5750200	12 - 16	PADDLE SPACER 300# RF LTCS A516 GR.70							
1580													
1590						***** BRANCHES *****							
1600						USE BRANCH CHART 150H23							
1610													
1620			Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 24						
1630			Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	2 - 24						
1640													
1650						***** CONSTRUCTION DETAILS *****							
1660													
1670						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
1680						PIPE BENDS ----- NONE							
1690						THREAD SEALANT ----- PTFE PASTE							
						CONSTRUCTION		FGP/WPMP					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 6 MM	DATE 17MAR17	BY KD	REV A	SHEET 4 OF 4	PA 5	CLASS 150H23

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
10						***** VALVES *****		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.								
20						*****BALL VALVES*****		810	NIPPLE TO BE CUT FROM SCH 160 PIPE								
30								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.								
40		VB2						FA3	FLAT FACE FLANGES & FULL FACE GASKETS SHALL ONLY BE USED WHEN MATING WITH FLAT FACE EQUIPMENT.								
50			Ball V	A01AC01	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO		FA4	SLIP ON FLANGE SHALL ONLY BE USED WHERE INSTRUMENT ENCLOSURES ARE INSTALLED.								
60			Ball V	A01AC02	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO		FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.								
70			Ball V	A01AC03	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO		FF5	FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY								
80			Ball V	A01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO		FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.								
90			Ball V	A01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.								
100			Ball V	A01AC08	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO		UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST								
110																	
120	D		Ball V	B01AC01	1/2 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO											
130																	
140	D		Ball V	B01AC02	1/2 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO											
150																	
160		VA5	Ball V	B01AC01	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO											
170			Ball V	B01AC02	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO											
180																	
190	D	VA5	Ball V	B01AC03	4 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO											
200																	
210	D	VA5	Ball V	B01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO											
220																	
230	D	VA5	Ball V	B01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO											
240																	
250	D	VA5	Ball V	B01AC08	8 - 24	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO											
260																	
270		VA5	Ball V	B01AC07	10 - 24	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO											
280																	
290																	
300	D	FS1	Ball V	B03AC01	3/4 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO											
310																	
320	D	FS1	Ball V	B03AC02	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO											
330																	
340	C	FS1	Ball V	A03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO											
350																	
360	C	FS1	Ball V	A03AC02	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO											
370																	
380						*****BUTTERFLY VALVE*****											
390																	
400			Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO											
410																	
420																	
430																	
440																	
450																	
460																	
470																	
480																	
490																	
500																	
510	C		Butter	T01JC61	30 - 30	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A											
520																	
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES -BUTTWELD PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS		CORR. ALLOW.		DATE		BY	REV	FGP/WPMP SHEET		PA	CLASS
								3 MM		17MAR17		SMS	E	1 OF 5		5	150H25

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
520						*****GATE VALVES*****		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
530	D		Gate V	G01AC48	1/2 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
540			Gate V	G01AC41	2 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW		VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY					
550	D	VAR	Gate V	G01AC41	10 - 10	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW		VAF	FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY.					
560						*****GLOBE VALVES*****		VAR	ONLY TO BE USED WHEN SPECIFIED ON THE P&ID.					
570								VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
580								VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
590			Globe	L01AC48	3/4 - 1 1/2	GLOBE 150# RF LTCS A350 LF2 TRIM 16 OS&Y HW		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
600			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
610			Globe	L01AC42	8 - 10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO								
620						*****CHECK VALVES*****								
630														
640														
650	D	VC1	Check	C01AC50	1/2 - 1 1/2	CHECK PISTON 150# RF LTCS A350 LF2 TRIM 16								
660		VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED								
670		VC2	Check	C01AC52	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16								
680						***** PIPE *****								
700														
710														
720	D		Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE								
730	C		Piping	5358514	2 - 2 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 BE								
740			Piping	5358505	3 - 14	PIPE SCH STD SMLS LTCS A333 GR.6 BE								
750			Piping	5358501	16 - 16	PIPE SCH 20 SMLS LTCS A333 GR.6 BE								
760	C		Piping	5358597	18 - 30	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE								
770	C		Piping	5358577	40 - 40	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE								
780														
790		810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE								
800						*****FITTINGS*****								
810														
820														
830	D		90 Deg	5392600	3/4- 2 1/2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
840			90 Deg	5392438	3 - 14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6								
850			90 Deg	5392276	16 - 16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6								
860	C		90 Deg	5392280	18 - 30	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W								
870	C		90 Deg	5392497	40 - 40	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W								
880														
890			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
900			45 Deg	5392444	3 - 14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6								
910			45 Deg	5392282	16 - 16	ELL BW 45 DEG SCH 20 LTCS A420 WPL6								
920	C		45 Deg	5392281	18 - 30	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W								
930	C		45 Deg	5392608	40 - 40	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W								
940														
950	C		30 Deg	5800095	30 - 30	INDUCTION BEND 30 DEG. X 5D RADIUS SCH 20 ASTM A671 CC60 CL.22 - C/W 1.5D TANGENT ENDS BE								
960														
970			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6								
980			Tee	5392447	3 - 14	TEE BW SCH STD LTCS A420 WPL6								
990			Tee	5392285	16 - 16	TEE BW SCH 20 LTCS A420 WPL6								
1000	D		Tee	5392295	18 - 30	TEE BW SCH 20 LTCS A420 WPL6-W								
1010														
1020			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2							
1030			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 4							
1040			Reduci	5392445	3 - 12	TEE RED BW SCH STD LTCS A420 WPL6	4 - 14							
1050			Reduci	5392289	6 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6	16 - 16							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES -BUTTWELD PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS		CORR. ALLOW.	FGP/WPMP					
								3 MM	DATE	BY	REV	SHEET	PA	CLASS
									17MAR17	SMS	D	2 OF 5	5	150H25

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1060	C		Reduci	5392296	8 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6-W	18 - 30	SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BB, BV, CC, CCS, CI , DCS, DHC, DL, DLC, DSL, NL, PHC, SL, TL, TM, VE, VU, WD, WDM, WF, WHR, WHS,WRA, WU MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 343 °C					
1070	D		Reduci	5392292	16 - 24	TEE RED BW SCH 20 LTCS A420 WPL6-W	18 - 30						
1080													
1090			Cross	5416601	4 - 4	CROSS SCH STD LTCS ASTM A420 WPL6							
1100													
1110			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6							
1120			Cap	5392448	3 - 14	CAP BW SCH STD LTCS A420 WPL6							
1130	D		Cap	5392286	16 - 30	CAP BW SCH 20 LTCS A420 WPL6							
1140													
1150			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2						
1160	D		Concen	5392452	1 1/2 - 2 1/2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4						
1170			Concen	5392449	3 - 12	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 14						
1180			Concen	5392290	8 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6	16 - 16						
1190			Concen	5392297	10 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18 - 20						
1200	C		Concen	5392287	16 - 24	REDUCER CONC SCH 20 LTCS A420 WPL6-W	18 - 30						
1210	C		Concen	5392512	30 - 30	REDUCER CONC SCH XS/20 LTCS A420 WPL6-W	40 - 40						
1220													
1230			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2						
1240	D		Eccent	5392453	1 1/2 - 2 1/2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4						
1250			Eccent	5392450	3 - 12	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 14						
1260			Eccent	5392291	8 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6	16 - 16						
1270			Eccent	5392298	10 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18 - 20						
1280	C		Eccent	5392288	16 - 24	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18 - 30						
1290	C		Eccent	5392513	30 - 30	REDUCER ECC SCH XS/20 LTCS A420 WPL6-W	40 - 40						
1300													
1310			Nipofl	5504001	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	2 - 2						
1320			Nipofl	5504002	3/4 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	3 - 14						
1330			Nipofl	5504016	3/4 - 2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2	16 - 24						
1340													
1350	C		Nipofl	5504020	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	1 1/2 - 2						
1360	D		Nipofl	5504021	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 14						
1370													
1380		UJ2	Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14						
1390		UJ2	Nipofl	5504019	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24						
1400													
1410	D	FF5	Endole	5504500	3/4 - 1 1/2	NIPOFLANGE FLAT BOTTOM 150# RF SCH XS LTCS A350 LF2							
1420													
1430			Weldol	5505335	3/4 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2						
1440	C		Weldol	5505179	3/4 - 2 1/2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 14						
1450	D		Weldol	5505572	3/4 - 2	WELDOLET SCH 20/XS LTCS A350 LF2	16 - 30						
1460			Weldol	5505164	3 - 4	WELDOLET SCH STD LTCS A350 LF2	8 - 14						
1470	C		Weldol	5505571	3 - 8	WELDOLET SCH 20/STD LTCS A350 LF2	16 - 30						
1480													
1490		UJ2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14						
1500		UJ2	Elbole	5516619	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24						
1510													
1520	D					***** OLETS *****							
1530													
1540	D		Sockol	5533072	3/4 - 1 1/2	SOCKOLET 3000# LTCS A350 LF2	2 - 24						
1550	D		Endole	5533232	3/4 - 1 1/2	FLAT SOCKOLET 3000# LTCS A350 LF2							
1560	D	UD7	Thredo	5514772	3/4 - 3/4	THREDOLET 3000# LTCS A350 LF2	2 - 24						
1570													
1580	D					****COUPLINGS , UNIONS & PLUGS****							
1590													
1600	D		Coupli	5478705	3/4 - 1 1/2	COUPLING 3000# SW LTCS A 350 LF2							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES -BUTTWELD PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS	CORR. ALLOW. 3 MM						DATE 17MAR17

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1610	D		Reduci	5478709	3/4 - 1	COUPLING RED 3000# SW LTCS A350 LF2	1 - 1 1/2							
1620														
1630	D		Union	5478706	3/4 - 1 1/2	UNION 3000# SW LTCS A350 LF2								
1640	D	UD7	Plug	5491303	3/4 - 3/4	PLUG ROUND HEAD SCRD LTCS A350 LF2								
1650														
1660		FS1				***** PIPE FLANGES *****								
1670														
1680		FA8	Reduci	5599692	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2	3/4 - 1 1/2							
1690														
1700	D	FA8	Reduci	5599703	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 2							
1710														
1720	C		Flange	5851200	1/2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE								
1730	C		Flange	5851203	1/2 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE								
1740														
1750	C		Blind	5615566	1/2 - 24	FLG BLIND 150# RF LTCS A350 LF2								
1760	D		Blind	5615570	30 - 30	FLG BLIND 150# RF LTCS A350 LF2 B16.47 SERIES A								
1770	C		Blind	5615576	1/2 - 24	FLG BLIND 300# RF LTCS A350 LF2								
1780														
1790	C		Flange	5556264	1/2 - 2 1/2	FLG WN 150# RF LTCS A350 LF2 SCH XS								
1800	D		Flange	5557393	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH XS								
1810			Flange	5556476	1/2 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS								
1820	C		Flange	5557036	2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS								
1830	D		Flange	5556217	2 - 2	FLG WN 150# FF LTCS A350 LF2 SCH XS								
1840														
1850			Flange	5556261	3 - 14	FLG WN 150# RF LTCS A350 LF2 SCH STD								
1860	D		Flange	5556216	3 - 3	FLG WN 150# FF LTCS A350 LF2 SCH STD								
1870			Flange	5556473	3 - 14	FLG WN 300# RF LTCS A350 LF2 SCH STD								
1880	D		Flange	5556456	8 - 10	FLG WN 300# FF LTCS A350 LF2 SCH STD								
1890	C		Flange	5557033	3 - 14	FLG WN 600# RF LTCS A350 LF2 SCH STD								
1900														
1910			Flange	5556265	16 - 24	FLG WN 150# RF LTCS A350 LF2 SCH 20								
1920			Flange	5556474	16 - 24	FLG WN 300# RF LTCS A350 LF2 SCH 20								
1930														
1940	C		Flange	5556468	30 - 30	FLG WN 150# RF LTCS A350 LF2 SCH 20 B16.47 SERIES A								
1950	C		Flange	5556469	30 - 30	FLG WN 300# RF LTCS A350 LF2 SCH 20 B16.47 SERIES A								
1960														
1970	C		Flange	5556266	40 - 40	FLG WN 150# RF LTCS A350 LF2 SCH XS B16.47 SERIES A								
1980														
1990	D	FS1	Flange	5556459	1 1/2 - 2 1/2	FLG WN RF PN16 SCH XS LTCS A350 LF2								
2000						***** ORIFICE FLANGES *****								
2010														
2020														
2030			Orific	5623947	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS								
2040			Orific	5623944	3 - 14	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD								
2050			Orific	5623941	16 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 20								
2060														
2070						***** PIPE NIPPLES *****								
2080														
2090		810				NIPPLE PBE CUT FROM PIPE								
2100														
2110		FS1				***** GASKETS *****								
2120														
2130			GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR								
2140	C		GASFRS	5674011	30 - 40	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES -BUTTWELD PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS		CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 4 OF 5	PA 5	CLASS 150H25

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2150	D	FA3	GASFF	5664562	2 - 3	GASKET 150# FF RING EPDM 3MM THK	3 - 40 3 - 40						
2160													
2170	D		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR							
2180	C		GASFRS	5674032	30 - 30	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2190	D	FA3	GASFF	5664563	8 - 10	GASKET 300# FF RING EPDM 3MM THICK							
2200													
2210	D		GASFRS	5673103	4 - 4	GASKET 600# 316SS SP WND GRAPHITE IR							
2220													
2230	D		GASFRS	5664564	1 1/2 - 2 1/2	GASKET PN16 316SS SP WND GRAPHITE IR							
2240													
2250		B1				***** BOLTING *****							
2260													
2270													
2280	B		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
2290		SB6	CPSBOL	5675675	12 - 24	JACK SCREW HEX HEAD A320 GR L7M							
2300													
2310													
2320		FS1				**** BLINDS & SPACERS ****							
2330													
2340	D		Specta	5749580	3/4 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70							
2350			Specta	5749581	1 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
2360													
2370			Paddle	5750023	16 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70							
2380			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70							
2390	C		Paddle	5750028	30 - 30	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A							
2400													
2410			Paddle	5750240	16 - 24	PADDLE SPACER 150# RF LTCS A516 GR.70							
2420			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70							
2430	C		Paddle	5750198	30 - 30	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A							
2440													
2450						***** BRANCHES *****							
2460													
2470						USE BRANCH CHART 150H25							
2480													
2490													
2500	C		Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN		3 - 40 3 - 40	SERVICE				
2510	C		Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN			DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BB, BV, CC, CCS, CI , DCS, DHC, DL, DLC, DSL, NL, PHC, SL, TL, TM, VE, VU, WD, WDM, WF, WHR, WHS,WRA, WU				
2520						***** CONSTRUCTION DETAILS *****							
2530													
2540													
2550	D	ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
2560						PIPE BENDS ----- NONE							
2570						THREAD SEALANT ----- PTFE PASTE							
						CONSTRUCTION							
						ALL SIZES -BUTTWELD PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							3 MM	17MAR17	SMS	D	5 OF 5	5	150H25
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN													

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10	C	PP9	Ball V	M01AC02	3/4 - 2	***** VALVES *****		032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.				
20						810		NIPPLE TO BE CUT FROM SCH 160 PIPE					
30						816		COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's					
40								ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.					
50								MITRE BENDS SHALL BE FABRICATED FROM THE FOLLOWING WALL WALL THICKNESS PIPE:					
60						36" NB THRU 44" NB - 10.0MM							
70						48" NB THRU 56" NB - 12.0MM							
80						60" NB THRU 80" NB - 14.0MM							
90			84" NB THRU 92" NB - 16.0MM										
100			VB2	BALFLG	A01AC01	3 - 4		BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO					
110								BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO					
120				BALFLG	A01AC02	3 - 4		BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO					
130													
140			VB2	BALFGO	A01AC03	6 - 6		BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB GO					
150													
160			BALFGO	A01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB GO							
170													
180		VB2	BALFGO	A01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO							
190													
200													
210		BALFGO	A01AC08	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
220													
230		VB2	BALFGO	B01AC01	3/4 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
240													
250			BALFGO	B01AC02	3/4 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
260													
270		VA5	BALFGO	B01AC01	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
280													
290			BALFGO	B01AC02	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
300													
310		C	VA5	BALFGO	B01AC03	4 - 4		BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB GO					
320													
330			VA5	BALFGO	B01AC04	4 - 4		BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB GO					
340													
350		C	VA5	BALFGO	B01AC07	6 - 24		BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO					
360													
370			VA5	BALFGO	B01AC08	6 - 24		BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO					
380													
390		C	VA5	BALFGO	B01JC07	26 - 60		BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A					
400													
410			VA5	BALFGO	B01JC08	26 - 60		BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A					
420													
430								*****BUTTERFLY VALVE*****					
440													
450													
460													
470													
480													
490													
500													
510													
520													
		FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD		CORR. ALLOW. 3 MM	DATE 11JUL13	BY MLA	REV C	SHEET 1 OF 6	PA 5	CLASS 150H26

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
530			Butter	T01JC61	26 - 60	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A		NR4	USE SWAGES WHERE SMALL END IS 2" OR SMALLER AND LARGE END IS 4" OR SMALLER. USE REDUCERS FOR SMALL END LARGER THAN 2".					
540								PJ9	FOR USE IN NON-JACKETED SULPHUR LINES ONLY.					
550								PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
560						*****GATE VALVES*****		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
570			Gate V	G01AC48	3/4 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW		SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.					
580			Gate V	G01AC41	2 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW		UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST					
590								UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.					
600						*****GLOBE VALVES*****		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
610			Globe	L01AC48	3/4 - 1 1/2	GLOBE 150# RF LTCS A350 LF2 TRIM 16 OS&Y HW		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
620			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW		VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY					
630			Globe	L01AC42	8 - 10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO		VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
640						*****CHECK VALVES*****								
650														
660														
670														
680														
690														
700														
710		VC1	Check	C01AC50	3/4 - 1 1/2	CHECK PISTON 150# RF LTCS A350 LF2 TRIM 16								
720		VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED								
730		VC2	Check	C01AC52	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16								
740		VC2	Check	C01JC52	26 - 60	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A								
760														
770						***** PIPE *****								
780														
790														
800			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE								
810			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE								
820			Piping	5358505	3 - 14	PIPE SCH STD SMLS LTCS A333 GR.6 BE								
830			Piping	5358501	16 - 16	PIPE SCH 20 SMLS LTCS A333 GR.6 BE								
840			Piping	5358597	18 - 24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE								
850			Piping	5358605	30 - 30	PIPE SCH 10 EFW LTCS A671 CC70 CL.22 BE								
860			Piping	5358607	36 - 92	PIPE WT 10.0MM EFW LTCS SHOP FABRICATED A671 CC70 CL.22 BE								
870														
880														
890		810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE								
900														
920						***** FITTINGS *****								
930														
940			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
950			90 Deg	5392438	3 - 14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6								
960			90 Deg	5392276	16 - 16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6								
970			90 Deg	5392280	18 - 24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W								
980			90 Deg	5392265	30 - 30	ELL BW 90 DEG LR SCH 10 LTCS A420 WPL6-W								
990			90 DEG	5392240	30 - 30	ELL BW 90 DEG SR SCH 10 LTCS A420 WPL6-W								
1000		BE3	90 DEG	5393010	36 - 92	90 DEG THREE CUT MITRE BEND SHORT RADIUS SHOP FABRICATED ASTM A671 CC70 CL2								
1010														
1020		NB6												
1030		NN7												
		BE3	90 Deg	5393006	36 - 92	90 DEG THREE CUT MITRE BEND LONG RADIUS SHOP FABRICATED ASTM A671 CC70 CL2								
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES - BUTTWELD		FGP/WPMP						
						CORR. ALLOW. 3 MM		DATE 11JUL13	BY MLA	REV C	SHEET 2 OF 6	PA 5	CLASS 150H26	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040		NB6						VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.				
1050		NN7											
1060													
1070			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1080			45 Deg	5392444	3 - 14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6							
1090			45 Deg	5392282	16 - 16	ELL BW 45 DEG SCH 20 LTCS A420 WPL6							
1100			45 Deg	5392281	18 - 24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W							
1110			45 Deg	5392275	30 - 30	ELL BW 45 DEG LR SCH 10 LTCS A420 WPL6-W							
1120		BE3	45 Deg	5393026	36 - 92	45 DEG TWO CUT MITRE BEND LONG RADIUS SHOP FABRICATED ASTM A671 CC70 CL2							
1130		NB6											
1140		NN7											
1150													
1160			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6							
1170			Tee	5392447	3 - 14	TEE BW SCH STD LTCS A420 WPL6							
1180			Tee	5392285	16 - 16	TEE BW SCH 20 LTCS A420 WPL6							
1190			Tee	5392295	18 - 24	TEE BW SCH 20 LTCS A420 WPL6-W							
1200		NN7	Tee	5392284	30 - 30	TEE BW SCH 10 LTCS A420 WPL6-W							
1210		NN7	Tee	5393032	36 - 56	TEE WT 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22							
1220		NN7	Tee	5393036	60 - 68	TEE WT 19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22							
1230		NN7	Tee	5393038	76 - 92	TEE WT 25.0MM LTCS SHOP FABRICATED A671 CC70 CL 22							
1240													
1250			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2						
1260			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 4						
1270			Reduci	5392445	3 - 12	TEE RED BW SCH STD LTCS A420 WPL6	4 - 14						
1280			Reduci	5392289	6 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6	16 - 16						
1290			Reduci	5392296	8 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6-W	18 - 24						
1300			Reduci	5392292	16 - 20	TEE RED BW SCH 20 LTCS A420 WPL6-W	18 - 24						
1310			Reduci	5393053	18 - 18	TEE REDUCING WT 15.0MM/SCH 20 LTCS SHOP FABRICATED A671 CC70 CL 22	36 - 36						
1320			Reduci	5393072	24 - 24	TEE REDUCING WT 15.0MM/SCH 20 LTCS SHOP FABRICATED A671 CC70 CL 22	52 - 52						
1330													
1340			Reduci	5393053	24 - 24	TEE REDUCING WT 15.0MM/SCH 20 LTCS SHOP FABRICATED A671 CC70 CL 22	54 - 54						
1350													
1360			Reduci	5392274	30 - 32	TEE RED BW SCH 10 LTCS A420 WPL6-W	32 - 34						
1370			Reduci	5393067	28 - 28	TEE REDUCING WT 15.0MM/SCH XS LTCS SHOP FABRICATED A671 CC70 CL 22	52 - 52						
1380			Reduci	5393068	36 - 36	TEE REDUCING WT 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	54 - 54						
1390			Reduci	5393042	44 - 44	TEE REDUCING WT 19.0MM / 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	68 - 68						
1400		NN7	Reduci	5393042	52 - 52	TEE REDUCING WT 19.0MM / 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	60 - 60						
1410		NN7	Reduci	5393044	56 - 56	TEE REDUCING WT 25.0MM / 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	88 - 88						
1420		NN7	Reduci	5393035	60 - 60	TEE REDUCING WT 19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	68 - 68						
1430			Reduci	5393037	60 - 60	TEE REDUCING WT 25.0MM /19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	70 - 70						
1440		NN7	Reduci	5393037	60 - 60	TEE REDUCING WT 25.0MM /19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	76 - 76						
1450													
1460		NN7	Reduci	5393037	60 - 60	TEE REDUCING WT 25.0MM /19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	84 - 84						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD		FGP/WPMP					
						CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
						3 MM	11JUL13	MLA	C	3 OF 6	5	150H26	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1470		NN7	Reduci	5393041	70 - 70	TEE REDUCING WT 25.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	80 - 80							
1480		NN7	Reduci	5393041	76 - 76	TEE REDUCING WT 25.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	80 - 80							
1490														
1500			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6								
1510			Cap	5392448	3 - 14	CAP BW SCH STD LTCS A420 WPL6								
1520			Cap	5392286	16 - 24	CAP BW SCH 20 LTCS A420 WPL6								
1530			Cap	5392318	30 - 30	CAP SCH 10 LTCS A420 WPL6								
1540			Cap	5392614	36 - 92	CAP BW WT 10.0MM LTCS STL A420 WPL6								
1550														
1560			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2							
1570			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4							
1580			Concen	5392449	3 - 12	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 14							
1590			Concen	5392290	8 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6	16 - 16							
1600			Concen	5392297	10 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18 - 20							
1610			Concen	5392287	16 - 20	REDUCER CONC SCH 20 LTCS A420 WPL6-W REDU	18 - 24							
1620			Concen	5393050	30 - 30	CER CONC SCH 10 LTCS A420 WPL6-W	32 - 32							
1630			Concen	5393051	30 - 34	REDUCER CONC WT 10.0MM/SCH 10 LTCS A420 WPL6-W	36 - 36							
1640			Concen	5393055	36 - 90	REDUCER CONC WT10.0MM LTCS SHOP FABRICATED A671 CC70 CL22	40 - 92							
1650														
1660														
1670			Eccent	5392612	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2							
1680			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4							
1690			Eccent	5392450	3 - 12	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 14							
1700			Eccent	5392291	8 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6	16 - 16							
1710			Eccent	5392298	10 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18 - 20							
1720			Eccent	5392288	16 - 20	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18 - 24							
1730			Eccent	5393069	24 - 24	REDUCER ECC WT10.0MM/SCH 20 LTCS SHOP FABRICTED A671 CC70 CL22	36 - 36							
1740			Eccent	5393060	30 - 30	REDUCER ECC SCH 10 LTCS A420 WPL6-W	32 - 32							
1750			Eccent	5393061	30 - 34	REDUCER ECC WT 10.0MM/SCH 10 LTCS A420 WPL6-W	36 - 36							
1760			Eccent	5393062	36 - 90	REDUCER ECC WT10.0MM LTCS SHOP FABRICTED A671 CC70 CL22	40 - 92							
1770														
1780						***** OLETS *****								
1790														
1800														
1810			Nipofl	5504001	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	2 - 2							
1820			Nipofl	5504002	3/4 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	3 - 14							
1830			Nipofl	5504016	3/4 - 2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2	16 - 24							
1840			Nipofl	5504472	3/4 - 2	NIPOFLANGE 150# RF SCH 10/ XS LTCS A350 LF2	30 - 30							
1850			Nipofl	5504099	3/4 - 2	NIPOFLANGE 150# RF SCH 10.0MM/XS LTCS A350 LF2	36 - 92							
1860														
1870			Nipofl	5504021	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	8 - 8							
1880		UJ2	Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14							
1890		UJ2	Nipofl	5504019	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24							
1900		UJ2	Nipofl	5504015	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 10/160 LTCS A350 LF2	30 - 30							
1910		UJ2	Nipofl	5504045	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 10.0MM/160 LTCS A350 LF2	36 - 92							
1920														
1930			Weldol	5505335	3/4 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2							
1940			Weldol	5505179	3/4 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 14							
1950			Weldol	5505572	2 - 2	WELDOLET SCH 20/XS LTCS A350 LF2	16 - 24							
1960			Weldol	5505337	2 - 2	WELDOLET SCH 10/ XS LTCS A350 LF2	30 - 30							
1970			Weldol	5505628	2 - 2	WELDOLET WT 10.0MM/SCH XS LTCS A350 LF2	36 - 92							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD		CORR. ALLOW.	DATE	BY MLA	REV C	SHEET	PA	CLASS
								3 MM	11JUL13			4 OF 6	5	150H26

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1980			Weldol	5505164	3 - 4	WELDOLET SCH STD LTCS A350 LF2	8 - 14						
1990			Weldol	5505571	3 - 8	WELDOLET SCH 20/STD LTCS A350 LF2	16 - 24						
2000			Weldol	5505165	3 - 14	WELDOLET SCH 10/STD WT LTCS A350 LF2	30 - 30						
2010			Weldol	5505600	3 - 14	WELDOLET WT 10.0MM/STD WT LTCS A350 LF2	36 - 92						
2020													
2030													
2040			Sockol	5533072	3/4 - 1 1/2	SOCKOLET 3000# LTCS A350 LF2	2 - 92						
2050		UD7	Thredo	5514772	3/4 - 3/4	THREDOLET 3000# LTCS A350 LF2	2 - 9						
2060			Endole	5533232	3/4 - 1 1/2	FLAT SOCKOLET 3000# LTCS A350 LF2							
2070													
2080						****COUPLINGS , UNIONS & PLUGS****							
2090													
2100			Coupli	5478705	3/4 - 1 1/2	COUPLING 3000# SW LTCS A 350 LF2							
2110			Reduci	5478709	3/4 - 1	COUPLING RED 3000# SW LTCS A350 LF2	1 - 1 1/2						
2120			Union	5478706	3/4 - 1 1/2	UNION 3000# SW LTCS A350 LF2							
2130		UD7	Plug	5491303	3/4 - 3/4	PLUG ROUND HEAD SCRD LTCS A350 LF2							
2140													
2150		UI2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14						
2160		UI2	Elbole	5516619	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24						
2170		UI2	Elbole	5516615	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 10/160 LTCS A350 LF2	30 - 30						
2180		UI2	Elbole	5516645	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 10.0MM/SCH 160 LTCS A350 LF2	36 - 92						
2190													
2200													
2210		816				***** PIPE FLANGES *****							
2220													
2230													
2240		FA8	Reduci	5599692	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2	1 1/2 - 1 1/2						
2250													
2260			Flange	5556264	3/4 - 2	FLG WN 150# RF LTCS A350 LF2 SCH XS							
2270			Flange	5556261	3 - 14	FLG WN 150# RF LTCS A350 LF2 SCH STD							
2280			Flange	5556265	16 - 24	FLG WN 150# RF LTCS A350 LF2 SCH 20							
2290			Flange	5556267	30 - 30	FLG WN 150# RF LTCS A350 LF2 SCH 10 B16.47 SERIES A							
2300			Flange	5556268	36 - 60	FLG WN 150# RF LTCS A350 LF2 WT 10.0MM B16.47 SERIES A							
2310													
2320			Flange	5556476	2 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS							
2330			Flange	5556473	3 - 6	FLG WN 300# RF LTCS A350 LF2 SCH STD							
2340													
2350			Blind	5615566	3/4 - 24	FLG BLIND 150# RF LTCS A350 LF2							
2360			Blind	5615570	30 - 60	FLG BLIND 150# RF LTCS A350 LF2 B16.47 SERIES A							
2370													
2380			Flange	5619895	62 - 92	FLG COMPACT 150# RJ LTCS A350 LF2 WT 10.0MM							
2390													
2400			Blind	5619750	62 - 92	FLG COMPACT BLIND 150# RJ LTCS A350 LF2							
2410													
2420													
2430													
2440						***** PIPE NIPPLES *****							
2450		810				NIPPLE PBE CUT FROM PIPE							
2460													
2470		FS1				***** GASKETS *****							
2480													
2490			GASFRS	5673043	3/4 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
2500			GASFRS	5674011	26 - 60	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD	CORR. ALLOW. 3 MM	DATE 11JUL13	BY MLA	REV C	SHEET 5 OF 6	PA 5	CLASS 150H26

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2510 2520 2530 2540 2550 2560 2570 2580 2590 2600 2610 2620 2630 2640 2650 2660 2670 2680 2690 2700 2710 2720 2730 2740 2750 2760 2770 2780 2790 2800 2810 2820 2830 2840 2850 2860 2870		B1 SB6 B1	GASFRS BOLS Y Specta Paddle Paddle Paddle Paddle Paddle Paddle	5673053 5675770 5675675 5749580 5750023 5750240 5750026 5750196 5750027 5750197	2 - 6 1/4 - 99 12 - 99 1 1/2 - 14 16 - 24 16 - 24 36 - 60 36 - 60 68 - 92 68 - 92	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR 							

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10	D	VB2				***** VALVES *****		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.						
20						***** BALL VALVES *****		O69	ONLY TO BE USED FOR TYPICAL ISOLATION VALVE BOX DETAIL WITH DRAIN						
30								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
40								CA3	REDUCED CORROSION ALLOWANCES APPLIES TO THE PIPE SIZES LISTED: PIPE SIZE CORROSION ALLOWANCE 1/2" - 2.5 mm 2" - 2.5 mm						
50									FR5	USE FLAT FACE FLANGES WHEN CONNECTING TO FLAT FACE PLASTIC OR FRP FLANGES.					
60					Ball V	A01AC01		3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.				
70										GF2	USE WITH FLAT FACE FLANGE ONLY.				
80					Ball V	A01AC03		6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO	ND8	USE ONLY WHERE SPACE IS LIMITED AND WITH APPROVAL BY MATERIALS ENGINEER.				
90										SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.				
100					Ball V	A01AC07		8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST				
110	C	VA5	Ball V	BA6B2	3/4 - 2	BALL 150# SCRD FNPT BS1400 LG2 BODY AND TRIM FB LO									
120			Ball V	B01AC01	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
130															
140															
150															
160															
170															
180															
190															
200															
210			Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO									
220			Butter	T01JC61	26 - 36	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A									
230															
240															
250															
260			Gate V	GA6B1	3/4 - 2	GATE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW									
270			Gate V	G01AC41	3 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW									
280			Gate V	G01AC99	2 - 10	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 3300MM EXT									
290															
300															
310	C	VA5				***** GLOBE VALVES *****									
320															
330															
340															
350															
360															
370															
380															
390															
400															
410															
420	D	069				***** CHECK VALVES *****									
430															
440															
450															
460															
470															
480															

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
490	C	069	Piping	5358505	4 - 14	PIPE SCH STD SMLS LTCS A333 GR.6 BE	3/4 - 2 4 - 6 6 - 14 16 - 16 18 - 24 28 - 36 36 - 36	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
500			Piping	5358501	16 - 16	PIPE SCH 20 SMLS LTCS A333 GR.6 BE		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
510			Piping	5358597	18 - 20	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE		VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
520			Piping	5358597	24 - 24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE		VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
530			Piping	5358576	26 - 36	PIPE SCH STD EFW LTCS A671 CC60 CL.22 BE		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
540														
550														
560									***** FITTINGS *****					
570														
580					90 Deg	5491326		1/2 - 1/2	ELL 90 DEG 3000# SCRDR MNPT X FNPT LTCS A350 LF2 GALV A153					
590														
600					90 Deg	5491291		1/2 - 1/2	ELL 90 DEG 3000# SCRDR MNPT LTCS A350 LF2 GALV A153					
610					90 Deg	5491290		1/2 - 2	ELL 90 DEG 3000# SCRDR FNPT LTCS A350 LF2 GALV A153					
620					90 Deg	5392600		3 - 3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6					
630					90 Deg	5392438		4 - 14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6					
640					90 Deg	5392276		16 - 16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6					
650					90 Deg	5392280		18 - 20	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W					
660					90 Deg	5392280		24 - 24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W					
670					90 Deg	5392496		26 - 36	ELL 90 DEG LR SCH STD LTCS A420 WPL6-W					
680														
690					45 Deg	5491327		1/2 - 1/2	ELL 45 DEG 3000# SCRDR MNPT X FNPT LTCS A350 LF2 GALV A153					
700														
710					45 Deg	5491301		1/2 - 1/2	ELL 45 DEG 3000# SCRDR MNPT LTCS A350 LF2 GALV A153					
720					45 Deg	5491300		1/2 - 2	ELL 45 DEG 3000# SCRDR FNPT LTCS A350 LF2 GALV A153					
730					45 Deg	5392606		3 - 3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6					
740					45 Deg	5392444		4 - 14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6					
750					45 Deg	5392282		16 - 16	ELL BW 45 DEG SCH 20 LTCS A420 WPL6					
760					45 Deg	5392281		18 - 20	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W					
770					45 Deg	5392281		24 - 24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W					
780					46 Deg	5392607		26 - 36	ELL BW 45 DEG SCH STD LTCS A420 WPL6-W					
790														
800					Tee	5491310		3/4 - 2	TEE 3000# SCRDR FNPT LTCS A350 LF2 GALV A153					
810					Tee	5392609		3 - 3	TEE BW SCH XS LTCS A420 WPL6					
820					Tee	5392447		4 - 14	TEE BW SCH STD LTCS A420 WPL6					
830					Tee	5392285		16 - 16	TEE BW SCH 20 LTCS A420 WPL6					
840					Tee	5392295		18 - 20	TEE BW SCH 20 LTCS A420 WPL6-W					
850					Tee	5392295		24 - 24	TEE BW SCH 20 LTCS A420 WPL6-W					
860					Tee	5392405		26 - 36	TEE SCH STD LTCS A420 WPL6-W					
870														
880					Reduci	5491311		1/2 - 1 1/2	TEE RED 3000# SCRDR FNPT LTCS A350 LF2 GALV A153					
890					Reduci	5392454		3 - 3	TEE RED BW SCH STD/XS LTCS A420 WPL6					
900					Reduci	5392445		4 - 12	TEE RED BW SCH STD LTCS A420 WPL6					
910					Reduci	5392289		6 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6					
920					Reduci	5392296		8 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6-W					
930					Reduci	5392292		16 - 20	TEE RED BW SCH 20 LTCS A420 WPL6-W					
940					Reduci	5556912		16 - 16	TEE RED BW SCH STD/ SCH 20 LTCS A420 WPL6-W					
950					Reduci	5392428		28 - 28	TEE RED BW SCH STD LTCS A420 WPL6-W					
960														
970					Cap	5491330		3/4 - 2	CAP 3000# SCRDR FNPT LTCS A350 LF2 GALV A153					
980					Cap	5491330		4 - 4	CAP 3000# SCRDR FNPT LTCS A350 LF2 GALV A153					
990					Cap	5392610		3 - 3	CAP BW SCH XS LTCS A420 WPL6					
1000					Cap	5392448		4 - 14	CAP BW SCH STD LTCS A420 WPL6					
1010					Cap	5392286		16 - 20	CAP BW SCH 20 LTCS A420 WPL6					
1020					Cap	5392286		24 - 24	CAP BW SCH 20 LTCS A420 WPL6					
1030					Cap	5392448		26 - 36	CAP BW SCH STD LTCS A420 WPL6					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" & SMALLER - GALV'D, THREADED COUPLED 3" & ABOVE - BUTT WELD-GALV'D AFTER FABRICATION PWHT FOR WT > 19MM		CORR. ALLOW. 3 MM	DATE 17MAR17	BY MLA	REV D	SHEET 2 OF 6	PA 5	CLASS 150H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1040	D		Concen	5392452	3 - 3	REDUCER CONC SCH STD/XS LTCS A420 WPL6	4 - 6							
1050			Concen	5392449	4 - 12	REDUCER CONC SCH STD LTCS A420 WPL6	6 - 14							
1060			Concen	5392290	8 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6	16 - 16							
1070			Concen	5392297	10 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18 - 20							
1080			Concen	5392287	16 - 20	REDUCER CONC SCH 20 LTCS A420 WPL6-W	18 - 24							
1090			Concen	5392500	20 - 24	REDUCER CONC SCH STD/20 LTCS A420 WPL6-W	26 - 36							
1100	C		Concen	5392505	26 - 28	REDUCER CONC SCH STD LTCS A420 WPL6-W	30 - 36							
1110														
1120			Eccent	5392453	3 - 3	REDUCER ECC SCH STD/XS LTCS A420 WPL6	4 - 6							
1130			Eccent	5392450	4 - 12	REDUCER ECC SCH STD LTCS A420 WPL6	6 - 14							
1140			Eccent	5392291	8 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6	16 - 16							
1150			Eccent	5392298	10 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18 - 20							
1160			Eccent	5392288	16 - 20	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18 - 24							
1170	D	ND8	Eccent	5392508	16 - 16	REDUCER ECC SCH STD/20 LTCS A420 WPL6-W	28 - 28							
1180			Eccent	5392508	20 - 24	REDUCER ECC SCH STD/20 LTCS A420 WPL6-W	26 - 36							
1190	C		Eccent	5392509	26 - 28	REDUCER ECC SCH STD LTCS A420 WPL6-W	30 - 36							
1200														
1210						*****OLETS*****								
1220														
1230			Nipofl	5504002	3/4 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	4 - 14							
1240			Nipofl	5504016	3/4 - 2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2	16 - 24							
1250			Nipofl	5504002	3/4 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	26 - 36							
1260														
1270		UJ2	Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14							
1280		UJ2	Nipofl	5504019	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24							
1290		UJ2	Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	26 - 36							
1300														
1310		UD7	Thredo	5514772	1/2 - 2	THREDOLET 3000# LTCS A350 LF2	3 - 36							
1320			Endole	5514773	1/2 - 1	FLAT THREDOLET 3000# LTCS A350 LF2								
1330														
1340			Weldol	5505179	3 - 3	WELDOLET SCH STD/XS LTCS A350 LF2	4 - 14							
1350			Weldol	5505572	3 - 3	WELDOLET SCH 20/XS LTCS A350 LF2	16 - 24							
1360			Weldol	5505179	3 - 3	WELDOLET SCH STD/XS LTCS A350 LF2	26 - 36							
1370			Weldol	5505164	4 - 4	WELDOLET SCH STD LTCS A350 LF2	8 - 14							
1380			Weldol	5505571	4 - 8	WELDOLET SCH 20/STD LTCS A350 LF2	16 - 24							
1390			Weldol	5505164	4 - 12	WELDOLET SCH STD LTCS A350 LF2	26 - 36							
1400														
1410	C	UJ2	Elbole	5516623	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	3 - 3							
1420		UJ2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14							
1430		UJ2	Elbole	5516619	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24							
1440		UJ2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	26 - 36							
1450														
1460						**** COUPLINGS, UNIONS & PLUGS ****								
1470														
1480			Coupli	5491250	3/4 - 2	COUPLING 3000# SCRDR FNPT LTCS A350 LF2 GALV A153								
1490			Reduci	5491260	3/4 - 1 1/2	COUPLING RED 3000# SCRDR FNPT LTCS A350 LF2 GALV A153	1 - 2							
1500														
1510			Union	5491270	3/4 - 2	UNION 3000# SCRDR FNPT LTCS A350 LF2 GALV A153								
1520			Union	5491271	3/4 - 2	UNION 3000# SCRDR (MNPT/FNPT) LTCS A350 LF2 GALV A153								
1530														
1540		UD7	Plug	5491280	1/2 - 2	PLUG ROUND HEAD SCRDR MNPT LTCS A350 LF2 GALV A153								
1550														
1560			Bushin	5491325	1/2 - 1 1/2	BUSHING 3000# SCRDR (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153	3/4 - 2							
						CONSTRUCTION		FGP/WPMP						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			2" & SMALLER - GALV'D, THREADED COUPLED 3" & ABOVE - BUTT WELD-GALV'D AFTER FABRICATION PWHT FOR WT > 19MM		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								3 MM	17MAR17	MLA	D	3 OF 6	5	150H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1570	D	FS1				***** PIPE FLANGES *****							
1580													
1590			Reduci	5599691	1/2 - 1 1/2	FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2 GALV A153	3/4 - 2						
1600													
1610			Reduci	5599702	1/2 - 1 1/2	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 GALV A153	1 - 2						
1620													
1630		FR5	Flange	5599689	1/2 - 2	FLG SCRD (FNPT) 150# RF LTCS A350 LF2 GALV A153							
1640			Flange	5599693	3/4 - 2	FLG SCRD (FNPT) 150# FF LTCS A350 LF2 GALV A153							
1650													
1660	C		Flange	5556264	3 - 3	FLG WN 150# RF LTCS A350 LF2 SCH XS							
1670	C	FR5	Flange	5556217	3 - 3	FLG WN 150# FF LTCS A350 LF2 SCH XS							
1680			Flange	5556261	4 - 14	FLG WN 150# RF LTCS A350 LF2 SCH STD							
1690		FR5	Flange	5556216	4 - 14	FLG WN 150# FF LTCS A350 LF2 SCH STD							
1700			Flange	5556265	16 - 20	FLG WN 150# RF LTCS A350 LF2 SCH 20							
1710		FR5	Flange	5556220	16 - 16	FLG WN 150# FF LTCS A350 LF2 SCH 20							
1720			Flange	5556265	24 - 24	FLG WN 150# RF LTCS A350 LF2 SCH 20							
1730			Flange	5556262	26 - 36	FLG WN 150# RF LTCS A350 LF2 SCH STD B16.47 SERIES A							
1740													
1750		FS1	Flange	5599701	3/4 - 2	FLG SCRD (FNPT) 300# RF LTCS A350 LF2 GALV A153							
1760	C	FS1	Flange	5599694	3/4 - 2	FLG SCRD (FNPT) 300# FF LTCS A350 LF2 GALV A153							
1770													
1780		FS1	Flange	5556476	3 - 3	FLG WN 300# RF LTCS A350 LF2 SCH XS							
1790	C	FS1	Flange	5556455	3 - 3	FLG WN 300# FF LTCS A350 LF2 SCH XS							
1800		FS1	Flange	5556473	4 - 14	FLG WN 300# RF LTCS A350 LF2 SCH STD							
1810	C	FS1	Flange	5556456	4 - 14	FLG WN 300# FF LTCS A350 LF2 SCH STD							
1820		FS1	Flange	5556474	16 - 20	FLG WN 300# RF LTCS A350 LF2 SCH 20							
1830	C	FS1	Flange	5556457	16 - 20	FLG WN 300# FF LTCS A350 LF2 SCH 20							
1840		FS1	Flange	5556474	24 - 24	FLG WN 300# RF LTCS A350 LF2 SCH 20							
1850	C	FS1	Flange	5556475	36 - 36	FLG WN 300# RF LTCS A350 LF2 SCH STD B16.47 SERIES A							
1860													
1870	D	FS1	Flange	5556458	1 1/4 - 2	FLG SCRD (FNPT) PN16 RF LTCS A350 LF2 GALV A153							
1880													
1890			Blind	5615567	1/2 - 24	FLG BLIND 150# RF LTCS A350 LF2 GALV A153							
1900			Blind	5615575	26 - 36	FLG BLIND 150# RF LTCS A350 LF2 GALV A153 B16.47 SERIES A							
1910													
1920													
1930		FS1	Blind	5615487	3/4 - 24	FLG BLIND 150# FF LTCS A350 LF2 GALV A153							
1940		FS1	Blind	5615577	1/2 - 24	FLG BLIND 300# RF LTCS A350 LF2 GALV A153							
1950													
1960						***** ORIFICE FLANGES *****							
1970													
1980			Orific	5623948	2 - 2	ORIF SET 300# RF SCRD (FNPT) LTCS A350 LF2 GALV A153							
1990			Orific	5623947	3 - 3	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS							
2000			Orific	5623944	4 - 14	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD							
2010			Orific	5623941	16 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 20							
2020	C		Orific	5624801	36 - 36	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD ASME B16.47 SERIES A							
2030						***** PIPE NIPPLES *****							
2040													
2050													
2060			Nipple	5541200	1/2 - 1 1/2	NIPPLE SCH 160 LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153							
2070													
2080			Nipple	5541201	1/2 - 1 1/2	NIPPLE SCH 160 LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153							
2090													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" & SMALLER - GALV'D, THREADED COUPLED 3" & ABOVE - BUTT WELD-GALV'D AFTER FABRICATION PWHT FOR WT > 19MM	CORR. ALLOW. 3 MM	DATE 17MAR17	BY MLA	REV D	SHEET 4 OF 6	PA 5	CLASS 150H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2100	C	069	Nipple	5541214	2 - 2	NIPPLE SCH XS LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153	1 - 1 1/2							
2110	C		Nipple	5541204	2 - 2	NIPPLE SCH XS LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153								
2120	D		Nipple	5541193	4 - 4	NIPPLE SCH XS/160 LTCS A333-6 BOE/TOE 7" LG								
2130														
2140														
2150						***** SWAGE NIPPLES *****								
2160														
2170			Concen	5544150	3/4 - 1	SWAGE CONC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153								
2180														
2190			Concen	5544151	3/4 - 1 1/2	SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153								
2200							2 - 2							
2210		Concen	5544142	3/4 - 1 1/2	SWAGE CONC SCH XS/160 LTCS A420 WPL6 BLE/TSE		3 - 3							
2220		Concen	5544154	3/4 - 1 1/2	SWAGE CONC SCH STD/160 LTCS A420 WPL6 BLE/TSE		4 - 4							
2230		Concen	5544092	2 - 2	SWAGE CONC SCH XS LTCS A420 WPL6 BLE/TSE		3 - 3							
2240	C	Concen	5544155	2 - 2	SWAGE CONC SCH STD/XS LTCS A420 WPL6 BLE/TSE		4 - 4							
2250														
2260		Eccent	5549150	3/4 - 1	SWAGE ECC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153		1 - 1 1/2							
2270														
2280		Eccent	5549151	3/4 - 1 1/2	SWAGE ECC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153		2 - 2							
2290														
2300		Eccent	5549225	3/4 - 1 1/2	SWAGE ECC SCH XS/160 LTCS A420 WPL6 BLE/TSE		3 - 3							
2310		Eccent	5549154	3/4 - 1 1/2	SWAGE ECC SCH STD/160 LTCS A420 WPL6 BLE/TSE		4 - 4							
2320		Eccent	5549182	2 - 2	SWAGE ECC SCH XS LTCS A420 WPL6 BLE/TSE		3 - 3							
2330	C	Eccent	5549155	2 - 2	SWAGE ECC SCH STD/XS LTCS A420 WPL6 BLE/TSE		4 - 4							
2340														
2350		FS1				***** GASKETS *****								
2360														
2370			GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR								
2380			GASFRS	5674011	26 - 36	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A								
2390														
2400	D	FS1	GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR								
2410	C	FS1	GASFRS	5674032	36 - 36	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A								
2420														
2430														
2440		GF2	GASFF	5664562	3/4 - 16	GASKET 150# FF RING EPDM 3MM THK								
2450	C	GF2	GASFF	5664563	1/2 - 24	GASKET 300# FF RING EPDM 3MM THICK								
2460														
2470	D	FS1	GASFRS	5664564	1 1/4 - 2	GASKET PN16 316SS SP WND GRAPHITE IR								
2480		B1				***** BOLTING *****								
2490														
2500														
2510	B		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
2520		SB6	CPSBOL	5675675	12 - 24	JACK SCREW HEX HEAD A320 GR L7M								
2530		B1												
2540														
2550		FS1				**** BLINDS & SPACERS ****								
2560														
2570			Specta	5749582	3/4 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153								
2580														
2590		FS1	Specta	5749583	3/4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70 GALV A153								
2600														
2610			Paddle	5749942	16 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70 GALV A153								
2620														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" & SMALLER - GALV'D, THREADED COUPLED 3" & ABOVE - BUTT WELD-GALV'D AFTER FABRICATION PWHT FOR WT > 19MM		CORR. ALLOW. 3 MM	DATE 17MAR17	BY MLA	REV D	SHEET 5 OF 6	PA 5	CLASS 150H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2630	C		Paddle	5750025	28 - 36	PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A GALV A153							
2640													
2650													
2660													
2670	C		Paddle	5750194	28 - 36	PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A GALV A153							
2680													
2690													
2700													
2710	FS1		Paddle	5749853	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70 GALV A153							
2720													
2730													
2740													
2750	FS1		Paddle	5750201	12 - 24	PADDLE SPACER 300# RF LTCS A516-GR70 GALV A153							
2760													
2770													
2780													
2790	C		Reinfo	5358576_R	14 - 14	REINFORCING PAD FROM PIPE A671 CC60 STD WT	36 - 36						
2800							36 - 36						
2810													
2820													
2830	C		Reinfo	RWELD	3 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	4 - 36						
2840	C		Reinfo	RWELD	3 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	4 - 36						
2850													
2860													
2870	D					SEAL WELDING-----FOR THREADED CONNECTIONS ONLY							
2880						PIPE BENDS ----- NONE							
2890						THREAD SEALANT ----- PTFE PASTE							
						CONSTRUCTION	FGP/WPMP						
						2" & SMALLER - GALV'D, THREADED COUPLED	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
						3" & ABOVE - BUTT WELD-GALV'D AFTER FABRICATION PWHT FOR WT > 19MM							
							3 MM	17MAR17	MLA	D	6 OF 6	5	150H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10		VB2				***** BALL VALVES *****		810	NIPPLE TO BE CUT FROM SCH 160 PIPE ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. VA5 USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. VB2 SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
20														
30	C		Ball V	A01AC01	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO		B1						
40								FA8						
50	C		Ball V	A01AC02	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO		VA5						
60														
70	C		Ball V	B01AC01	3/4 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO		VB2						
80														
90	C	VA5	Ball V	B01AC02	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO		ZA6						
100														
110						***** GATE VALVES *****								
120														
130			Gate V	G01AC48	3/4 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW								
140	C		Gate V	G01AC41	2 - 3	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW								
150														
160	C		Condui	W01AC70	3 - 3	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW								
170														
180						***** PIPE ABOVE GROUND *****								
190														
200	D		Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE								
210	C		Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE								
220	C		Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE								
230														
240	C	810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE								
250														
260						***** PIPE BELOW GROUND *****								
270														
280			Piping	9001254CTD	6 - 8	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED								
290														
300						***** FITTINGS ABOVE GROUND*****								
310														
320	C		90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
330	C		90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6								
340	C		45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
350	C		45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6								
360														
370	C		Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6								
380	C		Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6								
390														
400	C		Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2	SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 SERVICE - FUEL GAS, VENT TO ATMOSPHERE MAX. DESIGN PARAMETER = 10 BARG. @ 75 DEG.C. (LIMITED BY PIPE WALL THICKNESS)						
410	C		Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3							
420														
430	C		Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6								
440	C		Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6								
450														
460	C		Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 75 °C COMMENT: LTCS						
470	C		Concen	5392452	1 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3							
480	C		Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2							
490	C		Eccent	5392453	1 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3							
500														
510						*****INDUCTION BENDS ABOVE GROUND*****								
520														
530	C		90 Deg	9001813	6 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY JCS	REV C	SHEET 1 OF 3	PA 5	CLASS 150H5E	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
540	C		45 Deg	9001823	6 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60							
550	C		22.5 D	9001833	6 - 8	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
560						*****INDUCTION BENDS BELOW GROUND*****							
570													
580													
590	C		90 Deg	9001813CTD	6 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60							
600	C		45 Deg	9001823CTD	6 - 8	C/W 1.5D TANGENT ENDS BW COATED INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60							
610	C		22.5 D	9001833CTD	6 - 8	C/W 1.5D TANGENT ENDS BW COATED INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
620						***** OLETS ABOVE & BELOW GROUND***** WELDOLET SCH XS LTCS A350 LF2							
630													
640													
650	C		Weldol	5505335	3/4 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2						
660	C		Weldol	5505179	3/4 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 3						
670													
680	C		Nipofl	5504001	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	2 - 2						
690	C		Nipofl	5504002	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	3 - 3						
700													
710	C		Nipofl	5504020	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	2 - 2						
720	C		Nipofl	5504021	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 3						
730													
740						***** PIPE FLANGES ABOVE & BELOW GROUND*****							
750													
760	C	FA8	Reduci	5599692	1/2 - 1/2	FLG SCRd (FNPT) REDUCING 150# RF LTCS A350 LF2	3/4 - 1 1/2						
770													
780	C	FA8	Reduci	5599703	1/2 - 1/2	FLG SCRd (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 2						
790													
800			Flange	5851200	3/4 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
810													
820	C		Flange	5556264	1/2 - 2	FLG WN 150# RF LTCS A350 LF2 SCH XS							
830	C		Flange	5556261	3 - 3	FLG WN 150# RF LTCS A350 LF2 SCH STD							
840													
850	C		Flange	5555935	3 - 3	FLG WN 900# RF LTCS A350 LF2 SCH STD							
860													
870	C		Flange	5557393	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH XS							
880													
890	C		Flange	9001473	6 - 8	FLG WN 150# RF LTCS A694 F60 WT 6.35MM							
900													
910	C		Blind	5615566	1/2 - 3	FLG BLIND 150# RF LTCS A350 LF2							
920			Blind	9001449	6 - 8	FLG BLIND 150# RF LTCS A694 F60							
930													
940	C		Blind	5615616	3/4 - 2	FLG BLIND 1500# RF LTCS A350 LF2							
950						***** GASKETS *****							
960													
970													
980	C		GASFRRS	5673043	3/4 - 8	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
990													
1000	C		GASFRRS	5672251	3/4 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
1010													
1020		B1				***** BOLTING *****							
1030													
1040	C		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 3	PA 5	CLASS 150H5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
1050 1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1170 1180 1190 1200 1210 1220 1230 1240 1250 1260	C C	810	Specta	5749580	3/4 - 10	***** BLINDS & SPACERS SPECTACLE BLIND 150# RF LTCS A 516 GR.70 ***** PIPE NIPPLES ***** NIPPLE PBE CUT FROM PIPE ***** BRANCHES ***** USE BRANCH CHART 150H5E	3 - 3 3 - 3										
			Reinfo Reinfo	RWELD RWELD	2 - 2 2 - 2	NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- REQUIRED PIPE BENDS ----- INDUCTION BENDS THREAD SEALANT ----- PTFE PASTE											
			ZA6					SERVICE									
								DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 SERVICE - FUEL GAS, VENT TO ATMOSPHERE MAX. DESIGN PARAMETER = 10 BARG. @ 75 DEG.C. (LIMITED BY PIPE WALL THICKNESS)									
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 75 °C COMMENT: LTCS									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY JCS	REV C	FGP/WPMP SHEET 3 OF 3	PA 5	CLASS 150H5E			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10		PP9						O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. 800 THE TEMPERATURE IS LIMITED TO 260°C IN CRUDE OIL SERVICES CONTAINING H2S /AND /OR ORGANIC SULPHUR COMPOUNDS, FOR EXAMPLE, IN THE CRUDE STABILIZATION UNIT. IN HYDROCARBON GAS SERVICE CONTAINING H2S, FOR EXAMPLE, SOUR GAS MOLECULAR SIEVE DEHYDRATION UNITS, USE IS LIMITED TO 230 C. EXCEPTIONS TO THIS ARE THE SULPHUR RECOVERY AND TAIL GAS UNITS WHERE CARBON STEEL SOUR SERVICE LINE CLASSES CAN BE USED UP TO 315°C BECAUSE OF LOWER H2S PARTIAL PRESSURES. 810 NIPPLE TO BE CUT FROM SCH 160 PIPE B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FA4 SLIP ON FLANGE SHALL ONLY BE USED WHERE INSTRUMENT ENCLOSURES ARE INSTALLED. FA8 USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. FS1 USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. PG1 USE FOR PRESSURE GAUGES WITH CHEMICAL SEALS. PJ9 FOR USE IN NON-JACKETED SULPHUR LINES ONLY. PP9 ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 SB6 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. SC1 USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTRORACE. UD6 USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST. UD7 USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST						
20		800													
30						***** VALVES *****									
40		VA5				***** MODULAR VALVES *****									
50															
60															
70	D		2BV	K01AC02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV									
80															
90	D	VAV	2BV	K03AC02	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV									
100															
110	D		2BVR	M01AC02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 316 SS TRIM METAL LO									
120															
130	D	FS1	2BVR	M03AC02	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR									
140															
150	D	VAW	R2BVR	M03EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2								
160															
170	D	VAW	R2BVR	M03EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3								
180															
190		VB2				***** BALL VALVES *****									
200															
210			Ball V	A01AC01	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO									
220															
230			Ball V	A01AC02	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO									
240															
250	D		Ball V	A01AC03	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO									
260															
270	D		Ball V	A01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO									
280															
290			Ball V	A01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO									
300															
310			Ball V	A01AC08	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO									
320															
330			Ball V	B01AC01	3/4 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
340															
350			Ball V	B01AC02	3/4 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO									
360															
370		VA5	Ball V	B01AC01	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
380															
390		VA5	Ball V	B01AC02	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO									
400															
410	D	VA5	Ball V	B01AC03	4 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
420															
430	D	VA5	Ball V	B01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO									
440															
450	D	VA5	Ball V	B01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
460															
470		VA5	Ball V	B01AC07	10 - 24	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
480															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES. FOR SERVICE CODES CDF, WE AND WSC - WT > 19MM	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 1 OF 8	PA 5	CLASS 150K21		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
490	D	VA5	Ball V	B01AC08	8 - 24	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE). FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY. FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY. FOR LEVEL TRANSMITTER CONNECTIONS ONLY USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. FOR USE IN SULPHUR SERVICE ONLY. QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS. TO BE USED FOR COMMISSIONING ONLY USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.				
500		VA5	Ball V	B01JC07	26 - 56	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A		VA5					
510		VA5	Ball V	B01JC08	26 - 56	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A		VAB					
520								VAD					
530		FS1	Ball V	B03AC01	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO		VAF					
540		FS1	Ball V	B03AC02	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO		VAG					
550		FS1	Ball V	A03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO		VAH					
560		FS1	Ball V	A03AC02	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO		VAV					
570								VAW					
580								VB2					
590						*****ORBIT VALVES*****		VC1					
600								VC2					
610								VJ2					
620	D	VA5	Ball V	R01AC06	3/4 - 6	BALL RISING STEM 150# RF LTCS A352 LCC 316SS TRIM METAL RB HO		VQ2					
630						***** BUTTERFLY VALVES *****		VQ3					
640			Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO		VU2					
650			Butter	T01JC61	26 - 56	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A		ZA6					
660						***** GATE VALVES *****							
670													
680													
690													
700		VQ2											
710			Gate V	G01AC48	3/4 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW							
720			Gate V	G01AC41	2 - 10	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW							
730	D	VQ3	Gate V	G01AC42	12 - 20	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE GO							
740			Gate V	G01JC42	42 - 42	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE GO B16.47 SERIES A							
750	D												
760													
770			Gate V	G03AC41	2 - 2	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW							
780						*****THROUGH CONDUIT GATE VALVES *****							
790		VQ2											
800													
810			Condui	W01AC70	3 - 10	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
820	D		Condui	W01AC71	12 - 24	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
830			Condui	W01JC71	26 - 56	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A							
840	D					***** GLOBE VALVES *****							
850													
860													
870													
880			Globe	L01AC48	3/4 - 1 1/2	GLOBE 150# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
890			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW							
900			Globe	L01AC42	8 - 10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO							
910						***** CHECK VALVES *****							
920													
930		VC1	Check	C01AC50	3/4 - 1 1/2	CHECK PISTON 150# RF LTCS A350 LF2 TRIM 16							
940		VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
950		VC2	Check	C01AC52	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16							
960		VC2	Check	C01JC52	26 - 56	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A							
970													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES. FOR SERVICE CODES CDF, WE AND WSC - WT > 19MM	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 2 OF 8	PA 5	CLASS 150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
980						***** PIPE *****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 REFER TO PIM-SU-5112 FOR ALL FLUID SERVICES MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 232 °C COMMENT: REFER NOTE 800				
990												
1000			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE						
1010			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE						
1020			Piping	5358505	3 - 14	PIPE SCH STD SMLS LTCS A333 GR.6 BE						
1030			Piping	5358501	16 - 16	PIPE SCH 20 SMLS LTCS A333 GR.6 BE						
1040			Piping	5358597	18 - 24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE						
1050	C		Piping	5358577	26 - 44	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE						
1060	C		Piping	5358540	46 - 48	PIPE WT 14.0MM EFW LTCS A671 CC60 CL.22 BE						
1070			Piping	5358610	56 - 56	PIPE WT 15.0MM EFW LTCS A671 CC60 CL.22 BE						
1080												
1090		810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE						
1100						***** FITTINGS *****						
1110												
1120												
1130			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6						
1140			90 Deg	5392438	3 - 14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6						
1150			90 Deg	5392276	16 - 16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6						
1160			90 Deg	5392280	18 - 24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W						
1170	C		90 Deg	5392497	26 - 44	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W						
1180	C		90 Deg	5392472	46 - 48	ELL BW 90 DEG LR WT 14.0MM LTCS A420 WPL6-W						
1190			90 Deg	5392499	56 - 56	ELL 90 DEG LR WT 15.0MM LTCS A420 WPL6-W						
1200												
1210			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6						
1220			45 Deg	5392444	3 - 14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6						
1230			45 Deg	5392282	16 - 16	ELL BW 45 DEG SCH 20 LTCS A420 WPL6						
1240			45 Deg	5392281	18 - 24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W						
1250	C		45 Deg	5392608	26 - 44	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W						
1260	C		45 Deg	5392590	46 - 48	ELL BW 45 DEG LR WT 14.0MM LTCS A420 WPL6-W						
1270			45 Deg	5392520	56 - 56	ELL BW 45 DEG LR WT 15.0MM LTCS A420 WPL6 -W						
1280												
1290			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6						
1300			Tee	5392447	3 - 14	TEE BW SCH STD LTCS A420 WPL6						
1310			Tee	5392285	16 - 16	TEE BW SCH 20 LTCS A420 WPL6						
1320			Tee	5392295	18 - 24	TEE BW SCH 20 LTCS A420 WPL6-W						
1330	C		Tee	5392406	26 - 44	TEE BW SCH XS LTCS A420 WPL6-W						
1340	C		Tee	5392251	46 - 48	TEE BW WT 14.0MM LTCS A420 WPL 6-W						
1350			Tee	5392407	56 - 56	TEE BW WT 15.0MM LTCS A420 WPL6-W						
1360												
1370			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2					
1380			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 4					
1390			Reduci	5392445	3 - 12	TEE RED BW SCH STD LTCS A420 WPL6	4 - 14					
1400			Reduci	5392289	6 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6	16 - 16					
1410			Reduci	5392296	8 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6-W	18 - 24					
1420			Reduci	5392476	14 - 14	TEE RED SCH XS/STD LTCS A420 WPL6-W	26 - 32					
1430			Reduci	5392292	16 - 20	TEE RED BW SCH 20 LTCS A420 WPL6-W	18 - 24					
1440			Reduci	5392475	16 - 24	TEE RED BW SCH XS/20 LTCS A420 WPL6-W	26 - 42					
1450	C		Reduci	5392413	26 - 42	TEE RED SCH XS LTCS A420 WPL6-W	28 - 44					
1460	C		Reduci	5392650	26 - 42	TEE RED WT 14.0MM/SCH XS LTCS A420 WPL6-W	46 - 48					
1470	D		Reduci	5392653	24 - 24	TEE RED WT 14.0MM/SCH 20 LTCS A420 WPL6-W	48 - 48					
1480	C		Reduci	5392321	42 - 44	TEE RED BW WT 15.0MM/XS LTCS A420 WPL6-W	56 - 56					
1490	C		Reduci	5392651	48 - 48	TEE RED BW WT 15.0MM/14.0MM LTCS A420 WPL6-W	56 - 56					
1500												
1510			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES. FOR SERVICE CODES CDF, WE AND WSC - WT > 19MM	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 3 OF 8 PA 5 CLASS 150K21				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1520			Cap	5392448	3 - 14	CAP BW SCH STD LTCS A420 WPL6								
1530			Cap	5392286	16 - 24	CAP BW SCH 20 LTCS A420 WPL6								
1540	C		Cap	5392610	26 - 44	CAP BW SCH XS LTCS A420 WPL6								
1550	C		Cap	5392652	46 - 48	CAP BW WT 14.0MM LTCS A420 WPL6								
1560			Cap	5392615	56 - 56	CAP BW WT 15.0MM LTCS STL A420 WPL6								
1570														
1580	C		Concen	5392458	3/4 - 1	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2							
1590			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4							
1600			Concen	5392449	3 - 12	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 14							
1610			Concen	5392290	8 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6	16 - 16							
1620			Concen	5392297	10 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18 - 20							
1630			Concen	5392287	16 - 20	REDUCER CONC SCH 20 LTCS A420 WPL6-W	18 - 24							
1640			Concen	5392512	18 - 24	REDUCER CONC SCH XS/20 LTCS A420 WPL6-W	26 - 30							
1650	C		Concen	5392507	26 - 42	REDUCER CONC SCH XS LTCS A420 WPL6-W	28 - 44							
1660	C		Concen	5392260	40 - 44	REDUCER CONC WT 14.0MM/ SCH XS LTCS A420 WPL6-W	46 - 48							
1670	C		Concen	5392315	42 - 44	REDUCER CONC WT 15.0MM/ XS LTCS A420 WPL6-W	56 - 56							
1680	C		Concen	5392261	46 - 48	REDUCER CONC WT 15.0MM/ 14.0MM LTCS A420 WPL6-W	56 - 56							
1690														
1700	C		Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2							
1710			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4							
1720			Eccent	5392450	3 - 12	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 14							
1730			Eccent	5392291	8 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6	16 - 16							
1740			Eccent	5392298	10 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18 - 20							
1750			Eccent	5392288	16 - 20	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18 - 24							
1760			Eccent	5392513	18 - 24	REDUCER ECC SCH XS/20 LTCS A420 WPL6-W	26 - 36							
1770	C		Eccent	5392514	26 - 42	REDUCER ECC SCH XS LTCS A420 WPL6-W	28 - 44							
1780	C		Eccent	5392262	40 - 44	REDUCER ECC WT 14.0MM/SCH XS LTCS A420 WPL6-W	46 - 48							
1790	C		Eccent	5392515	42 - 44	REDUCER ECC 15.0MM/ XS LTCS A420 WPL6-W	56 - 56							
1800	C		Eccent	5392263	46 - 48	REDUCER ECC WT 15.0MM/14.0MM LTCS A420 WPL6-W	56 - 56							
1810														
1820			Nipofl	5504001	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	2 - 2							
1830			Nipofl	5504002	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	3 - 14							
1840			Nipofl	5504016	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2	16 - 24							
1850	C		Nipofl	5504001	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	26 - 44							
1860	C		Nipofl	5504157	3/4 - 1 1/2	NIPOFLANGE 150# RF WT 14.0MM/SCH XS LTCS A350 LF2	46 - 48							
1870			Nipofl	5504100	3/4 - 1 1/2	NIPOFLANGE 150# RF WT 15.0MM/XS LTCS A350 LF2	56 - 56							
1880														
1890	D		Nipofl	5504020	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	1 1/2 - 2							
1900	C		Nipofl	5504021	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 3							
1910														
1920		UJ2	Nipofl	5504025	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14							
1930		UJ2	Nipofl	5504019	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24							
1940	C	UJ2	Nipofl	5504023	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	26 - 44							
1950	C	UJ2	Nipofl	5504158	3/4 - 1 1/2	NIPOFLANGE 300# RF WT 14.0MM/160 LTCS A350 LF2	46 - 48							
1960	C	UJ2	Nipofl	5504112	3/4 - 1 1/2	NIPOFLANGE 300# RF WT 15.0MM/160 LTCS A350 LF2	56 - 56							
1970														
1980			Nipofl	5504002	2 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	3 - 14							
1990			Nipofl	5504016	2 - 2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2	16 - 24							
2000	C		Nipofl	5504001	2 - 2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	26 - 44							
2010	C		Nipofl	5504157	2 - 2	NIPOFLANGE 150# RF WT 14.0MM/SCH XS LTCS A350 LF2	46 - 48							
2020			Nipofl	5504100	2 - 2	NIPOFLANGE 150# RF WT 15.0MM/XS LTCS A350 LF2	56 - 56							
2030														
2040			Weldol	5505335	3/4 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2							
2050			Weldol	5505179	3/4 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 14							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES. FOR SERVICE CODES CDF, WE AND WSC - WT > 19MM		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								3 MM	17MAR17	SMS	D	4 OF 8	5	150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2060			Weldol	5505572	3/4 - 2	WELDOLET SCH 20/XS LTCS A350 LF2	16 - 24						
2070	C		Weldol	5505335	3/4 - 2	WELDOLET SCH XS LTCS A350 LF2	26 - 44						
2080	C		Weldol	5505560	3/4 - 2	WELDOLET WT 14.0MM/SCH XS LTCS A350 LF2	46 - 48						
2090			Weldol	5505575	3/4 - 2	WELDOLET WT 15.0MM/XS LTCS A350 LF2	56 - 56						
2100			Weldol	5505164	3 - 4	WELDOLET SCH STD LTCS A350 LF2	8 - 14						
2110			Weldol	5505571	3 - 8	WELDOLET SCH 20/STD LTCS A350 LF2	16 - 24						
2120	C		Weldol	5505334	3 - 12	WELDOLET SCH XS/STD LTCS A350 LF2	26 - 44						
2130	C		Weldol	5505561	3 - 12	WELDOLET WT 14.0MM/STD LTCS A350 LF2	46 - 48						
2140			Weldol	5505576	3 - 6	WELDOLET WT 15.0MM/STD LTCS A350 LF2	56 - 56						
2150													
2160		UJ2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14						
2170		UJ2	Elbole	5516619	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24						
2180	C	UJ2	Elbole	5516623	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	26 - 44						
2190	C	UJ2	Elbole	5516650	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 14.0MM/160 LTCS A350 LF2	46 - 48						
2200		UJ2	Elbole	5516682	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 15.0MM/SCH 160 LTCS A350 LF2	56 - 56						
2210													
2220			90 Deg	5478701	3/4 - 1 1/2	ELL 90 DEG 3000# SW LTCS A350 LF2							
2230													
2240			45 Deg	5478702	3/4 - 1 1/2	ELL 45 DEG 3000# SW LTCS A350 LF2							
2250													
2260			Tee	5478703	3/4 - 1 1/2	TEE 3000# SW LTCS A350 LF2							
2270													
2280	D		Cap	5478704	3/4 - 1 1/2	CAP 3000# SW LTCS A350 LF2							
2290	D	UD6	Cap	5491312	3/4 - 3/4	CAP 3000# SCRDLTCS A350 LF2							
2300													
2310	D		Coupli	5478705	3/4 - 1 1/2	COUPLING 3000# SW LTCS A 350 LF2							
2320	D		Reduci	5478709	3/4 - 1	COUPLING RED 3000# SW LTCS A350 LF2	1 - 1 1/2						
2330													
2340	D		Union	5478706	3/4 - 1 1/2	UNION 3000# SW LTCS A350 LF2							
2350													
2360	D	UD7	Plug	5491303	3/4 - 3/4	PLUG ROUND HEAD SCRDLTCS A350 LF2							
2370													
2380	D		Sockol	5533072	3/4 - 1 1/2	SOCKOLET 3000# LTCS A350 LF2	2 - 56						
2390	D		Endole	5533232	3/4 - 1 1/2	FLAT SOCKOLET 3000# LTCS A350 LF2							
2400													
2410	D	UD7	Thredo	5514772	3/4 - 3/4	THREDOLET 3000# LTCS A350 LF2	2 - 56						
2420													
2430	D	SC1	Flange	5599690	3/4 - 3/4	FLG SCRDL (FNPT) 150# RF LTCS A350 LF2							
2440	D		Flange	5602303	3/4 - 1 1/2	FLG SW 150# RF LTCS A350 LF2 SCH XS							
2450	D		Flange	5602356	3/4 - 1 1/2	FLG SW 300# RF LTCS A350 LF2 SCH XS							
2460	D		Flange	5602484	3/4 - 3/4	FLG SW 600# RF LTCS A350 LF2 SCH XS							
2470													
2480	D	PJ9	Cross	5416600	2 - 2	CROSS SCH XS LTCS A420 WPL6							
2490	D	PJ9	Cross	5416601	3 - 6	CROSS SCH STD LTCS ASTM A420 WPL6							
2500	D	PJ9	Cross	5416601	8 - 10	CROSS SCH STD LTCS ASTM A420 WPL6							
2510	D		Cross	5416602	3 - 6	CROSS RED SCH STD LTCS ASTM A420 WPL6	4 - 8						
2520	D		Cross	5416602	8 - 8	CROSS RED SCH STD LTCS ASTM A420 WPL6	10 - 10						
2530													
2540	D		Nipple	5541184	1/2 - 1/2	NIPPLE SCH XS/160 LTCS A333-6 POE/TOE 3" LG							
2550	D		Nipple	5541183	1/2 - 1/2	NIPPLE SCH XS/160 LTCS A333-6 POE/TOE 7" LG							
2560	D		Nipple	5541172	3/4 - 1 1/2	NIPPLE SCH 160 LTCS A333 GR.6 TBE 3" LONG							
2570	D		Nipple	5541173	3/4 - 1 1/2	NIPPLE SCH 160 LTCS A333 GR.6 TBE 7" LONG							
2580													
2590													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES. FOR SERVICE CODES CDF, WE AND WSC - WT > 19MM	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 5 OF 8 PA 5 CLASS 150K21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2600						***** SWAGE NIPPLES *****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 REFER TO PIM-SU-5112 FOR ALL FLUID SERVICES MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 232 °C COMMENT: REFER NOTE 800					
2610													
2620													
2630	D		Concen	5544132	3/4 - 1	SWAGE CONC SCH 160 LTCS A420 WPL6 TBE	1 - 1 1/2						
2640	D		Concen	5544094	3/4 - 1	SWAGE CONC SCH XS/160 LTCS A420 WPL6 PLE/TSE	1 - 1 1/2						
2650	D		Concen	5544135	3/4 - 1	SWAGE CONC SCH 160/XS LTCS A420 WPL6 TLE/PSE	1 - 1 1/2						
2660	D		Concen	5544084	3/4 - 1	SWAGE CONC SCH XS LTCS A420 WPL6 BLE/PSE	2 - 2						
2670	D		Concen	5544156	3/4 - 1 1/2	SWAGE CONC SCH STD/XS LTCS A420 WPL6 BLE/PSE	3 - 4						
2680													
2690													
2700	D		Eccent	5549222	3/4 - 1	SWAGE ECC SCH 160 LTCS A420 WPL6 TBE	1 - 1 1/2						
2710	D		Eccent	5549185	3/4 - 1	SWAGE ECC SCH XS/160 LTCS A420 WPL6 PLE/TSE	1 - 1 1/2						
2720	D		Eccent	5549195	3/4 - 1	SWAGE ECC SCH 160/XS LTCS A420 WPL6 TLE/PSE	1 - 1 1/2						
2730	D		Eccent	5549174	3/4 - 1 1/2	SWAGE ECC SCH XS LTCS A420 WPL6 BLE/PSE	2 - 2						
2740	D		Eccent	5549156	3/4 - 1 1/2	SWAGE ECC SCH STD/XS LTCS A420 WPL6 BLE/PSE	3 - 4						
2750													
2760													
2770		FS1				***** PIPE FLANGES *****							
2780													
2790		FA8	Reduci	5599692	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2	3/4 - 1 1/2						
2800													
2810		FA8	Reduci	5599703	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 3/4						
2820													
2830	C		Flange	5851200	1/2 - 3/4	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
2840	C		Flange	5851200	2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
2850	C		Flange	5851203	3/4 - 3/4	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
2860													
2870	C		Flange	5556264	1/2 - 2	FLG WN 150# RF LTCS A350 LF2 SCH XS							
2880			Flange	5556261	2 1/2 - 14	FLG WN 150# RF LTCS A350 LF2 SCH STD							
2890			Flange	5556265	16 - 24	FLG WN 150# RF LTCS A350 LF2 SCH 20							
2900	C		Flange	5556266	26 - 44	FLG WN 150# RF LTCS A350 LF2 SCH XS B16.47 SERIES A							
2910	C		Flange	5556250	46 - 48	FLG WN 150# RF LTCS A350 LF2 WT 14.0MM B16.47 SERIES A							
2920			Flange	5556271	56 - 56	FLG WN 150# RF LTCS A350 LF2 WT 15.0MM B16.47 SERIES A							
2930													
2940	C		Flange	5556476	1/2 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS							
2950			Flange	5556473	3 - 14	FLG WN 300# RF LTCS A350 LF2 SCH STD							
2960			Flange	5556474	16 - 24	FLG WN 300# RF LTCS A350 LF2 SCH 20							
2970	C		Flange	5556477	26 - 44	FLG WN 300# RF LTCS A350 LF2 SCH XS B16.47 SERIES A							
2980	C		Flange	5556450	46 - 48	FLG WN 300# RF LTCS A350 LF2 WT 14.0MM B16.47 SERIES A							
2990			Flange	5556478	56 - 56	FLG WN 300# RF LTCS A350 LF2 WT 15.0MM B16.47 SERIES A							
3000													
3010			Flange	5557036	3/4 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS							
3020	D		Flange	5557033	3 - 12	FLG WN 600# RF LTCS A350 LF2 SCH STD							
3030	C		Flange	5557285	16 - 24	FLG WN 600# RF LTCS A350 LF2 SCH 20							
3040													
3050	D	FS1	Flange	5556557	26 - 26	FLG WN 600# RF LTCS A350 LF2 SCH XS B16.47 SERIES A							
3060													
3070	C	FS1	Flange	5555935	3 - 10	FLG WN 900# RF LTCS A350 LF2 SCH STD							
3080													
3090	D	FS1	Flange	5557393	1 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH XS							
3100													
3110	C		Blind	5615566	1/2 - 24	FLG BLIND 150# RF LTCS A350 LF2							
3120			Blind	5615570	26 - 56	FLG BLIND 150# RF LTCS A350 LF2 B16.47 SERIES A							
3130													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES. FOR SERVICE CODES CDF, WE AND WSC - WT > 19MM	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 6 OF 8	PA 5	CLASS 150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3140	C		Blind	5615576	1/2 - 24	FLG BLIND 300# RF LTCS A350 LF2		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 REFER TO PIM-SU-5112 FOR ALL FLUID SERVICES MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 232 °C COMMENT: REFER NOTE 800					
3150			Blind	5615579	26 - 56	FLG BLIND 300# RF LTCS A350 LF2 B16.47 SERIES A							
3160						***** ORIFICE FLANGES *****							
3170													
3180													
3190			Orific	5623947	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS							
3200			Orific	5623944	3 - 14	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD							
3210			Orific	5623941	16 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 20							
3220	C		Orific	5624800	26 - 44	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS ASME B16.47 SERIES A							
3230	C		Orific	5624828	46 - 48	WN ORIF SET 300# RF LTCS A350 LF2 WT 14.0MM ASME B16.47 SERIES A							
3240			Orific	5624804	56 - 56	WN ORIF SET 300# RF LTCS A350 LF2 15MM WT ASME B16.47 SERIES A							
3250													
3260						***** PIPE NIPPLES *****							
3270													
3280		810				NIPPLE PBE - CUT FROM PIPE							
3290													
3300		FS1				***** GASKETS *****							
3310													
3320	C		GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
3330			GASFRS	5674011	26 - 56	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
3340													
3350	D		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR							
3360			GASFRS	5674032	26 - 56	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
3370													
3380	D		GASFRS	5673103	3/4 - 24	GASKET 600# 316SS SP WND GRAPHITE IR							
3390	D		GASFRS	5674012	26 - 26	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
3400	C		GASFRS	5674012	30 - 36	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
3410													
3420	D		GASFRS	5673117	1 1/2 - 10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
3430													
3440	D		GASFRS	5672251	1 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
3450		B1				***** BOLTING *****							
3460													
3470													
3480	D		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3490		SB6	CPSBOL	5675675	12 - 56	JACK SCREW HEX HEAD A320 GR L7M							
3500													
3510						***** BLINDS AND SPACERS *****							
3520													
3530			Specta	5749580	3/4 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70							
3540			Specta	5749581	1 1/2 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
3550													
3560			Paddle	5750023	16 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70							
3570			Paddle	5750026	26 - 56	PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A							
3580													
3590			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70							
3600			Paddle	5750028	26 - 56	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A							
3610													
3620			Paddle	5750240	16 - 24	PADDLE SPACER 150# RF LTCS A516 GR.70							
3630			Paddle	5750196	26 - 56	PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A							
3640													
3650			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70							
3660			Paddle	5750198	26 - 56	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A							
3670													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES. FOR SERVICE CODES CDF, WE AND WSC - WT > 19MM	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 7 OF 8	PA 5	CLASS 150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3680						***** BRANCHES ***** USE BRANCH CHART 150K21		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 REFER TO PIM-SU-5112 FOR ALL FLUID SERVICES MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 232 °C COMMENT: REFER NOTE 800					
3690													
3700													
3710	C		Reinfo	5358577_R	14 - 14	REINFORCING PAD FROM PIPE A671 CC60 SCH XS	34 - 44						
3720	C		Reinfo	5358577_R	14 - 14	REINFORCING PAD FROM PIPE A671 CC60 SCH XS	34 - 44						
3730													
3740	C		Reinfo	5358540_R	14 - 20	REINFORCING PAD FROM PIPE A671 CC60 14.0MM WT	48 - 48						
3750	C		Reinfo	5358540_R	14 - 20	REINFORCING PAD FROM PIPE A671 CC60 14.0MM WT	48 - 48						
3760													
3770			Reinfo	5358610_R	14 - 14	REINFORCING PAD FROM PIPE A671 CC60 CL22 15MM WT	56 - 56						
3780	C		Reinfo	5358610_R	14 - 14	REINFORCING PAD FROM PIPE A671 CC60 CL22 15MM WT	56 - 56						
3790													
3800			Reinfo	5358577_R	16 - 16	REINFORCING PAD FROM PIPE A671 CC60 SCH XS	38 - 40						
3810	C		Reinfo	5358577_R	16 - 16	REINFORCING PAD FROM PIPE A671 CC60 SCH XS	38 - 40						
3820													
3830													
3840	D		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 56						
3850	D		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 56						
3860													
3870													
3880													
3890						***** CONSTRUCTION DETAILS *****							
3900													
3910	D	ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
3920						PIPE BENDS ----- NONE							
3930						THREAD SEALANT ----- PTFE PASTE							
3940													
3950													
3960						FOR FABRICATION AND WELDING DETAILS							
3970						SEE DOC. Nº L-ST-2024							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES. FOR SERVICE CODES CDF, WE AND WSC - WT > 19MM	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 7 OF 8	PA 5	CLASS 150K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																			
10						***** VALVES *****		812	NIPPLE TO BE CUT FROM SCH XXS PIPE																		
20						B1		ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.																			
30						FA8		USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.																			
40						FS1		USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.																			
50						2BV		K01AM10	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.															
60										UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.																
70										VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.																
80										VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT																
90						FS1		2BVR	M03AM10	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	2 - 2	3 - 3	L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.													
100											2BVR			M01AM10	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.									
110						VAW		R2BVR	M03EM10	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR			VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.												
120											VC2			DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.													
130						VAW		R2BVR	M03EM10	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR			3 - 3	3 - 3												
140											VB2					Ball V	A01AM09	3 - 3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO								
150																			VA5	Ball V	A01AR09	4 - 6	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT RB LO				
160																							VA5	Ball V	A01AR15	8 - 16	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT RB GO
170						VA5		Ball V	B01AM09	3/4 - 2																	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO
180											VA5					Ball V	B01AM09	3 - 3									BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO
190																			VA5	Ball V	B01AR11	4 - 6					BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT FB LO
200																							VA5	Ball V	B01AR15	8 - 16	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT FB GO
210						FS1		Ball V	B03AM09	3/4 - 2																	BALL 300# RF INC 825 INC 625 TRIM FLOATING SOFT FB LO
220											FS1					Ball V	A03AM09	3 - 3									BALL 300# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO
230																			Butter	T01AR63	10 - 20	***** BUTTERFLY VALVES *****					
240																						Globe	L01AM49	3/4 - 3	GLOBE 150# RF INC 825 INC TRIM 625 OS&Y HW		
250						Globe		L01AR49	4 - 6	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y HW																	
260										Globe	L01AR44					8 - 10	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y GO										
270	VC1	Check	C01AM51	3/4 - 1 1/2	***** CHECK VALVES *****																						
280					VC2		Check										C01AM46	2 - 3	CHECK PISTON 150# RF INC 825 INC TRIM 625								
290						VC2		Check	C01AM46										2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED							
300										VC2	Check					C01AM46				2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED						
310	VC2	Check	C01AM46	2 - 3																	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED						
320					VC2		Check										C01AM46	2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED						
330						VC2		Check	C01AM46										2 - 3		CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED						
340										VC2	Check					C01AM46				2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED						
350	VC2	Check	C01AM46	2 - 3								CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED															
360					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
370						VC2		Check	C01AM46								2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED									
380										VC2	Check					C01AM46		2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED								
390	VC2	Check	C01AM46	2 - 3										CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED													
400					VC2		Check					C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED													
410						VC2		Check	C01AM46					2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED												
420										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
430	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
440					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
450						VC2		Check	C01AM46					2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
460										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
						VC2		Check	C01AM46					2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
						VC2		Check	C01AM46					2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
						VC2		Check	C01AM46					2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
						VC2		Check	C01AM46					2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
						VC2		Check	C01AM46					2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
						VC2		Check	C01AM46					2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
						VC2		Check	C01AM46					2 - 3			CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
										VC2	Check				C01AM46	2 - 3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
	VC2	Check	C01AM46	2 - 3													CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										
					VC2		Check					C01AM46	2 - 3				CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED										

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
470		VC2	Check	C01AR46	4 - 6	CHECK DUAL WAFER LUG 150# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED							
480		VC2	Check	C01AR53	8 - 16	CHECK DUAL DBL FLG 150# RF LTCS + CLAD INC 825 INC TRIM 625							
490													
500													
510						***** PIPE *****							
520													
530		812	Piping	5358511	3/4 - 1 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE							
540			Piping	5358513	2 - 2	PIPE SCH XXS SMLS LTCS A333 GR.6 BE							
550			Piping	5358514	3 - 16	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
560			Piping	5358577	20 - 20	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE							
570													
580													
590						***** FITTINGS *****							
600													
610			90 Deg	5392870	3/4 - 2	ELL 90 DEG LR XXS LTCS A420 WPL6							
620			90 Deg	5392600	3 - 16	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
630													
640			45 Deg	5392876	3/4 - 2	ELL 45 DEG LR XXS LTCS A420 WPL6							
650			45 Deg	5392606	3 - 20	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
660													
670			Tee	5392879	3/4 - 2	TEE SCH XXS LTCS A420 WPL6							
680			Tee	5392609	3 - 16	TEE BW SCH XS LTCS A420 WPL6							
690													
700			Reduci	5392877	3/4 - 1 1/2	TEE RED SCH XXS LTCS A420 WPL6	1 - 2						
710			Reduci	5392897	1 1/2 - 2	TEE RED BW SCH XS/XXS LTCS A420 WPL6	3 - 4						
720			Reduci	5392661	3 - 14	TEE RED BW SCH XS LTCS A420 WPL6	4 - 16						
730			Reduci	5392413	16 - 16	TEE RED SCH XS LTCS A420 WPL6-W	20 - 20						
740													
750			Cap	5392880	3/4 - 2	CAP SCH XXS LTCS A420 WPL6							
760			Cap	5392610	3 - 16	CAP BW SCH XS LTCS A420 WPL6							
770													
780			Concen	5392881	3/4 - 1 1/2	REDUCER CONC SCH XXS LTCS A420 WPL6	1 - 2						
790			Concen	5392898	1 1/2 - 2	REDUCER CONC SCH XS/XXS LTCS A420 WPL6	3 - 4						
800			Concen	5392458	3 - 14	REDUCER CONC SCH XS LTCS A420 WPL6	4 - 16						
810													
820			Eccent	5392882	3/4 - 1 1/2	REDUCER ECC SCH XXS LTCS A420 WPL6	1 - 2						
830			Eccent	5392899	1 1/2 - 2	REDUCER ECC SCH XS/XXS LTCS A420 WPL6	3 - 4						
840			Eccent	5392461	3 - 14	REDUCER ECC SCH XS LTCS A420 WPL6	4 - 16						
850													
860			Nipofl	5504211	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XXS LTCS A350 LF2	2 - 2						
870			Nipofl	5504212	3/4 - 2	NIPOFLANGE 150# RF SCH XS/XXS LTCS A350 LF2	3 - 16						
880													
890			Nipofl	5504215	1 - 1 1/2	NIPOFLANGE 300# RF SCH XXS LTCS A350 LF2	1 1/2 - 2						
900			Nipofl	5504216	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2	3 - 3						
910			Nipofl	5504216	3/4 - 3/4	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2	3 - 16						
920													
930		UJ2	Nipofl	5504213	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2	4 - 16						
940													
950			Weldol	5505197	3/4 - 1	WELDOLET SCH XXS LTCS A350 LF2	2 - 2						
960			Weldol	5505180	3/4 - 2	WELDOLET SCH XS/XXS LTCS A350 LF2	3 - 16						
970			Weldol	5505335	3 - 4	WELDOLET SCH XS LTCS A350 LF2	8 - 16						
980													
990		UJ2	Elbole	5516762	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2	4 - 16						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES		FGP/WPMP CORR. ALLOW. DATE BY REV SHEET PA CLASS 6 MM 17MAR17 RRP A 2 OF 4 5 150K23					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES												
1000		FS1				***** PIPE FLANGES *****														
1010		FA8	Reduci	5599692	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 150# RF LTCS A350 LF2	3/4 - 1 1/2													
1020																				
1030		FA8	Reduci	5599703	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 3/4													
1040																				
1050			Flange	5851200	1/2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE														
1060			Flange	5851203	3/4 - 3/4	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE														
1070			Flange	5556221	3/4 - 2	FLG WN 150# RF LTCS A350 LF2 SCH XXS														
1080			Flange	5556264	3 - 20	FLG WN 150# RF LTCS A350 LF2 SCH XS														
1090			Flange	5556481	3/4 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XXS														
1100			Flange	5556476	3 - 16	FLG WN 300# RF LTCS A350 LF2 SCH XS														
1110			Flange	5556976	3/4 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XXS														
1120			Flange	5557036	3 - 16	FLG WN 600# RF LTCS A350 LF2 SCH XS														
1130			Flange	5557141	6 - 8	FLG WN 900# RF LTCS A350 LF2 SCH XS														
1140																				
1150			Blind	5615566	3/4 - 16	FLG BLIND 150# RF LTCS A350 LF2														
1160			Blind	5615576	3/4 - 16	FLG BLIND 300# RF LTCS A350 LF2														
1170																				
1180			Orific	5623950	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XXS														
1190			Orific	5623947	3 - 16	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS														
1200																				
1210																				
1220																				
1230																				
1240																				
1250																				
1260																				
1270																				
1280																				
1290																				
1300																				
1310																				
1320																				
1330		812																		
1340																				
1350		FS1																		
1360																				
1370																				
1380																				
1390			GASF RS	5673043	3/4 - 20	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR														
1400			GASF RS	5673033	3/4 - 16	GASKET 300# 316SS SP WND GRAPHITE IR														
1410			GASF RS	5673103	3/4 - 16	GASKET 600# 316SS SP WND GRAPHITE IR														
1420			GASF RS	5673117	6 - 8	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR														
1430																				
1440																				
1450		B1																		
1460																				
1470																				
1480																				
1490			BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS														
1500																				
1510		SB6	CPSBOL	5675675	12 - 16	JACK SCREW HEX HEAD A320 GR L7M														
1520																				
1530																				
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - ALL SIZES	CORR. ALLOW. 6 MM							DATE 17MAR17	BY RRP	REV A	SHEET 3 OF 4	PA 5	CLASS 150K23	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1540			Specta Specta Paddle Paddle Paddle Paddle	5749580 5749581 5750023 5750100 5750240 5750200	3/4 - 14 1 1/2 - 10 16 - 20 12 - 16 16 - 20 12 - 16	***** BLINDS AND SPACERS ***** SPECTACLE BLIND 150# RF LTCS A 516 GR.70 SPECTACLE BLIND 300# RF LTCS A516 GR.70 PADDLE BLANK 150# RF LTCS A516 GR.70 PADDLE BLANK 300# RF LTCS A516 GR.70 PADDLE SPACER 150# RF LTCS A516 GR.70 PADDLE SPACER 300# RF LTCS A516 GR.70 ***** BRANCHES ***** USE BRANCH CHART 150K23 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE	3 - 20 3 - 20						
1550													
1560													
1570													
1580													
1590													
1600													
1610													
1620													
1630													
1640													
1650													
1660													
1670													
1680													
1690													
1700													
1710													
1720													
1730													
1740													
1750													
1760													
									SERVICE				
									SOUR SERVICE - NACE MR0175 / ISO 15156				
									DESIGN: ASME B16.5 MG 1.1				
									CODE: ASME B31.3				
									BB, BV, DL, WE, WP, WT				
									MATERIAL: LOW TEMP CARBON STEEL				
									FLANGE RATING: 150 RF				
									TEMPERATURE RANGE: -46 / 200 °C				

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
10		PP9						032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.							
11		800						800	THE TEMPERATURE IS LIMITED TO 260°C IN CRUDE OIL SERVICES CONTAINING H2S /AND /OR ORGANIC SULPHUR COMPOUNDS, FOR EXAMPLE, IN THE CRUDE STABILIZATION UNIT.							
20			Ball V	M01AC02	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 316 SS TRIM METAL LO			IN HYDROCARBON GAS SERVICE CONTAINING H2S, FOR EXAMPLE, SOUR GAS MOLECULAR SIEVE DEHYDRATION UNITS, USE IS LIMITED TO 230 C. EXCEPTIONS TO THIS ARE THE SULPHUR RECOVERY AND TAIL GAS UNITS WHERE CARBON STEEL SOUR SERVICE LINE CLASSES CAN BE USED UP TO 315°C BECAUSE OF LOWER H2S PARTIAL PRESSURES.							
30								810	NIPPLE TO BE CUT FROM SCH 160 PIPE							
40								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.							
50		VB2						BE3	MITRE BENDS SHALL BE FABRICATED FROM THE FOLLOWING WALL THICKNESS PIPE:							
60			BALFLG	A01AC01	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO			36" NB THRU 44" NB - 10.0MM							
70									48" NB THRU 56" NB - 12.0MM							
80			BALFLG	A01AC02	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO			60" NB THRU 80" NB - 14.0MM							
90			BALFGO	A01AC03	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB GO			84" NB THRU 92" NB - 16.0MM							
100			BALFGO	A01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB GO			BUTT WELD ENDS OF BENDS TO BE BEVELLED TO SUIT 10.0MM PIPE.							
110			BALFGO	A01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO		FA4	SLIP ON FLANGE SHALL ONLY BE USED WHERE INSTRUMENT ENCLOSURES ARE INSTALLED.							
120			BALFGO	A01AC08	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO										
130																
140			BALFLG	B01AC01	3/4 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO										
150			BALFLG	B01AC02	3/4 - 2	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO										
160																
170			BALFLG	B01AC01	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO										
180																
190		VA5	BALFLG	B01AC02	3 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO										
200																
210			BALFGO	B01AC03	4 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB GO										
220	C															
230		VA5														
240																
250	C	VA5	BALFGO	B01AC04	4 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB GO			SERVICE							
260									SOUR SERVICE - NACE MR0175/ ISO 15156,							
270	C		BALFGO	B01AC07	6 - 24	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO			DESIGN : ASME B16.47, ASME VIII CODE : ASME B31.3 DESIGN PRESSURE - 5 BARG MAXIMUM GCL, GRC, GRI, GRK, GS, GST, VCB							
280		VA5														
290																
300	C	VA5	BALFGO	B01AC08	6 - 24	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO										
310																
320			BALFGO	B01JC07	30 - 60	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A			MATERIAL: LOW TEMP CARBON STEEL							
330		VA5							FLANGE RATING: 150 RF							
340									TEMPERATURE RANGE: -36 / 343 °C							
350		VA5	BALFGO	B01JC08	30 - 60	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A			COMMENT: REFER TO NOTE 800 FOR LEGACY							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP								
						ALL SIZES - BUTTWELD PWHT - ALL SIZES		CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS	
								3 MM		11JUL13	MLA	C	1 OF 9	5	150K26	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
360						*****BUTTERFLY VALVE*****		FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.						
370								FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.						
380			Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO		NN7	ENDS OF MAIN RUN AND BRANCH SHALL BE BEVELLED, SUITABLE FOR ATTACHMENT TO 10.0MM WT. PIPE						
390			Butter	T01JC61	30 - 60	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A		NR4	USE SWAGES WHERE SMALL END IS 2" OR SMALLER AND LARGE END IS 4" OR SMALLER. USE REDUCERS FOR SMALL END LARGER THAN 2".						
400			Gate V	G01AC48	3/4 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW		PA1	22IN FOR PIPE AND FITTINGS IS A NON-PREFERRED SIZE. CONTACT MATERIAL ENGINEER BEFORE USE FOR APPROVED SCHEDULE.						
410			Gate V	G01AC41	2 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW		PG1	USE FOR PRESSURE GAUGES WITH CHEMICAL SEALS.						
420						*****GLOBE VALVES*****		PJ9	FOR USE IN NON-JACKETED SULPHUR LINES ONLY.						
430								PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
440			Globe	L01AC48	3/4 - 1 1/2	GLOBE 150# RF LTCS A350 LF2 TRIM 16 OS&Y HW		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
450			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW		SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.						
460			Globe	L01AC42	8 - 10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO		UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.						
470						*****CHECK VALVES*****		SERVICE							
480								SOUR SERVICE - NACE MR0175/ ISO 15156,							
490								DESIGN : ASME B16.47, ASME VIII							
500								CODE : ASME B31.3							
510								DESIGN PRESSURE - 5 BARG MAXIMUM							
520								GCL, GRC, GRI, GRK, GS, GST, VCB							
530								MATERIAL: LOW TEMP CARBON STEEL							
540								FLANGE RATING: 150 RF							
550								TEMPERATURE RANGE: -36 / 343 °C							
560								COMMENT: REFER TO NOTE 800 FOR LEGACY							
570			VC1	Check	C01AC50	3/4 - 1 1/2	CHECK PISTON 150# RF LTCS A350 LF2 TRIM 16								
580			VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED								
590			VC2	Check	C01AC52	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16								
600			VC2	Check	C01JC52	30 - 60	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A								
610			PA1			**** PIPE ****									
620															
630			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE									
640			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE									
650			Piping	5358505	3 - 14	PIPE SCH STD SMLS LTCS A333 GR.6 BE									
660			Piping	5358501	16 - 16	PIPE SCH 20 SMLS LTCS A333 GR.6 BE									
670			Piping	5358597	18 - 24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE									
680			Piping	5358605	30 - 30	PIPE SCH 10 EFW LTCS A671 CC70 CL.22 BE									
690			Piping	5358607	36 - 92	PIPE WT 10.0MM EFW LTCS SHOP FABRICATED A671 CC70 CL.22 BE									
700															
710			810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE								
720						***** FITTINGS *****									
730															
740															
750			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6									
760			90 Deg	5392438	3 - 14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6									
770			90 Deg	5392276	16 - 16	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6									
780			90 Deg	5392280	18 - 24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W									
790			90 Deg	5392265	30 - 30	ELL BW 90 DEG LR SCH 10 LTCS A420 WPL6-W									
800			BE3	90 Deg	5393006	36 - 92	90 DEG THREE CUT MITRE BEND LONG RADIUS SHOP FABRICATED ASTM A671 CC70 CL2								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP							
						ALL SIZES - BUTTWELD PWHT - ALL SIZES		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
								3 MM	11JUL13	MLA	C	2 OF 9	5	150K26	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
810								UD7 USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST. UJ2 USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. VA5 USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. VAD FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE). VAF FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY. VAG FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY. VB2 SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. VC1 SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. VC2 DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. VU2 USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. ZA6 REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
820			90 DEG	5392240	30 - 30	ELL BW 90 DEG SR SCH 10 LTCS A420 WPL6-W								
830		BE3	90 Deg	5393010	36 - 92	90 DEG THREE CUT MITRE BEND SHORT RADIUS SHOP FABRICATED ASTM A671 CC70 CL2								
840														
850			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
860			45 Deg	5392444	3 - 14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6								
870			45 Deg	5392282	16 - 16	ELL BW 45 DEG SCH 20 LTCS A420 WPL6								
880			45 Deg	5392281	18 - 24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W								
890			45 Deg	5392275	30 - 30	ELL BW 45 DEG LR SCH 10 LTCS A420 WPL6-W								
900		BE3	45 Deg	5393026	36 - 92	45 DEG TWO CUT MITRE BEND LONG RADIUS SHOP FABRICATED ASTM A671 CC70 CL2								
910														
920			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6								
930			Tee	5392447	3 - 14	TEE BW SCH STD LTCS A420 WPL6								
940			Tee	5392285	16 - 16	TEE BW SCH 20 LTCS A420 WPL6								
950			Tee	5392295	18 - 24	TEE BW SCH 20 LTCS A420 WPL6-W								
960			Tee	5392284	30 - 30	TEE BW SCH 10 LTCS A420 WPL6-W								
970		NN7	Tee	5393032	36 - 56	TEE WT 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22								
980		NN7	Tee	5393036	60 - 68	TEE WT 19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22								
990		NN7	Tee	5393038	76 - 92	TEE WT 25.0MM LTCS SHOP FABRICATED A671 CC70 CL 22								
1000														
1010			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2							
1020			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 4							
1030			Reduci	5392445	3 - 12	TEE RED BW SCH STD LTCS A420 WPL6	4 - 14							
1040			Reduci	5392289	6 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6	16 - 16							
1050			Reduci	5392296	8 - 14	TEE RED BW SCH 20/STD LTCS A420 WPL6-W	18 - 24							
1060			Reduci	5392292	16 - 20	TEE RED BW SCH 20 LTCS A420 WPL6-W	18 - 24							
1070			Reduci	5392274	30 - 32	TEE RED BW SCH 10 LTCS A420 WPL6-W	32 - 34							
1080			Reduci	5393072	24 - 24	TEE REDUCING WT 15.0MM/SCH 20 LTCS SHOP FABRICATED A671 CC70 CL 22	44 - 44							
1090			Reduci	5393067	28 - 28	TEE REDUCING WT 15.0MM/SCH XS LTCS SHOP FABRICATED A671 CC70 CL 22	52 - 52							
1100			Reduci	5393068	36 - 36	TEE REDUCING WT 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	54 - 54							
1110		NN7	Reduci	5393042	52 - 52	TEE REDUCING WT 19.0MM / 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	60 - 60							
1120		NN7	Reduci	5393042	44 - 44	TEE REDUCING WT 19.0MM / 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	68 - 68							
1130		NN7	Reduci	5393035	60 - 60	TEE REDUCING WT 19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	68 - 68							
1140			Reduci	5393071	28 - 28	TEE REDUCING WT 25.0MM / SCH XS LTCS SHOP FABRICATED A671 CC70 CL 22	70 - 70							
1150			Reduci	5393037	60 - 60	TEE REDUCING WT 25.0MM /19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	70 - 70							
1160		NN7	Reduci	5393037	60 - 60	TEE REDUCING WT 25.0MM /19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	76 - 76							
1170		NN7	Reduci	5393041	76 - 76	TEE REDUCING WT 25.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	80 - 80							
1180		NN7	Reduci	5393037	60 - 60	TEE REDUCING WT 25.0MM /19.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	84 - 84							
1190		NN7	Reduci	5393044	56 - 56	TEE REDUCING WT 25.0MM / 15.0MM LTCS SHOP FABRICATED A671 CC70 CL 22	88 - 88							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES - BUTTWELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 11JUL13	BY MLA	REV C	SHEET 3 OF 9	PA 5	CLASS 150K26	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1200								SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN : ASME B16.47, ASME VIII CODE : ASME B31.3 DESIGN PRESSURE - 5 BARG MAXIMUM GCL, GRC, GRI, GRK, GS, GST, VCB MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -36 / 343 °C COMMENT: REFER TO NOTE 800 FOR LEGACY						
1210		PJ9	Cross	5416600	2 - 2	CROSS SCH XS LTCS A420 WPL6								
1220		PJ9	Cross	5416601	3 - 10	CROSS SCH STD LTCS ASTM A420 WPL6								
1230		PJ9	Cross	5416602	3 - 8	CROSS RED SCH STD LTCS ASTM A420 WPL6	4 - 10							
1240														
1250			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6								
1260			Cap	5392448	3 - 14	CAP BW SCH STD LTCS A420 WPL6								
1270			Cap	5392286	16 - 24	CAP BW SCH 20 LTCS A420 WPL6								
1280			Cap	5392318	30 - 30	CAP SCH 10 LTCS A420 WPL6								
1290			Cap	5392614	36 - 92	CAP BW WT 10.0MM LTCS STL A420 WPL6								
1300														
1310			Concen	5392452	2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4							
1320			Concen	5392449	2 1/2 - 12	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 14							
1330			Concen	5392290	8 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6	16 - 16							
1340			Concen	5392297	10 - 14	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18 - 20							
1350			Concen	5392287	16 - 20	REDUCER CONC SCH 20 LTCS A420 WPL6-W	18 - 24							
1360			Concen	5393050	30 - 30	REDUCER CONC SCH 10 LTCS A420 WPL6-W	32 - 32							
1370			Concen	5393070	24 - 24	REDUCER CONC WT 10.0MM/SCH 20 LTCS SHOP FABRICATED A671 CC70 CL22	36 - 36							
1380			Concen	5393051	30 - 34	REDUCER CONC WT 10.0MM/SCH 10 LTCS A420 WPL6-W	36 - 36							
1390			Concen	5393052	32 - 32	REDUCER CONC WT 10.0MM/SCH 10 LTCS SHOP FABRICATED A671 CC70 CL22	44 - 44							
1400			Concen	5393052	32 - 32	REDUCER CONC WT 10.0MM/SCH 10 LTCS SHOP FABRICATED A671 CC70 CL22	52 - 52							
1410			Concen	5393055	36 - 90	REDUCER CONC WT10.0MM LTCS SHOP FABRICATED A671 CC70 CL22	40 - 92							
1420														
1430			Eccent	5392453	2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4							
1440			Eccent	5392450	3 - 12	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 14							
1450			Eccent	5392291	8 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6	16 - 16							
1460			Eccent	5392298	10 - 14	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18 - 20							
1470			Eccent	5392288	16 - 20	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18 - 24							
1480			Eccent	5393069	24 - 24	REDUCER ECC WT10.0MM/SCH 20 LTCS SHOP FABRICTED A671 CC70 CL22	36 - 36							
1490			Eccent	5393060	30 - 30	REDUCER ECC SCH 10 LTCS A420 WPL6-W	32 - 32							
1500			Eccent	5393061	30 - 34	REDUCER ECC WT 10.0MM/SCH 10 LTCS A420 WPL6-W	36 - 36							
1510			Eccent	5393061	30 - 30	REDUCER ECC WT 10.0MM/SCH 10 LTCS A420 WPL6-W	40 - 40							
1520			Eccent	5393062	36 - 90	REDUCER ECC WT10.0MM LTCS SHOP FABRICTED A671 CC70 CL22	40 - 92							
1530						***** OLETS *****								
1540														
1550														
1560														
1570			Nipofl	5504001	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	2 - 2							
1580			Nipofl	5504002	3/4 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	3 - 14							
1590			Nipofl	5504016	3/4 - 2	NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2	16 - 24							
1600			Nipofl	5504472	3/4 - 2	NIPOFLANGE 150# RF SCH 10/ XS LTCS A350 LF2	30 - 30							
1610			Nipofl	5504099	3/4 - 2	NIPOFLANGE 150# RF SCH 10.0MM/XS LTCS A350 LF2	36 - 92							
1620														
1630			Nipofl	5504021	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	8 - 8							
1640		UJ2	Nipofl	5504025	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES - BUTTWELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 11JUL13	BY MLA	REV C	SHEET 4 OF 9	PA 5	CLASS 150K26	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1650		UJ2	Nipofl	5504019	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24							
1660		UJ2	Nipofl	5504015	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 10/160 LTCS A350 LF2	30 - 30							
1670			Nipofl	5504045	3/4 - 1 1/2	NIPOFLANGE 300# RF WT 10.0MM/160 LTCS A350 LF2	36 - 92							
1680														
1690		UJ2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14							
1700		UJ2	Elbole	5516619	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2	16 - 24							
1710		UJ2	Elbole	5516615	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 10/160 LTCS A350 LF2	30 - 30							
1720			Elbole	5516645	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 10.0MM/SCH 160 LTCS A350 LF2	36 - 92							
1730														
1740			Weldol	5505179	2 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	6 - 14							
1750			Weldol	5505572	2 - 2	WELDOLET SCH 20/XS LTCS A350 LF2	16 - 24							
1760			Weldol	5505337	2 - 2	WELDOLET SCH 10/ XS LTCS A350 LF2	30 - 30							
1770			Weldol	5505628	2 - 2	WELDOLET WT 10.0MM/SCH XS LTCS A350 LF2	36 - 92							
1780			Weldol	5505164	3 - 4	WELDOLET SCH STD LTCS A350 LF2	8 - 14							
1790			Weldol	5505571	3 - 8	WELDOLET SCH 20/STD LTCS A350 LF2	16 - 24							
1800			Weldol	5505165	3 - 12	WELDOLET SCH 10/STD WT LTCS A350 LF2	30 - 34							
1810			Weldol	5505600	3 - 12	WELDOLET WT 10.0MM/STD WT LTCS A350 LF2	36 - 92							
1820														
1830						***** PIPE FLANGES *****								
1840														
1850		FA8	Reduci	5599692	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 150# RF LTCS A350 LF2	3/4 - 1 1/2							
1860			Flange	5556264	3/4 - 2	FLG WN 150# RF LTCS A350 LF2 SCH XS								
1870			Flange	5556261	2 1/2 - 14	FLG WN 150# RF LTCS A350 LF2 SCH STD								
1880			Flange	5556265	16 - 24	FLG WN 150# RF LTCS A350 LF2 SCH 20								
1890			Flange	5556267	30 - 32	FLG WN 150# RF LTCS A350 LF2 SCH 10 B16.47 SERIES A								
1900			Flange	5556268	36 - 60	FLG WN 150# RF LTCS A350 LF2 WT 10.0MM B16.47 SERIES A								
1910														
1920														
1930			Blind	5615566	3/4 - 24	FLG BLIND 150# RF LTCS A350 LF2								
1940			Blind	5615570	30 - 60	FLG BLIND 150# RF LTCS A350 LF2 B16.47 SERIES A								
1950														
1960			Flange	5619988	62 - 92	FLG COMPACT 150# RJ LTCS A350 LF2 WT 10.0MM								
1970														
1980			Blind	5619750	62 - 92	FLG COMPACT BLIND 150# RJ LTCS A350 LF2								
1990														
2000						***** PIPE NIPPLES *****								
2010														
2020						NIPPLE PBE CUT FROM PIPE								
2030														
2040														
2050														
2060						***** GASKETS *****								
2070														
2080														
2090			GASFRS	5673043	3/4 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR								
2100			GASFRS	5674011	30 - 60	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A								
2110														
2120			GASRJ	5673046	62 - 92	GASKET 150# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE								
2130														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES - BUTTWELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 11JUL13	BY MLA	REV C	SHEET 5 OF 9	PA 5	CLASS 150K26	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2140	C	B1	BOLS Y	5675770 5675675	1/4 - 99 12 - 99	***** BOLTING *****	1 – 1 1/2	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN : ASME B16.47, ASME VIII CODE : ASME B31.3 DESIGN PRESSURE - 5 BARG MAXIMUM GCL, GRC, GRI, GRK, GS, GST, VCB MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -36 / 343 °C COMMENT: REFER TO NOTE 800 FOR LEGACY						
2150														
2160														
2170														
2180														
2190														
2200														
2210														
2220														
2230														
2240														
2250		Paddle	5750023	16 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70									
2260		Paddle	5750240	16 - 24	PADDLE SPACER 150# RF LTCS A516 GR.70									
2270														
2280		Paddle	5750026	32 - 60	PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A									
2290		Paddle	5750196	32 - 60	PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A									
2300														
2310		Paddle	5750027	68 - 92	PADDLE BLANK 150# RF LTCS A516 GR70									
2320		Paddle	5750197	68 – 92	PADDLE SPACER 150# RF LTCS A516 GR.70									
2330														
2340														
2350														
2360														
2370		UD7	Coupli	5478705	3/4 - 1 1/2	COUPLING 3000# SW LTCS A 350 LF2								
2380			Reduci	5478709	3/4 - 1	COUPLING RED 3000# SW LTCS A350 LF2								
2390			Union	5478706	3/4 - 1 1/2	UNION 3000# SW LTCS A350 LF2								
2400			Plug	5491303	3/4 - 3/4	PLUG ROUND HEAD SCRD LTCS A350 LF2								
2410														
2420														
2430		FS1	***** BRANCHES *****											
2440	USE BRANCH CHART 150K26													
2450	***** CONSTRUCTION DETAILS *****													
2460	SEAL WELDING ----- NONE													
2470	PIPE BENDS ----- NONE													
2480	THREAD SEALANT-----PTFE PASTE													
2490	FOR FABRICATION AND WELDING DETAILS													
2500	SEE DOCUMENT N° L-ST-2024													
2510														
2520														
2530														
2540														
2550														
2560														
2570														
2580														
2590														
2600														
2610														
2620														
2630														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP						
						ALL SIZES - BUTTWELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 11JUL13	BY MLA	REV C	SHEET 6 OF 9	PA 5	CLASS 150K26	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10	C	UDC				***** VALVES *****	16 - 18	B1 UDC	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FOR USE WITH PRESSURE CONNECTIONS ONLY						
20															
30															
40			BALFLG	B01AP39	3/4 - 3	BALL 150# RF LTCS PTFE LINED FLOATING SOFT FB LO									
50			BALFGO	B01AP40	16 - 18	BALL 150# RF LTCS PTFE LINED TRUNNION SOFT FB GO									
60															
70															***** PIPE*****
80															
90			Piping	5717204	16 - 18	PIPE SCH 20 SMLS LTCS A333 GR.6 PTFE LINED 150# FLANGED ENDS									
100															
110			Piping	5717211	16 - 18	PIPE SCH 20 SMLS LTCS A333 GR.6 PTFE LINED 150# 3M FLANGED ENDS									
120															
130			Piping	5717212	16 - 18	PIPE SCH 20 SMLS LTCS A333 GR.6 PTFE LINED 150# 2M FLANGED ENDS									
140															
150			Piping	5717213	16 - 18	PIPE SCH 20 SMLS LTCS A333 GR.6 PTFE LINED 150# 1M FLANGED ENDS									
160															
170			Piping	5717214	16 - 18	PIPE SCH 20 SMLS LTCS A333 GR.6 PTFE LINED 150# 0.5M FLANGED ENDS									
180															
190															**** FITTINGS *****
200															
210			90 Deg	5717630	16 - 18	ELL 90 DEG 150# FLANGED SCH 20 LTCS PTFE LINED									
220			45 Deg	5717631	16 - 18	ELL 45 DEG 150# FLANGED SCH 20 LTCS PTFE LINED									
230			Tee	5717632	16 - 18	TEE EQUAL 150# FLANGED SCH 20 LTCS PTFE LINED									
240			22.5 D	9004190	16 - 18	BEND 5D 22.5 DEG 150# FLANGED SCH 20 LTCS PTFE LINED									
250			Tee	5717627	3/4 - 2	INSTRUMENT TEE 150# FLANGED LTCS PTFE LINED									
260			Blind	5715115	3/4 - 18	FLG BLIND 150# PTFE LINED LTCS A350 LF2									
270															
280															***** GASKETS *****
290															
300			GASFR	5664201	3/4 - 18	GASKET 150# RF FLAT RING PTFE									
310															
320															***** BOLTING *****
330															
340			B1	BOLS	5675769	3/4 - 18									STUD-BOLT A320 GR L7M/A194 GR 7M NUTS + LTCS WASHERS
350															
360															***** BRANCHES *****
370															
380															USE TEES AND REDUCERS AS REQUIRED
390															
400															***** CONSTRUCTION DETAILS *****
410															
420															SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY
430															PIPE BENDS ----- LONG RADIUS
440															THREAD SEALANT ----- NONE
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - FLANGED	CORR. ALLOW. 6 MM	FGP/WPMP DATE 17NOV16 BY JCS REV C SHEET 1 OF 1 PA 5 CLASS 150K5E							

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
10		PP9	2BV	K01AC02	1/2 - 2	***** VALVES *****	2 - 2	B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. TO BE USED IN NON-PIGGED LINES ONLY ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.										
20						FA8													
30						FS1													
40																			
50						***** MODULAR VALVES *****		NT5											
60								PP9											
70								SB6											
80		FS1	2BVR	M01AC02	1/2 - 3	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 316 SS TRIM METAL LO		VA5											
90								VAW											
100						MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VB2											
110		VAW	R2BVR	M03AC02	3/4 - 3			VC1											
120						MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		VC2											
130		VAW	R2BVR	M03EC02	3/4 - 1														
140						MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR													
150		VB2			1 - 1														
160						***** BALL VALVES *****													
170		VA5	Ball V	A01AC01	3 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO		SERVICE											
180								SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 CODE: ASME B31.8											
190						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO													
200																			
210						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO		PIPING: BD, BF, BV, DHC, FGH, PHC. COMMENT: LTCS											
220																			
230						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO													
240																			
250		VA5	Ball V	A01AC04	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO		MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C											
260																			
270						BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO													
280																			
290						BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO													
300																			
310																			
320						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO		FGP/WPMP											
330																			
340						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO													
350																			
360						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO													
370																			
380						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO													
390																			
400						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO													
410																			
420						BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO													
430																			
440						BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO													
450																			
460						BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO													
470																			
480						BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO													
490																			
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV A	SHEET 1 OF 6	PA 5	CLASS 150K5F						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
500		VA5	Ball V	B03AC01	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
510													
520		VA5	Ball V	B03AC02	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
530													
540		VA5	Ball V	A03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO							
550													
560		VA5	Ball V	A03AC02	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO							
570													
580						*****ORBIT VALVES*****							
590													
600													
610		VA5	Ball V	R01AC06	3/4 - 10	BALL RISING STEM 150# RF LTCS A352 LCC 316SS TRIM METAL RB HO							
620													
630						***** BUTTERFLY VALVES *****							
640													
650													
660			Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO							
670													
680						*****GATE VALVES*****							
690													
700													
710			Gate V	G01AC48	3/4 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW							
720				G01AC41	2 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW							
730													
740						*****THROUGH CONDUIT GATE VALVES *****							
750													
760													
770			Condui	W01AC70	3 - 16	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
780			Condui	W01AC71	18 - 24	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
790													
800						***** GLOBE VALVES *****							
810													
820													
830			Globe	L01AC48	3/4 - 1 1/2	GLOBE 150# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
840			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW							
850			Globe	L01AC42	8 - 10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO							
860													
870						***** CHECK VALVES *****							
880													
890													
900		VC1	Check	C01AC50	3/4 - 1 1/2	CHECK PISTON 150# RF LTCS A350 LF2 TRIM 16							
910													
920		VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
930													
940		VC2	Check	C01AC52	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16							
950						***** PIPE *****							
960													
970													
980			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE							
990			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV A	SHEET 2 OF 6	PA 5	CLASS 150K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1000			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE		SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 CODE: ASME B31.8 PIPING: BD, BF, BV, DHC, FGH, PHC. MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: LTCS						
1010			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE								
1020			Piping	9001254	6 - 14	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE								
1030			Piping	9001311	16 - 18	LINEPIPE SMLS WT=7.92MM API-5L X60 BE								
1040			Piping	9001485	20 - 20	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE								
1050			Piping	9001493	24 - 24	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE								
1060														
1070														
1080														***** FITTINGS ABOVE GROUND*****
1090			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
1100			90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6								
1110			90 Deg	9006566	4 - 4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60								
1120			90 Deg	9002190	6 - 14	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60								
1130			90 Deg	9006478	16 - 18	ELL BW 90 DEG LR WT 7.92 MM A860 WPHY 60								
1140			90 Deg	9006481	20 - 20	ELL BW 90 DEG LR WT 8.74 MM A860 WPHY 60-W								
1150			90 Deg	9006472	24 - 24	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60-W								
1160														
1170			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
1180			45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6								
1190			45 Deg	9006573	4 - 4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60								
1200			45 Deg	9006571	6 - 14	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60								
1210			45 Deg	9006588	16 - 18	ELL BW 45 DEG LR WT 7.92MM A860 WPHY 60								
1220			45 Deg	9006593	20 - 20	ELL BW 45 DEG LR WT 8.74MM A860 WPHY 60								
1230			45 Deg	9006583	24 - 24	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60								
1240														
1250			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6								
1260			Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6								
1270			NT5	Tee	9301610	4 - 4							TEE BW WT 6.02 MM LTCS A860 WPHY-60	
1280			NT5	Tee	9301609	6 - 14							TEE BW WT 6.35 MM LTCS A860 WPHY-60	
1290			NT5	Tee	9301606	16 - 18							TEE BW WT 7.92 MM LTCS A860 WPHY-60	
1300			NT5	Tee	9006487	20 - 20							TEE BW WT 8.74MM A860 WPHY 60-W	
1310			NT5	Tee	9004210	24 - 24							TEE BW WT 9.53MM A860 WPHY 60-W	
1320														
1330			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6							1 - 2	
1340			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6							3 - 3	
1350			Reduci	9006664	2 - 2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60							4 - 4	
1360			Reduci	9006659	3 - 3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60							4 - 4	
1370			Reduci	9006658	3 - 3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60							6 - 6	
1380			Reduci	9006605	4 - 4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60							6 - 10	
1390			Reduci	9006014	6 - 12	TEE RED BW WT 6.35 MM LTCS A860 WPHY 60							8 - 14	
1400			Reduci	9006017	6 - 14	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60							16 - 18	
1410			Reduci	9006018	16 - 16	TEE RED BW WT 7.92 MM LTCS A860 WPHY 60							18 - 18	
1420			Reduci	9004143	8 - 14	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60-W							20 - 20	
1430			Reduci	9006019	16 - 18	TEE RED BW WT 8.74 MM/7.92 MM LTCS A860 WPHY 60-W							20 - 20	
1440			Reduci	9004149	10 - 14	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W							24 - 24	
1450			Reduci	9006483	16 - 18	TEE RED BW WT 9.53 MM/7.92 MM LTCS A860 WPHY 60-W							24 - 24	
1460			Reduci	9006484	20 - 20	TEE RED BW WT 9.53 MM/8.74 MM LTCS A860 WPHY 60-W							24 - 24	
1470														
1480			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6								
1490			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6								
1500			Cap	9002480	4 - 4	CAP BW WT 6.02 MM LTCS A860 WPHY-60								
1510			Cap	9005055	6 - 14	CAP BW WT 6.35MM LTCS A860 WPHY-60								
1520			Cap	9005058	16 - 18	CAP BW WT 7.92 MM LTCS A860 WPHY-60								
1530			Cap	9005059	20 - 20	CAP BW WT 8.74 MM LTCS A860 WPHY-60								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD							CORR. ALLOW. 3 MM	DATE 17MAR17

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1540			Cap	9005060	24 - 24	CAP BW WT 9.53 MM LTCS A860 WPHY-60							
1550													
1560			Concen	5392458	3/4 - 1	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2						
1570			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3						
1580			Concen	9006918	1 1/2 - 2	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1590			Concen	9006926	3 - 3	REDUCER CONC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4 - 4						
1600			Concen	9006915	3 - 3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6 - 6						
1610													
1620			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2						
1630			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3						
1640			Eccent	9001372	1 1/2 - 2	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1650			Eccent	9002281	3 - 3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4 - 4						
1660			Eccent	9002510	3 - 3	REDUCER ECC WT 6.35 MM/STD LTCS A420 WPL6	6 - 6						
1670			Eccent	9002312	4 - 4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6 - 14						
1680			Eccent	9002298	6 - 12	REDUCER ECC WT 6.35 MM LTCS A860 WPHY-60	8 - 14						
1690			Eccent	9002272	8 - 14	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	16 - 16						
1700			Eccent	9002278	16 - 16	REDUCER ECC WT 7.92MM LTCS A860 WPHY-60	18 - 18						
1710			Eccent	9004131	12 - 14	REDUCER ECC WT 8.74 MM / 6.35 LTCS A860 WPHY-60-W	20 - 20						
1720			Eccent	9004133	16 - 18	REDUCER ECC WT 8.74 MM / 7.92MM LTCS A860 WPHY-60-W	20 - 20						
1730			Eccent	9004135	16 - 18	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	24 - 24						
1740			Eccent	9004136	20 - 20	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	24 - 24						
1750													
1760			Nipofl	5504001	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2	2 - 2						
1770			Nipofl	5504002	3/4 - 2	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2	3 - 3						
1780			Nipofl	9006800	3/4 - 2	NIPOFLANGE 150# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4						
1790			Nipofl	9006801	3/4 - 2	NIPOFLANGE 150# RF 6.35 MM/XS WT LTCS A694 F60	6 - 14						
1800			Nipofl	9006804	3/4 - 2	NIPOFLANGE 150# RF 7.92 MM/XS WT LTCS A694 F60	16 - 18						
1810			Nipofl	9006805	3/4 - 2	NIPOFLANGE 150# RF 8.74 MM/XS WT LTCS A694 F60	20 - 20						
1820			Nipofl	9006806	3/4 - 2	NIPOFLANGE 150# RF 9.53 MM/XS WT LTCS A694 F60	24 - 24						
1830			Nipofl	9002639	3/4 - 2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4						
1840			Nipofl	9002643	3/4 - 2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6 - 14						
1850			Nipofl	9002645	3/4 - 2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	16 - 18						
1860			Nipofl	9002646	3/4 - 2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	20 - 20						
1870			Nipofl	9002647	3/4 - 2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	24 - 24						
1880													
1890			Weldol	5505335	3/4 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2	SERVICE					
1900			Weldol	5505179	3/4 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 3						
1910			Weldol	9006219	3/4 - 1 1/2	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60 IMPACT TESTED	4 - 4						
						AT -46 DEG C							
1920			Weldol	9006407	3/4 - 2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6 - 14	PIPING: BD, BF, BV, DHC, FGH, PHC.					
1930			Weldol	9006409	3 - 3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8 - 14						
1940			Weldol	9006408	4 - 4	WELDOLET WT 6.35 MM/6.02 MM LTCS A694 F60	12 - 14						
1950			Weldol	9002552	3/4 - 2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	16 - 18						
1960			Weldol	9006415	3 - 3	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	16 - 18	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: LTCS					
1970			Weldol	9006414	4 - 4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	16 - 18						
1980			Weldol	9004124	6 - 6	WELDOLET WT 7.92 MM/6.35 MM LTCS A694 F60	18 - 18						
1990			Weldol	9006423	3/4 - 2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	20 - 20						
2000			Weldol	9006422	3 - 3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	20 - 20						
2010			Weldol	9006421	4 - 4	WELDOLET WT 8.74 MM/6.02 MM LTCS A694 F60	20 - 20						
2020			Weldol	9002008	6 - 6	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	20 - 20						
2030			Weldol	9006429	3/4 - 2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	24 - 24						
2040			Weldol	9006428	3 - 3	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	24 - 24						
2050			Weldol	9006427	4 - 4	WELDOLET WT 9.53 MM/6.02 MM LTCS A694 F60	24 - 24						
2060			Weldol	9006424	6 - 8	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	24 - 24						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV A	SHEET 4 OF 6	PA 5	CLASS 150KSF

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2070		FA8	Reduci	5599692	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 150# RF LTCS A350 LF2	3/4 - 1 1/2 3/4 - 3/4							
2080			Reduci	5599703	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 300# RF LTCS A350 LF2								
2090														
2100			Flange	5851200	1/2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE								
2110			Flange	5851203	3/4 - 3/4	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE								
2120														
2130			Flange	5556264	1/2 - 2	FLG WN 150# RF LTCS A350 LF2 SCH XS								
2140			Flange	5556261	3 - 3	FLG WN 150# RF LTCS A350 LF2 SCH STD								
2150			Flange	9006702	4 - 4	FLG WN 150# RF LTCS A694 F60 WT 6.02MM								
2160			Flange	9001473	6 - 14	FLG WN 150# RF LTCS A694 F60 WT 6.35MM								
2170														
2180			Flange	5556476	1/2 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS								
2190			Flange	5556473	3 - 3	FLG WN 300# RF LTCS A350 LF2 SCH STD								
2200			Flange	9001480	4 - 4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM								
2210														
2220			Flange	5557036	3/4 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS								
2230			Flange	5557033	3 - 3	FLG WN 600# RF LTCS A350 LF2 SCH STD								
2240			Flange	9001723	4 - 4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM								
2250														
2260			FS1	Flange	5558036	2 - 2								FLG WN 600# RTJ LTCS A350 LF2 SCH XS
2270														
2280			FS1	Flange	5555935	3 - 3								FLG WN 900# RF LTCS A350 LF2 SCH STD
2290			FS1	Flange	9005389	4 - 4								FLG WN 900# RF LTCS A694 F60 WT 6.02MM
2300														
2310			FS1	Flange	5557393	1 1/2 - 2								FLG WN 1500# RF LTCS A350 LF2 SCH XS
2320														
2330			FS1	Flange	5557507	2 - 2								FLG WN 1500# RTJ LTCS A350 LF2 SCH XS
2340														
2350				Blind	5615566	1/2 - 24								FLG BLIND 150# RF LTCS A350 LF2
2360				Blind	5615576	1/2 - 24								FLG BLIND 300# RF LTCS A350 LF2
2370														
2380			Orific	5623947	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS								
2390			Orific	5623944	3 - 14	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD								
2400														
2410														
2420														
2430			GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 CODE: ASME B31.8 PIPING: BD, BF, BV, DHC, FGH, PHC.							
2440			GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR								
2450			GASFRS	5673103	3/4 - 20	GASKET 600# 316SS SP WND GRAPHITE IR								
2460			GASFRS	5673117	3 - 10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: LTCS							
2470			GASFRS	5672251	1 1/2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR								
2480														
2490			GASRJ	5673202	2 - 2	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING								
2500														
2510			GASRJ	5674491	3 - 3	GASKET 900# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING								
2520														
2530			GASRJ	5674492	2 - 2	GASKET 1500# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING								
2540														
2550														
2560														
2570														
2580		B1	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
2590														
2600		SB6	CPSBOL	5675675	12 - 24	JACK SCREW HEX HEAD A320 GR L7M								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 3 MM	FGP/WPMP DATE: 17MAR17 BY: JCS REV: A SHEET: 5 OF 6 PA: 5 CLASS: 150KSF						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2610						**** BLINDS & SPACERS ****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 CODE: ASME B31.8 PIPING: BD, BF, BV, DHC, FGH, PHC. MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: LTCS					
2620													
2630			Specta	5749580	3/4 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70							
2640			Specta	5749581	1 1/2 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
2650													
2660			Paddle	5750023	16 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70							
2670			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70							
2680													
2690			Paddle	5750240	16 - 24	PADDLE SPACER 150# RF LTCS A516 GR.70							
2700			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70							
2710													
2720			Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 24						
2730			Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 24						
2740													
2750													
2760						***** DO NOT USE *****TIE-INS ONLY*****							
2770													
2780			Reduci	9006750	3 - 3	FLG WN REDUCING 150# RF LTCS A694 F60 BORED TO SUIT WT 6.02MM	4 - 4						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV A	SHEET 6 OF 6	PA 5	CLASS 150K5F

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10						CAUSTIC SUMP HEATING COIL		B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.							
20						***** PIPE *****									
30															
40															
50															
51	C														
60	C		Piping	5722001	3/4 - 2	PIPE STD WT SMLS MONEL 400 B165 UNS N04400 BE									
70															
80						***** FITTINGS *****									
90															
100			90 Deg	5722002	3/4 - 3/4	ELL BW 90 DEG LR STD WT MONEL 400 B366 UNS N04400									
110															
120			Concen	5722004	3/4 - 1 1/2	REDUCER CONC STD WT MONEL 400 B366 UNS N04400	2 - 2								
130															
140						***** PIPE FLANGES *****									
150															
160			Flange	5722003	3/4 - 2	FLG WN 150# RF MONEL 400 B564 UNS N04400 STD WT									
170															
180						***** GASKETS *****									
190															
200			GASFRS	5673060	3/4 - 2	GASKET 150# MONEL 400 SPIRAL WOUND GRAPHITE FILLED									
210															
220						***** BOLTING *****									
230															
240	C	B1	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS									
250															
260															
270						*****CONSTRUCTION DETAILS*****									
280															
290															
300						SEAL WELDING -----FOR THREADED CONNECTIONS ONLY									
310						PIPE BENDS-----NONE									
320						THREAD SEALANT-----NONE									
330															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP							
						ALL SIZES BUTTWELD		CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS
								0 MM		11JUL13	MLA	B	1 OF 1	5	150M40

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES
10 20 30 50 60 61 70 80 81 82 83 84 85 86 87 88 170 180 190 200 210 220 230 240 250 260 270 280 290 300 310 320 330 340 350 360 370 380 390 400 410 420 430 440 450 460 470						*****MODULAR VALVES***** <		

SERVICE

DESIGN ASME B16.5 MG 2.8
CODE ASME B31.3
BV, DL, CCB, CCS, CFA, SL, SLL, WM, WE

MATERIAL: 6% MOLY SS FLANGE
RATING: 150 RF
TEMPERATURE RANGE: -36 / 200 °C
COMMENT:

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
480						***** FITTINGS *****		VC2 DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. ZA6 REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
490														
500	C		90 Deg	5721124	3/4 - 1 1/2	ELL 90 DEG LR SCH 80S 6MO A403 UNS S31254								
510			90 Deg	5721104	2 - 6	ELL BW 90 DEG LR SCH 40S 6MO A403 UNS S31254								
520			90 Deg	5721105	8 - 10	ELL BW 90 DEG LR SCH 30 6MO A403 UNS S31254								
530														
540	C		45 Deg	5721934	3/4 - 1 1/2	ELL 45 DEG SCH 80S SS 6MO A403 UNS S31254								
550			45 Deg	5721902	2 - 6	ELL BW 45 DEG LR SCH 40S 6MO A403 UNS S31254								
560			45 Deg	5721922	8 - 10	ELL BW 45 DEG LR SCH 30S 6 MO A403 UNS S31254								
570														
580	C		Tee	5721210	3/4 - 1 1/2	TEE SCH 80S 6MO A403 UNS S31254								
590			Tee	5721204	2 - 6	TEE BW SCH 40S 6MO A403 UNS S31254								
600			Tee	5721205	8 - 10	TEE BW SCH 30 6MO A403 UNS S31254								
610														
620	C		Reduci	5721211	3/4 - 1	TEE RED SCH 80S 6MO A403 UNS S31254	1 - 1 1/2							
630			Reduci	5721212	1 1/2 - 1 1/2	TEE RED SCH 40S/80S 6MO A403 UNS S31254	2 - 3							
640			Reduci	5721201	2 - 4	TEE RED SCH 40S 6MO A403 UNS S31254	3 - 6							
650			Reduci	5721206	4 - 6	TEE RED BW SCH 30/SCH 40S 6MO A403 UNS S31254	8 - 10							
660			Reduci	5721209	8 - 8	TEE RED BW SCH 30 6MO A403 UNS S31254	10 - 10							
670														
680	C		Cap	5721936	3/4 - 1 1/2	CAP SCH 80S 6MO A403 UNS S31254								
690			Cap	5721905	2 - 6	CAP BW SCH 40S 6MO A403 UNS S31254								
700			Cap	5721923	8 - 10	CAP BW SCH 30 6MO A403 UNS S31254								
710														
720	C		Concen	5721935	3/4 - 1	REDUCER CONC SCH 80S 6MO A403 UNS S31254	1 - 1 1/2							
730			Concen	5721938	3/4 - 1 1/2	REDUCER CONC SCH 40S/80S 6MO A403 UNS S31254	2 - 4							
740			Concen	5721924	2 - 4	REDUCER CONC SCH 40S 6 MO A403 UNS S31254	3 - 6							
750			Concen	5721925	4 - 6	REDUCER CONC SCH 30/SCH 40S 6MO A403 UNS S31254	8 - 10							
760			Concen	5721926	8 - 8	REDUCER CONC SCH 30 6MO A403 UNS S31254	10 - 10							
770														
780	C		Eccent	5721945	3/4 - 1	REDUCER ECC SCH 80S 6MO A403 UNS S31254	1 - 1 1/2							
790			Eccent	5721948	3/4 - 1 1/2	REDUCER ECC SCH 40S/80S 6MO A403 UNS S31254	2 - 4							
800			Eccent	5721927	2 - 4	REDUCER ECC SCH 40S 6 MO A403 UNS S31254	3 - 6							
810			Eccent	5721928	4 - 6	REDUCER ECC SCH 30/SCH 40S 6MO A403 UNS S31254	8 - 10							
820			Eccent	5721929	8 - 8	REDUCER ECC SCH 30 6MO A403 UNS S31254	10 - 10							
830														
840	C		Nipofl	5721616	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 40S/80S 6MO A182 F44	2 - 6	DESIGN ASME B16.5 MG 2.8 CODE ASME B31.3 BV, DL, CCB, CCS, CFA, SL, SLL, WM, WE						
850	C		Nipofl	5721617	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 30/80S 6MO A182 F44	8 - 10							
860			Nipofl	5721606	2 - 2	NIPOFLANGE 150# RF SCH 40S 6MO A182 F44	3 - 6							
870			Nipofl	5721607	2 - 2	NIPOFLANGE 150# RF SCH 30/ SCH 40S 6MO A182 F44	8 - 10							
880														
890	UJ2		Nipofl	5721610	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/160 6MO A182 F44	4 - 6							
900	UJ2		Nipofl	5721611	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 30/160 6MO A182 F44	8 - 10							
910														
920	UJ2		Elbole	5721612	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40S/160 6MO A182 F44	4 - 6							
930	UJ2		Elbole	5721613	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 30/160 6MO A182 F44	8 - 10							
940														
950	C		Weldol	5721654	3/4 - 1 1/2	WELDOLET SCH 40S/80S 6MO A182 F44	2 - 6	MATERIAL: 6% MOLY SS FLANGE RATING: 150 RF TEMPERATURE RANGE: -36 / 200 °C COMMENT:						
960	C		Weldol	5721655	3/4 - 1 1/2	WELDOLET SCH 30/80S 6MO A182 F44	8 - 10							
970			Weldol	5721608	2 - 2	WELDOLET SCH 40S/40S 6MO A182 F44	2 - 6							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP						
						ALL SIZES BUTT-WELD		CORR. ALLOW. 1 MM	DATE 31OCT12	BY SMS	REV C	SHEET 2 OF 4	PA 5	CLASS 150MO6

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
980			Weldol	5721609	2 - 3	WELDOLET SCH 30/40S 6MO A182 F44	8 - 10							
990														
1000														
1010		FS1				***** PIPE FLANGES *****								
1020														
1030		FA8	Reduci	5721712	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 150# RF 6MO A182 F44	1 1/2 - 1 1/2							
1040	C		Flange	5721744	3/4 - 1 1/2	FLG WN 150# RF 6MO A182 F44 SCH 80S BORE								
1050			Flange	5721714	2 - 6	FLG WN 150# RF 6MO A182 F44 SCH 40S								
1060			Flange	5721715	8 - 10	FLG WN 150# RF 6MO A182 F44 SCH 30								
1070														
1080	C	FA3	Flange	5721764	3/4 - 1 1/2	FLG WN 150# FF 6MO A182 F44 SCH 80S BORE								
1090		FA3	Flange	5721722	2 - 6	FLG WN 150# FF 6MO A182 F44 SCH 40S								
1100														
1110	C		Blind	5721717	3/4 - 10	FLG BLIND 150# RF 6MO A182 F44								
1120														
1130			Flange	5721814	1/2 - 1 1/2	FLG WN 300# RF 6MO A182 F44 SCH 80S BORE								
1140			Flange	5721709	2 - 6	FLG WN 300# RF 6MO A182 F44 SCH 40S								
1150			Flange	5721716	8 - 10	FLG WN 300# RF 6MO A182 F44 SCH 30								
1160														
1170			Blind	5721711	3/4 - 10	FLG BLIND 300# RF 6MO A182 F44								
1180														
1190						**** ORIFICE FLANGES ****								
1200														
1210			Orific	5721718	2 - 6	WN ORIF SET 300# RF 6MO A182 F44 SCH 40S								
1220			Orific	5721719	8 - 10	WN ORIF SET 300# RF 6MO A182 F44 SCH 30								
1230														
1240						*****PIPE NIPPLES*****								
1250														
1260		810				NIPPLE PBE CUT FROM PIPE								
1270														
1280														
1290						*****OLETS*****								
1300														
1310			Sockol	5721603	1/2 - 1 1/2	SOCKOLET 3000# 6MO A182 F44	2 - 10							
1320			Endole	5721614	1/2 - 1 1/2	FLAT SOCKOLET 3000# 6MO A182 F44								
1330		UD7	Thredo	5721604	3/4 - 3/4	THREDOLET 3000# 6MO A182 F44	2 - 10							
1340														
1350														
1360														
1370														
1380														
1390														
1400														
1410														
1420														
1430														
1440														
1450														
1460														
1470														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP						
						ALL SIZES BUTT-WELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								1 MM	31OCT12	SMS	C	3 OF 4	5	150MO6

SERVICE

DESIGN ASME B16.5 MG 2.8
CODE ASME B31.3
BV, DL, CCB, CCS, CFA, SL, SLL, WM, WE

MATERIAL: 6% MOLY SS FLANGE
RATING: 150 RF
TEMPERATURE RANGE: -36 / 200 °C
COMMENT:

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
1480	C	FS1	GASFRS GASFF GASFRS	5673059 5664562 5673056	1/2 - 10 3/4 - 6 1/2 - 10	***** GASKETS *****										
1490																
1500																
1510		B1	BOLS	5675770	1/4 - 99	GASKET 150# S31254 SPIRAL WOUND GRAPHITE FILLED IR										
1520						GASKET 150# FF RING EPDM 3MM THK										
1530						GASKET 300# S31254 SPIRAL WOUND GRAPHITE FILLED IR										
1540						***** BOLTING *****										
1550						Specta								5721720	1 1/2 - 10	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS
1560																
1570																
1580						***** SPECTACLE BLINDS *****										
1590						SPECTACLE BLIND 150# RF 6MO A240 UNS S31254										
1600						***** BRANCHES *****										
1610						USE BRANCH CHART 150MO6										
1620																
1630																
1640											***** CONSTRUCTION DETAILS *****					
1650											SEAL WELDING ----- NONE					
1660											PIPE BENDS ----- NONE					
1670											THREAD SEALANT-----PTFE TAPE					
1680																
1690																
1700																
1710																
1720																
1730																
1740																
1750																
1760																
1770																
1780																
1790																
1800																
1810																
1820																
1830	SERVICE															
1840	DESIGN ASME B16.5 MG 2.8															
1850	CODE ASME B31.3															
1860	BV, DL, CCB, CCS, CFA, SL, SLL, WM, WE															
1870																
1880																
1890																
1900																
1910																
1920																
1930																
1940																
1950																
1960																
1970																
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP								
						ALL SIZES BUTT-WELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS		
								1 MM	31OCT12	SMS	C	4 OF 4	5	150MO6		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 20 30 40 50 60	D	PP9	2BV	K01AS22	1/2 - 2	*****VALVES***** *****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 150# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		O32 810 B1 FA3 FA4	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. NIPPLE TO BE CUT FROM SCH 160 PIPE ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FLAT FACE FLANGES & FULL FACE GASKETS SHALL ONLY BE USED WHEN MATING WITH FLAT FACE EQUIPMENT. SLIP ON FLANGE SHALL ONLY BE USED WHERE INSTRUMENT ENCLOSURES ARE INSTALLED.					
70 80	D	VAV	2BV	K03AS22	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		FA8 FF5	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY.					
90 100	D		2BVR	M01AS22	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR		FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.					
110 120	D	FS1	2BVR	M03AS22	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR		PG1 PP9	USE FOR PRESSURE GAUGES WITH CHEMICAL SEALS. ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
130 140	D	VAW	R2BVR	M03ES22	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
150 160	D	VAW	R2BVR	M03ES22	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTRORACE.					
170 180 190 200 210 220 230 240 250 260 270 280 290	E 													

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																																																																																																																																																																																																																																																																																																																																											
460	D	VA5	Ball V	B01AS27	10 - 24	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT		UDB	FOR "HARD PIPED" VENT CONNECTIONS ON CLASS 10000 5-WAY MONOFLANGE VALVES (CLASS 10000K)																																																																																																																																																																																																																																																																																																																																										
470		VA5	Ball V	B01AS28	8 - 24	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.																																																																																																																																																																																																																																																																																																																																										
480		VA5	Ball V	B01JS28	26 - 42	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A 718 BOLT		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.																																																																																																																																																																																																																																																																																																																																										
490								VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY																																																																																																																																																																																																																																																																																																																																										
510	D	FS1	Ball V	B03AS21	1/2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT		VAF	FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY.																																																																																																																																																																																																																																																																																																																																										
520		FS1	Ball V	B03AS22	1/2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT		VAG	FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY.																																																																																																																																																																																																																																																																																																																																										
530	C	FS1	Ball V	A03AS21	3 - 3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO 718 BOLT		VAM	FOR INSTRUMENT CONNECTIONS IN SOUR SERVICE ONLY.																																																																																																																																																																																																																																																																																																																																										
540								VAN	FOR LEVEL TRANSMITTER CONNECTIONS IN SOUR SERVICE ONLY.																																																																																																																																																																																																																																																																																																																																										
550	C	FS1	Ball V	A03AS22	3 - 3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL RB LO 718 BOLT	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.																																																																																																																																																																																																																																																																																																																																											
560							VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055																																																																																																																																																																																																																																																																																																																																											
570	C					*****BUTTERFLY VALVES*****		VB2	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.																																																																																																																																																																																																																																																																																																																																										
580								VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.																																																																																																																																																																																																																																																																																																																																										
590									VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.																																																																																																																																																																																																																																																																																																																																									
600								C					*****GLOBE VALVES*****		VU2	USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY.																																																																																																																																																																																																																																																																																																																																			
610	ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.																																																																																																																																																																																																																																																																																																																																																	
620		Butter	T01AS65	10 - 24	BUTTERFLY 150# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO 718 BOLT																																																																																																																																																																																																																																																																																																																																														
630															T01JS65	26 - 42	BUTTERFLY 150# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A 718 BOLT																																																																																																																																																																																																																																																																																																																																		
640	C					*****CHECK VALVES*****			SERVICE																																																																																																																																																																																																																																																																																																																																										
650									VC1	Check	C01AS54	3/4 - 1 1/2	CHECK PISTON 150# RF SS A182 F316 TRIM 16 718 BOLT			SOUR SERVICE - NACE MR0175/ ISO 15156																																																																																																																																																																																																																																																																																																																																			
660																VC2	Check	C01AS45	2 - 6	CHECK DUAL WAFER LUG 150# RF SS A351 CF8M TRIM 16 THROU BOLTED			DESIGN: ASME B16.5 MG 2.2																																																																																																																																																																																																																																																																																																																												
670									VC2	Check	C01AS52	8 - 24	CHECK DUAL DBL FLG 150# RF SS A351 CF8M TRIM 16			CODE - ASME B31.3																																																																																																																																																																																																																																																																																																																																			
680	C					CHECK DUAL DBL FLG 150# RF SS A351 CF8M TRIM 16 B16.47 SERIES A																																																																																																																																																																																																																																																																																																																																													

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
970	C		90 Deg	5451100	26 - 36	ELL 90 DEG LR SCH STD 316/316L SS A403 WP316/316L-W							
980			90 Deg	5451100	42 - 42	ELL 90 DEG LR SCH STD 316/316L SS A403 WP316/316L-W							
990													
1000			45 Deg	5449146	3/4 - 1 1/2	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S							
1010			45 Deg	5449144	2 - 6	ELL 45 SCH 40S SS A403 WP316/316L-S							
1020			45 Deg	5449037	8 - 12	ELL 45 SCH 30 SS A403 WP316/316L-S							
1030			45 Deg	5448891	14 - 18	ELL 45 DEG SCH 10 A403 WP316/316L-S							
1040			45 Deg	5449036	20 - 24	ELL 45 DEG SCH 20 A403 WP316/316L-S							
1050			45 Deg	5451106	26 - 36	ELL 45 DEG SCH STD 316/316L SS A403 WP316/316L-W							
1060			45 Deg	5451106	42 - 42	ELL 45 DEG SCH STD 316/316L SS A403 WP316/316L-W							
1070	C		Tee	5449156	3/4 - 1 1/2	TEE SCH 80S SS A403 WP316/316L-S							
1080			Tee	5449147	2 - 6	TEE SCH 40S SS A403 WP316/316L-S							
1090			Tee	5449043	8 - 12	TEE SCH 30 SS A403 WP316/316L-S							
1100			Tee	5448892	14 - 18	TEE SCH 10 SS A403 WP316/316L-S							
1110			Tee	5449039	20 - 24	TEE SCH 20 SS A403 WP316/316L-S							
1120			Tee	5451109	26 - 36	TEE SCH STD 316/316L SS A403 WP316/316L-W							
1130			Tee	5451109	42 - 42	TEE SCH STD 316/316L SS A403 WP316/316L-W							
1140	C		Reduci	5448905	3/4 - 1	TEE RED SCH 80S SS A403 WP316/316L-S	1 - 1 1/2						
1160			Reduci	5449250	1 - 1 1/2	TEE RED SCH 40S/80S SS A403 WP316/316L-S							
1170			Reduci	5449184	2 - 4	TEE RED SCH 40S SS A403 WP316/316L-S							
1180			Reduci	5448889	4 - 6	TEE RED SCH 30/40S SS A403 WP316/316L-S							
1190			Reduci	5448880	6 - 6	TEE RED SCH 10/40S SS A403 WP316/316L-S							
1200			Reduci	5449053	8 - 10	TEE RED SCH 30 SS A403 WP316/316L-S							
1210			Reduci	5448896	8 - 12	TEE RED SCH 10/30 SS A403 WP316/316L-S							
1220			Reduci	5449167	8 - 12	TEE RED SCH 20/30 SS A403 WP316/316L-S							
1230			Reduci	5448898	14 - 16	TEE RED SCH 10 SS A403 316/316L-S							
1240			Reduci	5449168	14 - 18	TEE RED SCH 20/10 SS A403 WP316/316L-S							
1250	C		Reduci	5449041	20 - 20	TEE RED SCH 20 SS A403 WP316/316L-S	20 - 24						
1260			Reduci	5449131	16 - 16	TEE RED SCH STD/10 SS A403 WP316/316L-W							
1270			Reduci	5449169	20 - 24	TEE RED SCH STD/20 SS A403 WP316/316L-W							
1280			Reduci	5449042	26 - 36	TEE RED SCH STD SS A403 WP316/316L-W							
1290													
1300													
1310			Cap	5449049	3/4 - 1 1/2	CAP SCH 80S SS A403 WP316/316L-S							
1320			Cap	5449148	2 - 6	CAP SCH 40S SS A403 WP316/316L-S							
1330			Cap	5449044	8 - 12	CAP SCH 30 SS A403 WP316/316L-S							
1340			Cap	5448893	14 - 18	CAP SCH 10 SS ASTM A403 WP316/316L-S							
1350	Cap	5449040	20 - 24	CAP SCH 20 SS A403 WP316/316L-S									
1360	C		Cap	5449256	26 - 36	CAP SCH STD SS A403 WP316/316L-S	26 - 30						
1370			Cap	5449256	42 - 42	CAP SCH STD SS A403 WP316/316L-S							
1380													
1390			Concen	5449158	3/4 - 1	REDUCER CONC SCH 80S SS A403 WP316/316L-S							
1400			Concen	5549110	3/4 - 1 1/2	REDUCER CONC SCH 40S/80S SS A403 WP316/316L-S							
1410			Concen	5449149	2 - 4	REDUCER CONC SCH 40S SS A403 WP316/316L-S							
1420			Concen	5448899	4 - 6	REDUCER CONC SCH 30/40S SS A403 WP316/316L-S							
1430			Concen	5448881	6 - 6	REDUCER CONC SCH 10/40S SS A403 WP316/316L-S							
1440			Concen	5448900	8 - 10	REDUCER CONC SCH 30 SS A403 WP316/316L-S							
1450			Concen	5448901	8 - 12	REDUCER CONC SCH 10/30 SS A403 WP316/316L-S							
1460			Concen	5449173	12 - 12	REDUCER CONC SCH 20/30 SS A403 WP316/316L-S							
1470			Concen	5448894	14 - 16	REDUCER CONC SCH 10 SS A403 WP316/316L-S							
1480			Concen	5450720	18 - 18	REDUCER CONC SCH STD/10 SS A403 WP316/316L-W							
1490			Concen	5449174	14 - 18	REDUCER CONC SCH 20/10 SS A403 WP316/316L-S							
1500	Concen	5449175	20 - 20	REDUCER CONC SCH 20 SS A403 WP316/316L-S									
						CONSTRUCTION		FGP/WPMP					
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN							1 MM	17NOV17	MC	D	3 OF 7	5	150P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
1510	C		Concen	5450721	20 - 24	REDUCER CONC SCH STD/20 SS A403 WP316/316L-W	26 - 36					
1520			Concen	5450722	26 - 36	REDUCER CONC SCH STD SS A403 WP316/316L-W	28 - 42					
1530												
1540			Eccent	5449159	3/4 - 1	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1 - 1 1/2					
1550			Eccent	5549310	3/4 - 1 1/2	REDUCER ECC SCH 40S/80S SS A403 WP316/316L-S	2 - 4					
1560			Eccent	5449150	2 - 4	REDUCER ECC SCH 40S SS A403 WP316/316L-S	3 - 6					
1570			Eccent	5449151	4 - 6	REDUCER ECC SCH 30/40S SS A403 WP316/316L-S	8 - 12					
1580			Eccent	5448882	6 - 6	REDUCER ECC SCH 10/40S SS A403 WP316/316L-S	14 - 14					
1590			Eccent	5449152	8 - 10	REDUCER ECC SCH 30 SS A403 WP316/316L-S	10 - 12					
1600			Eccent	5449153	8 - 12	REDUCER ECC SCH 10/30 SS A403 WP316/316L-S	14 - 18					
1610			Eccent	5449170	12 - 12	REDUCER ECC SCH 20/30 SS A403 WP316/316L-S	20 - 20					
1620			Eccent	5448895	14 - 16	REDUCER ECC SCH 10 SS A403 WP316/316L-S	16 - 18					
1630			Eccent	5449171	14 - 18	REDUCER ECC SCH 20/10 SS A403 WP316/316L-S	20 - 24					
1640			Eccent	5450723	18 - 18	REDUCER ECC SCH STD/10 SS A403 WP316/316L-W	26 - 28					
1650			Eccent	5449172	20 - 20	REDUCER ECC SCH 20 SS A403 WP316/316L-S	24 - 24					
1660			Eccent	5450724	20 - 24	REDUCER ECC SCH STD/20 SS A403 WP316/316L-W	26 - 36					
1670			Eccent	5450725	26 - 36	REDUCER ECC SCH STD SS A403 WP316/316L-W	28 - 42					
1680												
1690	C	UJ2	Nipofl	5504311	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 40S/80S SS A182 F316/316L	2 - 6					
1700	C		Nipofl	5504331	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 30/80S SS A182 F316/316L	8 - 12					
1710	C		Nipofl	5504312	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 10/80S SS A182 F316/316L	14 - 18					
1720	C		Nipofl	5504334	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 20/80S SS A182 F316/316L	20 - 24					
1730	C		Nipofl	5504343	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH STD/80S SS A182 F316/316L	26 - 42					
1740												
1750			Nipofl	5504011	2 - 2	NIPOFLANGE 150# RF SCH 40S/40S SS A182 F316/316L	3 - 6					
1760			Nipofl	5504031	2 - 2	NIPOFLANGE 150# RF SCH 30/40S SS A182 F316/316L	8 - 12					
1770			Nipofl	5504012	2 - 2	NIPOFLANGE 150# RF SCH 10/40S SS A182 F316/316L	14 - 18					
1780			Nipofl	5504034	2 - 2	NIPOFLANGE 150# RF SCH 20/40S SS A182 F316/316L	20 - 24					
1790			Nipofl	5504043	2 - 2	NIPOFLANGE 150# RF SCH STD/40S SS A182 F316/316L	26 - 42					
1800												
1810	D		Nipofl	5504411	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/80S SS A182 F316/316L	2 - 3					
1820	D		Nipofl	5504412	1 - 1	NIPOFLANGE 300# RF SCH 80S SS A182 F316/316L	1 1/2 - 1 1/2					
1830			Nipofl	5504132	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4 - 6					
1840			Nipofl	5504145	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 30/160 SS A182 F316/316L	8 - 12					
1850			Nipofl	5504146	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 10/160 SS A182 F316/316L	14 - 18					
1860			Nipofl	5504147	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 SS A182 F316/316L	20 - 24					
1870		Nipofl	5504144	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 SS A182 F316/316L	26 - 42						
1880												
1890	C	UJ2	Weldol	5510354	3/4 - 1 1/2	WELDOLET SCH 40S/80S SS A182 F316/316L	2 - 6	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE - ASME B31.3 AA, BD, BF, BFH, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS.				
1900	C		Weldol	5510355	3/4 - 1 1/2	WELDOLET SCH 30/80S SS A182 F316/316L	8 - 12					
1910	C		Weldol	5510356	3/4 - 1 1/2	WELDOLET SCH 10/80S SS A182 F316/316L	14 - 18					
1920	C		Weldol	5510357	3/4 - 1 1/2	WELDOLET SCH 20/80S SS A182 F316/316L	20 - 24					
1930	C		Weldol	5510358	3/4 - 1 1/2	WELDOLET STD/80S SS A182 F316/316L	26 - 42					
1940												
1950			Weldol	5509631	2 - 3	WELDOLET SCH 40S/40S SS A182 F316/316L	4 - 6					
1960			Weldol	5509634	2 - 4	WELDOLET SCH 30/40S SS A182 F316/316L	8 - 12					
1970			Weldol	5509630	2 - 6	WELDOLET SCH 10/40S SS A182 F316/316L	14 - 18					
1980			Weldol	5509633	2 - 6	WELDOLET SCH 20/40S SS A182 F316/316L	20 - 24					
1990			Weldol	5509650	2 - 6	WELDOLET STD/40S SS A182 F316/316L	26 - 42					
2000			Weldol	5509641	8 - 8	WELDOLET SCH 20/30 SS A182 F316/316L	24 - 24					
2010			Weldol	5509642	8 - 12	WELDOLET STD/30 SS A182 F316/316L	26 - 42					
2020												
2030	UJ2		Elbole	5516152	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4 - 6					
2040	UJ2		Elbole	5516153	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 30/160 SS A182 316/316L	8 - 12					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM		FGP/WPMP DATE BY REV SHEET PA CLASS 17NOV17 MC D 4 OF 7 5 150P21			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
2050		UJ2	Elbole	5516154	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 10/160 SS A182 316/316L	14 - 18									
2060		UJ2	Elbole	5516155	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 SS A182 316/316L	20 - 24									
2070		UJ2	Elbole	5516157	1 1/2 - 1 1/2	ELBOFLANGE 300# RF STD/160 SS A182 316/316L	26 - 42									
2080																
2090	D		Coupli	5480142	1/2 - 1 1/2	COUPLING 3000# SW SS A182 F316/316L	1 - 1 1/2									
2100	D		Reduci	5480134	3/4 - 1	COUPLING RED 3000# SW SS A182 F316/316L										
2110																
2120	D		Union	5480146	3/4 - 1 1/2	UNION 3000# SW SS A182 F316/316L	2 - 42									
2130																
2140	D	UD7	Plug	5498383	3/4 - 3/4	PLUG ROUND HEAD SCRD SS A182 F316/316L										
2150							2 - 42									
2160	D		Sockol	5537071	3/4 - 1 1/2	SOCKOLET 3000# SS A182 F316/316L										
2170	D		Endole	5533235	3/4 - 1 1/2	FLAT SOCKOLET 3000# SS A182 F316/316L										
2180	D	UD7	Thredo	5515771	3/4 - 3/4	THREDOLET 3000# SS A182 F316/316L	2 - 42									
2190																
2200		FS1				***** PIPE FLANGES *****										
2210																
2220	C		Flange	5582724	1/2 - 1 1/2	FLG WN 150# RF SS A182 F316/316L SCH 80S BORE										
2230			Flange	5582725	2 - 6	FLG WN 150# RF SS A182 F316/316L SCH 40S BORE										
2240			Flange	5582726	8 - 12	FLG WN 150# RF SS A182 F316/316L SCH 30 BORE										
2250			Flange	5582722	14 - 18	FLG WN 150# RF SS A182 F316/316L SCH 10 BORE										
2260			Flange	5582723	20 - 24	FLG WN 150# RF SS A182 F316/316L SCH 20 BORE										
2270			Flange	5583005	26 - 36	FLG WN 150# RF SS A182 F316/316L STD B16.47 SERIES A										
2280			Flange	5583005	42 - 42	FLG WN 150# RF SS A182 F316/316L STD B16.47 SERIES A										
2290																
2300		FA3	Flange	5582729	2 - 6	FLG WN 150# FF SS A182 F316/316L SCH 40S BORE										
2310																
2320			Reduci	5556281	2 - 2	FLG WN REDUCING 150# RF A182 F316/316L BORED TO SUIT SCH 40S										
2330																
2340			Flange	5582991	1/2 - 1 1/2	FLG WN 300# RF SS A182 F316/316L SCH 80S BORE										
2350			Flange	5582995	2 - 6	FLG WN 300# RF SS A182 F316/316L SCH 40S BORE										
2360			Flange	5582996	8 - 12	FLG WN 300# RF SS A182 F316/316L SCH 30 BORE										
2370			Flange	5582993	14 - 18	FLG WN 300# RF SS A182 F316 SCH 10S BORE										
2380			Flange	5582992	20 - 24	FLG WN 300# RF SS A182 F316/316L SCH 20 BORE										
2390			Flange	5583010	26 - 36	FLG WN 300# RF SS A182 F316/316L STD B16.47 SERIES A										
2400			Flange	5583010	42 - 42	FLG WN 300# RF SS A182 F316/316L STD B16.47 SERIES A										
2410																
2420	D		Flange	5582998	1 - 1	FLG WN 600# RF SS A182 F316/316L SCH 80S BORE										
2430	C		Flange	5583535	2 - 6	FLG WN 600# RF SS A182 F316/316L SCH 40S BORE										
2440	C		Flange	5583534	10 - 12	FLG WN 600# RF SS A182 F316/316L SCH 30 BORE										
2450	C		Flange	5583537	18 - 18	FLG WN 600# RF SS A182 F316/316L SCH 10 BORE										
2460																
2470	C		Flange	5583021	6 - 6	FLG WN 900# RF SS A182 F316/316L SCH 40S BORE										
2480																
2490	C		Flange	5583970	2 - 2	FLG WN 1500# RF SS A182 F316/316L SCH 40S BORE										
2500																
2510	D		Flange	5584167	1 1/2 - 1 1/2	FLG WN 2500# RTJ SS A182 F316/316L SCH 80S										
2520	D		Flange	5584169	2 - 2	FLG WN 2500# RTJ SS A182 F316/316L SCH 40S										
2530																
2540																
2550	C		Blind	5616520	1/2 - 24	FLG BLIND 150# RF SS A182 F316/316L										
2560			Blind	5616521	26 - 36	FLG BLIND 150# RF SS A182 F316/316L B16.47 SERIES A										
2570			Blind	5616521	42 - 42	FLG BLIND 150# RF SS A182 F316/316L B16.47 SERIES A										
2580																
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17NOV17	BY MC	REV D	SHEET 5 OF 7	PA 5	CLASS 150P21			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2590	C		Blind	5616529	1/2 - 24	FLG BLIND 300# RF SS A182 F316/316L	3/4 - 1 1/2						
2600			Blind	5616530	26 - 36	FLG BLIND 300# RF SS A182 F316/316L B16.47 SERIES A							
2610			Blind	5616530	42 - 42	FLG BLIND 300# RF SS A182 F316/316L B16.47 SERIES A							
2620													
2630		FA8	Reduci	5600655	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 150# RF SS A182 F316/316L							
2640													
2650	C		Flange	5851200	1/2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
2660	C		Flange	5851203	3/4 - 3/4	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
2670			Flange	5851203	2 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
2680													
2690						***** ORIFICE FLANGES *****							
2700													
2710			Orific	5640598	2 - 6	WN ORIF SET 300# RF SS A182 F316/316L SCH 40S BORE							
2720			Orific	5640597	8 - 12	WN ORIF SET 300# RF SS A182 F316/316L SCH 30 BORE							
2730			Orific	5640596	14 - 18	WN ORIF SET 300# RF SS A182 F316/316L SCH 10 BORE							
2740			Orific	5640595	20 - 24	WN ORIF SET 300# RF SS A182 F316/316L SCH 20 BORE							
2750			Orific	5640610	26 - 36	WN ORIF SET 300# RF SS A182 F316/316L STD WT B16.47 SERIES A							
2760			Orific	5640610	42 - 42	WN ORIF SET 300# RF SS A182 F316/316L STD WT B16.47 SERIES A							
2770													
2780						***** PIPE NIPPLES *****							
2790													
2800		810				NIPPLE PBE CUT FROM PIPE							
2810													
2820		FS1				***** GASKETS *****							
2830													
2840			GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
2850			GASFRS	5674011	26 - 42	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2860													
2870		FA3	GASFF	5664562	3/4 - 16	GASKET 150# FF RING EPDM 3MM THK							
2880													
2890			11.25	5673034	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE LOW STRESS							
2900													
2910	D		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR							
2920			GASFRS	5674032	26 - 42	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2930													
2940	D		GASFRS	5673103	1 - 18	GASKET 600# 316SS SP WND GRAPHITE IR							
2950													
2960	C		GASFRS	5673117	6 - 6	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
2970													
2980	C		GASFRS	5672251	2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
2990													
3000	D		GASRJ	5674480	1 1/2 - 2	GASKET 2500# RTJ SS OCTAGONAL RING							
3010													
3020													
3030		B1				***** BOLTING *****							
3040													
3050			BOLS	5675776	1/4 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
3060													
3070	D	SB6	CPSBOL	5675676	10 - 42	JACK SCREW HEX HEAD A320 B8M - CLASS 2							
3080													
3090						***** BLINDS AND SPACERS *****							
3100													
3110			Specta	5750421	3/4 - 14	SPECTACLE BLIND 150# RF SS A240 GR 316							
3120			Specta	5750422	3/4 - 10	SPECTACLE BLIND 300# RF SS A240 GR 316							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17NOV17	BY MC	REV D	SHEET 6 OF 7	PA 5	CLASS 150P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES													
3130	C	ZA6	Paddle	5750521	16 - 24	PADDLE BLANK 150# RF SS A240 GR 316	34 - 36	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE - ASME B31.3 AA, BD, BF, BFH, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS. MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 150 RF TEMPERATURE RANGE: -120 / 180 °C COMMENT: LOW TEMP BLOW DOWN													
3140			Paddle	5750526	26 - 36	PADDLE BLANK 150# RF SS A240 GR 316 B16.47 SERIES A															
3150			Paddle	5750526	42 - 42	PADDLE BLANK 150# RF SS A240 GR 316 B16.47 SERIES A															
3160			Paddle	5750522	12 - 24	PADDLE BLANK 300# RF SS A240 GR 316															
3170			Paddle	5750527	26 - 36	PADDLE BLANK 300# RF SS A240 GR 316 B16.47 SERIES A															
3180																					
3190			Paddle	5750621	16 - 24	PADDLE SPACER 150# RF SS A240 GR 316															
3200			Paddle	5750626	26 - 36	PADDLE SPACER 150# RF SS A240 GR 316 B16.47 SERIES A															
3210			Paddle	5750626	42 - 42	PADDLE SPACER 150# RF SS A240 GR 316 B16.47 SERIES A															
3220			Paddle	5750622	12 - 24	PADDLE SPACER 300# RF SS A240 GR 316															
3230			Paddle	5750627	26 - 36	PADDLE SPACER 300# RF SS A240 GR 316 B16.47 SERIES A															
3240																					
3250																					
3260						***** BRANCHES *****															
3270																					
3280						USE BRANCH CHART 150P21															
3290																					
3300					Reinfo	5365193_R							14 - 14	REINFORCING PAD FROM PIPE EFW SS A358 316/316L SCH STD							
3310																					
3320					Reinfo	5365193_R							14 - 14	REINFORCING PAD FROM PIPE EFW SS A358 316/316L SCH STD							
3330																					
3340			D		Reinfo	RWELD							2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 36						
3350			D		Reinfo	RWELD							2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 36						
3360																					
3370													***** CONSTRUCTION DETAILS *****								
3380													SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY								
3390													PIPE BENDS ----- NONE								
3400													THREAD SEALANT ----- PTFE PASTE								
3410																					
3420													FOR FABRICATION AND WELDING DETAILS SEE DOC. № L-ST-2024								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD							CORR. ALLOW. 1 MM	DATE 17NOV17	BY MC	REV D	SHEET 7 OF 7	PA 5	CLASS 150P21		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
460		VA5	Ball V	B01AS07	10 - 24	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO							
470													
480	D	VA5	Ball V	B01AS08	8 - 24	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO							
490													
500		VA5	Ball V	B01JS07	26 - 42	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A							
510		VA5	Ball V	B01JS08	26 - 42	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A							
520													
530	D	FS1	Ball V	B03AS01	1/2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO							
540													
550	D	FS1	Ball V	B03AS02	1/2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO							
560													
570	C	FS1	Ball V	A03AS01	3 - 3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO							
580													
590	C	FS1	Ball V	A03AS02	3 - 3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL RB LO							
600													
610	D	FS1	Ball V	B03AS07	10 - 10	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO							
620													
630						*****BUTTERFLY VALVES*****							
640													
650			Butter	T01AS61	10 - 24	BUTTERFLY 150# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO							
660			Butter	T01JS61	26 - 42	BUTTERFLY 150# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A							
670	D	FS1	Butter	T03AS61	12 - 12	BUTTERFLY 300# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO							
680													
690													
700	C					*****GATE VALVES *****							
710													
720	C		Gate V	G01AS48	3/4 - 1 1/2	GATE 150# RF SS A182 F316 TRIM 16 WEDGE HW							
730	D		Gate V	G01AS41	2 - 8	GATE 150# RF SS A351 CF8M TRIM 16 WEDGE HW							
740	D	VAR	Gate V	G01AS41	10 - 10	GATE 150# RF SS A351 CF8M TRIM 16 WEDGE HW							
750													
760	D	VA5	Gate V	G03AS41	3 - 3	GATE 300# RF SS A351 CF8M TRIM 16 WEDGE HW							
770													
780													
790						*****GLOBE VALVES*****							
800													
810			Globe	L01AS48	3/4 - 1 1/2	GLOBE 150# RF SS A182 F316 TRIM 16 OS&Y HW							
820			Globe	L01AS41	2 - 6	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y HW							
830	C		Globe	L01AS42	8 - 10	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y GO							
840													
850													
860						*****CHECK VALVES*****							
870													
880		VC1	Check	C01AS50	3/4 - 1 1/2	CHECK PISTON 150# RF SS A182 F316 TRIM 16							
890		VC2	Check	C01AS45	2 - 6	CHECK DUAL WAFER LUG 150# RF SS A351 CF8M TRIM 16 THROU BOLTED							
900		VC2	Check	C01AS52	8 - 24	CHECK DUAL DBL FLG 150# RF SS A351 CF8M TRIM 16							
910		VC2	Check	C01JS52	26 - 42	CHECK DUAL DBL FLG 150# RF SS A351 CF8M TRIM 16 B16.47 SERIES A							
920													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17NOV17	BY KD	REV D	FGP/WPMP SHEET 2 OF 7	PA 5	CLASS 150P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES													
930	C		Piping	5365140	3/4 - 1 1/2	***** PIPE *****															
940						PIPE SCH 80S SMLS SS A312 TP316/316L PE															
950						PIPE SCH 40S SMLS SS A312 TP316/316L BE															
960						PIPE SCH 30 SMLS SS A312 TP316/316L BE P															
970						PIPE SCH 10 SMLS SS A312 TP316/316L BE															
980						PIPE SCH 10 SMLS SS A312 TP316/316L BE															
990						PIPE SCH 20 SMLS SS A312 TP316/316L BE															
1000						PIPE SCH STD EFW SS A358 316/316L BE															
1010						PIPE SCH STD EFW SS A358 316/316L BE															
1020																					
1030	810	Piping	5365207	1/2 - 1/2	PIPE SCH 160 SMLS SS A312 TP316/316L PE																
1040																					
1050					***** FITTINGS *****																
1060																					
1070					D						90 Deg	5449033	1/2 - 1 1/2	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S							
1080														ELL 90 DEG LR SCH 40S SS A403 WP316/316L-S							
1090														ELL 90 DEG LR SCH 30 SS A403 WP316/316L-S							
1100														ELL 90 DEG LR SCH 10 SS A403 WP316/316L-S							
1110														ELL 90 DEG LR SCH 20 SS A403 WP316/316L-S							
1120														ELL 90 DEG LR SCH STD 316/316L SS A403 WP316/316L-W							
1130	ELL 90 DEG LR SCH STD 316/316L SS A403 WP316/316L-W																				
1140																					
1150	C	45 Deg	5449146	3/4 - 1 1/2		ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S															
1160						ELL 45 SCH 40S SS A403 WP316/316L-S															
1170					ELL 45 SCH 30 SS A403 WP316/316L-S																
1180					ELL 45 DEG SCH 10 A403 WP316/316L-S																
1190					ELL 45 DEG SCH 20 A403 WP316/316L-S																
1200					ELL 45 DEG SCH STD 316/316L SS A403 WP316/316L-W																
1210					ELL 45 DEG SCH STD 316/316L SS A403 WP316/316L-W																
1220																					
1230					C	Tee					5449156	3/4 - 1 1/2	TEE SCH 80S SS A403 WP316/316L-S								
1240													TEE SCH 40S SS A403 WP316/316L-S								
1250	TEE SCH 30 SS A403 WP316/316L-S																				
1260	TEE SCH 10 SS A403 WP316/316L-S																				
1270	TEE SCH 20 SS A403 WP316/316L-S																				
1280	TEE SCH STD 316/316L SS A403 WP316/316L-W																				
1290	TEE SCH STD 316/316L SS A403 WP316/316L-W																				
1300																					
1310	D	Reduci	5448905	1/2 - 1			TEE RED SCH 80S SS A403 WP316/316L-S	3/4 - 1 1/2													
1320							TEE RED SCH 40S/80S SS A403 WP316/316L-S														
1330					C	Reduci	5449184				2 - 4	TEE RED SCH 40S SS A403 WP316/316L-S			2 - 3						
1340												TEE RED SCH 30/40S SS A403 WP316/316L-S				3 - 6					
1350												TEE RED SCH 10/40S SS A403 WP316/316L-S					8 - 12				
1360												TEE RED SCH 30 SS A403 WP316/316L-S						14 - 16			
1370												TEE RED SCH 10/30 SS A403 WP316/316L-S							10 - 12		
1380												TEE RED SCH 20/30 SS A403 WP316/316L-S								14 - 18	
1390												TEE RED SCH 10 SS A403 316/316L-S									20 - 24
1400												TEE RED SCH 20/10 SS A403 WP316/316L-S									
1410	TEE RED SCH 20 SS A403 WP316/316L-S	20 - 24																			
1420	TEE RED SCH STD/20 SS A403 WP316/316L-W		24 - 24																		
1430	TEE RED SCH STD SS A403 WP316/316L-W			26 - 26																	
1440					28 - 42																
1450	C					Cap	5449049	3/4 - 1 1/2	CAP SCH 80S SS A403 WP316/316L-S												
1460										5449148	2 - 6	CAP SCH 40S SS A403 WP316/316L-S									
 FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION					FGP/WPMP										
						ALL SIZES- BUTT WELD					CORR. ALLOW. DATE BY REV SHEET PA CLASS										
						1 MM					17NOV17 KD D 3 OF 7 5 150P22										

SEQ. NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1470	D		Cap	5449044	8 - 12	CAP SCH 30 SS A403 WP316/316L-S	1 - 1 1/2						
1480		Cap	5448893	14 - 18	CAP SCH 10 SS ASTM A403 WP316/316L-S								
1490		Cap	5449040	20 - 24	CAP SCH 20 SS A403 WP316/316L-S								
1500		Cap	5449256	26 - 36	CAP SCH STD SS A403 WP316/316L-S								
1510		Cap	5449256	42 - 42	CAP SCH STD SS A403 WP316/316L-S								
1520													
1530		Concen	5449158	1/2 - 1	REDUCER CONC SCH 80S SS A403 WP316/316L-S								
1540		Concen	5549110	3/4 - 1 1/2	REDUCER CONC SCH 40S/80S SS A403 WP316/316L-S								
1550		Concen	5449149	2 - 4	REDUCER CONC SCH 40S SS A403 WP316/316L-S								
1560		Concen	5448899	4 - 6	REDUCER CONC SCH 30/40S SS A403 WP316/316L-S								
1570		Concen	5448881	6 - 6	REDUCER CONC SCH 10/40S SS A403 WP316/316L-S								
1580		Concen	5448900	8 - 10	REDUCER CONC SCH 30 SS A403 WP316/316L-S								
1590		Concen	5448901	8 - 12	REDUCER CONC SCH 10/30 SS A403 WP316/316L-S								
1600		Concen	5449173	12 - 12	REDUCER CONC SCH 20/30 SS A403 WP316/316L-S								
1610		Concen	5448894	14 - 16	REDUCER CONC SCH 10 SS A403 WP316/316L-S								
1620		Concen	5450720	18 - 18	REDUCER CONC SCH STD/10 SS A403 WP316/316L-W								
1630	Concen	5449174	14 - 18	REDUCER CONC SCH 20/10 SS A403 WP316/316L-S									
1640	Concen	5449175	20 - 20	REDUCER CONC SCH 20 SS A403 WP316/316L-S									
1650	Concen	5450721	20 - 24	REDUCER CONC SCH STD/20 SS A403 WP316/316L-W									
1660	Concen	5450722	26 - 36	REDUCER CONC SCH STD SS A403 WP316/316L-W									
1670	C						1 - 1 1/2						
1680		Eccent	5449159	3/4 - 1	REDUCER ECC SCH 80S SS A403 WP316/316L-S								
1690		Eccent	5549310	3/4 - 1 1/2	REDUCER ECC SCH 40S/80S SS A403 WP316/316L-S								
1700		Eccent	5449150	2 - 4	REDUCER ECC SCH 40S SS A403 WP316/316L-S								
1710		Eccent	5449151	4 - 6	REDUCER ECC SCH 30/40S SS A403 WP316/316L-S								
1720		Eccent	5448882	6 - 6	REDUCER ECC SCH 10/40S SS A403 WP316/316L-S								
1730		Eccent	5449152	8 - 10	REDUCER ECC SCH 30 SS A403 WP316/316L-S								
1740		Eccent	5449153	8 - 12	REDUCER ECC SCH 10/30 SS A403 WP316/316L-S								
1750		Eccent	5449170	12 - 12	REDUCER ECC SCH 20/30 SS A403 WP316/316L-S								
1760		Eccent	5448895	14 - 16	REDUCER ECC SCH 10 SS A403 WP316/316L-S								
1770		Eccent	5449171	14 - 18	REDUCER ECC SCH 20/10 SS A403 WP316/316L-S								
1780		Eccent	5450723	18 - 18	REDUCER ECC SCH STD/10 SS A403 WP316/316L-W								
1790		Eccent	5449172	20 - 20	REDUCER ECC SCH 20 SS A403 WP316/316L-S								
1800		Eccent	5450724	20 - 24	REDUCER ECC SCH STD/20 SS A403 WP316/316L-W								
1810		Eccent	5450725	26 - 36	REDUCER ECC SCH STD SS A403 WP316/316L-W								
1820													
1830	C	Nipofl	5504311	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 40S/80S SS A182 F316/316L	2 - 6	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE - ASME B31.3 REFER TO PIM-SU-5112 FOR ALL FLUID SERVICES						
1840	C	Nipofl	5504331	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 30/80S SS A182 F316/316L	8 - 12							
1850	C	Nipofl	5504312	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 10/80S SS A182 F316/316L	14 - 18							
1860	C	Nipofl	5504334	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH 20/80S SS A182 F316/316L	20 - 24							
1870	C	Nipofl	5504343	3/4 - 1 1/2	NIPOFLANGE 150# RF SCH STD/80S SS A182 F316/316L	26 - 42							
1880													
1890		Nipofl	5504011	2 - 2	NIPOFLANGE 150# RF SCH 40S/40S SS A182 F316/316L	3 - 6	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 535 °C COMMENT:						
1900		Nipofl	5504031	2 - 2	NIPOFLANGE 150# RF SCH 30/40S SS A182 F316/316L	8 - 12							
1910		Nipofl	5504012	2 - 2	NIPOFLANGE 150# RF SCH 10/40S SS A182 F316/316L	14 - 18							
1920		Nipofl	5504034	2 - 2	NIPOFLANGE 150# RF SCH 20/40S SS A182 F316/316L	20 - 24							
1930		Nipofl	5504043	2 - 2	NIPOFLANGE 150# RF SCH STD/40S SS A182 F316/316L	26 - 42							
1940													
1950	D		Nipofl	5504412	1 - 1	NIPOFLANGE 300# RF SCH 80S SS A182 F316/316L	1 1/2 - 1 1/2						
1960	D		Nipofl	5504411	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/80S SS A182 F316/316L	2 - 6						
1970													
1980	UJ2	Nipofl	5504132	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4 - 6							
1990	UJ2	Nipofl	5504145	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 30/160 SS A182 F316/316L	8 - 12							
2000	UJ2	Nipofl	5504146	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 10/160 SS A182 F316/316L	14 - 18							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17NOV17	BY KD	REV D	SHEET 4 OF 7	PA 5	CLASS 150P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2010		UJ2	Nipofl	5504147	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 20/160 SS A182 F316/316L	20 - 24						
2020		UJ2	Nipofl	5504144	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 SS A182 F316/316L	26 - 42						
2030													
2040	C		Weldol	5510354	3/4 - 1 1/2	WELDOLET SCH 40S/80S SS A182 F316/316L	2 - 6						
2050	C		Weldol	5510355	3/4 - 1 1/2	WELDOLET SCH 30/80S SS A182 F316/316L	8 - 12						
2060	C		Weldol	5510356	3/4 - 1 1/2	WELDOLET SCH 10/80S SS A182 F316/316L	14 - 18						
2070	C		Weldol	5510357	3/4 - 1 1/2	WELDOLET SCH 20/80S SS A182 F316/316L	20 - 24						
2080	C		Weldol	5510358	3/4 - 1 1/2	WELDOLET STD/80S SS A182 F316/316L	26 - 42						
2090													
2100			Weldol	5509631	2 - 3	WELDOLET SCH 40S/40S SS A182 F316/316L	4 - 6						
2110			Weldol	5509634	2 - 4	WELDOLET SCH 30/40S SS A182 F316/316L	8 - 12						
2120			Weldol	5509630	2 - 6	WELDOLET SCH 10/40S SS A182 F316/316L	14 - 18						
2130			Weldol	5509633	2 - 6	WELDOLET SCH 20/40S SS A182 F316/316L	20 - 24						
2140			Weldol	5509650	2 - 6	WELDOLET STD/40S SS A182 F316/316L	26 - 42						
2150			Weldol	5509641	8 - 8	WELDOLET SCH 20/30 SS A182 F316/316L	24 - 24						
2160			Weldol	5509642	8 - 12	WELDOLET STD/30 SS A182 F316/316L	26 - 42						
2170													
2180		UJ2	Elbole	5516152	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4 - 6						
2190		UJ2	Elbole	5516153	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 30/160 SS A182 316/316L	8 - 12						
2200		UJ2	Elbole	5516154	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 10/160 SS A182 316/316L	14 - 18						
2210		UJ2	Elbole	5516155	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 20/160 SS A182 316/316L	20 - 24						
2220		UJ2	Elbole	5516157	1 1/2 - 1 1/2	ELBOFLANGE 300# RF STD/160 SS A182 316/316L	26 - 42						
2230													
2240		FS1				***** PIPE FLANGES *****							
2250													
2260	D	FA8	Reduci	5600655	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 150# RF SS A182 F316/316L	3/4 - 2						
2270	D	FA8	Reduci	5600656	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 300# RF SS A182 F316/316L	3/4 - 2						
2280													
2290			Reduci	5556281	2 - 2	FLG WN REDUCING 150# RF A182 F316/316L BORED TO SUIT SCH 40S	3 - 3						
2300													
2310	C		Flange	5851200	1/2 - 3/4	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
2320	C		Flange	5851209	1 - 1	FLANGE ADAPTER 150# RF 316 SS X 25 MM TUBE							
2330	C		Flange	5851200	2 - 2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE							
2340	C		Flange	5851203	1/2 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
2350													
2360	C		Flange	5582724	1/2 - 1 1/2	FLG WN 150# RF SS A182 F316/316L SCH 80S BORE							
2370			Flange	5582725	2 - 6	FLG WN 150# RF SS A182 F316/316L SCH 40S BORE							
2380			Flange	5582726	8 - 12	FLG WN 150# RF SS A182 F316/316L SCH 30 BORE F							
2390			Flange	5582722	14 - 18	FLG WN 150# RF SS A182 F316/316L SCH 10 BORE							
2400			Flange	5582723	20 - 24	FLG WN 150# RF SS A182 F316/316L SCH 20 BORE							
2410			Flange	5583005	26 - 36	FLG WN 150# RF SS A182 F316/316L STD B16.47 SERIES A							
2420			Flange	5583005	42 - 42	FLG WN 150# RF SS A182 F316/316L STD B16.47 SERIES A							
2430													
2440		FA3	Flange	5582729	2 - 6	FLG WN 150# FF SS A182 F316/316L SCH 40S BORE							
2450													
2460			Flange	5582991	1/2 - 1 1/2	FLG WN 300# RF SS A182 F316/316L SCH 80S BORE							
2470			Flange	5582995	2 - 6	FLG WN 300# RF SS A182 F316/316L SCH 40S BORE							
2480			Flange	5582996	8 - 12	FLG WN 300# RF SS A182 F316/316L SCH 30 BORE							
2490			Flange	5582993	14 - 18	FLG WN 300# RF SS A182 F316 SCH 10S BORE							
2500			Flange	5582992	20 - 24	FLG WN 300# RF SS A182 F316/316L SCH 20 BORE							
2510			Flange	5583010	26 - 36	FLG WN 300# RF SS A182 F316/316L STD B16.47 SERIES A							
2520			Flange	5583010	42 - 42	FLG WN 300# RF SS A182 F316/316L STD B16.47 SERIES A							
2530													
2540	D		Flange	5583535	2 - 6	FLG WN 600# RF SS A182 F316/316L SCH 40S BORE							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17NOV17	BY KD	REV D	SHEET 5 OF 7	PA 5	CLASS 150P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2550	C		Flange	5583021	4 - 6	FLG WN 900# RF SS A182 F316/316L SCH 40S BORE							
2560													
2570	C		Flange	5583979	1 1/2 - 1 1/2	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE							
2580	C		Flange	5583970	2 - 2	FLG WN 1500# RF SS A182 F316/316L SCH 40S BORE							
2590													
2600	C		Blind	5616520	1/2 - 24	FLG BLIND 150# RF SS A182 F316/316L							
2610			Blind	5616521	26 - 36	FLG BLIND 150# RF SS A182 F316/316L B16.47 SERIES A							
2620			Blind	5616521	42 - 42	FLG BLIND 150# RF SS A182 F316/316L B16.47 SERIES A							
2630													
2640	C		Blind	5616529	1/2 - 24	FLG BLIND 300# RF SS A182 F316/316L							
2650			Blind	5616530	26 - 36	FLG BLIND 300# RF SS A182 F316/316L B16.47 SERIES A							
2660			Blind	5616530	42 - 42	FLG BLIND 300# RF SS A182 F316/316L B16.47 SERIES A							
2670													
2680						***** ORIFICE FLANGES *****							
2690													
2700			Orific	5640598	2 - 6	WN ORIF SET 300# RF SS A182 F316/316L SCH 40S BORE							
2710			Orific	5640597	8 - 12	WN ORIF SET 300# RF SS A182 F316/316L SCH 30 BORE WN							
2720			Orific	5640596	14 - 18	WN ORIF SET 300# RF SS A182 F316/316L SCH 10 BORE WN							
2730			Orific	5640595	20 - 24	WN ORIF SET 300# RF SS A182 F316/316L SCH 20 BORE WN							
2740			Orific	5640610	26 - 36	WN ORIF SET 300# RF SS A182 F316/316L STD WT B16.47 SERIES A							
2750			Orific	5640610	42 - 42	WN ORIF SET 300# RF SS A182 F316/316L STD WT B16.47 SERIES A							
2760													
2770						***** PIPE NIPPLES *****							
2780													
2790		810				NIPPLE PBE CUT FROM PIPE							
2800													
2810		FS1				***** GASKETS *****							
2820													
2830			GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
2840			GASFRS	5674011	26 - 42	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2850													
2860		FA3	GASFF	5664562	3/4 - 16	GASKET 150# FF RING EPDM 3MM THK							
2870													
2880	D		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR							
2890			GASFRS	5674032	26 - 42	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2900													
2910			GASFRS	5673103	2 - 6	GASKET 600# 316SS SP WND GRAPHITE IR							
2920													
2930	C		GASFRS	5673117	4 - 6	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
2940													
2950	C		GASFRS	5672251	1 1/2 - 1 1/2	GASKET 1500# 316SS SPIRAL WOUNDGRAPHITE IR							
2960													
2970		B1				***** BOLTING *****							
2980													
2990	D		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3000													
3010		SB6	CPSBOL	5675675	10 - 42	JACK SCREW HEX HEAD A320 GR L7M							
3020													
3030													
3040						***** BLINDS AND SPACERS *****							
3050													
3060			Specta	5750421	3/4 - 14	SPECTACLE BLIND 150# RF SS A240 GR 316							
3070			Specta	5750422	3/4 - 10	SPECTACLE BLIND 300# RF SS A240 GR 316							
3080													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17NOV17	BY KD	REV D	SHEET 6 OF 7	PA 5	CLASS 150P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
3090	C		Paddle	5750521	16 - 24	PADDLE BLANK 150# RF SS A240 GR 316									
3100			Paddle	5750526	26 - 36	PADDLE BLANK 150# RF SS A240 GR 316 B16.47 SERIES A									
3110			Paddle	5750526	42 - 42	PADDLE BLANK 150# RF SS A240 GR 316 B16.47 SERIES A									
3120			Paddle	5750522	12 - 24	PADDLE BLANK 300# RF SS A240 GR 316									
3130			Paddle	5750527	26 - 36	PADDLE BLANK 300# RF SS A240 GR 316 B16.47 SERIES A									
3140															
3150			Paddle	5750621	16 - 24	PADDLE SPACER 150# RF SS A240 GR 316									
3160			Paddle	5750626	26 - 36	PADDLE SPACER 150# RF SS A240 GR 316 B16.47 SERIES A									
3170			Paddle	5750626	42 - 42	PADDLE SPACER 150# RF SS A240 GR 316 B16.47 SERIES A									
3180			Paddle	5750622	12 - 24	PADDLE SPACER 300# RF SS A240 GR 316									
3190			Paddle	5750627	26 - 36	PADDLE SPACER 300# RF SS A240 GR 316 B16.47 SERIES A									
3200															
3210															
3220						***** BRANCHES *****									
3230															
3240						USE BRANCH CHART 150P22									
3250															
3260			Reinfo	5365193_R	14 - 14	REINFORCING PAD FROM PIPE EFW SS A358 316/316L SCH STD								34 - 36	
3270															
3280			Reinfo	5365193_R	14 - 14	REINFORCING PAD FROM PIPE EFW SS A358 316/316L SCH STD								34 - 36	
3290															
3300	D	ZA6	Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 36								
3310	D		Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 36								
3320															
3330															
3340						***** CONSTRUCTION DETAILS *****									
3350						PIPE BENDS -----NONE									
3360						THREAD SEALANT ----- NONE									
3370						SEAL WELDING----- FOR THREADED CONNECTIONS ONLY									
3380															
3390						FOR FABRICATION AND WELDING DETAILS									
3400						SEE DOC. N° L-ST-2024									
								SERVICE							
								SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE - ASME B31.3 REFER TO PIM-SU-5112 FOR ALL FLUID SERVICES							
								MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 535 °C COMMENT:							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 1 MM		DATE 17NOV17	BY KD	REV D	SHEET 7 OF 7	PA 5	CLASS 150P22

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10								032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.						
20		PT1				ALL MATERIALS IN THIS PIPING CLASS TO BE SUITABLE FOR USE WITH OXYGEN SERVICE		811	NIPPLE TO BE CUT FROM SCH 80S PIPE						
30						SEE NOTE CS1 FOR SPECIAL REQUIREMENTS		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
40		CS1						CS1	CERTAIN LINES IN THIS SPEC REQUIRE INTERNAL CLEANING, TO BE PERFORMED ON SITE. FOR THE ACID CLEANING TO BE EFFECTIVE, THE PAINT AND COATING MUST BE REMOVED FROM ID OF FITTINGS AND FLANGES BY BLASTING OR OTHER MECHANICAL MEANS.						
50						***** VALVES*****									
60						***MODULAR VALVES*****									
70			Ball V	M01AS02	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 150# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR									
80						*****BALL VALVES*****									
90								PT1	ALL PIPEWORK USED FOR OXYGEN SERVICE SHALL BE BE PAINTED BLUE AS DEFINED IN SPECIFICATION FOR PAINT & PROTECTIVE COATING COM-SU-4743-TCO						
100								UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.						
110			BALTXT	BE5S3OX	1/2 - 3/4	BALL COMPRESSIBLE TUBE ENDS 1500# 316 SS BODY & TRIM STANDARD PORT TFE SEAT & SEALS.		VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.						
120								VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.						
130			BALFLG	B01AS01	3/4 - 2	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
140						*****GLOBE VALVES*****									
150															
160	C		Globe	L01AS48	3/4 - 1 1/2	GLOBE 150# RF SS A182 F316 TRIM 16 OS&Y HW									
170			Globe	L01AS41	2 - 2	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y HW									
180						*****CHECK VALVES*****									
190															
200	C														
210		VC1	Check	C01AS50	3/4 - 1 1/2	CHECK PISTON 150# RF SS A182 F316 TRIM 16									
220		VC2	Check	C01AS45	2 - 2	CHECK DUAL WAFER LUG 150# RF SS A351 CF8M TRIM 16 THROU BOLTED									
230						***** TUBES *****									
240															
250			TUBING	5851000	1/2 - 1/2	TUBE SMLS WT 0.065" SS A269 TP 316									
260			TUBING	5851001	3/4 - 3/4	TUBE SMLS WT 0.083" SS A269 TP 316									
270						***** FITTINGS *****									
280															
290															
300															
310															
320															
330															
340															
350			90 Deg	5851100	1/2 - 3/4	ELBOW 90 DEG TWIN FERRULE TYPE COMPRESSION TUBE UNION ENDS 316 STAINLESS STEEL.									
360			Tee	5851101	1/2 - 3/4	TEE TWIN FERRULE TYPE COMPRESSION TUBE UNION ENDS 316 STAINLESS STEEL.									
370			Union	5851102	1/2 - 3/4	UNION TWIN FERRULE TYPE COMPRESSION TUBE UNION ENDS 316 STAINLESS STEEL									
380			Cap	5851103	1/2 - 3/4	CAP TWIN FERRULE TYPE COMPRESSION TUBE UNION END 316 STAINLESS STEEL									
390			Union	5851104	1/2 - 1/2	REDUCING UNION TWIN FERRULE TYPE COMPRESSION TUBE UNION ENDS 316 STAINLESS STEEL	3/4 - 3/4								
400						***** PIPE FLANGES *****									
410															
420															
430															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP							
						TUBE - 3/4" & SMALLER - TUBE COMPRESSION FITTINGS PIPE - 3/4" & LARGER - BUTT WELD		CORR. ALLOW. 0 MM		DATE 11JUL13	BY SMS	REV C	SHEET 1 OF 2	PA 5	CLASS 150P23

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
440														
450			ADPOR	5851105	1/2 - 1/2	ADAPTOR TWIN FERRULE TYPE COMPRESSION TUBE UNION TO FEMALE NPT PIPE CONNECTION 316 STAINLESS STEEL	1/2 - 1/2							
460						***** PIPE, PIPE FITTINGS *****								
470						***** PIPE *****								
480														
490														
500														
510	C		Piping	5365140	3/4 - 2	PIPE SCH 80S SMLS SS A312 TP316/316L PE								
520		811	Piping	5365207	1/2 - 1/2	PIPE SCH 160 SMLS SS A312 TP316/316L PE								
530						***** FITTINGS *****								
540														
550														
560														
570	C		90 Deg	5449033	3/4 - 2	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S								
580	C		45 Deg	5449146	3/4 - 2	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S								
590	C		Tee	5449156	3/4 - 2	TEE SCH 80S SS A403 WP316/316L-S								
600	C		Cap	5449049	3/4 - 2	CAP SCH 80S SS A403 WP316/316L-S								
610														
620	C		Concen	5449158	3/4 - 1 1/2	REDUCER CONC SCH 80S SS A403 WP316/316L-S	1 - 2							
630	C		Eccent	5449159	3/4 - 1 1/2	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1 - 2							
640														
650	C		Weldol	5509637	3/4 - 3/4	WELDOLET SCH 80S SS A182 F316/316L	2 - 2							
660			Weldol	5852300	1/2 - 1/2	WELDOLET SCH 40S/40S SS A182 F316/316L	2 - 2							
670		UD7	Thredo	5852301	3/4 - 3/4	THREDOLET 3000# SS A182 F316/316L	2 - 2							
680		UD7	Plug	5852108	3/4 - 3/4	PLUG ROUND HEAD SCRD SS A182 F316/316L								
690						***** FLANGES *****								
700														
710														
720	C		Flange	5582724	3/4 - 2	FLG WN 150# RF SS A182 F316/316L SCH 80S BORE								
730	C		Blind	5852201	3/4 - 2	FLG BLIND 150# RF SS A182 F316/316L								
740						***** NIPPLES*****								
750						NIPPLE PBE - CUT FROM PIPE								
760		811				***** GASKETS*****								
770														
780														
790														
800	C		GASFRS	5673043	3/4 - 2	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR								
810						***** BOLTING *****								
820														
830														
840	C	B1	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
850						*****CONSTRUCTION DETAILS*****								
860						SEAL WELDING-----NONE								
870						PIPE BENDS-----NONE								
880						THREAD SEALANT-----PTFE PASTE								
890														
900														
910														
920														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						TUBE - 3/4" & SMALLER - TUBE COMPRESSION FITTINGS PIPE - 3/4" & LARGER - BUTT WELD	CORR. ALLOW. 0 MM	DATE 11JUL13	BY SMS	REV C	SHEET 2 OF 2	PA 5	CLASS 150P23	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10						***** VALVES*****		032 ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. SC1 USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTRORTRACE. VB2 SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. VC1 SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. ZA6 REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
20														
30														
40			Ball V	BE5S3	1/2 - 3/4	BALL COMPRESSIBLE TUBE ENDS 1500# 316 SS BODY & TRIM STANDARD PORT TFE SEAT & SEALS.								
50														
60			Needle	NE5S3	1/2 - 3/4	NEEDLE COMPRESSIBLE TUBE ENDS 1500# 316 SS BODY & TRIM.								
70						***** FLANGED VALVES*****								
80														
90						***MODULAR VALVES*****								
100														
110														
112														
120			Ball V	M01AS02	1/2 - 1	MODULAR VALVE DB&B BALL TYPE 150# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR								
130						*****BALL VALVES*****								
140														
150														
160	C	VB2	BALFLG	B01AS01	3/4 - 1	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO								
170						*****GLOBE VALVES*****								
180														
190														
200	C		Globe	L01AS48	3/4 - 1	GLOBE 150# RF SS A182 F316 TRIM 16 OS&Y HW								
210						*****CHECK VALVES*****								
220														
230														
240	C	VC1	Check	C01AS50	3/4 - 1	CHECK PISTON 150# RF SS A182 F316 TRIM 16								
250						***** TUBES *****								
260														
270														
280			TUBING	5365303	1/2 - 1/2	TUBE SMLS WT 0.065" ANNEALED/STRESS RELIEVED A269 TP 316 STAINLESS STEEL								
290			TUBING	5365304	3/4 - 3/4	TUBE SMLS WT 0.083" ANNEALED/STRESS RELIEVED A269 TP 316 STAINLESS STEEL								
300						***** FITTINGS *****								
310														
320														
330			90 Deg	5365400	1/2 - 3/4	ELBOW 90 DEG TWIN FERRULE TYPE COMPRESSION TUBE UNION ENDS 316 STAINLESS STEEL.								
340			Tee	5365401	1/2 - 3/4	TEE TWIN FERRULE TYPE COMPRESSION TUBE UNION ENDS 316 STAINLESS STEEL.								
350			Union	5365402	1/2 - 3/4	UNION TWIN FERRULE TYPE COMPRESSION TUBE UNION ENDS 316 STAINLESS STEEL								
360			Cap	5365403	1/2 - 3/4	CAP TWIN FERRULE TYPE COMPRESSION TUBE UNION END 316 STAINLESS STEEL								
370			Union	5365404	1/2 - 1/2	REDUCING UNION TWIN FERRULE TYPE COMPRESSION TUBE UNION ENDS 316 STAINLESS STEEL	3/4 - 3/4							
380						***** ADAPTOR *****								
390														
400														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						3/4" & SMALLER - TUBE COMPRESSION FITTINGS PIPE- ALL SIZES- BUTT WELD	CORR. ALLOW. 0 MM	DATE 11JUL13	BY MLA	REV C	SHEET 1 OF 2	PA 5	CLASS 150P24	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
410								SERVICE DESIGN - ASME B16.5 MG 2.2 CODE - ASME B31.3 CCS, CFA, CHM, CI, CPE, CSB, DL, VB, WDM, WP, WU MATERIAL: 316 SS TUBE/PIPE FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 200 °C COMMENT: FOR USE ON VENDOR PACKAGES ONLY						
420			ADPOR	5365450	1/2 - 1/2	ADAPTOR RED TWIN FERRULE TYPE COMPRESSION TUBE UNION TO FEMALE NPT PIPE CONNECTION 316 STAINLESS STEEL	3/4 - 1							
430			ADPOR	5365452	1/2 - 3/4	ADAPTOR TWIN FERRULE TYPE COMPRESSION TUBE UNION TO MALE NPT PIPE CONNECTION 316 STAINLESS STEEL								
440						***** PIPE, PIPE FTGS, & FLANGED VALVES*****								
450						***** PIPE *****								
460														
470														
480														
490			Piping	5365120	1 - 1	PIPE SCH 40S SMLS SS A312 TP316/316L PE								
500						***** FITTINGS *****								
510			90 Deg	5852100	1 - 1	ELL 90 DEG LR SCH 40S SS A403 WP316/316L-S								
520														
530			45 Deg	5852101	1 - 1	ELL 45 SCH 40S SS A403 WP316/316L-S								
540														
550			Tee	5852102	1 - 1	TEE SCH 40S SS A403 WP316/316L-S								
560			Cap	5852106	1 - 1	CAP SCH 40S SS A403 WP316/316L-S								
570						***** FLANGES *****								
580														
590			Flange	5582725	1 - 1	FLG WN 150# RF SS A182 F316/316L SCH 40S BORE								
600			Blind	5852201	1 - 1	FLG BLIND 150# RF SS A182 F316/316L								
610						***** NIPPLES *****								
620						NIPPLES PBE CUT FROM PIPE								
630						***** GASKETS *****								
640	C					GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR								
650						***** BOLTING *****								
660			GASFRS	5673043	1 - 1									
670														
680														
690														
700	C													
710														
720														
730														
740	C	B1	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
750						*****CONSTRUCTION DETAILS*****								
760						SEAL WELDING-----NONE								
770						PIPE BENDS-----NONE								
780						THREAD SEALANT-----NONE								
790														
800														
810														
820														
830														
840														
850														
860														
870		032												
880														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						3/4" & SMALLER - TUBE COMPRESSION FITTINGS PIPE- ALL SIZES- BUTT WELD	CORR. ALLOW. 0 MM	DATE 11JUL13	BY MLA	REV C	SHEET 2 OF 2	PA 5	CLASS 150P24	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10						***** VALVES *****		032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.						
20								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
30								FW3	VIKING JOHNSON COUPLINGS SHALL ONLY BE USED ON 'WF' FIREWATER SERVICE						
40								GA3	SPIRAL WOUND GASKETS SHALL ONLY BE USED BETWEEN CARBON STEEL FLANGES AND CARBON STEEL VALVES.						
50								NX1	9" NOM SIZE (225MM O.D.) STUB FLANGES SHALL BE SUPPLIED C/W 8" NOM SIZE BACKING RINGS.						
60			Gate V	GA6B1	1/2 - 1 1/2	GATE 150# SCR D FNPT BS1400 LG2 BODY AND TRIM HW		PA5	THE FOLLOWING PIPE NOM.SIZE/OD. CROSS REFERENCES SHALL BE USED FOR IDENTIFICATION AND ORDERING PURPOSES FOR SDR 11 SYSTEMS:						
70	C	VA5	Gate V	G01AC41	2 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW			NOM (IN.)	O.D. (MM)	I.D. (MM)				
80									2"	63	51.4				
90	D	VA5	Gate V	G01AC96	2 - 10	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 2800MM EXT			3"	90	73.6				
100	D	VA5	Gate V	G01AC99	2 - 10	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 3300MM EXT			4"	110	90				
110	D	VA5	Gate V	G01AC95	2 - 10	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW 3700MM EXT			5"	140	114.6				
120						*****GLOBE VALVES*****			6"	160	130.8				
130									7"	180	147.2				
140									8"	200	163.6				
150	C		Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW			9"	225	184				
160	C		Globe	L01AC42	8 - 8	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO			10"	250	204.6				
170						*****BUTTERFLY VALVES*****			12"	315	257.8				
180									14"	355	290.6				
190			Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO			16"	400	327.4				
200						***** PIPE *****			18"	450	368.2				
210									20"	500	409.2				
220									22"	560	458.4				
230									24"	630	515.6				
240		PA5	Piping	5713401	2 - 24	PIPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427			28"	710	574				
250									32"	800	656				
260	D		Piping	5713402	26 - 40	PIPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)			36"	900	738				
270		PA5				***** FITTINGS *****									
280															
290															
300	C		90 Deg	5713801	2 - 12	ELBOW 90 DEG SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF									
310			90 Deg	5713501	2 - 12	ELBOW 90 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
320	C		90 Deg	5713811	14 - 24	BEND 90 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF									
330	C		90 Deg	5713511	14 - 24	BEND 90 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
340															
350	C		45 Deg	5713802	2 - 12	ELBOW 45 DEG SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF									
360			45 Deg	5713502	2 - 12	ELBOW 45 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
370	C		45 Deg	5713810	14 - 24	BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF									
380	C		45 Deg	5713510	14 - 24	BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
390	D		45 Deg	5713540	28 - 40	BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)									
400															
410	C		Tee	5713803	2 - 24	TEE EQUAL SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF									
420	C		Tee	5713503	2 - 24	TEE EQUAL ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW.	FGP/WPMP						
								0 MM	DATE	BY	REV	SHEET	PA	CLASS	
									17MAR17	KD	D	1 OF 3	5	150PE2	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
430	C		Reduci	5713804	2 - 18	TEE REDUCING SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	4 - 24	UDC	FOR USE WITH PRESSURE CONNECTIONS ONLY USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. FOR USE IN POTABLE WATER AND FOAM CONCENTRATE SERVICE ONLY.					
440														
450	C		Reduci	5713504	2 - 18	TEE REDUCING ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	4 - 24	VG6						
460									FOR USE IN OILY SEWER DRAIN SYSTEM REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
470	D		Reduci	5713545	4 - 4	TEE REDUCING ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	24 - 24	VG8						
480														
490	D		Reduci	5713546	16 - 16	TEE REDUCING ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	24 - 24	ZA6						
500														
510														
520	C		Cap	5713805	2 - 24	CAP SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF								
530														
540			Cap	5713505	2 - 24	CAP ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427								
550														
560	C		Concen	5713806	2 - 14	REDUCER CONC SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	3 - 16							
570	C		Concen	5713506	2 - 14	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	3 - 16							
580														
590	D		Concen	5713535	6 - 6	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	12 - 12							
600														
610	D		Concen	5713535	8 - 8	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	16 - 16							
620														
630	D		Concen	5713536	16 - 16	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE)	24 - 24							
640														
650														
660	C		Eccent	5713807	2 - 10	REDUCER ECC SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427 EF	3 - 12							
670			Eccent	5713507	2 - 10	REDUCER ECC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	3 - 12							
680														
690			Coupli	5713509	2 - 24	COUPLING ELECTRO FUSION SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427			SERVICE					
700									DESIGN: 16 BAR G MAX.					
710		UDC	SADDLE	5713515	2 - 2	PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR	3 - 8		CCB, CF, DHC (ATM PRESS @ 70°C & BELOW), DL (ATM PRESS @ 70°C & BELOW), DLC (ATM PRESS @ 70°C & BELOW), DS, FCC, FM, WD, WDM, WE, WF, WRA, WSC (ATM PRESS @ 70°C & BELOW), WT, WU					
720		UDC	SADDLE	5713516	2 - 2	PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR	10 - 24							
730														
740		UDC	Concen	5544151	3/4 - 1 1/2	SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153	2 - 2							
750														
760		UDC	Bushin	5491325	1/2 - 1 1/2	BUSHING 3000# SCRDR (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153	3/4 - 2		MATERIAL: POLYETHYLENE (HDPE) FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 40 °C COMMENT: 70 DEG. C.- UPSET CONDITION @ ATM PRESS.					
770														
780			Coupli	5713518	1/2 - 2	PIPE ADAPTOR WITH GUNMETAL THREADED (NPT-MALE) OUTLET								
790														
800														
810	D	FW3	Coupli	5719243	3 - 24	MECHANICAL COUPLING VIKING JOHNSON DISMANTLING JOINT TYPE 150#FF								
820														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW. 0 MM	FGP/WPMP DATE 17MAR17 BY KD REV D SHEET 2 OF 3 PA 5 CLASS 150PE2					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
830 840 850 860 870 880 890 900 910 920 930 940 950 960 970 980 990 1000 1010 1020 1030 1040 1050 1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1170 1180 1190 1200 1210 1220 1230 1240 1250 1260 1270 1280 1290						***** FLANGES ***** STUB FLG ELONGATED TYPE SDR 11 HD POLYETHYLENE C/W BACKING RING PE 100 WATER ASTM-D3350 / ISO 4427 FLG BLIND FF 150# HDPE PE 100 WATER ASTM-D3350 / ISO 4427 WITH STEEL INSERT STUB FLG ELONGATED TYPE SDR 11 HD POLYETHYLENE C/W BACKING RING PE 100 WATER ASTM-D3350 / ISO 4427 (NON-STANDARD SIZE) SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND WITH 8" PIPE SDR 11 HD PE 100 ASTM-D3350/ISO 4427 SPOOL - PUDDLE (ANCHOR) FLANGE WITH STUB FLANGE C/W BACKING RING AND WITH 24" PIPE SDR 11 HD PE 100 ASTM-D3350/ISO 4427 ***** PUDDLE FLANGE ***** PUDDLE (ANCHOR) FLANGE HDPE FF ***** GASKETS ***** GASKET 150# RF FLAT RING PTFE GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR ***** BOLTING ***** STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS ***** BLINDS AND SPACERS ***** SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153 ***** BRANCHES ***** USE TEES AND REDUCERS AS REQUIRED ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE								
								SERVICE						
								DESIGN: 16 BAR G MAX. CCB, CF, DHC (ATM PRESS @ 70°C & BELOW), DL (ATM PRESS @ 70°C & BELOW), DLC (ATM PRESS @ 70°C & BELOW), DS, FCC, FM, WD, WDM, WE, WF, WRA, WSC (ATM PRESS @ 70°C & BELOW), WT, WU						
								MATERIAL: POLYETHYLENE (HDPE) FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 40 °C COMMENT: 70 DEG. C.- UPSET CONDITION @ ATM PRESS.						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW. 0 MM	DATE 17MAR17	BY KD	REV D	SHEET 3 OF 3	PA 5	CLASS 150PE2

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10						*****GATE VALVES*****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. SPIRAL WOUND GASKETS SHALL ONLY BE USED BETWEEN CARBON STEEL FLANGES AND CARBON STEEL VALVES. 9" NOM SIZE (225MM O.D.) STUB FLANGES SHALL BE SUPPLIED C/W 8" NOM SIZE BACKING RINGS. THE FOLLOWING PIPE NOM.SIZE/OD. CROSS REFERENCES SHALL BE USED FOR IDENTIFICATION AND ORDERING PURPOSES FOR SDR 11 SYSTEMS: NOM O.D. I.D. (IN.) (MM) (MM) 2" 63 51.4 3" 90 73.6 4" 110 90 5" 140 114.6 6" 160 130.8 7" 180 147.2 8" 200 163.6 9" 225 184 10" 250 204.6 12" 315 257.8 14" 355 290.6 16" 400 327.4 18" 450 368.2 20" 500 409.2 22" 560 458.4 24" 630 515.6 28" 710 574 32" 800 656 36" 900 738				
20						GA3							
30						NX1							
40						PA5							
50													
60													
70													
80													
90													
100													
110													
120													
130						Globe		L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW			
140						Globe		L01AC42	8 - 8	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO			
150													
160													
170													
180						Butter		T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO			
190													
200													
210													
220													
230						PA5		Piping	5713827	2 - 24	PIPE SDR 11 HD POLYETHYLENE - ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
240													
250						PA5							
260													
270						NB9		90 Deg	5713814	2 - 12	ELBOW 90 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE - ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
280													
290								90BND	5713815	14 - 24	BEND 90 DEG LONG RADIUS SDR 11 HD POLYETHYLENE - ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
300													
310								60 Deg	5713818	2 - 12	ELBOW 60 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE - ID YELLOW PE 100 GAS ASTM-D3350 / ISO4427		
320													
330								60BND	5713819	14 - 24	BEND 60 DEG LONG RADIUS SDR 11 HD POLYETHYLENE - ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
340													
350								45 Deg	5713816	2 - 12	ELBOW 45 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
360													
370								45BND	5713817	14 - 24	BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
380													
390								30 Deg	5713820	2 - 12	ELBOW 30 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
400													
410								30BND	5713821	14 - 24	BEND 30 DEG LONG RADIUS SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
420													
430								11.25	5713822	2 - 12	ELBOW 15 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427		
440													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION	CORR. ALLOW. 0 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 JCS C 1 OF 3 5 150PE3					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES												
430	C	NX1	11BND	5713823	14 - 24	BEND 15 DEG LONG RADIUS SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427	3 - 24	ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.											
440			Tee	5713830	2 - 24	TEE EQUAL ELONGATED TYPE SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427														
450																				
460			Reduci	5713831	2 - 20	TEE REDUCING ELONGATED TYPE SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427														
470																				
480			Cap	5713812	2 - 24	CAP ELONGATED TYPE SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427														
490																				
500			Concen	5713828	2 - 20	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427														
510																				
520			Eccent	5713829	2 - 20	REDUCER ECC ELONGATED TYPE SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427														
530																				
540			Coupli	5713813	2 - 24	COUPLING ELECTRO FUSION SDR 11 HD POLYETHYLENE ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427														
550																				
560			Flange	5713824	2 - 24	***** FLANGES ***** STUB FLG ELONGATED TYPE SDR 11 HD POLYETHYLENE - ID YELLOW C/W BACKING RING PE 100 GAS ASTM-D3350 / ISO 4427														
570																				
580																				
590																				
600			Blind	5713825	2 - 24	FLG BLIND FF 150# HDPE - ID YELLOW PE 100 GAS ASTM-D3350 / ISO 4427 WITH STEEL INSERT														
610																				
620																				
630																				
640			Specta	5750421	2 - 12	SPECTACLE BLIND 150# RF SS A240 GR 316 ***** PUDDLE FLANGE *****														
650																				
660																				
670																				
680			ANCHOR	5597702	2 - 12	PUDDLE (ANCHOR) FLANGE HDPE FF ***** GASKETS *****														
690																				
700																				
710																				
720			GASFR	5664201	2 - 24	GASKET 150# RF FLAT RING PTFE														
730																				
740																				
750																				
760			GASFRS	5673043	2 - 18	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR ***** BOLTING *****														
770																				
780																				
790																				
800	C	B1	BOLS	5675768	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS ***** BRANCHES ***** USE TEES AND REDUCERS AS REQUIRED														
810																				
820																				
830																				
840																				
850																				
860																				
870																				
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW. 0 MM	FGP/WPMP DATE 17MAR17					BY JCS	REV C	SHEET 2 OF 3	PA 5	CLASS 150PE3		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
880 890 900 910 920 930 940	C	ZA6				***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- LONG RADIUS THREAD SEALANT ----- NONE ELECTRIC SOCKET WELDING OR BUTT WELDING AS APPLICABLE									
								SERVICE							
								DESIGN: 4.7 BAR G TEMPERATURE RANGE: -40 TO +40 DEG.C. FG - FUEL GAS (EARLY WORKS)							
								MATERIAL: POLYETHYLENE (HDPE) FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 40 °C COMMENT:							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW. 0 MM		DATE 17MAR17	BY JCS	REV C	FGP/WPMP SHEET 3 OF 3	PA 5	CLASS 150PE3

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10						*****GATE VALVES*****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. SPIRAL WOUND GASKETS SHALL ONLY BE USED BETWEEN CARBON STEEL FLANGES AND CARBON STEEL VALVES. 9" NOM SIZE (225MM O.D.) STUB FLANGES SHALL BE SUPPLIED C/W 8" NOM SIZE BACKING RINGS. THE FOLLOWING PIPE NOM.SIZE/OD. CROSS REFERENCES SHALL BE USED FOR IDENTIFICATION AND ORDERING PURPOSES FOR SDR 7.4 SYSTEMS: NOM (IN.) O.D. (MM) I.D. (MM) 1" 25 17.6 2" 63 44.8 3" 90 63.9 4" 125 88.7 6" 180 127.8 7" 225 159.7 8" 225 159.7 9" 225 159.7 10" 280 198.8 12" 315 223.6 14" 355 252.2 16" 400 284 18" 450 319.6 20" 500 355.2 24" 630 TBA						
20						GA3									
30															
40															
50															
60															
70															
80															
90															
100															
110															
120															
130															
140															
150															
160															
170															
180															
190															
200															
210															
220															
230															
240															
250															
260		UDC	FOR USE WITH PRESSURE CONNECTIONS ONLY USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.												
270	VA5														
280															
290															
300															
310															
320															
330															
340															
350															
360															
370															
380															
390															
400															
410															
420															
430															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP							
						ALL SIZES - BUTT FUSION		CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS
								0 MM		17MAR17	PDM	A	1 OF 2	5	150PE4

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
440			SADDLE	5713516	2 - 2	PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR	3 - 24						
450													
460													
470		UDC	Concen	5544151	3/4 - 1 1/2	SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153	2 - 2						
480													
490		UDC	Bushin	5491325	1/2 - 1 1/2	BUSHING 3000# SCRD (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153	3/4 - 2						
500													
510			Coupli	5713518	1/2 - 2	PIPE ADAPTOR WITH GUNMETAL THREADED (NPT-MALE) OUTLET							
520													
530													
540						***** FLANGES *****							
550													
560		NX1	Flange	5713598	2 - 24	STUB FLG ELONGATED TYPE SDR 7.4 HD POLYETHYLENE C/W BACKING RING PE 100 WATER ASTM-D3350 / ISO 4427							
580			Blind	5621091	2 - 24	FLG BLIND FF 150# HDPE PE 100 WATER ASTM-D3350 / ISO 4427 WITH STEEL INSERT							
590						***** PUDDLE FLANGE *****							
600													
610													
620			ANCHOR	5597701	2 - 12	PUDDLE (ANCHOR) FLANGE HDPE FF							
630													
640						***** GASKETS *****							
650													
660													
670			GASFR	5664201	2 - 24	GASKET 150# RF FLAT RING PTFE							
680		GA3	GASFRS	5673043	2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
690													
700						***** BOLTING *****							
710		B1											
720													
730			BOLS	5675768	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS							
740													
750													
760			Specta	5749582	2 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153							
770													
780													
790						***** BRANCHES *****							
800													
810						USE TEES AND REDUCERS AS REQUIRED							
820						***** CONSTRUCTION DETAILS *****							
830													
840													
850						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
860						PIPE BENDS ----- NONE							
870						THREAD SEALANT ----- NONE ELECTRIC							
880						SOCKET WELDING OR BUTT							
890						WELDING AS APPLICABLE							
900													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION	CORR. ALLOW. 0 MM	DATE 17MAR17	BY PDM	REV A	SHEET 2 OF 2	PA 5	CLASS 150PE4

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10	B	VC2	Gate V Gate V	GA6B1 G01AC41	1/2 - 1 1/2 2 - 8	*****VALVES*****	8 - 8	B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. SPIRAL WOUND GASKETS SHALL ONLY BE USED BETWEEN CARBON STEEL FLANGES AND CARBON STEEL VALVES. 9" NOM SIZE (225MM O.D.) STUB FLANGES SHALL BE SUPPLIED C/W 8" NOM SIZE BACKING RINGS. THE FOLLOWING PIPE NOM.SIZE/OD. CROSS REFERENCES SHALL BE USED FOR IDENTIFICATION AND ORDERING PURPOSES FOR SDR 7.4 SYSTEMS: NOM O.D. O.D. I.D. (IN.) (MM) (MM) 1" 25 17.6 2" 63 44.8 3" 90 63.9 4" 125 88.7 6" 180 127.8 7" FOR SGI/SGP CLASS ONLY - DO NOT USE 8" 225 159.7 9" FOR SGI/SGP CLASS ONLY - DO NOT USE 10" 280 198.8 12" 315 223.6 14" 355 252.2 16" 400 284 18" 450 319.6 20" 500 355.2 24" 630 TBA						
20						GA3									
30						NX1									
40						PA8									
50															
60															
70															
80															
90			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW									
100			Globe	L01AC42	8 - 8	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO									
110															
120								*****CHECK VALVES							
130															
140					Check	C01AC45		2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED						
150															
160					Check	C01AC52		8 - 8	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16						
170									***** PIPE *****						
180															
190															
200				PA8	Piping	5713411		6 - 8	PIPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350/ ISO 4427						
210															
220															
230				PA8					***** FITTINGS *****						
240															
250					90 Deg	5713572		6 - 8	ELBOW 90 DEG FABRICATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427						
260			60 Deg	5713836	6 - 8	ELBOW 60 DEG FABRICATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
270			45 Deg	5713570	6 - 8	ELBOW 45 DEG FABRICATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
280			30 Deg	5713835	6 - 8	ELBOW 30 DEG FABRICATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
290			Tee	5713591	6 - 8	TEE EQUAL FABRICATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
300			Reduci	5713592	6 - 6	TEE REDUCING FABRICATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
310								SERVICE							
320			Cap	5713562	6 - 6	CAP ELONGATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427		DESIGN: 14.5 BAR G TEMPERATURE RANGE: -40°C TO +25°C SERVICE: WRA TECHNICAL WATER							
330								MATERIAL: POLYETHYLENE (HDPE) FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 25 °C COMMENT:							
340			Concen	5713581	6 - 6	REDUCER CONC ELONGATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	8 - 8								
350			Eccent	5713585	6 - 6	REDUCER ECC ELONGATED TYPE SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	8 - 8								
360															
370			Coupli	5713565	6 - 8	COUPLING ELECTRO FUSION SDR 7.4 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
380															
390		UDC	SADDLE	5713516	2 - 2	PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR	6 - 8								
400															
410		UDC	Concen	5544151	3/4 - 1 1/2	SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153	2 - 2								
420															
430															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW. 0 MM	DATE 17MAR17	BY JCS	REV A	SHEET 1 OF 2	PA 5	CLASS 150PE6	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
440	B	UDC	Bushin	5491325	1/2 - 1 1/2	BUSHING 3000# SCRd (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153 PIPE ADAPTOR WITH GUNMETAL THREADED (NPT-MALE) OUTLET ***** FLANGES ***** STUB FLG ELONGATED TYPE SDR 7.4 HD POLYETHYLENE C/W BACKING RING PE 100 WATER ASTM-D3350 / ISO 4427 FLG BLIND FF 150# HDPE PE 100 WATER ASTM-D3350 / ISO 4427 WITH STEEL INSERT ***** GASKETS ***** GASKET 150# RF FLAT RING PTFE GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR ***** BOLTING ***** STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153 ***** BRANCHES ***** USE TEES AND REDUCERS AS REQUIRED ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- LONG RADIUS THREAD SEALANT ----- NONE ELECTRIC SOCKET WELDING OR BUTT WELDING AS APPLICABLE	3/4 - 2												
450																			
460			Coupli	5713518	1/2 - 2														
470																			
480																			
490			Flange	5713598	6 - 8														
500																			
510			Blind	5621090	6 - 8														
520																			
530																			
540		GA3	GASFR	5664201	2 - 8														
550																			
560		B1	GASFRS	5673043	2 - 8														
570																			
580			BOLS	5675768	1/4 - 99														
590																			
600			Specta	5749582	3/4 - 8														
610																			
620			B																
630																			
640																			
650																			
660																			
670																			
680																			
690																			
700																			
710																			
720																			
730																			
740																			
750																			
760																			
														SERVICE					
								DESIGN: 14.5 BAR G TEMPERATURE RANGE: -40°C TO +25°C SERVICE: WRA TECHNICAL WATER											
								MATERIAL: POLYETHYLENE (HDPE) FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 25 °C COMMENT:											
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW. 0 MM		DATE 17MAR17	BY JCS	REV A	SHEET 2 OF 2	PA 5	CLASS 150PE6				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES													
10	C	PA8	Piping	9004126	6 - 6	***** PIPE*****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. THE FOLLOWING PIPE NOM.SIZE/OD. CROSS REFERENCES SHALL BE USED FOR IDENTIFICATION AND ORDERING PURPOSES FOR SDR 7.4 SYSTEMS: NOM O.D. O.D. I.D. (IN.) (MM) (MM) 1" 25 17.6 2" 63 44.8 3" 90 63.9 4" 125 88.7 6" 180 127.8 7" FOR SGI/SGP CLASS ONLY - DO NOT USE 8" 225 159.7 9" FOR SGI/SGP CLASS ONLY - DO NOT USE 10" 280 198.8 12" 315 223.6 14" 355 252.2 16" 400 284 18" 450 319.6 20" 500 355.2 24" 630 TBA												
60				9004126	18 - 20	PIPE SDR 7.4 PE-RT POLYETHYLENE ASTM-D3350 / ISO 4427 PIPE SDR 7.4 PE-RT POLYETHYLENE ASTM-D3350 / ISO 4427															
70			90BND	9004127	18 - 20	BEND 90 DEG LONG RADIUS SDR 7.4 PE-RT POLYETHYLENE SOUR WATER ASTM-D3350 / ISO 4427															
80																					
90																					
100																					
110			45BND	9004128	18 - 20	BEND 45 DEG LONG RADIUS SDR 7.4 PE-RT POLYETHYLENE SOUR WATER ASTM-D3350 / ISO 4427															
120																					
130			Tee	9004173	18 - 20	TEE EQUAL ELONGATED TYPE SDR 7.4 PE-RT POLYETHYLENE SOUR WATER ASTM-D3350 / ISO 4427															
140																					
150																					
160																					
170	Coupli	9004174	18 - 20	COUPLING ELECTRO FUSION SDR 7.4 PE-RT POLYETHYLENE SOUR WATER ASTM-D3350 / ISO 4427																	
180																					
190																					
200																					
210	C		Flange	9004175	6 - 6	STUB FLG ELONGATED TYPE SDR 7.4 PE-RT POLYETHYLENE C/W BACKING RING PE SOUR WATER ASTM-D3350 / ISO 4427	UDC	FOR USE WITH PRESSURE CONNECTIONS ONLY													
220				Flange	9004175	18 - 20								STUB FLG ELONGATED TYPE SDR 7.4 PE-RT POLYETHYLENE C/W BACKING RING PE SOUR WATER ASTM-D3350 / ISO 4427							
230			Blind	9004176	18 - 20	FLG BLIND FF 150# PE-RT POLYETHYLENE SOUR WATER ASTM-D3350 / ISO 4427 WITH STEEL INSERT															
240			GASFR	5664201	18 - 20	***** GASKETS ***** GASKET 150# RF FLAT RING PTFE															
250																					
260																					
270																					
280			C	B1	BOLS	5675768								18 - 20	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS	SERVICE DESIGN: 7.1 BAR G TEMPERATURE RANGE: -40°C TO +80°C SERVICE: SOUR WATER MATERIAL : BELOW GROUND - POLYETHYLENE (PERT) MAX PRESSURE IS TEMPERATURE DEPENDENT - REF 040-0000-LLP-TCN-000-000-00005-00					
290						BOLS								9004177	18 - 20						
300																					
310																					
320																					
330																					
340																					
350																					
360																					
370	C	UDC							MATERIAL: POLYETHYLENE (PERT) FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 80 °C COMMENT:												
380																					
390																					
400																					
410																					
420																					
430																					
440																					
450																					
460																					
470																					
480																					
490																					
500																					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION ALL SIZES - ABOVE GROUND - FLANGED BELOW GROUND - BUTT FUSION OR ELECTRO-FUSION		CORR. ALLOW. 0 MM	DATE 17MAR17	BY JCS	REV C	SHEET 1 OF 1	PA 5	CLASS 150PE7								

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10	B	VC2	Gate V	GA6B1 G01AC41	1/2 - 1 1/2 2 - 8	*****VALVES*****	new	B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
20						GA3		SPIRAL WOUND GASKETS SHALL ONLY BE USED BETWEEN CARBON							
30								STEEL FLANGES AND CARBON STEEL VALVES.							
40								FOR 24" 90 DEG ELBOWS USE 2 X 45 DEG BENDS (CODE NO. 5713510)							
50								FUSION WELDED TOGETHER							
60								9" NOM SIZE (225MM O.D.) STUB FLANGES SHALL BE SUPPLIED C/W 8"							
70					Globe	L01AC41 L01AC42		2 - 6 8 - 8	*****GLOBE VALVES*****	NX1	NOM SIZE BACKING RINGS.				
80				PA5					THE FOLLOWING PIPE NOM.SIZE/OD. CROSS REFERENCES SHALL BE						
90									USED FOR IDENTIFICATION AND ORDERING PURPOSES FOR SDR 11						
100									SYSTEMS:						
110									NOM O.D. I.D.						
120									(IN.) (MM) (MM)						
130					Butter	T01AC61		10 - 24	*****BUTTERFLY VALVES*****	2"	63	51.4			
140				3"					90	73.6					
150				4"					110	90					
160				5"					140	114.6					
170				6"					160	130.8					
180				7"					180	147.2					
190			B	VC2	Check	C01AC45		2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16	8"	200	163.6			
200			B	VC2	Check	C01AC52		8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16	9"	225	184			
210									***** PIPE *****	10"	250	204.6			
220										12"	315	257.8			
230				PA5	Piping	5713401		2 - 24	PIPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	14"	355	290.6			
240									***** FITTINGS *****	16"	400	327.4			
250				PA5						18"	450	368.2			
260										20"	500	409.2			
270				NB9	90 Deg	5713501		2 - 12	ELBOW 90 DEG ELONGATED TYPE SDR 11 HD	22"	560	458.4			
						POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	24"	630	515.6						
280			90 Deg	5713511	14 - 24	BEND 90 DEG LONG RADIUS SDR 11 HD POLYETHYLENE PE	28"	710	574						
						100 WATER ASTM-D3350 / ISO 4427	32"	800	656						
290							36"	900	738						
300	C		60 Deg	5713839	2 - 12	ELBOW 60 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE		GENERAL, DESIGN & CODE SHALL BE CLARIFIED							
						PE 100 WATER ASTM-D3350 / ISO 4427									
320								SERVICE							
330			45 Deg	5713502	2 - 12	ELBOW 45 DEG ELONGATED TYPE SDR 11 HD		DESIGN: 9.3 BAR G							
340			45 Deg	5713510	14 - 24	POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427		TEMPERATURE RANGE: -40°C TO +25°C							
						BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE		SERVICE: WRA TECHNICAL WATER							
						PE 100 WATER ASTM-D3350 / ISO 4427									
350															
360	C		30 Deg	5713838	2 - 12	ELBOW 30 DEG ELONGATED TYPE SDR 11 HD	new								
						POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
370															
380			Tee	5713503	2 - 20	TEE EQUAL ELONGATED TYPE SDR 11 HD POLYETHYLENE PE									
						100 WATER ASTM-D3350 / ISO 4427									
390			Reduci	5713504	2 - 10	TEE REDUCING ELONGATED TYPE SDR 11 HD	4 - 12	MATERIAL:		POLYETHYLENE (HDPE)					
						POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427		FLANGE RATING:		150 RF					
400								TEMPERATURE RANGE:		-40 / 25 °C					
410			Cap	5713505	2 - 24	CAP ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100		COMMENT:							
						WATER ASTM-D3350 / ISO 4427									
420															
430			Concen	5713506	2 - 14	REDUCER CONC ELONGATED TYPE SDR 11 HD	3 - 16								
						POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
								0 MM	17MAR17	JCS	C	1 OF 2	5	150PE8	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
440			Eccent	5713507	2 - 10	REDUCER ECC ELONGATED TYPE SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427	3 - 12	UDC VC2	FOR USE WITH PRESSURE CONNECTIONS ONLY DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.				
450			Coupli	5713509	2 - 24	COUPLING ELECTRO FUSION SDR 11 HD POLYETHYLENE PE 100 WATER ASTM-D3350 / ISO 4427							
470								3 - 8					
480		UDC	SADDLE	5713515	2 - 2	PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR							
490		UDC	SADDLE	5713516	2 - 2	PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR	10 - 24						
500													
510		UDC	Concen	5544151	3/4 - 1 1/2	SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153	2 - 2						
		UDC	Bushin	5491325	1/2 - 1 1/2	BUSHING 3000# SCRD (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153	3/4 - 2						
520													
530			Coupli	5713518	1/2 - 2	PIPE ADAPTOR WITH GUNMETAL THREADED (NPT-MALE) OUTLET							
540						***** FLANGES *****							
550													
560													
570		NX1	Flange	5713508	2 - 24	STUB FLG ELONGATED TYPE SDR 11 HD POLYETHYLENE C/W BACKING RING PE 100 WATER ASTM-D3350 / ISO 4427							
580			Blind	5621090	2 - 24	FLG BLIND FF 150# HDPE PE 100 WATER ASTM-D3350 / ISO 4427 WITH STEEL INSERT							
590						***** PUDDLE FLANGE *****							
600													
610													
620			ANCHOR	5597702	2 - 12	PUDDLE (ANCHOR) FLANGE HDPE FF							
630						***** GASKETS *****							
640													
650													
660													
670			GASFR	5664201	2 - 24	GASKET 150# RF FLAT RING PTFE							
680		GA3	GASFRS	5673043	2 - 18	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
690						***** BOLTING *****							
700													
710													
720		C	B1	5675768	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS							
730													
740			Specta	5749582	3/4 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153							
750						***** BRANCHES *****							
760						USE TEES AND REDUCERS AS REQUIRED							
770													
780						***** CONSTRUCTION DETAILS *****							
790													
800													
810													
820													
830													
840		C	ZA6			SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
850						PIPE BENDS ----- LONG RADIUS							
860						THREAD SEALANT ----- NONE ELECTRIC							
870						SOCKET WELDING OR BUTT							
880						WELDING AS APPLICABLE							
890													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION	CORR. ALLOW. 0 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 2	PA 5	CLASS 150PE8

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES												
10	B		Gate V	GA6B1	1/2 - 1 1/2	*****VALVES*****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.											
20								GA3	SPIRAL WOUND GASKETS SHALL ONLY BE USED BETWEEN CARBON											
30									STEEL FLANGES AND CARBON STEEL VALVES.											
40								NX1	9" NOM SIZE (225MM O.D.) STUB FLANGES SHALL BE SUPPLIED C/W 8"											
50									NOM SIZE BACKING RINGS.											
60								PA5	THE FOLLOWING PIPE NOM.SIZE/OD. CROSS REFERENCES SHALL BE											
70									USED FOR IDENTIFICATION AND ORDERING PURPOSES FOR SDR 11											
80									SYSTEMS:											
90									NOM	O.D.	I.D.									
100									(IN.)	(MM)	(MM)									
110									2"	63	51.4									
120									3"	90	73.6									
130									4"	110	90									
140									5"	140	114.6									
150									6"	160	130.8									
160									7"	180	147.2									
170									8"	200	163.6									
180									9"	225	184									
190									10"	250	204.6									
200									12"	315	257.8									
210									14"	355	290.6									
220									16"	400	327.4									
230								B	VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16		18"	450	368.2			
240									VC2	Check	C01AC52	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16		20"	500	409.2			
250								22"							560	458.4				
260								24"							630	515.6				
270								28"							710	574				
280									PA5	Piping	9004100	2 - 24	PIPE SDR 11 HD POLYETHYLENE - ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427		32"	800	656			
290								36"							900	738				
300															GENERAL, DESIGN & CODE SHALL BE CLARIFIED					
310																				
320										90 Deg	9004101	2 - 12	ELBOW 90 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE - ID		SERVICE					
330													BLUE PE 100 WATER ASTM-D3350 / ISO 4427		DESIGN: 9.3 BAR G					
340										90 Deg	9004102	14 - 24	BEND 90 DEG LONG RADIUS SDR 11 HD POLYETHYLENE - ID		TEMPERATURE RANGE: -40 TO +40 DEG.C.					
350													BLUE PE 100 WATER ASTM-D3350 / ISO 4427		WD - POTABLE WATER					
360										60 Deg	9004116	2 - 12	ELBOW 60 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE - ID							
370													BLUE PE 100 WATER ASTM-D3350 / ISO 4427							
380										60 Deg	9004117	14 - 24	BEND 60 DEG LONG RADIUS SDR 11 HD POLYETHYLENE - ID							
390													BLUE PE 100 WATER ASTM-D3350 / ISO 4427							
400										45 Deg	9004103	2 - 12	ELBOW 45 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE ID		MATERIAL: POLYETHYLENE (HDPE)					
410													BLUE PE 100 WATER ASTM-D3350 / ISO 4427		FLANGE RATING: 150 RF					
420										45 Deg	9004104	14 - 24	BEND 45 DEG LONG RADIUS SDR 11 HD POLYETHYLENE ID		TEMPERATURE RANGE: -40 / 40 °C					
430													BLUE PE 100 WATER ASTM-D3350 / ISO 4427		COMMENT:					
440										30 Deg	9004118	2 - 12	ELBOW 30 DEG ELONGATED TYPE SDR 11 HD POLYETHYLENE ID							
440						BLUE PE 100 WATER ASTM-D3350 / ISO 4427														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP												
						ALL SIZES - BUTT FUSION	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS							
							0 MM	17MAR17	JCS	C	1 OF 3	5	150PE9							

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
450	B	B1	30 Deg	9004119	14 - 24	BEND 30 DEG LONG RADIUS SDR 11 HD POLYETHYLENE ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427	4 - 24	UDC	FOR USE WITH PRESSURE CONNECTIONS ONLY DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
460			Tee	9004105	2 - 24	TEE EQUAL ELONGATED TYPE SDR 11 HD POLYETHYLENE ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427		VC2						
470														
480			Reduci	9004106	2 - 18	TEE REDUCING ELONGATED TYPE SDR 11 HD POLYETHYLENE ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427		ZA6						
490														
500			Cap	9004107	2 - 24	CAP ELONGATED TYPE SDR 11 HD POLYETHYLENE ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427								
510														
520			Concen	9004108	2 - 14	REDUCER CONC ELONGATED TYPE SDR 11 HD POLYETHYLENE ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427								
530														
540			Eccent	9004109	2 - 10	REDUCER ECC ELONGATED TYPE SDR 11 HD POLYETHYLENE ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427								
550														
560			Coupli	9004110	2 - 24	COUPLING ELECTRO FUSION SDR 11 HD POLYETHYLENE ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427								
570														
580			UDC UDC	SADDLE	9004111	2 - 2		PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR						3 - 8
590				UDC UDC	SADDLE	9004112		2 - 2						PIPE SADDLE WITH GUNMETAL THREADED (NPT - FEMALE) OUTLET ADAPTOR
600			UDC UDC		Concen	5544151		3/4 - 1 1/2						SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153
610														
620			UDC UDC	Bushin	5491325	1/2 - 1 1/2		BUSHING 3000# SCRD (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153						3/4 - 2
630														
640			UDC UDC	Coupli	9004113	1/2 - 2		PIPE ADAPTOR WITH GUNMETAL THREADED (NPT-MALE) OUTLET						
650														
660			NX1	Flange	9004114	2 - 24		***** FLANGES ***** STUB FLG ELONGATED TYPE SDR 11 HD POLYETHYLENE - ID BLUE C/W BACKING RING PE 100 WATER ASTM-D3350 / ISO 4427						
670														
680														
690														
700														
710														
720														
730														
740														
750														
760			Blind	9004115	2 - 24	FLG BLIND FF 150# HDPE - ID BLUE PE 100 WATER ASTM-D3350 / ISO 4427 WITH STEEL INSERT								
770														
780	Specta	5750421	2 - 12	SPECTACLE BLIND 150# RF SS A240 GR 316										
790														
800	GA3	***** GASKETS *****												
810														
820														
830														
840														
850														
860														
870														
880														
890														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW. 0 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 2	PA 5	CLASS 150PE9

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
900 910 920 930 940 950 960 970 980 990 1000	C	ZA6				***** BRANCHES ***** USE TEES AND REDUCERS AS REQUIRED ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- LONG RADIUS THREAD SEALANT ----- NONE ELECTRIC SOCKET WELDING OR BUTT WELDING AS APPLICABLE									
								SERVICE							
								DESIGN: 16 BAR G TEMPERATURE RANGE: -40 TO +40 DEG.C. WD - POTABLE WATER							
								MATERIAL: POLYETHYLENE (HDPE) FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 40 °C COMMENT:							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT FUSION		CORR. ALLOW. 0 MM		DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 2	PA 5	CLASS 150PE9

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10						***** VALVES *****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.						
20						***** BALL VALVES *****		FS1							
30								FV5							
40															
50			Ball V	A01AC01	1 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO									
60			Ball V	A01AC03	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO									
70	B		Ball V	A01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO									
80								FV6							
90															
100		VU2	Ball V	BA6B2	1/2 - 1/2	BALL 150# SCR D FNPT BS1400 LG2 BODY AND TRIM FB LO									
110	B		Ball V	BA6B2	3/4 - 2	BALL 150# SCR D FNPT BS1400 LG2 BODY AND TRIM FB LO									
120								GS1	COMPONENTS AT NPD 12", 14" AND 16" ARE TO MEET MANUFACTURER'S MINIMUM WALL THICKNESS REQUIREMENTS OF STANDARD WT. USE FOR REPLACEMENT GASKET. SIZES SHOWN IN SIZE COLUMNS ARE THE SMALL END OF REDUCERS AND THE BRANCH SIZE OF OLETS AND TEES.						
130		VA5	Ball V	B01AC01	1 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
140	B	VA5	Ball V	B01AC03	4 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
150	B	VA5	Ball V	B01AC07	8- 12	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO		PA6							
160								UD7							
170						***** BUTTERFLY VALVES *****									
180								VA5							
190	B		Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO		VC1							
200															
210						***** GATE VALVES *****									
220								VC2							
230	B		Gate V	GA6B1	3/4 - 1	GATE 150# SCR D FNPT BS1400 LG2 BODY AND TRIM HW									
240			Gate V	G01AC48	1 1/2 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW		VU2							
250			Gate V	G01AC41	2 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW		ZA6							
260															
270						***** GLOBE VALVES *****									
280															
290															
300	B		Globe	LA6B1	3/4 - 1	GLOBE 150# SCR D FNPT BS1400 LG2 BODY AND TRIM HW									
310	B		Globe	L01AC48	1 1/2 - 1 1/2	GLOBE 150# RF LTCS A350 LF2 TRIM 16 OS&Y HW									
320			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW									
330			Globe	L01AC42	8 - 10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO									
340															
350						***** CHECK VALVES *****									
360															
370															
380	B	VC1	Check	CA6B1	3/4 - 1	CHECK 150# SCR D FNPT BS1400 LG2 BODY AND TRIM SPRING PISTON		SERVICE WE, WF, WP, WRA, WT, WU DESIGN: 15.8 BARG CODE: ASME B31.3							
390															
400	B	VC1	Check	C01AC50	1 1/2 - 1 1/2	CHECK PISTON 150# RF LTCS A350 LF2 TRIM 16									
410		VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED									
420															
430		VC2	Check	C01AC52	8 - 16	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16									
440						***** PIPE *****									
450															
460	B		Piping	5358601	3/4 - 1	PIPE SCH 160 SMLS LTCS A333 GR.6 GALV A153 T&C		MATERIAL: LOW TEMP CS-GROOVE FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 110 °C COMMENT:							
470	B	PA6	Piping	5358507_VC	1 1/2 - 3	PIPE SCH 80 SMLS LTCS A333 GR. 6 PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)									
480	B	PA6	Piping	5358505_VC	4 - 10	PIPE SCH STD SMLS LTCS A333 GR. 6 PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)									
490	B	PA6	Piping	5358502_VC	12 - 12	PIPE SCH 30 SMLS LTCS A333 GR. 6 PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)									
						CONSTRUCTION		FGP/WPMP							
			FGP / WPMP TENGIZCHEVROI - TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN			3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
								3MM	17MAR17	KD	B	1 OF 5	5	150VC01	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
500	B	PA6	Piping	5358501_VC	14 - 16	PIPE SCH 20 SMLS LTCS A333 GR. 6 PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)								
510	B	FV5	Piping	5358597	18 - 24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE								
520														
530		NR1				***** FITTINGS *****								
540														
550														
560	B		90 Deg	5491290	3/4 - 1	ELL 90 DEG 3000# SCRD FNPT LTCS A350 LF2 GALV A153								
570	B		45 Deg	5491300	3/4 - 1	ELL 45 DEG 3000# SCRD FNPT LTCS A350 LF2 GALV A153								
580	B		90 Deg	5392280	18 - 24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W								
590	B		45 Deg	5392281	18 - 24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W								
600														
610	B		Tee	5491310	3/4 - 1	TEE 3000# SCRD FNPT LTCS A350 LF2 GALV A153								
620	B		Tee	5392295	18 - 24	TEE BW SCH 20 LTCS A420 WPL6-W								
630														
640	B		Reduci	5491311	3/4 - 3/4	TEE RED 3000# SCRD FNPT LTCS A350 LF2 GALV A153								1 - 1
650	B		Reduci	5392296	8 - 10	TEE RED BW SCH 20/STD LTCS A420 WPL6-W								18 - 24
660	B		Reduci	5392244	12 - 12	TEE RED BW SCH 20/SCH 30 LTCS A420 WPL6-W								18 - 24
670	B		Reduci	5392296	14 - 16	TEE RED BW SCH 20/STD LTCS A420 WPL6-W								18 - 24
680	B		Reduci	5392292	18 - 20	TEE RED BW SCH 20 LTCS A420 WPL6-W								20 - 24
690														
700			Cap	5491330	3/4 - 1	CAP 3000# SCRD FNPT LTCS A350 LF2 GALV A153								
710														
720	B		Cap	5392286	18 - 24	CAP BW SCH 20 LTCS A420 WPL6								
730														
740	B		Concen	5392297	10 - 10	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W								18 - 18
750	B		Concen	5392243	12 - 12	REDUCER CONC SCH 20/SCH 30 LTCS A420 WPL6-W								18 - 20
760	B		Concen	5392287	14 - 20	REDUCER CONC SCH 20 LTCS A420 WPL6-W								18 - 24
770														
780	B		Eccent	5392298	10 - 10	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W								18 - 18
790	B		Eccent	5392242	12 - 12	REDUCER ECC SCH 20/SCH 30 LTCS A420 WPL6-W								18 - 20
800	B		Eccent	5392288	14 - 20	REDUCER ECC SCH 20 LTCS A420 WPL6-W								18 - 24
810														
820						***** FITTINGS - VICTAULIC *****								
830														
840														
850	B		90 Deg	5484860	1 1/2 - 3	ELL 90 DEG GRV SCH 80 LTCS A352 GR LCC GALV								
860			90 Deg	5484861	4 - 10	ELL 90 DEG GRV SCH STD LTCS A352 GR LCC GALV								
870	B	FV6	90 Deg	5484862	12 - 12	ELL 90 DEG GRV SCH 30 LTCS A352 GR LCC GALV								
880	B	FV6	90 Deg	5484863	14 - 16	ELL 90 DEG GRV SCH 20 LTCS A352 GR LCC GALV								
890														
900	B		45 Deg	5484865	1 1/2 - 3	ELL 45 DEG GRV SCH 80 LTCS A352 GR LCC GALV								
910			45 Deg	5484866	4 - 10	ELL 45 DEG GRV SCH STD LTCS A352 GR LCC GALV								
920	B	FV6	45 Deg	5484867	12 - 12	ELL 45 DEG GRV SCH 30 LTCS A352 GR LCC GALV								
930	B	FV6	45 Deg	5484868	14 - 16	ELL 45 DEG GRV SCH 20 LTCS A352 GR LCC GALV								
940														
950	B		Tee	5485120	1 1/2 - 3	TEE GRV SCH 80 LTCS A352 GR LCC GALV								
960			Tee	5485121	4 - 10	TEE GRV SCH STD LTCS A352 GR LCC GALV								
970	B	FV6	Tee	5485122	12 - 12	TEE GRV SCH 30 LTCS A352 GR LCC GALV								
980	B	FV6	Tee	5485123	14 - 16	TEE GRV SCH 20 LTCS A352 GR LCC GALV								
990														
1000	B		Reduci	5484880	1 1/2 - 2	TEE RED GRV SCH 80 LTCS A352 GR LCC GALV								2 - 3
1010	B		Reduci	5484884	1 1/2 - 3	TEE RED GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV								4 - 10
1020	B		Reduci	5484881	4 - 8	TEE RED GRV SCH STD LTCS A352 GR LCC GALV								6 - 10
			FGP / WPMP TENGIZCHEVOIL - TENGIZCHEVOIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED	CORR. ALLOW. 3MM	DATE 17MAR17	BY KD	REV B	SHEET 2 OF 5	PA 5	CLASS 150VC01	

SERVICE

WE, WF, WP, WRA, WT, WU
DESIGN: 15.8 BARG
CODE: ASME B31.3

MATERIAL: LOW TEMP CS-GROOVE
FLANGE RATING: 150 RF
TEMPERATURE RANGE: -46 / 110 °C
COMMENT:

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1030	B	FV6	Reduci	5484887	1 1/2 - 3	TEE RED GRV SCH 30/SCH 80 LTCS A352 GR LCC GALV	12 - 12						
1040	B	FV6	Reduci	5484882	4 - 10	TEE RED GRV SCH 30/STD LTCS A352 GR LCC GALV	12 - 12						
1050	B	FV6	Reduci	5484886	6 - 10	TEE RED GRV SCH 20/STD LTCS A352 GR LCC GALV	14 - 16						
1060	B	FV6	Reduci	5484885	12 - 12	TEE RED GRV SCH 20/SCH 30 LTCS A352 GR LCC GALV	14 - 16						
1070	B	FV6	Reduci	5484883	14 - 14	TEE RED GRV SCH 20 LTCS A352 GR LCC GALV	16 - 16						
1080													
1090	B		Cap	5485130	1 1/2 - 3	CAP GRV SCH 80 LTCS A352 GR LCC GALV							
1100			Cap	5485131	4 - 10	CAP GRV SCH STD LTCS A352 GR LCC GALV							
1110	B	FV6	Cap	5485132	12 - 12	CAP GRV SCH 30 LTCS A352 GR LCC GALV							
1120	B	FV6	Cap	5485133	14 - 16	CAP GRV SCH 20 LTCS A352 GR LCC GALV							
1130													
1140	B		Concen	5484890	1 1/2 - 2	REDUCER CONC GRV SCH 80 LTCS A352 GR LCC GALV	2 - 3						
1150			Concen	5484891	1 1/2 - 3	REDUCER CONC GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV	4 - 6						
1160													
1170			Concen	5484892	4 - 8	REDUCER CONC GRV SCH STD LTCS A352 GR LCC GALV	6 - 10						
1180	B	FV6	Concen	5484893	6 - 10	REDUCER CONC GRV SCH 30/SCH STD LTCS A352 GR LCC GALV	12 - 12						
1190	B	FV6	Concen	5484894	6 - 10	REDUCER CONC GRV SCH 20/SCH STD LTCS A352 GR LCC GALV	14 - 16						
1200	B	FV6	Concen	5484896	12 - 12	REDUCER CONC GRV SCH 20/SCH 30 LTCS A352 GR LCC GALV	14 - 16						
1210	B	FV6	Concen	5484895	14 - 14	REDUCER CONC GRV SCH 20 LTCS A352 GR LCC GALV	16 - 16						
1220													
1230	B		Eccent	5484830	1 1/2 - 2	REDUCER ECC GRV SCH 80 LTCS A352 GR LCC GALV	2 - 3						
1240			Eccent	5484831	1 1/2 - 3	REDUCER ECC GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV	4 - 6						
1250													
1260			Eccent	5484832	4 - 8	REDUCER ECC GRV SCH STD LTCS A352 GR LCC GALV	6 - 10						
1270	B	FV6	Eccent	5484833	6 - 10	REDUCER ECC GRV SCH 30/SCH STD LTCS A352 GR LCC GALV	12 - 12						
1280	B	FV6	Eccent	5484834	6 - 10	REDUCER ECC GRV SCH 20/SCH STD LTCS A352 GR LCC GALV	14 - 16						
1290	B	FV6	Eccent	5484836	12 - 12	REDUCER ECC GRV SCH 20/SCH 30 LTCS A352 GR LCC GALV	14 - 16						
1300	B	FV6	Eccent	5484835	14 - 14	REDUCER ECC GRV SCH 20 LTCS A352 GR LCC GALV	16 - 16						
1310													
1320						*****COUPLINGS, UNIONS, PLUGS & THREDOLETS*****							
1330													
1340	B		Coupli	5491250	3/4 - 1	COUPLING 3000# SCR D FNPT LTCS A350 LF2 GALV A153							
1350	B		Union	5491270	3/4 - 1	UNION 3000# SCR D FNPT LTCS A350 LF2 GALV A153							
1360													
1370		UD7	Plug	5491280	1/2 - 1	PLUG ROUND HEAD SCR D MNPT LTCS A350 LF2 GALV A153							
1380			Bushin	5491325	1/2 - 1 1/2	BUSHING 3000# SCR D (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153	3/4 - 2						
1390													
1400													
1410	B	UD7	Thredo	5514772	3/4 - 1	THREDOLET 3000# LTCS A350 LF2	18 - 24						
1420													
1430						***** COUPLINGS, BRANCH OUTLETS - VICTAULIC *****							
1440													
1450	B		Coupli	5484764	1 1/2 - 16	COUPLING RIGID GRV ASTM A352 GRADE LCC EPDM GASKET GALV							
1460													
1470													
1480	B		SADDLE	5485125	1/2 - 1	VICTAULIC BOLTED BRANCH OUTLET FNPT LTCS A352 GR LCC GALV	2 - 4						
1490													
1500	B		SADDLE	5485125	1 1/2 - 1 1/2	VICTAULIC BOLTED BRANCH OUTLET FNPT LTCS A352 GR LCC GALV	6 - 6						
1510													
1520	B		SADDLE	5485125	2 - 2	VICTAULIC BOLTED BRANCH OUTLET FNPT LTCS A352 GR LCC GALV	8 - 8						
1530													
1540													
1550	B		Thredo	5485127	3/4 - 3/4	VICTAULIC STRAPLESS OUTLET FNPT LTCS A352 GR LCC GALV	6 - 16						
1560													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED	CORR. ALLOW. 3MM	DATE 17MAR17	BY KD	REV B	SHEET 3 OF 5	PA 5	CLASS 150VC01

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																		
1570	B	FV6	Reduci	5599691	1/2 - 3/4	***** PIPE FLANGES ***** FLG SCRDR (FNPT) REDUCING 150# RF LTCS A350 LF2 GALV A153	3/4 - 2																			
1580			Flange	5599689	3/4 - 1	FLG SCRDR (FNPT) 150# RF LTCS A350 LF2 GALV A153																				
1590			Flange	5599701	3/4 - 1	FLG SCRDR (FNPT) 300# RF LTCS A350 LF2 GALV A153																				
1600	B		FV6	Flange	5556261	8 - 10	FLG WN 150# RF LTCS A350 LF2 SCH STD							1 - 1 1/2												
1610				Flange	5556274	12 - 12	FLG WN 150# RF LTCS A350 LF2 SCH 30																			
1620				Flange	5556265	14 - 16	FLG WN 150# RF LTCS A350 LF2 SCH 20																			
1630	B		Flange	5556265	18 - 24	FLG WN 150# RF LTCS A350 LF2 SCH 20																				
1640			Blind	5615567	1/2 - 24	FLG BLIND 150# RF LTCS A350 LF2 GALV A153																				
1650			Blind	5615577	1/2 - 1/2	FLG BLIND 300# RF LTCS A350 LF2 GALV A153																				
1660	B		***** PIPE FLANGES - VICTAULIC *****																							
1670			Flange	5754404	1 1/2 - 16	FLANGE ADAPTER NIPPLE 150# RF / GRV A352 GR LCC GALV																				
1680			Flange	5754405	1 1/2 - 12	FLANGE ADAPTER NIPPLE 300# RF / GRV A352 GR LCC GALV																				
1690			***** ORIFICE FLANGE *****																							
1700			Orific	5623941	18 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 20																				
1710			***** ORIFICE FLANGE - VICTAULIC *****																							
1720			Orific	5754195	2 - 16	METERING STATION ORIF TYPE GRV A352 GR LCC GALV																				
1730			***** PIPE NIPPLES *****																							
1740			Nipple	5541200	1/2 - 1	NIPPLE SCH 160 LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153																				
1750			Nipple	5541201	1/2 - 1	NIPPLE SCH 160 LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153																				
1760	B		***** SWAGE NIPPLES *****																							
1770			Concen	5544150	3/4 - 1	SWAGE CONC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153	1 - 1 1/2																			
1780			***** GASKETS *****																							
1790			GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR																				
1800			GASFRS	5673033	1/2 - 12	GASKET 300# 316SS SP WND GRAPHITE IR																				
1810			***** GASKETS - VICTAULIC *****																							
1820			GASFRS	NOMTO_G	1 1/2 - 16	GASKET FOR CLAMP FOR VICTAULIC SYSTEM - NON REPORTABLE																				
1830			***** BOLTING *****																							
1840			BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS																				
1850																										
1860																										
1870	B																									
1880																										
1890																										
1900																										
1910																										
1920																										
1930																										
1940																										
1950																										
1960																										
1970																										
1980	B																									
1990																										
2000																										
2010																										
2020																										
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2040																										
2050																										
2060																										
2070																										
2080																										
2090	B																									
2100																										
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED		CORR. ALLOW. 3MM	DATE 17MAR17	BY KD	REV B	FGP/WPMP SHEET 4 OF 5	PA 5	CLASS 150VC01												

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
2110	B	ZA6	Specta	5749582	3/4 - 14	**** BLINDS & SPACERS ****										
2120																
2130						Specta								5749583	3/4 - 10	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153
2140						Specta								5749583	3/4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70 GALV A153
2150						Paddle								5749942	16 - 16	PADDLE BLANK 150# RF LTCS A516 GR.70 GALV A153
2160																PADDLE SPACER 150# RF LTCS A516 GR.70 GALV A153
2170						Paddle								5750195	16 - 16	PADDLE SPACER 150# RF LTCS A516 GR.70 GALV A153
2180						Paddle								5750023	18 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70
2190																PADDLE SPACER 150# RF LTCS A516 GR.70
2200						Paddle								5750240	18 - 24	PADDLE SPACER 150# RF LTCS A516 GR.70
2210			Paddle	5749853	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70 GALV A153										
2220						PADDLE SPACER 300# RF LTCS A516-GR70 GALV A153										
2230						PADDLE SPACER 300# RF LTCS A516-GR70 GALV A153										
2240						***** BRANCHES *****										
2250			Reinfo	RWELD	2 - 16	USE BRANCH CHART 150VC01										
2260						NO-MTO REINFORCING WELD FOR STUB-IN										
2270						NO-MTO REINFORCING WELD FOR STUB-IN										
2280						***** CONSTRUCTION DETAILS *****										
2290						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY										
2300						PIPE BENDS ----- NONE										
2310	THREAD SEALANT ----- TEFLON TAPE															
2320	PWHT ----- NONE															
2330	Reduci	5484880				1 - 2	*****DO NOT USE FOLLOWING ITEMS*****									
2340							TEE RED GRV SCH 80 LTCS A352 GR LCC GALV									
2350			SWAGE CONC SCH 80/ SCH 160 STL A420 WPL6 MGE X MNPT GALV													
2360			SWAGE CONC SCH 80 STL A420 WPL6 MGE X MNPT GALV													
2370			TEE RED GRV X MNPT OUTLET SCH 80/ SCH 160 LTCS A352 GR LCC GALV													
2380			VICTAULIC BOLTED BRANCH OUTLET GRV LTCS A352 GR LCC GALV													
2390			GASKET C SHAPE, GRADE E EPDM													
2400			Concen	5543782	1 - 1		1 1/2 - 3									
2410							1 1/2 - 2									
2420							3 - 3									
2430	1 1/2 - 1 1/2															
2440	2 - 4															
2450	Reduci	5484888				3/4 - 1	WE, WF, WP, WRA, WT, WU									
2460							DESIGN: 15.8 BARG									
2470							CODE: ASME B31.3									
2480							Reduci	5485124	1 1/2 - 3	MATERIAL: LOW TEMP CS-GROOVE						
2490										FLANGE RATING: 150 RF						
2500			TEMPERATURE RANGE: -46 / 110 °C													
2510			COMMENT:													
2520			GS1	11.25	5664445					1 - 16	GASKET C SHAPE, GRADE E EPDM					

			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED			CORR. ALLOW. 3MM		FGP/WPMP DATE 17MAR17 BY KD REV B SHEET 5 OF 5 PA 5 CLASS 150VC01					
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SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES												
10		VB2				***** VALVES *****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. WHERE THERE IS A SIZE CHANGE FROM 18" - 24" CARBON STEEL TO VICTAULIC SYSTEM, USE REDUCER AND STANDARD RF WNK FLANGE TO CONNECT TO FLANGE ADAPTER NIPPLE ON VICTAULIC END.											
20						***** BALL VALVES *****		FS1												
30															FV5					
40																				
50									COMPONENTS AT NPD 12", 14" AND 16" ARE TO MEET MANUFACTURER'S MINIMUM WALL THICKNESS REQUIREMENTS OF STANDARD WT. USE FOR REPLACEMENT GASKET. SIZES SHOWN IN SIZE COLUMNS ARE THE SMALL END OF REDUCERS AND THE BRANCH SIZE OF OLETS AND TEES.											
60			Ball V	A01AC01	1 - 4	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO		FV6												
70	B		Ball V	A01AC03	6 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO		GS1												
80	B		Ball V	A01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO		NR1												
90									PIPE IS SUPPLIED WITH STANDARD PLAIN END/ BW END AND GROOVE TO BE CUT DURING FABRICATION.											
100		VU2	Ball V	BA6B2	1/2 - 1/2	BALL 150# SCRD FNPT BS1400 LG2 BODY AND TRIM FB LO		UD7												
110			Ball V	BA6B2	3/4 - 1	BALL 150# SCRD FNPT BS1400 LG2 BODY AND TRIM FB LO		VA5												
120								VB2												
130									SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.											
140	B	VA5	Ball V	B01AC01	1 - 3	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO		VC1												
150	B	VA5	Ball V	B01AC03	4 - 6	BALL 150# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO		VC2												
160	B	VA5	Ball V	B01AC07	8 - 8	BALL 150# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO		VU2												
170								ZA6	USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.											
180						***** BUTTERFLY VALVES *****														
190	B		Butter	T01AC61	10 - 24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO														
200																				
210						***** GATE VALVES *****														
220																				
230	B		Gate V	GA6B1	3/4 - 1	GATE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW														
240			Gate V	G01AC48	1 1/2 - 1 1/2	GATE 150# RF LTCS A350 LF2 TRIM 16 WEDGE HW														
250			Gate V	G01AC41	2 - 8	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW														
260																				
270																				
280						***** GLOBE VALVES *****														
290																				
300	B		Globe	LA6B1	3/4 - 1	GLOBE 150# SCRD FNPT BS1400 LG2 BODY AND TRIM HW														
310	B		Globe	L01AC48	1 1/2 - 1 1/2	GLOBE 150# RF LTCS A350 LF2 TRIM 16 OS&Y HW														
320			Globe	L01AC41	2 - 6	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW														
330			Globe	L01AC42	8 - 10	GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO														
340																				
350																				
360						***** CHECK VALVES *****														
370									WE, WF, WP, WRA, WT, WU DESIGN: 15.8 BARG CODE: ASME B31.3											
380	B	VC1	Check	CA6B1	3/4 - 1	CHECK 150# SCRD FNPT BS1400 LG2 BODY AND TRIM SPRING PISTON														
390																				
400	B	VC1	Check	C01AC50	1 1/2 - 1 1/2	CHECK PISTON 150# RF LTCS A350 LF2 TRIM 16														
410		VC2	Check	C01AC45	2 - 6	CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED														
420	B	VC2	Check	C01AC52	8 - 24	CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16														
430																				
440						***** PIPE *****														
450									MATERIAL: CS GALV - GROOVED FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 110 °C COMMENT:											
460	B		Piping	5358601	3/4 - 1	PIPE SCH 160 SMLS LTCS A333 GR.6 GALV A153 T&C														
470	B	PA6	Piping	5358167_VC	1 1/2 - 3	PIPE SCH 80 SMLS LTCS A333 GR. 6 PE GALV (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)														
480		PA6	Piping	5358164_VC	4 - 10	PIPE SCH STD SMLS LTCS A333 GR. 6 PE GALV (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)														
490		PA6	Piping	5358162_VC	12 - 12	PIPE SCH 30 SMLS LTCS A333 GR. 6 PE GALV (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS						
								3MM	17MAR17	KD	B	1 OF 6	5	150VG01						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
500		PA6	Piping	5358161_VC	14 - 16	PIPE SCH 20 SMLS LTCS A333 GR. 6 PE GALV (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)							
510	B		Piping	5358597	18 - 24	PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE							
520	B		Piping	5358576	36 - 36	PIPE SCH STD EFW LTCS A671 CC60 CL.22 BE							
530													
540													
550		NR1				***** FITTINGS *****							
560		FV5											
570	B		90 Deg	5491290	3/4 - 1	ELL 90 DEG 3000# SCRDR FNPT LTCS A350 LF2 GALV A153							
580	B		45 Deg	5491300	3/4 - 1	ELL 45 DEG 3000# SCRDR FNPT LTCS A350 LF2 GALV A153							
590	B		90 Deg	5392280	18 - 24	ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W							
600	B		45 Deg	5392281	18 - 24	ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W							
610	B		90 Deg	5392496	36 - 36	ELL 90 DEG LR SCH STD LTCS A420 WPL6-W							
620													
630	B		Tee	5491310	3/4 - 1	TEE 3000# SCRDR FNPT LTCS A350 LF2 GALV A153							
640	B		Tee	5392295	18 - 24	TEE BW SCH 20 LTCS A420 WPL6-W							
650	B		Tee	5392405	36 - 36	TEE SCH STD LTCS A420 WPL6-W							
660													
670	B		Reduci	5491311	3/4 - 3/4	TEE RED 3000# SCRDR FNPT LTCS A350 LF2 GALV A153	1 - 1						
680	B		Reduci	5392296	8 - 10	TEE RED BW SCH 20/STD LTCS A420 WPL6-W	18 - 24						
690	B		Reduci	5392244	12 - 12	TEE RED BW SCH 20/SCH 30 LTCS A420 WPL6-W	18 - 24						
700	B		Reduci	5392292	14 - 20	TEE RED BW SCH 20 LTCS A420 WPL6-W	18 - 24						
710	B		Reduci	5556912	16 - 18	TEE RED BW SCH STD/ SCH 20 LTCS A420 WPL6-W	36 - 36						
720													
730													
740	B		Cap	5491330	3/4 - 1	CAP 3000# SCRDR FNPT LTCS A350 LF2 GALV A153							
750	B		Cap	5392286	18 - 24	CAP BW SCH 20 LTCS A420 WPL6							
760	B		Cap	5392448	36 - 36	CAP BW SCH STD LTCS A420 WPL6							
770													
780	B		Concen	5392297	10 - 10	REDUCER CONC SCH 20/STD LTCS A420 WPL6-W	18 - 18						
790	B		Concen	5392243	12 - 12	REDUCER CONC SCH 20/SCH 30 LTCS A420 WPL6-W	18 - 20						
800	B		Concen	5392287	14 - 20	REDUCER CONC SCH 20 LTCS A420 WPL6-W	18 - 24						
810													
820	B		Eccent	5392298	10 - 10	REDUCER ECC SCH 20/STD LTCS A420 WPL6-W	18 - 18						
830	B		Eccent	5392242	12 - 12	REDUCER ECC SCH 20/SCH 30 LTCS A420 WPL6-W	18 - 20						
840	B		Eccent	5392288	14 - 20	REDUCER ECC SCH 20 LTCS A420 WPL6-W	18 - 24						
850													
860						***** FITTINGS - VICTAULIC *****							
870													
880													
890	B		90 Deg	5484860	1 1/2 - 3	ELL 90 DEG GRV SCH 80 LTCS A352 GR LCC GALV							
900			90 Deg	5484861	4 - 10	ELL 90 DEG GRV SCH STD LTCS A352 GR LCC GALV							
910	B	FV6	90 Deg	5484862	12 - 12	ELL 90 DEG GRV SCH 30 LTCS A352 GR LCC GALV							
920	B	FV6	90 Deg	5484863	14 - 16	ELL 90 DEG GRV SCH 20 LTCS A352 GR LCC GALV							
930													
940	B		45 Deg	5484865	1 1/2 - 3	ELL 45 DEG GRV SCH 80 LTCS A352 GR LCC GALV							
950			45 Deg	5484866	4 - 10	ELL 45 DEG GRV SCH STD LTCS A352 GR LCC GALV							
960	B	FV6	45 Deg	5484867	12 - 12	ELL 45 DEG GRV SCH 30 LTCS A352 GR LCC GALV							
970	B	FV6	45 Deg	5484868	14 - 16	ELL 45 DEG GRV SCH 20 LTCS A352 GR LCC GALV							
980													
990	B		Tee	5485120	1 1/2 - 3	TEE GRV SCH 80 LTCS A352 GR LCC GALV							
1000			Tee	5485121	4 - 10	TEE GRV SCH STD LTCS A352 GR LCC GALV							
1010	B	FV6	Tee	5485122	12 - 12	TEE GRV SCH 30 LTCS A352 GR LCC GALV							
1020	B	FV6	Tee	5485123	14 - 16	TEE GRV SCH 20 LTCS A352 GR LCC GALV							
			FGP / WPMP TENGIZCHEVOIL - TENGIZCHEVOIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4" - 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED	CORR. ALLOW. 3MM	DATE 17MAR17	BY KD	REV B	SHEET 2 OF 6	PA 5	CLASS 150VG01

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1030	B		Reduci	5484879	3/4 - 3/4	TEE RED GRV SCH 80/SCH 160 LTCS A352 GR LCC GALV	2 - 2						
1040	B		Reduci	5484880	1 1/2 - 2 1/2	TEE RED GRV SCH 80 LTCS A352 GR LCC GALV	2 - 3						
1050	B		Reduci	5484884	1 1/2 - 3	TEE RED GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV	4 - 10						
1060	B		Reduci	5484881	4 - 8	TEE RED GRV SCH STD LTCS A352 GR LCC GALV	6 - 10						
1070	B	FV6	Reduci	5484887	1 1/2 - 3	TEE RED GRV SCH 30/SCH 80 LTCS A352 GR LCC GALV	12 - 12						
1080	B	FV6	Reduci	5484882	4 - 10	TEE RED GRV SCH 30/STD LTCS A352 GR LCC GALV	12 - 12						
1090	B	FV6	Reduci	5484886	4 - 10	TEE RED GRV SCH 20/STD LTCS A352 GR LCC GALV	14 - 16						
1100	B	FV6	Reduci	5484885	12 - 12	TEE RED GRV SCH 20/SCH 30 LTCS A352 GR LCC GALV	14 - 16						
1110		FV6	Reduci	5484883	14 - 14	TEE RED GRV SCH 20 LTCS A352 GR LCC GALV	16 - 16						
1120													
1130	B		Cap	5485130	1 1/2 - 3	CAP GRV SCH 80 LTCS A352 GR LCC GALV							
1140			Cap	5485131	4 - 10	CAP GRV SCH STD LTCS A352 GR LCC GALV							
1150	B	FV6	Cap	5485132	12 - 12	CAP GRV SCH 30 LTCS A352 GR LCC GALV							
1160	B	FV6	Cap	5485133	14 - 16	CAP GRV SCH 20 LTCS A352 GR LCC GALV							
1170													
1180	B		Concen	5484890	1 1/2 - 2	REDUCER CONC GRV SCH 80 LTCS A352 GR LCC GALV	2 - 3						
1190			Concen	5484891	1 1/2 - 3	REDUCER CONC GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV	4 - 6						
1200													
1210			Concen	5484892	4 - 8	REDUCER CONC GRV SCH STD LTCS A352 GR LCC GALV	6 - 10						
1220	B	FV6	Concen	5484893	6 - 10	REDUCER CONC GRV SCH 30/SCH STD LTCS A352 GR LCC GALV	12 - 12						
1230	B	FV6	Concen	5484894	6 - 10	REDUCER CONC GRV SCH 20/SCH STD LTCS A352 GR LCC GALV	14 - 16						
1240	B	FV6	Concen	5484896	12 - 12	REDUCER CONC GRV SCH 20/SCH 30 LTCS A352 GR LCC GALV	14 - 16						
1250	B	FV6	Concen	5484895	14 - 14	REDUCER CONC GRV SCH 20 LTCS A352 GR LCC GALV	16 - 16						
1260													
1270	B		Eccent	5484830	1 1/2 - 2	REDUCER ECC GRV SCH 80 LTCS A352 GR LCC GALV	2 - 3						
1280			Eccent	5484831	1 1/2 - 3	REDUCER ECC GRV SCH STD/SCH 80 LTCS A352 GR LCC GALV	4 - 6						
1290													
1300			Eccent	5484832	4 - 8	REDUCER ECC GRV SCH STD LTCS A352 GR LCC GALV	6 - 10						
1310	B	FV6	Eccent	5484833	6 - 10	REDUCER ECC GRV SCH 30/SCH STD LTCS A352 GR LCC GALV	12 - 12						
1320	B	FV6	Eccent	5484834	6 - 10	REDUCER ECC GRV SCH 20/SCH STD LTCS A352 GR LCC GALV	14 - 16						
1330	B	FV6	Eccent	5484836	12 - 12	REDUCER ECC GRV SCH 20/SCH 30 LTCS A352 GR LCC GALV	14 - 16						
1340	B	FV6	Eccent	5484835	14 - 14	REDUCER ECC GRV SCH 20 LTCS A352 GR LCC GALV	16 - 16						
1350													
1360													
1370						*****COUPLINGS, UNIONS & PLUGS*****							
1380								SERVICE					
1390	B		Coupli	5491250	3/4 - 1	COUPLING 3000# SCRD FNPT LTCS A350 LF2 GALV A153		WE, WF, WP, WRA, WT, WU DESIGN: 15.8 BARG CODE: ASME B31.3					
1400	B		Union	5491270	3/4 - 1	UNION 3000# SCRD FNPT LTCS A350 LF2 GALV A153							
1410	B		Plug	5491280	1/2 - 2	PLUG ROUND HEAD SCRD MNPT LTCS A350 LF2 GALV A153							
1420			Bushin	5491325	1/2 - 1 1/2	BUSHING 3000# SCRD (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153	3/4 - 2						
1430													
1440								MATERIAL: CS GALV - GROOVED FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 110 °C COMMENT:					
1450	B	UD7	Thredo	5514772	3/4 - 2	THREDOLET 3000# LTCS A350 LF2	18 - 36						
1460			Weldol	5505571	4 - 6	WELDOLET SCH 20/STD LTCS A350 LF2	24 - 24						
1470			Weldol	5505164	6 - 6	WELDOLET SCH STD LTCS A350 LF2	36 - 36						
1480						***** COUPLINGS - VICTAULIC *****							
1490								FGP/WPMP					
1500													
1510	B		Coupli	5484764	1 1/2 - 16	COUPLING RIGID GRV LTCS A350 LF2 EPDM GASKET GALV							
1520													
1530													
1540	B		SADDLE	5485125	1/2 - 1	VICTAULIC BOLTED BRANCH OUTLET FNPT LTCS A352 GR LCC GALV	2 - 4						
1550								CORR. ALLOW. 3MM					
1560			SADDLE	5485125	1 1/2 - 1 1/2	VICTAULIC BOLTED BRANCH OUTLET FNPT LTCS A352 GR LCC GALV	6 - 6						
			FGP / WPMP TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED		DATE 17MAR17 BY KD REV B SHEET 3 OF 6 PA 5 CLASS 150VG01					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1570	B		SADDLE	5485125	2 - 2	VICTAULIC BOLTED BRANCH OUTLET FNPT LTCS A352 GR LCC GALV	6 - 8								
1580			Thredo	5485127	3/4 - 3/4	VICTAULIC STRAPLESS OUTLET FNPT LTCS A352 GR LCC GALV	6 - 16								
1590						***** PIPE FLANGES *****									
1600															
1610	B	FV6	Reduci	5599691	1/2 - 3/4	FLG SCRDR (FNPT) REDUCING 150# RF LTCS A350 LF2 GALV A153	3/4 - 2								
1620															
1630			Flange	5599689	3/4 - 1	FLG SCRDR (FNPT) 150# RF LTCS A350 LF2 GALV A153									
1640			Flange	5599701	3/4 - 1	FLG SCRDR (FNPT) 300# RF LTCS A350 LF2 GALV A153									
1650	B		Flange	5556261	4 - 10	FLG WN 150# RF LTCS A350 LF2 SCH STD									
1660	B		Flange	5556274	12 - 12	FLG WN 150# RF LTCS A350 LF2 SCH 30									
1670	B		Flange	5556265	14 - 16	FLG WN 150# RF LTCS A350 LF2 SCH 20									
1680	B		Flange	5556265	18 - 24	FLG WN 150# RF LTCS A350 LF2 SCH 20									
1690	B		Flange	5556262	36 - 36	FLG WN 150# RF LTCS A350 LF2 SCH STD B16.47 SERIES A									
1700															
1710															
1720															
1730			Blind	5615567	1/2 - 24	FLG BLIND 150# RF LTCS A350 LF2 GALV A153									
1740			Blind	5615577	1/2 - 1/2	FLG BLIND 300# RF LTCS A350 LF2 GALV A153									
1750															
1760															
1770						***** PIPE FLANGES - VICTAULIC *****									
1780															
1790	B		Flange	5754404	1 1/2 - 16	FLANGE ADAPTER NIPPLE 150# RF / GRV A352 GR LCC GALV									
1800	B		Flange	5754405	1 1/2 - 12	FLANGE ADAPTER NIPPLE 300# RF / GRV A352 GR LCC GALV									
1810	B	FS1	Flange	5754407	2 - 2	FLANGE ADAPTER NIPPLE 150# FF / GRV A352 GR LCC GALV									
1820															
1830															
1840															
1850	B		Orific	5754195	3 - 16	METERING STATION ORIF TYPE GRV A352 GR LCC GALV									
1860															
1870															
1880															
1890						***** ORIFICE FLANGE - VICTAULIC *****									
1900	B		Orific	5623941	18 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 20									
1910															
1920															
1930			Nipple	5541200	1/2 - 1	NIPPLE SCH 160 LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153									
1940	B		Nipple	5541201	1/2 - 1	NIPPLE SCH 160 LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153									
1950	B		Nipple	5541214	2 - 2	NIPPLE SCH XS LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153									
1960															
1970						***** SWAGE NIPPLES *****									
1980															
1990															
2000															
2010			Concen	5544150	3/4 - 1	SWAGE CONC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153	1 - 1 1/2								
2020	B														
2030	B		Eccent	5549150	3/4 - 1	SWAGE ECC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153	1 - 1 1/2								
2040															
2050						***** GASKETS *****									
2060															
2070															
2080	B		GASFRS	5673043	1/2 - 24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR									
2090	B		GASFRS	5674011	36 - 36	GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED		CORR. ALLOW. 3MM	FGP/WPMP DATE 17MAR17 BY KD REV B SHEET 4 OF 6 PA 5 CLASS 150VG01						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
2100	B	FS1	GASFF	5664562	2 - 2	GASKET 150# FF RING EPDM 3MM THK						
2110												
2120	B	FS1	GASFRS	5673033	1/2 - 12	GASKET 300# 316SS SP WND GRAPHITE IR						
2130												
2140												
2150						***** GASKETS - VICTAULIC *****						
2160												
2170	B		GASFRS	NOMTO_G	1 1/2 - 16	GASKET FOR CLAMP FOR VICTAULIC SYSTEM - NON REPORTABLE						
2180												
2190												
2200		B1				***** BOLTING *****						
2210												
2220	B		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS						
2230												
2240						**** BLINDS & SPACERS ****						
2250												
2260			Specta	5749582	3/4 - 14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70 GALV A153						
2270												
2280			Specta	5749583	3/4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70 GALV A153						
2290												
2300			Paddle	5749942	16 - 16	PADDLE BLANK 150# RF LTCS A516 GR.70 GALV A153						
2310			Paddle	5750195	16 - 16	PADDLE SPACER 150# RF LTCS A516 GR.70 GALV A153						
2320												
2330	B		Paddle	5749942	18 - 24	PADDLE BLANK 150# RF LTCS A516 GR.70 GALV A153						
2340	B		Paddle	5750195	18 - 24	PADDLE SPACER 150# RF LTCS A516 GR.70 GALV A153						
2350	B		Paddle	5750025	36 - 36	PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A GALV A153						
2360												
2370	B		Paddle	5750194	36 - 36	PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A GALV A153						
2380												
2390			Paddle	5749853	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70 GALV A153						
2400			Paddle	5750201	12 - 24	PADDLE SPACER 300# RF LTCS A516-GR70 GALV A153						
2410												
2420												
2430						***** BRANCHES *****						
2440												
2450						USE BRANCH CHART 150VG01						
2460												
2470	B		Reinfo	RWELD	24 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	36 - 36					
2480	B		Reinfo	RWELD	24 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	36 - 36					
2490												
2500												
2510						***** CONSTRUCTION DETAILS *****						
2520												
2530	B	ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY						
2540	B					PIPE BENDS ----- NONE						
2550						THREAD SEALANT ----- TEFLON						
2560						PASTE PWHT ----- NONE						
2570												
2580						*****DO NOT USE FOLLOWING ITEMS*****						
2590												
2600												
2610	B		Reduci	5485128	2 - 2	DO NO USE - BRANCH OUTLET GRV LTCS A352 GR LCC GALV	10 - 24					
2620			Reduci	5484880	1 - 1	TEE RED GRV SCH 80 LTCS A352 GR LCC GALV	3 - 3					
2630			Reducn	5484889	1 - 1	TEE RED GRV X MNPT OUTLET SCH 80 LTCS A352 GR LCC GALV	3 - 3					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED	CORR. ALLOW. 3MM	FGP/WPMP DATE 17MAR17 BY KD REV B SHEET 5 OF 6 PA 5 CLASS 150VG01				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
2650	B		Concen	5543782	3/4 - 3/4	SWAGE CONC SCH 80/ SCH 160 STL A420 WPL6 MGE X MNPT GALV	1 1/2 - 2								
2660															
2670	B		Concen	5543783	1 - 1	SWAGE CONC SCH 80 STL A420 WPL6 MGE X MNPT GALV	2 - 3								
2680	B		Concen	5543784	1 - 1	SWAGE CONC SCH STD/ SCH 80 STL A420 WPL6 MGE X MNPT GALV	4 - 6								
2690															
2700	B		Reduci	5484888	3/4 - 1	TEE RED GRV X MNPT OUTLET SCH 80/ SCH 160 LTCS A352 GR LCC GALV	1 1/2 - 1 1/2								
2710	B		Reduci	5485124	1 1/2 - 4	VICTAULIC BOLTED BRANCH OUTLET GRV LTCS A352 GR LCC GALV	2 - 6								
2720															
2730															
2740	B		GS1	GASFR	5664445	1 - 16	GASKET C SHAPE, GRADE E EPDM								
								SERVICE							
								WE, WF, WP, WRA, WT, WU DESIGN: 15.8 BARG CODE: ASME B31.3							
								MATERIAL: CS GALV - GROOVED FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 110 °C COMMENT:							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP							
						3/4"- 1" - SCREWED 1 1/2" - 16" - VICTAULIC GROOVE & COUPLING 18" & ABOVE BUTTWELDED		CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS
								3MM		17MAR17	KD	B	6 OF 6	5	150VG01

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10	B	PP9				***** VALVES*****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.					
20		VB2				*****BALL VALVES*****		FS1	USE FOR REPLACEMENT GASKET. SIZES SHOWN IN SIZE COLUMNS ARE THE SMALL END OF REDUCERS AND THE BRANCH SIZE OF OLETS AND TEES.					
30								GS1	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
40								NR1	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
50									PP9	ONLY TO BE USED WHEN SPECIFIED ON THE P&ID. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.				
60			Ball V	A01AS01	3 - 4	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO		VA5	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
70			Ball V	A01AS03	6 - 6	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO		VAR	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
80			Ball V	A01AS07	8 - 8	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO		VB2						
90								VC1						
100			Ball V	B01AS01	1/2 - 2	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO		VC2						
110			Ball V	B01AS01	3 - 3	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO								
120	B	VA5	Ball V	B01AS03	4 - 6	BALL 150# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO								
130	B	VA5	Ball V	B01AS07	8 - 8	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO								
140		VA5	Ball V	B01AS07	10 - 16	BALL 150# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO								
150														
160			Ball V	B03AS01	3/4 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO								
170														
180						*****BUTTERFLY VALVES*****								
190														
200			Butter	T01AS61	10 - 16	BUTTERFLY 150# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO								
210														
220						*****GATE VALVES *****								
230														
240			Gate V	G01AS48	3/4 - 1 1/2	GATE 150# RF SS A182 F316 TRIM 16 WEDGE HW								
250			Gate V	G01AS41	2 - 8	GATE 150# RF SS A351 CF8M TRIM 16 WEDGE HW								
260			Gate V	G01AS41	10 - 16	GATE 150# RF SS A351 CF8M TRIM 16 WEDGE HW								
270														
280														
290						*****GLOBE VALVES*****								
300														
310			Globe	L01AS48	3/4 - 1 1/2	GLOBE 150# RF SS A182 F316 TRIM 16 OS&Y HW								
320			Globe	L01AS41	2 - 6	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y HW								
330			Globe	L01AS42	8 - 10	GLOBE 150# RF SS A351 CF8M TRIM 16 OS&Y GO								
340														
350						*****CHECK VALVES*****								
360														
370														
380		VC1	Check	C01AS50	3/4 - 1 1/2	CHECK PISTON 150# RF SS A182 F316 TRIM 16		WDM						
390		VC2	Check	C01AS45	2 - 6	CHECK DUAL WAFER LUG 150# RF SS A351 CF8M TRIM 16 THROU BOLTED		DESIGN: 15.8 BARG						
400		VC2	Check	C01AS52	8 - 16	CHECK DUAL DBL FLG 150# RF SS A351 CF8M TRIM 16		CODE: ASME B31.3						
410														
420														
430						***** PIPE *****								
440														
450			Piping	5365117_VC	1 - 16	PIPE SCH 10S SMLS SS A312 TP316/316L PE (GROOVING NOT BY SUPPLIER - TO BE DONE DURING FABRICATION)								
460														
470														
480						***** FITTINGS - VICTAULIC *****		MATERIAL:	LOW TEMP SS - GROOVE					
490								FLANGE RATING:	150 RF					
500								TEMPERATURE RANGE:	-46 / 110 °C					
510								COMMENT:						
520														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 1" - 16" - VICTAULIC GROOVE & COUPLING		FGP/WPMP						
								CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								3MM	17MAR167	PDM	B	1 OF 3	5	150VS01

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
530			Tee	5485113	1 - 16	TEE GRV SCH 10S 316 SS A351 CF8M		SERVICE WDM DESIGN: 15.8 BARG CODE: ASME B31.3 MATERIAL: LOW TEMP SS - GROOVE FLANGE RATING: 150 RF TEMPERATURE RANGE: -46 / 110 °C COMMENT:					
540							2 - 16						
550			Reduci	5485114	1 1/2 - 14	TEE RED GRV SCH 10S 316 SS A351 CF8M							
560													
570			Cap	5485119	1 - 16	CAP GRV SCH 10S 316 SS A351 CF8M							
580													
590			Concen	5485115	1 - 14	REDUCER CONC GRV SCH 10S 316 SS A351 CF8M	1 1/2 - 16						
600			Eccent	5485118	1 - 14	REDUCER ECC GRV SCH 10S 316 SS A351 CF8M	1 1/2 - 16						
610													
620						*****COUPLINGS, UNIONS & PLUGS *****							
630													
640													
650			Bushin	5498221	1/2 - 1 1/2	BUSHING 3000# SCR D (MNPT/FNPT) HEX HEAD 316 SS	3/4 - 2						
660													
670						***** VICTAULIC COUPLINGS *****							
680													
690													
700			Coupli	5485021	1 - 16	COUPLING RIGID GRV 316SS EPDM GRADE E							
710													
720			Thredo	5485137	3/4 - 3/4	VICTAULIC STRAPLESS OUTLET FNPT 316 SS	6 - 16						
730													
740			SADDLE	5485138	1/2 - 1	VICTAULIC BOLTED BRANCH OUTLET FNPT SS316 A351 CF8M	2 - 4						
750			SADDLE	5485138	1 1/2 - 1 1/2	VICTAULIC BOLTED BRANCH OUTLET FNPT SS316 A351 CF8M	6 - 6						
760			SADDLE	5485138	2 - 2	VICTAULIC BOLTED BRANCH OUTLET FNPT SS316 A351 CF8M	8 - 8						
770													
780						***** PIPE FLANGES *****							
790													
800			Flange	5600657	1/2 - 3/4	FLG SCR D (FNPT) 150# RF SS A182 F316/316L							
810			Flange	5600657	1 - 1	FLG SCR D (FNPT) 150# RF SS A182 F316/316L							
820			Flange	5600658	3/4 - 3/4	FLG SCR D (FNPT) 300# RF SS A182 F316/316L							
830													
840			Blind	5616520	3/4 - 16	FLG BLIND 150# RF SS A182 F316/316L							
850			Blind	5616529	1 1/2 - 16	FLG BLIND 300# RF SS A182 F316/316L							
860													
870						***** PIPE FLANGES - VICTAULIC *****							
880													
890													
900			Flange	5754196	1 - 16	FLANGE ADAPTER NIPPLE 150# RF SCH 10S/GRV A182 316 SS							
910			Flange	5754197	1 1/2 - 16	FLANGE ADAPTER NIPPLE 300# RF SCH 10S/GRV A182 316 SS							
920													
930						***** ORIFICE FLANGE *****							
940													
950			Orific	5754198	3 - 10	METERING STATION ORIF TYPE GRV 316 SS							
960													
970						***** PIPE NIPPLES - VICTAULIC *****							
980													
990			Nipple	5540198	1/2 - 3/4	NIPPLE SCH 40S 316 SS TBE 3" LG (MNPT)							
1000													
1010													
1020						***** GASKETS *****							
1030													
1040			GASFRS	5673043	1 1/2 - 16	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR							
1050		FS1	GASFRS	5673033	1 1/2 - 16	GASKET 300# 316SS SP WND GRAPHITE IR							
1060													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 1" - 16" - VICTAULIC GROOVE & COUPLING	CORR. ALLOW. 3MM	DATE 17MAR167	BY PDM	REV B	FGP/WPMP SHEET 2 OF 3	PA 5	CLASS 150VS01

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1170 1180 1190 1200 1210 1220 1230 1240 1250 1260 1270 1280 1290 1300 1310 1320 1330 1340 1350 1360 1370 1380 1390 1400 1410 1420 1430 1440 1450 1460 1470 1480	B	B1	GASFRS	NOMTO_G	1 - 16	***** GASKETS - VICTAULIC ***** GASKET FOR CLAMP FOR VICTAULIC SYSTEM - NON REPORTABLE ***** BOLTING ***** STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS *****BLINDS AND SPACERS***** SPECTACLE BLIND 150# RF SS A240 GR 316 PADDLE BLANK 150# RF SS A240 GR 316 PADDLE SPACER 150# RF SS A240 GR 316 ***** BRANCHES ***** USE BRANCH CHART 150VS01 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING -----FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- TEFLON PASTE PWHT ----- NONE *****DO NOT USE*****	3 - 16 3 - 16 <						

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
470		VC2	Check	C03AR46	4 - 10	CHECK DUAL WAFER LUG 300# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED							
480		VC2	Check	C03AR53	12 - 24	CHECK DUAL DBL FLG 300# RF LTCS + CLAD INC 825 TRIM 625							
490						*****PIPES*****							
500													
510													
520			Piping	5732001	3/4 - 1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE							
530			Piping	5361603	1 1/2 - 3	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 BE							
540			Piping	5358505_C80	4 - 14	PIPE SCH STD SMLS LTCS A333 GR.6 WITH 3MM INC 825							
						INTERNAL CLADDING							
550			Piping	5358514_C80	16 - 16	PIPE SCH XS SMLS LTCS A333 GR.6 WITH 3 MM INC 825 INTERNAL							
						CLADDING							
560			Piping	5358577_C80	18 - 20	PIPE SCH XS EFW LTCS A671 CC60 CL.22 WITH 3 MM INC 825 INTERNAL							
						CLADDING							
570			Piping	5358385_C80	24 - 24	PIPE SCH 30 EFW LTCS A671 CC60 CL.22 WITH 3 MM INC 825 INTERNAL							
						CLADDING							
580			Piping	5358578_C80	42 - 42	PIPE WT 24.0MM EFW LTCS A671 CC60 CL.22 WITH 3MM INC 825							
						INTERNAL CLADDING							
590													
600		810	Piping	5732003	1/2 - 1/2	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE							
610						*****FITTINGS*****							
620													
630													
640			90 Deg	5425111	3/4 - 1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825							
650			90 Deg	5424722	1 1/2 - 3	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPS UNS N08825							
660			90 Deg	5392438_C80	4 - 14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6 WITH 3MM INC							
						825 INTERNAL CLADDING							
670			90 Deg	5392600_C80	16 - 16	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC							
						825 INTERNAL CLADDING							
680			90 Deg	5392497_C80	18 - 20	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W WITH 3 MM INC							
						825 INTERNAL CLADDING							
690			90 Deg	5392331_C80	24 - 24	ELL 90 DEG LR SCH 30 LTCS A420 WPL6-W WITH 3 MM INC 825							
						INTERNAL CLADDING							
700			90 Deg	5392551_C80	42 - 42	ELL 90 DEG LR WT 24.0MM LTCS A420 WPL6-W WITH 3MM INC							
						825 INTERNAL CLADDING							
710													
720			45 Deg	5425113	3/4 - 1	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825							
730			45 Deg	5424727	1 1/2 - 3	ELL 45 DEG SCH 40S ALLOY 825 B366-WPS UNS N08825							
740			45 Deg	5392444_C80	4 - 14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6 WITH 3MM INC 825							
						INTERNAL CLADDING							
750			45 Deg	5392606_C80	16 - 16	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 825							
						INTERNAL CLADDING							
760			45 Deg	5392608_C80	18 - 20	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825							
						INTERNAL CLADDING							
770			45 Deg	5392337_C80	24 - 24	ELL 45 DEG SCH 30 LTCS A420 WPL6 -W WITH 3 MM INC 825							
						INTERNAL CLADDING							
780													
790			Tee	5425119	3/4 - 1	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825							
800			Tee	5424725	1 1/2 - 3	TEE SCH 40S ALLOY 825 B366-WPS UNS N08825							
810			Tee	5392447_C80	4 - 14	TEE BW SCH STD LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL							
						CLADDING							
820			Tee	5392609_C80	16 - 16	TEE BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL							
						CLADDING							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZE -BUTTWELD PWHT FOR SIZES 4" AND LARGER	CORR. ALLOW. 1 MM	DATE 17NOV16	BY KD	REV B	SHEET 2 OF 6	PA 5	CLASS 300C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
830			Tee	5392406_C80	18 - 20	TEE BW SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING							
840			Tee	5392412_C80	24 - 24	TEE SCH 30 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING							
850													
860			Reduci	5425126	3/4 - 3/4	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1 - 1						
870			Reduci	5425124	3/4 - 1	TEE RED SCH 40S/80S ALLOY 825 B366-WPS UNS N08825	1 1/2 - 2						
880			Reduci	5424724	1 1/2 - 2	TEE RED SCH 40S ALLOY 825 B366-WPS UNS N08825	2 - 3						
890			Reduci	5425128	1 1/2 - 3	TEE RED SCH STD/SCH 40S ALLOY 825 B366-WPS UNS N08825	4 - 6						
900			Reduci	5392445_C80	4 - 12	TEE RED BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 14						
910			Reduci	5392460_C80	6 - 14	TEE RED BW SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	16 - 16						
920			Reduci	5392476_C80	8 - 14	TEE RED SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18 - 20						
930			Reduci	5393023_C80	10 - 14	TEE RED SCH 30/ SCH STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24 - 24						
940			Reduci	5393022_C80	16 - 20	TEE RED SCH 30/ SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24 - 24						
950													
960			Cap	5424270	3/4 - 1	CAP SCH 80S ALLOY 825 B366 UNS N08825							
970			Cap	5424219	1 1/2 - 3	CAP SCH 40S ALLOY 825 B366 UNS N08825							
980			Cap	5392448_C80	4 - 14	CAP BW SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING							
990			Cap	5392610_C80	16 - 20	CAP BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING							
1000			Cap	5392340_C80	24 - 24	CAP SCH 30 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING							
1010													
1020			Concen	5425122	3/4 - 3/4	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1						
1030			Concen	5425143	3/4 - 1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2						
1040			Concen	5424733	1 1/2 - 2	REDUCER CONC SCH 40S ALLOY 825 B366-WPS N08825	2 - 3						
1050			Concen	5425146	1 1/2 - 3	REDUCER CONC SCH STD/40S ALLOY 825 B366-WPS N08825	4 - 6						
1060			Concen	5392449_C80	4 - 12	REDUCER CONC SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 14						
1070			Concen	5392457_C80	8 - 14	REDUCER CONC SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	16 - 16						
1080			Concen	5392506_C80	10 - 14	REDUCER CONC SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18 - 20						
1090			Concen	5392507_C80	16 - 16	REDUCER CONC SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18 - 20						
1100			Concen	5392523_C80	16 - 20	REDUCER CONC SCH 30/ SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24 - 24						
1110													
1120			Eccent	5425123	3/4 - 3/4	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1						
1130			Eccent	5425145	3/4 - 1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2						
1140			Eccent	5424734	1 1/2 - 2	REDUCER ECC SCH 40S ALLOY 825 B366-WPS N08825	2 - 3						
1150			Eccent	5425148	1 1/2 - 3	REDUCER ECC SCH STD/40S ALLOY 825 B366-WPS N08825	4 - 6						
1160			Eccent	5392450_C80	4 - 12	REDUCER ECC SCH STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 14						
1170			Eccent	5392459_C80	8 - 14	REDUCER ECC SCH XS/STD LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	16 - 16						
1180			Eccent	5392510_C80	10 - 14	REDUCER ECC SCH XS/STD LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18 - 20						
1190			Eccent	5392514_C80	16 - 18	REDUCER ECC SCH XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	18 - 20						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZE -BUTTWELD PWHT FOR SIZES 4" AND LARGER	CORR. ALLOW. 1 MM						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1200			Eccent	5392397_C80	16 - 20	REDUCER ECC SCH 30/XS LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	24 - 24						
1210													
1220			Nipofl	5513491	3/4 - 1	NIPOFLANGE 300# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2 - 3						
1230			Nipofl	5513492	3/4 - 1	NIPOFLANGE 300# RF SCH STD/80S ALLOY 825 B564 UNS N08825	4 - 14						
1240			Nipofl	5513493	3/4 - 1	NIPOFLANGE 300# RF SCH XS/80S ALLOY 825 B564 UNS N08825	16 - 20						
1250			Nipofl	5513494	3/4 - 1	NIPOFLANGE 300# RF SCH SCH 30/80S ALLOY 825 B564 UNS N08825	24 - 24						
1260			Nipofl	5513495	1 1/2 - 2	NIPOFLANGE 300# RF SCH 40S ALLOY 825 B564 UNS N08825	3 - 3						
1270			Nipofl	5513496	1 1/2 - 2	NIPOFLANGE 300# RF SCH STD/ 40S ALLOY 825 B564 UNS N08825	4 - 14						
1280			Nipofl	5513497	1 1/2 - 2	NIPOFLANGE 300# RF SCH XS/40S ALLOY 825 B564 UNS N08825	16 - 20						
1290			Nipofl	5513498	1 1/2 - 2	NIPOFLANGE 300# RF SCH 30/40S ALLOY 825 B564 UNS N08825	24 - 24						
1300			Nipofl	5513499	1 1/2 - 2	NIPOFLANGE 300# RF WT 24.0MM/40S ALLOY 825 B564 UNS N08825	42 - 42						
1310													
1320		FF5	Endole	5504518	3/4 - 1	NIPOFLANGE FLAT BOTTOM 300# RF SCH 80S ALLOY 825 B564 UNS N08825							
1330		FF5	Endole	5504519	1 1/2 - 1 1/2	NIPOFLANGE FLAT BOTTOM 300# RF SCH 40S ALLOY 825 B564 UNS N08825							
1340													
1350		UJ2	Nipofl	5513479	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 ALLOY 825 B564 UNS N08825	4 - 14						
1360		UJ2	Nipofl	5513478	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH XS/160 ALLOY 825 B564 UNS N08825	16 - 20						
1370		UJ2	Nipofl	5513477	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 30/160 ALLOY 825 B564 UNS N08825	24 - 24						
1380													
1390			Weldol	5513413	3/4 - 1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	2 - 3						
1400			Weldol	5513417	3/4 - 1	WELDOLET SCH STD/80S ALLOY 825 B564 UNS N08825	4 - 14						
1410			Weldol	5513415	3/4 - 1	WELDOLET SCH XS/80S ALLOY 825 B564 UNS N08825	16 - 20						
1420			Weldol	5513416	3/4 - 1	WELDOLET SCH 30/80S ALLOY 825 B564 UNS N08825	24 - 24						
1430			Weldol	5513418	1 1/2 - 3	WELDOLET SCH STD/ 40S ALLOY 825 B564 UNS N08825	4 - 14						
1440			Weldol	5513419	1 1/2 - 3	WELDOLET SCH XS/ 40S ALLOY 825 B564 UNS N08825	16 - 20						
1450			Weldol	5513410	1 1/2 - 3	WELDOLET SCH 30/ 40S ALLOY 825 B564 UNS N08825	24 - 24						
1460			Weldol	5505164_C80	4 - 4	WELDOLET SCH STD LTCS A350 LF2 WITH 3MM INC 825 INTERNAL CLADDING	12 - 14						
1470			Weldol	5505334_C80	4 - 6	WELDOLET SCH XS/STD LTCS A350 LF2 WITH 3MM INC 825 INTERNAL CLADDING	16 - 20						
1480			Weldol	5505336_C80	4 - 8	WELDOLET SCH 30/STD LTCS A350 LF2 WITH 3MM INC 825 INTERNAL CLADDING	24 - 24						
1490		new	Weldol	5505338_C80	6 - 6	WELDOLET WT 24.0MM/STD LTCS A350 LF2 WITH 3MM INC 826 INTERNAL CLADDING	42 - 42						
1500													
1510			Elbole	5516626	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 ALLOY 825 B564 UNS N08825	4 - 14						
1520			Elbole	5516627	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH XS/160 ALLOY 825 B564 UNS N08825	16 - 20						
1530			Elbole	5516628	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 30/160 ALLOY 825 B564 UNS N08825	24 - 24						
1540													
1550													
1560						*****FLANGES*****							
1570			Flange	5569290	3/4 - 1	FLG WN 300# RF ALLOY 825 SCH 80S B564 UNS N08825							
1580			Flange	5569276	1 1/2 - 3	FLG WN 300# RF ALLOY 825 SCH 40S B564 UNS N08825							
1590			Flange	5556473_C80	4 - 14	FLG WN 300# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE							
1600			Flange	5556476_C80	16 - 20	FLG WN 300# RF LTCS A350 LF2 SCH XS WITH 3MM INC 825 CLAD FACE & BORE							
1610			Flange	5556471_C80	24 - 24	FLG WN 300# RF LTCS A350 LF2 SCH 30 WITH 3MM INC 825 CLAD FACE & BORE							
1620			Flange	5556488_C80	42 - 42	FLG WN 300# RF LTCS A350 LF2 WT 24.0MM B16.47 SERIES A WITH 3MM INC 825 CLAD FACE & BORE							
1630													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZE -BUTTWELD PWHT FOR SIZES 4" AND LARGER	CORR. ALLOW. 1 MM	DATE 17NOV16	BY KD	REV B	FGP/WPMP SHEET 4 OF 6	PA 5	CLASS 300C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1640		FS1	Blind	5617159	1/2 - 3	FLG BLIND 300# RF ALLOY 825 B564 UNS N08825	3/4 - 1 1/2							
1650			Blind	5615576_C80	4 - 24	FLG BLIND 300# RF LTCS A350 LF2 WITH 3 MM INC 825 CLAD FACE								
1660			Flange	5557033_C80	8 - 10	FLG WN 600# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE								
1670														
1680		FA8	Reduci	5614310	1/2 - 1/2	FLG SCRd (FNPT) REDUCING 300# RF ALLOY 825 B564 UNS N08825								
1690			Flange	5851203	1/2 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE								
1700														
1710														
1720			Orific	5650830	2 - 3	WN ORIF SET 300# RF ALLOY 825 B564 UNS N08825 SCH 40S								
1730														
1740														
1750														
1760			Orific	5623944_C80	4 - 14	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD WITH 3MM INC 825 CLAD FACE & BORE								
1770														
1780														
1790														
1800			Orific	5623947_C80	16 - 20	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS WITH 3MM INC 825 CLAD FACE & BORE								
1810														
1820														
1830														
1840			Orific	5623942_C80	24 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 30 WITH 3MM INC 825 CLAD FACE & BORE								
1850														
1860														
1870														
1880						***** PIPE NIPPLES *****								
1890														
1900														
1910														
1920					***** GASKETS *****									
1930														
1940														
1950														
1960					***** BLINDS & SPACERS *****									
1970														
1980														
1990														
2000					***** BOLTING *****									
2010														
2020														
2030														
2040					***** BLINDS & SPACERS *****									
2050														
2060														
2070														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZE -BUTTWELD PWHT FOR SIZES 4" AND LARGER		CORR. ALLOW. 1 MM	DATE 17NOV16	BY KD	REV B	FGP/WPMP SHEET 5 OF 6	PA 5	CLASS 300C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2080 2090 2100 2110 2120 2130 2140 2150 2160 2170 2180 2190 2200 2210 2220	B 												



SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
460		VA5	Gate V	G03AC41	2 - 8	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW		VAE	FOR INSTRUMENT CONNECTIONS ON HYDROCARBON LINES ONLY						
470			Gate V	G03AC42	10 - 24	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE GO		VAF	FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY.						
480								VAG	FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY.						
490								*****GLOBE VALVES*****	VAH	FOR LEVEL TRANSMITTER CONNECTIONS ONLY					
500									VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.					
510				Globe	L03AC48	3/4 - 1 1/2		GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055					
520				Globe	L03AC41	2 - 6		GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW		- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.					
530				Globe	L03AC42	8 - 10		GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
540								*****CHECK VALVES*****	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
550									VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
570		VC1	Check	C03AC50	3/4 - 1 1/2	CHECK PISTON 300# RF LTCS A350 LF2 TRIM 16		VU2	DELETED						
580		VC2	Check	C03AC45	2 - 10	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
590		VC2	Check	C03AC52	12 - 24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16									
600		VC2	Check	C03JC52	26 - 48	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A									
610						*****PIPES*****									
620															
630															
640			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE									
650			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE									
660			Piping	5358505	3 - 14	PIPE SCH STD SMLS LTCS A333 GR.6 BE									
670			Piping	5358504	16 - 16	PIPE SCH 40 SMLS LTCS A333GR.6 BE									
680			Piping	5358598	18 - 24	PIPE SCH 40 EFW LTCS A671 CC60 CL.22 BE									
690			Piping	5358611	26 - 30	PIPE WT 18MM EFW LTCS A671 CC60 CL.22 BE									
700	C		Piping	5358612	32 - 32	PIPE WT 20.0MM EFW LTCS A671 CC60 CL.22 BE									
710			Piping	5358614	34 - 38	PIPE WT 22MM EFW LTCS A671 CC60 CL.22 BE									
720	C		Piping	5358617	40 - 42	PIPE WT 25MM EFW LTCS A671 CC60 CL.22 BE									
730	C		Piping	5358619	44 - 46	PIPE WT 27MM EFW LTCS A671 CC60 CL.22 BE									
740	C		Piping	5358620	48 - 48	PIPE WT 29MM EFW LTCS A671 CC60 CL.22 BE									
750															
760		810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE									
770						*****FITTINGS*****									
780															
790															
800			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6			SERVICE						
810			90 Deg	5392438	3 - 14	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6			DESIGN: ASME B16.5 MG 1.1						
820			90 Deg	5392384	16 - 16	ELL 90 DEG LR SCH 40 LTCS A420 WPL6			CODE: ASME B31.3						
830			90 Deg	5392386	18 - 24	ELL 90 DEG LR SCH 40 LTCS A420 WPL6-W			BB, BD, BF, BFH, BV, CDM, CS, DHC, DL, DSL, FM,						
840	C		90 Deg	5392530	26 - 30	ELL 90 DEG LR WT 18.0MM LTCS A420 WPL6-W			FG, FGH, GE, GHC, GRC, GRI, GSL, NH, NL, PB, PBT,						
850	C		90 Deg	5392535	32 - 32	ELL 90 DEG LR WT 20.0MM LTCS A420 WPL6-W			PE, PGC, PHC, PP, PSE, SL, SM, SVL, TL, TM, TR, VI,						
860			90 Deg	5392531	34 - 38	ELL 90 DEG LR WT 22.0MM LTCS A420 WPL6-W			VP, VU, WBL, WCS, WHS, WM, WP, WPS, WRA, WT,						
870	C		90 Deg	5392532	40 - 42	ELL 90 DEG LR WT 25.0MM LTCS A420 WPL6-W			WU						
880	C		90 Deg	5392533	44 - 46	ELL 90 DEG LR WT 27.0MM LTCS A420 WPL6-W									
890	C		90 Deg	5392534	48 - 48	ELL 90 DEG LR WT 29.0MM LTCS A420 WPL6-W									
900															
910			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6			MATERIAL: LOW TEMP CARBON STEEL						
920			45 Deg	5392444	3 - 14	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6			FLANGE RATING: 300 RF						
930			45 Deg	5392390	16 - 16	ELL 45 SCH 40 LTCS A420 WPL6			TEMPERATURE RANGE: -46 / 343 °C						
940			45 Deg	5392389	18 - 24	ELL 45 DEG SCH 40 LTCS A420 WPL6 -W									
950	C		45 Deg	5392540	26 - 30	ELL BW 45 DEG LR WT 18.0MM LTCS A420 WPL6 -W									
960	C		45 Deg	5392545	32 - 32	ELL 45 DEG LR WT 20.0MM LTCS A420 WPL6-W									
970			45 Deg	5392541	34 - 38	ELL BW 45 DEG LR WT 22.0MM LTCS A420 WPL6 -W									
980	C		45 Deg	5392542	40 - 42	ELL BW 45 DEG LR WT 25.0MM LTCS A420 WPL6 -W									
990	C		45 Deg	5392543	44 - 46	ELL BW 45 DEG LR WT 27.0MM LTCS A420 WPL6 -W									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM ALL SIZES FOR CDM		CORR. ALLOW. 1.5 MM	DATE 17NOV16	BY SMS	REV D	SHEET 2 OF 6	PA 5	CLASS 300H21	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1000	C		45 Deg	5392544	48 - 48	ELL 45 DEG LR WT 29.0MM LTCS A420 WPL6-W		SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BB, BD, BF, BFH, BV, CDM, CS, DHC, DL, DSL, FM, FG, FGH, GE, GHC, GRC, GRI, GSL, NH, NL, PB, PBT, PE, PGC, PHC, PP, PSE, SL, SM, SVL, TL, TM, TR, VI, VP, VU, WBL, WCS, WHS, WM, WP, WPS, WRA, WT, WU MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 343 °C					
1010													
1020			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6	3/4 - 2						
1030			Tee	5392447	3 - 14	TEE BW SCH STD LTCS A420 WPL6	3 - 3						
1040			Tee	5392393	16 - 16	TEE SCH 40 LTCS A420 WPL6	3 - 4						
1050			Tee	5392400	18 - 24	TEE SCH 40 LTCS A420 WPL6-W	4 - 14						
1060	C		Tee	5392408	26 - 30	TEE WT 18.0MM LTCS A420 WPL6-W	16 - 16						
1070	C		Tee	5392415	32 - 32	TEE BW WT 20.0MM LTCS A420 WPL6-W	10 - 14						
1080			Tee	5392409	34 - 38	TEE WT 22.0MM LTCS A420 WPL6-W	18 - 24						
1090	C		Tee	5392410	40 - 42	TEE WT 25.0MM LTCS A420 WPL6-W	20 - 24						
1100	C		Tee	5392411	44 - 46	TEE WT 27.0MM LTCS A420 WPL6-W	26 - 30						
1110	C		Tee	5392414	48 - 48	TEE BW WT 29.0MM LTCS A420 WPL6-W	34 - 38						
1120													
1130			Reduci	5392661	1/2 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	28 - 30						
1140			Reduci	5392454	1 1/2 - 1 1/2	TEE RED BW SCH STD/XS LTCS A420 WPL6	32 - 32						
1150			Reduci	5392454	2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	34 - 38						
1160			Reduci	5392445	3 - 6	TEE RED BW SCH STD LTCS A420 WPL6	36 - 38						
1170			Reduci	5392293	6 - 14	TEE RED SCH 40/STD LTCS A420 WPL6	40 - 42						
1180			Reduci	5392445	8 - 12	TEE RED BW SCH STD LTCS A420 WPL6	42 - 42						
1190	C		Reduci	5392401	8 - 14	TEE RED SCH 40/STD LTCS A420 WPL6-W	44 - 46						
1200	C		Reduci	5392474	18 - 20	TEE RED SCH 40 LTCS A420 WPL6-W	44 - 46						
1210	C		Reduci	5393013	16 - 24	TEE RED WT 18MM/SCH 40 LTCS A420 WPL6-W	46 - 46						
1220			Reduci	5392350	18 - 24	TEE RED WT 22.0MM/SCH 40 LTCS A420 WPL6-W	48 - 48						
1230	C		Reduci	5392201	26 - 28	TEE RED WT 18.0MM LTCS A420 WPL6-W							
1240	C		Reduci	5392203	26 - 30	TEE RED WT 20.0MM/18.0MM LTCS A420 WPL6-W							
1250	C		Reduci	5392204	32 - 32	TEE RED WT 22.0MM/ 20.0MM LTCS A420 WPL6-W							
1260			Reduci	5392323	34 - 36	TEE RED WT 22MM LTCS A420 WPL6-W							
1270	C		Reduci	5392326	34 - 38	TEE RED WT 25.0MM/22.0MM LTCS A420 WPL6-W							
1280	C		Reduci	5392324	40 - 40	TEE RED WT 25.0MM LTCS A420 WPL6-W							
1290	C		Reduci	5392207	34 - 38	TEE RED WT 27.0MM/22.0MM LTCS A420 WPL6-W							
1300	C		Reduci	5392327	40 - 42	TEE RED WT 27.0MM/25MM LTCS A420 WPL6-W							
1310	C		Reduci	5392325	44 - 44	TEE RED WT 27.0MM LTCS A420 WPL6-W							
1320	C		Reduci	5392697	40 - 42	TEE RED WT 29.0MM/25.0MM LTCS A420 WPL6-W							
1330	C		Reduci	5392205	44 - 46	TEE RED WT 29.0MM/ 27.0MM LTCS A420 WPL6-W							
1340													
1350			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6							
1360			Cap	5392448	3 - 14	CAP BW SCH STD LTCS A420 WPL6							
1370			Cap	5392394	16 - 24	CAP SCH 40 LTCS STL A420 WPL6							
1380	C		Cap	5392616	26 - 30	CAP WT 18.0MM LTCS A420 WPL6							
1390	C		Cap	5392625	32 - 32	CAP WT 20.0MM LTCS A420 WPL6							
1400			Cap	5392617	34 - 38	CAP WT 22.0MM LTCS A420 WPL6							
1410	C		Cap	5392618	40 - 42	CAP WT 25.0MM LTCS A420 WPL6							
1420	C		Cap	5392619	44 - 46	CAP WT 27.0MM LTCS A420 WPL6							
1430	C		Cap	5392624	48 - 48	CAP WT 29.0MM LTCS A420 WPL6							
1440													
1450			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2						
1460			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4						
1470			Concen	5392449	3 - 12	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 14						
1480			Concen	5392300	8 - 14	REDUCER CONC SCH 40/STD LTCS A420 WPL6	16 - 16						
1490			Concen	5392301	10 - 14	REDUCER CONC SCH 40/STD LTCS A420 WPL6-W	18 - 20						
1500			Concen	5392395	16 - 20	REDUCER CONC SCH 40 LTCS A420 WPL6-W	18 - 24						
1510	C		Concen	5392359	20 - 24	REDUCER CONC 18.0MM/SCH 40 LTCS A420 WPL6-W	26 - 30						
1520	C		Concen	5392971	24 - 24	REDUCER CONC WT 20.0MM/SCH 40 LTCS A420 WPL6-W	32 - 32						
1530	C		Concen	5392360	24 - 24	REDUCER CONC 22.0MM/SCH 40 LTCS A420 WPL6-W	34 - 36						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM ALL SIZES FOR CDM	CORR. ALLOW. 1.5 MM	DATE	BY	REV	SHEET	PA	CLASS
								17NOV16	SMS	D	3 OF 6	5	300H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1540	C		Concen	5392345	26 - 30	REDUCER CONC WT 20.0MM/18.0MM LTCS A420 WPL6-W	32 - 32	SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BB, BD, BF, BFH, BV, CDM, CS, DHC, DL, DSL, FM, FG, FGH, GE, GHC, GRC, GRI, GSL, NH, NL, PB, PBT, PE, PGC, PHC, PP, PSE, SL, SM, SVL, TL, TM, TR, VI, VP, VU, WBL, WCS, WHS, WM, WP, WPS, WRA, WT, WU MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 343 °C					
1550	C		Concen	5392355	26 - 30	REDUCER CONC 22.0MM/18.0MM LTCS A420 WPL6-W	34 - 38						
1560	C		Concen	5392356	30 - 30	REDUCER CONC 25.0MM/18.0MM LTCS A420 WPL6-W	40 - 42						
1570	C		Concen	5392357	34 - 38	REDUCER CONC 25.0MM/22.0MM LTCS A420 WPL6-W	40 - 42						
1580	C		Concen	5392347	36 - 38	REDUCER CONC WT 27.0MM/22.0MM LTCS A420 WPL6-W	44 - 46						
1590			Concen	5392358	40 - 42	REDUCER CONC WT 27.0MM/25.0MM LTCS A420 WPL6-W	44 - 46						
1600	C		Concen	5392997	40 - 42	REDUCER CONC WT 29.0MM/25.0MM LTCS A420 WPL6-W	48 - 48						
1610	C												
1620			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2						
1630			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4						
1640			Eccent	5392450	3 - 12	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 14						
1650			Eccent	5392302	8 - 14	REDUCER ECC SCH 40/STD LTCS A420 WPL6	16 - 16						
1660			Eccent	5392303	10 - 14	REDUCER ECC SCH 40/STD LTCS A420 WPL6-W	18 - 20						
1670			Eccent	5392396	16 - 20	REDUCER ECC SCH 40 LTCS A420 WPL6-W	18 - 24						
1680			Eccent	5392517	20 - 24	REDUCER ECC 18.0MM/SCH 40 LTCS A420 WPL6-W	26 - 30						
1690	C		Eccent	5392972	24 - 24	REDUCER ECC WT 20.0MM/SCH 40 LTCS A420 WPL6-W	32 - 32						
1700	C		Eccent	5392518	24 - 24	REDUCER ECC 22.0MM/SCH 40 LTCS A420 WPL6-W	34 - 38						
1710			Eccent	5392868	26 - 30	REDUCER ECC WT 20.0MM/18.0MM LTCS A420 WPL6-W	32 - 32						
1720	C		Eccent	5392525	26 - 30	REDUCER ECC 22.0MM/18.0MM LTCS A420 WPL6-W	34 - 38						
1730	C		Eccent	5392526	30 - 30	REDUCER ECC 25.0MM/18.0MM LTCS A420 WPL6-W	40 - 42						
1740	C		Eccent	5392527	34 - 38	REDUCER ECC 25.0MM/22.0MM LTCS A420 WPL6-W	40 - 42						
1750	C		Eccent	5392565	36 - 38	REDUCER ECC WT 27.0MM/22.0MM LTCS A420 WPL6-W	44 - 46						
1760	C		Eccent	5392528	40 - 42	REDUCER ECC WT 27.0MM/25.0MM LTCS A420 WPL6-W	44 - 46						
1770	C		Eccent	5392998	40 - 42	REDUCER ECC WT 29.0MM/25.0MM LTCS A420 WPL6-W	48 - 48						
1780	C												
1790				Nipofl	5504020	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2						2 - 2
1800				Nipofl	5504021	3/4 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2						3 - 14
1810				Nipofl	5504022	3/4 - 2	NIPOFLANGE 300# RF SCH 40/XS LTCS A350 LF2						16 - 24
1820				Nipofl	5504104	3/4 - 2	NIPOFLANGE 300# RF WT 18.0MM/XS LTCS A350 LF2						26 - 30
1830	C			Nipofl	5504105	3/4 - 2	NIPOFLANGE 300# RF 20.0MM/XS LTCS A350 LF2						32 - 32
1840	C			Nipofl	5504106	3/4 - 2	NIPOFLANGE 300# RF WT 22MM/XS LTCS A350 LF2						34 - 38
1850				Nipofl	5504107	3/4 - 2	NIPOFLANGE 300# RF WT 25MM/XS LTCS A350 LF2						40 - 42
1860	C			Nipofl	5504108	3/4 - 2	NIPOFLANGE 300# RF WT 27MM/XS LTCS A350 LF2						44 - 46
1870	C			Nipofl	5504109	3/4 - 2	NIPOFLANGE 300# RF WT 29MM/XS LTCS A350 LF2						48 - 48
1880	C												
1890			FF5	Endole	5504501	3/4 - 1 1/2	NIPOFLANGE FLAT BOTTOM 300# RF SCH XS LTCS A350 LF2						
1900													
1910			UJ2	Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2						4 - 14
1920			UJ2	Nipofl	5504018	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2						16 - 24
1930	C		UJ2	Nipofl	5504114	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 18.0MM/160 LTCS A350 LF2						26 - 30
1940	C	UJ2	Nipofl	5504115	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 20.0MM/160 LTCS A350 LF2	32 - 32						
1950		UJ2	Nipofl	5504116	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 22.0MM/160 LTCS A350 LF2	34 - 38						
1960	C	UJ2	Nipofl	5504117	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 25MM/160 LTCS A350 LF2	40 - 42						
1970	C	UJ2	Nipofl	5504118	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 27MM/160 LTCS A350 LF2	44 - 46						
1980	C	UJ2	Nipofl	5504119	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 29.0MM/160 LTCS A350 LF2	48 - 48						
1990													
2000			Weldol	5505335	3/4 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2						
2010			Weldol	5505179	3/4 - 1	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 14						
2020			Weldol	5505179	1 1/2 - 1 1/2	WELDOLET SCH STD/XS LTCS A350 LF2	4 - 14						
2030			Weldol	5505179	2 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	6 - 14						
2040			Weldol	5505574	3/4 - 2	WELDOLET SCH 40/XS LTCS A350 LF2	16 - 24						
2050	C		Weldol	5505579	3/4 - 2	WELDOLET WT 18MM/ SCH XS LTCS A350 LF2	26 - 30						
2060	C		Weldol	5505582	3/4 - 2	WELDOLET WT 20.0MM/XS LTCS A350 LF2	32 - 32						
2070			Weldol	5505585	3/4 - 2	WELDOLET WT 22MM/XS LTCS A350 LF2	34 - 38						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM ALL SIZES FOR CDM	CORR. ALLOW. 1.5 MM	DATE 17NOV16	BY SMS	REV D	SHEET 4 OF 6	PA 5	CLASS 300H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
2080			Weldol	5505587	3/4 - 2	WELDOLET WT 25MM/XS LTCS A350 LF2	40 - 42					
2090	C		Weldol	5505589	3/4 - 2	WELDOLET WT 27MM/XS LTCS A350 LF2	44 - 46					
2100	C		Weldol	5505597	3/4 - 2	WELDOLET WT 29.0MM/XS LTCS A350 LF2	48 - 48					
2110	C		Weldol	5505164	3 - 4	WELDOLET SCH STD LTCS A350 LF2	8 - 14					
2120			Weldol	5505554	3 - 8	WELDOLET SCH 40/STD LTCS A350 LF2	16 - 24					
2130			Weldol	5505580	3 - 12	WELDOLET 18.0MM/ STD LTCS A350 LF2	26 - 30					
2140	C		Weldol	5505583	3 - 12	WELDOLET WT 20.0MM/STD LTCS A350 LF2	32 - 32					
2150	C		Weldol	5505586	3 - 12	WELDOLET SCH 22.0MM/STD LTCS A350 LF2	34 - 38					
2160			Weldol	5505588	3 - 12	WELDOLET WT 25.0MM/STD LTCS A350 LF2	40 - 42					
2170	C		Weldol	5505590	3 - 12	WELDOLET SCH 27.0MM/STD LTCS A350 LF2	44 - 46					
2180	C		Weldol	5505598	3 - 12	WELDOLET WT 29.0MM/ STD LTCS A350 LF2	48 - 48					
2190	C											
2200			Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 14					
2210			Elbole	5516618	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40/160 LTCS A350 LF2	16 - 24					
2220			Elbole	5516684	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 18.0MM/160 LTCS A350 LF2	26 - 30					
2230	C		Elbole	5516685	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 20.0MM/160 LTCS A350 LF2	32 - 32					
2240	C		Elbole	5516686	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 22.0MM/160 LTCS A350 LF2	34 - 38					
2250			Elbole	5516687	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 25.0MM/160 LTCS A350 LF2	40 - 42					
2260	C		Elbole	5516688	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 27.0MM/160 LTCS A350 LF2	44 - 46					
2270	C		Elbole	5516689	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 29.0MM/160 LTCS A350 LF2	48 - 48					
2280	C											
2290		FS1				*****FLANGES*****						
2300												
2310			Flange	5556476	1/2 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS						
2320	C		Flange	5556473	3 - 14	FLG WN 300# RF LTCS A350 LF2 SCH STD						
2330			Flange	5556472	16 - 20	FLG WN 300# RF LTCS A350 LF2 SCH 40 BORE						
2340			Flange	5556472	24 - 24	FLG WN 300# RF LTCS A350 LF2 SCH 40 BORE						
2350			Flange	5556479	26 - 30	FLG WN 300# RF LTCS A350 LF2 WT 18.0MM B16.47 SERIES A						
2360	C		Flange	5556492	32 - 32	FLG WN 300# RF LTCS A350 LF2 WT 20.0MM B16.47 SERIES A						
2370	C		Flange	5556483	34 - 38	FLG WN 300# RF LTCS A350 LF2 WT 22.0MM B16.47 SERIES A						
2380			Flange	5556484	40 - 42	FLG WN 300# RF LTCS A350 LF2 WT 25.0MM B16.47 SERIES A						
2390	C		Flange	5556485	44 - 46	FLG WN 300# RF LTCS A350 LF2 WT 27.0MM B16.47 SERIES A						
2400	C		Flange	5556498	48 - 48	FLG WN 300# RF LTCS A350 LF2 WT 29.0MM B16.47 SERIES A						
2410	C											
2420			Blind	5615576	1/2 - 24	FLG BLIND 300# RF LTCS A350 LF2						
2430	C		Blind	5615579	26 - 48	FLG BLIND 300# RF LTCS A350 LF2 B16.47 SERIES A						
2440												
2450			Flange	5557036	2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS						
2460			Flange	5557033	3 - 4	FLG WN 600# RF LTCS A350 LF2 SCH STD						
2470	C						3/4 - 1 1/2					
2480		FA8	Reduci	5599703	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 300# RF LTCS A350 LF2						
2490												
2500			Flange	5851203	1/2 - 3	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE						
2510	C					***** ORIFICE FLANGES *****						
2520												
2530												
2540			Orific	5623947	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS						
2550			Orific	5623944	3 - 14	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD						
2560			Orific	5623943	16 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 40						
2570			Orific	5624831	26 - 28	WN ORIF SET 300# RF LTCS A350 LF2 18.0MM WT ASME B16.47 SERIES A						
2580	C		Orific	5624806	30 - 30	WN ORIF SET 300# RF LTCS A350 LF2 18.0MM WT ASME B16.47 SERIES A						
2590			Orific	5624832	32 - 32	WN ORIF SET 300# RF LTCS A350 LF2 20.0MM WT ASME B16.47 SERIES A						
2600	C		Orific	5624808	34 - 34	WN ORIF SET 300# RF LTCS A350 LF2 22.0MM WT ASME B16.47 SERIES A						
2610			Orific	5624809	36 - 36	WN ORIF SET 300# RF LTCS A350 LF2 22.0MM WT ASME B16.47 SERIES A						
2620			Orific	5624810	38 - 38	WN ORIF SET 300# RF LTCS A350 LF2 22.0MM WT ASME B16.47 SERIES A						
2630			Orific	5624811	40 - 40	WN ORIF SET 300# RF LTCS A350 LF2 25.0MM WT ASME B16.47 SERIES A						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM ALL SIZES FOR CDM	CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 17NOV16 BY SMS REV D SHEET 5 OF 6 PA 5 CLASS 300H21				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2640	C	810	Orific	5624833	44 - 44	WN ORIF SET 300# RF LTCS A350 LF2 27.0MM WT ASME B16.47 SERIES A							
2650			Orific	5624813	46 - 46	WN ORIF SET 300# RF LTCS A350 LF2 27.0MM WT ASME B16.47 SERIES A							
2660			Orific	5624834	48 - 48	WN ORIF SET 300# RF LTCS A350 LF2 29.0MM WT ASME B16.47 SERIES A							
2670													
2680	C	FS1				***** PIPE NIPPLES *****							
2690						NIPPLE PBE CUT FROM PIPE							
2700													
2710						***** GASKETS *****							
2720	D		GASFRS	5673053	1/2 - 24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR							
2730			GASFRS	5674032	26 - 48	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2740													
2750													
2760	D		GASFRS	5673102	2 - 4	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR							
2770	D		GASFRS	5673103	16 - 16	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR							
2780													
2790						***** BOLTING *****							
2800	D	SB6											
2810			BOLS	5675669	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M NUTS							
2820			CPSBOL	5675675	12 - 48	JACK SCREW HEX HEAD A320 GR L7M							
2830													
2840						**** BLINDS & SPACERS ****							
2850													
2860													
2870			Specta	5749581	3/4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
2880			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70							
2890			Paddle	5750028	26 - 48	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A							
2900			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70							
2910			Paddle	5750198	26 - 48	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A							
2920						***** BRANCHES *****							
2930													
2940													
2950						USE BRANCH CHART 300H21							
2960	C		Reinfo	5358614_R	14 - 16	REINFORCING PAD FROM PIPE A671 CC60 CL22 22MM WT	34 - 38						
2970													
2980			Reinfo	5358617_R	14 - 16	REINFORCING PAD FROM PIPE A671 CC60 CL22 25MM WT	40 - 42						
2990													
3000	C		Reinfo	5358619_R	14 - 20	REINFORCING PAD FROM PIPE A671 CC60 CL22 27MM WT	44 - 46	SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BB, BD, BF, BFH, BV, CDM, CS, DHC, DL, DSL, FM, FG, FGH, GE, GHC, GRC, GRI, GSL, NH, NL, PB, PBT, PE, PGC, PHC, PP, PSE, SL, SM, SVL, TL, TM, TR, VI, VP, VU, WBL, WCS, WHS, WM, WP, WPS, WRA, WT, WU MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 343 °C					
3010	C												
3020	C		Reinfo	5358620_R	14 - 20	REINFORCING PAD FROM PIPE A671 CC60 CL22 29MM WT	48 - 48						
3030													
3040													
3050													
3060	D		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 48						
3070	D		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 48						
3080							***** CONSTRUCTION DETAILS *****						
3090													
3100					SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY								
3110					PIPE BENDS ----- NONE								
3120	D	ZA6				THREAD SEALANT----- REQUIRED							
3130													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM ALL SIZES FOR CDM	CORR. ALLOW. 1.5 MM	DATE 17NOV16	BY SMS	REV D	SHEET 6 OF 6	PA 5	CLASS 300H21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																			
10		PP9				***** VALVES*****		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.																		
20		VB2				812		NIPPLE TO BE CUT FROM SCH XXS PIPE																			
30						B1		ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.																			
40						FA4		DELETED																			
50						FA8		USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.																			
60						FF5		FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY.																			
70		Ball V				A03AC01		3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO	PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156																
80		Ball V				A03AC02		3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.																
90		Ball V				A03AC07		6 - 8	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO	SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.																
100		Ball V				A03AC08		6 - 8	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO	UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.																
110		D				VA5		Ball V	B03AC01	1/2 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.														
120								Ball V	B03AC02	1/2 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.														
130								Ball V	B03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO	VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.														
140								Ball V	B03AC02	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO	VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY														
150								Ball V	B03AC03	4 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO																
160		D				VA5		Ball V	B03AC04	4 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO																
170								Ball V	B03AC07	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO																
180								Ball V	B03AC08	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO																
190								Butter	T03AC61	10 - 24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	*****GATE VALES*****		SERVICE													
200														Gate V	G03AC48	G03AC41	1/2 - 1 1/2	2 - 8	GATE 300# RF LTCS A350 LF2 TRIM 16 WEDGE HW	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW	*****GLOBE VALVES*****		DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 CC, CCS, DCS, SL, SM, ST, TL, TM, WD				
210		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
220								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
230														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
240		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
250								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
260														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
270		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
280								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
290														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
300		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
310								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
320														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
330		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
340								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
350														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
360		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
370								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
380														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
390		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
400								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
410														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
420		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
430								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
440														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
450		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
460								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
470														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
480		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
490								D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2														
500														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
510		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
	D		Globe	L03AC48	L03AC41		L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO														
														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
	D		Globe	L03AC48	L03AC41		L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO														
														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
	D		Globe	L03AC48	L03AC41		L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO														
														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
	D		Globe	L03AC48	L03AC41		L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO														
														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
	D		Globe	L03AC48	L03AC41		L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO														
														D	Globe	L03AC48	L03AC41	L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW				
		D				Globe																		L03AC48	L03AC41	L03AC42	3/4 - 1 1/2
	D		Globe	L03AC48	L03AC41		L03AC42	3/4 - 1 1/2	2 - 6	8 - 10	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO														
														D	Globe												

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
520			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE		VAF	FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY. FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY. FOR INSTRUMENT CONNECTIONS ON LINES WITH HIGH CAUSTIC CONCENTRATIONS ONLY. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
530			Piping	5358505	3 - 8	PIPE SCH STD SMLS LTCS A333 GR.6 BE		VAG						
540			Piping	5358514	10 - 14	PIPE SCH XS SMLS LTCS A333 GR.6 BE		VAL						
550			Piping	5358504	16 - 16	PIPE SCH 40 SMLS LTCS A333GR.6 BE								
560			Piping	5358598	18 - 24	PIPE SCH 40 EFW LTCS A671 CC60 CL.22 BE								
570														
580		812	Piping	5358511	1/2 - 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE		VB2						
590								VC1						
600						***** FITTINGS *****		VC2						
610														
620			90 Deg	5392816	3/4 - 3/4	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6			REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
630			90 Deg	5392600	1 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
640			90 Deg	5392438	3 - 8	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6								
650			90 Deg	5392600	10 - 14	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
660			90 Deg	5392384	16 - 16	ELL 90 DEG LR SCH 40 LTCS A420 WPL6								
670			90 Deg	5392386	18 - 24	ELL 90 DEG LR SCH 40 LTCS A420 WPL6-W								
680														
690			45 Deg	5392822	3/4 - 3/4	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6								
700			45 Deg	5392606	1 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
710			45 Deg	5392444	3 - 8	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6								
720			45 Deg	5392606	10 - 14	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
730			45 Deg	5392390	16 - 16	ELL 45 SCH 40 LTCS A420 WPL6								
740			45 Deg	5392389	18 - 24	ELL 45 DEG SCH 40 LTCS A420 WPL6 -W								
750														
760			90 Deg	5478803	3/4 - 1 1/2	ELL 90 DEG 6000# SW LTCS A350 LF2			1 - 1 1/2					
770	D		45 Deg	5478804	3/4 - 1 1/2	ELL 45 DEG 6000# SW LTCS A350 LF2								
780	D		Tee	5478805	3/4 - 1 1/2	TEE 6000# SW LTCS A350 LF2								
790	D		Reduci	5478811	3/4 - 1	TEE RED 6000# SW LTCS A350 LF2								
800	D	UD6	Cap	5491312	3/4 - 1	CAP 3000# SCRDR LTCS A350 LF2								
810	D		Cap	5478806	3/4 - 1 1/2	CAP 6000# SW LTCS A350 LF2								
820														
830						***** OLETS *****								
840														
850	D		Sockol	5533071	3/4 - 1 1/2	SOCKOLET 6000# LTCS A350 LF2		2 - 24						
860	D		Endole	5533233	3/4 - 1 1/2	FLAT SOCKOLET 6000# LTCS A350 LF2			SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 CC, CCS, DCS, SL, SM, ST, TL, TM, WD					
870	D	UD7	Thredo	5514772	3/4 - 1	THREDOLET 3000# LTCS A350 LF2		2 - 24						
880														
890						****COUPLINGS , UNIONS & PLUGS****								
900														
910	D		Coupli	5478807	3/4 - 1 1/2	COUPLING 6000# SW LTCS A350 LF2								
920	D		Reduci	5478812	3/4 - 1	COUPLING RED 6000# SW LTCS A350 LF2								
930	D		Union	5478809	3/4 - 1 1/2	UNION 6000# SW LTCS STL A350 LF2		1 - 1 1/2						
940	D	UD7	Plug	5491303	3/4 - 1	PLUG ROUND HEAD SCRDR LTCS A350 LF2								
950														
960			Tee	5392825	1/2 - 3/4	TEE BW SCH 160 LTCS A420 WPL6								
970			Tee	5392609	1 - 2	TEE BW SCH XS LTCS A420 WPL6								
980			Tee	5392447	3 - 8	TEE BW SCH STD LTCS A420 WPL6								
990			Tee	5392609	10 - 14	TEE BW SCH XS LTCS A420 WPL6								
1000			Tee	5392393	16 - 16	TEE SCH 40 LTCS A420 WPL6								
1010			Tee	5392400	18 - 24	TEE SCH 40 LTCS A420 WPL6-W								
1020														
1030			Reduci	5392469	3/4 - 3/4	TEE RED BW SCH XS/160 LTCS A420 WPL6		1 - 1 1/2						
1040			Reduci	5392661	1 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6		1 1/2 - 2						
1050			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6		3 - 4						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS	CORR. ALLOW. 3 MM	DATE 17MAR17						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1060	D		Reduci	5392445	3 - 6	TEE RED BW SCH STD LTCS A420 WPL6	4 - 8	SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 CC, CCS, DCS, SL, SM, ST, TL, TM, WD MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 343 °C							
1070			Reduci	5392460	4 - 8	TEE RED BW SCH XS/STD LTCS A420 WPL6	10 - 14								
1080			Reduci	5392293	6 - 8	TEE RED SCH 40/STD LTCS A420 WPL6	16 - 16								
1090			Reduci	5392299	8 - 8	TEE RED SCH 40/STD LTCS A420 WPL6-W	18 - 20								
1100			Reduci	5392661	10 - 12	TEE RED BW SCH XS LTCS A420 WPL6	12 - 14								
1110			Reduci	5392294	10 - 14	TEE RED SCH 40/XS LTCS A420 WPL6	16 - 16								
1120			Reduci	5392304	10 - 14	TEE RED BW SCH 40/XS LTCS A420 WPL6-W	18 - 24								
1130			Reduci	5392474	16 - 20	TEE RED SCH 40 LTCS A420 WPL6-W	18 - 24								
1140															
1150			Cap	5392826	3/4 - 3/4	CAP BW SCH 160 LTCS A420 WPL6									
1160			Cap	5392610	1 - 2	CAP BW SCH XS LTCS A420 WPL6									
1170			Cap	5392448	3 - 8	CAP BW SCH STD LTCS A420 WPL6									
1180			Cap	5392610	10 - 14	CAP BW SCH XS LTCS A420 WPL6									
1190			Cap	5392394	16 - 24	CAP SCH 40 LTCS STL A420 WPL6									
1200															
1210			Concen	5392462	3/4 - 3/4	REDUCER CONC SCH XS/160 LTCS A420 WPL6	1 - 2								
1220			Concen	5392458	1 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 1/2 - 2								
1230			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4								
1240			Concen	5392449	3 - 6	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 8								
1250			Concen	5392457	4 - 8	REDUCER CONC SCH XS/STD LTCS A420 WPL6	10 - 14								
1260			Concen	5392300	8 - 8	REDUCER CONC SCH 40/STD LTCS A420 WPL6	16 - 16								
1270			Concen	5392458	10 - 12	REDUCER CONC SCH XS LTCS A420 WPL6	12 - 14								
1280			Concen	5392305	10 - 14	REDUCER CONC SCH 40/XS LTCS A420 WPL6	16 - 16								
1290			Concen	5392306	10 - 14	REDUCER CONC SCH 40/XS LTCS A420 WPL6-W	18 - 20								
1300			Concen	5392395	16 - 20	REDUCER CONC SCH 40 LTCS A420 WPL6-W	18 - 24								
1310															
1320			Eccent	5392465	3/4 - 3/4	REDUCER ECC SCH XS/160 LTCS A420 WPL6	1 - 2								
1330			Eccent	5392461	1 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 1/2 - 2								
1340			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4								
1350			Eccent	5392450	3 - 6	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 8								
1360			Eccent	5392459	4 - 8	REDUCER ECC SCH XS/STD LTCS A420 WPL6	10 - 14								
1370			Eccent	5392302	8 - 8	REDUCER ECC SCH 40/STD LTCS A420 WPL6	16 - 16								
1380			Eccent	5392461	10 - 12	REDUCER ECC SCH XS LTCS A420 WPL6	12 - 14								
1390			Eccent	5392307	10 - 14	REDUCER ECC SCH 40/XS LTCS A420 WPL6	16 - 16								
1400			Eccent	5392264	10 - 14	REDUCER ECC SCH 40/XS LTCS A420 WPL6-W	18 - 20								
1410			Eccent	5392396	16 - 20	REDUCER ECC SCH 40 LTCS A420 WPL6-W	18 - 24								
1420															
1430			Nipofl	5504450	3/4 - 3/4	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	2 - 2								
1440			Nipofl	5504020	1 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	2 - 2								
1450			Nipofl	5504451	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/SCH 160 LTCS A350 LF2	3 - 8								
1460			Nipofl	5504021	1 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 8								
1470			Nipofl	5504450	3/4 - 3/4	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	10 - 14								
1480			Nipofl	5504020	1 - 2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	10 - 14								
1490			Nipofl	5504452	3/4 - 3/4	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2	16 - 24								
1500			Nipofl	5504022	1 - 2	NIPOFLANGE 300# RF SCH 40/XS LTCS A350 LF2	16 - 24								
1510	UJ2		Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 8	FGP/ WPMP CORR. ALLOW. DATE 17MAR17 BY SMS REV D SHEET 3 OF 5 PA 5 CLASS 300H25							
1520			Nipofl	5504023	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	10 - 14								
1530			Nipofl	5504018	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2	16 - 24								
1540															
1550	D	FF5	Endole	5504515	3/4 - 3/4	NIPOFLANGE FLAT BOTTOM 300# RF SCH 160 LTCS A350 LF2									
1560															
1570															
1580			Weldol	5505393	3/4 - 3/4	WELDOLET SCH XS/160 LTCS A350 LF2	2 - 2								
1590			Weldol	5505192	3/4 - 3/4	WELDOLET STD/160 LTCS A350 LF2	3 - 8								
1600	Weldol	5505393	3/4 - 3/4	WELDOLET SCH XS/160 LTCS A350 LF2	10 - 14										
1610	Weldol	5505333	1 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2										
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELD PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 3 OF 5	PA 5	CLASS 300H25		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1620			Weldol	5505591	3/4 - 3/4	WELDOLET SCH 40/ SCH 160 LTCS A350 LF2	16 - 24						
1630			Weldol	5505179	1 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 8						
1640			Weldol	5505335	1 - 2	WELDOLET SCH XS LTCS A350 LF2	10 - 14						
1650			Weldol	5505574	1 - 2	WELDOLET SCH 40/XS LTCS A350 LF2	16 - 24						
1660			Weldol	5505164	3 - 3	WELDOLET SCH STD LTCS A350 LF2	8 - 8						
1670			Weldol	5505334	3 - 4	WELDOLET SCH XS/STD LTCS A350 LF2	10 - 14						
1680			Weldol	5505554	3 - 8	WELDOLET SCH 40/STD LTCS A350 LF2	16 - 24						
1690													
1700		UJ2	Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 8						
1710		UJ2	Elbole	5516623	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	10 - 14						
1720		UJ2	Elbole	5516618	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40/160 LTCS A350 LF2	16 - 24						
1730													
1740						***** PIPE FLANGES *****							
1750													
1760		FA8	Reduci	5599703	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 1 1/2						
1770													
1780	C		Flange	5556480	1/2 - 3/4	FLG WN 300# RF LTCS A350 LF2 SCH 160							
1790			Flange	5556476	1 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS							
1800			Flange	5556473	3 - 8	FLG WN 300# RF LTCS A350 LF2 SCH STD							
1810			Flange	5556476	10 - 14	FLG WN 300# RF LTCS A350 LF2 SCH XS							
1820			Flange	5556472	16 - 24	FLG WN 300# RF LTCS A350 LF2 SCH 40 BORE							
1830													
1840	C		Flange	5557141	10 - 10	FLG WN 900# RF LTCS A350 LF2 SCH XS							
1850													
1860			Blind	5615576	1/2 - 24	FLG BLIND 300# RF LTCS A350 LF2							
1870													
1880	C		Flange	5851203	1/2 - 3	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
1890						***** ORIFICE FLANGES *****							
1900													
1910													
1920			Orific	5623947	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS							
1930			Orific	5623944	3 - 8	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD							
1940			Orific	5623947	10 - 14	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS							
1950			Orific	5623943	16 - 24	WN ORIF SET 300# RF LTCS A350 LF2 SCH 40							
1960						***** PIPE NIPPLES *****							
1970						NIPPLE PBE TO BE CUT FROM PIPE							
1980		812				***** GASKETS *****							
1990													
2000													
2010													
2020	D		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WOUND GRAPHITE IR							
2030													
2040	C		GASFRS	5673117	10 - 10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
2050						***** BOLTING *****							
2060		B1											
2070													
2080	D		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
2090													
2100		SB6	CPSBOL	5675675	12 - 24	JACK SCREW HEX HEAD A320 GR L7M							
2110						***** BLINDS & SPACERS ****							
2120													
2130													
2140			Specta	5749581	3/4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
2150			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70							
2160			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELD PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 4 OF 5	PA 5	CLASS 300H25

SEQ. NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2170 2180 2190 2200 2210 2220 2230 2240 2250 2260 2270 2280	D D D	 ZA6	Reinfo Reinfo	RWELD RWELD	2 - 16 2 - 16	***** BRANCHES ***** USE BRANCH CHART 300H25 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE	3 - 24 3 - 24							
								SERVICE						
								DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 CC, CCS, DCS, SL, SM, ST, TL, TM, WD						
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 343 °C						
						CONSTRUCTION		FGP/WPMP						
						ALL SIZES BUTT-WELD PWHT FOR WT > 19MM. ALL SIZES FOR CC, CCS, DCS		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN								3 MM	17MAR17	SMS	D	5 OF 5	5	300H25

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10						*****BALL VALVES*****		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.				
20								B1					
30								VA5					
40			Ball V	BB6B2	3/4 - 2	BALL 300# SCR D FNPT BS1400 LG2 BODY AND TRIM FB LO		VC1					
50			BALFLG	B03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO		VC2					
60						*****GATE VALVES*****							
70													
80													
90													
100			Gate V	GB6B1	1/2 - 2	GATE 300# SCR D FNPT BS1400 LG2 BODY AND TRIM HW							
110			Gate V	G03AC41	3 - 3	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW							
120						*****GLOBE VALVES*****							
130													
140													
150													
160			Globe	LB6B1	3/4 - 2	GLOBE 300# SCR D FNPT BS1400 LG2 BODY AND TRIM HW							
170			Globe	L03AC41	3 - 3	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW							
180						*****CHECK VALVES*****							
190													
200													
210		VC1	Check	CB6B1	3/4 - 2	CHECK 300# SCR D FNPT BS1400 LG2 BODY AND TRIM SPRING PISTON							
220													
230		VC2	Check	C03AC45	3 - 3	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
240													
250						***** PIPE *****							
260													
270													
280													
290			Piping	5358601	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 GALV A153 T&C							
300			Piping	5358602	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 GALV A153 T&C							
310			Piping	5358514	3 - 3	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
320						***** FITTINGS *****							
330													
340													
350													
360			90 Deg	5491290	1/2 - 2	ELL 90 DEG 3000# SCR D FNPT LTCS A350 LF2 GALV A153							
370			90 Deg	5392600	3 - 3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
380													
390			45 Deg	5491300	1/2 - 2	ELL 45 DEG 3000# SCR D FNPT LTCS A350 LF2 GALV A153							
400			45 Deg	5392606	3 - 3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
410													
420			Tee	5491310	3/4 - 2	TEE 3000# SCR D FNPT LTCS A350 LF2 GALV A153							
430			Tee	5392609	3 - 3	TEE BW SCH XS LTCS A420 WPL6							
440													
450			Reduci	5491311	1/2 - 1 1/2	TEE RED 3000# SCR D FNPT LTCS A350 LF2 GALV A153	1 - 2						
460			Reduci	5392469	1 1/2 - 1 1/2	TEE RED BW SCH XS/160 LTCS A420 WPL6	3 - 3						
470			Reduci	5392661	2 - 2	TEE RED BW SCH XS LTCS A420 WPL6	3 - 3						
480													
490			Cap	5491330	3/4 - 2	CAP 3000# SCR D FNPT LTCS A350 LF2 GALV A153							
500			Cap	5392610	3 - 3	CAP BW SCH XS LTCS A420 WPL6							
510						***** COUPLINGS, UNIONS, PLUGS & BUSHINGS *****							
520													
530													
540			Bushin	5491325	1/2 - 1 1/2	BUSHING 3000# SCR D (MNPT/FNPT) HEX HEAD LTCS A350 LF2 GALV A153	3/4 - 2						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" & SMALLER - GALV'D, THREADED & COUPLED 3" & ABOVE - BUTT WELD-GALV AFTER FABRICATION	CORR. ALLOW. 1.5 MM	DATE 11JUL13	BY MLA	REV B	FGP/WPMP SHEET 1 OF 3	PA 5	CLASS 300H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
550	B	B1	Coupli	5491250	3/4 - 2	COUPLING 3000# SCRD FNPT LTCS A350 LF2 GALV A153	1 - 2							
560			Reduci	5491260	3/4 - 1 1/2	COUPLING RED 3000# SCRD FNPT LTCS A350 LF2 GALV A153								
570			Union	5491270	3/4 - 2	UNION 3000# SCRD FNPT LTCS A350 LF2 GALV A153								
580			Union	5491271	3/4 - 2	UNION 3000# SCRD (MNPT/FNPT) LTCS A350 LF2 GALV A153								
590			Plug	5491280	3/4 - 2	PLUG ROUND HEAD SCRD MNPT LTCS A350 LF2 GALV A153								
600						**** PIPE FLANGES ****								
610														
620			Reduci	5599702	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 GALV A153	3/4 - 3/4							
630			Flange	5599701	3/4 - 2	FLG SCRD (FNPT) 300# RF LTCS A350 LF2 GALV A153								
640			Flange	5556476	3 - 3	FLG WN 300# RF LTCS A350 LF2 SCH XS								
650														
660			Blind	5615577	3/4 - 3	FLG BLIND 300# RF LTCS A350 LF2 GALV A153								
670						***** ORIFICE FLANGES *****								
680														
690			Orific	5623948	2 - 2	ORIF SET 300# RF SCRD (FNPT) LTCS A350 LF2 GALV A153								
700			Orific	5623947	3 - 3	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS								
710						***** PIPE NIPPLES *****								
720			Nipple	5541200	1/2 - 1 1/2	NIPPLE SCH 160 LTCS A333 GR.6 TBE 3" LG (MNPT) GALV A153								
730			Nipple	5541201	1/2 - 1 1/2	NIPPLE SCH 160 LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153								
740			Nipple	5541204	2 - 2	NIPPLE SCH XS LTCS A333 GR.6 TBE 7" LG (MNPT) GALV A153								
750						***** SWAGE NIPPLES *****								
760			Concen	5544150	3/4 - 1	SWAGE CONC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153		1 - 1 1/2						
770			Concen	5544151	3/4 - 1 1/2	SWAGE CONC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153		2 - 2						
780			Concen	5544142	3/4 - 1 1/2	SWAGE CONC SCH XS/160 LTCS A420 WPL6 BLE/TSE		3 - 3						
790			Concen	5544092	2 - 2	SWAGE CONC SCH XS LTCS A420 WPL6 BLE/TSE		3 - 3						
800			Eccent	5549150	3/4 - 1	SWAGE ECC SCH 160 LTCS A420 WPL6 TBE (MNPT) GALV A153		1 - 1 1/2						
810			Eccent	5549151	3/4 - 1 1/2	SWAGE ECC SCH XS/160 LTCS A420 WPL6 TBE (MNPT) GALV A153	2 - 2							
820			Eccent	5549225	3/4 - 1 1/2	SWAGE ECC SCH XS/160 LTCS A420 WPL6 BLE/TSE	3 - 3							
830			Eccent	5549182	2 - 2	SWAGE ECC SCH XS LTCS A420 WPL6 BLE/TSE	3 - 3							
840						***** GASKETS *****								
850			GASFRS	5673033	3/4 - 3	GASKET 300# 316SS SP WND GRAPHITE IR								
860						***** BOLTING *****								
870			BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
880														
890														
900														
910														
920														
930														
940														
950														
960														
970														
980														
990														
1000														
1010														
1020														
1030														
1040														
1050														
1060														
1070														
1080														
1090														
1100														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" & SMALLER - GALV'D, THREADED & COUPLED 3" & ABOVE - BUTT WELD-GALV AFTER FABRICATION	CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 11JUL13 BY MLA REV B SHEET 2 OF 3 PA 5 CLASS 300H2G						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1110 1120 1130 1140 1150 1160 1170 1180 1190 1200 1210 1220 1230			Specta	5749583	3/4 - 3	**** BLINDS & SPACERS **** SPECTACLE BLIND 300# RF LTCS A516 GR.70 GALV A153 ***** BRANCHES ***** USE BRANCH CHART 300H2G ***** CONSTRUCTION DETAILS ***** SEAL WELDING-----FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE								
								SERVICE						
								DESIGN: 32 BARG CODE: ASME B31.3 VI, WF (ABOVE GROUND), WRA (ABOVE GROUND)						
								MATERIAL: LOW TEMP CARBON STEEL - GALV'D FLANGE RATING: 300 RF TEMPERATURE RANGE: -29 / 150 °C COMMENT: LTCS						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" & SMALLER - GALV'D, THREADED & COUPLED 3" & ABOVE - BUTT WELD-GALV AFTER FABRICATION		CORR. ALLOW. 1.5 MM	DATE 11JUL13	BY MLA	REV B	FGP/WPMP SHEET 3 OF 3	PA 5	CLASS 300H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10						***** MODULAR VALVES *****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
20								SB6							
30	C		2BV	K03AC02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		VA5							
40								VAV							
50	C	VAV	2BV	K03AC02	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		VAW							
60								VB2							
70	C		2BVR	M03AC02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VC1							
80							2 - 2								
90	C	VAW	R2BVR	M03EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR									
100							3 - 3	VC2							
110	C	VAW	R2BVR	M03EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		ZA6							
120						***** BALL VALVES *****			SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 47 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - DSL - DIESEL MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS						
130		VB2													
140															
150	C		Ball V	A03AC01	3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO									
160															
170	C		Ball V	A03AC02	3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO									
180															
190	C		Ball V	A03AC07	6 - 8	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO									
200															
210	C		Ball V	A03AC08	6 - 8	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO									
220															
230															
240	C		Ball V	B03AC01	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
250															
260	C		Ball V	B03AC02	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO									
270															
280	C	VA5	Ball V	B03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
290															
300	C	VA5	Ball V	B03AC02	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO									
310															
320	D	VA5	Ball V	B03AC03	4 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
330															
340	D	VA5	Ball V	B03AC04	4 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO									
350															
360	C	VA5	Ball V	B03AC07	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
370															
380	C	VA5	Ball V	B03AC08	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO									
390															
400	C	VA5	Ball V	B03JC07	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A									
410	C	VA5	Ball V	B03JC08	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A									
420						***** ORBIT VALVES *****									
430															
440															
450	D	VA5	Ball V	R03AC06	2 - 4	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB HO									
460															
470	D	VA5	Ball V	R03AC08	6 - 10	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB GO									
480															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	ALL SIZES: BUTTWELD	CORR. ALLOW.							
								3 MM	17MAR17	JCS	C	1 OF 6	5	300H5F	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
490						*****BUTTERFLY VALVES*****							
500													
510													
520	C		Butter	T03AC61	10 - 24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORPAT GO							
530	C		Butter	T03JC61	28 - 28	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A							
540													
550													
560						***** GATE VALVES *****							
570													
580	C		Gate V	G03AC48	3/4 - 1 1/2	GATE 300# RF LTCS A350 LF2 TRIM 16 WEDGE HW							
590	C		Gate V	G03AC41	2 - 2	GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW							
600	D		Condui	W03AC70	3 - 8	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
610	D		Condui	W03AC71	10 - 20	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
620													
630													
640						*****GLOBE VALVES*****							
650													
660	C		Globe	L03AC48	3/4 - 1 1/2	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
670	C		Globe	L03AC41	2 - 6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW							
680	C		Globe	L03AC42	8 - 10	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO							
690													
700						*****CHECK VALVES*****							
710													
720	C	VC1	Check	C03AC50	3/4 - 1 1/2	CHECK PISTON 300# RF LTCS A350 LF2 TRIM 16							
730	C	VC2	Check	C03AC45	2 - 10	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
740	C	VC2	Check	C03AC52	12 - 24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16							
750	C	VC2	Check	C03JC52	28 - 28	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A							
760													
770													
780						***** PIPE ABOVE GROUND *****							
790													
800	C		Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE							
810			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
820			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE							
830			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE							
840			Piping	9001254	6 - 10	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE							
850			Piping	9001499	12 - 12	LINEPIPE SMLS WT = 7.09 MM API-5L X60 BE							
860			Piping	9001311	14 - 14	LINEPIPE SMLS WT=7.92MM API-5L X60 BE							
870			Piping	9001485	16 - 18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE							
880			Piping	9001493	20 - 20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE							
890													
900						***** PIPE BELOW GROUND *****							
910													
920			Piping	9001257CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED							
930			Piping	9001254CTD	6 - 10	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED							
940			Piping	9001499CTD	12 - 12	LINEPIPE SMLS WT= 7.09 MM API-5L X60 BE COATED							
950			Piping	9001311CTD	14 - 14	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED							
960			Piping	9001485CTD	16 - 18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED							
970			Piping	9001493CTD	20 - 20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 6	PA 5	CLASS 300H5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
980						***** FITTINGS ABOVE & BELOW GROUND*****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 47 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - DSL - DIESEL MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS					
990													
1000			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
1010			90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6							
1020			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1030			45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6							
1040													
1050			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6							
1060			Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6							
1070													
1080			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2						
1090			Reduci	5392454	3/4 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3						
1100			Reduci	9006200	4 - 4	TEE RED BW WT 7.09 MM / 6.02MM LTCS A860 WPHY 60	12 - 12						
1110			Reduci	9004123	6 - 6	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	16 - 16						
1120													
1130			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6							
1140			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6							
1150			Cap	9005055	6 - 10	CAP BW WT 6.35MM LTCS A860 WPHY-60							
1160			Cap	9005063	12 - 12	CAP BW WT 7.09 MM LTCS A860 WPHY-60							
1170			Cap	9005058	14 - 14	CAP BW WT 7.92 MM LTCS A860 WPHY-60							
1180			Cap	9005059	16 - 18	CAP BW WT 8.74 MM LTCS A860 WPHY-60							
1190			Cap	9005060	20 - 20	CAP BW WT 9.53 MM LTCS A860 WPHY-60							
1200													
1210			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2						
1220			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2						
1230			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3						
1240			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3						
1250			Eccent	9002281	3 - 3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4 - 4						
1260			Eccent	9006103	3 - 3	REDUCER ECC WT 6.35 MM / STD LTCS A860 WPHY-60	6 - 8						
1270			Eccent	9002312	4 - 4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6 - 10						
1280			Eccent	9002266	6 - 8	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8 - 10						
1290			Eccent	9006000	4 - 4	REDUCER ECC WT 7.09MM / 6.02 MM LTCS A860 WPHY-60	12 - 12						
1300			Eccent	9002416	6 - 10	REDUCER ECC WT 7.09 MM / 6.35 MM A860 WPHY 60	12 - 12						
1310			Eccent	9002272	6 - 10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	14 - 14						
1320			Eccent	9002292	8 - 10	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	16 - 16						
1330			Eccent	9002250	14 - 14	REDUCER ECC WT 8.74 MM / 7.92 MM LTCS A860 WPHY-60	16 - 18						
1340			Eccent	9004135	14 - 14	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	20 - 20						
1350			Eccent	9004136	16 - 18	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	20 - 20						
1360													
1370													
1380													
1390													
1400			90 Deg	9001810	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1410			90 Deg	9001813	6 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1420			90 Deg	9005430	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1430			90 Deg	9001815	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1440			90 Deg	9001816	16 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1450			90 Deg	9001817	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1460													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 3 OF 6	PA 5	CLASS 300H5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1470			45 Deg	9001820	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1480			45 Deg	9001823	6 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1490			45 Deg	9005431	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1500			45 Deg	9001825	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1510			45 Deg	9001826	16 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1520			45 Deg	9001827	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1530			22.5 D	9001830	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1540			22.5 D	9001833	6 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1550			22.5 D	9005432	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1560			22.5 D	9001835	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1570			22.5 D	9001835	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1580			22.5 D	9001836	16 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1590			22.5 D	9001837	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1600						*****INDUCTION BENDS BELOW GROUND*****							
1610													
1620													
1630			90 Deg	9001810CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1640			90 Deg	9001813CTD	6 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1650			90 Deg	9005430CTD	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1660													
1670			90 Deg	9001815CTD	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1680			90 Deg	9001816CTD	16 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1690			90 Deg	9001817CTD	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1700			45 Deg	9001820CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1710			45 Deg	9001823CTD	6 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1720			45 Deg	9001823CTD	6 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1730			45 Deg	9005431CTD	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1740			45 Deg	9001825CTD	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1750			45 Deg	9001826CTD	16 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1760			45 Deg	9001827CTD	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 4 OF 6 PA 5 CLASS 300HSF					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
1770	C		22.5 D	9001830CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
1780			22.5 D	9001833CTD	6 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
1790			22.5 D	9005432CTD	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
1800			22.5 D	9001835CTD	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
1810			22.5 D	9001836CTD	16 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
1820			22.5 D	9001837CTD	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
1830												
1840												
1850												
1860			Weldol	9006493	3/4 - 2	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60	4 - 4					
1870			Weldol	9006494	3 - 3	WELDOLET WT 6.02MM/SCH STD LTCS A694 F60	4 - 4					
1880			Weldol	9006407	3/4 - 2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6 - 10					
1890			Weldol	9006409	3 - 3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	6 - 10					
1900			Weldol	9006490	3/4 - 2	WELDOLET WT 7.09 MM/XS WT LTCS A694 F60	12 - 12					
1910			Weldol	9006491	3 - 3	WELDOLET WT 7.09MM/STD WT LTCS A694 F60	12 - 12					
1920			Weldol	9002552	3/4 - 2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	14 - 14					
1930			Weldol	9006415	3 - 3	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	14 - 14					
1940			Weldol	9006414	4 - 4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	14 - 14					
1950			Weldol	9006423	3/4 - 2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	16 - 18					
1960			Weldol	9006422	3 - 3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	16 - 18					
1970			Weldol	9006421	4 - 4	WELDOLET WT 8.74 MM/6.02 MM LTCS A694 F60	16 - 18					
1980			Weldol	9002008	6 - 10	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	16 - 18					
1990												
2000			Flange	5504021	3/4 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 3					
2010			Flange	9002639	3/4 - 2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4					
2020			Flange	9002643	3/4 - 2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6 - 10					
2030			Flange	9006492	3/4 - 2	NIPOFLANGE 300# RF 7.09 MM/XS WT LTCS A694 F60	12 - 12					
2040			Flange	9002645	3/4 - 2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	14 - 14					
2050			Flange	9002646	3/4 - 2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	16 - 18					
2060			Flange	9002647	3/4 - 2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	20 - 20					
2070												
2080												
2090												
2100			Flange	5556476	3/4 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS		SERVICE				
2110			Flange	5556473	3 - 3	FLG WN 300# RF LTCS A350 LF2 SCH STD		SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 47 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - DSL - DIESEL				
2120			Flange	9001480	4 - 4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM						
2130			Flange	9001483	6 - 10	FLG WN 300# RF LTCS A694 F60 WT 6.35MM						
2140			Flange	9006225	12 - 12	FLG WN 300# RF LTCS A694 F60 WT 7.09 MM						
2150			Flange	9001455	14 - 14	FLG WN 300# RF LTCS A694 F60 WT 7.92MM						
2160			Flange	9001457	16 - 18	FLG WN 300# RF LTCS A694 F60 WT 8.74MM						
2170			Flange	9001458	20 - 20	FLG WN 300# RF LTCS A694 F60 WT 9.53MM						
2180			Blind	5615576	3/4 - 3	FLG BLIND 300# RF LTCS A350 LF2						
2190			Blind	9001450	4 - 20	FLG BLIND 300# RF LTCS A694 F60						
2200												
2210												
2220						***** GASKETS *****						
2230												
2240	C		GASFRS	5673033	3/4 - 20	GASKET 300# 316SS SP WND GRAPHITE IR						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 5 OF 6 PA 5 CLASS 300H5F				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
2260	C	B1	BOLS	5675770	1/4 - 99	***** BOLTING *****						
2270						STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS						
2280												
2290												
2300		SB6				JACK SCREW HEX HEAD A320 GR L7M						
2310												
2320												
2330						***** BLINDS & SPACERS						
2340												
2350						Specta						
2360		Paddle	5750100	12 - 20	PADDLE BLANK 300# RF LTCS A516 GR.70							
2370		Paddle	5750200	12 - 20	PADDLE SPACER 300# RF LTCS A516 GR.70							
2380												
2390												
2400												
2410						***** BRANCHES *****						
2420												
2430						USE BRANCH CHART 300H5F						
2440												
2450	C		Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 20					
2460	C		Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 20					
2470												
2480												
2490						***** CONSTRUCTION DETAILS *****						
2500												
2510		ZA6				SEAL WELDING ----- REQUIRED						
2520						PIPE BENDS ----- INDUCTION BENDS						
2530						THREAD SEALANT ----- PTFE PASTE						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 20 30	B		2BV	K03AC02	1/2 - 2	***** MODULAR VALVES ***** MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		810 B1 NT5 SB6	NIPPLE TO BE CUT FROM SCH 160 PIPE ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED IN NON-PIGGED LINES ONLY USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
40 50	B	VAV	2BV	K03AC02	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
60 70	B		2BVR	M03AC02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VAV VAW	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.					
80 90	B	VAW	R2BVR	M03EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
100 110	B	VAW	R2BVR	M03EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING					
120 140 150 160 170 180 190	B	VB2	Ball V	A03AC01	3 - 4	***** BALL VALVES ***** BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO								
200 210 220 230 240	B	VB2	Ball V	A03AC02	3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO								
250 260 270 280	B	VB2	Ball V	A03AC07	6 - 8	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
290 300 310 320 330 340 350 360	B	VB2	Ball V	A03AC08	6 - 8	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
370 380 390 400 410	B	VB2 VA5	Ball V	B03AC01	1/2 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
420 430 440 450 460 470 480 490 500	B	VB2 VA5 VA5	Ball V	B03AC02	1/2 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
	B	VB2 VA5	Ball V	B03AC03	6 - 8	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
	B	VB2 VA5	Ball V	B03AC04	6 - 8	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
	B	VB2 VA5	Ball V	B03AC07	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
	B	VB2 VA5	Ball V	B03AC08	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
	B	VB2 VA5 VA5	Ball V	B03JC07	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A								
	B	VB2 VA5 VA5	Ball V	B03JC08	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A								
	B	VA5	Ball V	R03AC06	2 - 4	*****ORBIT VALVES***** BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB HO								
	B					*****BUTTERFLY VALVES*****								
	B		Butter	T03AC61	24 - 24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD		CORR. ALLOW. 1.5 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 JCS B 1 OF 6 5 300H5P					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
510	B		Butter	T03JC61	28 - 28	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A		SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - FGH - FUEL GAS MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C COMMENT: LTCS					
520						***** GATE VALVES *****							
530													
540			Condui	W03AC70	3 - 6	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
550													
560			Condui	W03AC71	24 - 24	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
570													
580			Condui	W03JC71	28 - 28	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A							
590						***** GLOBE VALVES *****							
600													
610													
620													
630	B		Globe	L03AC48	3/4 - 1 1/2	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
640	B		Globe	L03AC41	2 - 6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW							
650													
660						***** CHECK VALVES *****							
670													
680													
690	B		Check	C03AC50	3/4 - 1 1/2	CHECK PISTON 300# RF LTCS A350 LF2 TRIM 16							
700	B		Check	C03AC45	2 - 6	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
710													
720	B		Check	C03AC52	24 - 24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16							
730	B		Check	C03JC52	28 - 28	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A							
740						***** PIPE ABOVE GROUND *****							
750													
760													
770			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE							
780			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
790			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE							
800			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE							
810	B		Piping	9001254	6 - 16	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE							
820	B		Piping	9001256	18 - 20	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE							
830			Piping	9001311	24 - 24	LINEPIPE SMLS WT=7.92MM API-5L X60 BE							
840			Piping	9001493	28 - 28	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE							
850													
860	B	810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE							
870						***** PIPE BELOW GROUND *****							
880													
890													
900			Piping	9001257CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED							
910	B		Piping	9001254CTD	6 - 16	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED							
920	B		Piping	9001256CTD	18 - 20	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE COATED							
930			Piping	9001311CTD	24 - 24	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED							
940			Piping	9001493CTD	28 - 28	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED							
950						***** FITTINGS ABOVE & BELOW GROUND*****							
960													
970													
980			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
990			90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6							
1000	B		90 Deg	9006566	4 - 4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60							
1010	B		90 Deg	9002190	6 - 6	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60							
1020													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 6	PA 5	CLASS 300H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1030			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1040			45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6							
1050	B	NT5	45 Deg	9006573	4 - 4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60							
1060	B	NT5	45 Deg	9006571	6 - 6	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60							
1070													
1080			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6							
1090			Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6							
1100	B	NT5	Tee	9301610	4 - 4	TEE BW WT 6.02 MM LTCS A860 WPHY-60							
1110	B	NT5	Tee	9301609	6 - 6	TEE BW WT 6.35 MM LTCS A860 WPHY-60							
1120													
1130			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2						
1140			Reduci	5392454	3/4 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3						
1150	B	NT5	Reduci	9006664	2 - 2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60	4 - 4						
1160	B	NT5	Reduci	9006659	3 - 3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60	4 - 4						
1170	B	NT5	Reduci	9006658	3 - 3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60	6 - 6						
1180	B	NT5	Reduci	9006605	4 - 4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60	6 - 6						
1190													
1200			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6							
1210			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6							
1220	B		Cap	9005062	4 - 4	CAP BW WT 6.02 MM LTCS A860 WPHY-60							
1230	B		Cap	9005055	6 - 16	CAP BW WT 6.35MM LTCS A860 WPHY-60							
1240	B		Cap	9005057	18 - 20	CAP BW WT 7.14 MM LTCS A860 WPHY-60							
1250			Cap	9005058	24 - 24	CAP BW WT 7.92 MM LTCS A860 WPHY-60							
1260			Cap	9005060	28 - 28	CAP BW WT 9.53 MM LTCS A860 WPHY-60							
1270													
1280			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2						
1290			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3						
1300	B		Concen	9006918	1 1/2 - 3	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1310	B		Concen	9006915	3 - 3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6 - 6						
1320													
1330			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2						
1340			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3						
1350	B		Eccent	9001372	1 1/2 - 3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1360	B		Eccent	9002510	3 - 3	REDUCER ECC WT 6.35 MM/STD LTCS A420 WPL6	6 - 6						
1370	B		Eccent	9002312	4 - 4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6 - 10						
1380	B		Eccent	9002266	6 - 14	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8 - 16						
1390	B		Eccent	9002269	10 - 16	REDUCER ECC WT 7.14 MM / 6.35 MM LTCS A860 WPHY-60	18 - 18						
1400	B		Eccent	9006501	12 - 16	REDUCER ECC WT 7.14 MM / 6.35MM LTCS A860 WPHY-60-W	20 - 20						
1410	B		Eccent	9006502	18 - 18	REDUCER ECC WT 7.14 MM LTCS A860 WPHY-60-W	20 - 20						
1420	B		Eccent	9004138	16 - 16	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60-W	24 - 24						
1430	B		Eccent	9004139	18 - 20	REDUCER ECC WT 7.92 MM / 7.14 MM LTCS A860 WPHY-60-W	24 - 24						
1440	B		Eccent	9004134	18 - 20	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60-W	28 - 28						
1450			Eccent	9004135	24 - 24	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	28 - 28						
1460						*****INDUCTION BENDS ABOVE GROUND*****							
1470													
1480													
1490			90 Deg	9001810	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1500	B		90 Deg	9001813	6 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1510	B		90 Deg	9001814	18 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1520			90 Deg	9001815	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 1.5 MM	FGP/WPMP					
								DATE	BY	REV	SHEET	PA	CLASS
								17MAR17	JCS	B	3 OF 6	5	300H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES										
1530	B		90 Deg	9001817	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1540			45 Deg	9001820	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1550																		
1560			45 Deg	9001823	6 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1570			45 Deg	9001824	18 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1580			45 Deg	9001825	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1590			45 Deg	9001827	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1600			22.5 D	9001830	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1610																		
1620	B		22.5 D	9001833	6 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1630	B		22.5 D	9001834	18 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1640	B		22.5 D	9001835	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1650			22.5 D	9001837	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
1660			90 Deg	9001810CTD	4 - 4	*****INDUCTION BENDS BELOW GROUND***** INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED												
1670																		
1680																		
1690																		
1700																		
1710																		
1720																		
1730																		
1740																		
1750			45 Deg	9001820CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED												
1760	B		45 Deg	9001823CTD	6 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED												
1770	B		45 Deg	9001824CTD	18 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED												
1780	45 Deg	9001825CTD	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED														
1790	45 Deg	9001827CTD	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED														
1800	22.5 D	9001830CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED														
1810																		
1820	B		22.5 D	9001833CTD	6 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED												
1830	B		22.5 D	9001834CTD	18 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED												
 FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION		FGP/WPMP										
						ALL SIZES: BUTTWELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS					
							1.5 MM	17MAR17	JCS	B	4 OF 6	5	300H5P					

SERVICE				
DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - FGH - FUEL GAS				

MATERIAL:	LOW TEMP CARBON STEEL
FLANGE RATING:	300 RF
TEMPERATURE RANGE:	-40 / 80 °C
COMMENT:	LTCS

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1840			22.5 D	9001835CTD	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L								
1850			22.5 D	9001837CTD	28 - 28	X60 C/W 1.5D TANGENT ENDS BW COATED INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1860						***** OLETS ABOVE & BELOW GROUND*****								
1870														
1880														
1890	B		Weldol	9006219	3/4 - 1 1/2	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60 IMPACT TESTED AT -46 DEG C	4 - 4							
1900	B		Weldol	9006407	3/4 - 2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6 - 6							
1910			Weldol	9006408	4 - 4	WELDOLET WT 6.35 MM/6.02 MM LTCS A694 F60	12 - 16							
1920	B		Weldol	9006411	4 - 4	WELDOLET WT 7.14 MM/6.02 MM LTCS A694 F60	18 - 20							
1930	B		Weldol	9006701	6 - 6	WELDOLET WT 7.14 MM/6.35 MM LTCS A694 F60	18 - 20							
1940	B		Weldol	9006414	4 - 4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	24 - 24							
1950			Weldol	9004124	6 - 6	WELDOLET WT 7.92 MM/6.35 MM LTCS A694 F60	24 - 24							
1960			Weldol	9006427	4 - 4	WELDOLET WT 9.53 MM/6.02 MM LTCS A694 F60	28 - 28							
1970			Weldol	9006424	6 - 10	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	28 - 28							
1980														
1990														
2000	B		Nipofl	5504020	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	2 - 2							
2010			Nipofl	5504021	3/4 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 3							
2020			Nipofl	9002639	3/4 - 2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4							
2030			Nipofl	9002643	3/4 - 2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6 - 16							
2040	B		Nipofl	9002644	3/4 - 2	NIPOFLANGE 300# RF 7.14 MM/XS WT LTCS A694 F60	18 - 20							
2050			Nipofl	9002645	3/4 - 2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	24 - 24							
2060			Nipofl	9002647	3/4 - 2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	28 - 28							
2070						***** PIPE FLANGES ABOVE & BELOW GROUND*****								
2080														
2090														
2100	B		Flange	5556476	1/2 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS								
2110			Flange	5556473	3 - 3	FLG WN 300# RF LTCS A350 LF2 SCH STD								
2120			Flange	9001480	4 - 4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM								
2130	B		Flange	9001483	6 - 16	FLG WN 300# RF LTCS A694 F60 WT 6.35MM								
2140	B		Flange	9001484	18 - 20	FLG WN 300# RF LTCS A694 F60 WT 7.14MM								
2150			Flange	9001455	24 - 24	FLG WN 300# RF LTCS A694 F60 WT 7.92MM								
2160			Flange	9001461	28 - 28	FLG WN 300# RF LTCS A694 F60 WT 9.53 MM B16.47 SERIES A								
2170	B		Blind	5615576	1/2 - 3	FLG BLIND 300# RF LTCS A350 LF2								
2180			Blind	9001450	4 - 24	FLG BLIND 300# RF LTCS A694 F60								
2190			Blind	9001451	28 - 28	FLG BLIND 300# RF LTCS A694 F60 SERIES A								
2200														
2210			Reduci	5599703	1/2 - 1/2	FLG SCRd (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 1 1/2							
2220														
2230	B		Reduci	9006752	3 - 3	FLG WN REDUCING 300# RF LTCS A694 F60 BORED TO SUIT	4 - 4							
2240														
2250	B		Flange	5557036	2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS								
2260	B		Flange	5557033	3 - 3	FLG WN 600# RF LTCS A350 LF2 SCH STD								
2270	B		Flange	9001723	4 - 4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM								
2280	B		Flange	9001471	6 - 6	FLG WN 600# RF LTCS A694 F60 WT 6.35MM								
2290	B		Flange	9005389	4 - 4	FLG WN 900# RF LTCS A694 F60 WT 6.02MM								
2300														
2310	B		Flange	5851203	1/2 - 1/2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						ALL SIZES: BUTTWELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								1.5 MM	17MAR17	JCS	B	5 OF 6	5	300H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2320 2330 2340 2350	B	B1	GASFRS	5673033	1/2 - 24	***** GASKETS ***** GASKET 300# 316SS SP WND GRAPHITE IR								
			GASFRS	5674032	28 - 28	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A								
2360 2370 2380	B B		GASFRS	5673103	2 - 6	GASKET 600# 316SS SP WND GRAPHITE IR								
			GASFRS	5673117	4 - 4	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR								
2390 2400 2410 2420	B		BOLS	5675770	1/4 - 99	***** BOLTING ***** STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
2430 2440 2450 2460	SB6		CPSBOL	5675675	12 - 28	JACK SCREW HEX HEAD A320 GR L7M								
						***** BLINDS & SPACERS								
2470 2480 2490			Specta Paddle Paddle	5749581 5750100 5750028	3/4 - 6 24 - 24 28 - 28	SPECTACLE BLIND 300# RF LTCS A516 GR.70 PADDLE BLANK 300# RF LTCS A516 GR.70 PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A								
2500 2510 2520			Paddle Paddle	5750200 5750198	24 - 24 28 - 28	PADDLE SPACER 300# RF LTCS A516 GR.70 PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A								
2530 2540 2550 2560						***** BRANCHES ***** USE BRANCH CHART 300H5P								
2570 2580 2590 2600			B B	Reinfo Reinfo	RWELD RWELD	2 - 4 2 - 4							NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN	3 - 6 3 - 6
2610 2620 2630 2640 2650 2660 2670 2680 2690			B B	Reinfo Reinfo	RWELD RWELD	16 - 20 16 - 20							NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN	24 - 28 24 - 28
		ZA6				***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- REQUIRED PIPE BENDS ----- INDUCTION BENDS THREAD SEALANT ----- PTFE PASTE								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY JCS	REV B	FGP/WPMP SHEET 6 OF 6	PA 5	CLASS 300H5P	

[illegible]



SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
990						***** FITTINGS *****		VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS. DELETED REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
1000								VU2						
1010			90 Deg	5392816	3/4 - 3/4	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6		ZA6						
1020			90 Deg	5392600	1 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
1030			90 Deg	5392438	3 - 8	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6								
1040			90 Deg	5392600	10 - 14	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
1050			90 Deg	5392384	16 - 16	ELL 90 DEG LR SCH 40 LTCS A420 WPL6								
1060			90 Deg	5392386	18 - 24	ELL 90 DEG LR SCH 40 LTCS A420 WPL6-W								
1070			90 Deg	5392535	26 - 30	ELL 90 DEG LR WT 20.0MM LTCS A420 WPL6-W								
1080			90 Deg	5392532	32 - 40	ELL 90 DEG LR WT 25.0MM LTCS A420 WPL6-W								
1090			90 Deg	5392534	42 - 48	ELL 90 DEG LR WT 29.0MM LTCS A420 WPL6-W								
1100	C		90 Deg	5392566	56 - 56	ELL 90 DEG LR WT 34.0MM LTCS A420 WPL6-W								
1110	C		90 Deg	5392587	60 - 60	ELL 90 DEG LR WT 36.0MM LTCS A420 WPL6-W								
1120														
1130	C		90 Deg	5556911	6 - 6	INDUCTION BEND 90 DEG. X 5D RADIUS SCH STD ASTM A671								
						CC60 CL.22 - C/W 1.5D TANGENT ENDS BE								
1140	C		90 Deg	5800058	60 - 60	INDUCTION BEND 90 DEG. X 5D RADIUS WT 36.0MM ASTM A671 CC60 CL.22 - C/W 1.5D TANGENT ENDS BE								
1150														
1160			45 Deg	5392822	3/4 - 3/4	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6								
1170			45 Deg	5392606	1 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
1180			45 Deg	5392444	3 - 8	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6								
1190			45 Deg	5392606	10 - 14	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
1200			45 Deg	5392390	16 - 16	ELL 45 SCH 40 LTCS A420 WPL6								
1210			45 Deg	5392389	18 - 24	ELL 45 DEG SCH 40 LTCS A420 WPL6 -W								
1220			45 Deg	5392545	26 - 30	ELL 45 DEG LR WT 20.0MM LTCS A420 WPL6-W								
1230			45 Deg	5392542	32 - 40	ELL BW 45 DEG LR WT 25.0MM LTCS A420 WPL6 -W								
1240			45 Deg	5392544	42 - 48	ELL 45 DEG LR WT 29.0MM LTCS A420 WPL6-W								
1250	C		45 Deg	5392569	56 - 56	ELL 45 DEG LR WT 34.0MM LTCS A420 WPL6-W								
1260														
1270	C		45 Deg	5800087	60 - 60	INDUCTION BEND 45 DEG. X 5D RADIUS WT 36.0MM ASTM A671 CC60 CL.22 - C/W 1.5D TANGENT ENDS BE								
1280														
1290			Tee	5392825	3/4 - 3/4	TEE BW SCH 160 LTCS A420 WPL6								
1300			Tee	5392609	1 - 2	TEE BW SCH XS LTCS A420 WPL6								
1310			Tee	5392447	3 - 8	TEE BW SCH STD LTCS A420 WPL6								
1320			Tee	5392609	10 - 14	TEE BW SCH XS LTCS A420 WPL6								
1330			Tee	5392393	16 - 16	TEE SCH 40 LTCS A420 WPL6								
1340			Tee	5392400	18 - 24	TEE SCH 40 LTCS A420 WPL6-W								
1350			Tee	5392415	26 - 30	TEE BW WT 20.0MM LTCS A420 WPL6-W								
1360			Tee	5392410	32 - 40	TEE WT 25.0MM LTCS A420 WPL6-W								
1370			Tee	5392414	42 - 48	TEE BW WT 29.0MM LTCS A420 WPL6-W								
1380	C		Tee	5392553	56 - 56	TEE BW WT 34.0MM LTCS A420 WPL6-W								
1390							1 - 1 1/2							
1400			Reduci	5392469	3/4 - 3/4	TEE RED BW SCH XS/160 LTCS A420 WPL6								
1410			Reduci	5392661	1 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 1/2 - 2							
1420			Reduci	5392454	1 1/2 - 1 1/2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3							
1430			Reduci	5392454	2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 4							
1440			Reduci	5392445	3 - 6	TEE RED BW SCH STD LTCS A420 WPL6	4 - 8							
1450			Reduci	5392460	4 - 8	TEE RED BW SCH XS/STD LTCS A420 WPL6	10 - 14							
1460			Reduci	5392293	6 - 8	TEE RED SCH 40/STD LTCS A420 WPL6	16 - 16							
1470			Reduci	5392299	8 - 8	TEE RED SCH 40/STD LTCS A420 WPL6-W	18 - 20							
1480			Reduci	5392661	10 - 12	TEE RED BW SCH XS LTCS A420 WPL6	12 - 14							
1490			Reduci	5392294	10 - 14	TEE RED SCH 40/XS LTCS A420 WPL6	16 - 16							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD PWHT FOR ALL SIZES. FOR SERVICE CODE CDF, PWHT FOR WT > 19MM		CORR. ALLOW. 3 MM						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
1500			Reduci	5392304	10 - 14	TEE RED BW SCH 40/XS LTCS A420 WPL6-W	18 - 24					
1510			Reduci	5392474	16 - 20	TEE RED SCH 40 LTCS A420 WPL6-W	18 - 24					
1520			Reduci	5392672	16 - 24	TEE RED WT 20.0MM/SCH 40 LTCS A420 WPL6-W	26 - 30					
1530	D		Reduci	5392671	14 - 14	TEE RED WT 20.0MM/XS LTCS A420 WPL6-W	30 - 30					
1540			Reduci	5392670	26 - 28	TEE RED WT 20.0MM LTCS A420 WPL6-W	28 - 30					
1550	D		Reduci	5393017	16 - 24	TEE RED WT 25.0MM/ SCH 40 LTCS A420 WPL6-W	36 - 36					
1560			Reduci	5392687	26 - 30	TEE RED WT 25.0MM/20.0MM LTCS A420 WPL6-W	32 - 40					
1570			Reduci	5392324	32 - 38	TEE RED WT 25.0MM LTCS A420 WPL6-W	34 - 40					
1580	C		Reduci	5392673	18 - 24	TEE RED WT 29.0MM/ SCH 40 LTCS A420 WPL6-W	42 - 48					
1590	C		Reduci	5393021	26 - 30	TEE RED WT 29.0MM/20.0MM LTCS A420 WPL6-W	42 - 48					
1600			Reduci	5392697	32 - 40	TEE RED WT 29.0MM/25.0MM LTCS A420 WPL6-W	42 - 48					
1610			Reduci	5392690	42 - 46	TEE RED WT 29.0MM LTCS A420 WPL6-W	44 - 48					
1620	C		Reduci	5392581	30 - 30	TEE RED WT 34.0MM/20.0MM LTCS A420 WPL6-W	56 - 56					
1630	D	NT6	Reduci	5392588	20 - 20	TEE RED WT 34.0MM/SCH 40 LTCS A420 WPL6-W	56 - 56					
1640												
1650			Cap	5392826	3/4 - 3/4	CAP BW SCH 160 LTCS A420 WPL6						
1660			Cap	5392610	1 - 2	CAP BW SCH XS LTCS A420 WPL6						
1670			Cap	5392448	3 - 8	CAP BW SCH STD LTCS A420 WPL6						
1680			Cap	5392610	10 - 14	CAP BW SCH XS LTCS A420 WPL6						
1690			Cap	5392394	16 - 24	CAP SCH 40 LTCS STL A420 WPL6						
1700			Cap	5392625	26 - 30	CAP WT 20.0MM LTCS A420 WPL6						
1710			Cap	5392618	32 - 40	CAP WT 25.0MM LTCS A420 WPL6						
1720			Cap	5392624	42 - 48	CAP WT 29.0MM LTCS A420 WPL6						
1730												
1740			Concen	5392462	3/4 - 3/4	REDUCER CONC SCH XS/160 LTCS A420 WPL6	1 - 2					
1750			Concen	5392458	1 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 1/2 - 2					
1760			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4					
1770			Concen	5392449	3 - 6	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 8					
1780			Concen	5392300	8 - 8	REDUCER CONC SCH 40/STD LTCS A420 WPL6	16 - 16					
1790			Concen	5392458	10 - 12	REDUCER CONC SCH XS LTCS A420 WPL6	12 - 14					
1800			Concen	5392457	4 - 8	REDUCER CONC SCH XS/STD LTCS A420 WPL6	10 - 14					
1810			Concen	5392305	10 - 14	REDUCER CONC SCH 40/XS LTCS A420 WPL6	16 - 16					
1820			Concen	5392306	10 - 14	REDUCER CONC SCH 40/XS LTCS A420 WPL6-W	18 - 20					
1830			Concen	5392395	16 - 20	REDUCER CONC SCH 40 LTCS A420 WPL6-W	18 - 24					
1840			Concen	5392971	18 - 24	REDUCER CONC WT 20.0MM/SCH 40 LTCS A420 WPL6-W	26 - 30					
1850			Concen	5392981	24 - 24	REDUCER CONC WT 25.0MM/SCH 40 LTCS A420 WPL6-W	32 - 36					
1860			Concen	5392969	26 - 28	REDUCER CONC WT 20MM LTCS A420 WPL6-W	28 - 30					
1870			Concen	5392987	26 - 30	REDUCER CONC WT 25.0MM/20.0MM LTCS A420 WPL6-W	32 - 40					
1880			Concen	5392995	30 - 30	REDUCER CONC WT 29.0MM/20.0MM LTCS A420 WPL6-W	42 - 42					
1890			Concen	5392979	32 - 38	REDUCER CONC WT 25.0MM LTCS A420 WPL6-W	34 - 40					
1900			Concen	5392997	32 - 40	REDUCER CONC WT 29.0MM/25.0MM LTCS A420 WPL6-W	42 - 48					
1910			Concen	5392989	42 - 46	REDUCER CONC WT 29.0MM LTCS A420 WPL6-W	44 - 48					
1920	C		Concen	5392582	36 - 36	REDUCER CONC WT 34.0MM/25.0MM LTCS A420 WPL6-W	56 - 56					
1930	C		Concen	5392585	42 - 42	REDUCER CONC WT 34.0MM/29.0MM LTCS A420 WPL6-W	56 - 56					
1940	C		Concen	5392965	48 - 48	REDUCER CONC WT 36.0MM/29.0MM LTCS A420 WPL6-W	60 - 60					
1950												
1960			Eccent	5392465	3/4 - 3/4	REDUCER ECC SCH XS/160 LTCS A420 WPL6	1 - 2					
1970			Eccent	5392461	1 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 1/2 - 2					
1980			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4					
1990			Eccent	5392450	3 - 6	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 8					
2000			Eccent	5392459	4 - 8	REDUCER ECC SCH XS/STD LTCS A420 WPL6	10 - 14					
2010			Eccent	5392302	8 - 8	REDUCER ECC SCH 40/STD LTCS A420 WPL6	16 - 16					
2020			Eccent	5392612	10 - 12	REDUCER ECC SCH XS LTCS A420 WPL6	12 - 14					
2030			Eccent	5392307	10 - 14	REDUCER ECC SCH 40/XS LTCS A420 WPL6	16 - 16					
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION All SIZES - BUTTWELD PWHT FOR ALL SIZES. FOR SERVICE CODE CDF, PWHT FOR WT > 19MM	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 4 OF 8 PA 5 CLASS 300K21				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2040	D		Eccent	5392264	10 - 14	REDUCER ECC SCH 40/XS LTCS A420 WPL6-W	18 - 20						
2050			Eccent	5392396	16 - 20	REDUCER ECC SCH 40 LTCS A420 WPL6-W	18 - 24						
2060			Eccent	5392972	18 - 24	REDUCER ECC WT 20.0MM/SCH 40 LTCS A420 WPL6-W	26 - 30						
2070			Eccent	5392982	24 - 24	REDUCER ECC WT 25.0MM/SCH 40 LTCS A420 WPL6-W	32 - 36						
2080			Eccent	5392970	26 - 28	REDUCER ECC WT 20MM LTCS A420 WPL6-W	28 - 30						
2090			Eccent	5392988	26 - 30	REDUCER ECC WT 25.0MM/20.0MM LTCS A420 WPL6-W	32 - 40						
2100			Eccent	5392996	30 - 30	REDUCER ECC WT 29.0MM/20.0MM LTCS A420 WPL6-W	42 - 42						
2110			Eccent	5392980	32 - 38	REDUCER ECC WT 25.0MM LTCS A420 WPL6-W	34 - 40						
2120			Eccent	5392998	32 - 40	REDUCER ECC WT 29.0MM/25.0MM LTCS A420 WPL6-W	42 - 48						
2130			Eccent	5392990	42 - 46	REDUCER ECC WT 29.0MM LTCS A420 WPL6-W	44 - 48						
2140	C		Eccent	5392583	36 - 36	REDUCER ECC WT 34.0MM/25.0MM LTCS A420 WPL6-W	56 - 56						
2150	C		Eccent	5392584	42 - 42	REDUCER ECC WT 34.0MM/29.0MM LTCS A420 WPL6-W	56 - 56						
2160	C		Eccent	5392966	48 - 48	REDUCER ECC WT 36.0MM/29.0MM LTCS A420 WPL6-W	60 - 60						
2170													
2180			Nipofl	5504450	3/4 - 3/4	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	2 - 2						
2190			Nipofl	5504451	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/SCH 160 LTCS A350 LF2	3 - 8						
2200			Nipofl	5504450	3/4 - 3/4	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	10 - 14						
2210			Nipofl	5504452	3/4 - 3/4	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2	16 - 24						
2220			Nipofl	5504453	3/4 - 3/4	NIPOFLANGE 300# RF WT 20.0MM/160 LTCS A350 LF2	26 - 30						
2230			Nipofl	5504454	3/4 - 3/4	NIPOFLANGE 300# RF WT 25.0MM/SCH 160 LTCS A350 LF2	32 - 40						
2240			Nipofl	5504455	3/4 - 3/4	NIPOFLANGE 300# RF WT 29.0MM/SCH 160 LTCS A350 LF2	42 - 48						
2250	C		Nipofl	5504473	3/4 - 3/4	NIPOFLANGE 300# RF WT 34.0MM/SCH 160 LTCS A350 LF2	56 - 56						
2260	C		Nipofl	5504476	3/4 - 3/4	NIPOFLANGE 300# RF WT 36.0MM/SCH 160 LTCS A350 LF2	60 - 60						
2270													
2280	D	FF5	Endole	5504515	3/4 - 3/4	NIPOFLANGE FLAT BOTTOM 300# RF SCH 160 LTCS A350 LF2							
2290	D	FF5	Endole	5504501	1 - 1 1/2	NIPOFLANGE FLAT BOTTOM 300# RF SCH XS LTCS A350 LF2							
2300			Nipofl	5504020	1 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	2 - 2						
2310			Nipofl	5504021	1 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 8						
2320			Nipofl	5504020	1 - 2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	10 - 14						
2330			Nipofl	5504022	1 - 2	NIPOFLANGE 300# RF SCH 40/XS LTCS A350 LF2	16 - 24						
2340			Nipofl	5504105	1 - 2	NIPOFLANGE 300# RF 20.0MM/XS LTCS A350 LF2	26 - 30						
2350			Nipofl	5504107	1 - 2	NIPOFLANGE 300# RF WT 25MM/XS LTCS A350 LF2	32 - 40						
2360			Nipofl	5504109	1 - 2	NIPOFLANGE 300# RF WT 29MM/XS LTCS A350 LF2	42 - 48						
2370	C		Nipofl	5504103	1 - 2	NIPOFLANGE 300# RF WT 34.0MM/XS LTCS A350 LF2	56 - 56						
2380	C		Nipofl	5504140	1 - 2	NIPOFLANGE 300# RF WT 36.0MM/XS LTCS A350 LF2	60 - 60						
2390													
2400		UJ2	Nipofl	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 8	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BD, BF, BFH, BFL, CCD, CCS, CDE (BELOW 90 DEG C), CDF, CDM (BELOW 70 DEG C), CS, DCS, DHC, DL, DMD, FG, FGH, FM, FX, GCL, GE, GHC, GRC, GS, GST, NL, PHC, SL, SM, VE, WM, WP, WPS, WSC, WSR, WU					
2410		UJ2	Nipofl	5504023	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	10 - 14						
2420		UJ2	Nipofl	5504018	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40/160 LTCS A350 LF2	16 - 24						
2430		UJ2	Nipofl	5504115	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 20.0MM/160 LTCS A350 LF2	26 - 30						
2440		UJ2	Nipofl	5504117	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 25MM/160 LTCS A350 LF2	32 - 40						
2450		UJ2	Nipofl	5504119	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 29.0MM/160 LTCS A350 LF2	42 - 48						
2460	C	UJ2	Nipofl	5504120	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 34.0MM/160 LTCS A350 LF2	56 - 56						
2470	C	UJ2	Nipofl	5504139	1 1/2 - 1 1/2	NIPOFLANGE 300# RF WT 36.0MM/160 LTCS A350 LF2	60 - 60						
2480													
2490						***** OLETS *****							
2500													
2510			Weldol	5505393	3/4 - 3/4	WELDOLET SCH XS/160 LTCS A350 LF2	2 - 2	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 260 °C COMMENT: REFER NOTE 800					
2520			Weldol	5505192	3/4 - 3/4	WELDOLET STD/160 LTCS A350 LF2	3 - 8						
2530			Weldol	5505393	3/4 - 3/4	WELDOLET SCH XS/160 LTCS A350 LF2	10 - 14						
2540			Weldol	5505333	1 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 2						
2550			Weldol	5505179	1 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 8						
2560			Weldol	5505333	1 - 2	WELDOLET SCH XS LTCS A350 LF2	10 - 14						
2570			Weldol	5505574	1 - 2	WELDOLET SCH 40/XS LTCS A350 LF2	16 - 24						
2580			Weldol	5505582	1 - 2	WELDOLET WT 20.0MM/XS LTCS A350 LF2	26 - 30						
2590			Weldol	5505587	1 - 2	WELDOLET WT 25MM/XS LTCS A350 LF2	32 - 40						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION							
						All SIZES - BUTTWELD PWHT FOR ALL SIZES. FOR SERVICE CODE CDF, PWHT FOR WT > 19MM		CORR. ALLOW.	DATE	BY	REV	SHEET	PA
						3 MM		17MAR17	SMS	D	5 OF 8	5	300K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2600	D		Weldol	5505597	1 - 2	WELDOLET WT 29.0MM/XS LTCS A350 LF2	42 - 48							
2610			Weldol	5505164	3 - 3	WELDOLET SCH STD LTCS A350 LF2	8 - 8							
2620			Weldol	5505334	3 - 4	WELDOLET SCH XS/STD LTCS A350 LF2	10 - 14							
2630			Weldol	5505554	3 - 8	WELDOLET SCH 40/STD LTCS A350 LF2	16 - 24							
2640			Weldol	5505583	3 - 8	WELDOLET WT 20.0MM/STD LTCS A350 LF2	26 - 30							
2650			Weldol	5505588	3 - 8	WELDOLET WT 25.0MM/STD LTCS A350 LF2	32 - 40							
2660			Weldol	5505598	3 - 8	WELDOLET WT 29.0MM/ STD LTCS A350 LF2	42 - 48							
2670			Weldol	5505599	3 - 3	WELDOLET WT 36.0MM/ STD LTCS A350 LF2	60 - 60							
2680			Weldol	5505582	10 - 12	WELDOLET WT 20.0MM/XS LTCS A350 LF2	26 - 30							
2690			Weldol	5505587	10 - 12	WELDOLET WT 25MM/XS LTCS A350 LF2	32 - 40							
2700			Weldol	5505597	10 - 12	WELDOLET WT 29.0MM/XS LTCS A350 LF2	42 - 48							
2710	UJ2		Elbole	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 8							
2720			UJ2	Elbole	5516623	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH XS/160 LTCS A350 LF2							10 - 14
2730			UJ2	Elbole	5516618	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40/160 LTCS A350 LF2							16 - 24
2740			UJ2	Elbole	5516685	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 20.0MM/160 LTCS A350 LF2							26 - 30
2750			UJ2	Elbole	5516687	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 25.0MM/160 LTCS A350 LF2							32 - 40
2760			UJ2	Elbole	5516689	1 1/2 - 1 1/2	ELBOFLANGE 300# RF WT 29.0MM/160 LTCS A350 LF2							42 - 48
2770	UD7		Sockol	5533071	3/4 - 1 1/2	SOCKOLET 6000# LTCS A350 LF2	2 - 48							
2780			Endole	5533233	3/4 - 1 1/2	FLAT SOCKOLET 6000# LTCS A350 LF2	2 - 48							
2810			Thredo	5514772	3/4 - 1	THREDOLET 3000# LTCS A350 LF2	2 - 48							
2820			***** COUPLINGS, UNIONS & PLUGS *****											
2830			Coupli	5478807	3/4 - 1 1/2	COUPLING 6000# SW LTCS A350 LF2	1 - 1 1/2							
2840														
2850	Reduci	5478812	3/4 - 1	COUPLING RED 6000# SW LTCS A350 LF2	1 - 1 1/2									
2860														
2870	Union	5478809	3/4 - 1 1/2	UNION 6000# SW LTCS STL A350 LF2	1 - 1 1/2									
2880														
2890	Plug	5491303	3/4 - 1	PLUG ROUND HEAD SCRD LTCS A350 LF2	1 - 1 1/2									
2900														
2910	**** FLANGES ****					3/4 - 1 1/2								
2920														
2930	FA8		Reduci	5599703	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 1 1/2							
2940														
2950	C		Flange	5556480	1/2 - 3/4	FLG WN 300# RF LTCS A350 LF2 SCH 160		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BD, BF, BFH, BFL, CCD, CCS, CDE (BELOW 90 DEG C), CDF, CDM (BELOW 70 DEG C), CS, DCS, DHC, DL, DMD, FG, FGH, FM, FX, GCL, GE, GHC, GRC, GS, GST, NL, PHC, SL, SM, VE, WM, WP, WPS, WSC, WSR, WU						
2960			Flange	5556476	1 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS								
2970			Flange	5556473	3 - 8	FLG WN 300# RF LTCS A350 LF2 SCH STD								
2980			Flange	5556476	10 - 14	FLG WN 300# RF LTCS A350 LF2 SCH XS								
2990			Flange	5556472	16 - 24	FLG WN 300# RF LTCS A350 LF2 SCH 40 BORE								
3000			Flange	5556492	26 - 30	FLG WN 300# RF LTCS A350 LF2 WT 20.0MM B16.47 SERIES A								
3010	C		Flange	5556484	32 - 40	FLG WN 300# RF LTCS A350 LF2 WT 25.0MM B16.47 SERIES A								
3020			Flange	5556498	42 - 48	FLG WN 300# RF LTCS A350 LF2 WT 29.0MM B16.47 SERIES A								
3030			Flange	5556496	56 - 56	FLG WN 300# RF LTCS A350 LF2 WT 34.0MM B16.47 SERIES A								
3040			Flange	5556464	60 - 60	FLG WN 300# RF LTCS A350 LF2 WT 36.0MM B16.47 SERIES A								
3050			C	FS1	Flange	5557036		2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS					
3060					Flange	5557033		3 - 8	FLG WN 600# RF LTCS A350 LF2 SCH STD					
3070	C	FS1	Flange	5557036	10 - 12	FLG WN 600# RF LTCS A350 LF2 SCH XS								
3080			Flange	5556967	16 - 24	FLG WN 600# RF LTCS A350 LF2 SCH 40								
3090	C	FS1	Flange	5556558	42 - 42	FLG WN 600# RF LTCS A350 LF2 WT 29.0MM								
3100														
3110	C	FS1	Flange	5555935	3 - 8	FLG WN 900# RF LTCS A350 LF2 SCH STD								
3120			D	FS1	Flange	5557141		10 - 10	FLG WN 900# RF LTCS A350 LF2 SCH XS					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD PWHT FOR ALL SIZES. FOR SERVICE CODE CDF, PWHT FOR WT > 19MM		CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 6 OF 8 PA 5 CLASS 300K21					



SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3520		FS1	GASFRS	5672251	1 - 1	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
3530		B1				***** BOLTING *****							
3540													
3550													
3560													
3570	D		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3580													
3590	C	SB6	CPSBOL	5675675	12 - 60	JACK SCREW HEX HEAD A320 GR L7M							
3600						***** BLINDS AND SPACERS *****							
3610													
3620													
3630													
3640			Specta	5749581	3/4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
3650													
3660			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70							
3670	C		Paddle	5750028	26 - 60	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A							
3680													
3690			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70							
3700	C		Paddle	5750198	26 - 60	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A							
3710						***** BRANCHES *****							
3720													
3730						USE BRANCH CHART 300K21							
3740													
3750													
3760			Reinfo	5358617_R	14 - 16	REINFORCING PAD FROM PIPE A671 CC60 CL22 25MM WT	32 - 40						
3770	C		Reinfo	5358617_R	14 - 16	REINFORCING PAD FROM PIPE A671 CC60 CL22 25MM WT	32 - 40						
3780													
3790			Reinfo	5358620_R	14 - 20	REINFORCING PAD FROM PIPE A671 CC60 CL22 29MM WT	42 - 48						
3800	C		Reinfo	5358620_R	14 - 20	REINFORCING PAD FROM PIPE A671 CC60 CL22 29MM WT	42 - 48						
3810													
3820			Reinfo	5358612_R	14 - 26	REINFORCING PAD FROM PIPE A671 CC60 CL22 20MM WT	26 - 30						
3830	C		Reinfo	5358612_R	14 - 26	REINFORCING PAD FROM PIPE A671 CC60 CL22 20MM WT	26 - 30						
3840													
3850	C		Reinfo	5358616_R	18 - 18	REINFORCING PAD FROM PIPE A671 CC60 CL22 34MM WT	56 - 56						
3860	C		Reinfo	5358616_R	18 - 18	REINFORCING PAD FROM PIPE A671 CC60 CL22 34MM WT	56 - 56						
3870													
3880													
3890	D		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 60						
3900	D		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 60						
3910						***** CONSTRUCTION DETAILS *****							
3920						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
3930						PIPE BENDS ----- NONE							
3940						THREAD SEALANT ----- PTFE PASTE							
3950		ZA6											
3960													
3970													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD PWHT FOR ALL SIZES. FOR SERVICE CODE CDF, PWHT FOR WT > 19MM	CORR. ALLOW. 3 MM						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 20 30 40	C		2BV	K03AC02	3/4 - 2	***** MODULAR VALVES ***** MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		812 B1 FF5 FS1	NIPPLE TO BE CUT FROM SCH XXS PIPE ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.					
50 60	C	VAV	2BV	K03AC02	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		NT5 SB6	TO BE USED IN NON-PIGGED LINES ONLY USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
70 80	C		2BVR	M03AC02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
90 100	C	VAW	R2BVR	M03EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	VAV VAW	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.					
110 120	C	VAW	R2BVR	M03EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	VB2 VC1	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
130 140 150		VB2				***** BALL VALVES *****		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
160 170	C		Ball V	A03AC01	3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
180 190 200 210	C		Ball V	A03AC02	3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO								
220 230 240			Ball V	A03AC07	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
250 260 270 280			Ball V	A03AC08	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
290 300 310 320	C	VA5	Ball V	B03AC01	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
330 340 350 360	C	VA5	Ball V	B03AC02	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
370 380 390 400	D	VA5	Ball V	B03AC03	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
410 420 430 440 450 460 470	D	VA5	Ball V	B03AC04	4 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
480 490 500 510 520 530 540		VA5	Ball V	B03AC07	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
	C	VA5	Ball V	B03AC08	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
		VA5	Ball V	B03JC07	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A								
		VA5	Ball V	B03JC08	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A								
						*****ORBIT VALVES*****								
	C	VA5	Ball V	R03AC06	2 - 4	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB HO			MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 88 °C COMMENT: LTCS					
	C	VA5	Ball V	R03AC08	6 - 10	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB GO								
						*****BUTTERFLY VALVES*****								
	C		Butter	T03AC61	10 - 24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 1 OF 8	PA 5	CLASS 300K5A

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
550			Butter	T03JC61	28 - 28	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A							
560													
570													
580						***** GATE VALVES *****							
590													
600	C		Condui	W03AC70	3 - 8	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
610													
620	C		Condui	W03AC71	10 - 24	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
630													
640			Condui	W03JC71	28 - 28	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A							
650													
660													
670						***** CHECK VALVES *****							
680													
690	C	VC1	Check	C03AC50	3/4 - 1 1/2	CHECK PISTON 300# RF LTCS A350 LF2 TRIM 16							
700	C	VC2	Check	C03AC45	2 - 10	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
710	C	VC2	Check	C03AC52	12 - 24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16							
720	C	VC2	Check	C03JC52	28 - 28	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A							
730													
740													
750						***** GLOBE VALVES *****							
760													
770	C		Globe	L03AC48	3/4 - 1 1/2	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
780	C		Globe	L03AC41	2 - 6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW							
790	C		Globe	L03AC42	8 - 10	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO							
800													
810						***** PIPE ABOVE GROUND *****							
820													
830	C		Piping	5358512	3/4 - 3/4	PIPE SCH 160 SMLS LTCS A333 GR.6 PE							
840	C		Piping	5358508	1 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE							
850	C		Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
860			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE							
870			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE							
880	C		Piping	9001254	6 - 10	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE							
890	C		Piping	9001256	12 - 14	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE							
900			Piping	9001311	16 - 18	LINEPIPE SMLS WT=7.92MM API-5L X60 BE							
910			Piping	9001485	20 - 20	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE							
920			Piping	9001493	24 - 24	LINEPIPE SMLS WT LS WT = 9.53 MM API-5L X60 BE							
930			Piping	9001494	28 - 28	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE							
940													
950	C	812	Piping	5358513	1/2 - 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 BE							
960						***** PIPE BELOW GROUND *****							
970													
980													
990			Piping	9001257CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED							
1000	C		Piping	9001254CTD	6 - 10	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED							
1010	C		Piping	9001256CTD	12 - 14	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE COATED							
1020			Piping	9001311CTD	16 - 18	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED							
1030			Piping	9001485CTD	20 - 20	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED							
1040			Piping	9001493CTD	24 - 24	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED							
1050			Piping	9001494CTD	28 - 28	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE COATED							
1060													
1070													
1080													
1090						***** FITTINGS ABOVE & BELOW GROUND*****							
1100	C		90 Deg	5392816	3/4 - 3/4	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6							
1110	C		90 Deg	5392600	1 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
1120			90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 8	PA 5	CLASS 300K5A

SEQ.NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1130			90 Deg	9006566	4 - 4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60							
1140	C		90 Deg	9002190	6 - 10	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60							
1150	C		90 Deg	9006477	12 - 14	ELL BW 90 DEG LR WT 7.14 MM A860 WPHY 60							
1160			90 Deg	9006478	16 - 18	ELL BW 90 DEG LR WT 7.92 MM A860 WPHY 60							
1170			90 Deg	9006481	20 - 20	ELL BW 90 DEG LR WT 8.74 MM A860 WPHY 60-W							
1180			90 Deg	9006472	24 - 24	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60-W							
1190			90 Deg	9006473	28 - 28	ELL BW 90 DEG LR WT 10.31 MM A860 WPHY 60-W							
1200	C		45 Deg	5392822	3/4 - 3/4	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6							
1210	C		45 Deg	5392606	1 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1220			45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6							
1230	D		45 Deg	9006571	6 - 10	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60							
1240	D		45 Deg	9006589	12 - 14	ELL BW 45 DEG LR WT 7.14MM A860 WPHY 60							
1250	C		45 Deg	9006588	16 - 18	ELL BW 45 DEG LR WT 7.92MM A860 WPHY 60							
1260			45 Deg	9006570	24 - 24	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60-W							
1270													
1280	C		Tee	5392825	3/4 - 3/4	TEE BW SCH 160 LTCS A420 WPL6							
1290	C		Tee	5392609	1 - 2	TEE BW SCH XS LTCS A420 WPL6							
1300			Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6							
1310		NT5	Tee	9301610	4 - 4	TEE BW WT 6.02 MM LTCS A860 WPHY-60							
1320	C	NT5	Tee	9301609	6 - 10	TEE BW WT 6.35 MM LTCS A860 WPHY-60							
1330	C	NT5	Tee	9301605	12 - 14	TEE BW WT 7.14 MM LTCS A860 WPHY-60							
1340		NT5	Tee	9301606	16 - 18	TEE BW WT 7.92 MM LTCS A860 WPHY-60							
1350		NT5	Tee	9006487	20 - 20	TEE BW WT 8.74MM A860 WPHY 60-W							
1360		NT5	Tee	9004210	24 - 24	TEE BW WT 9.53MM A860 WPHY 60-W							
1370													
1380													
1390	C	NT5	Reduci	5392469	3/4 - 3/4	TEE RED BW SCH XS/160 LTCS A420 WPL6	1 - 1 1/2						
1400	C	NT5	Reduci	5392661	1 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2						
1410	C	NT5	Reduci	5392454	1 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3						
1420	C	NT5	Reduci	9006664	2 - 2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60	4 - 4						
1430	C	NT5	Reduci	9006659	3 - 3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60	4 - 4						
1440	C	NT5	Reduci	9006658	3 - 3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60	6 - 6						
1450		NT5	Reduci	9006605	4 - 4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60	6 - 10						
1460	C	NT5	Reduci	9006014	6 - 8	TEE RED BW WT 6.35 MM LTCS A860 WPHY 60	8 - 10						
1470	C	NT5	Reduci	9006015	6 - 10	TEE RED BW WT 7.14 MM/6.35 MM LTCS A860 WPHY 60	12 - 14						
1480	C	NT5	Reduci	9006637	12 - 12	TEE RED BW WT 7.14 MM LTCS A860 WPHY 60	14 - 14						
1490			Reduci	9006017	6 - 6	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	16 - 16						
1500	C	NT5	Reduci	9006017	8 - 10	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	16 - 18						
1510	C	NT5	Reduci	9006016	12 - 14	TEE RED BW WT 7.92 MM/7.14 MM LTCS A860 WPHY 60	16 - 18						
1520		NT5	Reduci	9006018	16 - 16	TEE RED BW WT 7.92 MM LTCS A860 WPHY 60	18 - 18						
1530			Reduci	9004143	8 - 8	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60-W	20 - 20						
1540	C	NT5	Reduci	9004143	10 - 10	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60-W	20 - 20						
1550	C	NT5	Reduci	9006020	12 - 14	TEE RED BW WT 8.74 MM/7.14 MM LTCS A860 WPHY 60-W	20 - 20						
1560		NT5	Reduci	9006019	16 - 18	TEE RED BW WT 8.74 MM/7.92 MM LTCS A860 WPHY 60-W	20 - 20						
1570		NT5	Reduci	9004149	10 - 10	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W	24 - 24						
1580		NT5	Reduci	9006606	12 - 14	TEE RED BW WT 9.53 MM/7.14 MM LTCS A860 WPHY 60-W	24 - 24						
1590		NT5	Reduci	9006483	16 - 18	TEE RED BW WT 9.53 MM/7.92 MM LTCS A860 WPHY 60-W	24 - 24						
1600		NT5	Reduci	9006484	20 - 20	TEE RED BW WT 9.53 MM/8.74 MM LTCS A860 WPHY 60-W	24 - 24						
1610		NT5	Reduci	9006635	20 - 20	TEE RED BW WT 10.31 MM/8.74 MM LTCS A860 WPHY 60-W	28 - 28						
1620		NT5	Reduci	9006636	24 - 24	TEE RED BW WT 10.31 MM/9.53 MM LTCS A860 WPHY 60-W	28 - 28						
1630													
1640	C		Cap	5392826	3/4 - 3/4	CAP BW SCH 160 LTCS A420 WPL6							
1650	C		Cap	5392610	1 - 2	CAP BW SCH XS LTCS A420 WPL6							
1660			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6							
1670	C		Cap	9005055	6 - 10	CAP BW WT 6.35MM LTCS A860 WPHY-60							
1680	C		Cap	9005057	12 - 14	CAP BW WT 7.14 MM LTCS A860 WPHY-60							
1690			Cap	9005058	16 - 18	CAP BW WT 7.92 MM LTCS A860 WPHY-60							
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 3 OF 8 PA 5 CLASS 300K5A					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1700			Cap	9005059	20 - 22	CAP BW WT 8.74 MM LTCS A860 WPHY-60		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 88 °C COMMENT: LTCS					
1710			Cap	9005060	24 - 24	CAP BW WT 9.53 MM LTCS A860 WPHY-60							
1720			Cap	9005061	28 - 28	CAP BW WT 10.31 MM LTCS A860 WPHY-60							
1730													
1740	C		Concen	5392462	3/4 - 3/4	REDUCER CONC SCH XS/160 LTCS A420 WPL6	1 - 2						
1750	C		Concen	5392458	1 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 1/2 - 2						
1760	C		Ecccent	5392465	3/4 - 3/4	REDUCER ECC SCH XS/160 LTCS A420 WPL6	1 - 2						
1770	C		Ecccent	5392461	1 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 1/2 - 2						
1780			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3						
1790			Ecccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3						
1800	C		Concen	9006918	1 1/2 - 3	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1810	C		Ecccent	9001372	1 1/2 - 3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1820	C		Concen	9006915	3 - 3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6 - 6						
1830	C		Ecccent	9002510	3 - 3	REDUCER ECC WT 6.35 MM/STD LTCS A420 WPL6	6 - 6						
1840			Ecccent	9002312	4 - 4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6 - 10						
1850	C		Ecccent	9002266	6 - 8	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8 - 10						
1860	C		Ecccent	9002269	8 - 10	REDUCER ECC WT 7.14 MM / 6.35 MM LTCS A860 WPHY-60	12 - 14						
1870	C		Ecccent	9006908	12 - 12	REDUCER ECC WT 7.14 MM LTCS A860 WPHY-60	14 - 14						
1880	C		Ecccent	9002272	10 - 10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	16 - 18						
1890	C		Ecccent	9002270	12 - 14	REDUCER ECC WT 7.92 MM / 7.14 MM LTCS A860 WPHY-60	16 - 18						
1900	C		Ecccent	9002278	16 - 16	REDUCER ECC WT 7.92MM LTCS A860 WPHY-60	18 - 18						
1910	C		Ecccent	9004132	12 - 14	REDUCER ECC WT 8.74 MM / 7.14 LTCS A860 WPHY-60-W	20 - 20						
1920			Ecccent	9004133	16 - 18	REDUCER ECC WT 8.74 MM / 7.92MM LTCS A860 WPHY-60-W	20 - 20						
1930			Ecccent	9004134	16 - 16	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60-W	24 - 24						
1940			Concen	9006901	18 - 18	REDUCER CONC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	24 - 24						
1950			Ecccent	9004135	16 - 18	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPH Y-60-W	24 - 24						
1960			Concen	9006902	20 - 20	REDUCER CONC WT 9.53 MM / 8.74MM LTCS A860 WPHY-60-W	24 - 24						
1970			Ecccent	9004136	20 - 20	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	24 - 24						
1980			Concen	9006903	24 - 24	REDUCER CONC WT 10.31 MM / 9.53 MM LTCS A860WPHY-60-W	28 - 28						
1990			Ecccent	9004137	24 - 24	REDUCER ECC WT 10.31 MM / 9.53 MM LTCS A860 WPHY-60-W	28 - 28						
2000													
2010													
2020						*****INDUCTION BENDS ABOVE GROUND*****							
2030													
2040			90 Deg	9001810	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60							
2050	C		90 Deg	9001813	6 - 10	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60							
2060	C		90 Deg	9001814	12 - 14	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60							
2070			90 Deg	9001815	16 - 18	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60							
2080			90 Deg	9001816	20 - 20	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60							
2090			90 Deg	9001817	24 - 24	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60							
2100			90 Deg	9001818	28 - 28	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60							
2110													
2120			45 Deg	9001820	4 - 4	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60							
2130	C		45 Deg	9001823	6 - 10	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60							
2140	C		45 Deg	9001824	12 - 14	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60							
2150			45 Deg	9001825	16 - 18	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	FGP/WPMP SHEET 4 OF 8	PA 5	CLASS 300K5A

SEQ.NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2160			45 Deg	9001826	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2170			45 Deg	9001827	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2180			45 Deg	9001828	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2190			22.5 D	9001830	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2200			22.5 D	9001833	6 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2210			22.5 D	9001834	12 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2220			22.5 D	9001835	16 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2230			22.5 D	9001836	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2240			22.5 D	9001837	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2250			22.5 D	9001838	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2260			22.5 D	9001838	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2270						*****INDUCTION BENDS BELOW GROUND*****							
2280													
2290													
2300			90 Deg	9001810CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2310			90 Deg	9001813CTD	6 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2320			90 Deg	9001814CTD	12 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2330			90 Deg	9001815CTD	16 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2340			90 Deg	9001816CTD	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2350			90 Deg	9001817CTD	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2360			90 Deg	9001818CTD	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2370			45 Deg	9001820CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2380			45 Deg	9001823CTD	6 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2390			45 Deg	9001824CTD	12 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2400			45 Deg	9001825CTD	16 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2410			45 Deg	9001826CTD	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2420			45 Deg	9001827CTD	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2430			45 Deg	9001828CTD	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2440			45 Deg	9001830CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2450			22.5 D	9001830CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2460													
2470													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 5 OF 8	PA 5	CLASS 300K5A

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2480			22.5 D	9001833CTD	6 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2490			22.5 D	9001834CTD	12 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2500			22.5 D	9001835CTD	16 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2510			22.5 D	9001836CTD	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2520			22.5 D	9001837CTD	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2530			22.5 D	9001838CTD	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2540													
2550													
2560													
2570						***** OLETS ABOVE & BELOW GROUND*****							
2580	C		Weldol	9006400	3/4 - 3/4	WELDOLET WT 6.02 MM / SCH 160 LTCS A694 F60	4 - 4						
2590	C		Weldol	9006219	1 - 1 1/2	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60 IMPACT TESTED AT -46 DEG C	4 - 4						
2600	C		Weldol	9002450	3/4 - 3/4	WELDOLET WT 6.35 MM/SCH 160 WT LTCS A694 F60	6 - 10						
2610	C		Weldol	9006407	1 - 2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6 - 10						
2620	C		Weldol	9006409	3 - 3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8 - 10						
2630	C		Weldol	9006817	3/4 - 3/4	WELDOLET WT 7.14 MM/ SCH 160 WT LTCS A694 F60	12 - 14						
2640	C		Weldol	9006413	1 - 2	WELDOLET WT 7.14 MM/XS WT LTCS A694 F60	12 - 14						
2650	C		Weldol	9006412	3 - 3	WELDOLET WT 7.14 MM/STD WT LTCS A694 F60	12 - 14						
2660			Weldol	9006411	4 - 4	WELDOLET WT 7.14 MM/6.02 MM LTCS A694 F60	12 - 14						
2670	C		Weldol	9006818	3/4 - 3/4	WELDOLET WT 7.92 MM/ SCH 160 WT LTCS A694 F60	16 - 18						
2680	C		Weldol	9002552	1 - 2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	16 - 18						
2690	C		Weldol	9006415	3 - 3	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	16 - 18						
2700			Weldol	9006414	4 - 4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	16 - 18						
2710			Weldol	9004124	6 - 6	WELDOLET WT 7.92 MM/6.35 MM LTCS A694 F60	18 - 18						
2720	C		Weldol	9002455	3/4 - 3/4	WELDOLET WT 8.74MM/ SCH 160 LTCS A694 F60	20 - 20						
2730	C		Weldol	9006423	1 - 2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	20 - 20						
2740	C		Weldol	9006422	3 - 3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	20 - 20						
2750			Weldol	9006421	4 - 4	WELDOLET WT 8.74 MM/6.02 MM LTCS A694 F60	20 - 20						
2760			Weldol	9002008	6 - 6	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	20 - 20						
2770	C		Weldol	9002457	3/4 - 3/4	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	24 - 24						
2780	C		Weldol	9006429	1 - 2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	24 - 24						
2790	C		Weldol	9006428	3 - 3	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	24 - 24						
2800			Weldol	9006427	4 - 4	WELDOLET WT 9.53 MM/6.02 MM LTCS A694 F60	24 - 24						
2810			Weldol	9006424	6 - 8	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	24 - 24						
2820			Weldol	9006433	4 - 4	WELDOLET WT 10.31 MM/6.02 MM LTCS A694 F60	28 - 28						
2830			Weldol	9006430	6 - 10	WELDOLET WT 10.31 MM/6.35 MM LTCS A694 F60	28 - 28						
2840			Weldol	9006816	12 - 12	WELDOLET WT 10.31 MM/7.14 MM LTCS A694 F60	28 - 28						
2850													
2860	C		Nipofl	5504450	3/4 - 3/4	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	2 - 2						
2870	C		Nipofl	5504451	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/SCH 160 LTCS A350 LF2	3 - 3						
2880	C		Nipofl	9006807	3/4 - 3/4	NIPOFLANGE 300# RF 6.02 MM/SCH 160 WT LTCS A694 F60	4 - 4						
2890	C		Nipofl	9006808	3/4 - 3/4	NIPOFLANGE 300# RF 6.35 MM/SCH 160 WT LTCS A694 F60	6 - 10						
2900	C		Nipofl	9006809	3/4 - 3/4	NIPOFLANGE 300# RF 7.14 MM/SCH 160 WT LTCS A694 F60	12 - 14						
2910	C		Nipofl	9006810	3/4 - 3/4	NIPOFLANGE 300# RF 7.92 MM/SCH 160 WT LTCS A694 F60	16 - 18						
2920	C		Nipofl	9006811	3/4 - 3/4	NIPOFLANGE 300# RF 8.74 MM/SCH 160 WT LTCS A694 F60	20 - 20						
2930	C		Nipofl	9006812	3/4 - 3/4	NIPOFLANGE 300# RF 9.53 MM/SCH 160 WT LTCS A694 F60	24 - 24						
2940	C		Nipofl	9006813	3/4 - 3/4	NIPOFLANGE 300# RF 10.31 MM/SCH 160 WT LTCS A694 F60	28 - 28						
2950	C		Nipofl	5504020	1 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	2 - 2						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP						
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
						3 MM	17MAR17	JCS	C	6 OF 8	5	300K5A	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2960			Nipofl	5504021	1 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 3							
2970			Nipofl	9002639	1 - 2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4							
2980			Nipofl	9002643	1 - 2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6 - 10							
2990			Nipofl	9002644	1 - 2	NIPOFLANGE 300# RF 7.14 MM/XS WT LTCS A694 F60	12 - 14							
3000			Nipofl	9002645	1 - 2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	16 - 18							
3010			Nipofl	9002646	1 - 2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	20 - 20							
3020			Nipofl	9002647	1 - 2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	24 - 24							
3030			Nipofl	9002648	1 - 2	NIPOFLANGE 300# RF 10.31 MM/XS WT LTCS A694 F60	28 - 28							
3040														
3050	C	FF5	Endole	9504524	1 - 1	NIPOFLANGE FLAT BOTTOM 300# RF SCH XS LTCS A694 F60	new							
3060														
3070														
3080						***** PIPE FLANGES ABOVE & BELOW GROUND*****								
3090														
3100	C		Flange	5556480	1/2 - 3/4	FLG WN 300# RF LTCS A350 LF2 SCH 160								
3110			Flange	5556476	1 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS								
3120			Flange	5556473	3 - 3	FLG WN 300# RF LTCS A350 LF2 SCH STD								
3130			Flange	9001480	4 - 4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM								
3140	C		Flange	9001483	6 - 10	FLG WN 300# RF LTCS A694 F60 WT 6.35MM								
3150	C		Flange	9001484	12 - 14	FLG WN 300# RF LTCS A694 F60 WT 7.14MM								
3160			Flange	9001455	16 - 18	FLG WN 300# RF LTCS A694 F60 WT 7.92MM								
3170			Flange	9001457	20 - 20	FLG WN 300# RF LTCS A694 F60 WT 8.74MM								
3180			Flange	9001458	24 - 24	FLG WN 300# RF LTCS A694 F60 WT 9.53MM								
3190			Flange	9001459	28 - 28	FLG WN 300# RF LTCS A694 F60 WT 10.31MM SERIES A								
3200														
3210			Blind	5615576	3/4 - 3	FLG BLIND 300# RF LTCS A350 LF2								
3220			Blind	9001450	4 - 24	FLG BLIND 300# RF LTCS A694 F60								
3230			Blind	9001451	28 - 28	FLG BLIND 300# RF LTCS A694 F60 SERIES A								
3240														
3250	C	FS1	Flange	5557036	3 - 3	FLG WN 600# RF LTCS A350 LF2 SCH XS								
3260	C	FS1	Flange	9001723	4 - 4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM								
3270	C	FS1	Flange	9001471	6 - 10	FLG WN 600# RF LTCS A694 F60 WT 6.35MM								
3280	C	FS1	Flange	9003493	12 - 14	FLG WN 600# RF LTCS A694 F60 WT 7.14 MM								
3290	C	FS1	Flange	9003494	16 - 16	FLG WN 600# RF LTCS A694 F60 WT 7.92 MM								
3300	C	FS1	Flange	9003485	24 - 24	FLG WN 600# RF LTCS A694 F60 WT 9.53 MM								
3310														
3320	C	FS1	Flange	9006757	12 - 12	FLG WN 600# RTJ A694 F60 WT 7.14MM								
3330	C	FS1	Flange	9006711	18 - 18	FLG WN 600# RTJ A694 F60 WT 7.92MM								
3340														
3350			Reduci	5599703	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 1 1/2							
3360														
3370														
3380	C		Orific	5623944	3 - 3	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD								
3390														
3400	C		Flange	5851203	1/2 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE								
3410														
3420						***** GASKETS *****								
3430														
3440	C		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR								
3450			GASFRS	5674032	28 - 28	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A								
3460														
3470	C	FS1	GASFRS	5673103	2 - 24	GASKET 600# 316SS SP WND GRAPHITE IR								
3480														
3490	C	FS1	GASRJ	5673202	18 - 18	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 7 OF 8	PA 5	CLASS 300K5A

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
3500	C	B1	BOLS	5675770	1/4 - 99	***** BOLTING *****																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10		800						032ENGINEERING SME’S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. 800THE TEMPERATURE IS LIMITED TO 260°C IN CRUDE OIL SERVICES CONTAINING H2S /AND /OR ORGANIC SULPHUR COMPOUNDS, FOR EXAMPLE, IN THE CRUDE STABILIZATION UNIT. IN HYDROCARBON GAS SERVICE CONTAINING H2S, FOR EXAMPLE, SOUR GAS MOLECULAR SIEVE DEHYDRATION UNITS, USE IS LIMITED TO 230 C. EXCEPTIONS TO THIS ARE THE SULPHUR RECOVERY AND TAIL GAS UNITS WHERE CARBON STEEL SOUR SERVICE LINE CLASSES CAN BE USED UP TO 315°C BECAUSE OF LOWER H2S PARTIAL PRESSURES. 812NIPPLE TO BE CUT FROM SCH XXS PIPE B1ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FA4SLIP ON FLANGE SHALL ONLY BE USED WHERE INSTRUMENT ENCLOSURES ARE INSTALLED. FA8USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. SC1USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE. UD6USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST. UD7USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST. UJ2USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. VA5USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. VABTO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 PHC MATERIAL:LOW TEMP CARBON STEEL FLANGE RATING:300 RF TEMPERATURE RANGE:-40 / 343 °C COMMENT: REFER TO NOTE 800						
20														
30														
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50														
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110														
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430														
440														
450														
460														
470														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV C	SHEET 1 OF 3	PA 5	CLASS 300K5D	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
480	C		Piping	5358512	3/4 - 3/4	***** PIPE *****		VAD	FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE). FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY. FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY. FOR LEVEL TRANSMITTER CONNECTIONS ONLY SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. DELETED REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
490														
500														
510	C		Piping	5358508	1 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE		VAF						
520	C		Piping	5358514	1 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE		VAG						
530			Piping	5358505	3 - 6	PIPE SCH STD SMLS LTCS A333 GR.6 BE		VAH						
540	812							VB2						
550		Piping	5358513	1/2 - 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 BE		VC1							
560														
570														
580						***** FITTINGS *****		VC2						
590														
600			90 Deg	5392816	3/4 - 3/4	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6		VU2						
610			90 Deg	5392600	1 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6		ZA6						
620			90 Deg	5392438	3 - 6	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6								
630														
640			45 Deg	5392822	3/4 - 3/4	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6								
650			45 Deg	5392606	1 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
660			45 Deg	5392444	3 - 6	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6								
670														
680			Tee	5392825	3/4 - 3/4	TEE BW SCH 160 LTCS A420 WPL6								
690			Tee	5392609	1 - 2	TEE BW SCH XS LTCS A420 WPL6								
700			Tee	5392447	3 - 6	TEE BW SCH STD LTCS A420 WPL6								
710														
720			Reduci	5392469	3/4 - 3/4	TEE RED BW SCH XS/160 LTCS A420 WPL6	1 - 2							
730			Reduci	5392661	1 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 1/2 - 2							
740			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3							
750			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	4 - 4							
760			Reduci	5392445	3 - 4	TEE RED BW SCH STD LTCS A420 WPL6	4 - 6							
770														
780			Cap	5392826	3/4 - 3/4	CAP BW SCH 160 LTCS A420 WPL6								
790			Cap	5392610	1 - 2	CAP BW SCH XS LTCS A420 WPL6								
800			Cap	5392448	3 - 6	CAP BW SCH STD LTCS A420 WPL6								
810														
820			Concen	5392579	3/4 - 3/4	REDUCER CONC SCH XS/160 LTCS A420 WPL6	1 - 2							
830			Concen	5392458	1 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 1/2 - 2							
840			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 4							
850			Concen	5392449	3 - 4	REDUCER CONC SCH STD LTCS A420 WPL6	4 - 6							
860														
870			Eccent	5392465	3/4 - 3/4	REDUCER ECC SCH XS/160 LTCS A420 WPL6	1 - 2							
880			Eccent	5392461	1 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 1/2 - 2							
890			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 4							
900			Eccent	5392450	3 - 4	REDUCER ECC SCH STD LTCS A420 WPL6	4 - 6							
910														
920			Nipofl	5504450	3/4 - 3/4	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	2 - 2							
930			Nipofl	5504451	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/SCH 160 LTCS A350 LF2	3 - 6							
940			Nipofl	5504020	1 - 1 1/2	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	2 - 2							
950			Nipofi	5504021	1 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 6							
960														
970			UJ2	5504025	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 6							
980														
990			Weldol	5505192	3/4 - 3/4	WELDOLET STD/160 LTCS A350 LF2	3 - 6							
1000			Weldol	5505179	1 - 2	WELDOLET SCH STD/XS LTCS A350 LF2	3 - 6							
1010														
1020			UJ2	5516625	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2	4 - 6							
1030														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT WELD PWHT - ALL SIZES		CORR. ALLOW. 3 MM						
								FGP/WPMP DATE 17MAR17 BY SMS REV C SHEET 2 OF 3 PA 5 CLASS 300K5D						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1050						***** PIPE FLANGES *****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 PHC MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 343 °C COMMENT: REFER TO NOTE 800						
1060														
1070	C	FA8	Reduci	5599703	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2	3/4 - 1 1/2							
1080														
1090	C		Flange	5556480	1/2 - 3/4	FLG WN 300# RF LTCS A350 LF2 SCH 160								
1100			Flange	5556476	1 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS								
1110			Flange	5556473	3 - 6	FLG WN 300# RF LTCS A350 LF2 SCH STD								
1120														
1130			Blind	5615576	3/4 - 6	FLG BLIND 300# RF LTCS A350 LF2								
1140														
1150	C		Flange	5851203	1/2 - 1 1/2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE								
1160														
1170						***** ORIFICE FLANGES*****								
1180														
1190			Orific	5623947	2 - 2	WN ORIF SET 300# RF LTCS A350 LF2 SCH XS								
1200			Orific	5623944	3 - 6	WN ORIF SET 300# RF LTCS A350 LF2 SCH STD								
1210														
1220														
1230						***** PIPE NIPPLES *****								
1240														
1250		812				BBE - TO BE CUT FROM PIPE								
1260														
1270														
1280						***** GASKETS *****								
1290														
1300	C		GASFRS	5673033	1/2 - 6	GASKET 300# 316SS SP WND GRAPHITE IR								
1310						***** BOLTING *****								
1320		B1												
1330														
1340	C		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
1350														
1360														
1370						**** BLINDS & SPACERS ****								
1380			Specta	5749581	3/4 - 6	SPECTACLE BLIND 300# RF LTCS A516 GR.70								
1390														
1400														
1410						***** BRANCHES *****								
1420														
1430						USE BRANCH CHART 300K5D								
1440														
1450						***** CONSTRUCTION DETAILS *****								
1460														
1470														
1480						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY								
1490						PIPE BENDS ----- NONE								
1500														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV C	FGP/WPMP SHEET 3 OF 3		PA 5	CLASS 300K5D

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10						***** VALVES *****		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. TO BE USED IN NON-PIGGED LINES ONLY USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
20						***** MODULAR VALVES *****		FF5						
30								FS1						
40								NT5						
50	C		2BV	K03AC02	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		SB6						
60								VA5						
70	C	VAV	2BV	K03AC02	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		VAV						
80								VAW						
90	C		2BVR	M03AC02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VB2						
100							2 - 2	VC1						
110	C	VAW	R2BVR	M03EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		VC2						
150							3 - 3	ZA6						
160	C	VAW	R2BVR	M03EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR								
170						***** BALL VALVES *****								
180		VB2												
190														
200	C		Ball V	A03AC01	3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT RB LO								
210														
220	C		Ball V	A03AC02	3 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL RB LO								
230														
240	C		Ball V	A03AC07	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
250														
260	C		Ball V	A03AC08	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
270														
280														
290			Ball V	B03AC01	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
300														
310	C		Ball V	B03AC02	3/4 - 2	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
320														
330			Ball V	B03AC01	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
340														
350	C	VA5	Ball V	B03AC02	3 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
360														
370	C	VA5	Ball V	B03AC03	4 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
380														
390	C	VA5	Ball V	B03AC04	4 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
400														
410	C	VA5	Ball V	B03AC07	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
420														
430	C	VA5	Ball V	B03AC08	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
440														
450	C	VA5	Ball V	B03JC07	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A								
460														
470	C	VA5	Ball V	B03JC08	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A								
480														
490						*****ORBIT VALVES*****								
500														
510														
520	C	VA5	Ball V	R03AC06	3/4 - 4	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB HO								
530														
540	C	VA5	Ball V	R03AC08	6 - 10	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB GO								
550														
560														
570	C					*****BUTTERFLY VALVES*****								
580														
590	C		Butter	T03AC61	10 - 24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 JCS C 1 OF 8 5 300K5E						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
600	C		Butter	T03JC61	28 - 28	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A							
610													
620													
630						***** GATE VALVES *****							
640													
650	C		Condui	W03AC70	3 - 8	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
660													
670	C		Condui	W03AC71	10 - 24	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
680													
690			Condui	W03JC71	28 - 28	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A							
700													
710						*****GLOBE VALVES*****							
720													
730													
740	C		Globe	L03AC48	3/4 - 1 1/2	GLOBE 300# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
750	C		Globe	L03AC41	2 - 6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW							
760	C		Globe	L03AC42	8 - 10	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO							
770													
780						*****CHECK VALVES*****							
790													
800													
810	C		Check	C03AC50	3/4 - 1 1/2	CHECK PISTON 300# RF LTCS A350 LF2 TRIM 16							
820	C	VC2	Check	C03AC45	2 - 10	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
830	C	VC2	Check	C03AC52	12 - 24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16							
840	C		Check	C03JC52	28 - 28	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A							
850													
860						***** PIPE ABOVE GROUND *****							
870													
880													
890			Piping	5358512	3/4 - 3/4	PIPE SCH 160 SMLS LTCS A333 GR.6 PE							
900	C		Piping	5358508	1 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE							
910			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
920			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE							
930			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE							
940			Piping	9001254	6 - 10	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE							
950			Piping	9001256	12 - 12	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE							
960			Piping	9001311	14 - 14	LINEPIPE SMLS WT=7.92MM API-5L X60 BE							
970			Piping	9001485	16 - 18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE							
980			Piping	9001493	20 - 20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE							
990	C		Piping	9001488	24 - 24	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE							
1000			Piping	9001490	28 - 28	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE							
1010													
1020	C		Piping	5358513	1/2 - 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 BE							
1030													
1040						***** PIPE BELOW GROUND *****							
1050													
1060													
1070			Piping	9001257CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED							
1080			Piping	9001254CTD	6 - 10	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED							
1090			Piping	9001256CTD	12 - 12	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE COATED							
1100			Piping	9001311CTD	14 - 14	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED							
1110			Piping	9001485CTD	16 - 18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED							
1120			Piping	9001493CTD	20 - 20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED							
1130	C		Piping	9001488CTD	24 - 24	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED							
1140			Piping	9001490CTD	28 - 28	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE COATED							
1150													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17					
								BY JCS	REV C	SHEET 2 OF 8	PA 5	CLASS 300K5E	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1160						***** FITTINGS ABOVE & BELOW GROUND*****							
1170													
1180			90 Deg	5392816	3/4 - 3/4	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6							
1190			90 Deg	5392600	1 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
1200			90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6							
1210	C		90 Deg	9006566	4 - 4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60							
1220	C		90 Deg	9002190	6 - 10	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60							
1230			90 Deg	9006477	12 - 12	ELL BW 90 DEG LR WT 7.14 MM A860 WPHY 60							
1240			90 Deg	9002157	16 - 18	ELL BW 90 DEG LR WT 8.74MM A860 WPHY 60							
1250			90 Deg	9006472	20 - 20	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60-W							
1260	C		90 Deg	9006575	24 - 24	ELL BW 90 DEG LR WT 11.13 MM A860 WPHY 60-W							
1270			45 Deg	5392822	3/4 - 3/4	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6							
1280			45 Deg	5392606	1 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1290			45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6							
1300	C		45 Deg	9006573	4 - 4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60							
1310	C		45 Deg	9006593	16 - 18	ELL BW 45 DEG LR WT 8.74MM A860 WPHY 60							
1320	C		45 Deg	9006570	20 - 20	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60-W							
1330	C		45 Deg	9006577	24 - 24	ELL BW 45 DEG LR WT 11.13MM A860 WPHY 60-W							
1340													
1350			Tee	5392825	3/4 - 3/4	TEE BW SCH 160 LTCS A420 WPL6							
1360			Tee	5392609	1 - 2	TEE BW SCH XS LTCS A420 WPL6							
1370			Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6							
1380	C	NT5	Tee	9301610	4 - 4	TEE BW WT 6.02 MM LTCS A860 WPHY-60							
1390		NT5	Tee	9301609	8 - 8	TEE BW WT 6.35 MM LTCS A860 WPHY-60							
1400	C	NT5	Tee	9301609	10 - 10	TEE BW WT 6.35 MM LTCS A860 WPHY-60							
1410		NT5	Tee	9301605	12 - 12	TEE BW WT 7.14 MM LTCS A860 WPHY-60							
1420		NT5	Tee	9002113	16 - 18	TEE BW WT 8.7MM A860 WPHY 60							
1430		NT5	Tee	9004210	20 - 20	TEE BW WT 9.53MM A860 WPHY 60-W							
1440	C	NT5	Tee	9006638	24 - 24	TEE BW WT 11.13MM A860 WPHY 60-W							
1450													
1460			Reduci	5392469	3/4 - 3/4	TEE RED BW SCH XS/160 LTCS A420 WPL6	1 - 1 1/2						
1470			Reduci	5392661	1 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 1/2 - 2						
1480			Reduci	5392454	1 1/2 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3						
1490	C	NT5	Reduci	9006664	2 - 2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60	4 - 4						
1500	C	NT5	Reduci	9006659	3 - 3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60	4 - 4						
1510	C	NT5	Reduci	9006658	3 - 3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60	6 - 6						
1520	C	NT5	Reduci	9006605	4 - 4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60	6 - 10						
1530	C	NT5	Reduci	9006014	6 - 6	TEE RED BW WT 6.35 MM LTCS A860 WPHY 60	8 - 10						
1540			Reduci	9004123	6 - 6	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	16 - 16						
1550	C	NT5	Reduci	9004123	8 - 8	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	16 - 18						
1560			Reduci	9004149	8 - 8	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W	20 - 20						
1570	C	NT5	Reduci	9004149	10 - 10	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W	20 - 20						
1580		NT5	Reduci	9006484	16 - 18	TEE RED BW WT 9.53 MM/8.74 MM LTCS A860 WPHY 60-W	20 - 20						
1590	C	NT5	Reduci	9006652	10 - 10	TEE RED BW WT 11.13 MM/6.35 MM LTCS A860 WPHY 60-W	24 - 24						
1600	C	NT5	Reduci	9006639	16 - 18	TEE RED BW WT 11.13 MM/8.74 MM LTCS A860 WPHY 60-W	24 - 24						
1610	C	NT5	Reduci	9006640	20 - 20	TEE RED BW WT 11.13 MM/9.53 MM LTCS A860 WPHY 60-W	24 - 24						
1620													
1630			Cap	5392826	3/4 - 3/4	CAP BW SCH 160 LTCS A420 WPL6							
1640			Cap	5392610	1 - 2	CAP BW SCH XS LTCS A420 WPL6							
1650			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6							
1660			Cap	9005062	4 - 4	CAP BW WT 6.02 MM LTCS A860 WPHY-60							
1670			Cap	9005055	6 - 10	CAP BW WT 6.35MM LTCS A860 WPHY-60							
1680			Cap	9005057	12 - 12	CAP BW WT 7.14 MM LTCS A860 WPHY-60							
1690			Cap	9005058	14 - 14	CAP BW WT 7.92 MM LTCS A860 WPHY-60							
1700			Cap	9005059	16 - 18	CAP BW WT 8.74 MM LTCS A860 WPHY-60							
1710			Cap	9005060	20 - 20	CAP BW WT 9.53 MM LTCS A860 WPHY-60							
1720	C		Cap	9005064	24 - 24	CAP BW WT 11.13 MM LTCS A860 WPHY-60							
1730			Cap	9005065	28 - 28	CAP BW WT 11.91 MM LTCS A860 WPHY-60							
1740													
1750			Concen	5392462	3/4 - 3/4	REDUCER CONC SCH XS/160 LTCS A420 WPL6	1 - 2						
1760			Concen	5392458	1 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 1/2 - 2						
1770			Eccent	5392465	3/4 - 3/4	REDUCER ECC SCH XS/160 LTCS A420 WPL6	1 - 2						
						CONSTRUCTION	FGP/WPMP						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							3 MM	17MAR17	JCS	C	3 OF 8	5	300K5E



FGP / WPMP
TENGIZCHEVROIL - TENGIZCHEVROIL
TCO
TENGIZ, KAZAKHSTAN

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
1780	C		Eccent	5392461	1 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 1/2 - 2												
1790			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3												
1800			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3												
1810			Concen	9006918	1 1/2 - 3	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4												
1820			Eccent	9001372	1 1/2 - 3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4												
1830			Concen	9006915	3 - 3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6 - 6												
1840			Eccent	9002510	3 - 3	REDUCER ECC WT 6.35 MM/STD LTCS A420 WPL6	6 - 6												
1850			Eccent	9002312	4 - 4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6 - 10												
1860			Eccent	9002266	6 - 8	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8 - 10												
1870			Eccent	9002269	6 - 10	REDUCER ECC WT 7.14 MM / 6.35 MM LTCS A860 WPHY-60	12 - 12												
1880			Eccent	9002272	6 - 10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	14 - 14												
1890			Eccent	9002270	12 - 12	REDUCER ECC WT 7.92 MM / 7.14 MM LTCS A860 WPHY-60	14 - 14												
1900			Eccent	9002292	8 - 8	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	16 - 16												
1910			Eccent	9002292	10 - 10	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	16 - 18												
1920			Eccent	9002294	12 - 12	REDUCER ECC WT 8.74 MM / 7.14 LTCS A860 WPHY-60	16 - 18												
1930			Eccent	9002293	14 - 14	REDUCER ECC WT 8.74 MM / 7.92 LTCS A860 WPHY-60	16 - 18												
1940			Eccent	9002291	16 - 16	REDUCER ECC WT 8.74 MM LTCS A860 WPHY-60	18 - 18												
1950			Eccent	9004134	12 - 12	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60-W	20 - 20												
1960			Eccent	9004135	14 - 14	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	20 - 20												
1970			Eccent	9004136	16 - 16	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	20 - 20												
1980			Eccent	9004136	18 - 18	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	20 - 20												
1990			Eccent	9004161	16 - 18	REDUCER ECC WT 11.13 MM / 8.74 MM LTCS A860 WPHY-60-W	24 - 24												
2000			Eccent	9004162	20 - 20	REDUCER ECC WT 11.13 MM / 9.53 MM LTCS A860 WPHY-60-W	24 - 24												
2010			Eccent	9004146	18 - 18	REDUCER ECC WT 11.91MM / 8.74MM LTCS A860 WPHY-60-W	28 - 28												
2020			Eccent	9004147	20 - 20	REDUCER ECC WT 11.91 MM / 9.53 LTCS A860 WPHY-60-W	28 - 28												
2030			Eccent	9004172	24 - 24	REDUCER ECC WT 11.91MM / 11.13MM LTCS A860 WPHY-60-W	28 - 28												
2040																			
2050							***** INDUCTION BENDS ABOVE GROUND*****												
2060																			
2070					90 Deg	9001810	4 - 4							INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS			
2080					90 Deg	9001813	6 - 10							INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW					
2090					90 Deg	9001814	12 - 12							INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW					
2100					90 Deg	9001815	14 - 14							INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW					
2110					90 Deg	9001816	16 - 18							INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW					
2120					90 Deg	9001817	20 - 20							INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW					
2130				90 Deg	9001899	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2140				90 Deg	9001397	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2150																			
2160				45 Deg	9001820	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2170				45 Deg	9001823	6 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2180				45 Deg	9001824	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2190				45 Deg	9001825	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2200				45 Deg	9001826	16 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2210				45 Deg	9001827	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2220				45 Deg	9005326	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2230				45 Deg	9001500	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
2240																			
2250				22.5 D	9001830	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 4 OF 8	PA 5	CLASS 300K5E					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																
2260	C		22.5 D	9001833	6 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW																		
2270			22.5 D	9001834	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW																		
2280			22.5 D	9001835	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW																		
2290			22.5 D	9001836	16 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW																		
2300			22.5 D	9001837	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW																		
2310			22.5 D	9005331	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW																		
2320			22.5 D	9002001	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW																		
2330						***** INDUCTION BENDS BELOW GROUND*****																		
2340																								
2350																								
2360	C		90 Deg	9001810CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2370			90 Deg	9001813CTD	6 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2380			90 Deg	9001814CTD	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2390			90 Deg	9001815CTD	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2400			90 Deg	9001816CTD	16 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2410			90 Deg	9001817CTD	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2420			90 Deg	9001899CTD	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2430			90 Deg	9001397CTD	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2440																								
2450			45 Deg	9001820CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2460	C		45 Deg	9001823CTD	6 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2470			45 Deg	9001824CTD	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2480			45 Deg	9001825CTD	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2490			45 Deg	9001826CTD	16 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2500			45 Deg	9001827CTD	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2510			45 Deg	9005326CTD	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2520			45 Deg	9004001CTD	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2530																								
2540			22.5 D	9001830CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2550			22.5 D	9001833CTD	6 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2560	C		22.5 D	9001834CTD	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2570			22.5 D	9001835CTD	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2580			22.5 D	9001836CTD	16 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2590			22.5 D	9001837CTD	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
2600			22.5 D	9005331CTD	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED																		
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION									CORR. ALLOW.	FGP/WPMP								
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND										DATE	BY	REV	SHEET	PA	CLASS			
															3 MM	17MAR17	JCS	C	5 OF 8	5	300K5E			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2610			22.5 D	9002001CTD	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2620														
2630						***** OLETS ABOVE AND BELOW GROUND*****								
2640														
2650	C		Weldol	9006400	3/4 - 3/4	WELDOLET WT 6.02 MM / SCH 160 LTCS A694 F60	4 - 4							
2660	C		Weldol	9006219	1 - 1 1/2	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60 IMPACT TESTED AT -46 DEG C	4 - 4							
2670	C		Weldol	9002450	3/4 - 3/4	WELDOLET WT 6.35 MM/SCH 160 WT LTCS A694 F60	6 - 10							
2680	C		Weldol	9006407	1 - 2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6 - 10							
2690	C		Weldol	9006409	3 - 3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8 - 10							
2700	C		Weldol	9006408	4 - 4	WELDOLET WT 6.35 MM/6.02 MM LTCS A694 F60	10 - 10							
2710	C		Weldol	9006817	3/4 - 3/4	WELDOLET WT 7.14 MM/ SCH 160 WT LTCS A694 F60	12 - 12							
2720	C		Weldol	9006413	1 - 2	WELDOLET WT 7.14 MM/XS WT LTCS A694 F60	12 - 12							
2730	C		Weldol	9006412	3 - 3	WELDOLET WT 7.14 MM/STD WT LTCS A694 F60	12 - 12							
2740			Weldol	9006411	4 - 4	WELDOLET WT 7.14 MM/6.02 MM LTCS A694 F60	12 - 12							
2750	C		Weldol	9006818	3/4 - 3/4	WELDOLET WT 7.92 MM/ SCH 160 WT LTCS A694 F60	14 - 14							
2760	C		Weldol	9002552	1 - 2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	14 - 14							
2770	C		Weldol	9006415	3 - 3	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	14 - 14							
2780			Weldol	9006417	4 - 4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	14 - 14							
2790	C		Weldol	9002455	3/4 - 3/4	WELDOLET WT 8.74MM/ SCH 160 LTCS A694 F60	16 - 18							
2800	C		Weldol	9006423	1 - 2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	16 - 18							
2810	C		Weldol	9006422	3 - 3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	16 - 18							
2820			Weldol	9006421	4 - 4	WELDOLET WT 8.74 MM/6.02 MM LTCS A694 F60	16 - 18							
2830			Weldol	9002008	6 - 6	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	18 - 18							
2840	C		Weldol	9002457	3/4 - 3/4	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	20 - 20							
2850	C		Weldol	9006429	1 - 2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	20 - 20							
2860	C		Weldol	9006428	3 - 3	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	20 - 20							
2870			Weldol	9006427	4 - 4	WELDOLET WT 9.53 MM/6.02 MM LTCS A694 F60	20 - 20							
2880			Weldol	9006424	6 - 6	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	20 - 20							
2890	C		Weldol	9002076	3/4 - 3/4	WELDOLET WT 11.13MM/ SCH 160 LTCS A694 F60	24 - 24							
2900	C		Weldol	9005119	1 - 2	WELDOLET WT 11.13 MM/ SCH XS WT LTCS A694 F60	24 - 24							
2910	C		Weldol	9005086	3 - 3	WELDOLET WT 11.13 MM/STD WT LTCS A694 F60	24 - 24							
2920	C		Weldol	9006802	4 - 4	WELDOLET WT 11.13 MM/ 6.02 MM WT LTCS A694 F60	24 - 24							
2930	C		Weldol	9005120	6 - 8	WELDOLET WT 11.13 MM/ 6.35 MM WT LTCS A694 F60	24 - 24							
2940			Weldol	9002004	4 - 4	WELDOLET WT 11.91MM/6.02MM LTCS A694 F60	28 - 28							
2950			Weldol	9002009	6 - 10	WELDOLET WT 11.91MM/6.35MM LTCS A694 F60	28 - 28							
2960														
2970	C		Nipofl	5504450	3/4 - 3/4	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2	2 - 2							
2980	C		Nipofl	5504451	3/4 - 3/4	NIPOFLANGE 300# RF SCH STD/SCH 160 LTCS A350 LF2	3 - 3							
2990	C		Nipofl	9006807	3/4 - 3/4	NIPOFLANGE 300# RF 6.02 MM/SCH 160 WT LTCS A694 F60	4 - 4							
3000	C		Nipofl	9002643	3/4 - 3/4	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6 - 10							
3010	C		Nipofl	9006809	3/4 - 3/4	NIPOFLANGE 300# RF 7.14 MM/SCH 160 WT LTCS A694 F60	12 - 12							
3020	C		Nipofl	9006810	3/4 - 3/4	NIPOFLANGE 300# RF 7.92 MM/SCH 160 WT LTCS A694 F60	14 - 14							
3030	C		Nipofl	9006811	3/4 - 3/4	NIPOFLANGE 300# RF 8.74 MM/SCH 160 WT LTCS A694 F60	16 - 18							
3040	C		Nipofl	9006812	3/4 - 3/4	NIPOFLANGE 300# RF 9.53 MM/SCH 160 WT LTCS A694 F60	20 - 20							
3050	C		Nipofl	9006815	3/4 - 3/4	NIPOFLANGE 300# RF 11.13 MM/SCH 160 WT LTCS A694 F60	24 - 24							
3060	C		Nipofl	9006814	3/4 - 3/4	NIPOFLANGE 300# RF 11.91 MM/SCH 160 WT LTCS A694 F60	28 - 28							
3070			Nipofl	5504020	1 - 1	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2	2 - 2							
3080			Nipofl	5504021	1 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 3							
3090			Nipofl	9002639	1 - 2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4							
3100			Nipofl	9002643	1 - 2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6 - 10							
3110			Nipofl	9002644	1 - 2	NIPOFLANGE 300# RF 7.14 MM/XS WT LTCS A694 F60	12 - 12							
3120			Nipofl	9002645	1 - 2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	14 - 14							
3130			Nipofl	9002646	1 - 2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	16 - 18							
3140			Nipofl	9002647	1 - 2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	20 - 20							
3150	C		Nipofl	9002649	1 - 2	NIPOFLANGE 300# RF 11.13 MM/XS WT LTCS A694 F60	24 - 24							
3160			Nipofl	9002010	1 - 2	NIPOFLANGE 300# RF 11.91 MM/XS WT LTCS A694 F60	28 - 28							
3170														
3180	C	FF5	Endole	9504524	1 - 1	NIPOFLANGE FLAT BOTTOM 300# RF SCH XS LTCS A694 F60								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 6 OF 8 PA 5 CLASS 300K5E					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3190	C		Flange	5556480	1/2 - 3/4	***** PIPE FLANGES ABOVE & BELOW GROUND*****	3/4 - 1 1/2						
3200						FLG WN 300# RF LTCS A350 LF2 SCH 160							
3210						FLG WN 300# RF LTCS A350 LF2 SCH XS							
3220						FLG WN 300# RF LTCS A350 LF2 SCH STD							
3230						FLG WN 300# RF LTCS A694 F60 WT 6.02MM							
3240						FLG WN 300# RF LTCS A694 F60 WT 6.35MM							
3250						FLG WN 300# RF LTCS A694 F60 WT 7.14MM							
3260						FLG WN 300# RF LTCS A694 F60 WT 7.92MM							
3270						FLG WN 300# RF LTCS A694 F60 WT 8.74MM							
3280						FLG WN 300# RF LTCS A694 F60 WT 9.53MM							
3290	C		Flange	9001458	24 - 24	FLG WN 300# RF LTCS A694 F60 WT 11.13MM	3 - 28						
3300						FLG WN 300# RF LTCS A694 F60 WT 11.91MM SERIES A							
3310						FLG WN 300# RF LTCS A694 F60 WT 11.91MM SERIES A							
3320						FLG BLIND 300# RF LTCS A350 LF2							
3330						FLG BLIND 300# RF LTCS A694 F60							
3340						FLG BLIND 300# RF LTCS A694 F60 SERIES A							
3350													
3360													
3370													
3380													
3390	C	FS1	Flange	9003484	16 - 16	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM	3 - 28						
3400	C	FS1	Flange	9003485	20 - 20	FLG WN 600# RF LTCS A694 F60 WT 9.53 MM							
3410	C	FS1	Flange	9003487	24 - 24	FLG WN 600# RF LTCS A694 F60 WT 11.13 MM							
3420	C		Flange	5851203	1/2 - 1/2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE							
3430													
3440						***** GASKETS *****							
3450													
3460	C		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR							
3470			GASFRS	5674032	28 - 28	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
3480													
3490	C		GASFRS	5673103	16 - 24	GASKET 600# 316SS SP WND GRAPHITE IR	3 - 28						
3500													
3510		B1				***** BOLTING *****							
3520													
3530	C		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3540													
3550		SB6	CPSBOL	5675675	12 - 28	JACK SCREW HEX HEAD A320 GR L7M							
3560													
3570													
3580						***** BLINDS & SPACERS							
3590							3 - 28						
3600			Specta	5749581	3/4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70							
3610			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70							
3620			Paddle	5750028	28 - 28	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A							
3630			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70							
3640			Paddle	5750198	28 - 28	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A							
3650													
3660						***** BRANCHES *****							
3670													
3680						USE BRANCH CHART 300K5E							
3690							3 - 28						
3700	C		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN							
3710	C		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN							
3720													
3730													
3740													
3750													
3760													
3770													
3780													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 7 OF 8	PA 5	CLASS 300K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3790	C		BOLS	5675669	1/4 - 99	*****DO NOT USE*****							
3800						STUD-BOLT A320 GR L7M/A194 GR 7M NUTS							
3810													
3820													
3830													
3840													
3850	C		Reduci	9006752	3 - 3	*****NOTE***** THE BELOW ITEMS HAVE BEEN REMOVED FROM THE LINE CLASS DUE TO INSTRUCTION NOT TO WELD DISSIMILAR METALS OR WALL THICKNESSES BEING UPDATED AND ITEMS ALREADY PLACED *****NOTE*****	4 - 4						
3860													
3870													
3880													
3890													
3900	C		Weldol	9006434	3 - 3	WELDOLET WT 10.31 MM/STD WT LTCS A694 F60	24 - 24						
3910													
3920	C		Nipofl	9002648	1 - 2	NIPOFLANGE 300# RF 10.31 MM/XS WT LTCS A694 F60	24 - 24						
3930													
3940													
3950													
3960													

DATE	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10 20 30	C		2BV	K03AC02	3/4 - 2	***** MODULAR VALVES ***** MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		B1 SB6	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
40 50	C	VAV	2BV	K03AC02	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		VAV VAW	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055						
60 70	C		2BVR	M03AC02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		ZA6	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
80 90	C	VAW	R2BVR	M03EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2								
100 110	C	VAW	R2BVR	M03EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3								
120 130 140 150 160 170 180 190 200 210 220 230 240 250 260						***** BALL VALVES ***** BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
170 180 190 200 210 220 230 240 250 260	C		Ball V	B03AC01	3/4 - 3	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
180 190 200 210 220 230 240 250 260	C		Ball V	B03AC03	4 - 4	BALL 300# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
190 200 210 220 230 240 250 260	C		Ball V	B03AC07	6 - 24	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
220 230 240 250 260	C		Ball V	B03JC07	28 - 28	BALL 300# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A									
270 280 290 300 310	C		Condui	W03AC70	3 - 8	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW									
280 290 300 310	C		Condui	W03AC71	10 - 24	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO									
300 310 320 330 340			Condui	W03JC71	28 - 28	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A									
320 330 340						***** PIPE ABOVE GROUND*****									
350 360 370 380 390 400 410 420 430 440 450 460 470 480	C		Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE			SERVICE						
360 370 380 390 400 410 420 430 440 450 460 470 480			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE			SOUR SERVICE - NACE MR0175/ ISO 15156,						
370 380 390 400 410 420 430 440 450 460 470 480			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE			DESIGN: ASME B16.5 MG 1.1						
380 390 400 410 420 430 440 450 460 470 480			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE			CODE: ASME B31.8						
390 400 410 420 430 440 450 460 470 480			Piping	9001254	6 - 10	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE			ABOVE GROUND FACTOR = 0.5						
400 410 420 430 440 450 460 470 480			Piping	9001256	12 - 12	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE			BELOW GROUND FACTOR = 0.6						
410 420 430 440 450 460 470 480			Piping	9001311	14 - 16	LINEPIPE SMLS WT=7.92MM API-5L X60 BE			MAX. DESIGN PARAMETER = 40 BARG. @ 80						
420 430 440 450 460 470 480			Piping	9001485	18 - 18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE			DEG.C. (LIMITED BY PIPE WALL THICKNESS)						
430 440 450 460 470 480			Piping	9001493	20 - 20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE			SERVICE - PHC - MULTIPHASE						
440 450 460 470 480			Piping	9001494	24 - 24	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE									
450 460 470 480			Piping	9001490	28 - 28	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE									
460 470 480						*****PIPE BELOW GROUND*****									
490 500 510 520 530 540			Piping	9001257CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED			MATERIAL: LOW TEMP CARBON STEEL						
500 510 520 530 540			Piping	9001254CTD	6 - 10	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED			FLANGE RATING: 300 RF						
510 520 530 540			Piping	9001256CTD	12 - 12	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE COATED			TEMPERATURE RANGE: -40 / 80 °C						
520 530 540			Piping	9001311CTD	14 - 16	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED			COMMENT: LTCS						
530 540			Piping	9001487CTD	18 - 18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED									
540			Piping	9001493CTD	20 - 20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 1 OF 4 PA 5 CLASS 300K5F						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10	C		Piping	9001493CTD	24 - 24	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED									
560			Piping	9001490CTD	28 - 28	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE COATED									
570															
580						*****FITTINGS ABOVE AND BELOW GROUND*****									
590															
600			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6									
610			90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6									
620			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6									
630			45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6									
640															
650			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6									
660			Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6									
670															
680			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2								
690			Reduci	5392454	3/4 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3								
700			Reduci	9006017	6 - 6	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	16 - 16								
710			Reduci	9004149	8 - 8	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W	20 - 20								
720															
730			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6									
740			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6									
750			Cap	9005062	4 - 4	CAP BW WT 6.02 MM LTCS A860 WPHY-60									
760			Cap	9005055	6 - 10	CAP BW WT 6.35MM LTCS A860 WPHY-60									
770			Cap	9005057	12 - 12	CAP BW WT 7.14 MM LTCS A860 WPHY-60									
780			Cap	9005058	14 - 16	CAP BW WT 7.92 MM LTCS A860 WPHY-60									
790			Cap	9005059	18 - 18	CAP BW WT 8.74 MM LTCS A860 WPHY-60									
800			Cap	9005060	20 - 20	CAP BW WT 9.53 MM LTCS A860 WPHY-60									
810			Cap	9005064	24 - 24	CAP BW WT 11.13 MM LTCS A860 WPHY-60									
820			Cap	9005065	28 - 28	CAP BW WT 11.91 MM LTCS A860 WPHY-60									
830															
840			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2								
850			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2								
860			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3								
870			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3								
880			Eccent	9001372	1 1/2 - 2	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4								
890			Eccent	9002281	3 - 3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4 - 4								
900			Eccent	9002012	3 - 3	REDUCER ECC WT 6.35 MM / 5.49 MM LTCS A860 WPHY-60	6 - 6								
910			Eccent	9002312	4 - 4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6 - 10								
920			Eccent	9002266	6 - 8	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8 - 10								
930			Eccent	9002269	6 - 10	REDUCER ECC WT 7.14 MM / 6.35 MM LTCS A860 WPHY-60	12 - 12								
940			Eccent	9002272	6 - 10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	14 - 14								
950			Eccent	9002272	8 - 10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	16 - 16								
960			Eccent	9002278	14 - 14	REDUCER ECC WT 7.92MM LTCS A860 WPHY-60	16 - 16								
970			Eccent	9002292	10 - 10	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	18 - 18								
980			Eccent	9002294	12 - 12	REDUCER ECC WT 8.74 MM / 7.14 LTCS A860 WPHY-60	18 - 18								
990			Eccent	9002250	14 - 16	REDUCER ECC WT 8.74 MM / 7.92 MM LTCS A860 WPHY-60	18 - 18								
1000			Eccent	9004134	12 - 12	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60-W	20 - 20								
1010			Eccent	9004135	14 - 16	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	20 - 20								
1020			Eccent	9004136	18 - 18	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	20 - 20								
1030			Eccent	9004144	16 - 16	REDUCER ECC WT 10.31 MM / 7.92 MM LTCS A860 WPHY-60-W	24 - 24								
1040			Eccent	9004145	18 - 18	REDUCER ECC WT 10.31 MM / 8.74 MM LTCS A860 WPHY-60-W	24 - 24								
1050			Eccent	9004137	20 - 20	REDUCER ECC WT 10.31 MM / 9.53 MM LTCS A860 WPHY-60-W	24 - 24								
1060			Eccent	9004146	18 - 18	REDUCER ECC WT 11.91MM / 8.74MM LTCS A860 WPHY-60-W	28 - 28								
1070			Eccent	9004147	20 - 20	REDUCER ECC WT 11.91 MM / 9.53 LTCS A860 WPHY-60-W	28 - 28								
1080			Eccent	9004148	24 - 24	REDUCER ECC WT 11.91 MM / 10.31 LTCS A860 WPHY-60-W	28 - 28								
1090															
1100															
1110															
1120															
1130															
					90 Deg	9001810	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM	DATE 17MAR17		BY JCS	REV C	SHEET 2 OF4	PA 5	CLASS 300K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10	C		90 Deg	9001813	6 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1150	C		90 Deg	9001814	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1160	C		90 Deg	9001815	14 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1170	C		90 Deg	9001816	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1180	C		90 Deg	9001817	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1190	C		90 Deg	9001818	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1200	C		90 Deg	9001397	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1210													
1220	C		45 Deg	9001820	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1230	C		45 Deg	9001823	6 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1240	C		45 Deg	9001824	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1250	C		45 Deg	9001825	14 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1260	C		45 Deg	9001826	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1270	C		45 Deg	9001827	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1280	C		45 Deg	9001828	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1290	C		45 Deg	9001500	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1300													
1310	C		22BND	9001830	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1320	C		22BND	9001833	6- 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1330	C		22BND	9001834	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1340	C		22BND	9001835	14 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1350	C		22BND	9001836	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1360	C		22BND	9001837	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1370	C		22BND	9001838	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1380	C		22BND	9002001	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L							
1390						***** INDUCTION BENDS BELOW GROUND*****							
1400													
1410													
1420	C		90 Deg	9001810CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1430	C		90 Deg	9001813CTD	6- 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1440	C		90 Deg	9001814CTD	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BWCOATED							
1450	C		90 Deg	9001815CTD	14 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BWCOATED							
1460	C		90 Deg	9001816CTD	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1470	C		90 Deg	9001817CTD	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP						
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							3 MM	17MAR17	JCS	C	3 OF 4	5	300K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10	C		90 Deg	9001818CTD	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1490	C		90 Deg	9001397CTD	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1500													
1510	C		45 Deg	9001820CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1520	C		45 Deg	9001823CTD	6 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1530	C		45 Deg	9001824CTD	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1540	C		45 Deg	9001825CTD	14 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1550	C		45 Deg	9001826CTD	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1560	C		45 Deg	9001827CTD	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1570	C		45 Deg	9001828CTD	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1580	C		45 Deg	9001500CTD	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1590													
1600	C		22BND	9001830CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1610	C		22BND	9001833CTD	6- 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1620	C		22BND	9001834CTD	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1630	C		22BND	9001835CTD	14 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1640	C		22BND	9001836CTD	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1650	C		22BND	9001837CTD	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1660	C		22BND	9001838CTD	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1670	C		22BND	9002001CTD	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L							
1680						*****OLETS ABOVE & BELOW GROUND WT*****							
1690													
1700			Weldol	9006407	3/4 - 2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6 - 10	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX DESIGN PARAMETER = 47.5 BARG. @ 80 DEG.C (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE					
1710			Weldol	9006409	3 - 3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8 - 10						
1720			Weldol	9006413	3/4 - 2	WELDOLET WT 7.14 MM/XS WT LTCS A694 F60	12 - 12						
1730			Weldol	9006412	3 - 4	WELDOLET WT 7.14 MM/STD WT LTCS A694 F60	12 - 12						
1740			Weldol	9002552	3/4 - 2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	14 - 16						
1750			Weldol	9006415	3 - 4	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	14 - 16						
1760			Weldol	9006423	3/4 - 2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	18 - 18						
1770			Weldol	9006422	3 - 4	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	18 - 18						
1780			Weldol	9002008	6 - 6	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	18 - 18						
1790			Weldol	9006429	3/4 - 2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	20 - 20						
1800			Weldol	9006428	3 - 4	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	20 - 20						
1810			Weldol	9006424	6 - 6	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	20 - 20						
1820			Weldol	9002551	3/4 - 2	WELDOLET WT 10.31MM/ XS WT LTCS A694 F60	24 - 24						
1830			Weldol	9006434	3 - 4	WELDOLET WT 10.31 MM/STD WT LTCS A694 F60	24 - 24						
1840			Weldol	9006430	6 - 8	WELDOLET WT 10.31 MM/6.35 MM LTCS A694 F60	24 - 24						
1850			Weldol	9002003	3/4 - 2	WELDOLET WT 11.91MM/SCH XS LTCS A694 F60	28 - 28						
1860			Weldol	9002014	3 - 3	WELDOLET WT 11.91MM/5.49MM LTCS A694 F60	28 - 28						
1870			Weldol	9002004	4 - 4	WELDOLET WT 11.91MM/6.02MM LTCS A694 F60	28 - 28						
1880			Weldol	9002009	6 - 10	WELDOLET WT 11.91MM/6.35MM LTCS A694 F60	28 - 28						
1890													
1900			Flange	5504021	3/4 - 2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2	3 - 3	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80°C COMMENT: LTCS					
1910			Flange	9002639	3/4 - 2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4						
1920			Flange	9002643	3/4 - 2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6 - 10						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 4 OF 6 PA 5 CLASS 300K5F					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10	C		Flange	9002644	3/4 - 2	NIPOFLANGE 300# RF 7.14 MM/XS WT LTCS A694 F60	12 - 12							
1940			Flange	9002645	3/4 - 2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	14 - 16							
1950			Flange	9002646	3/4 - 2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	18 - 18							
1960			Flange	9002647	3/4 - 2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	20 - 20							
1970			Flange	9002648	3/4 - 2	NIPOFLANGE 300# RF 10.31 MM/XS WT LTCS A694 F60	24 - 24							
1980			Flange	9002010	3/4 - 2	NIPOFLANGE 300# RF 11.91 MM/XS WT LTCS A694 F60	28 - 28							
1990														
2000						*****PIPE FLANGES ABOVE GROUND*****								
2010														
2020			Flange	5556476	1/2 - 2	FLG WN 300# RF LTCS A350 LF2 SCH XS								
2030			Flange	5556473	3 - 3	FLG WN 300# RF LTCS A350 LF2 SCH STD								
2040			Flange	9001480	4 - 4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM								
2050			Flange	9001483	6 - 10	FLG WN 300# RF LTCS A694 F60 WT 6.35MM								
2060			Flange	9001484	12 - 12	FLG WN 300# RF LTCS A694 F60 WT 7.14MM								
2070			Flange	9001455	14 - 16	FLG WN 300# RF LTCS A694 F60 WT 7.92MM								
2080			Flange	9001457	18 - 18	FLG WN 300# RF LTCS A694 F60 WT 8.74MM								
2090			Flange	9001458	20 - 20	FLG WN 300# RF LTCS A694 F60 WT 9.53MM								
2100			Flange	9001460	24 - 24	FLG WN 300# RF LTCS A694 F60 WT 10.31MM								
2110			Flange	9002011	28 - 28	FLG WN 300# RF LTCS A694 F60 WT 11.91MM SERIES A								
2120			Blind	5615576	3/4 - 3	FLG BLIND 300# RF LTCS A350 LF2								
2130			Blind	9001450	4 - 24	FLG BLIND 300# RF LTCS A694 F60								
2140			Blind	9001451	28 - 28	FLG BLIND 300# RF LTCS A694 F60 SERIES A								
2150														
2160				*****PIPE FLANGES BELOW GROUND*****										
2170														
2180														
2190	Flange	9001480	4 - 4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM										
2200	Flange	9001483	6 - 12	FLG WN 300# RF LTCS A694 F60 WT 6.35MM										
2210	Flange	9001484	14 - 16	FLG WN 300# RF LTCS A694 F60 WT 7.14MM										
2220	Flange	9001455	18 - 18	FLG WN 300# RF LTCS A694 F60 WT 7.92MM										
2230	Flange	9001457	20 - 20	FLG WN 300# RF LTCS A694 F60 WT 8.74MM										
2240	Flange	9001458	24 - 24	FLG WN 300# RF LTCS A694 F60 WT 9.53MM										
2250	Flange	9001459	28 - 28	FLG WN 300# RF LTCS A694 F60 WT 10.31MM SERIES A										
2260														
2270				***** GASKETS *****										
2280														
2290														
2300	C		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR								
2310			GASFRS	5674032	28 - 28	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A								
2320	B1					***** BOLTING *****	SERVICE							
2330							SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE							
2340	C													
2350			BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
2360	SB6													
2370			BOLS	5675675	12 - 28	JACK SCREW HEX HEAD A320 GR L7M								
2380						***** BLINDS & SPACERS								
2390														
2400														
2410														
2420			Specta	5749581	3/4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70								
2430			Paddle	5750100	12 - 24	PADDLE BLANK 300# RF LTCS A516 GR.70								
2440			Paddle	5750028	28 - 28	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A								
2450			Paddle	5750200	12 - 24	PADDLE SPACER 300# RF LTCS A516 GR.70								
2460	Paddle	5750198	28 - 28	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A										
2470														
2480				***** BRANCHES *****										
2490														
2500				USE BRANCH CHART 300K5F										
2510														
2520														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 5 OF 6	PA 5	CLASS 300K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 2540 2550 2560 2570 2580 2590 2600 2610	C C	ZA6	Reinfo Reinfo	RWELD RWELD	2 - 20 2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- REQUIRED PIPE BENDS ----- INDUCTION BENDS THREAD SEALANT ----- PTFE PASTE	3 - 28 3 - 28							
								SERVICE						
								SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE						
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C COMMENT: LTCS						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	FGP/WPMP SHEET 6 OF 6	PA 5	CLASS 300K5F

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10	C	PP9	Ball V	M03AM10	3/4 - 2	*****VALVES*****	8 - 8	B1 FA8 PP9	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
20						*****MODULAR VALVES*****								
30						MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM								
40						625 METAL LO - CONTROLLED BORE TYPE 2BVR								
50						***** BALL VALVES *****								
60														
70														
80														
90														
100						Ball V								B03AM10
110			BALFLG	A03AM10	BALL 300# RF INC 825 INC TRIM 625 FLOATING METAL RB LO									
120			Ball V	A03AR16	BALL 300# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION METAL RB GO									
130														
140						*****THROUGH CONDUIT GATE VALVES*****								
150														
160			C	Condui	W03AM71	10 - 10								GATE 300# RF INCONEL 825 N08825 THROUGH COND SOLID SLAB GO
170														
180				Condui	W03AM71	16 - 16								GATE 300# RF INCONEL 825 N08825 THROUGH COND SOLID SLAB GO
190														
200														
210														
220						*****GATE VALVES*****								
230														
240			C	Gate V	G03AM49	3/4 - 2								GATE 300# RF INC 825 INC 625 TRIM WEDGE HW
250														***** CHECK VALVES *****
260														
270				Check	C03AR46	8 - 8								CHECK DUAL WAFER LUG 300# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED
280														
290														
300			C											***** PIPE *****
310														
320				Piping	5361607	3/4 - 3								PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 BE
330				Piping	9002671	6 - 8								PIPE STD WT SMLS API 5L X60 WITH 3 MM INCONEL 825 INTERNAL CLADDING
340				Piping	9002671	16 - 16								PIPE STD WT SMLS API 5L X60 WITH 3 MM INCONEL 825 INTERNAL CLADDING
350														
360														
370														
380				90 Deg	5425111	3/4 - 3								ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825
390				90 Deg	9002672	6 - 8								ELL BW 90 DEG LR STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING
400				90 Deg	9002672	16 - 16								ELL BW 90 DEG LR STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING
410														
420				45 Deg	5425113	3/4 - 2								ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825
430				45 Deg	9002673	8 - 8								ELL BW 45 DEG LR STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING
440														
450				Tee	9002674	8 - 8								TEE BW STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING
460				Tee	9002674	16 - 16								TEE BW STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING
470				Reduci	9002698	6 - 6								TEE RED BW SCH STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 0 MM	DATE 17MAR17	BY TC	REV C	SHEET 1 OF 2	PA 5	CLASS 300K5G	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
480	C	FA8	Reduci	9002698	6 - 8	TEE RED BW SCH STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING	16 - 16							
490			Cap	9002675	8 - 8	CAP BW STD WT A860 WPHY60 WITH 3 MM INCONEL 825 INTERNAL CLADDING								
500														
510						***** OLETS *****								
520														
530			Flange	9002690	3/4 - 2	NIPOFLANGE 300# RF SCH STD/80S INCONEL 825 N08825	8 - 8							
540														
550			Flange	9002690	3/4 - 2	NIPOFLANGE 300# RF SCH STD/80S INCONEL 825 N08825	16 - 16							
560			Weldol	9002677	3/4 - 3	WELDOLET STD WT/80S INCONEL 825 N08825	8 - 8							
570														
580			Weldol	9002677	3/4 - 2	WELDOLET STD WT/80S INCONEL 825 N08825	16 - 16							
590						**** FLANGES ****								
600														
610			Reduci	5600656	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF SS A182 F316/316L	3/4 - 1 1/2							
620														
630			Flange	9002691	1/2 - 3	FLG WN 300# RF SCH 80S INCONEL 825 UNS N08825								
640			Flange	9002692	6 - 8	FLG WN 300# RF STD WT A694 F60 WITH 3 MM INCONEL 825 CLAD FACE & BORE								
650			Flange	9002692	16 - 16	FLG WN 300# RF STD WT A694 F60 WITH 3 MM INCONEL 825 CLAD FACE & BORE								
660						FLG BLIND 300# RF A694 F60 WITH 3 MM INCONEL 825 CLAD FACE								
670														
680			Blind	9002693	1/2 - 16									
690														
700						***** PIPE NIPPLES *****								
710														
720						NIPPLE PBE - CUT FROM PIPE								
730														
740						***** GASKETS *****								
750														
760			GASFRS	5673657	1/2 - 16	GASKET 300# ALLOY 825 SPIRAL WOUND GRAPHITE IR								
770														
780						***** BOLTING *****								
790														
800	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS										
810														
820				***** BLINDS AND SPACERS *****										
830														
840	Specta	5749683	3/4 - 3	SPECTACLE BLIND 300# RF ALLOY 825 B424 UNS N08825										
850						Specta	5749581_C80	4 - 10	SPECTACLE BLIND 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE					
860	Paddle	5750100_C80	16 - 16	PADDLE BLANK 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE										
870	Paddle	5750200_C80	16 - 16	PADDLE SPACER 300# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE										
880														
890				*****BRANCHES*****										
900														
910				USE BRANCH CHART 300K5G										
920														
930				***** CONSTRUCTION DETAILS *****										
940														
950				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY										
960														
970	C			PIPE BENDS ----- NONE										
980						THREAD SEALANT ----- PTFE PASTE								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 0 MM	DATE 17MAR17	BY TC	REV C	FGP/WPMP SHEET 2 OF 2		PA 5	CLASS 300K5G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
10 20 30 40	D	PP9	2BV	K03AS22	1/2 - 2	*****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. DELETED										
50 60	D	VAV	2BV	K03AS22	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		800	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.										
70 80	D		2BVR	M03AS22	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR		814 B1 FA4 FA8 FF5	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. DELETED USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY.										
90 100	D	VAW	R2BVR	M03ES22	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.										
110 120	D	VAW	R2BVR	M03ES22	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	PP9 SB6	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.										
140 150 160 170						*****BALL VALVES*****													
180		VB2	Ball V	A03AS21	3 - 4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO 718 BOLT													
190			Ball V	A03AS22	3 - 4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL RB LO 718 BOLT													
200 210		VB2	Ball V	A03AS27	6 - 8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT													
220 230 240 250 260			Ball V	A03AS28	6 - 8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT													
270	D	VB2	Ball V	B03AS21	1/2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT													
280 290 300	D		Ball V	B03AS22	1/2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT													
310 320		VB2 VA5	Ball V	B03AS21	3 - 3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT													
330 340		VA5	Ball V	B03AS22	3 - 3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT													
350 360	D	VB2 VA5	Ball V	B03AS23	4 - 4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT			SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3										
370 380	D	VA5	Ball V	B03AS24	4 - 4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT			AA, BD, BF, BFC, BV, CCD, CDE (90 DEG C AND ABOVE), CDM (70 DEG C AND ABOVE), CFA, CI, DL, FG, FGH, FX, GE, GS, NL, PHC, PL, VA, VI, VP, WDM, WRA, WSR										
390 400		VB2 VA5	Ball V	B03AS27	6 - 8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT													
410 420		VB2 VA5	Ball V	B03AS27	10 - 24	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT													
430 440		VA5	Ball V	B03AS28	6 - 24	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT													
450 460 470	C		Butter	T03AS65	10 - 24	*****BUTTERFLY VALVES***** BUTTERFLY 300# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO 718 BOLT			MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 300 RF TEMPERATURE RANGE: -100 / 180 °C COMMENT: LOW TEMP BLOW DOWN										
480 490 500 510 520			Globe	L03AS55	3/4 - 1 1/2	*****GLOBE VALVES***** GLOBE 300# RF SS A182 F316 TRIM 16 OS&Y HW 718 BOLT													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD		CORR. ALLOW. 1 MM		FGP/WPMP DATE 17MAR17					BY MC	REV D	SHEET 1 OF 4	PA 5	CLASS 300P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
530			Globe	L03AS56	2 - 6	GLOBE 300# RF SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT		SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.					
540			Globe	L03AS57	8 - 10	GLOBE 300# RF SS A351 CF8M TRIM 16 OS&Y GO 718 BOLT		UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.					
550						*****CHECK VALVES*****		UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST					
560								UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FORDBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
570								VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
580								VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY					
590	VC1		Check	C03AS54	3/4 - 1 1/2	CHECK PISTON 300# RF SS A182 F316 TRIM 16 718 BOLT		VAF	FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY.					
600	VC2		Check	C03AS45	2 - 10	CHECK DUAL WAFER LUG 300# RF SS A351 CF8M TRIM 16 THROU BOLTED		VAG	FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY.					
610								VAM	FOR INSTRUMENT CONNECTIONS IN SOUR SERVICE ONLY.					
620								VAN	FOR LEVEL TRANSMITTER CONNECTIONS IN SOUR SERVICE ONLY.					
630								VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.					
640								VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055					
650								VB2	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
660								VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
670								VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
680														
690														
700	814		Piping	5365207	1/2 - 1/2	PIPE SCH 160 SMLS SS A312 TP316/316L PE								
710						***** FITTINGS *****								
720														
730														
740	C		90 Deg	5449033	3/4 - 1 1/2	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S								
750			90 Deg	5449138	2 - 12	ELL 90 DEG LR SCH 40S SS A403 WP316/316L-S								
760			90 Deg	5449032	14 - 24	ELL 90 DEG LR SCH 40 SS A403 WP316/316L-S								
770														
780	C		45 Deg	5449146	3/4 - 1 1/2	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S								
790			45 Deg	5449144	2 - 12	ELL 45 SCH 40S SS A403 WP316/316L-S								
800			45 Deg	5449038	14 - 24	ELL 45 SCH 40 SS A403 WP316/316L-S								
810														
820	C		Tee	5449156	3/4 - 1 1/2	TEE SCH 80S SS A403 WP316/316L-S								
830			Tee	5449147	2 - 12	TEE SCH 40S SS A403 WP316/316L-S								
840			Tee	5449045	14 - 24	TEE SCH 40 SS A403 WP316/316L-S								
850														
860	C		Reduci	5448905	3/4 - 1	TEE RED SCH 80S SS A403 WP316/316L-S	1 - 1 1/2							
870			Reduci	5449250	1 1/2 - 1 1/2	TEE RED SCH 40S/80S SS A403 WP316/316L-S	2 - 3							
880			Reduci	5449184	2 - 10	TEE RED SCH 40S SS A403 WP316/316L-S	3 - 12							
890			Reduci	5448897	6 - 12	TEE RED SCH 40/40S SS A403 WP316/316L-S	14 - 24							
900			Reduci	5449046	14 - 20	TEE RED SCH 40 SS A403 WP316/316L-S	16 - 24							
910														
920	C		Cap	5449049	3/4 - 1 1/2	CAP SCH 80S SS A403 WP316/316L-S								
930			Cap	5449148	2 - 12	CAP SCH 40S SS A403 WP316/316L-S								
940			Cap	5449047	14 - 24	CAP SCH 40 SS A403 WP316/316L-S								
950														
960	C		Concen	5449158	3/4 - 1	REDUCER CONC SCH 80S SS A403 WP316/316L-S	1 - 1 1/2							
970			Concen	5549110	3/4 - 1 1/2	REDUCER CONC SCH 40S/80S SS A403 WP316/316L-S	2 - 4							
980			Concen	5449149	2 - 10	REDUCER CONC SCH 40S SS A403 WP316/316L-S	3 - 12							
990			Concen	5448902	6 - 12	REDUCER CONC SCH 40/40S SS A403 WP316/316L-S	14 - 20							
1000			Concen	5448903	14 - 20	REDUCER CONC SCH 40 SS A403 WP316/316L-S	16 - 24							
1010														
1020	C		Eccent	5449159	3/4 - 1	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1 - 1 1/2							
1030			Eccent	5549310	3/4 - 1 1/2	REDUCER ECC SCH 40S/80S SS A403 WP316/316L-S	2 - 4							
1040			Eccent	5449150	2 - 10	REDUCER ECC SCH 40S SS A403 WP316/316L-S	3 - 12							
1050			Eccent	5449154	6 - 12	REDUCER ECC SCH 40/40S SS A403 WP316/316L-S	14 - 20							
1060			Eccent	5449155	14 - 20	REDUCER ECC SCH 40 SS A403 WP316/316L-S	16 - 24							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD		CORR. ALLOW. 1 MM	FGP/WPMP DATE 17MAR17 BY MC REV D SHEET 2 OF 4 PA 5 CLASS 300P21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1070	C	FF5	Flange	5504515	3/4 - 1 1/2	NIPOFLANGE FLAT BOTTOM 300# RF SCH 160 LTCS A350 LF2		VU2 ZA6	USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
1080														
1090	C		Nipofl	5504411	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/80S SS A182 F316/316L	2 - 12							
1100	C		Nipofl	5504421	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 40/80S SS A182 F316/316L	14 - 24							
1110			Nipofl	5504032	2 - 2	NIPOFLANGE 300# RF SCH 40S/40S SS A182 F316/316L	2 - 12							
1120			Nipofl	5504033	2 - 2	NIPOFLANGE 300# RF SCH 40/40S SS A182 F316/316L	14 - 24							
1130														
1140		UJ2	Nipofl	5504132	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4 - 12							
1150		UJ2	Nipofl	5504142	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40/160 SS A182 F316/316L	14 - 24							
1160														
1170	C		Weldol	5510354	3/4 - 1 1/2	WELDOLET SCH 40S/80S SS A182 F316/316L	2 - 12							
1180	C		Weldol	5510364	3/4 - 1 1/2	WELDOLET SCH 40/80S SS A182 F316/316L	14 - 24							
1190			Weldol	5509631	2 - 4	WELDOLET SCH 40S/40S SS A182 F316/316L	6 - 12							
1200			Weldol	5509636	2 - 8	WELDOLET SCH 40/40S SS A182 F316/316L	14 - 24							
1210														
1220														
1230		UJ2	Elbole	5516152	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4 - 12							
1240		UJ2	Elbole	5516158	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40/160 SS A182 316/316L	14 - 24							
1250														
1260	D		Coupli	5480142	3/4 - 1 1/2	COUPLING 3000# SW SS A182 F316/316L								
1270	D		Reduci	5480134	3/4 - 1	COUPLING RED 3000# SW SS A182 F316/316L	1 - 1 1/2							
1280														
1290	D	UD7	Plug	5498383	3/4 - 3/4	PLUG ROUND HEAD SCRD SS A182 F316/316L								
1300														
1310	D		Sockol	5537071	3/4 - 1 1/2	SOCKOLET 3000# SS A182 F316/316L	2 - 24							
1320	D		Endole	5533235	3/4 - 1 1/2	FLAT SOCKOLET 3000# SS A182 F316/316L								
1330														
1340	D	UD7	Thredo	5515771	3/4 - 3/4	THREDOLET 3000# SS A182 F316/316L	2 - 24							
1350														
1360						***** PIPE FLANGES *****								
1370														
1380	C		Flange	5851203	1/2 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE								
1390														
1400			Flange	5582991	1/2 - 1 1/2	FLG WN 300# RF SS A182 F316/316L SCH 80S BORE								
1410			Flange	5582995	2 - 12	FLG WN 300# RF SS A182 F316/316L SCH 40S BORE								
1420			Flange	5582997	14 - 24	FLG WN 300# RF SS A182 F316/316L SCH 40 BORE								
1430														
1440	C	FS1	Flange	5583535	3 - 6	FLG WN 600# RF SS A182 F316/316L SCH 40S BORE								
1450														
1460	C	FA8	Reduci	5600656	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 300# RF SS A182 F316/316L	3/4 - 1 1/2							
1470														
1480	C		Blind	5616529	1/2 - 24	FLG BLIND 300# RF SS A182 F316/316L								
1490														
1500						***** ORIFICE FLANGES *****								
1510														
1520			Orific	5640598	2 - 12	WN ORIF SET 300# RF SS A182 F316/316L SCH 40S BORE								
1530			Orific	5640599	14 - 24	WN ORIF SET 300# RF SS A182 F316/316L SCH 40 BORE								
1540														
1550						***** PIPE NIPPLES *****								
1560														
1570		814				NIPPLE PBE CUT FROM PIPE								
1580														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD		CORR. ALLOW. 1 MM	FGP/WPMP DATE 17MAR17 BY MC REV D SHEET 3 OF 4 PA 5 CLASS 300P21					

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 20 30 40	D	PP9	2BV	K03AS02	1/2 - 2	*****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	2 - 2	814 B1 FA8 FF5	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE. ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY.					
50 60	D	VAV	2BV	K03AS02	3 - 3	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.					
70 80	D		2BVR	M03AS02	1/2 - 2	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
90 100	D	VAW	R2BVR	M03ES02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
110 120	D	VAW	R2BVR	M03ES02	1 - 1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FORDBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
130 140								VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
150 160 170		VB2				*****BALL VALVES*****		VAV VAW	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.					
180 190 200			Ball V	A03AS01	2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO		VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
210 220 230			Ball V	A03AS01	3 - 4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO		VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
240 250 260 270			Ball V	A03AS02	3 - 4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL RB LO								
280 290 300	D		Ball V	A03AS07	6 - 8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO								
310 320 330		VA5	Ball V	A03AS08	6 - 8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO								
340 350 360	D	VA5	Ball V	B03AS01	1/2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO								
370 380 390	D	VA5	Ball V	B03AS02	1/2 - 2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO								
400 410 420		VA5	Ball V	B03AS01	3 - 3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO		SERVICE						
430 440 450 460 470		VA5	Ball V	B03AS02	3 - 3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO		SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 AA, BD, BF, BFC, BV, CCD, CDE (90 DEG C AND ABOVE), CDM (70 DEG C AND ABOVE), CDS, CFA, CI, DL, FG, FGH, FX, GE, GS, NL, PHC, PL, VA, VI, VP, WDM, WRA, WSR						
480 490 500 510 520	D	ZA8	Gate V	B03AS03	4 - 4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO								
			Ball V	B03AS04	4 - 4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO								
			Ball V	B03AS07	6 - 8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO								
			Ball V	B03AS07	10 - 24	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO								
			Ball V	B03AS08	6 - 24	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO		MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 350 °C COMMENT:						
			Butter	T03AS61	10 - 24	*****BUTTERFLY VALVES***** BUTTERFLY 300# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO								
	D					*****GATE VALVES*****								
	D													
	D	ZA8	Gate V	G03AS48	1/2 - 1 1/2	GATE 300# RF SS A182 F316 TRIM 16 WEDGE HW								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						All SIZES - BUTTWELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								1 MM	17MAR17	KD	D	1 OF 4	5	300P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
530	D	ZA8	Gate V	G03AS41	2 - 4	GATE 300# RF SS A351 CF8M TRIM 16 WEDGE HW		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING. *****TO BE USED IN STEAM AND CONDENSATE SERVICE ONLY*****				
540	D	ZA8	Gate V	G03AS41	8 - 8	GATE 300# RF SS A351 CF8M TRIM 16 WEDGE HW							
550						*****GLOBE VALVES*****							
560													
570													
580			Globe	L03AS48	3/4 - 1 1/2	GLOBE 300# RF SS A182 F316 TRIM 16 OS&Y HW							
590			Globe	L03AS41	2 - 6	GLOBE 300# RF SS A351 CF8M TRIM 16 OS&Y HW							
600			Globe	L03AS42	8 - 10	GLOBE 300# RF SS A351 CF8M TRIM 16 OS&Y GO							
610													
620						*****CHECK VALVES*****							
630													
640													
650	D	VC1	Check	C03AS50	1/2 - 1.1/2	CHECK PISTON 300# RF SS A182 F316 TRIM 16							
660		VC2	Check	C03AS45	2 - 10	CHECK DUAL WAFER LUG 300# RF SS A351 CF8M TRIM 16 THROU BOLTED							
670						CHECK DUAL DBL FLG 300# RF SS A351 CF8M TRIM 16							
680						*****PIPE*****							
690													
700													
710													
720			Piping	5365140	3/4 - 1 1/2	PIPE SCH 80S SMLS SS A312 TP316/316L PE							
730			Piping	5365136	2 - 12	PIPE SCH 40S SMLS SS A312 TP316/316L BE							
740			Piping	5365137	14 - 24	PIPE SCH 40 SMLS SS A312 TP316/316L BE							
750													
760		814	Piping	5365207	1/2 - 1/2	PIPE SCH 160 SMLS SS A312 TP316/316L PE							
770						***** FITTINGS *****							
780													
790													
800	D		90 Deg	5449033	1/2 - 1.1/2	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S							
810			90 Deg	5449138	2 - 12	ELL 90 DEG LR SCH 40S SS A403 WP316/316L-S							
820			90 Deg	5449032	14 - 24	ELL 90 DEG LR SCH 40 SS A403 WP316/316L-S							
830													
840	C		45 Deg	5449146	3/4 - 1 1/2	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S							
850			45 Deg	5449144	2 - 12	ELL 45 SCH 40S SS A403 WP316/316L-S	1 - 1 1/2 2 - 3 3 - 12 14 - 24 16 - 24		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 AA, BD, BF, BFC, BV, CCD, CDE (90 DEG C AND ABOVE), CDM (70 DEG C AND ABOVE), CDS, CFA, CI, DL, FG, FGH, FX, GE, GS, NL, PHC, PL, VA, VI, VP, WDM, WRA, WSR				
860			45 Deg	5449038	14 - 24	ELL 45 SCH 40 SS A403 WP316/316L-S							
870													
880	C		Tee	5449156	3/4 - 1 1/2	TEE SCH 80S SS A403 WP316/316L-S							
890			Tee	5449147	2 - 12	TEE SCH 40S SS A403 WP316/316L-S							
900			Tee	5449045	14 - 24	TEE SCH 40 SS A403 WP316/316L-S							
910													
920	C		Reduci	5448905	3/4 - 1	TEE RED SCH 80S SS A403 WP316/316L-S							
930	C		Reduci	5449250	1 - 1 1/2	TEE RED SCH 40S/80S SS A403 WP316/316L-S							
940			Reduci	5449184	2 - 10	TEE RED SCH 40S SS A403 WP316/316L-S							
950			Reduci	5448897	6 - 12	TEE RED SCH 40/40S SS A403 WP316/316L-S	1 - 1 1/2 2 - 4 3 - 12 14 - 20 16 - 24		MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 350 °C COMMENT:				
960			Reduci	5449046	14 - 20	TEE RED SCH 40 SS A403 WP316/316L-S							
970													
980	C		Cap	5449049	3/4 - 1 1/2	CAP SCH 80S SS A403 WP316/316L-S							
990			Cap	5449148	2 - 12	CAP SCH 40S SS A403 WP316/316L-S							
1000			Cap	5449047	14 - 24	CAP SCH 40 SS A403 WP316/316L-S							
1010													
1020	C		Concen	5449158	3/4 - 1	REDUCER CONC SCH 80S SS A403 WP316/316L-S							
1030			Concen	5549110	3/4 - 1 1/2	REDUCER CONC SCH 40S/80S SS A403 WP316/316L-S							
1040			Concen	5449149	2 - 10	REDUCER CONC SCH 40S SS A403 WP316/316L-S							
1050			Concen	5448902	6 - 12	REDUCER CONC SCH 40/40S SS A403 WP316/316L-S							
1060			Concen	5448903	14 - 20	REDUCER CONC SCH 40 SS A403 WP316/316L-S							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY KD	REV D	SHEET 2 OF 4	PA 5	CLASS 300P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1070	C		Eccent	5449159	3/4 - 1	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1 - 1 1/2							
1080			Eccent	5549310	3/4 - 1 1/2	REDUCER ECC SCH 40S/80S SS A403 WP316/316L-S	2 - 4							
1090			Eccent	5449150	2 - 10	REDUCER ECC SCH 40S SS A403 WP316/316L-S	3 - 12							
1100			Eccent	5449154	6 - 12	REDUCER ECC SCH 40/40S SS A403 WP316/316L-S	14 - 20							
1110			Eccent	5449155	14 - 20	REDUCER ECC SCH 40 SS A403 WP316/316L-S	16 - 24							
1120														
1130	C	FF5	Flange	5504515	3/4 - 1 1/2	NIPOFLANGE FLAT BOTTOM 300# RF SCH 160 LTCS A350 LF2								
1140														
1150	C		Nipofl	5504411	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/80S SS A182 F316/316L	2 - 12							
1190	C		Nipofl	5504421	3/4 - 1 1/2	NIPOFLANGE 300# RF SCH 40/80S SS A182 F316/316L	14 - 24							
1200			Nipofl	5504032	2 - 2	NIPOFLANGE 300# RF SCH 40S/40S SS A182 F316/316L	2 - 12							
1210			Nipofl	5504033	2 - 2	NIPOFLANGE 300# RF SCH 40/40S SS A182 F316/316L	14 - 24							
1220														
1230		UJ2	Nipofl	5504132	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4 - 12							
1240		UJ2	Nipofl	5504142	1 1/2 - 1 1/2	NIPOFLANGE 300# RF SCH 40/160 SS A182 F316/316L	14 - 24							
1250														
1260	C		Weldol	5510354	3/4 - 1 1/2	WELDOLET SCH 40S/80S SS A182 F316/316L	2 - 12							
1270	C		Weldol	5510364	3/4 - 1 1/2	WELDOLET SCH 40/80S SS A182 F316/316L	14 - 24							
1280			Weldol	5509631	2 - 4	WELDOLET SCH 40S/40S SS A182 F316/316L	6 - 12							
1290			Weldol	5509636	2 - 8	WELDOLET SCH 40/40S SS A182 F316/316L	14 - 24							
1300														
1310														
1320		UJ2	Elbole	5516152	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4 - 12							
1330		UJ2	Elbole	5516158	1 1/2 - 1 1/2	ELBOFLANGE 300# RF SCH 40/160 SS A182 316/316L	14 - 24							
1340														
1350						***** PIPE FLANGES *****								
1360														
1370	C		Flange	5851203	1/2 - 2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE								
1380														
1390			Flange	5582991	1/2 - 1 1/2	FLG WN 300# RF SS A182 F316/316L SCH 80S BORE								
1400			Flange	5582995	2 - 12	FLG WN 300# RF SS A182 F316/316L SCH 40S BORE								
1410			Flange	5582997	14 - 24	FLG WN 300# RF SS A182 F316/316L SCH 40 BORE								
1420														
1430	C	FS1	Flange	5583535	3 - 6	FLG WN 600# RF SS A182 F316/316L SCH 40S BORE								
1440														
1450	D	FS1	Flange	5583021	3 - 3	FLG WN 900# RF SS A182 F316/316L SCH 40S BORE								
1460														
1470	C	FA8	Reduci	5600656	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 300# RF SS A182 F316/316L	3/4 - 2							
1480														
1490	C		Blind	5616529	1/2 - 24	FLG BLIND 300# RF SS A182 F316/316L								
1500														
1510														
1520						***** ORIFICE FLANGES *****								
1530														
1540			Orific	5640598	2 - 12	WN ORIF SET 300# RF SS A182 F316/316L SCH 40S BORE								
1550			Orific	5640599	14 - 24	WN ORIF SET 300# RF SS A182 F316/316L SCH 40 BORE								
1560														
1570														
1580						***** PIPE NIPPLES *****								
1590														
1600						NIPPLE PBE CUT FROM PIPE								
1610		814												
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD		CORR. ALLOW. 1 MM						
								FGP/WPMP DATE 17MAR17 BY KD REV D SHEET 3 OF 4 PA 5 CLASS 300P22						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1620						***** GASKETS *****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 AA, BD, BF, BFC, BV, CCD, CDE (90 DEG C AND ABOVE), CDM (70 DEG C AND ABOVE), CDS, CFA, CI, DL, FG, FGH, FX, GE, GS, NL, PHC, PL, VA, VI, VP, WDM, WRA, WSR MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 300 RF TEMPERATURE RANGE: -46 / 350 °C COMMENT:					
1630													
1640	D		GASFRS	5673033	1/2 - 24	GASKET 300# 316SS SP WND GRAPHITE IR							
1650													
1660	D		GASFRS	5673103	3 - 6	GASKET 600# 316SS SP WND GRAPHITE IR							
1670													
1680		B1				***** BOLTING *****							
1690													
1700	D		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1710													
1720		SB6	CPSBOL	5675675	12 - 24	JACK SCREW HEX HEAD A320 GR L7M							
1730													
1740													
1750						***** BLINDS AND SPACERS *****							
1760													
1770			Specta	5750422	3/4 - 10	SPECTACLE BLIND 300# RF SS A240 GR 316							
1780			Paddle	5750522	12 - 24	PADDLE BLANK 300# RF SS A240 GR 316							
1790			Paddle	5750622	12 - 24	PADDLE SPACER 300# RF SS A240 GR 316							
1800													
1810						***** BRANCHES *****							
1820													
1830						USE BRANCH CHART 300P22							
1840													
1850	D		Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 24						
1860	D		Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 24						
1870													
1880													
1890						***** CONSTRUCTION DETAILS *****							
1900						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
1910						PIPE BENDS ----- NONE							
1920						THREAD SEALANT ----- NONE							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION All SIZES - BUTTWELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY KD	REV D	SHEET 4 OF 4	PA 5	CLASS 300P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180 190 200 210 220 230 240 250 260 270 280 290 300 310 320 330 340 350			Piping	5718004	2 - 3	***** PIPE ***** PIPE ASTM D2996 RTRP 11F FEMALE TAPER X MALE TAPER SUPPLIED IN 5.95M LENGTHS ***** FITTINGS ***** ELL 90 DEG ASTM D2996 RTRP 11F MALE TAPER X MALE TAPER BEND 90 DEG 10D ASTM D2996 RTRP 11F MALE TAPER X MALE TAPER BEND 45 DEG 10D ASTM D2996 RTRP 11F MALE TAPER X MALE TAPER COUPLI 5719269 2 - 3 ***** PIPE FLANGES ***** FLG 300 FF RTRP 11F FLG FEMALE TAPER Blind 5719268 2 - 3 ***** GASKETS ***** GASKET 300# FF RING EPDM 3MM THICK ***** BOLTING ***** BOLS 5675768 1/4 - 99 ***** JOINT LUBRICANT **** CEMENT 5719629 2 - 3 CEMENT 5719787 2 - 3 CEMENT 5719788 2 - 3		B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. <					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
10		PP9				*****VALVES*****		810 816 B1 BA1 BA5 FF5 PP9 SB6 UJ2 VA5 VAV VAV VC1 VC2 NIPPLE TO BE CUT FROM SCH 160 PIPE COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED WITH RAISED FACE FLANGES ONLY. TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY. ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.								
20						*****MODULAR VALVES*****										
30																
40																
50																
60	B		2BV	K06AM10	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV										
70																
80	B		2BV	K06AM14	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV										
90																
100	B	VAV	2BV	K06AM14	3 - 3	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV										
110																
120	B		2BVR	M06AM10	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR										
130																
140	B		2BVR	M06AM14	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR										
150																
160	B	VAW	R2BVR	M06EM10	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2									
170																
180	B	VAW	R2BVR	M06EM10	1 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3									
190																
200																
210						***** BALL VALVES *****										
220																
230			Ball V	A06AM14	2 - 4	BALL 600# RF 825 INCONEL 625 TRIM TRUNNION METAL RB LO										
240			Ball V	A06AM16	6 - 24	BALL 600# RF INC 825 INC 625 TRIM TRUNNION METAL RB GO										
250																
260			Ball V	B06AM10	3/4 - 1 1/2	BALL 600# RF 825 INCONEL 625 TRIM FLOATING METAL FB LO										
270	B	VA5	Ball V	B06AM14	2 - 4	BALL 600# RF 825 INCONEL 625 TRIM TRUNNION METAL FB LO										
280	B	VA5	Ball V	B06AM16	6 - 24	BALL 600# RF INC 825 INC 625 TRIM TRUNNION METAL FB GO										
290																
300						***** GLOBE VALVES *****										
310																
320																
330			Globe	L06AM43	3/4 - 4	GLOBE 600# RF INC 825 INC TRIM 625 OS&Y HW										
340			Globe	L06AM44	6 - 10	GLOBE 600# RF INC 825 INC 625 TRIM TRIM OS&Y GO										
350																
360						***** CHECK VALVES *****										
370																
380																
390		VC1	Check	C06AM51	3/4 - 1 1/2	CHECK PISTON 600# RF INC 825 TRIM 625										
400		VC2	Check	C06AM46	2 - 8	CHECK DUAL WAFER LUG 600# RF INCONEL 825 N08825 THROU BOLTED										
410		VC2	Check	C06AM53	10 - 24	CHECK DUAL DBL FLG 600# RF INC 825 INC 625 TRIM										
420																
430																
440						*****BUTTERFLY VALVES*****										
450																
460			Butter	T06AM63	10 - 24	BUTTERFLY 600# RF INC 825 TRIM 625 TOV METAL DBL FLG SHORT PAT GO										
470			Butter	T06CM63	30 - 30	BUTTERFLY 600# CF INC 825 TRIM 625 TOV METAL DBL FLG SHORT PAT GO										
480																
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP									
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS			
							1 MM	17MAR17	SMS	B	1 OF 6	5	600A80			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
490						***** PIPE *****							
500													
510	B		Piping	5732001	3/4 - 1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE							
520	B		Piping	5732002	1 1/2 - 1 1/2	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 PE							
530	B		Piping	5361603	2 - 4	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 BE							
540			Piping	5361607	6 - 8	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 BE							
550			Piping	5361635	10 - 10	PIPE SCH 80S WLD ALLOY 825 B705 CL.2 N08825 BE							
560			Piping	5361637	12 - 24	PIPE SCH 60 WLD ALLOY 825 B705 CL.2 N08825 BE							
570			Piping	5361640	30 - 30	PIPE NOM. WT 29.0MM WLD ALLOY 825 B705 CL.2 N08825 BE							
580													
590	B	810	Piping	5361608	1/2 - 1/2	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 BE							
600		810	Piping	5361608	3/4 - 3/4	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 BE							
610													
620						***** FITTINGS *****							
630													
640													
650			90 Deg	5425111	3/4 - 1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825							
660			90 Deg	5424722	1 1/2 - 4	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPS UNS N08825							
670			90 Deg	5425111	6 - 8	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825							
680			90 Deg	5425109	10 - 10	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPW UNS N08825							
690			90 Deg	5425130	12 - 24	ELL 90 DEG LR SCH 60 ALLOY 825 B366-WPW UNS N08825							
700			90 Deg	5425144	30 - 30	ELL 90 DEG LR WT 29.0 MM ALLOY 825 B366-WPW UNS N08825							
710													
720			45 Deg	5425113	3/4 - 1	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825							
730			45 Deg	5424727	1 1/2 - 4	ELL 45 DEG SCH 40S ALLOY 825 B366-WPS UNS N08825							
740			45 Deg	5425113	6 - 8	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825							
750			45 Deg	5425115	10 - 10	ELL 45 DEG SCH 80S ALLOY 825 B366-WPW UNS N08825							
760			45 Deg	5425131	12 - 24	ELL 45 DEG SCH 60 ALLOY 825 B366-WPW UNS N08825							
770			45 Deg	5425114	30 - 30	ELL 45 DEG WT 29.0MM ALLOY 825 B366-WPW UNS N08825							
780													
790			Tee	5425119	3/4 - 1	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825							
800			Tee	5424725	1 1/2 - 4	TEE SCH 40S ALLOY 825 B366-WPS UNS N08825							
810			Tee	5425119	6 - 8	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825							
820			Tee	5425118	10 - 10	TEE SCH 80S ALLOY 825 B366-WPW UNS N08825							
830			Tee	5425132	12 - 24	TEE SCH 60 ALLOY 825 B366-WPW UNS N08825							
840			Tee	5425151	30 - 30	TEE WT 29.0MM ALLOY 825 B366-WPW UNS N08825							
850													
860			Reduci	5425126	3/4 - 3/4	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1 - 1						
870			Reduci	5425124	3/4 - 1	TEE RED SCH 40S/80S ALLOY 825 B366-WPS UNS N08825	1 1/2 - 2						
880			Reduci	5424724	1 1/2 - 3	TEE RED SCH 40S ALLOY 825 B366-WPS UNS N08825	2 - 4						
890			Reduci	5425125	3 - 4	TEE RED SCH 80S/40S ALLOY 825 B366-WPS UNS N08825	6 - 8						
900			Reduci	5425126	6 - 6	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	8 - 8						
910			Reduci	5425134	3 - 4	TEE RED SCH 80S/40S ALLOY 825 B366-WPW UNS N08825	10 - 10						
920			Reduci	5425117	6 - 8	TEE RED SCH 80S ALLOY 825 B366-WPW UNS N08825	10 - 10						
930			Reduci	5425135	6 - 10	TEE RED SCH 60/80S ALLOY 825 B366-WPW UNS N08825	12 - 24						
940			Reduci	5425133	12 - 20	TEE RED SCH 60 ALLOY 825 B366-WPW UNS N08825	14 - 24						
950			Reduci	5425153	10 - 10	TEE RED WT 29.0MM/SCH 80S ALLOY 825 B366-WPW UNS N08825	30 - 30						
960			Reduci	5425154	12 - 24	TEE RED WT 29.0MM/SCH 60 ALLOY 825 B366-WPW UNS N08825	30 - 30						
970													
980			Cap	5424270	3/4 - 1	CAP SCH 80S ALLOY 825 B366 UNS N08825							
990			Cap	5424219	1 1/2 - 4	CAP SCH 40S ALLOY 825 B366 UNS N08825							
1000			Cap	5424270	6 - 10	CAP SCH 80S ALLOY 825 B366 UNS N08825							
1010			Cap	5425136	12 - 24	CAP SCH 60 ALLOY 825 B366 UNS N08825							
1020			Cap	5425150	30 - 30	CAP WT 29.0MM ALLOY 825 B366 UNS N08825							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP						
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							1 MM	17MAR17	SMS	B	2 OF 6	5	600A80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
1030	B		Concen	5425122	3/4 - 3/4	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1												
1040		Concen	5425143	3/4 - 1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2													
1050		Concen	5424733	1 1/2 - 3	REDUCER CONC SCH 40S ALLOY 825 B366-WPS N08825	2 - 4													
1060		Concen	5425142	3 - 4	REDUCER CONC SCH 80S/40S ALLOY 825 B366-WPS N08825	6 - 8													
1070		Concen	5425122	6 - 6	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	8 - 8													
1080		Concen	5425100	4 - 4	REDUCER CONC SCH 80S/40S ALLOY 825 B366-WPW N08825	10 - 10													
1090		Concen	5425120	6 - 8	REDUCER CONC SCH 80S ALLOY 825 B366-WPW N08825	10 - 10													
1100		Concen	5425141	6 - 10	REDUCER CONC SCH 60/80S ALLOY 825 B366-WPW N08825	12 - 18													
1110		Concen	5425137	12 - 20	REDUCER CONC SCH 60 ALLOY 825 B366-WPW N08825	14 - 24													
1120		Concen	5425155	20 - 24	REDUCER CONC WT 29MM/SCH 60 ALLOY 825 B366-WPW N08825	30 - 30													
1130		Concen	5425157	30 - 30	REDUCER CONC WT 31MM/29 MM ALLOY 825 B366-WPW N08825	36 - 36													
1140																			
1150				Eccent	5425123	3/4 - 3/4	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825							1 - 1					
1160				Eccent	5425145	3/4 - 1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPS N08825							1 1/2 - 2					
1170				Eccent	5424734	1 1/2 - 3	REDUCER ECC SCH 40S ALLOY 825 B366-WPS N08825							2 - 4					
1180				Eccent	5425101	3 - 4	REDUCER ECC SCH 80S/40S ALLOY 825 B366-WPS N08825							6 - 8					
1190				Eccent	5425121	6 - 6	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825							8 - 8					
1200				Eccent	5425139	4 - 4	REDUCER ECC SCH 80S/40S ALLOY 825 B366-WPW N08825							10 - 10					
1210				Eccent	5424732	6 - 8	REDUCER ECC SCH 80S ALLOY 825 B366-WPW N08825							10 - 10					
1220				Eccent	5425140	6 - 10	REDUCER ECC SCH 60/80S ALLOY 825 B366-WPW N08825							12 - 18					
1230				Eccent	5425138	12 - 20	REDUCER ECC SCH 60 ALLOY 825 B366-WPW N08825							14 - 24					
1240				Eccent	5425156	20 - 24	REDUCER ECC WT 29MM/SCH 60 ALLOY 825 B366-WPW N08825							30 - 30					
1250																			
1260																			
1270							***** OLETS *****												
1280																			
1290				Nipofl	5513485	3/4 - 1	NIPOFLANGE 600# RF SCH 40S/80S ALLOY 825 B564 UNS N08825							2 - 4					
1300				Nipofl	5513481	1 1/2 - 2	NIPOFLANGE 600# RF SCH 40S ALLOY 825 B564 UNS N08825							2 - 4					
1310																			
1320				Nipofl	5513486	3/4 - 1	NIPOFLANGE 600# RF SCH 80S ALLOY 825 B564 UNS N08825							6 - 10					
1330				Nipofl	5513482	1 1/2 - 2	NIPOFLANGE 600# RF SCH 80S/40S ALLOY 825 B564 UNS N08825							6 - 10					
1340																			
1350				Nipofl	5513487	3/4 - 1	NIPOFLANGE 600# RF SCH 60/80S ALLOY 825 B564 UNS N08825							12 - 24					
1360				Nipofl	5513483	1 1/2 - 2	NIPOFLANGE 600# RF SCH 60/40S ALLOY 825 B564 UNS N08825							12 - 24					
1370																			
1380				Nipofl	5513488	3/4 - 1	NIPOFLANGE 600# RF WT 29.0MM/SCH 80S ALLOY 825 B564 UNS N08825							30 - 30					
1390				Nipofl	5513484	1 1/2 - 2	NIPOFLANGE 600# RF WT 29.0MM/SCH 40S ALLOY 825 B564 UNS N08825							30 - 30					
1400																			
1410		UJ2	Nipofl	5516485	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 40S/160 ALLOY 825 B564 UNS N08825	4 - 4												
1420																			
1430		UJ2	Nipofl	5516486	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 80S/160 ALLOY 825 B564 UNS N08825	6 - 10												
1440																			
1450		UJ2	Nipofl	5516487	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 60/160 ALLOY 825 B564 UNS N08825	12 - 24												
1460																			
1470		UJ2	Nipofl	5516488	1 1/2 - 1 1/2	NIPOFLANGE 600# RF WT 29.0MM/SCH 160 ALLOY 825 B564 UNS N08825	30 - 30												
1480																			
1490																			
1500		FF5	Endole	5516491	3/4 - 1	NIPOFLANGE FLAT BOTTOM 600# RF SCH 80S ALLOY 825 B564 UNS N08825													
1510		FF5	Endole	5516490	1 1/2 - 1 1/2	NIPOFLANGE FLAT BOTTOM 600# RF SCH 40S ALLOY 825 B564 UNS N08825													
1520																			
1530			Weldol	5513437	3/4 - 1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	2 - 4												
1540			Weldol	5513411	1 1/2 - 1 1/2	WELDOLET SCH 40S ALLOY 825 B564 UNS N08825	3 - 4												
1550			Weldol	5513438	3/4 - 1	WELDOLET SCH 80S ALLOY 825 B564 UNS N08825	6 - 10												
1560			Weldol	5513412	1 1/2 - 4	WELDOLET SCH 80S/40S ALLOY 825 B564 UNS N08825	6 - 10												
						CONSTRUCTION	FGP/WPMP												
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS						
							1 MM	17MAR17	SMS	B	3 OF 6	5	600A80						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1570	B		Weldol	5513433	3/4 - 1	WELDOLET SCH 60/80S ALLOY 825 B564 UNS N08825	12 - 24	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 3.8 CODE: ASME B31.3 GS, PHC, WSR MATERIAL: ALLOY 825 (NACE) FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 230 °C							
1580			Weldol	5513432	1 1/2 - 4	WELDOLET SCH 60/40S ALLOY 825 B564 UNS N08825	12 - 24								
1590			Weldol	5513433	6 - 8	WELDOLET SCH 60/80S ALLOY 825 B564 UNS N08825	12 - 24								
1600			Weldol	5513434	3/4 - 1	WELDOLET WT 29.0MM/SCH 80S ALLOY 825 B564 UNS N08825	30 - 30								
1610			Weldol	5513435	1 1/2 - 4	WELDOLET WT 29.0MM/SCH 40S ALLOY 825 B564 UNS N08825	30 - 30								
1620			Weldol	5513434	6 - 8	WELDOLET WT 29.0MM/SCH 80S ALLOY 825 B564 UNS N08825	30 - 30								
1630															
1640			UJ2	Elbole	5516662	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 40S/ SCH 160 ALLOY 825 B564 UNS N08825							4 - 4	
1650															
1660			UJ2	Elbole	5516663	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 80S/ SCH 160 ALLOY 825 B564 UNS N08825							6 - 10	
1670															
1680			UJ2	Elbole	5516671	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 60/ SCH 160 ALLOY 825 B564 UNS N08825							12 - 24	
1690															
1700			UJ2	Elbole	5516672	1 1/2 - 1 1/2	ELBOFLANGE 600# RF WT 29.0MM/ SCH 160 ALLOY 825 B564 UNS N08825							30 - 30	
1710															
1720															
1730							***** FLANGES *****								
1740															
1750					Reduci	5614300	1/2 - 1/2							FLG SCRD (FNPT) REDUCING 600# RF ALLOY 825 B564 UNS N08825	3/4 - 3/4
1760															
1770			B		Flange	5851204	1/2 - 2							FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE	
1780															
1790	B		Flange	5569391	1/2 - 1	FLG WN 600# RF ALLOY 825 SCH 80S B564 UNS N08825									
1800			Flange	5569392	1 1/2 - 4	FLG WN 600# RF ALLOY 825 SCH 40S B564 UNS N08825									
1810			Flange	5569391	6 - 10	FLG WN 600# RF ALLOY 825 SCH 80S B564 UNS N08825									
1820			Flange	5569393	12 - 24	FLG WN 600# RF ALLOY 825 SCH 60 B564 UNS N08825									
1830															
1840			Flange	5569394	30 - 30	FLG WN 600# RF ALLOY 825 WT 29.0MM B564 UNS N08825									
1850															
1860		816	Flange	5619812	30 - 30	FLG COMPACT 600# RJ ALLOY 825 WT 29.0MM B564 UNS N08825									
1870															
1880	B		Flange	5619969	36 - 36	FLG COMPACT 600# RJ ALLOY 825 WT 31.0MM B564 UNS N08825									
1890															
1900															
1910			Blind	5617167	1/2 - 24	FLG BLIND 600# RF ALLOY 825 B564 UNS N08825									
1920			Blind	5617168	30 - 30	FLG BLIND 600# RF ALLOY 825 B564 UNS N08825 B16.47 SERIES A									
1930															
1940	B	816	Blind	5619745	30 - 36	FLG COMPACT BLIND 600# RJ ALLOY 825 B564 UNS N08825 PIPE CLASS 600A80									
1950						***** ORIFICE FLANGES *****									
1960															
1970															
1980			Orific	5650840	2 - 4	WN ORIF SET 600# RF ALLOY 825 B564 UNS N08825 SCH 40S									
1990			Orific	5650841	6 - 10	WN ORIF SET 600# RF ALLOY 825 B564 UNS N08825 SCH 80S									
2000			Orific	5650842	12 - 24	WN ORIF SET 600# RF ALLOY 825 B564 UNS N08825 SCH 60									
2010															
2020						***** NIPPLES *****									
2030															
2040	B					NIPPLE CUT FROM PIPE									
2050															
2060						***** GASKETS *****									
2070															
2080															
2090			GASFRS	5673661	1/2 - 24	GASKET 600# ALLOY 825 SPIRAL WOUND GRAPHITE IR									
2100			GASFRS	5674014	30 - 30	GASKET 600# ALLOY 825 KAMMPROFILE GRAPHITE ASME B16.47 SERIES A									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP								
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS		
						1 MM	17MAR17	SMS	B	4 OF 6	5	600A80			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2110	B	816	GASRJ	5673400	30 - 36	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS								
2120														
2130														
2140		B1				***** BOLTING *****								
2150														
2160	B	BA1	BOLS	5675770	1/2 - 28	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
2170	B	BA5	BOLS	5675778	30 - 99	STUD-BOLT/COMPACT NUT API-6A718								
2180														
2190	B	BA1	BOLS	5675770	30 - 36	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
2200														
2210		SB6	CPSBOL	5675675	10 - 30	JACK SCREW HEX HEAD A320 GR L7M								
2220														
2230						***** BLINDS & SPACERS *****								
2240														
2250														
2260			Specta	5750230	3/4 - 8	SPECTACLE BLIND 600# RF ALLOY 825 B424 UNS N08825								
2270			Paddle	5750231	10 - 24	PADDLE BLANK 600# RF ALLOY 825 B424 UNS N08825								
2280			Paddle	5750232	10 - 24	PADDLE SPACER 600# RF ALLOY 825 B424 UNS N08825								
2290			Paddle	5750229	30 - 30	PADDLE BLANK 600# RF ALLOY 825 B424 UNS N08825 B16.47 SERIES A								
2300			Paddle	5750241	30 - 30	PADDLE SPACER 600# RF ALLOY 825 B424 UNS N08825 B16.47 SERIES A								
2310						***** BRANCHES *****								
2320														
2330						USE BRANCH CHART 600A80								
2340														
2350														
2360	B		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 30							
2370	B		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 30							
2380														
2390						***** CONSTRUCTION DETAILS *****								
2400						SEAL WELDING-----FOR THREADED CONNECTIONS ONLY								
2410						PIPE BENDS ----- NONE								
2420						THREAD SEALANT-----PTFE TAPE								
2430														
2440														
2450						*****NOTE*****								
2460						THE FOLLOWING ITEMS								
2470						WERE USED ON CAR PROJECT								
2480						THESE ITEMS SHALL NOT BE USED ON FGP								
2490														
2500														
2510														
2520														
2530														
2540														
2550														
2560						***** VALVES *****								
2570														
2580														
2590			Ball V	MC1L00258	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF ALLOY 825 B564 UNS N08825 ALLOY 825 TRIM								
2600														
2610			Gate V	GC1L00204	3/4 - 2	GATE 600# RF ALLOY 825 B564 UNS N08825, ALLOY 825 TRIM BB								
2620														
2630			Gate V	GC1K00204	3 - 4	GATE 600# RF A352 LCC + ALLOY 625 CLAD, ALLOY 825 TRIM BB								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						ALL SIZES- BUTT WELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								1 MM	17MAR17	SMS	B	5 OF 6	5	600A80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2640	B	VA5	Gate V	GC1K00205	6 - 30	GATE 600# RF A352 LCC + ALLOY 625 CLAD, ALLOY 825 TRIM BB GO								
2650														
2660														
2670			Ball V	BC1L00039	3/4 - 2	BALL 600# RF ALLOY 825 B564 UNS N08825 ALLOY 825 TRIM FS RB LONG PATT								
2680														
2690			Ball V	BC1L00055	3/4 - 2	BALL 600# RF ALLOY 825 B564 UNS N08825 ALLOY 825 TRIM FS FB LONG PATT								
2700														
2710			Ball V	BC1K00039	3 - 4	BALL 600# RF A352 LCC + ALLOY 625 CLAD ALLOY 825 TRIM FS RB LONG PATT								
2720														
2730			Ball V	BC1K00055	3 - 4	BALL 600# RF A352 LCC + ALLOY 625 CLAD ALLOY 825 TRIM FS FB LONG PATT								
2740														
2750			Ball V	BC1K00048	6 - 24	BALL 600# RF A352 LCC + ALLOY 625 CLAD ALLOY 825 TRIM FS RB LONG PATT GO								
2760														
2770			Ball V	BC1K00032	6 - 24	BALL 600# RF A352 LCC + ALLOY 625 CLAD ALLOY 825 TRIM FS FB LONG PATT GO								
2780														
2790														
2800			Check	CC1L00212	3/4 - 2	CHECK 600# RF ALLOY 825 B564 UNS N08825 ALLOY 825 TRIM PISTON TYPE								
2810														
2820			Check	CC1K00222	3 - 30	CHECK 600# RF A352 LCC + ALLOY 625 CLAD ALLOY 825 TRIM SWING TYPE								
2830														
2840														
2850			Globe	LC1L00204	3/4 - 2	GLOBE 600# RF ALLOY 825 B564 UNS N08825 ALLOY 825 TRIM								
2860			Globe	LC1K00204	3 - 4	GLOBE 600# RF A352 LCC + ALLOY 625 CLAD ALLOY 825 TRIM								
2870			Globe	LC1K00205	6 - 8	GLOBE 600# RF A352 LCC + ALLOY 625 CLAD ALLOY 825 TRIM GO								
2880														
2890														
2900														
2910														
2920						***** FLANGES *****								
2930			Flange	5557303	3/4 - 2	FLG WN 1500# RTJ ALLOY 825 SCH 40S B564 UNS N08825								
2940			Flange	5557301	3 - 4	FLG WN 900# RTJ ALLOY 825 SCH 40S B564 UNS N08825								
2950			Flange	5569486	6 - 10	FLG WN 900# RTJ ALLOY 825 SCH 80S BORE B564 UNS N08825								
2960			Flange	5557300	12 - 24	FLG WN 900# RTJ ALLOY 825 SCH 60 B564 UNS N08825								
2970			Flange	5557302	30 - 30	FLG WN 900# RTJ ALLOY 825 WT 29.0MM B564 UNS N08825 B16.47 SERIES A								
2980														
2990						***** NIPPLES *****								
3000														
3010														
3020			Nipple	5541484	3/4 - 1 1/2	NIPPLE SCH 40S ALLOY 825 B423 TBE 3" LONG								
3030			Nipple	5541486	3/4 - 1 1/2	NIPPLE SCH 40S ALLOY 825 B423 TOE/POE 3" LONG								
3040														
3050						***** GASKETS *****								
3060														
3070			11.25	5675300	3 - 24	GASKET 900# RTJ ALLOY 825 OCTAGONAL RING								
3080			11.25	5675301	30 - 30	GASKET 900# RTJ ALLOY 825 OCTAGONAL RING ASME B16.47 SERIES A								
3090			11.25	5675280	3/4 - 2	GASKET 1500# RTJ ALLOY 825 OCTAGONAL RING								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP							
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
						1 MM	17MAR17	SMS	B	6 OF 6	5	600A80		



SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
510						*****CHECK VALVES*****								
520														
530		VC1	Check	C06AM51	3/4 - 1 1/2	CHECK PISTON 600# RF INC 825 TRIM 625								
540		VC2	Check	C06AM46	2 - 3	CHECK DUAL WAFER LUG 600# RF INCONEL 825 N08825 THROU BOLTED								
550		VC2	Check	C06AR46	4 - 8	CHECK DUAL WAFER LUG 600# RF LTCS + CLAD INC 825 TRIM 625 THROU BOLTED								
560		VC2	Check	C06AR53	10 - 16	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC + 825 CLAD TRIM 625								
570														
580														
590						*****PIPES*****								
600														
610			Piping	5732001	3/4 - 1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE								
620			Piping	5732002	1 1/2 - 1 1/2	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 PE								
630			Piping	5361603	2 - 3	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 BE								
640			Piping	5358514_C80	4 - 8	PIPE SCH XS SMLS LTCS A333 GR.6 WITH 3 MM INC 825 INTERNAL CLADDING								
650			Piping	5358507_C80	10 - 16	PIPE SCH 80 SMLS LTCS A333 GR.6 WITH 3MM INC 825 INTERNAL CLADDING								
660														
670		810	Piping	5732003	1/2 - 1/2	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE								
680		810	Piping	5732003	3/4 - 3/4	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE								
690														
700						***** FITTINGS *****								
710														
720														
730			90 Deg	5425111	3/4 - 1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825								
740			90 Deg	5424722	1 1/2 - 3	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPS UNS N08825								
750			90 Deg	5392600_C80	4 - 8	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING								
760			90 Deg	5392546_C80	10 - 16	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING								
770														
780			45 Deg	5425113	3/4 - 1	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825								
790			45 Deg	5424727	1 1/2 - 3	ELL 45 DEG SCH 40S ALLOY 825 B366-WPS UNS N08825								
800			45 Deg	5392606_C80	4 - 8	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING								
810			45 Deg	5392552_C80	10 - 16	ELL BW 45 DEG LR SCH 80 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING								
820														
830			Tee	5425119	3/4 - 1	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825								
840			Tee	5424725	1 1/2 - 3	TEE SCH 40S ALLOY 825 B366-WPS UNS N08825								
850			Tee	5392609_C80	4 - 8	TEE BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING								
860			Tee	5392555_C80	10 - 16	TEE BW SCH 80 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING								
870														
880			Reduci	5425126	3/4 - 3/4	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1 - 1							
890			Reduci	5425124	3/4 - 1	TEE RED SCH 40S/80S ALLOY 825 B366-WPS UNS N08825	1 1/2 - 2							
900			Reduci	5424724	1 1/2 - 2	TEE RED SCH 40S ALLOY 825 B366-WPS UNS N08825	2 - 3							
910			Reduci	5392413_C80	4 - 6	TEE RED BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 8							
920			Reduci	5392463_C80	4 - 8	TEE RED BW SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10 - 16							
930			Reduci	5392464_C80	10 - 14	TEE RED BW SCH 80 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	12 - 16							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELD PWHT FOR SIZES 4" AND LARGER		CORR. ALLOW. 1 MM	DATE 17MAR17	BY YS	REV A	FGP/WPMP SHEET 2 OF 5	PA 5	CLASS 600C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
940			Cap	5424270	3/4 - 1	CAP SCH 80S ALLOY 825 B366 UNS N08825							
950			Cap	5424219	1 1/2 - 3	CAP SCH 40S ALLOY 825 B366 UNS N08825							
960			Cap	5392610_C80	4 - 8	CAP BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING							
970			Cap	5392556_C80	10 - 16	CAP BW SCH 80 LTCS STL A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING							
980			Concen	5425122	3/4 - 3/4	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1						
990				5425143	3/4 - 1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2						
1000				5424733	1 1/2 - 2	REDUCER CONC SCH 40S ALLOY 825 B366-WPS N08825	2 - 3						
1010				5425149	1 1/2 - 3	REDUCER CONC SCH XS/40S ALLOY 825 B366-WPS N08825	4 - 6						
1020				5392458_C80	4 - 6	REDUCER CONC SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 8						
1030			Concen	5392309_C80	8 - 8	REDUCER CONC SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10 - 16						
1040			Concen	5392559_C80	10 - 14	REDUCER CONC SCH 80 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	12 - 16						
1050													
1060			Eccent	5425123	3/4 - 3/4	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1						
1070				5425145	3/4 - 1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2						
1080				5424734	1 1/2 - 2	REDUCER ECC SCH 40S ALLOY 825 B366-WPS N08825	2 - 3						
1090				5425160	1 1/2 - 3	REDUCER ECC SCH XS/40S ALLOY 825 B366-WPS N08825	4 - 6						
1100				5392461_C80	4 - 6	REDUCER ECC SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6 - 8						
1110			Eccent	5392311_C80	8 - 8	REDUCER ECC SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10 - 16						
1120				5392313_C80	10 - 14	REDUCER ECC SCH 80 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	12 - 16						
1130			Nipofl	5513489	3/4 - 1	NIPOFLANGE 600# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2 - 3						
1140				5513490	3/4 - 1	NIPOFLANGE 600# RF SCH XS/80S ALLOY 825 B564 UNS N08825	4 - 8						
1150			Nipofl	5513480	1 1/2 - 2	NIPOFLANGE 600# RF SCH XS/40S ALLOY 825 B564 UNS N08825	4 - 8						
1160				5513500	3/4 - 1	NIPOFLANGE 600# RF SCH 80/80S ALLOY 825 B564 UNS N08825	10 - 16						
1170			Nipofl	5513501	1 1/2 - 2	NIPOFLANGE 600# RF SCH 80/40S ALLOY 825 B564 UNS N08825	10 - 16						
1180				5513510	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH XS/160 ALLOY 825 B564 UNS N08825	4 - 8						
1190			Nipofl	5513512	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 80/160 ALLOY 825 B564 UNS N08825	10 - 16						
1200				5516491	3/4 - 1	NIPOFLANGE FLAT BOTTOM 600# RF SCH 80S ALLOY 825 B564 UNS N08825							
1210			Endole										
1220													
1230			Weldol	5513413	3/4 - 1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	2 - 3						
1240				5513415	3/4 - 1	WELDOLET SCH XS/80S ALLOY 825 B564 UNS N08825	4 - 8						
1250			Weldol	5513419	1 1/2 - 3	WELDOLET SCH XS/ 40S ALLOY 825 B564 UNS N08825	4 - 8						
1260				5513445	3/4 - 1	WELDOLET SCH 80/80S ALLOY 825 B564 UNS N08825	10 - 16						
1270			Weldol	5513446	1 1/2 - 3	WELDOLET SCH 80/40S ALLOY 825 B564 UNS N08825	10 - 16						
1280				5505262_C80	4 - 6	WELDOLET SCH 80/XS LTCS A350 LF2 WITH 3MM INC 825 INTERNAL CLADDING	10 - 16						
1290			Weldol										
1300													
1310			Weldol										
1320													
1330			Weldol										
1340													
1350			Weldol										
1360													
1370			Weldol										
1380													
1390			Weldol										
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELD PWHT FOR SIZES 4" AND LARGER	CORR. ALLOW. 1 MM	DATE 17MAR17	BY YS	REV A	SHEET 3 OF 5	PA 5	CLASS 600C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1400		UJ2	Elbole	5516673	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH XS/ SCH 160 ALLOY 825 B564 UNS N08825	4 - 8						
1410		UJ2	Elbole	5516674	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 80/ SCH 160 ALLOY 825 B564 UNS N08825	10 - 16						
1420													
1430													
1440													
1450						***** PIPE FLANGES *****							
1460													
1470		FA8	Reduci	5614300	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 600# RF ALLOY 825 B564 UNS N08825	3/4 - 1 1/2						
1480													
1490													
1500			Flange	5851204	1/2 - 2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE							
1510													
1520			Flange	5569391	3/4 - 1	FLG WN 600# RF ALLOY 825 SCH 80S B564 UNS N08825							
1530			Flange	5569392	1 1/2 - 3	FLG WN 600# RF ALLOY 825 SCH 40S B564 UNS N08825							
1540			Flange	5557036_C80	4 - 8	FLG WN 600# RF LTCS A350 LF2 SCH XS WITH 3MM INC 825 CLAD FACE & BORE							
1550			Flange	5557040_C80	10 - 16	FLG WN 600# RF LTCS A350 LF2 SCH 80 WITH 3MM INC 825 CLAD FACE & BORE							
1560													
1570			Blind	5617167	1/2 - 3	FLG BLIND 600# RF ALLOY 825 B564 UNS N08825							
1580			Blind	5615598_C80	4 - 16	FLG BLIND 600# RF LTCS A350 LF2 WITH 3 MM INC 825 CLAD FACE							
1590													
1600			Flange	5517251_C80	6 - 6	FLG WN 10000 API 6A TYPE 6BX 5 1/8" NS (6"NB) RTJ BORE 131MM LTCS A350 LF2 SCH XS WITH 3MM INC 825 CLAD FACE & BORE							
1610													
1620						***** ORIFICE FLANGES *****							
1630													
1640			Orific	5650840	2 - 3	WN ORIF SET 600# RF ALLOY 825 B564 UNS N08825 SCH 40S							
1650			Orific	5624352_C80	4 - 8	WN ORIF SET 600# RF LTCS A350 LF2 SCH XS WITH 3MM INC 825 CLAD FACE & BORE							
1660			Orific	5624353_C80	10 - 16	WN ORIF SET 600# RF LTCS A350 LF2 SCH 80 WITH 3MM INC 825 CLAD FACE & BORE							
1670		810				***** PIPE NIPPLES *****							
1680													
1690						NIPPLE PBE TO BE CUT FROM PIPE							
1700													
1710						***** GASKETS *****							
1720													
1730													
1740			GASFRS	5673661	1/2 - 16	GASKET 600# ALLOY 825 SPIRAL WOUND GRAPHITE IR							
1750													
1760			GASRJ	5674469	6 - 6	GASKET 10000 API 6A BX 169 5 1/8" NS ALLOY 625 RING							
1770													
1780		B1				***** BOLTING *****							
1790													
1800			BOLS	5675770	1/4 - 16	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1810													
1820													
1830		SB6	CPSBOL	5675675	10 - 16	JACK SCREW HEX HEAD A320 GR L7M							
1840		B1											
1850													
1860		SB5				***** BLINDS & SPACERS *****							
1870													
1880			Specta	5750230	3/4 - 3	SPECTACLE BLIND 600# RF ALLOY 825 B424 UNS N08825							
1890			Specta	5749587_C80	4 - 8	SPECTACLE BLIND 600# RF LTCS A516-GR70 WITH 3MM INC 825 CLAD FACE & BORE							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELD PWHT FOR SIZES 4" AND LARGER	CORR. ALLOW. 1 MM	DATE 17MAR17	BY YS	REV A	SHEET 4 OF 5	PA 5	CLASS 600C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																			
1900			Paddle	5750040_C80	10 - 16	PADDLE BLANK 600# RF LTCS A516-GR70 WITH 3MM INC 825 CLAD FACE	3 - 16 3 - 16																				
1910			Paddle	5750251_C80	10 - 16	PADDLE SPACER 600# RF LTCS A516-GR70 WITH 3MM INC 825 CLAD FACE & BORE																					
1920			Reinfo Reinfo	RWELD RWELD	2 - 12 2 - 12	***** BRANCHES *****																					
1930						USE BRANCH CHART 600C80																					
1940						NO-MTO REINFORCING WELD FOR STUB-IN																					
1950						NO-MTO REINFORCING WELD FOR STUB-IN																					
1960						***** CONSTRUCTION DETAILS *****																					
1970						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY																					
1980						PIPE BENDS ----- NONE																					
1990						THREAD SEALANT ----- PTFE PASTE																					
2000																											
2010																											
2020																											
2030																											
2040																											
2050																											
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION								CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS							
						ALL SIZES BUTT-WELD PWHT FOR SIZES 4" AND LARGER								1 MM	17MAR17	YS	A	5 OF 5	5	600C80							

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 20 30 40	D	PP9	2BV	K06AC02	1/2 - 1 1/2	*****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.					
50 60	D		2BV	K06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		810	NIPPLE TO BE CUT FROM SCH 160 PIPE					
70 80	D	VAV	2BV	K06AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		815	TO BE USED WITH COMPACT FLANGES ONLY.					
90 100	D		2BVR	M06AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's					
110 120	D		2BVR	M06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		817	USE ONLY FOR TIE IN					
130 140	D	VAW	R2BVR	M06EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.					
150 160	D	VAW	R2BVR	M06EC02	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.					
170 180 190 200		VB2						FF5	FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY.					
210 220	D		Ball V	A06AC05	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
230 240	D		Ball V	A06AC06	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
250 260	D		Ball V	A06AC07	6 - 8	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO		SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.					
270 280	D		Ball V	A06AC08	6 - 8	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO		UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.					
290 300 310 320 330			Ball V	B06AC01	3/4 - 1 1/2	BALL 600# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO		UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.					
340 350	D	VA5	Ball V	B06AC05	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO		UDA						
360 370	D	VA5	Ball V	B06AC06	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO								
380 390	D	VA5	Ball V	B06AC07	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
400 410	D	VA5	Ball V	B06AC08	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
420 430	D	VA5	Ball V	B06CC07	30 - 30	BALL 600# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
440 450 460 470 480	D	VA5	Ball V	B06CC08	30 - 30	BALL 600# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
						*****ORBIT VALVES*****								
	C	VA5	Ball V	R06AC06	3/4 - 6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 1.5 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 SMS D 1 OF 6 5 600H21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
490	D	VA5	Ball V	R06AC08	8 - 10	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB GO		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
500						*****BUTTERFLY VALVES*****		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
510								VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY					
520	D		Butter	T06AC61	8 - 24	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO		VAD	FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE).					
530								VAG	FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY.					
540	D		Butter	T06CC61	30 - 30	BUTTERFLY 600# CF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO		VAK	FOR ALL INLINE APPLICATIONS AND LEVEL TRANSMITTER CONNECTIONS					
550						*****GATE VALVES*****		VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
560								VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
570	D		Gate V	G06AC48	3/4 - 1 1/2	GATE 600# RF LTCS A350 LF2 TRIM 16 WEDGE HW		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
580	D		Gate V	G06AC41	2 - 4	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE HW		VU2	USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY.					
590			Gate V	G06AC42	6 - 24	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE GO		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
600						*****GLOBE VALVES*****								
610														
620														
630														
640														
650														
660														
670														
680			Globe	L06AC48	3/4 - 1 1/2	GLOBE 600# RF LTCS A350 LF2 TRIM 16 OS&Y HW								
690			Globe	L06AC41	2 - 4	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW								
700			Globe	L06AC42	6 - 10	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y GO								
710						*****CHECK VALVES*****								
720														
730														
740														
750		VC1	Check	C06AC50	3/4 - 1 1/2	CHECK PISTON 600# RF LTCS A350 LF2 TRIM 16								
760		VC2	Check	C06AC45	2 - 8	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED								
770														
780	D	VC2	Check	C06AC52	10 - 24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16								
790			Check	C06CC52	30 - 30	CHECK DUAL 600# CF LTCS A352 LCC TRIM 16								
800						*****PIPE*****								
810														
820														
830			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE								
840			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE								
850	C		Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE								
860	C		Piping	5358514	4 - 8	PIPE SCH XS SMLS LTCS A333 GR.6 BE								
870	C		Piping	5358507	10 - 16	PIPE SCH 80 SMLS LTCS A333 GR.6 BE								
880	C		Piping	5358599	18 - 24	PIPE SCH 80 EFW LTCS A671 CC60 CL.22 BE								
890			Piping	5358616	30 - 30	PIPE WT 34.0MM EFW LTCS A671 CC60 CL.22 BE								
900														
910	D	810	Piping	5358512	1/2 - 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE								
920	D	810	Piping	5358512	3/4 - 3/4	PIPE SCH 160 SMLS LTCS A333 GR.6 PE								
930						***** FITTINGS *****								
940														
950														
960														
970			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
980	C		90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6								
990	C		90 Deg	5392600	4 - 8	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
1000	C		90 Deg	5392546	10 - 16	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6								
1010			90 Deg	5392387	18 - 24	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6-W								
1020														
1030														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 2 OF 6 PA 5 CLASS 600H21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1050	C		45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6							
1060	C		45 Deg	5392606	4 - 8	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1070	C		45 Deg	5392552	10 - 16	ELL BW 45 DEG LR SCH 80 LTCS A420 WPL6							
1080			45 Deg	5392391	18 - 24	ELL 45 DEG SCH 80 LTCS A420 WPL6-W							
1090													
1100			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6							
1110	C		Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6							
1120	C		Tee	5392609	4 - 8	TEE BW SCH XS LTCS A420 WPL6							
1130	C		Tee	5392555	10 - 16	TEE BW SCH 80 LTCS A420 WPL6							
1140			Tee	5392402	18 - 24	TEE SCH 80 LTCS A420 WPL6-W							
1150													
1160			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2						
1170	C		Reduci	5392454	1 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3						
1180	C		Reduci	5392661	2 - 2	TEE RED BW SCH XS LTCS A420 WPL6	4 - 4						
1190	C		Reduci	5392460	3 - 3	TEE RED BW SCH XS/STD LTCS A420 WPL6	4 - 8						
1200	C		Reduci	5392661	4 - 6	TEE RED BW SCH XS LTCS A420 WPL6	6 - 8						
1210	C		Reduci	5392463	4 - 8	TEE RED BW SCH 80/XS LTCS A420 WPL6	10 - 16						
1220	C		Reduci	5392403	8 - 8	TEE RED BW SCH 80/XS LTCS A420 WPL6-W	18 - 24						
1230	C		Reduci	5392464	10 - 14	TEE RED BW SCH 80 LTCS A420 WPL6	12 - 16						
1240	C		Reduci	5392353	10 - 20	TEE RED BW SCH 80 LTCS A420 WPL6-W	18 - 24						
1250													
1260	C		Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6							
1270	C		Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6							
1280	C		Cap	5392610	4 - 8	CAP BW SCH XS LTCS A420 WPL6							
1290	C		Cap	5392556	10 - 24	CAP SCH 80 LTCS STL A420 WPL6							
1300													
1310			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2						
1320	C		Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3						
1330	C		Concen	5392458	1 1/2 - 2	REDUCER CONC SCH XS LTCS A420 WPL6	4 - 4						
1340	C		Concen	5392455	3 - 3	REDUCER CONC SCH XS/STD LTCS A420 WPL6	4 - 8						
1350	C		Concen	5392458	4 - 6	REDUCER CONC SCH XS LTCS A420 WPL6	6 - 8						
1360	C		Concen	5392309	4 - 8	REDUCER CONC SCH 80/XS LTCS A420 WPL6	10 - 16						
1370	C		Concen	5392557	10 - 14	REDUCER CONC SCH 80 LTCS A420 WPL6	12 - 16						
1380	C		Concen	5392559	10 - 20	REDUCER CONC SCH 80 LTCS A420 WPL6-W	18 - 24						
1390	C		Concen	5392419	20 - 20	REDUCER CONC WT 34.0MM/SCH 80 LTCS A420 WPL6-W	30 - 30						
1400								SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BB, BD, BFH, BV, DHC, DL, DMD, FG, FGH, FM, GE, GHC, GRC, GRI, GRK, GS, GSL, GW, NH, NL, PHC, PL, PME, SG, SHM, SL, SM, THM, TL, TM, WBH, WBL, WE, WRA, WT MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 343 °C					
1410			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2						
1420	C		Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3						
1430	C		Eccent	5392461	1 1/2 - 2	REDUCER ECC SCH XS LTCS A420 WPL6	4 - 4						
1440	C		Eccent	5392456	3 - 3	REDUCER ECC SCH XS/STD LTCS A420 WPL6	4 - 8						
1450	C		Eccent	5392612	4 - 6	REDUCER ECC SCH XS LTCS A420 WPL6	6 - 8						
1460	C		Eccent	5392311	4 - 8	REDUCER ECC SCH 80/XS LTCS A420 WPL6	10 - 16						
1470	C		Eccent	5392558	10 - 14	REDUCER ECC SCH 80 LTCS A420 WPL6	12 - 16						
1480	C		Eccent	5392313	10 - 20	REDUCER ECC SCH 80 LTCS A420 WPL6-W	18 - 24						
1490	C		Eccent	5392613	20 - 20	REDUCER ECC WT 34.0MM/SCH 80 LTCS A420 WPL6-W	30 - 30						
1500													
1510			Nipofl	5504006	3/4 - 1 1/2	NIPOFLANGE 600# RF SCH XS LTCS A350 LF2	2 - 2						
1520	C		Nipofl	5504007	3/4 - 2	NIPOFLANGE 600# RF SCH STD/XS LTCS A350 LF2	3 - 3						
1530	C		Nipofl	5504006	3/4 - 2	NIPOFLANGE 600# RF SCH XS LTCS A350 LF2	4 - 8						
1540	C		Nipofl	5504026	3/4 - 2	NIPOFLANGE 600# RF SCH 80/XS LTCS A350 LF2	10 - 24						
1550	C		Nipofl	5504095	3/4 - 2	NIPOFLANGE 600# RF SCH 34.0MM/XS LTCS A350 LF2	30 - 30						
1560													
1570	D	FF5	Endole	5504502	3/4 - 3/4	NIPOFLANGE FLAT BOTTOM 600# RF SCH XS LTCS A350 LF2							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 3 OF 6 PA 5 CLASS 600H21				

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
2670 2680 2690 2700 2710 2720 2730 2740 2750 2760 2770 2780 2790 2800	D D D		Reinfo Reinfo	RWELD RWELD	2 - 20 2 - 20	***** BRANCHES ***** USE BRANCH CHART 600H21 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE	3 - 30 3 - 30										
								SERVICE									
								DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BB, BD, BFH, BV, DHC, DL, DMD, FG, FGH, FM, GE, GHC, GRC, GRI, GRK, GS, GSL, GW, NH, NL, PHC, PL, PME, SG, SHM, SL, SM, THM, TL, TM, WBH, WBL, WE, WRA, WT									
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 343 °C									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 1.5 MM		DATE 17MAR17		BY SMS	REV D	SHEET 6 OF 6	PA 5	CLASS 600H21	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
10		PP9				*****VALVES*****		812 816	NIPPLE TO BE CUT FROM SCH XXS PIPE COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. PP9 ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 SB6 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. UJ2 USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. VA5 USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. VAV USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VAV VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 VC1 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. VC2 DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. ZA6 REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
500	B B	VC2	Check	C06AC45	2 - 8	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
510		VC2	Check	C06AC52	10 - 24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16							
520													
530													
540						***** PIPE *****							
550													
560		Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE								
570		Piping	5358515	2 - 2	PIPE SCH 160 SMLS LTCS A333 GR.6 BE								
580		Piping	5358514	3 - 8	PIPE SCH XS SMLS LTCS A333 GR.6 BE								
590		Piping	5358507	10 - 16	PIPE SCH 80 SMLS LTCS A333 GR.6 BE								
600		Piping	5358599	18 - 24	PIPE SCH 80 EFW LTCS A671 CC60 CL.22 BE								
610													
620		812	Piping	5358511	1/2 - 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE							
630		812	Piping	5358511	3/4 - 3/4	PIPE SCH XXS SMLS LTCS A333 GR.6 PE							
640													
650						***** FITTINGS *****							
660													
670													
680		90 Deg	5392816	3/4 - 2	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6								
690		90 Deg	5392600	3 - 8	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
700		90 Deg	5392546	10 - 16	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6								
710		90 Deg	5392387	18 - 24	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6-W								
720													
730		45 Deg	5392822	3/4 - 2	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6								
740		45 Deg	5392606	3 - 8	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
750		45 Deg	5392552	10 - 16	ELL BW 45 DEG LR SCH 80 LTCS A420 WPL6								
760		45 Deg	5392391	18 - 24	ELL 45 DEG SCH 80 LTCS A420 WPL6-W								
770													
780		Tee	5392825	3/4 - 2	TEE BW SCH 160 LTCS A420 WPL6								
790		Tee	5392609	3 - 8	TEE BW SCH XS LTCS A420 WPL6								
800		Tee	5392555	10 - 16	TEE BW SCH 80 LTCS A420 WPL6								
810		Tee	5392402	18 - 24	TEE SCH 80 LTCS A420 WPL6-W								
820													
830	Reduci	5392821	3/4 - 1 1/2	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 2								
840	Reduci	5392469	1 1/2 - 2	TEE RED BW SCH XS/160 LTCS A420 WPL6	3 - 4								
850	Reduci	5392661	3 - 6	TEE RED BW SCH XS LTCS A420 WPL6	4 - 8								
860	Reduci	5392463	4 - 8	TEE RED BW SCH 80/XS LTCS A420 WPL6	10 - 16								
870	Reduci	5392403	8 - 8	TEE RED BW SCH 80/XS LTCS A420 WPL6-W	18 - 20								
880	Reduci	5392464	10 - 14	TEE RED BW SCH 80 LTCS A420 WPL6	12 - 16								
890	Reduci	5392353	10 - 20	TEE RED BW SCH 80 LTCS A420 WPL6-W	18 - 24								
900													
910	Cap	5392826	3/4 - 2	CAP BW SCH 160 LTCS A420 WPL6									
920	Cap	5392610	3 - 8	CAP BW SCH XS LTCS A420 WPL6									
930	Cap	5392556	10 - 24	CAP SCH 80 LTCS STL A420 WPL6									
940													
950	Concen	5392827	3/4 - 1 1/2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 2								
960	Concen	5392462	1 1/2 - 2	REDUCER CONC SCH XS/160 LTCS A420 WPL6	3 - 4								
970	Concen	5392458	3 - 6	REDUCER CONC SCH XS LTCS A420 WPL6	4 - 8								
980	Concen	5392309	4 - 8	REDUCER CONC SCH 80/XS LTCS A420 WPL6	10 - 16								
990	Concen	5392557	10 - 14	REDUCER CONC SCH 80 LTCS A420 WPL6	12 - 16								
1000	Concen	5392559	10 - 20	REDUCER CONC SCH 80 LTCS A420 WPL6-W	18 - 24								
1010													
1020	Eccent	5392828	3/4 - 1 1/2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 2								
				FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION	FGP/WPMP						
						ALL SIZES- BUTT WELD PWHT FOR WT > 19MM	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							3 MM	17MAR17	KD	B	2 OF 4	5	600H25

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1030	B		Eccent	5392465	1 1/2 - 2	REDUCER ECC SCH XS/160 LTCS A420 WPL6	3 - 4							
1040			Eccent	5392612	3 - 6	REDUCER ECC SCH XS LTCS A420 WPL6	4 - 8							
1050			Eccent	5392311	4 - 8	REDUCER ECC SCH 80/XS LTCS A420 WPL6	10 - 16							
1060			Eccent	5392558	10 - 14	REDUCER ECC SCH 80 LTCS A420 WPL6	12 - 16							
1070			Eccent	5392313	10 - 20	REDUCER ECC SCH 80 LTCS A420 WPL6-W	18 - 24							
1080														
1090						***** OLETS *****								
1100														
1110			Nipofl	5504460	3/4 - 1 1/2	NIPOFLANGE 600# RF SCH 160 LTCS A350 LF2	2 - 2							
1120			Nipofl	5504461	3/4 - 2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2	3 - 8							
1130			Nipofl	5504462	3/4 - 2	NIPOFLANGE 600# RF SCH 80/160 LTCS A350 LF2	10 - 24							
1140														
1150		UJ2	Nipofl	5504028	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2	4 - 8							
1160		UJ2	Nipofl	5504029	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 80/160 LTCS A350 LF2	10 - 24							
1170														
1180			Weldol	5505392	3/4 - 1	WELDOLET SCH 160 LTCS A350 LF2	2 - 2							
1190			Weldol	5505393	3/4 - 2	WELDOLET SCH XS/160 LTCS A350 LF2	3 - 8							
1200			Weldol	5505263	3/4 - 2	WELDOLET SCH 80/160 LTCS A350 LF2	10 - 24							
1210														
1220			Weldol	5505335	3 - 3	WELDOLET SCH XS LTCS A350 LF2	8 - 8							
1230			Weldol	5505262	3 - 8	WELDOLET SCH 80/XS LTCS A350 LF2	10 - 24							
1240														
1250		UJ2	Elbole	5516659	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH XS/160 LTCS A350 LF2	4 - 8							
1260		UJ2	Elbole	5516660	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 80/160 LTCS A350 LF2	10 - 24							
1270														
1280		816				**** FLANGES ****								
1290														
1300		FA8	Reduci	5599705	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 600# RF A350 LF2	3/4 - 1 1/2							
1310														
1320			Flange	5851204	1/2 - 2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE								
1330			Flange	5557038	1/2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH 160								
1340			Flange	5557036	3 - 8	FLG WN 600# RF LTCS A350 LF2 SCH XS								
1350			Flange	5557040	10 - 24	FLG WN 600# RF LTCS A350 LF2 SCH 80								
1360														
1370			Blind	5615598	1/2 - 24	FLG BLIND 600# RF LTCS A350 LF2								
1380														
1390						***** ORIFICE FLANGES *****								
1400														
1410			Orific	5624351	2 - 2	WN ORIF SET 600# RF LTCS A350 LF2 SCH 160								
1420			Orific	5624352	3 - 8	WN ORIF SET 600# RF LTCS A350 LF2 SCH XS BORE								
1430			Orific	5624353	10 - 24	WN ORIF SET 600# RF LTCS A350 LF2 SCH 80 BORE								
1440														
1450														
1460						***** PIPE NIPPLES *****								
1470														
1480		812				NIPPLE PBE - CUT FROM PIPE								
1490														
1500														
1510						***** GASKETS *****								
1520														
1530		B	GASFRS	5673103	1/2 - 24	GASKET 600# 316SS SP WND GRAPHITE IR								
1540														
1550														
1560		B1				***** BOLTING *****								
1570														
1580	B		BOLS	5675770	1/4 - 24	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP							
						ALL SIZES- BUTT WELD PWHT FOR WT > 19MM	CORR. ALLOW. 3 MM	DATE 17MAR17	BY KD	REV B	SHEET 3 OF 4	PA 5	CLASS 600H25	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES												
1590	B B	SB6 B1	CPSBOL	5675675	10 - 42	JACK SCREW HEX HEAD A320 GR L7M	3 - 24 3 - 24													
1600																				
1610																				
1620																				
1630																				
1640			Specta	5749587	3/4 - 8	SPECTACLE BLIND 600# RF LTCS A516-GR70														
1650			Paddle	5750040	10 - 24	PADDLE BLANK 600# RF LTCS A516-GR70														
1660			Paddle	5750251	10 - 24	PADDLE SPACER 600# RF LTCS A516-GR70														
1670																				
1680																				
1690																				
1700																				
1710																				
1720			Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN														
1730			Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN														
1740																				
1750																				
1760																				
1770																				
1780																				
1790																				
1800																				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
10		PP9	BALFLG	A06AC05	2 - 3	*****BALL VALVES***** BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.										
20								810	NIPPLE TO BE CUT FROM SCH 160 PIPE										
30								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.										
40								PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156										
50								VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.										
60								VAD	FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE).										
70								VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.										
80								VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.										
90								VU2	USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY.										
100																			
110																			
120																			
130																			
140										Gate V	G06AC48	3/4 - 1 1/2	GATE 600# RF LTCS A350 LF2 TRIM 16 WEDGE HW						
150										Gate V	G06AC41	2 - 3	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE HW						
160																			
170																			
180																			
190																			
200				Globe	L06AC48	3/4 - 1 1/2	GLOBE 600# RF LTCS A350 LF2 TRIM 16 OS&Y HW												
210				Globe	L06AC41	2 - 3	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW												
220																			
230																			
240																			
250																			
260		VC1 VC2	Check Check	C06AC50 C06AC45	3/4 - 1 1/2 2 - 3	CHECK PISTON 600# RF LTCS A350 LF2 TRIM 16 CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED													
270																			
280																			
290																			
300																			
310																			
320																			
330																			
340																			
350																			
360																			
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510																			
520																			
530																			
540																			
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" AND SMALLER-GALVANIZED THREADED AND COUPLED 3" - BUTT WELD,GALVANIZED AFTER FABRICATION		CORR. ALLOW. 1.5 MM		DATE 17NOV16		BY SMS	REV B	FGP/WPMP SHEET 1 OF 3		PA 5	CLASS 600H2G		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
550	B	810	Coupli Reduci	5491250 5491260	3/4 - 2 3/4 - 1 1/2	***** COUPLINGS, UNIONS, PLUGS & BUSHINGS *****	1 - 2						
560													
570													
580													
590													
600													
610													
620													
630													
640													
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990													
1000													
1010													
1020													
1030													
1040													
1050													
1060													
1070													
1080													
1090													
B	B1	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" AND SMALLER-GALVANIZED THREADED AND COUPLED 3" - BUTT WELD,GALVANIZED AFTER FABRICATION	CORR. ALLOW. 1.5 MM	DATE 17NOV16	BY SMS	REV B	SHEET 2 OF 3	PA 5	CLASS 600H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1100 1110 1120 1130 1140 1150 1160 1170 1180 1190 1200 1210 1220			Specta	5749588	3/4 - 3	**** BLINDS & SPACERS **** SPECTACLE BLIND 600# RF LTCS A516 GR.70 GALV A153 **** BRANCHES **** USE BRANCH CHART 600H2G **** CONSTRUCTION DETAILS **** SEAL WELDING-----NONE PIPE BENDS-----NONE THREAD SEALANT ----- PTFE PASTE								
								SERVICE						
								DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 CO						
								MATERIAL: LOW TEMP CARBON STEEL - GALV'D FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 38 °C						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION 2" AND SMALLER-GALVANIZED THREADED AND COUPLED 3" - BUTT WELD,GALVANIZED AFTER FABRICATION		CORR. ALLOW. 1.5 MM	DATE 17NOV16	BY SMS	REV B	SHEET 3 OF 3	PA 5	CLASS 600H2G

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
10 20 30	C		2BV	K06AC02	1/2 - 1 1/2	***** MODULAR VALVES ***** MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		B1 VA5	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VAV VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. VB2 SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. ZA6 REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.								
40 50	C		2BV	K06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV											
60 70	C	VAV	2BV	K06AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV											
80 90	C		2BVR	M06AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR											
100 110	C		2BVR	M06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR											
130 140	C	VAW	R2BVR	M06EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2										
150 160	C	VAW	R2BVR	M06EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3										
170 180 190 200		VB2				***** BALL VALVES *****											
210			Ball V	B06AC01	3/4 - 1 1/2	BALL 600# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO											
220																	
230	C		Ball V	B06AC05	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO											
240																	
250	C		Ball V	B06AC07	6 - 6	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO											
260 270																	
280 290						***** ORBIT VALVES *****											
300	C	VA5	Ball V	R06AC06	2 - 4	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO											
310 320 330 340						***** GATE VALVES *****		SERVICE									
350			Gate V	G06AC48	3/4 - 1 1/2	GATE 600# RF LTCS A350 LF2 TRIM 16 WEDGE HW		DESIGN: ASME B16.5 MG 1.1									
360			Gate V	G06AC41	2 - 2	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE HW		CODE: ASME B31.8									
370								ABOVE GROUND FACTOR = 0.5									
380			Condui	W06AC70	3 - 4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW		BELOW GROUND FACTOR = 0.6									
390 400 410 420						***** PIPE ABOVE GROUND *****		MAX. DESIGN PARAMETER = 67.5 BARG. @ 75									
430			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE		DEG.C. (LIMITED BY PIPE WALL THICKNESS)									
440			Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE		SERVICE - PP - PROPANE (MIN DESIGN									
450			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE		TEMPERATURE -43 DEG. C)									
460			Piping	9006210	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE IMPACT TESTED AT -46 DEG C		MATERIAL: LOW TEMP CARBON STEEL									
470								FLANGE RATING: 600 RF									
480			Piping	9001255	8 - 8	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE		TEMPERATURE RANGE: -46 / 75 °C									
490								COMMENT: LTCS									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17									
								BY JCS						REV C	SHEET 1 OF 3	PA 5	CLASS 600H5M

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
500						***** PIPE BELOW GROUND *****							
510													
520			Piping	9006210CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED IMPACT TESTED AT -46 DEG C							
530													
540													
550						***** FITTINGS ABOVE GROUND *****							
560													
570			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
580			90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6							
590			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
600			45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6							
610													
620			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6							
630			Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6							
640													
650			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2						
660			Reduci	5392454	3/4 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3						
670													
680			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6							
690			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6							
700			Cap	9006215	4 - 4	CAP BW WT 6.02 MM LTCS A860 WPHY-60 IMPACT TESTED AT -46 DEG C							
710													
720													
730			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2						
740			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2						
750			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3						
760			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3						
770			Eccent	9006216	2 - 2	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60 IMPACT TESTED AT -46 DEG C	4 - 4						
780			Eccent	9006217	3 - 3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60 IMPACT TESTED AT -46 DEG C	4 - 4						
790			Eccent	9001373	4 - 4	REDUCER ECC WT 7.11 MM / 6.02 MM LTCS A860 WPHY-60	8 - 8						
800													
810						*****FITTINGS BELOW GROUND*****							
820													
830			Eccent	9006216	2 - 2	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60 IMPACT TESTED AT -46 DEG C	4 - 4						
840			Eccent	9006217	3 - 3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60 IMPACT TESTED AT -46 DEG C	4 - 4						
850													
860						***** INDUCTION BENDS ABOVE GROUND*****							
870													
880			90BND	9006212	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW IMPACT TESTED AT -46 DEG C							
890													
900			45BND	9006213	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW IMPACT TESTED AT -46 DEG C							
910													
920			22BND	9006214	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW IMPACT TESTED AT -46 DEG C							
930													
940						***** INDUCTION BENDS BELOW GROUND*****							
950													
960			90BND	9006212CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED IMPACT TESTED AT -46							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 2 OF 3 PA 5 CLASS 600H5M					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES										
970	C	B1	45BND	9006213CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED IMPACT TESTED AT -46 DEG C	3 - 3 4 - 4											
980			22BND	9006214CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED IMPACT TESTED AT -46 DEG C												
990																		
1000																		
1010			Flange	5504007 9006220	3/4 - 2	NIPOFLANGE 600# RF SCH STD/XS LTCS A350 LF2 NIPOFLANGE 600# RF 6.02 MM/XS WT LTCS A694 F60 IMPACT TESTED AT -46 DEG C												
1020					3/4 - 2													
1030																		
1040			Flange	5557036 5557033 9006218	1/2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS FLG WN 600# RF LTCS A350 LF2 SCH STD FLG WN 600# RF LTCS A694 F60 WT 6.02MM IMPACT TESTED AT -46 DEG C												
1050					3 - 3													
1060					4 - 4													
1070			Blind	9006222	3/4 - 4	FLG BLIND 600# RF LTCS A694 F60 IMPACT TESTED AT -46 DEG C												
1080																		
1090																		
1100			Flange	9006218	4 - 4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM IMPACT TESTED AT -46 DEG C												
1110																		
1120																		
1130			GASFRS	5673103	1/2 - 4	GASKET 600# 316SS SP WND GRAPHITE IR												
1140																		
1150																		
1160			BOLS	5675770	1/4 - 4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS												
1170																		
1180																		
1190	Specta	5749587	3/4 - 4	SPECTACLE BLIND 600# RF LTCS A516-GR70														
1200																		
1210																		
1220	Reinfo	RWELD	2 - 2	NO-MTO REINFORCING WELD FOR STUB-IN														
1230																		
1240																		
1250	Reinfo	RWELD	2 - 2	NO-MTO REINFORCING WELD FOR STUB-IN														
1260																		
1270																		
1280	Reinfo	RWELD	6 - 6	NO-MTO REINFORCING WELD FOR STUB-IN														
1290																		
1300																		
1310	Reinfo	RWELD	6 - 6	NO-MTO REINFORCING WELD FOR STUB-IN														
1320																		
1330																		
1340	Reinfo	RWELD	6 - 6	NO-MTO REINFORCING WELD FOR STUB-IN														
1350																		
1360																		
1370	Reinfo	RWELD	6 - 6	NO-MTO REINFORCING WELD FOR STUB-IN														
1380																		
1390																		
1400	Reinfo	RWELD	6 - 6	NO-MTO REINFORCING WELD FOR STUB-IN														
1410																		
1420																		
1430	Reinfo	RWELD	6 - 6	NO-MTO REINFORCING WELD FOR STUB-IN														
1440																		
1450																		
1460	Reinfo	RWELD	6 - 6	NO-MTO REINFORCING WELD FOR STUB-IN														
1470																		
1480																		
1490	Reinfo	RWELD	6 - 6	NO-MTO REINFORCING WELD FOR STUB-IN														
1500																		
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD		FGP/WPMP CORR. ALLOW. DATE BY REV SHEET PA CLASS 3 MM 17NOV16 JCS C 3 OF 3 5 600H5M										

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
510	C		Butter	T06AC61	24 - 24	*****BUTTERFLY VALVES*****							
520						BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO							
530													
540													
550	C		Condui	W06AC70	3 - 4	***** GATE VALVES *****							
560						GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
570													
580						GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
590	C		Condui	W06AC71	6 - 6	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
600													
610													
620						GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
630	C		Condui	W06AC71	24 - 24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
640													
650													
660													
670	C		Globe	L06AC48	3/4 - 1 1/2	GLOBE 600# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
680	C		Globe	L06AC41	2 - 4	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW							
690	C		Globe	L06AC42	6 - 6	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y GO							
700													
710	C		Check	C06AC50	3/4 - 1 1/2	***** CHECK VALVES *****							
720													
730													
740						CHECK PISTON 600# RF LTCS A350 LF2 TRIM 16							
750	C		Check	C06AC45	2 - 6	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
760													
770													
780													
790	C		Check	C06AC52	24 - 24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16							
800													
810													
820													
830	C		Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE							
840	C		Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
850			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE							
860			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE							
870	C		Piping	9001254	6 - 6	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE							
880	C		Piping	9001491	24 - 24	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE							
890			Piping	9005244	28 - 28	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE							
900													
910	C		Piping	5358512	1/2 - 1/2	***** PIPE ABOVE GROUND *****							
920													
930													
940													
950	C		Piping	9001257CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED							
960				Piping	9001254CTD	6 - 6							LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED
970				Piping	9001491CTD	24 - 24							LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE COATED
980													
990	C		90 Deg	5392600	3/4 - 2	***** PIPE BELOW GROUND *****							
1000													
1010													
1020													
1030			90 Deg	5392438	3 - 3	***** FITTINGS ABOVE & BELOW GROUND *****							
			90 Deg	9006566	4 - 4	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
						ELL BW 90 DEG LR SCH STD LTCS A420 WPL6							
						ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60							
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION	CORR. ALLOW.	FGP/WPMP					
						ALL SIZES: BUTTWELD PWHT - PIPE SIZE 24"	1.5 MM	DATE	BY	REV	SHEET	PA	CLASS
								17MAR17	JCS	C	2 OF 6	5	600H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1040	C		90 Deg	9002190	6 - 6	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60								
1050		45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6									
1060		45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6									
1070		45 Deg	9006573	4 - 4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60									
1080	C		Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6								
1090		Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6									
1110		Tee	9301610	4 - 4	TEE BW WT 6.02 MM LTCS A860 WPHY-60									
1120		Tee	9301609	6 - 6	TEE BW WT 6.35 MM LTCS A860 WPHY-60									
1130	C		Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6								
1140	C		Reduci	5392454	1 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6								
1150	C		Reduci	9006664	2 - 2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60								
1160	C		Reduci	9006659	3 - 3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60								
1170	C		Reduci	9006658	3 - 3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60								
1180	C		Reduci	9006605	4 - 4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60								
1190	C		Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6								
1210			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6								
1220			Cap	9005062	4 - 4	CAP BW WT 6.02 MM LTCS A860 WPHY-60								
1230			Cap	9005055	6 - 6	CAP BW WT 6.35MM LTCS A860 WPHY-60								
1240	C		Cap	9005068	24 - 24	CAP BW WT 14.27 MM LTCS A860 WPHY-60								
1250	C		Cap	9005069	28 - 28	CAP BW WT 15.88 MM LTCS A860 WPHY-60								
1260	C		Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6								
1270			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6								
1280	C		Concen	9006918	1 1/2 - 3	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60								
1290	C		Concen	9006915	3 - 3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60								
1300	C		Concen	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6								
1310	C		Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6								
1320	C		Eccent	9001372	1 1/2 - 3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60								
1330	C		Eccent	9002510	3 - 3	REDUCER ECC WT 6.35 MM/STD LTCS A420 WPL6								
1340	C		Eccent	9002312	4 - 4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60								
1350	C		Eccent	9001386	24 - 24	REDUCER ECC WT 15.88 MM / 14.27MM LTCS A860 WPHY-60-W								
1360	C					***** INDUCTION BENDS ABOVE GROUND *****								
1370														
1380														
1390														
1400	C		90 Deg	9001810	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1410			90 Deg	9001813	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1420			90 Deg	9005322	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1430			45 Deg	9001820	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1440	C		45 Deg	9001823	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1450			45 Deg	9005328	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1460			45 Deg											
1470			22.5 D	9001830	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1480			22.5 D	9001833	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1490														
1500														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - PIPE SIZE 24"		CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY JCS	REV C	SHEET 3 OF 6	PA 5	CLASS 600H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1510	C		22.5 D	9005334	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
1520						***** INDUCTION BENDS BELOW GROUND *****							
1530													
1540													
1550			90 Deg	9001810CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1560			90 Deg	9001813CTD	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1570	C		90 Deg	9005322CTD	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1580													
1590			45 Deg	9001820CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1600			45 Deg	9001823CTD	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1610	C		45 Deg	9005328CTD	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1620													
1630			22.5 D	9001830CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1640			22.5 D	9001833CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1650	C		22.5 D	9005334CTD	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
1660						***** OLETS ABOVE & BELOW GROUND *****							
1670													
1680													
1690	C		Weldol	9006219	3/4 - 1 1/2	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60 IMPACT TESTED AT -46 DEG C	4 - 4	SERVICE					
1700	C		Weldol	9006407	3/4 - 2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6 - 6	DESIGN: ASME B16.5 CODE: ASME B31.8 MAX. DESIGN PARAMETERS (LIMITED BY PIPE WALL THICKNESS) - 93 BARG @ 80 DEG. C - 75 BARG. @ 100 DEG.C SERVICE PIPELINES/PIPING - FG, FGH					
1710	C		Weldol	9006820	3 - 3	WELDOLET WT 19.05 MM/STD WT LTCS A694 F60	24 - 24						
1720	C		Weldol	9005044	4 - 4	WELDOLET WT 14.27 MM/ 6.02 MM LTCS A694 F60	24 - 24						
1730	C		Weldol	9002473	6 - 6	WELDOLET WT 14.27 MM/ 6.35 MM LTCS A694 F60	24 - 24						
1740	C		Weldol	9002819	8 - 8	WELDOLET WT 14.27 MM/ 7.92 MM LTCS A694 F60	24 - 24						
1750			Weldol	9005046	4 - 4	WELDOLET WT 15.88 MM/ 6.02 MM LTCS A694 F60	28 - 28						
1760			Weldol	9002088	6 - 6	WELDOLET WT 15.88 MM/ 6.35 MM LTCS A694 F60	28 - 28						
1770			Weldol	9005047	8 - 8	WELDOLET WT 15.88 MM/ 7.92 MM LTCS A694 F60	28 - 28						
1780	C		Weldol	9002086	10 - 10	WELDOLET WT 15.88 MM/ 8.74 MM LTCS A694 F60	28 - 28						
1790													
1800	C		Nipofl	5504006	3/4 - 1 1/2	NIPOFLANGE 600# RF SCH XS LTCS A350 LF2	2 - 2	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 100 °C COMMENT: 93 BARG @ 80 DEGREES C					
1810	C		Nipofl	5504007	3/4 - 2	NIPOFLANGE 600# RF SCH STD/XS LTCS A350 LF2	3 - 3						
1820			Nipofl	9005030	3/4 - 2	NIPOFLANGE 600# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4						
1830			Nipofl	9004270	3/4 - 2	NIPOFLANGE 600# RF 6.35 MM/SCH XS WT LTCS A694 F60	6 - 6						
1840	C		Nipofl	9005105	3/4 - 2	NIPOFLANGE 600# RF 14.27 MM/XS WT LTCS A694 F60	24 - 24						
1850	C		Nipofl	9005034	3/4 - 2	NIPOFLANGE 600# RF 15.88 MM/XS WT LTCS A694 F60	28 - 28						
1860													
1870	C		Nipofl	9006821	2 - 2	NIPOFLANGE 1500# RF WT 6.35 MM/ SCH 160 LTCS A694 F60	6 - 6						
1880						***** PIPE FLANGES ABOVE & BELOW GROUND*****							
1890													
1900													
1910	C		Flange	5557036	1/2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS							
1920	C		Flange	5557033	3 - 3	FLG WN 600# RF LTCS A350 LF2 SCH STD							
1930			Flange	9001723	4 - 4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM							
			FGP / WPMP TENGINZCHEVROIL - TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - PIPE SIZE 24"	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY JCS	REV C	SHEET 4 OF 6	PA 5	CLASS 600H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES												
1940	C	B1	Flange	9001471	6 - 6	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	3/4 - 1 1/2													
1950			Flange	9001515	24 - 24	FLG WN 600# RF LTCS A694 F60 WT 14.27MM														
1960																				
1970	C		Blind	5615598	1/2 - 3	FLG BLIND 600# RF LTCS A350 LF2														
1980			Blind	9001518	4 - 24	FLG BLIND 600# RF LTCS A694 F60														
1990	C		Reduci	5599705	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 600# RF A350 LF2														
2000																				
2010	C		817	Flange	5558036	1 1/2 - 2								FLG WN 600# RTJ LTCS A350 LF2 SCH XS						
2040				Flange	9006724	4 - 4								FLG WN 600# RTJ A694 F60 WT 6.02MM						
2050				Flange	9006704	6 - 6								FLG WN 600# RTJ A694 F60 WT 6.35MM						
2060	C		817	Blind	5615698	1/2 - 3								FLG BLIND 600# RTJ LTCS A350 LF2						
2070																				
2080																				
2090	C			Flange	5557393	2 - 2								FLG WN 1500# RF LTCS A350 LF2 SCH XS						
2100	C			Flange	9005389	4 - 4								FLG WN 900# RF LTCS A694 F60 WT 6.02MM						
2110	C			Flange	9006755	6 - 6								FLG WN 900# RF LTCS A694 F60 WT 6.35MM						
2120	C		Orific	9001556	6 - 6	WN ORIF SET 600# RF A694 F60 WT 6.35MM														
2130																				
2140																				
2150	C					***** GASKETS*****														
2160																				
2170																				
2180	C		GASFRS	5673103	1/2 - 24	GASKET 600# 316SS SP WND GRAPHITE IR														
2190	C		GASRJ	5673202	1 1/2 - 6	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING														
2200																				
2210																				
2220						*****BOLTING*****														
2230																				
2240																				
2250	C		BOLS	5675770	3/4 - 24	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS														
2260			CPSBOL	5675675	3/4 - 24	JACK SCREW HEX HEAD A320 GR L7M														
2270																				
2280																				
2290						***** BLINDS & SPACERS														
2300																				
2310																				
2320			Specta	5749587	3/4 - 6	SPECTACLE BLIND 600# RF LTCS A516-GR70														
2330														Paddle	5750040	24 - 24	PADDLE BLANK 600# RF LTCS A516-GR70			
2340														Paddle	5750251	24 - 24	PADDLE SPACER 600# RF LTCS A516-GR70			
2350						***** BRANCHES *****														
2360																				
2370																				
2380						USE BRANCH CHART 600H5P														
2390																				
2400																				
2410	C	B1	Reinfo	RWELD	2 - 4	NO-MTO REINFORCING WELD FOR STUB-IN								3 - 6	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 100 °C COMMENT: 93 BARG @ 80 DEGREES C					
2420	C		Reinfo	RWELD	2 - 4	NO-MTO REINFORCING WELD FOR STUB-IN								3 - 6						
2430	C		Reinfo	RWELD	16 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	24 - 26													
2440																				
2450	C		Reinfo	RWELD	16 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	24 - 26													
2460																				
2470																				
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - PIPE SIZE 24"		CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY JCS	REV C	SHEET 5 OF 6	PA 5	CLASS 600H5P						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2480	C	26	Flange	SP-2648	2 - 2	*****SP-ITEMS*****								
2490						FLANGE TRANSITION 600H5P/10000H5A COMPACT FLG RJ								
2500						A350 LF2 SCH XS								
2510						***** CONSTRUCTION DETAILS *****								
2520														
2530						SEAL WELDING ----- REQUIRED								
2540						PIPE BENDS ----- INDUCTION BENDS								
2550						THREAD SEALANT ----- PTFE PASTE								
2560														
2570						*****DO NOT USE*****								
2580	C		Eccent	9001372	1 1/2 - 3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4							
2590														
2600														
2610														
2620														
2630														
								SERVICE						
								DESIGN: ASME B16.5 CODE: ASME B31.8 MAX. DESIGN PARAMETERS (LIMITED BY PIPE WALL THICKNESS) - 93 BARG @ 80 DEG. C - 75 BARG. @ 100 DEG.C SERVICE PIPELINES/PIPING - FG, FGH						
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 100 °C COMMENT: 93 BARG @ 80 DEGREES C						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						ALL SIZES: BUTTWELD PWHT - PIPE SIZE 24"	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
							1.5 MM	17MAR17	JCS	C	6OF 6	5	600H5P	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10		800						O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. THE TEMPERATURE IS LIMITED TO 260°C IN CRUDE OIL SERVICES CONTAINING H2S /AND /OR ORGANIC SULPHUR COMPOUNDS, FOR EXAMPLE, IN THE CRUDE STABILIZATION UNIT. IN HYDROCARBON GAS SERVICE CONTAINING H2S, FOR EXAMPLE, SOUR GAS MOLECULAR SIEVE DEHYDRATION UNITS, USE IS LIMITED TO 230 C. EXCEPTIONS TO THIS ARE THE SULPHUR RECOVERY AND TAIL GAS UNITS WHERE CARBON STEEL SOUR SERVICE LINE CLASSES CAN BE USED UP TO 315°C BECAUSE OF LOWER H2S PARTIAL PRESSURES.					
20		PP9				*****MODULAR VALVES*****		800						
30														
40														
50	D		2BV	K06AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV								
60														
70	D		2BV	K06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV								
80														
90	D	VAV	2BV	K06AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV								
100														
110	D		2BVR	M06AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		812	NIPPLE TO BE CUT FROM SCH XXS PIPE					
120								815	TO BE USED WITH COMPACT FLANGES ONLY.					
130	D		2BVR	M06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME'S					
140								817	USE ONLY FOR TIE IN					
150	D	VAW	R2BVR	M06EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.					
170								BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.					
180	D	VAW	R2BVR	M06EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.					
190								FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.					
200								FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.					
210								ID1	DELETED					
220						***** BALL VALVES *****		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
230	D	VB2	Ball V	A06AC05	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
240								SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTRORACE.					
250	D		Ball V	A06AC06	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO								
260														
270	D		Ball V	A06AC07	6 - 8	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
280														
290	D		Ball V	A06AC08	6 - 8	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
300														
310														
320			Ball V	B06AC01	3/4 - 1 1/2	BALL 600# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
330														
340			Ball V	B06AC02	3/4 - 1 1/2	BALL 600# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
350														
360	D	VA5	Ball V	B06AC05	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO								
370														
380	D	VA5	Ball V	B06AC06	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO								
390														
400	D	VA5	Ball V	B06AC07	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
410														
420	D	VA5	Ball V	B06AC08	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
430														
440	D	VA5	Ball V	B06CC07-K21	26 - 42	BALL 600# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
450														
460	D	VA5	Ball V	B06CC08-K21	26 - 42	BALL 600# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
470														
480														
490						*****ORBIT VALVES*****								
500														
510	D	VA5	Ball V	R06AC06	3/4 - 6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO								
520	D	VA5	Ball V	R06AC08	8 - 10	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB GO								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES		CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 1 OF 9 PA 5 CLASS 600K21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
530	D	VQ2	Butter	T06AC61	10 - 24	***** BUTTERFLY VALVES *****		UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.							
540						UD7		USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.								
550			Butter	T06CC61-K21	26 - 42	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO		UDA	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.							
560								UJ2	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.							
570			C	Condui	W06AC70	3 - 4		GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	VA5	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY						
580									VAB	FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE).						
590			C	Condui	W06CC71	26 - 42		GATE 600# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	VAD	FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY.						
600									VAG	FOR ALL INLINE APPLICATIONS AND LEVEL TRANSMITTER CONNECTIONS						
610			VQ2						*****GATE VALVE*****	VAK	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055					
620										VAV	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
630	C	817	Gate V	G06AC48	3/4 - 3/4	GATE 600# RF LTCS A350 LF2 TRIM 16 WEDGE HW	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.								
640							VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.								
650	D	817	Gate V	G06AC41	4 - 4	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE HW	VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.								
660																
670	D	VQ2				***** GLOBE VALVES *****		SERVICE								
680						SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3										
690						BD, BF, BFH, BFL, CCD, CDE, CDM, DCS, DHC, DMD, FG, FGH, FM, GE, GHP, GSL, GRC, GS, NL, PHC, SL, WM, WSR, WRA										
700																
710																
720																
730																
740																
750																
760																
770	D	VC1	Check	C06AC50	3/4 - 1 1/2	CHECK PISTON 600# RF LTCS A350 LF2 TRIM 16		MATERIAL: LOW TEMP CARBON STEEL								
780								FLANGE RATING: 600 RF								
790								TEMPERATURE RANGE: -46 / 260 °C								
800								COMMENT: REFER NOTE 800								
810																
820																
830																
840																
850																
860								D	VC2	Check	C06AC52	10 - 24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16			
870																
880																
890																
900																
910																
920																
930																
940																
950	D	VC2	Check	C06CC52-K21	26 - 42	CHECK DUAL 600# CF LTCS A352 LCC TRIM 16										
960																
970																
980																
990																
1000																
1010																
1020																
1030																
1040								D	812	Piping	5358511	1/2 - 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE			
1050	D	812	Piping	5358511	3/4 - 3/4	PIPE SCH XXS SMLS LTCS A333 GR.6 PE										
1060																
1070																
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 2 OF 9	PA 5	CLASS 600K21			

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1620	D		Reduci	5392482	34 - 34	TEE RED WT 41.0MM/39.0MM LTCS A420 WPL6-W	36 - 36						
1630	D		Reduci	5392592	20 - 24	TEE RED WT 48.0MM/SCH 80 LTCS A420 WPL6-W	42 - 42						
1640	D		Reduci	5392593	30 - 30	TEE RED WT 48.0MM/35.0MM LTCS A420 WPL6-W	42 - 42						
1650	D		Reduci	5392595	36 - 36	TEE RED WT 48.0MM/41.0MM LTCS A420 WPL6-W	42 - 42						
1660													
1670			Cap	5392826	3/4 - 2	CAP BW SCH 160 LTCS A420 WPL6							
1680			Cap	5392610	3 - 8	CAP BW SCH XS LTCS A420 WPL6							
1690			Cap	5392556	10 - 24	CAP SCH 80 LTCS STL A420 WPL6							
1700			Cap	5392630	26 - 26	CAP WT 31.0MM LTCS A420 WPL6							
1710			Cap	5392631	28 - 28	CAP WT 33.0MM LTCS A420 WPL6							
1720			Cap	5392632	30 - 30	CAP WT 35.0MM LTCS A420 WPL6							
1730			Cap	5392633	32 - 32	CAP WT 37.0MM LTCS A420 WPL6							
1740			Cap	5392634	34 - 34	CAP WT 39.0MM LTCS A420 WPL6							
1750			Cap	5392635	36 - 36	CAP WT 41.0MM LTCS A420 WPL6							
1760			Cap	5392636	42 - 42	CAP WT 48.0MM LTCS A420 WPL6							
1770													
1780	C		Concen	5392827	3/4 - 1 1/2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 2						
1790	C		Concen	5392462	1 1/2 - 2	REDUCER CONC SCH XS/160 LTCS A420 WPL6	3 - 4						
1800			Concen	5392458	3 - 6	REDUCER CONC SCH XS LTCS A420 WPL6	4 - 8						
1810			Concen	5392309	4 - 8	REDUCER CONC SCH 80/XS LTCS A420 WPL6	10 - 16						
1820			Concen	5392557	10 - 14	REDUCER CONC SCH 80 LTCS A420 WPL6	12 - 16						
1830			Concen	5392559	10 - 20	REDUCER CONC SCH 80 LTCS A420 WPL6-W	18 - 24						
1840			Concen	5392362	18 - 24	REDUCER CONC WT 31MM/SCH 80 LTCS A420 WPL6-W	26 - 26						
1850			Concen	5392361	18 - 24	REDUCER CONC WT 33MM/SCH 80 LTCS A420 WPL6-W	28 - 28						
1860			Concen	5392369	20 - 24	REDUCER CONC WT 35.0MM/SCH 80 LTCS A420 WPL6-W	30 - 30						
1870			Concen	5392373	24 - 24	REDUCER CONC WT 37.0MM/SCH 80 LTCS A420 WPL6-W	32 - 32						
1880			Concen	5392363	24 - 24	REDUCER CONC WT 41.0MM/SCH 80 LTCS A420 WPL6-W	36 - 36						
1890			Concen	5392364	26 - 26	REDUCER CONC WT 33.0MM/31.0MM LTCS A420 WPL6-W	28 - 28						
1900			Concen	5392367	26 - 26	REDUCER CONC WT 35.0MM/31.0MM LTCS A420 WPL6-W	30 - 30						
1910			Concen	5392372	26 - 26	REDUCER CONC WT 37.0MM/31.0MM LTCS A420 WPL6-W	32 - 32						
1920			Concen	5392377	26 - 26	REDUCER CONC WT 39.0MM/31.0MM LTCS A420 WPL6-W	34 - 34						
1930			Concen	5392382	26 - 26	REDUCER CONC WT 41.0MM/31.0MM LTCS A420 WPL6-W	36 - 36						
1940			Concen	5392366	28 - 28	REDUCER CONC WT 35.0MM/33.0MM LTCS A420 WPL6-W	30 - 30						
1950			Concen	5392371	28 - 28	REDUCER CONC WT 37.0MM/33.0MM LTCS A420 WPL6-W	32 - 32						
1960			Concen	5392376	28 - 28	REDUCER CONC WT 39.0MM/33.0MM LTCS A420 WPL6-W	34 - 34						
1970			Concen	5392381	28 - 28	REDUCER CONC WT 41.0MM/33.0MM LTCS A420 WPL6-W	36 - 36						
1980			Concen	5392370	30 - 30	REDUCER CONC WT 37.0MM/35.0MM LTCS A420 WPL6-W	32 - 32						
1990			Concen	5392375	30 - 30	REDUCER CONC WT 39.0MM/35.0MM LTCS A420 WPL6-W	34 - 34						
2000			Concen	5392380	30 - 30	REDUCER CONC WT 41.0MM/35.0MM LTCS A420 WPL6-W	36 - 36						
2010			Concen	5392431	30 - 30	REDUCER CONC WT 48.0MM/35.0MM LTCS A420 WPL6-W	42 - 42						
2020			Concen	5392374	32 - 32	REDUCER CONC WT 39.0MM/37.0MM LTCS A420 WPL6-W	34 - 34						
2030			Concen	5392379	32 - 32	REDUCER CONC WT 41.0MM/37.0MM LTCS A420 WPL6-W	36 - 36						
2040			Concen	5392430	32 - 32	REDUCER CONC WT 48.0MM/37.0MM LTCS A420 WPL6-W	42 - 42						
2050			Concen	5392378	34 - 34	REDUCER CONC WT 41.0MM/39.0MM LTCS A420 WPL6-W	36 - 36						
2060			Concen	5392429	34 - 34	REDUCER CONC WT 48.0MM/39.0MM LTCS A420 WPL6-W	42 - 42						
2070			Concen	5392383	36 - 36	REDUCER CONC WT 48.0MM/41.0MM LTCS A420 WPL6-W	42 - 42						
2080													
2090	C		Eccent	5392828	3/4 - 1 1/2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 2						
2100	C		Eccent	5392465	1 1/2 - 2	REDUCER ECC SCH XS/160 LTCS A420 WPL6	3 - 4						
2110			Eccent	5392612	3 - 6	REDUCER ECC SCH XS LTCS A420 WPL6	4 - 8						
2120			Eccent	5392311	4 - 8	REDUCER ECC SCH 80/XS LTCS A420 WPL6	10 - 16						
2130			Eccent	5392558	10 - 14	REDUCER ECC SCH 80 LTCS A420 WPL6	12 - 16						
2140			Eccent	5392313	10 - 20	REDUCER ECC SCH 80 LTCS A420 WPL6-W	18 - 24						
2150			Eccent	5392975	18 - 24	REDUCER ECC WT 31.0MM/SCH 80 LTCS A420 WPL6-W	26 - 26						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP						
						ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							3 MM	17MAR17	SMS	D	4 OF 9	5	600K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2160			Eccent	5392519	18 - 24	REDUCER ECC WT 33MM/SCH 80 LTCS A420 WPL6-W	28 - 28						
2170			Eccent	5392640	20 - 24	REDUCER ECC WT 35.0MM/SCH 80 LTCS A420 WPL6-W	30 - 30						
2180			Eccent	5392492	24 - 24	REDUCER ECC WT 37.0MM/SCH 80 LTCS A420 WPL6-W	32 - 32						
2190			Eccent	5392529	24 - 24	REDUCER ECC WT 41.0MM/SCH 80 LTCS A420 WPL6-W	36 - 36						
2200			Eccent	5392999	26 - 26	REDUCER ECC WT 33MM/31.0MM LTCS A420 WPL6-W	28 - 28						
2210			Eccent	5392641	26 - 26	REDUCER ECC WT 35.0MM/31.0MM LTCS A420 WPL6-W	30 - 30						
2220			Eccent	5392493	26 - 26	REDUCER ECC WT 37.0MM/31.0MM LTCS A420 WPL6-W	32 - 32						
2230			Eccent	5392488	26 - 26	REDUCER ECC WT 39.0MM/31.0MM LTCS A420 WPL6-W	34 - 34						
2240			Eccent	5392437	26 - 26	REDUCER ECC WT 41.0MM/31.0MM LTCS A420 WPL6-W	36 - 36						
2250			Eccent	5392642	28 - 28	REDUCER ECC WT 35.0MM/33.0MM LTCS A420 WPL6-W	30 - 30						
2260			Eccent	5392494	28 - 28	REDUCER ECC WT 37.0MM/33.0MM LTCS A420 WPL6-W	32 - 32						
2270			Eccent	5392489	28 - 28	REDUCER ECC WT 39.0MM/33.0MM LTCS A420 WPL6-W	34 - 34						
2280			Eccent	5392484	28 - 28	REDUCER ECC WT 41.0MM/33.0MM LTCS A420 WPL6-W	36 - 36						
2290			Eccent	5392495	30 - 30	REDUCER ECC WT 37.0MM/35.0MM LTCS A420 WPL6-W	32 - 32						
2300			Eccent	5392490	30 - 30	REDUCER ECC WT 39.0MM/35.0MM LTCS A420 WPL6-W	34 - 34						
2310			Eccent	5392485	30 - 30	REDUCER ECC WT 41.0MM/35.0MM LTCS A420 WPL6-W	36 - 36						
2320			Eccent	5392432	30 - 30	REDUCER ECC WT 48.0MM/35.0MM LTCS A420 WPL6-W	42 - 42						
2330			Eccent	5392491	32 - 32	REDUCER ECC WT 39.0MM/37.0MM LTCS A420 WPL6-W	34 - 34						
2340			Eccent	5392486	32 - 32	REDUCER ECC WT 41.0MM/37.0MM LTCS A420 WPL6-W	36 - 36						
2350			Eccent	5392433	32 - 32	REDUCER ECC WT 48.0MM/37.0MM LTCS A420 WPL6-W	42 - 42						
2360			Eccent	5392487	34 - 34	REDUCER ECC WT 41.0MM/39.0MM LTCS A420 WPL6-W	36 - 36						
2370			Eccent	5392434	34 - 34	REDUCER ECC WT 48.0MM/39.0MM LTCS A420 WPL6-W	42 - 42						
2380			Eccent	5392435	36 - 36	REDUCER ECC WT 48.0MM/41.0MM LTCS A420 WPL6-W	42 - 42						
2390						***** OLETS *****							
2400													
2410													
2420			Nipofl	5504460	3/4 - 1 1/2	NIPOFLANGE 600# RF SCH 160 LTCS A350 LF2	2 - 2						
2430			Nipofl	5504461	3/4 - 2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2	3 - 8						
2440			Nipofl	5504462	3/4 - 2	NIPOFLANGE 600# RF SCH 80/160 LTCS A350 LF2	10 - 24						
2450			Nipofl	5504463	3/4 - 2	NIPOFLANGE 600# RF WT 31.0MM/SCH 160 LTCS A350 LF2	26 - 26						
2460			Nipofl	5504464	3/4 - 2	NIPOFLANGE 600# RF WT 33.0MM/SCH 160 LTCS A350 LF2	28 - 28						
2470			Nipofl	5504465	3/4 - 2	NIPOFLANGE 600# RF WT 35.0MM/SCH 160 LTCS A350 LF2	30 - 30						
2480			Nipofl	5504466	3/4 - 2	NIPOFLANGE 600# RF WT 37.0MM/SCH 160 LTCS A350 LF2	32 - 32						
2490			Nipofl	5504467	3/4 - 2	NIPOFLANGE 600# RF WT 39.0MM/SCH 160 LTCS A350 LF2	34 - 34						
2500			Nipofl	5504468	3/4 - 2	NIPOFLANGE 600# RF WT 41.0MM/SCH 160 LTCS A350 LF2	36 - 36						
2510			Nipofl	5504469	3/4 - 2	NIPOFLANGE 600# RF WT 48.0MM/SCH 160 LTCS A350 LF2	42 - 42						
2520								SERVICE					
2530	UJ2		Nipofl	5504028	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2	4 - 8	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BD, BF, BFH, BFL, CCD, CDE, CDM, DCS, DHC, DMD, FG, FGH, FM, GE, GHP, GSL, GRC, GS, NL, PHC, SL, WM, WSR, WRA					
2540	UJ2		Nipofl	5504029	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 80/160 LTCS A350 LF2	10 - 24						
2550	UJ2		Nipofl	5504127	1 1/2 - 1 1/2	NIPOFLANGE 600# RF WT 31.0MM/160 LTCS A350 LF2	26 - 26						
2560	UJ2		Nipofl	5504126	1 1/2 - 1 1/2	NIPOFLANGE 600# RF WT 33.0MM/160 LTCS A350 LF2	28 - 28						
2570	UJ2		Nipofl	5504128	1 1/2 - 1 1/2	NIPOFLANGE 600# RF WT 35.0MM/160 LTCS A350 LF2	30 - 30						
2580	UJ2		Nipofl	5504129	1 1/2 - 1 1/2	NIPOFLANGE 600# RF WT 37.0MM/160 LTCS A350 LF2	32 - 32						
2590	UJ2		Nipofl	5504130	1 1/2 - 1 1/2	NIPOFLANGE 600# RF WT 39.0MM/160 LTCS A350 LF2	34 - 34						
2600	UJ2		Nipofl	5504131	1 1/2 - 1 1/2	NIPOFLANGE 600# RF WT 41.0MM/160 LTCS A350 LF2	36 - 36						
2610	UJ2		Nipofl	5504133	1 1/2 - 1 1/2	NIPOFLANGE 600# RF WT 48.0MM/160 LTCS A350 LF2	42 - 42						
2620								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 260 °C COMMENT: REFER NOTE 800					
2630			Weldol	5505392	3/4 - 1	WELDOLET SCH 160 LTCS A350 LF2	2 - 2						
2640			Weldol	5505393	3/4 - 2	WELDOLET SCH XS/160 LTCS A350 LF2	3 - 8						
2650			Weldol	5505263	3/4 - 2	WELDOLET SCH 80/160 LTCS A350 LF2	10 - 24						
2660			Weldol	5505618	3/4 - 2	WELDOLET WT 31.0MM/SCH 160 LTCS A350 LF2	26 - 26						
2670			Weldol	5505635	3/4 - 2	WELDOLET WT 33.0MM/SCH 160 LTCS A350 LF2	28 - 28						
2680			Weldol	5505605	3/4 - 2	WELDOLET WT 35.0MM/SCH 160 LTCS A350 LF2	30 - 30						
2690			Weldol	5505608	3/4 - 2	WELDOLET WT 37.0MM/SCH 160 LTCS A350 LF2	32 - 32						
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY SMS REV D SHEET 5 OF 9 PA 5 CLASS 600K21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2700			Weldol	5505621	3/4 - 2	WELDOLET WT 39.0MM/SCH 160 LTCS A350 LF2	34 - 34							
2710			Weldol	5505624	3/4 - 2	WELDOLET WT 41.0MM/SCH 160 LTCS A350 LF2	36 - 36							
2720			Weldol	5505614	3/4 - 2	WELDOLET WT 48.0MM/SCH 160 LTCS A350 LF2	42 - 42							
2730														
2740			Weldol	5505335	3 - 3	WELDOLET SCH XS LTCS A350 LF2	8 - 8							
2750			Weldol	5505262	3 - 8	WELDOLET SCH 80/XS LTCS A350 LF2	10 - 24							
2760			Weldol	5505617	3 - 8	WELDOLET WT 31.0MM/SCH XS LTCS A350 LF2	26 - 26							
2770			Weldol	5505602	3 - 8	WELDOLET WT 33.0MM/SCH XS LTCS A350 LF2	28 - 28							
2780			Weldol	5505606	3 - 8	WELDOLET WT 35.0MM/SCH XS LTCS A350 LF2	30 - 30							
2790			Weldol	5505619	3 - 8	WELDOLET WT 37.0MM/SCH XS LTCS A350 LF2	32 - 32							
2800			Weldol	5505622	3 - 8	WELDOLET WT 39.0MM/SCH XS LTCS A350 LF2	34 - 34							
2810			Weldol	5505611	3 - 8	WELDOLET WT 41.0MM/SCH XS LTCS A350 LF2	36 - 36							
2820			Weldol	5505601	3 - 8	WELDOLET WT 48.0MM/SCH XS LTCS A350 LF2	42 - 42							
2830														
2840			Weldol	5505609	10 - 12	WELDOLET WT 31.0MM/SCH 80 LTCS A350 LF2	26 - 26							
2850			Weldol	5505603	10 - 12	WELDOLET WT 33.0MM/SCH 80 LTCS A350 LF2	28 - 28							
2860			Weldol	5505607	10 - 12	WELDOLET WT 35.0MM/SCH 80 LTCS A350 LF2	30 - 30							
2870			Weldol	5505620	10 - 12	WELDOLET WT 37.0MM/SCH 80 LTCS A350 LF2	32 - 32							
2880			Weldol	5505623	10 - 12	WELDOLET WT 39.0MM/SCH 80 LTCS A350 LF2	34 - 34							
2890			Weldol	5505610	10 - 12	WELDOLET WT 41.0MM/SCH 80 LTCS A350 LF2	36 - 36							
2900			Weldol	5505627	10 - 12	WELDOLET WT 48.0MM/SCH 80 LTCS A350 LF2	42 - 42							
2910														
2920			UJ2	Elbole	5516659	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH XS/160 LTCS A350 LF2							4 - 8
2930			UJ2	Elbole	5516660	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 80/160 LTCS A350 LF2							10 - 24
2940			UJ2	Elbole	5516664	1 1/2 - 1 1/2	ELBOFLANGE 600# RF WT 31.0MM/160 LTCS A350 LF2							26 - 26
2950			UJ2	Elbole	5516665	1 1/2 - 1 1/2	ELBOFLANGE 600# RF WT 33.0MM/160 LTCS A350 LF2							28 - 28
2960			UJ2	Elbole	5516666	1 1/2 - 1 1/2	ELBOFLANGE 600# RF WT 35.0MM/160 LTCS A350 LF2							30 - 30
2970			UJ2	Elbole	5516667	1 1/2 - 1 1/2	ELBOFLANGE 600# RF WT 37.0MM/160 LTCS A350 LF2							32 - 32
2980			UJ2	Elbole	5516668	1 1/2 - 1 1/2	ELBOFLANGE 600# RF WT 39.0MM/160 LTCS A350 LF2							34 - 34
2990			UJ2	Elbole	5516669	1 1/2 - 1 1/2	ELBOFLANGE 600# RF WT 41.0MM/160 LTCS A350 LF2							36 - 36
3000			UJ2	Elbole	5516670	1 1/2 - 1 1/2	ELBOFLANGE 600# RF WT 48.0MM/160 LTCS A350 LF2							42 - 42
3010														
3020														
3030						***** OLETS *****								
3040														
3050	D	UD7	Sockol	5533071	3/4 - 1 1/2	SOCKOLET 6000# LTCS A350 LF2	2 - 42	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BD, BF, BFH, BFL, CCD, CDE, CDM, DCS, DHC, DMD, FG, FGH, FM, GE, GHP, GSL, GRC, GS, NL, PHC, SL, WM, WSR, WRA MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 260°C COMMENT: REFER NOTE 800						
3060	D		Endole	5533233	3/4 - 1 1/2	FLAT SOCKOLET 6000# LTCS A350 LF2								
3070	D		Thredo	5514772	3/4 - 1	THREDOLET 3000# LTCS A350 LF2	2 - 42							
3080														
3090						***** COUPLINGS, UNIONS & PLUGS *****								
3100														
3110	D		Coupli	5478807	3/4 - 1 1/2	COUPLING 6000# SW LTCS A350 LF2								
3120	D		Reduci	5478812	3/4 - 1	COUPLING RED 6000# SW LTCS A350 LF2	1 - 1 1/2							
3130	D		Union	5478809	3/4 - 1 1/2	UNION 6000# SW LTCS STL A350 LF2								
3140	D		Plug	5491303	3/4 - 1	PLUG ROUND HEAD SCR D LTCS A350 LF2								
3150														
3160		816				**** FLANGES ****								
3170														
3180		FA8	Reduci	5599705	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 600# RF A350 LF2	3/4 - 1 1/2							
3190														
3200	D		Flange	5851204	1/2 - 2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE								
3210														
3220	D		Flange	5557038	1/2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH 160								
3230			Flange	5557036	3 - 8	FLG WN 600# RF LTCS A350 LF2 SCH XS								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES		CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	FGP/WPMP SHEET 6 OF 9 PA 5 CLASS 600K21		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3240			Flange	5557040	10 - 24	FLG WN 600# RF LTCS A350 LF2 SCH 80		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BD, BF, BFH, BFL, CCD, CDE, CDM, DCS, DHC, DMD, FG, FGH, FM, GE, GHP, GSL, GRC, GS, NL, PHC, SL, WM, WSR, WRA MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 260 °C COMMENT: REFER NOTE 800					
3250			Flange	5556550	26 - 26	FLG WN 600# RF LTCS A350 LF2 WT 31.0MM B16.47 SERIES A							
3260			Flange	5556551	28 - 28	FLG WN 600# RF LTCS A350 LF2 WT 33.0MM B16.47 SERIES A							
3270			Flange	5556552	30 - 30	FLG WN 600# RF LTCS A350 LF2 WT 35.0MM B16.47 SERIES A							
3280			Flange	5556553	32 - 32	FLG WN 600# RF LTCS A350 LF2 WT 37.0MM B16.47 SERIES A							
3290			Flange	5556554	34 - 34	FLG WN 600# RF LTCS A350 LF2 WT 39.0MM B16.47 SERIES A							
3300			Flange	5556555	36 - 36	FLG WN 600# RF LTCS A350 LF2 WT 41.0MM B16.47 SERIES A							
3310			Flange	5556556	42 - 42	FLG WN 600# RF LTCS A350 LF2 WT 48.0MM B16.47 SERIES A							
3320													
3330			Flange	5619801	26 - 26	FLG COMPACT 600# RJ LTCS A350 LF2 WT 31.0MM							
3340			Flange	5619970	28 - 28	FLG COMPACT 600# RJ LTCS A350 LF2 WT 33.0MM							
3350			Flange	5619802	30 - 30	FLG COMPACT 600# RJ LTCS A350 LF2 WT 35.0MM							
3360			Flange	5619803	32 - 32	FLG COMPACT 600# RJ LTCS A350 LF2 WT 37.0MM							
3370			Flange	5619804	34 - 34	FLG COMPACT 600# RJ LTCS A350 LF2 WT 39.0MM							
3380			Flange	5619805	36 - 36	FLG COMPACT 600# RJ LTCS A350 LF2 WT 41.0MM							
3390			Flange	5619806	42 - 42	FLG COMPACT 600# RJ LTCS A350 LF2 WT 48.0MM							
3400													
3410	D	817	Flange	5558040	16 - 16	FLG WN 600# RTJ LTCS A350 LF2 SCH 80							
3420													
3430	C	FS1	Flange	5557141	3 - 8	FLG WN 900# RF LTCS A350 LF2 SCH XS							
3440	D	FS1	Flange	5557147	10 - 10	FLG WN 900# RF LTCS A350 LF2 SCH 80							
3450	D	FS1	Flange	5557147	24 - 24	FLG WN 900# RF LTCS A350 LF2 SCH 80							
3460	D	FS1	Flange	5556542	30 - 30	FLG WN 900# RF LTCS A350 LF2 WT 35.0MM B16.47 SERIES A							
3470													
3480	C	FS1	Flange	5557397	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH 160 BORE							
3490													
3500	C		Blind	5615598	1/2 - 24	FLG BLIND 600# RF LTCS A350 LF2							
3510			Blind	5615580	26 - 36	FLG BLIND 600# RF LTCS A350 LF2 B16.47 SERIES A							
3520			Blind	5615581	42 - 42	FLG BLIND 600# RF LTCS A350 LF2 B16.47 SERIES A							
3530													
3540			Blind	5619764	26 - 42	FLG COMPACT BLIND 600# RJ LTCS A350 LF2 PIPE CLASS 600K21							
3550													
3560													
3570	D		Blind	5615605	3 - 3	FLG BLIND 900# RF LTCS A350 LF2							
3580													
3590						***** ORIFICE FLANGES *****							
3600													
3610			Orific	5624351	2 - 2	WN ORIF SET 600# RF LTCS A350 LF2 SCH 160							
3620			Orific	5624352	3 - 8	WN ORIF SET 600# RF LTCS A350 LF2 SCH XS BORE							
3630			Orific	5624353	10 - 24	WN ORIF SET 600# RF LTCS A350 LF2 SCH 80 BORE							
3640			Orific	5624355	26 - 26	WN ORIF SET 600# RF LTCS A350 LF2 31.0MM WT ASME B16.47 SERIES A							
3650			Orific	5624356	28 - 28	WN ORIF SET 600# RF LTCS A350 LF2 33.0MM WT ASME B16.47 SERIES A							
3660			Orific	5624357	30 - 30	WN ORIF SET 600# RF LTCS A350 LF2 35.0MM WT ASME B16.47 SERIES A							
3670			Orific	5624358	32 - 32	WN ORIF SET 600# RF LTCS A350 LF2 37.0MM WT ASME B16.47 SERIES A							
3680			Orific	5624359	34 - 34	WN ORIF SET 600# RF LTCS A350 LF2 39.0MM WT ASME B16.47 SERIES A							
3690			Orific	5624360	36 - 36	WN ORIF SET 600# RF LTCS A350 LF2 41.0MM WT ASME B16.47 SERIES A							
3700			Orific	5624361	42 - 42	WN ORIF SET 600# RF LTCS A350 LF2 48.0MM WT ASME B16.47 SERIES A							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 17MAR17	BY SMS	REV D	SHEET 7 OF 9	PA 5	CLASS 600K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3710 3720 3730 3740 3750 3760 3770 3780 3790	D	812	GASFRS GASFRS	5673103 5674012	1/2 - 24 26 - 42	***** PIPE NIPPLES ***** NIPPLE PBE - CUT FROM PIPE ***** GASKETS ***** GASKET 600# 316SS SP WND GRAPHITE IR GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BD, BF, BFH, BFL, CCD, CDE, CDM, DCS, DHC, DMD, FG, FGH, FM, GE, GHP, GSL, GRC, GS, NL, PHC, SL, WM, WSR, WRA MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 260 °C COMMENT: REFER NOTE 800					
3800 3810		815	GASRJ	5673414	26 - 42	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 600K21							
3820 3830 3840	D	817	GASRJ	5673202	16 - 16	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING							
3850	D		GASFRS	5673117	3 - 10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
3860	D		GASFRS	5673117	24 - 24	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
3870			GASFRS	5674033	30 - 30	GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
3880													
3890	D		GASFRS	5672251	2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
3900		B1				***** BOLTING *****							
3910													
3920													
3930	D		BOLS	5675770	1/4 - 34	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3940													
3950	D	BA1	BOLS	5675776	36 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
3960	D	BA5	BOLS	5675770	36 - 42	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3970													
3980		SB6	CPSBOL	5675675	10 - 42	JACK SCREW HEX HEAD A320 GR L7M							
3990													
4000						***** BLINDS AND SPACERS *****							
4010													
4020													
4030			Specta	5749587	3/4 - 8	SPECTACLE BLIND 600# RF LTCS A516-GR70							
4040													
4050			Paddle	5750040	10 - 24	PADDLE BLANK 600# RF LTCS A516-GR70							
4060			Paddle	5750030	26 - 42	PADDLE BLANK 600# RF LTCS A516 GR70 B16.47 SERIES A							
4070													
4080			Paddle	5750251	10 - 24	PADDLE SPACER 600# RF LTCS A516-GR70							
4090			Paddle	5750252	26 - 42	PADDLE SPACER 600# RF LTCS A516 GR70 B16.47 SERIES A							
4100													
4110	C		Paddle	5750263	26 - 42	PADDLE BLANK COMPACT 600# RJ LTCS A516 GR.70							
4120	C		Paddle	5750268	26 - 42	PADDLE SPACER COMPACT 600# RJ LTCS A516 GR.70							
4130						***** BRANCHES *****							
4140													
4150	C		Reinfo	5358514_R	6 - 6	REINFORCING PAD FROM PIPE SCH XS SMLS LTCS A333 GR.6 BE	8 - 8						
4160	C		Reinfo	5358514_R	6 - 6	REINFORCING PAD FROM PIPE SCH XS SMLS LTCS A333 GR.6 BE	8 - 8						
4170													
4180													
4190			Reinfo	5358626_R	14 - 14	REINFORCING PAD FROM PIPE EFW LTCS A671 CC60 CL.22 WT 39.0MM	34 - 34						
4200	C		Reinfo	5358626_R	14 - 14	REINFORCING PAD FROM PIPE EFW LTCS A671 CC60 CL.22 WT 39.0MM	34 - 34						
4210													
4220													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 SMS D 8 OF 9 5 600K21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
4230	C		Reinfo	5358628_R	14 - 14	REINFORCING PAD FROM PIPE EFW LTCS A671 CC60 CL.22 WT 41.0MM	36 - 36					
4240			Reinfo	5358628_R	13 - 14	REINFORCING PAD FROM PIPE EFW LTCS A671 CC60 CL.22 WT 41.0MM	35 - 36					
4250												
4260	C	Reinfo	5358635_R	14 - 14	REINFORCING PAD FROM PIPE EFW LTCS A671 CC60 CL.22 WT 48.0MM	42 - 42						
4270		Reinfo	5358635_R	14 - 14	REINFORCING PAD FROM PIPE EFW LTCS A671 CC60 CL.22 WT 48.0MM	42 - 42						
4280												
4290												
4300												
4310												
4320	D		Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 42					
4330	D		Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 42					
4340												
4350												
4360												
4370												
4380	D											
4390												
4400												

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10						***** VALVES *****		B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. SB6 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. VA5 USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.							
11						***** MODULAR VALVES *****									
12															
20															
30															
40	C		Ball V	M06AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR									
50															
51	C		Ball V	M06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR									
52															
53															
60						***** BALL VALVES *****									
70															
80			Ball V	B06AC01	3/4 - 1 1/2	BALL 600# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
90			Ball V	B06AC05	2 - 3	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO									
100			Ball V	B06AC07	4 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
120															
121															
122						***** ORBIT VALVES *****									
123															
130	C	VA5	Ball V	R06AC06	2 - 6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO									
140	C	VA5	Ball V	R06AC08	8 - 10	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB GO									
150															
151															
160						***** GATE VALVES *****									
170															
180	C		Condui	W06AC70	3 - 4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW									
190	C		Condui	W06AC71	6 - 24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO									
200			Condui	W06CC71	28 - 28	GATE 600# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO									
210															
211															
212															
220						***** PIPE ABOVE GROUND & BELOW GROUND*****									
230															
231	C		Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6 PE									
240	C		Piping	5358514	2 - 2	PIPE SCH XS SMLS LTCS A333 GR.6 BE									
250			Piping	5358505	3 - 3	PIPE SCH STD SMLS LTCS A333 GR.6 BE									
260			Piping	9001208CTD	4 - 4	LINEPIPE SMLS WT= 5.16MM API-5L X60BE COATED									
270			Piping	9001254CTD	6 - 8	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED									
280			Piping	9001496CTD	10 - 10	LINEPIPE SMLS WT = 7.80 MM API-5L X60 BE COATED									
290			Piping	9001498CTD	12 - 12	LINEPIPE SMLS WT= 8.38 MM API-5L X60 BE COATED									
300			Piping	9001487CTD	14 - 14	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED									
310			Piping	9001493CTD	16 - 16	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP							
						ALL SIZES: BUTTWELD		CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS
						PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		3 MM		29JUL13	TN	B	1 OF 7	5	600K5D

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
320			Piping	9001494CTD	18 - 18	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE COATED										
330			Piping	9001488CTD	20 - 20	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED										
340			Piping	9001495CTD	24 - 24	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE COATED										
350			Piping	9001329CTD	28 - 28	LINEPIPE DSAW WT= 14.27 MM API-5L X60 BE COATED										
360																
370			Piping	9001498	12 - 12	LINEPIPE SMLS WT= 8.38 MM API-5L X60 BE										
380																
390						*****FITTINGS ABOVE & BELOW GROUND GROUND*****										
400																
410			90 Deg	5392600	3/4 - 2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6										
420			90 Deg	5392438	3 - 3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6										
430			45 Deg	5392606	3/4 - 2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6										
440			45 Deg	5392444	3 - 3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6										
450																
460			Tee	5392609	3/4 - 2	TEE BW SCH XS LTCS A420 WPL6										
470			Tee	5392447	3 - 3	TEE BW SCH STD LTCS A420 WPL6										
480																
490			Reduci	5392661	3/4 - 1 1/2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 2									
500			Reduci	5392454	3/4 - 2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3 - 3									
510																
520			Cap	5392610	3/4 - 2	CAP BW SCH XS LTCS A420 WPL6										
530			Cap	5392448	3 - 3	CAP BW SCH STD LTCS A420 WPL6										
540			Cap	9005074	4 - 4	CAP BW WT 5.16 MM LTCS A860 WPHY-60										
550			Cap	9005055	6 - 8	CAP BW WT 6.35MM LTCS A860 WPHY-60										
560			Cap	9005075	10 - 10	CAP BW WT 7.80 MM LTCS A860 WPHY-60										
570			Cap	9005076	12 - 12	CAP BW WT 8.38 MM LTCS A860 WPHY-60										
580			Cap	9005059	14 - 14	CAP BW WT 8.74 MM LTCS A860 WPHY-60										
590			Cap	9005060	16 - 16	CAP BW WT 9.53 MM LTCS A860 WPHY-60										
600			Cap	9005061	18 - 18	CAP BW WT 10.31 MM LTCS A860 WPHY-60										
610			Cap	9005064	20 - 20	CAP BW WT 11.13 MM LTCS A860 WPHY-60										
620			Cap	9005065	22 - 22	CAP BW WT 11.91 MM LTCS A860 WPHY-60										
630			Cap	9005067	24 - 24	CAP BW WT 12.70 MM LTCS A860 WPHY-60										
640			Cap	9005068	26 - 28	CAP BW WT 14.27 MM LTCS A860 WPHY-60										
650																
660			Concen	5392458	3/4 - 1 1/2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 2									
670			Eccent	5392461	3/4 - 1 1/2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 2									
680			Concen	5392452	1 1/2 - 2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3 - 3									
690			Eccent	5392453	1 1/2 - 2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3 - 3									
700			Eccent	9005077	3 - 3	REDUCER ECC WT 5.16 MM / SCH STD A860 WPHY 60	4 - 4									
710			Eccent	9006103	3 - 3	REDUCER ECC WT 6.35 MM / STD LTCS A860 WPHY-60	6 - 6									
720			Eccent	9002385	4 - 4	REDUCER ECC WT 6.35 MM / 5.16 MM LTCS A860 WPHY-60	6 - 8									
730			Eccent	9002266	6 - 6	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8 - 8									
740			Eccent	9002386	4 - 4	REDUCER ECC WT 7.80 MM / 5.16 MM LTCS A860 WPHY-60	10 - 10									
750			Eccent	9002387	6 - 8	REDUCER ECC WT 7.80 MM / 6.35 MM LTCS A860 WPHY-60	10 - 10									
760			Eccent	9002388	6 - 8	REDUCER ECC WT 8.38 MM / 6.35 MM LTCS A860 WPHY-60	12 - 12									
770			Eccent	9002266	10 - 10	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	12 - 12									
780			Eccent	9002390	6 - 8	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	14 - 14									
790			Eccent	9002391	10 - 10	REDUCER ECC WT 8.74 MM / 7.80 MM LTCS A860 WPHY-60	14 - 14									
800			Eccent	9002392	12 - 12	REDUCER ECC WT 8.74 MM / 8.38 MM LTCS A860 WPHY-60	14 - 14									
810			Eccent	9002393	8 - 8	REDUCER ECC WT 9.53 MM / 6.35 MM LTCS A860 WPHY-60	16 - 16									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP								
						ALL SIZES: BUTTWELD		CORR. ALLOW.		DATE		BY	REV	SHEET	PA	CLASS
						PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		3 MM		29JUL13		TN	B	2 OF 7		5

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
820			Eccent	9002394	10 - 10	REDUCER ECC WT 9.53 MM / 7.80 MM LTCS A860 WPHY-60	16 - 16								
830			Eccent	9002395	12 - 12	REDUCER ECC WT 9.53 MM / 8.38 MM LTCS A860 WPHY-60	16 - 16								
840			Eccent	9002396	14 - 14	REDUCER ECC WT 9.53 MM / 8.74 MM LTCS A860 WPHY-60	16 - 16								
850			Eccent	9002397	10 - 10	REDUCER ECC WT 10.31 MM / 7.80 MM LTCS A860 WPHY-60	18 - 18								
860			Eccent	9002398	12 - 12	REDUCER ECC WT 10.31 MM / 8.38 MM LTCS A860 WPHY-60	18 - 18								
870			Eccent	9002399	14 - 14	REDUCER ECC WT 10.31 MM / 8.74 MM LTCS A860 WPHY-60	18 - 18								
880			Eccent	9002400	16 - 16	REDUCER ECC WT 10.31 MM / 9.53 MM LTCS A860 WPHY-60	18 - 18								
890			Eccent	9004160	12 - 12	REDUCER ECC WT 11.13 MM / 8.38 MM LTCS A860 WPHY-60-W	20 - 20								
900			Eccent	9004161	14 - 14	REDUCER ECC WT 11.13 MM / 8.74 MM LTCS A860 WPHY-60-W	20 - 20								
910			Eccent	9004162	16 - 16	REDUCER ECC WT 11.13 MM / 9.53 MM LTCS A860 WPHY-60-W	20 - 20								
920			Eccent	9004163	18 - 18	REDUCER ECC WT 11.13 MM / 10.31 MM LTCS A860 WPHY-60-W	20 - 20								
930			Eccent	9004164	16 - 16	REDUCER ECC WT 12.70 MM / 9.53 MM LTCS A860 WPHY-60-W	24 - 24								
940			Eccent	9004165	18 - 18	REDUCER ECC WT 12.70 MM / 10.31 MM LTCS A860 WPHY-60-W	24 - 24								
950			Eccent	9004166	20 - 20	REDUCER ECC WT 12.70 MM / 11.13 MM LTCS A860 WPHY-60-W	24 - 24								
960			Eccent	9002408	18 - 18	REDUCER ECC WT 14.27 MM / 10.31 MM LTCS A860 WPHY-60-W	28 - 28								
970			Eccent	9004167	20 - 20	REDUCER ECC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60-W	28 - 28								
980			Eccent	9004168	24 - 24	REDUCER ECC WT 14.27 MM / 12.70 MM LTCS A860 WPHY-60-W	28 - 28								
990															
1000															
1010						***** INDUCTION BENDS ABOVE GROUND*****									
1020															
1030			90 Deg	9001819	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1040			90 Deg	9001813	6 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1050			90 Deg	9001829	10 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1060			90 Deg	9001839	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1070			90 Deg	9001816	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1080			90 Deg	9001817	16 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1090			90 Deg	9001818	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1100			90 Deg	9001899	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1110			90 Deg	9005321	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1120			90 Deg	9005322	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1130															
1140			45 Deg	9005323	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1150			45 Deg	9001823	6 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
1160			45 Deg	9005324	10 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW									
 FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN							CONSTRUCTION		FGP / WPMP						
							ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM		DATE 29JUL13	BY TN	REV B	SHEET 3 OF 7	PA 5

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1170			45 Deg	9005325	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1180			45 Deg	9001826	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1190			45 Deg	9001827	16 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1200			45 Deg	9001828	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1210			45 Deg	9005326	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1220			45 Deg	9005327	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1230			45 Deg	9005328	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1240														
1250			22BND	9005329	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1260			22BND	9001833	6 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1270			22BND	9005330	10 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1280			22.5 D	9003260	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1290			22BND	9001836	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1300			22BND	9001837	16 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1310			22BND	9001838	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1320			22BND	9001836	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1330			22BND	9005333	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1340			22BND	9005334	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1341						***** INDUCTION BENDS BELOW GROUND*****								
1342														
1343														
1344			90 Deg	9001819CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1345			90 Deg	9001813CTD	6 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1346			90 Deg	9001829CTD	10 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1347			90 Deg	9001839CTD	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1348			90 Deg	9001816CTD	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1349			90 Deg	9001817CTD	16 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1350			90 Deg	9001818CTD	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1360			90 Deg	9001899CTD	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 29JUL13	BY TN	REV B	SHEET 4 OF 7	PA 5	CLASS 600K5D	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1370			90 Deg	9005321CTD	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	6 - 8 8 - 8 10 - 10 10 - 10 12 - 12 12 - 12 14 - 14	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 60 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE						
1380			90 Deg	9005322CTD	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1381														
1382			45 Deg	9005323CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1383			45 Deg	9001823CTD	6 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1384			45 Deg	9005324CTD	10 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.80 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1385			45 Deg	9005325CTD	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1386			45 Deg	9001826CTD	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1387			45 Deg	9001827CTD	16 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1388			45 Deg	9001828CTD	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1389			45 Deg	9005326CTD	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1390			45 Deg	9005327CTD	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1400			45 Deg	9005328CTD	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1410														
1411			22BND	9005329CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 5.16 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1412			22BND	9001833CTD	6 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1413														
1414			22.5 D	9003260CTD	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.38 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1415			22BND	9001836CTD	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1416			22BND	9001837CTD	16 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1417			22BND	9001838CTD	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1418			22BND	9001836CTD	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1419														
1420			22BND	9005334CTD	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1421						*****OLETS ABOVE & BELOW GROUND*****								
1430														
1440														
1450			Weldol	9006407	3/4 - 2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60		MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS						
1460			Weldol	9006409	3 - 3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60								
1470			Weldol	9002452	3/4 - 2	WELDOLET WT 7.80 MM/SCH XS WT LTCS A694 F60								
1480			Weldol	9005084	3 - 3	WELDOLET WT 7.80 MM/STD WT LTCS A694 F60								
1490			Weldol	9002453	3/4 - 2	WELDOLET WT 8.38 MM/SCH XS WT LTCS A694 F60								
1500			Weldol	9005085	3 - 3	WELDOLET WT 8.38 MM/STD WT LTCS A694 F60								
1510			Weldol	9006423	3/4 - 2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP						
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM		DATE 29JUL13		BY TN	REV B	SHEET 5 OF 7

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1520			Weldol	9006422	3 - 3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	14 - 14							
1530			Weldol	9002456	4 - 4	WELDOLET WT 8.74MM/ 5.16 MM LTCS A694 F60	14 - 14							
1540			Weldol	9006429	3/4 - 2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	16 - 16							
1550			Weldol	9006428	3 - 3	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	16 - 16							
1560			Weldol	9002458	4 - 4	WELDOLET WT 9.53 MM/5.16 MM WT LTCS A694 F60	16 - 16							
1570			Weldol	9002459	3/4 - 2	WELDOLET WT 10.31 MM/ SCH XS WT LTCS A694 F60	18 - 18							
1580			Weldol	9006434	3 - 3	WELDOLET WT 10.31 MM/STD WT LTCS A694 F60	18 - 18							
1590			Weldol	9002461	4 - 4	WELDOLET WT 10.31 MM/ 5.16 WT LTCS A694 F60	18 - 18							
1600			Weldol	9005119	3/4 - 2	WELDOLET WT 11.13 MM/ SCH XS WT LTCS A694 F60	20 - 20							
1610			Weldol	9005086	3 - 3	WELDOLET WT 11.13 MM/STD WT LTCS A694 F60	20 - 20							
1620			Weldol	9005089	4 - 4	WELDOLET WT 11.13 MM/ 5.16 MM WT LTCS A694 F60	20 - 20							
1630			Weldol	9005120	6 - 6	WELDOLET WT 11.13 MM/ 6.35 MM WT LTCS A694 F60	20 - 20							
1640			Weldol	9002467	3/4 - 2	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	24 - 24							
1650			Weldol	9005087	3 - 3	WELDOLET WT 12.70 MM/STD WT LTCS A694 F60	24 - 24							
1660			Weldol	9002468	4 - 4	WELDOLET WT 12.70 MM/ 5.16 MM WT LTCS A694 F60	24 - 24							
1670			Weldol	9002469	6 - 8	WELDOLET WT 12.70 MM/ 6.35 MM WT LTCS A694 F60	24 - 24							
1680			Weldol	9002471	3/4 - 2	WELDOLET WT 14.27 MM/ SCH XS WT LTCS A694 F60	28 - 28							
1690			Weldol	9005087	3 - 3	WELDOLET WT 12.70 MM/STD WT LTCS A694 F60	28 - 28							
1700			Weldol	9002472	4 - 4	WELDOLET WT 14.27 MM/ 5.16 MM LTCS A694 F60	28 - 28							
1710			Weldol	9002473	6 - 8	WELDOLET WT 14.27 MM/ 6.35 MM LTCS A694 F60	28 - 28							
1720			Weldol	9002474	10 - 10	WELDOLET WT 14.27 MM/ 7.80 MM LTCS A694 F60	28 - 28							
1730														
1740			Flange	5504207	3/4 - 2	NIPOFLANGE 600# RTJ SCH STD/XS LTCS A350 LF2	3 - 3							
1750			Flange	9005079	3/4 - 2	NIPOFLANGE 600# RF 5.16 MM/XS WT LTCS A694 F60	4 - 4							
1760			Flange	9004270	3/4 - 2	NIPOFLANGE 600# RF 6.35 MM/SCH XS WT LTCS A694 F60	6 - 8							
1770			Flange	9005080	3/4 - 2	NIPOFLANGE 600# RF 7.80 MM/XS WT LTCS A694 F60	10 - 10							
1780			Flange	9005081	3/4 - 2	NIPOFLANGE 600# RF 8.38 MM/XS WT LTCS A694 F60	12 - 12							
1790			Flange	9005032	3/4 - 2	NIPOFLANGE 600# RF 8.74 MM/XS WT LTCS A694 F60	14 - 14							
1800			Flange	9005082	3/4 - 2	NIPOFLANGE 600# RF 9.53 MM/XS WT LTCS A694 F60	16 - 16							
1810			Flange	9005033	3/4 - 2	NIPOFLANGE 600# RF 10.31 MM/XS WT LTCS A694 F60	18 - 18							
1820			Flange	9005102	3/4 - 2	NIPOFLANGE 600# RF 11.13 MM/XS WT LTCS A694 F60	20 - 20							
1830			Flange	9005104	3/4 - 2	NIPOFLANGE 600# RF 12.70 MM/XS WT LTCS A694 F60	24 - 24							
1840			Flange	9005105	3/4 - 2	NIPOFLANGE 600# RF 14.27 MM/XS WT LTCS A694 F60	28 - 28							
1850														
1860						***** PIPE FLANGES ABOVE & BELOW GROUND*****								
1870														
1880			Flange	5557036	3/4 - 2	FLG WN 600# RF LTCS A350 LF2 SCH XS								
1890			Flange	5557033	3 - 3	FLG WN 600# RF LTCS A350 LF2 SCH STD FLG								
1900			Flange	9003480	4 - 4	WN 600# RF LTCS A694 F60 WT 5.16MM FLG								
1910			Flange	9001471	6 - 8	WN 600# RF LTCS A694 F60 WT 6.35MM FLG								
1920			Flange	9003482	10 - 10	WN 600# RF LTCS A694 F60 WT 7.80 MM FLG								
1930			Flange	9003483	12 - 12	WN 600# RF LTCS A694 F60 WT 8.38 MM FLG								
1940			Flange	9003484	14 - 14	WN 600# RF LTCS A694 F60 WT 8.74 MM FLG								
1950			Flange	9003485	16 - 16	WN 600# RF LTCS A694 F60 WT 9.53 MM FLG								
1960			Flange	9003486	18 - 18	WN 600# RF LTCS A694 F60 WT 10.31 MM FLG								
1970			Flange	9003487	20 - 20	WN 600# RF LTCS A694 F60 WT 11.13 MM FLG								
1980			Flange	9001512	24 - 24	WN 600# RF LTCS A694 F60 WT 12.70MM FLG								
1990			Flange	9001753	28 - 28	COMPACT 600# RJ A694 F60 WT 14.27 MM FLG								
2000			Blind	5615598	3/4 - 3	BLIND 600# RF LTCS A350 LF2								
2010			Blind	9001518	4 - 24	FLG BLIND 600# RF LTCS A694 F60								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES: BUTTWELD								
						PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 29JUL13	BY TN	REV B	SHEET 6 OF 7	PA 5	CLASS 600K5D	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES														
2030	C		Flange	9003489	28 - 28	FLG WN 600# RF LTCS A694 F60 WT 14.27 MM SERIES A																
2040						***** GASKETS *****																
2050																						
2060																						
2070			GASFRS	5673103	3/4 - 24	GASKET 600# 316SS SP WND GRAPHITE IR																
2080			GASRJ	5673300	28 - 28	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 600K5D																
2090			GASFRS	5674012	28 - 28	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A																
2100																						
2110			B1			***** BOLTING *****																
2120																						
2130	C		BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS																
2131	SB6																					
2140		BOLS	5675675	10 - 28	JACK SCREW HEX HEAD A320 GR L7M																	
2150																						
2151																						
2160					***** BLINDS & SPACERS																	
2170																						
2180		Specta	5749587	3/4 - 8	SPECTACLE BLIND 600# RF LTCS A516-GR70																	
2190		Paddle	5750040	10 - 24	PADDLE BLANK 600# RF LTCS A516-GR70																	
2200		Paddle	5750251	10 - 24	PADDLE SPACER 600# RF LTCS A516-GR70																	
2210		Paddle	5750030	28 - 28	PADDLE BLANK 600# RF LTCS A516 GR70 B16.47 SERIES A																	
2220	Paddle	5750252	28 - 28	PADDLE SPACER 600# RF LTCS A516 GR70 B16.47 SERIES A																		
2230																						
2240				***** BRANCHES *****																		
2250																						
2260				USE BRANCH CHART 600K5D																		
2270																						
2280				***** CONSTRUCTION DETAILS *****																		
2290																						
2300				SEAL WELDING ----- REQUIRED																		
2310				PIPE BENDS ----- INDUCTION BENDS																		
2320				THREAD SEALANT ----- PTFE PASTE																		
2330																						
													SERVICE									
													SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 60 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE									
													MATERIAL: LOW TEMP CARBON STEEL									
													FLANGE RATING: 600 RF									
													TEMPERATURE RANGE: -40 / 90 °C									
													COMMENT: LTCS									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN										CONSTRUCTION		FGP / WPMP							
													ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM		DATE 29JUL13	BY TN	REV B	SHEET 7 OF 7	PA 5	CLASS 600K5D



SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
510						***** BUTTERFLY VALVES *****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE PIPING: BB, BD, BV, DHC, FGH, PHC, GS MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 88 °C COMMENT: LTCS					
520													
530	C		Butter	T06AC61	10 - 24	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO							
540													
550	C		Butter	T06CC61-K5E	26 - 28	BUTTERFLY 600# CF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO							
560													
570													
580						***** CHECK VALVES *****							
590													
600	C	VC1	Check	C06AC50	3/4 - 1 1/2	CHECK PISTON 600# RF LTCS A350 LF2 TRIM 16							
610		VC2	Check	C06AC45	2 - 8	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
620		VC2	Check	C06AC52	10 - 24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16							
630	C	VC2	Check	C06CC52-K5E	26 - 28	CHECK DUAL 600# CF LTCS A352 LCC TRIM 16							
640													
650						***** GLOBE VALVES *****							
660													
670													
680	C		Globe	L06AC48	3/4 - 1 1/2	GLOBE 600# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
690	C		Globe	L06AC41	2 - 4	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW							
700	C		Globe	L06AC42	6 - 10	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y GO							
710													
720						***** GATE VALVES *****							
730													
740													
750	C		Condui	W06AC70	3 - 4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
760													
770	C		Condui	W06AC71	6 - 24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
780													
790	C		Condui	W06CC71-K5E	26 - 28	GATE 600# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
800						*****PIPE ABOVE GROUND*****							
810													
820													
830	C		Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE							
840	C		Piping	5358515	2 - 2	PIPE SCH 160 SMLS LTCS A333 GR.6 BE							
850			Piping	5358514	3 - 3	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
860			Piping	9001211	4 - 4	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE							
870			Piping	9001255	6 - 6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE							
880			Piping	9001487	8 - 8	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE							
890			Piping	9001488	10 - 12	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE							
900			Piping	9001490	14 - 14	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE							
910			Piping	9001495	16 - 16	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE							
920			Piping	9001491	18 - 18	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE							
930			Piping	9005241	20 - 20	LINEPIPE SMLS WT= 15.09 MM API-5L X60 BE							
940			Piping	9005242	24 - 24	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE							
950	C		Piping	9005243	26 - 26	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE							
960			Piping	9005254	28 - 28	LINEPIPE SMLS WT= 20.62 MM API-5L X60 BE							
970													
980	C	812	Piping	5358511	1/2 - 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE							
990						***** PIPE BELOW GROUND *****							
1000													
1010													
1020			Piping	9001254CTD	4 - 4	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED							
1030			Piping	9001255CTD	6 - 6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE COATED							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 10	PA 5	CLASS 600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040	C		Piping	9001487CTD	8 - 8	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED							
1050			Piping	9001488CTD	10 - 12	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED							
1060			Piping	9001490CTD	14 - 14	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE COATED							
1070			Piping	9001495CTD	16 - 16	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE COATED							
1080			Piping	9001491CTD	18 - 18	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE COATED							
1090			Piping	9005241CTD	20 - 20	LINEPIPE SMLS WT= 15.09 MM API-5L X60 BE COATED							
1100			Piping	9005242CTD	24 - 24	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE COATED							
1110			Piping	9005242CTD	26 - 26	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE COATED							
1120			Piping	9005242CTD	28 - 28	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE COATED							
1130													
1140						***** FITTINGS ABOVE & BELOW GROUND *****							
1150													
1160			90 Deg	5392816	3/4 - 2	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6							
1170			90 Deg	5392600	3 - 3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
1180			90 Deg	9002190	4 - 4	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60							
1190			90 Deg	9006470	6 - 6	ELL BW 90 DEG LR WT 7.11 MM A860 WPHY 60							
1200			90 Deg	9002157	8 - 8	ELL BW 90 DEG LR WT 8.74MM A860 WPHY 60							
1210			90 Deg	9006468	10 - 12	ELL BW 90 DEG LR WT 11.13 MM A860 WPHY 60							
1220			90 Deg	9006479	16 - 16	ELL BW 90 DEG LR WT 12.70 MM A860 WPHY 60							
1230			90 Deg	9006480	18 - 18	ELL BW 90 DEG LR WT 14.27 MM A860 WPHY 60							
1240			90 Deg	9006482	20 - 20	ELL BW 90 DEG LR WT 15.09 MM A860 WPHY 60-W							
1250			90 Deg	9006474	24 - 24	ELL BW 90 DEG LR WT 17.48 MM A860 WPHY 60-W							
1260			90 Deg	5392822	3/4 - 2	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6							
1270	C		90 Deg	5392606	3 - 3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1280			90 Deg	9006569	12 - 12	ELL BW 45 DEG LR WT 11.13MM A860 WPHY 60							
1290													
1300			Tea	5392825	3/4 - 2	TEE BW SCH 160 LTCS A420 WPL6							
1310			Tee	5392609	3 - 3	TEE BW SCH XS LTCS A420 WPL6							
1320		NT5	Tee	9301609	4 - 4	TEE BW WT 6.35 MM LTCS A860 WPHY-60							
1330		NT5	Tee	9006600	6 - 6	TEE BW WT 7.11MM A860 WPHY 60							
1340		NT5	Tee	9301607	8 - 8	TEE BW WT 8.74 MM LTCS A860 WPHY-60							
1350		NT5	Tee	9006604	10 - 12	TEE BW WT 11.13MM A860 WPHY 60							
1360		NT5	Tee	9006632	14 - 14	TEE BW WT 11.91MM A860 WPHY 60							
1370		NT5	Tee	9006633	16 - 16	TEE BW WT 12.7 MM A860 WPHY 60							
1380		NT5	Tee	9006486	18 - 18	TEE BW WT 14.27MM A860 WPHY 60							
1390		NT5	Tee	9006488	20 - 20	TEE BW WT 15.09MM A860 WPHY 60-W							
1400		NT5	Tee	9006489	24 - 24	TEE BW WT 17.48MM A860 WPHY 60-W							
1410													
1420			Reduci	5392821	3/4 - 1 1/2	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 2	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE PIPING: BB, BD, BV, DHC, FGH, PHC, GS					
1430		Reduci	5392469	2 - 2	TEE RED BW SCH XS/160 LTCS A420 WPL6	3 - 3							
1440	C	NT5	Reduci	9006608	2 - 3	TEE RED BW WT 6.35 MM / SCH XS LTCS A860 WPHY 60	4 - 4						
1450	C	NT5	Reduci	9006609	3 - 3	TEE RED BW WT 7.11 MM/SCH XS LTCS A860 WPHY 60	6 - 6						
1460		NT5	Reduci	9004208	4 - 4	TEE RED BW WT 7.11 MM/6.35 MM LTCS A860 WPHY 60	6 - 6						
1470		NT5	Reduci	9004123	4 - 4	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	8 - 8						
1480		NT5	Reduci	9006611	6 - 6	TEE RED BW WT 8.74 MM/7.11 MM LTCS A860 WPHY 60	8 - 8						
1490		NT5	Reduci	9006612	4 - 4	TEE RED BW WT 11.13 MM/6.35 MM LTCS A860 WPHY 60	10 - 10						
1500		NT5	Reduci	9004209	6 - 6	TEE RED BW WT 11.13 MM/7.11 MM LTCS A860 WPHY 60	10 - 12						
1510		NT5	Reduci	9006485	8 - 8	TEE RED BW WT 11.13 MM/8.74 MM LTCS A860 WPHY 60	10 - 12						
1520		NT5	Reduci	9006613	10 - 10	TEE RED BW WT 11.13 MM LTCS A860 WPHY 60	12 - 12						
1530		NT5	Reduci	9006028	6 - 6	TEE RED BW WT 11.91 MM/7.11 MM LTCS A860 WPHY 60	14 - 14						
1540		NT5	Reduci	9006614	8 - 8	TEE RED BW WT 11.91 MM/8.74 MM LTCS A860 WPHY 60	14 - 14						
1550		NT5	Reduci	9006615	10 - 12	TEE RED BW WT 11.91 MM/11.13 MM LTCS A860 WPHY 60	14 - 14						
1560			Reduci	9006602	6 - 6	TEE RED BW WT 12.70 MM/7.11 MM LTCS A860 WPHY 60	16 - 16						
1570		NT5	Reduci	9006616	8 - 8	TEE RED BW WT 12.70 MM/8.74 MM LTCS A860 WPHY 60	16 - 16						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 3 OF 10	PA 5	CLASS 600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1580		NT5	Reduci	9006617	10 - 12	TEE RED BW WT 12.70 MM/11.13 MM LTCS A860 WPHY 60	16 - 16						
1590		NT5	Reduci	9006618	14 - 14	TEE RED BW WT 12.70 MM/11.91 MM LTCS A860 WPHY 60	16 - 16						
1600		NT5	Reduci	9006619	8 - 8	TEE RED BW WT 14.27 MM/8.74 MM LTCS A860 WPHY 60-W	18 - 18						
1610		NT5	Reduci	9006620	10 - 12	TEE RED BW WT 14.27 MM/11.13 MM LTCS A860 WPHY 60-W	18 - 18						
1620		NT5	Reduci	9006621	14 - 14	TEE RED BW WT 14.27 MM/11.91 MM LTCS A860 WPHY 60-W	18 - 18						
1630		NT5	Reduci	9006622	16 - 16	TEE RED BW WT 14.27 MM/12.7 MM LTCS A860 WPHY 60-W	18 - 18						
1640			Reduci	9006603	8 - 8	TEE RED BW WT 15.09 MM/8.74 MM LTCS A860 WPHY 60-W	20 - 20						
1650		NT5	Reduci	9006623	10 - 12	TEE RED BW WT 15.09 MM/11.13 MM LTCS A860 WPHY 60-W	20 - 20						
1660		NT5	Reduci	9006624	14 - 14	TEE RED BW WT 15.09 MM/11.91 MM LTCS A860 WPHY 60-W	20 - 20						
1670		NT5	Reduci	9006625	16 - 16	TEE RED BW WT 15.09 MM/12.7 MM LTCS A860 WPHY 60-W	20 - 20						
1680		NT5	Reduci	9006626	18 - 18	TEE RED BW WT 15.09 MM/14.27 MM LTCS A860 WPHY 60-W	20 - 20						
1690		NT5	Reduci	9006627	10 - 12	TEE RED BW WT 17.48 MM/11.13 MM LTCS A860 WPHY 60-W	24 - 24						
1700		NT5	Reduci	9006628	14 - 14	TEE RED BW WT 17.48 MM/11.91 MM LTCS A860 WPHY 60-W	24 - 24						
1710		NT5	Reduci	9006629	16 - 16	TEE RED BW WT 17.48 MM/12.7 MM LTCS A860 WPHY 60-W	24 - 24						
1720		NT5	Reduci	9006630	18 - 18	TEE RED BW WT 17.48 MM/14.27 MM LTCS A860 WPHY 60-W	24 - 24						
1730		NT5	Reduci	9006631	20 - 20	TEE RED BW WT 17.48 MM/15.09 MM LTCS A860 WPHY 60-W	24 - 24						
1740													
1750			Cap	5392826	3/4 - 2	CAP BW SCH 160 LTCS A420 WPL6							
1760			Cap	5392610	3 - 3	CAP BW SCH XS LTCS A420 WPL6							
1770			Cap	9005071	4 - 4	CAP BW WT 6.35 MM LTCS A860 WPHY-60							
1780			Cap	9005056	6 - 6	CAP BW WT 7.11MM LTCS A860 WPHY-60							
1790			Cap	9005059	8 - 8	CAP BW WT 8.74 MM LTCS A860 WPHY-60							
1800			Cap	9005064	10 - 12	CAP BW WT 11.13 MM LTCS A860 WPHY-60							
1810			Cap	9005065	14 - 14	CAP BW WT 11.91 MM LTCS A860 WPHY-60							
1820			Cap	9005067	16 - 16	CAP BW WT 12.70 MM LTCS A860 WPHY-60							
1830			Cap	9005068	18 - 18	CAP BW WT 14.27 MM LTCS A860 WPHY-60							
1840			Cap	9005091	20 - 20	CAP BW WT 15.09 MM LTCS A860 WPHY-60							
1850			Cap	9005070	24 - 24	CAP BW WT 17.48 MM LTCS A860 WPHY-60							
1860			Cap	9005073	28 - 28	CAP BW WT 20.62 MM LTCS A860 WPHY-60							
1870													
1880			Concen	5392827	3/4 - 1 1/2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 2						
1890			Concen	5392462	2 - 2	REDUCER CONC SCH XS/160 LTCS A420 WPL6	3 - 3						
1900			Eccent	5392828	3/4 - 1 1/2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 2						
1910			Eccent	5392465	2 - 2	REDUCER ECC SCH XS/160 LTCS A420 WPL6	3 - 3						
1920	C		Concen	9006923	1 1/2 - 2	REDUCER CONC WT 6.35MM / SCH 160 A860 WPHY 60	4 - 4						
1930	C		Concen	9006919	3 - 3	REDUCER CONC WT 6.35 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1940	C		Concen	9006920	3 - 3	REDUCER CONC WT 7.11 MM / SCH XS MM LTCS A860 WPHY-60	6 - 6						
1950								SERVICE					
1960	C		Eccent	9002042	1 1/2 - 2	REDUCER ECC WT 6.35 MM / SCH 160 LTCS A860 WPHY-60	4 - 4	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE PIPING: BB, BD, BV, DHC, FGH, PHC, GS					
1970	C		Eccent	9001581	3 - 3	REDUCER ECC WT 6.35 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1980	C		Eccent	9001582	3 - 3	REDUCER ECC WT 7.11 MM / SCH XS MM LTCS A860 WPHY-60	6 - 6						
1990			Eccent	9001583	4 - 4	REDUCER ECC WT 7.11 MM / 6.35 MM LTCS A860 WPHY-60	6 - 6						
2000			Eccent	9002292	4 - 4	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	8 - 8						
2010			Eccent	9001377	6 - 6	REDUCER ECC WT 8.74 MM / 7.11 MM LTCS A860 WPHY-60	8 - 8						
2020			Eccent	9001586	4 - 4	REDUCER ECC WT 11.13 MM / 6.35MM LTCS A860 WPHY-60	10 - 10						
2030			Eccent	9002021	6 - 6	REDUCER ECC WT 11.13MM / 7.11MM LTCS A860 WPHY-60	10 - 12						
2040			Eccent	9002402	8 - 8	REDUCER ECC WT 11.13 MM / 8.74 MM LTCS A860 WPHY-60	10 - 12						
2050			Eccent	9002022	10 - 10	REDUCER ECC WT 11.13MM LTCS A860 WPHY-60	12 - 12						
2060			Eccent	9002023	6 - 6	REDUCER ECC WT 11.91MM / 7.11MM LTCS A860 WPHY-60	14 - 14						
2070			Eccent	9001380	8 - 8	REDUCER ECC WT 11.91MM / 8.74MM LTCS A860 WPHY-60	14 - 14						
2080			Eccent	9001393	10 - 12	REDUCER ECC WT 11.91MM / 11.13MM LTCS A860 WPHY-60	14 - 14						
2090			Eccent	9005116	8 - 8	REDUCER ECC WT 12.70 MM / 8.74 MM LTCS A860 WPHY-60	16 - 16						
2100			Eccent	9002407	10 - 12	REDUCER ECC WT 12.70 MM / 11.13 MM LTCS A860 WPHY-60	16 - 16						
2110			Eccent	9001590	14 - 14	REDUCER ECC WT 12.70 MM / 11.91 MM LTCS A860 WPHY-60	16 - 16						
			FGP / WPMP TENGINZCHEVROIL - TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 4 OF 10	PA 5	CLASS 600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2120			Eccent	9002409	10 - 12	REDUCER ECC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60	18 - 18						
2130			Eccent	9001591	14 - 14	REDUCER ECC WT 14.27 MM / 11.91 MM LTCS A860 WPHY-60	18 - 18						
2140			Eccent	9002410	16 - 16	REDUCER ECC WT 14.27 MM / 12.70 MM LTCS A860 WPHY-60	18 - 18						
2150			Eccent	9006301	12 - 12	REDUCER ECC WT 15.09 MM / 11.13MM LTCS A860 WPHY-60-W	20 - 20						
2160			Eccent	9001593	14 - 14	REDUCER ECC WT 15.09 MM / 11.91 MM LTCS A860 WPHY-60-W	20 - 20						
2170			Eccent	9001594	16 - 16	REDUCER ECC WT 15.09 MM / 12.70 MM LTCS A860 WPHY-60-W	20 - 20						
2180			Eccent	9001595	18 - 18	REDUCER ECC WT 15.09 MM / 14.27 MM LTCS A860 WPHY-60-W	20 - 20						
2190			Eccent	9001387	16 - 16	REDUCER ECC WT 17.48 MM / 12.70 MM LTCS A860 WPHY-60-W	24 - 24						
2200			Eccent	9001596	20 - 20	REDUCER ECC WT 17.48 MM / 15.09 MM LTCS A860 WPHY-60-W	24 - 24						
2210			Eccent	9002024	18 - 18	REDUCER ECC WT 20.62MM / 14.27MM LTCS A860 WPHY-60-W	28 - 28						
2220			Eccent	9001597	20 - 20	REDUCER ECC WT 20.62MM / 15.09 MM LTCS A860 WPHY-60-W	28 - 28						
2230			Eccent	9001598	24 - 24	REDUCER ECC WT 20.62MM / 17.48 MM LTCS A860 WPHY-60-W	28 - 28						
2240													
2250						***** INDUCTION BENDS ABOVE GROUND*****							
2260													
2270			90 Deg	9001534	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2280			90 Deg	9001396	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2290			90 Deg	9001816	8 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2300			90 Deg	9001899	10 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2310			90 Deg	9001397	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2320			90 Deg	9005321	16 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2330			90 Deg	9005322	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2340			90 Deg	9001572	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2350			90 Deg	9001573	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2360			90 Deg	9001574	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2370													
2380			45 Deg	9001575	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2390			45 Deg	9001399	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2400			45 Deg	9001826	8 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2410			45 Deg	9005326	10 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2420			45 Deg	9001500	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2430			45 Deg	9005327	16 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2440			45 Deg	9005328	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2450			45 Deg	9001577	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2460			45 Deg	9005040	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	FGP/WPMP SHEET 5 OF 10	PA 5	CLASS 600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2470	C		45 Deg	9002026	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2480														
2490			22.5 D	9001580	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2500			22.5 D	9002000	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2510			22.5 D	9001836	8 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2520			22.5 D	9005331	10 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2530			22.5 D	9002001	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2540			22.5 D	9005333	16 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2550			22.5 D	9005334	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2560			22.5 D	9002811	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2570			22.5 D	9001579	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2580			22.5 D	9002062	26 - 26	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2590			22.5 D	9002027	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 20.62MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
2600														
2610														
2620														***** INDUCTION BENDS BELOW GROUND*****
2630														
2640			90 Deg	9001534CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2650			90 Deg	9001396CTD	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2660			90 Deg	9001816CTD	8 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2670			90 Deg	9001899CTD	10 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2680			90 Deg	9001397CTD	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2690			90 Deg	9005321CTD	16 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2700			90 Deg	9005322CTD	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2710			90 Deg	9001572CTD	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
2720	C	90 Deg	9001573CTD	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED									
2730	C	90 Deg	9001574CTD	28 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED									
2740														
2750	45 Deg	9001575CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
2760	45 Deg	9001399CTD	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	FGP/WPMP SHEET 6 OF 10		PA 5	CLASS 600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2770			45 Deg	9001826CTD	8 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2780			45 Deg	9005326CTD	10 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2790			45 Deg	9004001CTD	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2800			45 Deg	9005327CTD	16 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2810			45 Deg	9005328CTD	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2820			45 Deg	9001577CTD	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2830	C		45 Deg	9005040CTD	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2840	C		45 Deg	9002026CTD	28 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2850													
2860			22.5 D	9001580CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2870			22.5 D	9002000CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2880			22.5 D	9001836CTD	8 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2890			22.5 D	9005331CTD	10 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2900			22.5 D	9002001CTD	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2910			22.5 D	9004000CTD	16 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2920			22.5 D	9005334CTD	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2930			22.5 D	9002811CTD	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.09 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2940	C		22.5 D	9001579CTD	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2950	C		22.5 D	9002062CTD	26 - 26	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2960	C		22.5 D	9002027CTD	28 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 20.62MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
2970													
2980						***** OLETS ABOVE & BELOW GROUND*****							
2990													
3000	C		Weldol	5505392	3/4 - 1	WELDOLET SCH 160 LTCS A350 LF2	2 - 2	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE PIPING: BB, BD, BV, DHC, FGH, PHC, GS MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 88 °C COMMENT: LTCS					
3010	C		Weldol	5505393	3/4 - 1	WELDOLET SCH XS/160 LTCS A350 LF2	3 - 3						
3020	C		Weldol	9002450	3/4 - 1 1/2	WELDOLET WT 6.35 MM/SCH 160 WT LTCS A694 F60	4 - 4						
3030	C		Weldol	9002072	3/4 - 2	WELDOLET WT 7.11MM/ SCH 160 LTCS A694 F60	6 - 6						
3040	C		Weldol	9002455	3/4 - 2	WELDOLET WT 8.74MM/ SCH 160 LTCS A694 F60	8 - 8						
3050	C		Weldol	9006423	3 - 3	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	8 - 8						
3060	C		Weldol	9002076	3/4 - 2	WELDOLET WT 11.13MM/ SCH 160 LTCS A694 F60	10 - 12						
3070	C		Weldol	9005119	3 - 3	WELDOLET WT 11.13 MM/ SCH XS WT LTCS A694 F60	10 - 12						
3080			Weldol	9005120	4 - 4	WELDOLET WT 11.13 MM/ 6.35 MM WT LTCS A694 F60	12 - 12						
3090	C		Weldol	9002080	3/4 - 2	WELDOLET WT 11.91MM/ SCH 160 LTCS A694 F60	14 - 14						
3100	C		Weldol	9002003	3 - 3	WELDOLET WT 11.91MM/SCH XS LTCS A694 F60	14 - 14						
3110			Weldol	9002009	4 - 4	WELDOLET WT 11.91MM/6.35MM LTCS A694 F60	14 - 14						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 7 OF 10	PA 5	CLASS 600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3120	C		Weldol	9002466	3/4 - 2	WELDOLET WT 12.70 MM/ SCH 160 WT LTCS A694 F60	16 - 16	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE PIPING: BB, BD, BV, DHC, FGH, PHC, GS MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 88 °C COMMENT: LTCS					
3130	C		Weldol	9002467	3 - 3	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	16 - 16						
3140			Weldol	9002469	4 - 4	WELDOLET WT 12.70 MM/ 6.35 MM WT LTCS A694 F60	16 - 16						
3150	C		Weldol	9002470	3/4 - 2	WELDOLET WT 14.27 MM/ SCH 160 WT LTCS A694 F60	18 - 18						
3160	C		Weldol	9002471	3 - 3	WELDOLET WT 14.27 MM/ SCH XS WT LTCS A694 F60	18 - 18						
3170			Weldol	9002473	4 - 4	WELDOLET WT 14.27 MM/ 6.35 MM LTCS A694 F60	18 - 18						
3180			Weldol	9005121	6 - 6	WELDOLET WT 14.27 MM/ 7.11 MM LTCS A694 F60	18 - 18						
3190	C		Weldol	9002820	3/4 - 2	WELDOLET WT 15.09 MM/ SCH 160 LTCS A694 F60	20 - 20						
3200	C		Weldol	9005124	3 - 3	WELDOLET WT 15.09 MM/ SCH XS WT LTCS A694 F60	20 - 20						
3210			Weldol	9005122	4 - 4	WELDOLET WT 15.09 MM/ 6.35 MM LTCS A694 F60	20 - 20						
3220			Weldol	9005123	6 - 6	WELDOLET WT 15.09 MM/ 7.11 MM LTCS A694 F60	20 - 20						
3230	C		Weldol	9004336	3/4 - 2	WELDOLET WT 17.48 MM/ SCH 160 LTCS A694 F60	24 - 24						
3240	C		Weldol	9005053	3 - 3	WELDOLET WT 17.48 MM/ SCH XS MM LTCS A694 F60	24 - 24						
3250			Weldol	9005049	4 - 4	WELDOLET WT 17.48 MM/ 6.35 MM LTCS A694 F60	24 - 24						
3260			Weldol	9005050	6 - 6	WELDOLET WT 17.48 MM/ 7.11 MM LTCS A694 F60	24 - 24						
3270			Weldol	9005052	8 - 8	WELDOLET WT 17.48 MM/ 8.74 MM LTCS A694 F60	24 - 24						
3280	C		Weldol	9002089	3/4 - 2	WELDOLET WT 19.05 MM/ SCH 160 LTCS A694 F60	26 - 26						
3290	C		Weldol	9006252	3 - 3	WELDOLET WT 19.05 MM/ SCH XS LTCS A694 F60	26 - 26						
3300			Weldol	9002028	4 - 4	WELDOLET WT 20.62 MM/ 6.35 MM LTCS A694 F60	28 - 28						
3310			Weldol	9002029	6 - 6	WELDOLET WT 20.62 MM/ 7.11 MM LTCS A694 F60	28 - 28						
3320			Weldol	9002095	8 - 8	WELDOLET WT 20.62 MM/ 8.74 MM LTCS A694 F60	28 - 28						
3330			Weldol	9002030	10 - 10	WELDOLET WT 20.62 MM/ 11.13 MM LTCS A694 F60	28 - 28						
3340													
3350	C		Nipofl	5504460	3/4 - 1 1/2	NIPOFLANGE 600# RF SCH 160 LTCS A350 LF2	2 - 2						
3360	C		Nipofl	5504461	3/4 - 2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2	3 - 3						
3370			Nipofl	9002070	3/4 - 2	NIPOFLANGE 600# RF WT 6.35MM/ SCH 160 LTCS A694 F60	4 - 4						
3380			Nipofl	9002065	3/4 - 2	NIPOFLANGE 600# RF WT 7.11 MM/ SCH 160 LTCS A694 F60	6 - 6						
3390			Nipofl	9003454	3/4 - 2	NIPOFLANGE 600# RF WT 8.74 MM/ SCH 160 LTCS A694 F60	8 - 8						
3400			Nipofl	9003457	3/4 - 2	NIPOFLANGE 600# RF WT 11.13 MM/ SCH 160 WT LTCS A694 F60	10 - 12						
3410			Nipofl	9002066	3/4 - 2	NIPOFLANGE 600# RF WT 11.91MM/ SCH 160 LTCS A694 F60	14 - 14						
3420			Nipofl	9003458	3/4 - 2	NIPOFLANGE 600# RF WT 12.70 MM/ SCH 160 WT LTCS A694 F60	16 - 16						
3430			Nipofl	9003459	3/4 - 2	NIPOFLANGE 600# RF WT 14.27 MM/ SCH 160 WT LTCS A694 F60	18 - 18						
3440			Nipofl	9006495	3/4 - 2	NIPOFLANGE 600# RF 15.09 MM/ SCH 160 LTCS A694 F60	20 - 20						
3450			Nipofl	9006496	3/4 - 2	NIPOFLANGE 600# RF 17.48 MM/ SCH 160 LTCS A694 F60	24 - 24						
3460			Nipofl	9002069	3/4 - 2	NIPOFLANGE 600# RF WT 20.62MM/ SCH 160 LTCS A694 F60	28 - 28						
3470													
3480													
3490						***** PIPE FLANGES ABOVE & BELOW GROUND *****							
3500													
3510	C		Flange	5851204	1/2 - 2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE							
3520													
3530	C		Flange	5557038	1/2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH 160							
3540			Flange	5557036	3 - 3	FLG WN 600# RF LTCS A350 LF2 SCH XS							
3550			Flange	9001471	4 - 4	FLG WN 600# RF LTCS A694 F60 WT 6.35MM							
3560			Flange	9001462	6 - 6	FLG WN 600# RF LTCS A694 F60 WT 7.11MM							
3570			Flange	9003484	8 - 8	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM							
3580			Flange	9001464	10 - 12	FLG WN 600# RF LTCS A694 F60 WT 11.13MM							
3590			Flange	9001511	14 - 14	FLG WN 600# RF LTCS A694 F60 WT 11.91MM							
3600			Flange	9001512	16 - 16	FLG WN 600# RF LTCS A694 F60 WT 12.70MM							
3610			Flange	9001515	18 - 18	FLG WN 600# RF LTCS A694 F60 WT 14.27MM							
3620			Flange	9001514	20 - 20	FLG WN 600# RF LTCS A694 F60 WT 15.09MM							
3630			Flange	9001513	24 - 24	FLG WN 600# RF LTCS A694 F60 WT 17.48MM							
3640			Flange	9003496	28 - 28	FLG WN 600# RF LTCS A694 F60 WT 20.62 MM SERIES A							
3650													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 8 OF 10	PA 5	CLASS 600K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
4180						***** CONSTRUCTION DETAILS *****									
4190						SEAL WELDING ----- REQUIRED									
4200						PIPE BENDS ----- INDUCTION BENDS									
4210						THREAD SEALANT ----- PTFE PASTE									
4220															
4230						***** DO NOT USE *****TIE-INS ONLY*****									
4240															
4250	C		Reduci	9006751	3 - 3	FLG WN REDUCING 600# RF LTCS A694 F60 BORED TO SUIT WT 6.35MM	4 - 4								
4260															
4270	C														
								SERVICE							
								SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 88 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE PIPING: BB, BD, BV, DHC, FGH, PHC, GS							
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 88 °C COMMENT: LTCS							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 3 MM		DATE 17MAR17	BY JCS	REV C	SHEET 10 OF 10	PA 5	CLASS 600KSE

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10						***** VALVES *****		B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. VB2 SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.						
11						***** MODULAR VALVES *****								
12														
20														
30														
40			Ball V	M06AC02	3/4 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR								
50														
51														
60		VB2				***** BALL VALVES *****								
70														
80			Ball V	B06AC01	3/4 - 1 1/2	BALL 600# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
90			Ball V	B06AC05	2 - 3	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO								
100														
101														
140														
150						***** GATE VALVES *****								
160			Condui	W06AC70	3 - 4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW								
170			Condui	W06AC71	6 - 24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO								
190														
191														
192														
200						*****PIPE ABOVE GROUND*****								
210														
220			Piping	5358514	3/4 - 3	PIPE SCH XS SMLS LTCS A333 GR.6 BE								
230			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE								
240			Piping	9001255	6 - 6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE								
250			Piping	9001311	8 - 8	LINEPIPE SMLS WT=7.92MM API-5L X60 BE								
260			Piping	9001487	10 - 10	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE								
270			Piping	9001494	12 - 14	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE								
280			Piping	9001490	16 - 16	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE								
290			Piping	9001495	18 - 18	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE								
300			Piping	9005244	20 - 20	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE								
310			Piping	9005243	24 - 24	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE								
311			Piping	9001306	28 - 28	LINEPIPE DSAW WT= 19.05MM API-5L X60 BE								
320														
330						***** PIPE BELOW GROUND *****								
340														
350			Piping	9001257CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED								
360			Piping	9001255CTD	6 - 6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE COATED								
370			Piping	9001311CTD	8 - 8	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED								
380			Piping	9001487CTD	10 - 10	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED								
390			Piping	9001494CTD	12 - 14	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE COATED								
400			Piping	9001490CTD	16 - 16	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE COATED								
410			Piping	9001495CTD	18 - 18	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE COATED								
420			Piping	9005244CTD	20 - 20	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE COATED								
430			Piping	9005243CTD	24 - 24	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE COATED								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 24" AND ABOVE	CORR. ALLOW. 3 MM	DATE 29JUL13	BY TN	REV B	SHEET 1 OF 6	PA 5	CLASS 600K5F	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
431			Piping	9001306CTD	28 - 28	LINEPIPE DSAW WT= 19.05MM API-5L X60 BE COATED								
440														
450														
460						***** FITTINGS ABOVE & BELOW GROUND *****								
470														
480			90 Deg	5392600	3/4 - 3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6								
490			45 Deg	5392606	3/4 - 3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6								
500														
510			Tee	5392609	3/4 - 3	TEE BW SCH XS LTCS A420 WPL6								
520														
530			Reduci	5392661	3/4 - 2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 3							
540			Reduci	9006028	6 - 6	TEE RED BW WT 11.91 MM/7.11 MM LTCS A860 WPHY 60	16 - 16							
550			Reduci	9006029	8 - 8	TEE RED BW WT 15.88 MM/7.92 MM LTCS A860 WPHY 60-W	20 - 20							
560														
570			Cap	5392610	3/4 - 3	CAP BW SCH XS LTCS A420 WPL6								
580			Cap	9005062	4 - 4	CAP BW WT 6.02 MM LTCS A860 WPHY-60								
590			Cap	9005056	6 - 6	CAP BW WT 7.11MM LTCS A860 WPHY-60								
600			Cap	9005058	8 - 8	CAP BW WT 7.92 MM LTCS A860 WPHY-60								
610			Cap	9005059	10 - 10	CAP BW WT 8.74 MM LTCS A860 WPHY-60								
620			Cap	9005061	12 - 14	CAP BW WT 10.31 MM LTCS A860 WPHY-60								
630			Cap	9005065	16 - 16	CAP BW WT 11.91 MM LTCS A860 WPHY-60								
640			Cap	9005067	18 - 18	CAP BW WT 12.70 MM LTCS A860 WPHY-60								
650			Cap	9005069	20 - 20	CAP BW WT 15.88 MM LTCS A860 WPHY-60								
660			Cap	9005072	24 - 28	CAP BW WT 19.05 MM LTCS A860 WPHY-60								
670														
680			Concen	5392458	3/4 - 2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 3							
690			Eccent	5392461	3/4 - 2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 3							
700			Eccent	9001372	1 1/2 - 3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4 - 4							
710			Eccent	9001582	3 - 3	REDUCER ECC WT 7.11 MM / SCH XS MM LTCS A860 WPHY-60	6 - 6							
720			Eccent	9001373	4 - 4	REDUCER ECC WT 7.11 MM / 6.02 MM LTCS A860 WPHY-60	6 - 6							
730			Eccent	9001374	4 - 4	REDUCER ECC WT 7.92 MM / 6.02 MM LTCS A860 WPHY-60	8 - 8							
740			Eccent	9001375	6 - 6	REDUCER ECC WT 7.92 MM / 7.11 MM LTCS A860 WPHY-60	8 - 8							
750			Eccent	9001376	4 - 4	REDUCER ECC WT 8.74 MM / 6.02 MM LTCS A860 WPHY-60	10 - 10							
760			Eccent	9001377	6 - 6	REDUCER ECC WT 8.74 MM / 7.11 MM LTCS A860 WPHY-60	10 - 10							
770			Eccent	9002250	8 - 8	REDUCER ECC WT 8.74 MM / 7.92 MM LTCS A860 WPHY-60	10 - 10							
780			Eccent	9001378	6 - 6	REDUCER ECC WT 10.31 MM / 7.11 MM LTCS A860 WPHY-60	12 - 14							
790			Eccent	9001379	8 - 8	REDUCER ECC WT 10.31 MM / 7.92 MM LTCS A860 WPHY-60	12 - 14							
800			Eccent	9002399	10 - 10	REDUCER ECC WT 10.31 MM / 8.74 MM LTCS A860 WPHY-60	12 - 14							
810			Eccent	9005035	12 - 12	REDUCER ECC WT 10.31 MM LTCS A860 WPHY-60	14 - 14							
820			Eccent	9005036	8 - 8	REDUCER ECC WT 11.91 MM / 7.92 MM LTCS A860 WPHY-60	16 - 16							
830			Eccent	9001380	10 - 10	REDUCER ECC WT 11.91MM / 8.74MM LTCS A860 WPHY-60	16 - 16							
840			Eccent	9001382	12 - 14	REDUCER ECC WT 11.91 MM / 10.31 LTCS A860 WPHY-60	16 - 16							
850			Eccent	9005116	10 - 10	REDUCER ECC WT 12.70 MM / 8.74 MM LTCS A860 WPHY-60	18 - 18							
860			Eccent	9002406	12 - 14	REDUCER ECC WT 12.70 MM / 10.31 MM LTCS A860 WPHY-60	18 - 18							
870			Eccent	9001590	16 - 16	REDUCER ECC WT 12.70 MM / 11.91 MM LTCS A860 WPHY-60	18 - 18							
880			Eccent	9006250	12 - 14	REDUCER ECC WT 15.88 MM / 10.31 MM LTCS A860 WPHY-60-W	20 - 20							
890			Eccent	9004151	16 - 16	REDUCER ECC WT 15.88 MM / 11.91 MM LTCS A860 WPHY-60-W	20 - 20							
900			Eccent	9001384	18 - 18	REDUCER ECC WT 15.88 MM / 12.70MM LTCS A860 WPHY-60-W	20 - 20							
910			Eccent	9004301	18 - 18	REDUCER ECC WT 19.05MM / 12.70MM LTCS A860 WPHY-60-W	24 - 28							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 24" AND ABOVE	CORR. ALLOW. 3 MM	DATE 29JUL13	BY TN	REV B	SHEET 2 OF 6	PA 5	CLASS 600K5F	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
920			Eccent	9004302	20 - 20	REDUCER ECC WT 19.05MM / 15.88MM LTCS A860 WPHY-60-W	24 - 28							
930			Eccent	9006251	24 - 24	REDUCER ECC WT 19.05 MM LTCS A860 WPHY-60-W	28 - 28							
940						***** INDUCTION BENDS ABOVE GROUND *****								
950														
960														
970			90 Deg	9001810	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
980			90 Deg	9001396	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
990			90 Deg	9001815	8 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1000			90 Deg	9001816	10 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1010			90 Deg	9001818	12 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1020			90 Deg	9001397	16 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1030			90 Deg	9005321	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1040			90 Deg	9001398	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1050			90 Deg	9002060	24 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1060														
1070			45 Deg	9001820	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1080			45 Deg	9001399	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1090			45 Deg	9001825	8 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1100			45 Deg	9001826	10 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1110			45 Deg	9001828	12 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1120			45 Deg	9001500	16 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1130			45 Deg	9005327	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1140			45 Deg	9001200	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1150			45 Deg	9002061	24 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 19.05 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1160														
1170			22BND	9001830	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1180			22BND	9002000	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1190			22BND	9001835	8 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1200			22BND	9001836	10 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1210			22BND	9005041	12 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1220			22BND	9002001	16 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES: BUTTWELD								
						PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 24" AND ABOVE	CORR. ALLOW. 3 MM	DATE 29JUL13	BY TN	REV B	SHEET 3 OF 6	PA 5	CLASS 600K5F	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1230			22BND	9005333	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 75 BARG. @ 100 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: LTCS						
1240			22BND	9002002	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1250			22BND	9002062	24 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1260						***** INDUCTION BENDS BELOW GROUND *****								
1270														
1280														
1290			90 Deg	9001810CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1300			90 Deg	9001396CTD	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1310			90 Deg	9001815CTD	8 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1320			90 Deg	9001816CTD	10 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1330			90 Deg	9001818CTD	12 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1340			90 Deg	9001397CTD	16 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1350			90 Deg	9005321CTD	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1360			90 Deg	9001398CTD	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1370			90 Deg	9002060CTD	24 - 28	INDUCTION BEND 90 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1380														
1390			45 Deg	9001820CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1400			45 Deg	9001399CTD	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1410			45 Deg	9001825CTD	8 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1420			45 Deg	9001826CTD	10 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1430			45 Deg	9001828CTD	12 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1440			45 Deg	9004001CTD	16 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1450			45 Deg	9005327CTD	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1460			45 Deg	9001200CTD	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1470			45 Deg	9002061CTD	24 - 28	INDUCTION BEND 45 DEG X 5D RADIUS WT 19.05 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1480														
1490			22BND	9001830CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1500			22BND	9002000CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1510			22BND	9001835CTD	8 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1520			22BND	9001836CTD	10 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 24" AND ABOVE	CORR. ALLOW. 3 MM	DATE 29JUL13	BY TN	REV B	SHEET 4 OF 6	PA 5	CLASS 600K5F	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1530			22BND	9005041CTD	12 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 75 BARG. @ 100 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: LTCS						
1540			22BND	9002001CTD	16 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1550			22BND	9004000CTD	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1560			22BND	9002002CTD	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1570			22BND	9002062CTD	24 - 28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1580						***** OLETS ABOVE & BELOW GROUND *****								
1590														
1600														
1610			Weldol	9005117	3/4 - 2	WELDOLET WT 7.11MM/ SCH XS LTCS A694 F60	6 - 6							
1620			Weldol	9002552	3/4 - 3	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	8 - 8							
1630			Weldol	9006423	3/4 - 3	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	10 - 10							
1640			Weldol	9002551	3/4 - 3	WELDOLET WT 10.31MM/ XS WT LTCS A694 F60	12 - 14							
1650			Weldol	9006433	4 - 4	WELDOLET WT 10.31 MM/6.02 MM LTCS A694 F60	12 - 14							
1660			Weldol	9002003	3/4 - 3	WELDOLET WT 11.91MM/SCH XS LTCS A694 F60	16 - 16							
1670			Weldol	9002004	4 - 4	WELDOLET WT 11.91MM/6.02MM LTCS A694 F60	16 - 16							
1680			Weldol	9002467	3/4 - 3	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	18 - 18							
1690			Weldol	9005042	4 - 4	WELDOLET WT 12.70 MM/ 6.02 MM WT LTCS A694 F60	18 - 18							
1700			Weldol	9005043	6 - 6	WELDOLET WT 12.70 MM/ 7.11 MM WT LTCS A694 F60	18 - 18							
1710			Weldol	9005045	3/4 - 3	WELDOLET WT 15.88 MM/ SCH XS LTCS A694 F60	20 - 20							
1720			Weldol	9005046	4 - 4	WELDOLET WT 15.88 MM/ 6.02 MM LTCS A694 F60	20 - 20							
1730			Weldol	9002005	6 - 6	WELDOLET WT 15.88 MM/ 7.11 MM LTCS A694 F60	20 - 20							
1740			Weldol	9005047	8 - 8	WELDOLET WT 15.88 MM/ 7.92 MM LTCS A694 F60	20 - 20							
1750			Weldol	9006252	3/4 - 3	WELDOLET WT 19.05 MM/ SCH XS LTCS A694 F60	24 - 28							
1760			Weldol	9005386	4 - 4	WELDOLET WT 19.05 MM/ 6.02MM LTCS A694 F60	24 - 28							
1770			Weldol	9002093	6 - 6	WELDOLET WT 19.05 MM/ 7.11 MM LTCS A694 F60	24 - 28							
1780			Weldol	9002094	8 - 8	WELDOLET WT 19.05 MM/ 7.92 MM LTCS A694 F60	24 - 28							
1790			Weldol	9002090	10 - 10	WELDOLET WT 19.05 MM/ 8.74 MM LTCS A694 F60	28 - 28							
1800														
1810			Nipofl	5504006	3/4 - 2	NIPOFLANGE 600# RF SCH XS LTCS A350 LF2	3 - 3							
1820			Nipofl	9005030	3/4 - 2	NIPOFLANGE 600# RF 6.02 MM/XS WT LTCS A694 F60	4 - 4							
1830			Nipofl	9005100	3/4 - 2	NIPOFLANGE 600# RF 7.11 MM/XS WT LTCS A694 F60	6 - 6							
1840			Nipofl	9005031	3/4 - 2	NIPOFLANGE 600# RF 7.92 MM/XS WT LTCS A694 F60	8 - 8							
1850			Nipofl	9005032	3/4 - 2	NIPOFLANGE 600# RF 8.74 MM/XS WT LTCS A694 F60	10 - 10							
1860			Nipofl	9005033	3/4 - 2	NIPOFLANGE 600# RF 10.31 MM/XS WT LTCS A694 F60	12 - 14							
1870			Nipofl	9005103	3/4 - 2	NIPOFLANGE 600# RF 11.91 MM/XS WT LTCS A694 F60	16 - 16							
1880			Nipofl	9005104	3/4 - 2	NIPOFLANGE 600# RF 12.70 MM/XS WT LTCS A694 F60	18 - 18							
1890			Nipofl	9005034	3/4 - 2	NIPOFLANGE 600# RF 15.88 MM/XS WT LTCS A694 F60	20 - 20							
1900			Nipofl	9003000	3/4 - 2	NIPOFLANGE 600# RF WT 19.05MM/SCH XS LTCS A694 F60	24 - 28							
1910						***** PIPE FLANGES ABOVE & BELOW GROUND*****								
1920														
1930														
1940			Flange	5557036	3/4 - 3	FLG WN 600# RF LTCS A350 LF2 SCH XS								
1950			Flange	9001723	4 - 4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM								
1960			Flange	9001462	6 - 6	FLG WN 600# RF LTCS A694 F60 WT 7.11MM								
1970			Flange	9003494	8 - 8	FLG WN 600# RF LTCS A694 F60 WT 7.92 MM								
1980			Flange	9003484	10 - 10	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP							
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 24" AND ABOVE	CORR. ALLOW. 3 MM	DATE 29JUL13	BY TN	REV B	SHEET 5 OF 6	PA 5	CLASS 600K5F	

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1990			Flange	9003486	12 - 14	FLG WN 600# RF LTCS A694 F60 WT 10.31 MM		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 75 BARG. @ 100 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 100 °C COMMENT: LTCS							
2000			Flange	9001511	16 - 16	FLG WN 600# RF LTCS A694 F60 WT 11.91MM									
2010			Flange	9001512	18 - 18	FLG WN 600# RF LTCS A694 F60 WT 12.70MM									
2020			Flange	9005128	20 - 20	FLG WN 600# RF LTCS A694 F60 WT 15.88 MM									
2030			Flange	9002064	24 - 24	FLG WN 600# RF LTCS A694 F60 WT 19.05 MM									
2040			Flange	9003498	28 - 28	FLG WN 600# RF LTCS A694 F60 WT 19.05 MM SERIES A									
2050			Flange	9006169	28 - 28	FLG COMPACT 600# RJ A694 F60 WT 19.05 MM									
2060															
2070			Blind	5615598	3/4 - 3	FLG BLIND 600# RF LTCS A350 LF2									
2080			Blind	9001518	4 - 24	FLG BLIND 600# RF LTCS A694 F60									
2090			Blind	9001519	28 - 28	FLG BLIND 600# RF LTCS A694 F60 SERIES A									
2110															
2112															
2120		B1				***** GASKETS*****									
2130															
2140	C		GASFRS	5673103	3/4 - 24	GASKET 600# 316SS SP WND GRAPHITE IR									
2150			GASFRS	5673055	28 - 28	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR B16.47 SERIES A									
2161	C		GASRJ	5673417	28 - 28	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 600K5F									
2170															
2180						*****BOLTING*****									
2190															
2201															
2202	C		BOLS	5675770	3/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS									
2203															
2209			CPSBOL	5675675	3/4 - 99	JACK SCREW HEX HEAD A320 GR L7M									
2220															
2221						***** BLINDS & SPACERS									
2230															
2240															
2250			Specta	5749587	3/4 - 8	SPECTACLE BLIND 600# RF LTCS A516-GR70									
2260			Paddle	5750040	10 - 28	PADDLE BLANK 600# RF LTCS A516-GR70									
2270			Paddle	5750251	10 - 28	PADDLE SPACER 600# RF LTCS A516-GR70									
2280															
2290															
2300															
2310															
2320															
2330						***** BRANCHES *****									
2340															
2350						USE BRANCH CHART 600K5F									
2360															
2370						***** CONSTRUCTION DETAILS *****									
2380															
2390						SEAL WELDING ----- REQUIRED									
2400						PIPE BENDS ----- INDUCTION BENDS									
2410						THREAD SEALANT ----- PTFE PASTE									
2420															
2430															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP / WPMP								
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 24" AND ABOVE	CORR. ALLOW. 3 MM	DATE 29JUL13	BY TN	REV B	SHEET 6 OF 6	PA 5	CLASS 600K5F		

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
20	C	PP9	Ball V	M06AM10	3/4 - 1 1/2	***** MODULAR VALVES *****		B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FA8 USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. PP9 ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
40														
50														
60														
70														
71														
72														
73														
90														
100														
101	Ball V	B06AM10 A06AM14 A06AR16	3/4 - 1 1/2 2 - 2 8 - 8	BALL 600# RF 825 INCONEL 625 TRIM FLOATING METAL FB LO BALL 600# RF 825 INCONEL 625 TRIM TRUNNION METAL RB LO BALL 600# RF LTCS + CLAD 825 INCONEL 625 TRIM TRUNNION METAL RB GO										
120														
121														
123	C					*****THROUGH CONDUIT GATE VALVES*****								
320														
410														
420														
440														
460														
470														
480														
481														
570														
580	Gate V	G06AM48	3/4 - 2	GATE 600# RF INCONEL 825 N08825 WEDGE HW ***** CHECK VALVES *****										
590														
640														
650	C					***** PIPE *****								
670														
680														
690														
710														
711														
830														
840														
850														
860														
861	90 Deg	5425111 9002672 9002699	3/4 - 3 6 - 8 16 - 16	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825 ELL BW 90 DEG LR STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING ELL BW 90 DEG LR SCH XS WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING										
870														
871														
970	45 Deg 45 Deg	5425113 9002673	3/4 - 2 8 - 8	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825 ELL BW 45 DEG LR STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING										
980														
990														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN		CONSTRUCTION ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		FGP / WPMP CORR. ALLOW. DATE BY REV SHEET PA CLASS 0 MM 14NOV13 TC B 1 OF 3 5 600K5G							

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
991			45 Deg	9006576	16 - 16	ELL BW 45 DEG LR SCH XS WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 PHC MATERIAL: CARBON STEEL / CRA FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT:							
1090															
1091	C		Tee	5425119	2 - 2	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825									
1110			Tee	9002674	8 - 8	TEE BW STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING									
1111			Tee	9002767	16 - 16	TEE BW SCH XS A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING									
1112			Reduci	9002698	6 - 6	TEE RED BW SCH STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING	8 - 8								
1113			Reduci	9002766	6 - 8	TEE RED BW SCH XS/STD WT A860 WPHY 60 WITH 3 MM INCONEL 825 INTERNAL CLADDING	16 - 16								
1360															
1380			Cap	9002675	8 - 8	CAP BW STD WT A860 WPHY60 WITH 3 MM INCONEL 825 INTERNAL CLADDING									
2070															
2072															
2073															
2074															
2080						***** OLETS *****									
2090															
2110			Flange	9002676	3/4 - 2	NIPOFLANGE 600# RF STD WT/80S INCONEL 825 N08825	8 - 8								
2111			Flange	9002762	3/4 - 2	NIPOFLANGE 600# RF SCH XS WT/80S INCONEL 825 N08825	16 - 16								
2300															
2320			Weldol	9002677	3/4 - 2	WELDOLET STD WT/80S INCONEL 825 NO8825	8 - 8								
2321			Weldol	9002761	3/4 - 2	WELDOLET SCH XS/80S INCONEL 825 NO8825	16 - 16								
2700															
2710						**** FLANGES ****									
2720															
2721		FA8	Reduci	5600660	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 600# RF SS A182 F316/316L	3/4 - 1 1/2								
2722															
2740			Flange	5569391	3/4 - 3	FLG WN 600# RF ALLOY 825 SCH 80S B564 UNS N08825									
2750			Flange	9002678	6 - 8	FLG WN 600# RF STD WT A694 F60 WITH 3MM INCONEL 825 INTERNAL CLAD BORE & FACE									
2751			Flange	9002760	16 - 16	FLG WN 600# RF SCH XS WT A694 F60 WITH 3 MM INCONEL 825 CLAD FACE & BORE									
2752															
2930															
2940			Blind	9002682	3/4 - 16	FLG BLIND 600# RF A694 F60 WITH 3 MM INCONEL 825 CLAD FACE									
2980						***** PIPE NIPPLES *****									
3120															
3130						NIPPLE PBE - CUT FROM PIPE									
3140															
3150						***** GASKETS *****									
3160															
3170															
3180			GASFRS	5673661	3/4 - 16	GASKET 600# ALLOY 825 SPIRAL WOUND GRAPHITE IR									
3220						***** BOLTING *****									
3230															
3240															
3250	C	B1	BOLS	5675770	1/4 - 99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL T9CO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP / WPMP							
						ALL SIZES - BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW. 0 MM		DATE 14NOV13	BY TC	REV B	SHEET 2 OF 3	PA 5	CLASS 600K5G

SEQ NO.	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
3280			Specta	5749587_C80	4 - 8	***** BLINDS AND SPACERS *****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 PHC MATERIAL: CARBON STEEL / CRA FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT:							
3290															
3300															
3310															
3420															
3421															
3422															
3430															
3440															
3450															
3460															
3470															
3480															
3490															
3500															

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 20 30 40	C		2BV	K06AC02	1/2 - 1 1/2	***** MODULAR VALVES ***** MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		817 B1 FF5	USE ONLY FOR TIE IN ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY. TO BE USED IN NON-PIGGED LINES ONLY					
50 60	C		2BV	K06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		NT5 SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
70 80	C	VAV	2BV	K06AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		VA5 VAV VAW	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055					
90 100	C		2BVR	M06AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VB2	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
110 120	C		2BVR	M06AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VC1 VC2	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
130 140	C	VAW	R2BVR	M06EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2							
150 160	C	VAW	R2BVR	M06EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3							
170 180 190 200		VB2				***** BALL VALVES *****								
210	C		Ball V	A06AC05	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO								
220	C		Ball V	A06AC06	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO								
230	C		Ball V	A06AC07	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
240	C		Ball V	A06AC08	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
250 260														
270	C		Ball V	A06AC08	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
280														
290			Ball V	B06AC01	3/4 - 1 1/2	BALL 600# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
300														
310	C		Ball V	B06AC02	3/4 - 1 1/2	BALL 600# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
320														
330	D	VA5	Ball V	B06AC05	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO								
340	D	VA5	Ball V	B06AC07	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
350														
360	D	VA5	Ball V	B06AC07	6 - 24	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
370														
380	D	VA5	Ball V	B06AC06	2 - 4	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO								
390														
400	D	VA5	Ball V	B06AC08	6- 16	BALL 600# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
410														
420														
430						*****ORBIT VALVES*****								
440														
450	C	VA5	Ball V	R06AC06	2 - 6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO								
460														
470	C	VA5	Ball V	R06AC08	8 - 12	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB GO								
480														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
							3 MM	17MAR17	JCS	C	1 OF 10	5	600K5H	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
490	C					***** BUTTERFLY VALVES *****							
500													
510	C		Butter	T06AC61	10 - 24	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO							
520													
530													
540						***** GATE VALVES *****							
550													
560	C		Gate V	G06AC48	3/4 - 3/4	GATE 600# RF LTCS A350 LF2 TRIM 16 WEDGE HW							
570													
580	C		Condui	W06AC70	3 - 4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
590													
600	C		Condui	W06AC71	6 - 24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
610													
620													
630						***** GLOBE VALVES *****							
640													
650			Globe	L06AC48	3/4 - 1 1/2	GLOBE 600# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
660	C		Globe	L06AC41	2 - 4	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW							
670	C		Globe	L06AC42	6 - 10	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y GO							
680													
690													
700						***** CHECK VALVES *****							
710													
720	C	VC1	Check	C06AC50	3/4 - 1	CHECK PISTON 600# RF LTCS A350 LF2 TRIM 16							
730	C	VC2	Check	C06AC45	2 - 8	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
740	C	VC2	Check	C06AC52	10 - 24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16							
750													
760													
770						*****PIPE ABOVE GROUND*****							
780													
790	C		Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE							
800	C		Piping	5358515	2 - 2	PIPE SCH 160 SMLS LTCS A333 GR.6 BE							
810			Piping	5358514	3 - 3	PIPE SCH XS SMLS LTCS A333 GR.6 BE							
820			Piping	9001211	4 - 4	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE							
830	C		Piping	9001311	6 - 6	LINEPIPE SMLS WT=7.92MM API-5L X60 BE							
840			Piping	9001493	8 - 8	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE							
850	C		Piping	9001488	10 - 10	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE							
860	C		Piping	9001495	12 - 12	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE							
870			Piping	9001491	14 - 14	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE							
880			Piping	9005244	16 - 16	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE							
890			Piping	9005242	18 - 18	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE							
900			Piping	9005243	20 - 20	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE							
910			Piping	9005254	24 - 24	LINEPIPE SMLS WT= 20.62 MM API-5L X60 BE							
920													
930	C		Piping	5358511	1/2 - 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE							
940													
950						*****PIPE BELOW GROUND*****							
960													
970			Piping	9001254CTD	4 - 4	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED							
980	C		Piping	9001311CTD	6 - 6	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED							
990			Piping	9001493CTD	8 - 8	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED							
1000	C		Piping	9001488CTD	10 - 10	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED							
1010	C		Piping	9001495CTD	12 - 12	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE COATED							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 10	PA 5	CLASS 600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1020			Piping	9001491CTD	14 - 14	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE COATED							
1030			Piping	9005244CTD	16 - 16	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE COATED							
1040			Piping	9005242CTD	18 - 18	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE COATED							
1050			Piping	9005243CTD	20 - 20	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE COATED							
1060			Piping	9005254CTD	24 - 24	LINEPIPE SMLS WT= 20.62MM API-5L X60 BE COATED							
1070													
1080													
1090						*****FITTINGS ABOVE & BELOW GROUND*****							
1100													
1110			90 Deg	5392816	3/4 - 2	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6							
1120			90 Deg	5392600	3 - 3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6							
1130	C		90 Deg	9002190	4 - 4	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60							
1140	C		90 Deg	9006478	6 - 6	ELL BW 90 DEG LR WT 7.92 MM A860 WPHY 60							
1150			90 Deg	9006471	8 - 8	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60							
1160	C		90 Deg	9006468	10 - 10	ELL BW 90 DEG LR WT 11.13 MM A860 WPHY 60							
1170	C		90 Deg	9006479	12 - 12	ELL BW 90 DEG LR WT 12.70 MM A860 WPHY 60							
1180	C		90 Deg	9006480	14 - 14	ELL BW 90 DEG LR WT 14.27 MM A860 WPHY 60							
1190			90 Deg	9006469	16 - 16	ELL BW 90 DEG LR WT 15.88 MM A860 WPHY 60							
1200			90 Deg	9006474	18 - 18	ELL BW 90 DEG LR WT 17.48 MM A860 WPHY 60-W							
1210			90 Deg	9006475	20 - 20	ELL BW 90 DEG LR WT 19.05 MM A860 WPHY 60-W							
1220			90 Deg	9006476	24 - 24	ELL BW 90 DEG LR WT 20.62 MM A860 WPHY 60-W							
1230													
1240			45 Deg	5392822	3/4 - 2	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6							
1250			45 Deg	5392606	3 - 3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1260			45 Deg	9006571	4 - 4	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60							
1270	C		45 Deg	9006588	6 - 6	ELL BW 45 DEG LR WT 7.92MM A860 WPHY 60							
1280	C		45 Deg	9006583	8 - 8	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60							
1290	C		45 Deg	9006569	10 - 10	ELL BW 45 DEG LR WT 11.13MM A860 WPHY 60							
1300	C		45 Deg	9006590	12 - 12	ELL BW 45 DEG LR WT 12.70MM A860 WPHY 60							
1310	C		45 Deg	9006595	14 - 14	ELL BW 45 DEG LR WT 14.27MM A860 WPHY 60							
1320	C		45 Deg	9006581	16 - 16	ELL BW 45 DEG LR WT 15.88MM A860 WPHY 60							
1330	C		45 Deg	9006582	18 - 18	ELL BW 45 DEG LR WT 17.48MM A860 WPHY 60							
1340			45 Deg	9006572	20 - 20	ELL BW 45 DEG LR WT 19.05MM A860 WPHY 60-W							
1350			45 Deg	9006578	24 - 24	ELL BW 45 DEG LR WT 20.62MM A860 WPHY 60-W							
1360													
1370			Tee	5392825	3/4 - 2	TEE BW SCH 160 LTCS A420 WPL6							
1380			Tee	5392609	3 - 3	TEE BW SCH XS LTCS A420 WPL6							
1390		NT5	Tee	9301609	4 - 4	TEE BW WT 6.35 MM LTCS A860 WPHY-60							
1400	C	NT5	Tee	9301606	6 - 6	TEE BW WT 7.92 MM LTCS A860 WPHY-60							
1410		NT5	Tee	9301612	8 - 8	TEE BW WT 9.53 MM LTCS A860 WPHY-60							
1420	C	NT5	Tee	9006604	10 - 10	TEE BW WT 11.13MM A860 WPHY 60							
1430	C	NT5	Tee	9006633	12 - 12	TEE BW WT 12.7 MM A860 WPHY 60							
1440	C	NT5	Tee	9006486	14 - 14	TEE BW WT 14.27MM A860 WPHY 60							
1450		NT5	Tee	9004200	16 - 16	TEE BW WT 15.88MM A860 WPHY 60							
1460		NT5	Tee	9004201	18 - 18	TEE BW WT 17.48MM A860 WPHY 60							
1470		NT5	Tee	9004212	20 - 20	TEE BW WT 19.05MM A860 WPHY 60-W							
1480		NT5	Tee	9004213	24 - 24	TEE BW WT 20.62MM A860 WPHY 60-W							
1490													
1500			Reduci	5392821	3/4 - 1 1/2	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 2						
1510			Reduci	5392469	1 1/2 - 2	TEE RED BW SCH XS/160 LTCS A420 WPL6	3 - 3						
1520	C	NT5	Reduci	9006608	2 - 3	TEE RED BW WT 6.35 MM / SCH XS LTCS A860 WPHY 60	4 - 4						
1530	C	NT5	Reduci	9006657	3 - 3	TEE RED BW WT 7.92 MM/SCH XS LTCS A860 WPHY 60	6 - 6						
1540	C	NT5	Reduci	9006017	4 - 4	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	6 - 6						
1550	C	NT5	Reduci	9006100	4 - 4	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60	8 - 8						
1560	C	NT5	Reduci	9006612	4 - 4	TEE RED BW WT 11.13 MM/6.35 MM LTCS A860 WPHY 60	10 - 10						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.	CORR. ALLOW. 3 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 JCS C 3 OF 10 5 600K5H					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1570	C	NT5	Reduci	9006649	6 - 6	TEE RED BW WT 11.13 MM/7.92 MM LTCS A860 WPHY 60	10 - 10						
1580	C	NT5	Reduci	9006022	6 - 6	TEE RED BW WT 9.53 MM/7.92 MM LTCS A860 WPHY 60	8 - 8						
1590	C	NT5	Reduci	9006641	6 - 6	TEE RED BW WT 12.70 MM/7.92 MM LTCS A860 WPHY 60	12 - 12						
1600	C	NT5	Reduci	9006642	6 - 6	TEE RED BW WT 14.27 MM/7.92 MM LTCS A860 WPHY 60	14 - 14						
1610	C	NT5	Reduci	9006643	8 - 8	TEE RED BW WT 12.70 MM/9.53 MM LTCS A860 WPHY 60	12 - 12						
1620	C		Reduci	9006650	6 - 6	TEE RED BW WT 15.88 MM/7.92 MM LTCS A860 WPHY 60	16 - 16						
1630		NT5	Reduci	9004207	8 - 8	TEE RED BW WT 15.88 MM/9.53 MM LTCS A860 WPHY 60	16 - 16						
1640	C	NT5	Reduci	9006644	10 - 10	TEE RED BW WT 15.88 MM/11.13 MM LTCS A860 WPHY 60	16 - 16						
1650	C	NT5	Reduci	9006646	12 - 12	TEE RED BW WT 15.88 MM/12.70 MM LTCS A860 WPHY 60	16 - 16						
1660	C	NT5	Reduci	9006651	14 - 14	TEE RED BW WT 15.88 MM/14.27 MM LTCS A860 WPHY 60	16 - 16						
1670		NT5	Reduci	9301690	8 - 8	TEE RED BW WT 17.48 MM/9.53MM LTCS A860 WPHY-60	18 - 18						
1680	C	NT5	Reduci	9006629	12 - 12	TEE RED BW WT 17.48 MM/12.7 MM LTCS A860 WPHY 60-W	18 - 18						
1690	C	NT5	Reduci	9006656	16 - 16	TEE RED BW WT 17.48 MM/15.88 MM LTCS A860 WPHY 60-W	18 - 18						
1700			Reduci	9006202	8 - 8	TEE RED BW WT 19.05 MM/9.53 MM LTCS A860 WPHY 60-W	20 - 20						
1710	C	NT5	Reduci	9006660	12 - 12	TEE RED BW WT 19.05 MM/12.70 MM LTCS A860 WPHY 60-W	20 - 20						
1720	C	NT5	Reduci	9006648	16 - 16	TEE RED BW WT 19.05 MM/15.88 MM LTCS A860 WPHY 60-W	20 - 20						
1730	C	NT5	Reduci	9004202	10 - 10	TEE RED BW WT 20.62 MM/11.13 MM LTCS A860 WPHY 60-W	24 - 24						
1740	C	NT5	Reduci	9006647	12 - 12	TEE RED BW WT 20.62 MM/12.70 MM LTCS A860 WPHY 60-W	24 - 24						
1750		NT5	Reduci	9004203	16 - 16	TEE RED BW WT 20.62 MM/15.88 MM LTCS A860 WPHY 60-W	24 - 24						
1760		NT5	Reduci	9004204	18 - 18	TEE RED BW WT 20.62 MM/17.48 MM LTCS A860 WPHY 60-W	24 - 24						
1770		NT5	Reduci	9004205	20 - 20	TEE RED BW WT 20.62 MM/19.05 MM LTCS A860 WPHY 60-W	24 - 24						
1780													
1790			Cap	5392826	3/4 - 2	CAP BW SCH 160 LTCS A420 WPL6							
1800			Cap	5392610	3 - 3	CAP BW SCH XS LTCS A420 WPL6							
1810			Cap	9005071	4 - 4	CAP BW WT 6.35 MM LTCS A860 WPHY-60							
1820	C		Cap	9005058	6 - 6	CAP BW WT 7.92 MM LTCS A860 WPHY-60							
1830			Cap	9005060	8 - 8	CAP BW WT 9.53 MM LTCS A860 WPHY-60							
1840			Cap	9005064	10 - 10	CAP BW WT 11.13 MM LTCS A860 WPHY-60							
1850	C		Cap	9005067	12 - 12	CAP BW WT 12.70 MM LTCS A860 WPHY-60							
1860			Cap	9005068	14 - 14	CAP BW WT 14.27 MM LTCS A860 WPHY-60							
1870			Cap	9005069	16 - 16	CAP BW WT 15.88 MM LTCS A860 WPHY-60							
1880			Cap	9005070	18 - 18	CAP BW WT 17.48 MM LTCS A860 WPHY-60							
1890			Cap	9005072	20 - 20	CAP BW WT 19.05 MM LTCS A860 WPHY-60							
1900			Cap	9005073	24 - 24	CAP BW WT 20.62 MM LTCS A860 WPHY-60							
1910													
1920			Concen	5392827	3/4 - 1 1/2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 2						
1930			Eccent	5392828	3/4 - 1 1/2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 2						
1940			Concen	5392462	1 1/2 - 2	REDUCER CONC SCH XS/160 LTCS A420 WPL6	3 - 3						
1950			Eccent	5392465	1 1/2 - 2	REDUCER ECC SCH XS/160 LTCS A420 WPL6	3 - 3						
1960	C		Concen	9006923	1 1/2 - 2	REDUCER CONC WT 6.35MM / SCH 160 A860 WPHY 60	4 - 4						
1970	C		Eccent	9002042	1 1/2 - 2	REDUCER ECC WT 6.35 MM / SCH 160 LTCS A860 WPHY-60	4 - 4						
1980	C		Concen	9006919	3 - 3	REDUCER CONC WT 6.35 MM / XS WT LTCS A860 WPHY-60	4 - 4						
1990	C		Eccent	9001581	3 - 3	REDUCER ECC WT 6.35 MM / XS WT LTCS A860 WPHY-60	4 - 4						
2000	C		Concen	9006921	3 - 3	REDUCER CONC WT 7.92 MM / SCH XS MM LTCS A860 WPHY-60	6 - 6						
2010	C		Eccent	9006922	3 - 3	REDUCER ECC WT 7.92 MM / SCH XS MM LTCS A860 WPHY-60	6 - 6						
2020	C		Concen	9006909	4 - 4	REDUCER CONC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	6 - 6						
2030													
2040	C		Eccent	9002272	4 - 4	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	6 - 6						
2050			Eccent	9002393	4 - 4	REDUCER ECC WT 9.53 MM / 6.35 MM LTCS A860 WPHY-60	8 - 8						
2060	C		Eccent	9002251	6 - 6	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60	8 - 8						
2070			Eccent	9001586	4 - 4	REDUCER ECC WT 11.13 MM / 6.35MM LTCS A860 WPHY-60	10 - 10						
2080	C		Eccent	9002053	6 - 6	REDUCER ECC WT 11.13 MM / 7.92 MM LTCS A860 WPHY-60	10 - 10						
2090	C		Eccent	9002403	8 - 8	REDUCER ECC WT 11.13 MM / 9.53 MM LTCS A860 WPHY-60	10 - 10						
2100	C		Eccent	9002098	6 - 6	REDUCER ECC WT 12.70 MM / 7.92 MM LTCS A860 WPHY-60	12 - 12						
2110	C		Eccent	9002405	8 - 8	REDUCER ECC WT 12.70 MM / 9.53 MM LTCS A860 WPHY-60	12 - 12						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 4 OF 10	PA 5	CLASS 600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2120			Eccent	9002407	10 - 10	REDUCER ECC WT 12.70 MM / 11.13 MM LTCS A860 WPHY-60	12 - 12						
2130			Eccent	9002025	8 - 8	REDUCER ECC WT 14.27 MM / 9.53 MM LTCS A860 WPHY-60	14 - 14						
2140	C		Concen	9006905	10 - 10	REDUCER CONC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60	14 - 14						
2150	C		Eccent	9002409	10 - 10	REDUCER ECC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60	14 - 14						
2160	C		Concen	9006900	12 - 12	REDUCER CONC WT 14.27MM / 12.7MM LTCS A860 WPHY 60	14 - 14						
2170	C		Eccent	9002410	12 - 12	REDUCER ECC WT 14.27 MM / 12.70 MM LTCS A860 WPHY-60	14 - 14						
2180			Eccent	9004278	8 - 8	REDUCER ECC WT 15.88 MM / 9.53 MM LTCS A860 WPHY-60	16 - 16						
2190	C		Eccent	9004279	10 - 10	REDUCER ECC WT 15.88 MM / 11.13 MM LTCS A860 WPHY-60	16 - 16						
2200	C		Eccent	9004280	12 - 12	REDUCER ECC WT 15.88 MM / 12.70 MM LTCS A860 WPHY-60	16 - 16						
2210			Eccent	9004299	14 - 14	REDUCER ECC WT 15.88 MM / 14.27MM LTCS A860 WPHY-60	16 - 16						
2220	C		Eccent	9004282	10 - 10	REDUCER ECC WT 17.48 MM / 11.13 MM LTCS A860 WPHY-60	18 - 18						
2230	C		Eccent	9004283	12 - 12	REDUCER ECC WT 17.48 MM / 12.70 MM LTCS A860 WPHY-60	18 - 18						
2240			Eccent	9004300	14 - 14	REDUCER ECC WT 17.48 MM / 14.27MM LTCS A860 WPHY-60	18 - 18						
2250	C		Concen	9006906	16 - 16	REDUCER CONC WT 17.48 MM / 15.88MM LTCS A860 WPHY-60	18 - 18						
2260			Eccent	9004284	16 - 16	REDUCER ECC WT 17.48 MM / 15.88MM LTCS A860 WPHY-60	18 - 18						
2270			Eccent	9004302	16 - 16	REDUCER ECC WT 19.05MM / 15.88MM LTCS A860 WPHY-60-W	20 - 20						
2280			Eccent	9004303	18 - 18	REDUCER ECC WT 19.05MM / 17.48 MM LTCS A860 WPHY-60-W	20 - 20						
2290													
2300	C		Concen	9006913	16 - 16	REDUCER CONC WT 20.62 MM / 15.88 MM LTCS A860 WPHY-60-W	24 - 24						
2310			Eccent	9002049	16 - 16	REDUCER ECC WT 20.62MM / 15.88MM LTCS A860 WPHY-60-W	24 - 24						
2320	C		Concen	9006911	18 - 18	REDUCER CONC WT 20.62 MM / 17.48 MM LTCS A860 WPHY-60-W	24 - 24						
2330			Eccent	9001598	18 - 18	REDUCER ECC WT 20.62MM / 17.48 MM LTCS A860 WPHY-60-W	24 - 24						
2340	C		Concen	9009610	20 - 20	REDUCER CONC WT 20.62 MM / 19.05 MM LTCS A860 WPHY-60-W	24 - 24						
2350			Eccent	9002050	20 - 20	REDUCER ECC WT 20.62MM / 19.05MM LTCS A860 WPHY-60-W	24 - 24						
2360													
2370													
2380						***** INDUCTION BENDS ABOVE GROUND*****							
2390													
2400			90 Deg	9001534	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2410	C		90 Deg	9001815	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2420			90 Deg	9001817	8 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2430	C		90 Deg	9001899	10 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2440	C		90 Deg	9005321	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2450			90 Deg	9005322	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2460			90 Deg	9001398	16 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2470			90 Deg	9001573	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2480			90 Deg	9002060	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2490			90 Deg	9001574	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2500													
2510			45 Deg	9001575	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2520	C		45 Deg	9001825	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
2530			45 Deg	9001827	8 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	CORR. ALLOW.						
ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.							3 MM	DATE	BY	REV	SHEET	PA	CLASS
								17MAR17	JCS	C	5 OF 10	5	600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
2540	C		45 Deg	9005326	10 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2550	C		45 Deg	9005327	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2560			45 Deg	9005328	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2570			45 Deg	9001200	16 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2580			45 Deg	9005040	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2590			45 Deg	9002061	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 19.05 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2600			45 Deg	9002026	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2610			22.5 D	9001580	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2620																
2630	C		22.5 D	9001835	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2640			22.5 D	9001837	8 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2650	C		22.5 D	9005331	10 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2660	C		22.5 D	9005333	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2670			22.5 D	9005334	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2680			22.5 D	9002002	16 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2690			22.5 D	9001579	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2700			22.5 D	9002062	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2710			22.5 D	9002027	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 20.62MM API 5L X60 C/W 1.5D TANGENT ENDS BW										
2720																
2730																
2740																
2750																
2760			90 Deg	9001534CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 95 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE PIPING: BD, BV, CI, DHC, FGH, PHC, WRA. MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS									
2770	C		90 Deg	9001815CTD	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
2780			90 Deg	9001817CTD	8 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
2790	C		90 Deg	9001899CTD	10 - 10	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
2800	C		90 Deg	9005321CTD	12 - 12	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
2810			90 Deg	9005322CTD	14 - 14	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
2820			90 Deg	9001398CTD	16 - 16	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
2830			90 Deg	9001573CTD	18 - 18	INDUCTION BEND 90 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED										
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.D		CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 6 OF 10	PA 5	CLASS 600K5H		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
2840			90 Deg	9002060CTD	20 - 20	INDUCTION BEND 90 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2850			90 Deg	9001574CTD	24 - 24	INDUCTION BEND 90 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2860			45 Deg	9001575CTD	4 - 4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2870			45 Deg	9001825CTD	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2880	C		45 Deg	9001827CTD	8 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2890			45 Deg	9005326CTD	10 - 10	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2900	C		45 Deg	9005327CTD	12 - 12	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2910	C		45 Deg	9005328CTD	14 - 14	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2920			45 Deg	9001200CTD	16 - 16	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2930			45 Deg	9005040CTD	18 - 18	INDUCTION BEND 45 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2940			45 Deg	9002061CTD	20 - 20	INDUCTION BEND 45 DEG X 5D RADIUS WT 19.05 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2950			45 Deg	9002026CTD	24 - 24	INDUCTION BEND 45 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2960			22.5 D	9001580CTD	4 - 4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2970			22.5 D	9001835CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2980	C		22.5 D	9001837CTD	8 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
2990			22.5 D	9005331CTD	10 - 10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
3000	C		22.5 D	9004000CTD	12 - 12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
3010	C		22.5 D	9005334CTD	14 - 14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
3020			22.5 D	9002002CTD	16 - 16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
3030			22.5 D	9001579CTD	18 - 18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
3040			22.5 D	9002062CTD	20 - 20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
3050			22.5 D	9002027CTD	24 - 24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 20.62MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED						
3060						*****OLETS ABOVE & BELOW GROUND*****						
3070												
3080												
3090												
3100												
3110												
3120	C		Weldol	5505392	3/4 - 3/4	WELDOLET SCH 160 LTCS A350 LF2	2 - 2					
3130	C		Weldol	5505393	3/4 - 1	WELDOLET SCH XS/160 LTCS A350 LF2	3 - 3					
3140	C		Weldol	9002450	3/4 - 1 1/2	WELDOLET WT 6.35 MM/SCH 160 WT LTCS A694 F60	4 - 4					
3150	C		Weldol	9006818	3/4 - 2	WELDOLET WT 7.92 MM/ SCH 160 WT LTCS A694 F60	6 - 6					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 7 OF 10 PA 5 CLASS 600K5H				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3160	C		Weldol	9002457	3/4 - 2	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	8 - 8						
3170	C		Weldol	9006429	3 - 3	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	8 - 8						
3180	C		Weldol	9002076	3/4 - 2	WELDOLET WT 11.13MM/ SCH 160 LTCS A694 F60	10 - 10						
3190	C		Weldol	9005119	3 - 3	WELDOLET WT 11.13 MM/ SCH XS WT LTCS A694 F60	10 - 10						
3200			Weldol	9002079	4 - 4	WELDOLET WT 11.13MM/ 6.35MM LTCS A694 F60	10 - 10						
3210	C		Weldol	9002466	3/4 - 2	WELDOLET WT 12.70 MM/ SCH 160 WT LTCS A694 F60	12 - 12						
3220	C		Weldol	9002467	3 - 3	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	12 - 12						
3230			Weldol	9002469	4 - 4	WELDOLET WT 12.70 MM/ 6.35 MM WT LTCS A694 F60	12 - 12						
3240	C		Weldol	9002470	3/4 - 2	WELDOLET WT 14.27 MM/ SCH 160 WT LTCS A694 F60	14 - 14						
3250	C		Weldol	9002471	3 - 3	WELDOLET WT 14.27 MM/ SCH XS WT LTCS A694 F60	14 - 14						
3260			Weldol	9002473	4 - 4	WELDOLET WT 14.27 MM/ 6.35 MM LTCS A694 F60	14 - 14						
3270	C		Weldol	9002085	3/4 - 2	WELDOLET WT 15.88 MM/ SCH 160 LTCS A694 F60	16 - 16						
3280	C		Weldol	9005045	3 - 3	WELDOLET WT 15.88 MM/ SCH XS LTCS A694 F60	16 - 16						
3290			Weldol	9002088	4 - 4	WELDOLET WT 15.88 MM/ 6.35 MM LTCS A694 F60	16 - 16						
3300	C		Weldol	9004336	3/4 - 2	WELDOLET WT 17.48 MM/SCH 160 LTCS A694 F60	18 - 18						
3310	C		Weldol	9005053	3 - 3	WELDOLET WT 17.48 MM/SCH XS MM LTCS A694 F60	18 - 18						
3320			Weldol	9005049	4 - 4	WELDOLET WT 17.48 MM/6.35 MM LTCS A694 F60	18 - 18						
3330	C		Weldol	9005051	6 - 6	WELDOLET WT 17.48 MM/7.92 MM LTCS A694 F60	18 - 18						
3340	C		Weldol	9002089	3/4 - 2	WELDOLET WT 19.05 MM/ SCH 160 LTCS A694 F60	20 - 20						
3350	C		Weldol	9006252	3 - 3	WELDOLET WT 19.05 MM/ SCH XS LTCS A694 F60	20 - 20						
3360			Weldol	9002092	4 - 4	WELDOLET WT 19.05 MM/ 6.35 MM LTCS A694 F60	20 - 20						
3370	C		Weldol	9002094	6 - 6	WELDOLET WT 19.05 MM/ 7.92 MM LTCS A694 F60	20 - 20						
3380	C		Weldol	9004339	8 - 8	WELDOLET WT 19.05 MM/ 9.53 MM LTCS A694 F60	20 - 20						
3390	C		Weldol	9002096	3/4 - 2	WELDOLET WT 20.62 MM/ SCH 160 LTCS A694 F60	24 - 24						
3400	C		Weldol	9005126	3 - 3	WELDOLET WT 20.62 MM/ SCH XS LTCS A694 F60	24 - 24						
3410			Weldol	9002028	4 - 4	WELDOLET WT 20.62 MM/ 6.35 MM LTCS A694 F60	24 - 24						
3420	C		Weldol	9002824	6 - 6	WELDOLET WT 20.62 MM/ 7.92 MM LTCS A694 F60	24 - 24						
3430			Weldol	9005392	8 - 8	WELDOLET WT 20.62 MM/ 9.53 MM LTCS A694 F60	24 - 24						
3440													
3450	C		Nipofl	5504460	3/4 - 1	NIPOFLANGE 600# RF SCH 160 LTCS A350 LF2	2 - 2						
3460			Nipofl	5504461	3/4 - 2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2	3 - 3						
3470			Nipofl	9002070	3/4 - 2	NIPOFLANGE 600# RF WT 6.35MM/SCH 160 LTCS A694 F60	4 - 4						
3480	C		Nipofl	9006803	3/4 - 2	NIPOFLANGE 600# RF WT 7.92 MM/SCH 160 LTCS A694 F60	6 - 6						
3490			Nipofl	9003455	3/4 - 2	NIPOFLANGE 600# RF 9.53 MM/SCH 160 WT LTCS A694 F60	8 - 8						
3500	C		Nipofl	9003457	3/4 - 2	NIPOFLANGE 600# RF WT 11.13 MM/SCH 160 WT LTCS A694 F60	10 - 10						
3510													
3520	C		Nipofl	9003458	3/4 - 2	NIPOFLANGE 600# RF WT 12.70 MM/SCH 160 WT LTCS A694 F60	12 - 12						
3530													
3540			Nipofl	9003459	3/4 - 2	NIPOFLANGE 600# RF WT 14.27 MM/SCH 160 WT LTCS A694 F60	14 - 14						
3550													
3560			Nipofl	9002067	3/4 - 2	NIPOFLANGE 600# RF WT 15.88MM/SCH 160 LTCS A694 F60	16 - 16						
3570			Nipofl	9006496	3/4 - 2	NIPOFLANGE 600# RF 17.48 MM/SCH 160 LTCS A694 F60	18 - 18						
3580			Nipofl	9002068	3/4 - 2	NIPOFLANGE 600# RF WT 19.05MM/SCH 160 LTCS A694 F60	20 - 20						
3590			Nipofl	9002069	3/4 - 2	NIPOFLANGE 600# RF WT 20.62MM/SCH 160 LTCS A694 F60	24 - 24						
3600													
3610	C	FF5	Endole	9504522	3/4 - 3/4	NIPOFLANGE FLAT BOTTOM 600# RF SCH 160 LTCS A694 F60							
3620													
3630													
3640						*****PIPE FLANGES ABOVE & BELOW GROUND*****							
3650													
3660	C		Flange	5851204	1/2 - 2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE							
3670													
3680	C		Flange	5557038	1/2 - 2	FLG WN 600# RF LTCS A350 LF2 SCH 160							
3690			Flange	5557036	3 - 3	FLG WN 600# RF LTCS A350 LF2 SCH XS							
			FGP / WPMP TENGINZCHEVROIL - TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 8 OF 10	PA 5	CLASS 600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3700			Flange	9001471	4 - 4	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	3/4 - 1 1/2						
3710	C		Flange	9003494	6 - 6	FLG WN 600# RF LTCS A694 F60 WT 7.92 MM							
3720			Flange	9003485	8 - 8	FLG WN 600# RF LTCS A694 F60 WT 9.53 MM							
3730	C		Flange	9001464	10 - 10	FLG WN 600# RF LTCS A694 F60 WT 11.13MM							
3740	C		Flange	9001512	12 - 12	FLG WN 600# RF LTCS A694 F60 WT 12.70MM							
3750			Flange	9001515	14 - 14	FLG WN 600# RF LTCS A694 F60 WT 14.27MM							
3760			Flange	9005128	16 - 16	FLG WN 600# RF LTCS A694 F60 WT 15.88 MM							
3770			Flange	9001513	18 - 18	FLG WN 600# RF LTCS A694 F60 WT 17.48MM							
3780			Flange	9002064	20 - 20	FLG WN 600# RF LTCS A694 F60 WT 19.05 MM							
3790			Flange	9001517	24 - 24	FLG WN 600# RF LTCS A694 F60 WT 20.62MM							
3800													
3810	C		Blind	5615598	1/2 - 3	FLG BLIND 600# RF LTCS A350 LF2							
3820			Blind	9001518	4 - 24	FLG BLIND 600# RF LTCS A694 F60							
3830													
3840			Reduci	5599705	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 600# RF A350 LF2							
3850													
3860	C	817	Flange	5558038	2 - 2	FLG WN 600# RTJ LTCS A350 LF2 SCH 160							
3870	C	817	Flange	9006704	4 - 4	FLG WN 600# RTJ A694 F60 WT 6.35MM							
3880	C	817	Flange	9006711	6 - 6	FLG WN 600# RTJ A694 F60 WT 7.92MM							
3890	C	817	Flange	9006723	8 - 8	FLG WN 600# RTJ A694 F60 WT 9.53MM							
3900	C	817	Flange	9006706	10 - 10	FLG WN 600# RTJ A694 F60 WT 11.13MM							
3910	C	817	Flange	9006707	12 - 12	FLG WN 600# RTJ A694 F60 WT 12.70MM							
3920	C	817	Flange	9006714	14 - 14	FLG WN 600# RTJ A694 F60 WT 14.27MM							
3930	C	817	Flange	9006712	16 - 16	FLG WN 600# RTJ A694 F60 WT 15.88MM							
3940	C	817	Flange	9006709	18 - 18	FLG WN 600# RTJ A694 F60 WT 17.48MM							
3950	C	817	Flange	9006713	20 - 20	FLG WN 600# RTJ A694 F60 WT 19.05MM							
3960													
3970			Blind	5615698	3/4 - 3	FLG BLIND 600# RTJ LTCS A350 LF2							
3980			Blind	9006710	4 - 24	FLG BLIND 600# RTJ LTCS A694 F60							
3990													
4000	C	817	Flange	5557281	2 - 3	FLG WN 900# RTJ LTCS A350 LF2 SCH 160 BORE							
4010	C	817	Flange	9006717	4 - 4	FLG WN 900# RTJ LTCS A694 F60 WT 6.35MM							
4020	C	817	Flange	9006756	6 - 6	FLG WN 900# RTJ LTCS A694 F60 WT 7.92MM							
4030	C	817	Flange	9006719	12 - 12	FLG WN 900# RTJ LTCS A694 F60 WT 12.70MM							
4040													
4050						***** GASKETS *****							
4060													
4070	C		GASFRS	5673103	1/2 - 24	GASKET 600# 316SS SP WND GRAPHITE IR							
4080	C		GASRJ	5673202	2 - 20	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING							
4090													
4100	C		GASRJ	5674491	2 - 20	GASKET 900# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING							
4110													
4120		B1				***** BOLTING *****							
4130													
4140	C		BOLS	5675770	1/4 - 24	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
4150													
4160		SB6	CPSBOL	5675675	12 - 24	JACK SCREW HEX HEAD A320 GR L7M							
4170													
4180						***** BLINDS & SPACERS							
4190													
4200													
4210			Specta	5749587	3/4 - 8	SPECTACLE BLIND 600# RF LTCS A516-GR70							
4220			Paddle	5750040	10 - 24	PADDLE BLANK 600# RF LTCS A516-GR70							
4230			Paddle	5750251	10 - 24	PADDLE SPACER 600# RF LTCS A516-GR70							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 9 OF 10	PA 5	CLASS 600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
4240	C		Specta	5749591	4 - 4	SPECTACLE BLIND 900# RTJ FEMALE LTCS A516-GR70							
4250						***** SP ITEMS *****							
4260													
4270													
4280			Tee	SP-9261	16 - 16	TEE BARRED EQUAL NPS 16 406.4MM OD X 15.88MM WT A860							
4290			C	Ball V	SP-1492	12 - 12							WPHY 60 BALL 600# RTJ LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO
4300													
4310			C	Ball V	SP-1499	12 - 12							BALL 600# RTJ LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO
4320													
4330						***** BRANCHES *****							
4340	C		Reinfo	RWELD RWELD	2 - 16	USE BRANCH CHART 600K5H	3 - 24						
4350						NO-MTO REINFORCING WELD FOR STUB-IN							3 - 24
4360						NO-MTO REINFORCING WELD FOR STUB-IN							
4370						***** CONSTRUCTION DETAILS *****							
4380						SEAL WELDING ----- REQUIRED							
4390						PIPE BENDS ----- INDUCTION BENDS							
4400						THREAD SEALANT ----- PTFE PASTE							
4410													
4420													
4430													
4440	C		Reduci	9006751	3 - 3	**** DO NOT USE****	4 - 4						
4450													
4460						FLG WN REDUCING 600# RF LTCS A694 F60 BORED TO SUIT WT 6.35MM							
4470													
4480													
4490													
4500													
4510													
4520													
4530													
4540	C		Weldol	9002078	3 - 3	WELDOLET WT 11.13MM/ 7.62MM LTCS A694 F60	10 - 10						
4550	C		Weldol	9002091	3 - 3	WELDOLET WT 19.05 MM/ 7.62 MM LTCS A694 F60	20 - 20						
4560	C		Weldol	9002097	3 - 3	WELDOLET WT 20.62 MM/ 7.62 MM LTCS A694 F60	24 - 24						
								SERVICE					
								SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 95 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - PHC - MULTIPHASE PIPING: BD, BV, CI, DHC, FGH, PHC, WRA. MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP					
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. BELOW GROUND, PIPE SIZE 20" AND ABOVE.	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							3 MM	17MAR17	JCS	C	10 OF 10	5	600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10 20 30 40	D	PP9	2BV	K06AS22	1/2 - 1 1/2	*****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 600# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		O32 814 815 816	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE. TO BE USED WITH COMPACT FLANGES ONLY. COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's						
50 60	D		2BV	K06AS26	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		B1 BA1 BA5	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED WITH RAISED FACE FLANGES ONLY. TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.						
70 80	D	VAV	2BV	K06AS26	3 - 3	MODULAR VALVE DB&B BALL TYPE 600# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		FA8 PP9	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
90 100	D		2BVR	M06AS22	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 600# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
110 120	D		2BVR	M06AS26	2 - 2	MODULAR VALVE DB&B BALL TYPE 600# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR		SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTRORTRACE.						
130 140	D	VAW	R2BVR	M06ES22	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.						
150 160	D	VAW	R2BVR	M06ES22	1 - 1	MODULAR VALVE DB&B BALL TYPE 600# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	UD7 UDA	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.						
170 180 190 200 210 220 230 240 250 260 270 280 290 300 310 320 330 340 350 360 370 380 390 400 410 420 430 440 450 460 470 480	D	VB2	Ball V	A06AS27	2 - 4	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB LO 718 BOLT									
	D		Ball V	A06AS28	2 - 4	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION METAL RB LO 718 BOLT									
	D		Ball V	A06AS27	6 - 10	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT									
			Ball V	A06AS28	6 - 8	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT									
			Ball V	B06AS21	3/4 - 1 1/2	BALL 600# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT									
			Ball V	B06AS22	3/4 - 1 1/2	BALL 600# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT									
	D	VA5	Ball V	B06AS27	2 - 4	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB LO 718 BOLT									
	D	VA5	Ball V	B06AS28	2 - 4	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION METAL FB LO 718 BOLT									
	D	VA5	Ball V	B06AS27	6 - 8	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT									
		VA5	Ball V	B06AS27	10 - 24	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT									
	D	VA5	Ball V	B06AS28	4 - 24	BALL 600# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT									
		VA5	Ball V	B06CS27	26 - 48	BALL 600# CF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT									
		VA5	Ball V	B06CS28	26 - 48	BALL 600# CF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 1 MM	DATE 17MAR17	BY MC	REV D	SHEET 1 OF 5	PA 5	CLASS 600P21	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES										
490	C		Butter	T06AS65	10 - 24	*****BUTTERFLY VALVES*****		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE). FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY. FOR ALL INLINE APPLICATIONS AND LEVEL TRANSMITTER CONNECTIONS USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.									
500																		
510																		
520	C	Butter	T06CS65	26 - 48	BUTTERFLY 600# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO 718 BOLT BUTTERFLY 600# CF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO 718 BOLT													
530			Globe	L06AS55	3/4 - 1 1/2	GLOBE 600# RF SS A182 F316 TRIM 16 OS&Y HW 718 BOLT		VAB										
540								VAD										
550								VAG										
560								VAK										
570								VAV										
580								VAW										
590								VB2										
600	VC1	Check	C06AS54	3/4 - 1 1/2	CHECK PISTON 600# RF SS A182 F316 TRIM 16 718 BOLT		VC1											
610							VC2	Check								C06AS45	2 - 8	CHECK DUAL WAFER LUG 600# RF SS A351 CF8M TRIM 16 THROU BOLTED
620							VC2	Check								C06AS52	10 - 24	CHECK DUAL DBL FLG 600# RF SS A351 CF8M TRIM 16
630	VC2	Check	C06CS52	26 - 48	CHECK DUAL 600# CF SS A351 CF8M TRIM 16		VC2											
640							VU2											
650							ZA6											
660	C		Piping	5365140	3/4 - 1 1/2	PIPE SCH 80S SMLS SS A312 TP316/316L PE												
670																		
680																		
690																		
700																		
710																		
720																		
730																		
740																		
750																		
760	814	814	Piping	5365200	30 - 30	PIPE WT 38.0MM EFW SS A358 316/316L BE												
770																		
780																		
790	814	814	Piping	5365213	48 - 48	PIPE WT 50.0MM EFW SS A358 316/316L BE												
800																		
810																		
820	D		Piping	5365207	1/2 - 1/2	PIPE SCH 160 SMLS SS A312 TP316/316L PE	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN - ASME B16.5 MG 2.2 CODE - ASME B31.3 AA, BD, BF, BFH, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 600 RF TEMPERATURE RANGE: -120 / 180 °C COMMENT: LOW TEMP BLOW DOWN											
830	D	Piping	5365207	3/4 - 3/4	PIPE SCH 160 SMLS SS A312 TP316/316L PE													
840	C		Piping	5449033	3/4 - 8	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S												
850																		
860																		
870																		
880																		
890																		
900																		
910																		
920																		
930																		
940	C		45 Deg	5449146	3/4 - 8	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S												
950																		
960																		
970	C		45 Deg	5451110	30 - 30	ELL 45 LR WT 38.0MM 316/316L SS A403 WP316/316L-W												
980																		
990																		
1000	C		Tee	5449156	3/4 - 8	TEE SCH 80S SS A403 WP316/316L-S												
1010									Tee	5449157	10 - 24	TEE SCH 80 SS A403 WP316/316L-S						
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	FGP/WPMP DATE 17MAR17 BY MC REV D SHEET 2 OF 5 PA 5 CLASS 600P21										

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
1020	C		Tee	5451113	30 - 30	TEE 38.0MM SS A403 WP316/316L-W	1 - 8 10 - 20 12 - 24 26 - 26 26 - 26												
1030			Tee	5451112	36 - 36	TEE 40.0MM SS A403 WP316/316L-W													
1040			Tee	5451115	48 - 48	TEE 50.0MM SS A403 WP316/316L-W													
1050																			
1060			Reduci	5448905	3/4 - 6	TEE RED SCH 80S SS A403 WP316/316L-S													
1070			Reduci	5448906	4 - 8	TEE RED SCH 80/80S SS A403 WP316/316L-S													
1080			Reduci	5448907	10 - 20	TEE RED SCH 80 SS A403 WP316/316L-S													
1090			Reduci	5448913	12 - 12	TEE RED WT 33.0MM X SCH 80 SS A403 WP316/316L-S													
1100			Reduci	5448913	20 - 20	TEE RED WT 33.0MM X SCH 80 SS A403 WP316/316L-S													
1110			C		Cap	5449049								3/4 - 8	CAP SCH 80S SS A403 WP316/316L-S	1 - 8 10 - 20 12 - 24 26 - 26 26 - 26			
1120	Cap	5449050			10 - 24	CAP SCH 80 SS A403 WP316/316L-S													
1130	Cap	5449261			26 - 26	CAP WT 33.0MM SS A403 WP316/316L-S													
1140	Cap	5449260			30 - 30	CAP WT 38.0MM SS A403 WP316/316L-S													
1150	Cap	5449262			36 - 36	CAP WT 40.0MM SS A403 WP316/316L-S													
1160	Cap	5449263			48 - 48	CAP WT 50.0MM SS A403 WP316/316L-S													
1170	C		Concen	5449158	3/4 - 6	REDUCER CONC SCH 80S SS A403 WP316/316L-S	1 - 8 10 - 20 12 - 24 30 - 30 36 - 36 36 - 36 42 - 42 42 - 42 48 - 48												
1180			Concen	5448904	4 - 8	REDUCER CONC SCH 80/80S SS A403 WP316/316L-S													
1190			Concen	5448908	10 - 20	REDUCER CONC SCH 80 SS A403 WP316/316L-S													
1200			Concen	5450730	20 - 24	REDUCER CONC 38.0MM/SCH 80 SS A403 WP316/316L-W													
1210			Concen	5450749	24 - 24	REDUCER CONC 40.0MM / SCH 80 SS A403 WP316/316L-W													
1220			Concen	5450742	30 - 30	REDUCER CONC 40.0MM/ 38.0MM SS A403 WP316/316L-W													
1230			Concen	5450741	30 - 30	REDUCER CONC 45.0MM/ 38.0MM SS A403 WP316/316L-W													
1240			Concen	5450746	36 - 36	REDUCER CONC 45.0MM/ 40.0MM SS A403 WP316/316L-W													
1250			Concen	5450744	42 - 42	REDUCER CONC 50.0MM/ 45.0MM SS A403 WP316/316L-W													
1260																			
1270	C		Eccent	5449159	3/4 - 6	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1 - 8 10 - 20 12 - 24 30 - 30 36 - 36 36 - 36 42 - 42 48 - 48												
1280			Eccent	5449160	4 - 8	REDUCER ECC SCH 80/80S SS A403 WP316/316L-S													
1290			Eccent	5449161	10 - 20	REDUCER ECC SCH 80 SS A403 WP316/316L-S													
1300			Eccent	5450728	20 - 24	REDUCER ECC 38.0MM/SCH 80 SS A403 WP316/316L-W													
1310			Eccent	5450739	24 - 24	REDUCER ECC 40.0MM / SCH 80 SS A403 WP316/316L-W													
1320			Eccent	5450729	30 - 30	REDUCER ECC 40.0MM / 38.0MM SS A403 WP316/316L-W													
1330			Eccent	5450754	36 - 36	REDUCER ECC 45.0MM / 40.0MM SS A403 WP316/316L-W													
1340			Eccent	5450752	42 - 42	REDUCER ECC 50.0MM / 45.0MM SS A403 WP316/316L-W													
1350																			
1360																			
1370	C				***** OLETS *****	SERVICE													
1380							SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN - ASME B16.5 MG 2.2 CODE - ASME B31.3 AA, BD, BF, BFH, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM												
1390							VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS												
1400			C	FF5	Endole	5504613	3/4 - 1 1/2							NIPOFLANGE FLAT BOTTOM 600# RF SCH 80S SS A182 F316/316L	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 600 RF TEMPERATURE RANGE: -120 / 180 °C COMMENT: LOW TEMP BLOW DOWN				
1410			C	UJ2	Nipofl	5504075	1 1/2 - 1 1/2							NIPOFLANGE 600# RF SCH 80S/160 SS A182 F316/316L	4 - 8				
1420			C	UJ2	Nipofl	5504076	1 1/2 - 1 1/2							NIPOFLANGE 600# RF SCH 80/160 SS A182 F316/316L	10 - 24				
1430	C	UJ2	Nipofl	5504083	1 1/2 - 1 1/2	NIPOFLANGE 600# RF 33.0MM WT/SCH 160 SS A182 F316/316L	26 - 26												
1440	C	UJ2	Nipofl	5504080	1 1/2 - 1 1/2	NIPOFLANGE 600# RF 38.0MM WT/SCH 160 SS A182 F316/316L	30 - 30												
1450	C	UJ2	Nipofl	5504081	1 1/2 - 1 1/2	NIPOFLANGE 600# RF 40.0MM WT/SCH 160 SS A182 F316/316L	36 - 36												
1460	C	UJ2	Nipofl	5504540	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 45.0MM/160 SS A182 F316/316L	48 - 48												
1470	C																		
1480																			
1490	C																		
1500																			
1510	C																		
1520																			
1530	C																		
1540																			
1550	C																		
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD								FGP/WPMP					
						CORR. ALLOW.								DATE	BY	REV	SHEET	PA	CLASS
						1 MM								17MAR17	MC	D	3 OF 5	5	600P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1560		UJ2	Nipofl	5504082	1 1/2 - 1 1/2	NIPOFLANGE 600# RF 50.0MM WT/SCH 160 SS A182 F316/316L	48 - 48								
1570															
1580	C		Weldol	5509637	3/4 - 3	WELDOLET SCH 80S SS A182 F316/316L	2 - 8								
1590	C		Weldol	5509638	3/4 - 8	WELDOLET SCH 80/80S SS A182 F316/316L	10 - 24								
1600	C		Weldol	5509655	3/4 - 8	WELDOLET 38.0MM WT/ SCH 80S SS A182 F316/316L	30 - 30								
1610	C		Weldol	5509656	3/4 - 8	WELDOLET 40.0MM WT/ SCH 80S SS A182 F316/316L	36 - 36								
1620	C		Weldol	5509657	3/4 - 8	WELDOLET 50.0MM WT/ SCH 80S SS A182 F316/316L	48 - 48								
1630			Weldol	5509660	10 - 12	WELDOLET 38.0MM WT/ SCH 80 SS A182 F316/316L	30 - 30								
1640			Weldol	5509661	10 - 12	WELDOLET 40.0MM WT/ SCH 80 SS A182 F316/316L	36 - 36								
1650			Weldol	5509665	10 - 12	WELDOLET 50.0MM WT/ SCH 80 SS A182 F316/316L	48 - 48								
1660															
1670		UJ2	Elbole	5516375	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 80S/160 SS A182 F316/316L	4 - 8								
1680		UJ2	Elbole	5516376	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 80 /160 SS A182 F316/316L	10 - 24								
1690	C	UJ2	Elbole	5516377	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 33.0MM WT/160 SS A182 F316/316L	26 - 26								
1700		UJ2	Elbole	5516380	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 38.0MM WT/160 SS A182 F316/316L	30 - 30								
1710		UJ2	Elbole	5516381	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 40.0MM WT/160 SS A182 F316/316L	36 - 36								
1720	C	UJ2	Elbole	5516383	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 45.0MM WT/160 SS A182 F316/316L	42 - 42								
1730		UJ2	Elbole	5516382	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 50.0MM WT/160 SS A182 F316/316L	48 - 48								
1740															
1750	D		Elbole	5509801	3/4 - 3/4	ELBOLET SCH 80/80S SS A182 F316/316L	16 - 16								
1760			Elbole	5509800	2 - 2	ELBOLET SCH 80S/80S SS A182 F316/316L	8 - 8								
1770															
1780															
1790						***** OLETS *****									
1800															
1810			Sockol	5537071	3/4 - 1 1/2	SOCKOLET 3000# SS A182 F316/316L	2 - 48								
1820			Endole	5533235	3/4 - 1 1/2	FLAT SOCKOLET 3000# SS A182 F316/316L									
1830		UD7	Thredo	5515771	3/4 - 3/4	THREDOLET 3000# SS A182 F316/316L	2 - 48								
1840															
1850		816				***** PIPE FLANGES *****									
1860															
1870															
1880	C	FA8	Reduci	5600660	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 600# RF SS A182 F316/316L	3/4 - 1 1/2								
1890															
1900	D		Flange	5851204	1/2 - 2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE									
1910															
1920	D		Flange	5582998	1/2 - 8	FLG WN 600# RF SS A182 F316/316L SCH 80S BORE									
1930			Flange	5582999	10 - 24	FLG WN 600# RF SS A182 F316/316L SCH 80 BORE									
1940			Flange	5583635	30 - 30	FLG WN 600# RF SS A182 F316/316L 38.0MM WT B16.47 SERIES A									
1950			Flange	5583632	36 - 36	FLG WN 600# RF SS A182 F316/316L 40.0MM WT B16.47 SERIES A									
1960			Flange	5583634	42 - 42	FLG WN 600# RF SS A182 F316/316L 45.0MM WT B16.47 SERIES A									
1970			Flange	5583633	48 - 48	FLG WN 600# RF SS A182 F316/316L 50.0MM WT B16.47 SERIES A									
1980															
1990			Flange	5619807	26 - 26	FLG COMPACT 600# RJ SS A182 F316/316L WT 33.0MM									
2000			Flange	5619808	30 - 30	FLG COMPACT 600# RJ SS A182 F316/316L WT 38.0MM									
2010			Flange	5619809	36 - 36	FLG COMPACT 600# RJ SS A182 F316/316L WT 40.0MM									
2020			Flange	5619810	42 - 42	FLG COMPACT 600# RJ SS A182 F316/316L WT 45.0MM									
2030			Flange	5619811	48 - 48	FLG COMPACT 600# RJ SS A182 F316/316L WT 50.0MM									
2040															
2050	D		Flange	5583979	2 - 2	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE									
2060															
2070	C		Blind	5616544	1/2 - 24	FLG BLIND 600# RF SS A182 F316/316L									
2080			Blind	5616531	30 - 48	FLG BLIND 600# RF SS A182 F316/316L B16.47 SERIES A									
2090			Blind	5619715	26 - 48	FLG COMPACT BLIND 600# RJ SS A182 F316/316L PIPE CLASS 600P21									
2100															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY MC	REV D	FGP/WPMP SHEET 4 OF 5			PA 5	CLASS 600P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2110						***** ORIFICE FLANGES *****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN - ASME B16.5 MG 2.2 CODE - ASME B31.3 AA, BD, BF, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 600 RF TEMPERATURE RANGE: -120 / 180 °C COMMENT: LOW TEMP BLOW DOWN					
2120													
2130			Orific	5640600	2 - 8	WN ORIF SET 600# RF SS A182 F316/316L SCH 80S BORE							
2140			Orific	5640601	10 - 24	WN ORIF SET 600# RF SS A182 F316/316L SCH 80 BORE							
2150													
2160													
2170						***** PIPE NIPPLES *****							
2180													
2190						NIPPLE PBE CUT FROM PIPE							
2200													
2210						***** GASKETS *****							
2220													
2230													
2240	D		GASFRS	5673103	1/2 - 24	GASKET 600# 316SS SP WND GRAPHITE IR							
2250			GASFRS	5674012	26 - 48	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2260													
2270	D	815	GASRJ	5673420	26 - 48	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 600P21							
2280													
2290	D	815	GASFRS	5672251	2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
2300													
2310		B1				***** BOLTING *****							
2320													
2330	D	BA1	BOLS BOLS	5675776	1/4 - 24	STUD-BOLT WITH 2 NUTS API-6A718							
2340	D	BA5		5675778	26 - 99	STUD-BOLT/COMPACT NUT API-6A718							
2350	D	BA1	BOLS	5675776	26 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
2360													
2370	D	SB6	CPSBOL	5675676	10 - 48	JACK SCREW HEX HEAD A320 B8M - CLASS 2							
2380													
2390						***** BLINDS AND SPACERS *****							
2400													
2410			Specta	5750423	3/4 - 8	SPECTACLE BLIND 600# RF SS A240 GR 316							
2420													
2430			Paddle	5750523	10 - 24	PADDLE BLANK 600# RF SS A240 GR 316							
2440			Paddle	5750528	30 - 48	PADDLE BLANK 600# RF SS A240 GR 316 B16.47 SERIES A							
2450													
2460			Paddle	5750623	10 - 24	PADDLE SPACER 600# RF SS A240 GR 316							
2470			Paddle	5750628	30 - 48	PADDLE SPACER 600# RF SS A240 GR 316 B16.47 SERIES A							
2480													
2490						***** BRANCHES *****							
2500													
2510						USE BRANCH CHART 600P21							
2520													
2530													
2540	D		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 48						
2550	D		Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 48						
2560													
2570						***** CONSTRUCTION DETAILS *****							
2580													
2590						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
2600						PIPE BENDS ----- NONE							
2610						THREAD SEALANT ----- PTFE PASTE							
2620													
2630													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY MC	REV D	SHEET 5 OF 5	PA 5	CLASS 600P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
490	D		Butter	T06AS61	10 - 24	*****BUTTERFLY VALVES*****		VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.				
500						VC1		SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
510						VC2		DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
520								FOR USE IN MAKE-UP WATER ONLY					
530								Globe	L06AS48	3/4 - 1 1/2	GLOBE 600# RF SS A182 F316 TRIM 16 OS&Y HW		
540													
550													
560													
570						Globe		L06AS41	2 - 4	GLOBE 600# RF SS A351 CF8M TRIM 16 OS&Y HW			
580													
590													
600													
610	Globe	L06AS42	6 - 10	GLOBE 600# RF SS A351 CF8M TRIM 16 OS&Y GO									
620													
630													
640													
650	VC1	Check	C06AS50	3/4 - 1 1/2	CHECK PISTON 600# RF SS A182 F316 TRIM 16								
660	VC2	Check	C06AS45		2 - 8	CHECK DUAL WAFER LUG 600# RF SS A351 CF8M TRIM 16 THROU BOLTED							
670	D	VC2	Check	C06AS52	10 - 24	CHECK DUAL DBL FLG 600# RF SS A351 CF8M TRIM 16							
680		VC2	Check	C06CS52	30 - 30	CHECK DUAL 600# CF SS A351 CF8M TRIM 16							
690	D	VGA	Gate V	G06AS41	2 - 6	*****GATE VALVES*****							
700													
710													
720													
730													
740													
750													
760													
770													
780						C	Piping	5365140	3/4 - 1 1/2	PIPE SCH 80S SMLS SS A312 TP316/316L PE			
790	Piping	5365138	2 - 8	PIPE SCH 80S SMLS SS A312 TP316/316L-S BE									
800	Piping	5365139	10 - 24	PIPE SCH 80 SMLS SS A312 TP316/316L BE									
810	Piping	5365143	26 - 26	PIPE WT 33.0MM EFW SS A358 TP316/316L BE									
820	Piping	5365200	30 - 30	PIPE WT 38.0MM EFW SS A358 316/316L BE									
830	Piping	5365212	36 - 36	PIPE WT 40.0MM EFW SS A358 316/316L BE									
840	Piping	5365214	42 - 42	PIPE WT 45.0MM EFW SS A358 316/316L BE									
850	Piping	5365213	48 - 48	PIPE WT 50.0MM EFW SS A358 316/316L BE									
860													
870	D	814	Piping	5365207	1/2 - 1/2					PIPE SCH 160 SMLS SS A312 TP316/316L PE			
880	D	814	Piping	5365207	3/4 - 3/4	PIPE SCH 160 SMLS SS A312 TP316/316L PE							
890	C		90 Deg	5449033	3/4 - 8	***** FITTINGS *****							
900													
910													
920													
930													
940													
950													
960													
970													
980													
990	C	45 Deg	5449146	3/4 - 8	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S								
1000					45 Deg	5449145	10 - 24	ELL 90 DEG LR SCH 80 SS A403 WP316/316L-S					
1010					45 Deg	5451110	30 - 30	ELL 90 LR WT 38.0MM 316/316L SS A403 WP316/316L-W					
1020					45 Deg	5451102	36 - 36	ELL 90 LR WT 40.0MM 316/316L SS A403 WP316/316L-W					
1030					45 Deg	5451103	48 - 48	ELL 90 LR WT 50.0MM 316/316L SS A403 WP316/316L-W					
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	FGP/WPMP DATE 17MAR17 BY KD REV D SHEET 2 OF 5 PA 5 CLASS 600P22					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040	C		Tee	5449156	3/4 - 8	TEE SCH 80S SS A403 WP316/316L-S							
1050			Tee	5449157	10 - 24	TEE SCH 80 SS A403 WP316/316L-S							
1060			Tee	5451113	30 - 30	TEE 38.0MM SS A403 WP316/316L-W							
1070			Tee	5451112	36 - 36	TEE 40.0MM SS A403 WP316/316L-W							
1080			Tee	5451115	48 - 48	TEE 50.0MM SS A403 WP316/316L-W							
1090													
1100	C		Reduci	5448905	3/4 - 6	TEE RED SCH 80S SS A403 WP316/316L-S	1 - 8						
1110			Reduci	5448906	4 - 8	TEE RED SCH 80/80S SS A403 WP316/316L-S	10 - 20						
1120	C		Reduci	5448907	10 - 20	TEE RED SCH 80 SS A403 WP316/316L-S	12 - 24						
1130			Reduci	5448913	12 - 12	TEE RED WT 33.0MM X SCH 80 SS A403 WP316/316L-S	26 - 26						
1140			Reduci	5448913	20 - 20	TEE RED WT 33.0MM X SCH 80 SS A403 WP316/316L-S	26 - 26						
1150													
1160	C		Cap	5449049	3/4 - 8	CAP SCH 80S SS A403 WP316/316L-S							
1170			Cap	5449050	10 - 24	CAP SCH 80 SS A403 WP316/316L-S							
1180			Cap	5449261	26 - 26	CAP WT 33.0MM SS A403 WP316/316L-S							
1190			Cap	5449260	30 - 30	CAP WT 38.0MM SS A403 WP316/316L-S							
1200			Cap	5449262	36 - 36	CAP WT 40.0MM SS A403 WP316/316L-S							
1210			Cap	5449263	48 - 48	CAP WT 50.0MM SS A403 WP316/316L-S							
1220													
1230	C		Concen	5449158	3/4 - 6	REDUCER CONC SCH 80S SS A403 WP316/316L-S	1 - 8						
1240			Concen	5448904	4 - 8	REDUCER CONC SCH 80/80S SS A403 WP316/316L-S	10 - 20						
1250			Concen	5448908	10 - 20	REDUCER CONC SCH 80 SS A403 WP316/316L-S	12 - 24						
1260			Concen	5450730	20 - 24	REDUCER CONC 38.0MM/SCH 80 SS A403 WP316/316L-W	30 - 30						
1270			Concen	5450749	24 - 24	REDUCER CONC 40.0MM / SCH 80 SS A403 WP316/316L-W	36 - 36						
1280			Concen	5450742	30 - 30	REDUCER CONC 40.0MM/ 38.0MM SS A403 WP316/316L-W	36 - 36						
1290			Concen	5450741	30 - 30	REDUCER CONC 45.0MM/ 38.0MM SS A403 WP316/316L-W	42 - 42						
1300			Concen	5450746	36 - 36	REDUCER CONC 45.0MM/ 40.0MM SS A403 WP316/316L-W	42 - 42						
1310			Concen	5450744	42 - 42	REDUCER CONC 50.0MM/ 45.0MM SS A403 WP316/316L-W	48 - 48						
1320													
1330	C		Eccent	5449159	3/4 - 6	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1 - 8						
1340			Eccent	5449160	4 - 8	REDUCER ECC SCH 80/80S SS A403 WP316/316L-S	10 - 20						
1350			Eccent	5449161	10 - 20	REDUCER ECC SCH 80 SS A403 WP316/316L-S	12 - 24						
1360			Eccent	5450728	20 - 24	REDUCER ECC 38.0MM/SCH 80 SS A403 WP316/316L-W	30 - 30						
1370			Eccent	5450739	24 - 24	REDUCER ECC 40.0MM / SCH 80 SS A403 WP316/316L-W	36 - 36						
1380			Eccent	5450729	30 - 30	REDUCER ECC 40.0MM / 38.0MM SS A403 WP316/316L-W	36 - 36						
1390			Eccent	5450754	36 - 36	REDUCER ECC 45.0MM / 40.0MM SS A403 WP316/316L-W	42 - 42						
1400			Eccent	5450752	42 - 42	REDUCER ECC 50.0MM / 45.0MM SS A403 WP316/316L-W	48 - 48						
1410								SERVICE					
1420								SOUR SERVICE - NACE MR0175/ ISO 15156,					
1430								DESIGN - ASME B16.5 MG 2.2					
1440								CODE - ASME B31.3					
1450	C		Nipofl	5504504	3/4 - 2	NIPOFLANGE 600# RF SCH 80S SS A182 F316/316L	2 - 8						
1460	C		Nipofl	5504505	3/4 - 2	NIPOFLANGE 600# RF SCH 80/80S SS A182 F316/316L	10 - 24						
1470	C		Nipofl	5504416	3/4 - 2	NIPOFLANGE 600# RF WT 33.0MM/SCH 80S SS A182 F316/316L	26 - 26						
1480	C		Nipofl	5504506	3/4 - 2	NIPOFLANGE 600# RF WT 38.0MM/SCH 80S SS A182 F316/316L	30 - 30						
1490	C		Nipofl	5504539	3/4 - 2	NIPOFLANGE 600# RF SCH 40.0MM/80S SS A182 F316/316L	36 - 36						
1500	C		Nipofl	5504541	3/4 - 2	NIPOFLANGE 600# RF SCH 50.0MM/80S SS A182 F316/316L	48 - 48						
1510								MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 350 °C COMMENT:					
1520	C	FF5	Endole	5504613	3/4 - 1 1/2	NIPOFLANGE FLAT BOTTOM 600# RF SCH 80S SS A182 F316/316L							
1530													
1540		UJ2	Nipofl	5504075	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 80S/160 SS A182 F316/316L	4 - 8						
1550		UJ2	Nipofl	5504076	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 80/160 SS A182 F316/316L	10 - 24						
1560	C	UJ2	Nipofl	5504083	1 1/2 - 1 1/2	NIPOFLANGE 600# RF 33.0MM WT/SCH 160 SS A182 F316/316L	26 - 26						
1570		UJ2	Nipofl	5504080	1 1/2 - 1 1/2	NIPOFLANGE 600# RF 38.0MM WT/SCH 160 SS A182 F316/316L	30 - 30						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 1 MM		FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 KD D 3 OF 5 5 600P22			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
1580	C	UJ2	Nipofl	5504081	1 1/2 - 1 1/2	NIPOFLANGE 600# RF 40.0MM WT/SCH 160 SS A182 F316/316L	36 - 36												
1590		UJ2	Nipofl	5504540	1 1/2 - 1 1/2	NIPOFLANGE 600# RF SCH 45.0MM/160 SS A182 F316/316L	42 - 42												
1600		UJ2	Nipofl	5504082	1 1/2 - 1 1/2	NIPOFLANGE 600# RF 50.0MM WT/SCH 160 SS A182 F316/316L	48 - 48												
1610																			
1620	C		Weldol	5509637	3/4 - 3	WELDOLET SCH 80S SS A182 F316/316L	2 - 8												
1630	C		Weldol	5509638	3/4 - 8	WELDOLET SCH 80/80S SS A182 F316/316L	10 - 24												
1640	C		Weldol	5509655	3/4 - 8	WELDOLET 38.0MM WT/ SCH 80S SS A182 F316/316L	30 - 30												
1650	C		Weldol	5509656	3/4 - 8	WELDOLET 40.0MM WT/ SCH 80S SS A182 F316/316L	36 - 36												
1660	C		Weldol	5509657	3/4 - 8	WELDOLET 50.0MM WT/ SCH 80S SS A182 F316/316L	48 - 48												
1670			Weldol	5509660	10 - 12	WELDOLET 38.0MM WT/ SCH 80 SS A182 F316/316L	30 - 30												
1680			Weldol	5509661	10 - 12	WELDOLET 40.0MM WT/ SCH 80 SS A182 F316/316L	36 - 36												
1690			Weldol	5509665	10 - 12	WELDOLET 50.0MM WT/ SCH 80 SS A182 F316/316L	48 - 48												
1700																			
1710		UJ2	Elbole	5516375	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 80S/160 SS A182 F316/316L	4 - 8												
1720		UJ2	Elbole	5516376	1 1/2 - 1 1/2	ELBOFLANGE 600# RF SCH 80 /160 SS A182 F316/316L	10 - 24												
1730	C	UJ2	Elbole	5516377	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 33.0MM WT/160 SS A182 F316/316L	26 - 26												
1740		UJ2	Elbole	5516380	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 38.0MM WT/160 SS A182 F316/316L	30 - 30												
1750		UJ2	Elbole	5516381	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 40.0MM WT/160 SS A182 F316/316L	36 - 36												
1760	C	UJ2	Elbole	5516383	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 45.0MM WT/160 SS A182 F316/316L	42 - 42												
1770		UJ2	Elbole	5516382	1 1/2 - 1 1/2	ELBOFLANGE 600# RF 50.0MM WT/160 SS A182 F316/316L	48 - 48												
1780																			
1790	D		Elbole	5509801	3/4 - 3/4	ELBOLET SCH 80/80S SS A182 F316/316L	16 - 16												
1800			Elbole	5509800	2 - 2	ELBOLET SCH 80S/80S SS A182 F316/316L	8 - 8												
1810																			
1820																			
1830						***** PIPE FLANGES *****													
1840																			
1850	C	816	Reduci	5600660	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 600# RF SS A182 F316/316L	3/4 - 1 1/2												
1860																			
1870	D	FA8	Flange	5851204	1/2 - 2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE													
1880																			
1890	D		Flange	5582998	1/2 - 8	FLG WN 600# RF SS A182 F316/316L SCH 80S BORE													
1900			Flange	5582999	10 - 24	FLG WN 600# RF SS A182 F316/316L SCH 80 BORE													
1910			Flange	5583635	30 - 30	FLG WN 600# RF SS A182 F316/316L 38.0MM WT B16.47 SERIES A													
1920			Flange	5583632	36 - 36	FLG WN 600# RF SS A182 F316/316L 40.0MM WT B16.47 SERIES A													
1930			Flange	5583634	42 - 42	FLG WN 600# RF SS A182 F316/316L 45.0MM WT B16.47 SERIES A													
1940			Flange	5583633	48 - 48	FLG WN 600# RF SS A182 F316/316L 50.0MM WT B16.47 SERIES A													
1950																			
1960	C	FS1	Flange	5583006	3 - 3	FLG WN 900# RF SS A182 F316/316L SCH 80S BORE													
1970	C	FS1	Flange	5583979	2 - 2	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE													
1980																			
1990			Flange	5619808	30 - 30	FLG COMPACT 600# RJ SS A182 F316/316L WT 38.0MM													
2000																			
2010	C		Blind	5616544	1/2 - 24	FLG BLIND 600# RF SS A182 F316/316L													
2020			Blind	5616531	30 - 48	FLG BLIND 600# RF SS A182 F316/316L B16.47 SERIES A													
2030																			
2040			Blind	5619716	30 - 30	FLG COMPACT BLIND 600# RJ SS A182 F316/316L PIPE CLASS 600P22													
2050																			
2060																			
2070																			
2080						***** ORIFICE FLANGES *****													
2090			Orific	5640600	2 - 8	WN ORIF SET 600# RF SS A182 F316/316L SCH 80S BORE													
2100			Orific	5640601	10 - 24	WN ORIF SET 600# RF SS A182 F316/316L SCH 80 BORE													
2110																			
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP											
						ALL SIZES- BUTT WELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS					
								1 MM	17MAR17	KD	D	4 OF 5	5	600P22					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
2120	D	814				***** PIPE NIPPLES *****													
2130						NIPPLE PBE CUT FROM PIPE													
2140																			
2150																			
2160						***** GASKETS *****													
2170																			
2180																			
2190						GASKET 600# 316SS SP WND GRAPHITE IR													
2200						GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A													
2210																			
2220	D	815	GASRJ	5673421	26 - 48	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 600P22													
2230																			
2240	C					GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR													
2250	C					GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR													
2260																			
2270	B1					***** BOLTING *****													
2280																			
2290						D									BA1	BOLS	5675770	1/4 - 24	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS
2300						D									BA5	BOLS	5675778		26 - 99
2310																			
2320	D	BA1	BOLS	5675770	26 - 34	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS													
2330	D					BA1									BOLS	5675776	36 - 99	STUD-BOLT WITH 2 NUTS API-6A718	
2340																			
2350	C	SB6	CPSBOL	5675675	10 - 48	JACK SCREW HEX HEAD A320 GR L7M													
2360																			
2370																			
2380																			
2390																			
2400			Specta	5750423	3/4 - 8	SPECTACLE BLIND 600# RF SS A240 GR 316													
2410																			
2420			Paddle	5750523	10 - 24	PADDLE BLANK 600# RF SS A240 GR 316													
2430			Paddle	5750528	30 - 48	PADDLE BLANK 600# RF SS A240 GR 316 B16.47 SERIES A													
2440																			
2450		Paddle	5750269	30 - 48	PADDLE BLANK COMPACT 600# RJ SS A240 GR 316														
2460																			
2470		Paddle	5750623	10 - 24	PADDLE SPACER 600# RF SS A240 GR 316														
2480		Paddle	5750628	30 - 48	PADDLE SPACER 600# RF SS A240 GR 316 B16.47 SERIES A														
2490																			
2500	C	Paddle		5750258	30 - 48	PADDLE SPACER COMPACT 600# RJ SS A240 GR 316													
2510																			
2520																			
2530																			
2540																			
2550																			
2560																			
2570		D	Reinfo	Reinfo	RWELD	2 - 30	NO-MTO REINFORCING WELD FOR STUB-IN												
2580		D					NO-MTO REINFORCING WELD FOR STUB-IN												
2590																			
2600																			
2610																			
2620																			
2630																			
2640																			
2650																			
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION									FGP/WPMP				
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS						
							1 MM	17MAR17	KD	D	5 OF 5	5	600P22						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10		PP9				*****VALVES*****		810	NIPPLE TO BE CUT FROM SCH 160 PIPE				
20						*****MODULAR VALVES*****		816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's				
30								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.				
40								BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.				
50								BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.				
60	C	VA5	2BV	K15AM10	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV		FF5	FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY.				
70								PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156				
80	C		2BV	K15AM14	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.				
90								UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.				
100	C	VAV	2BV	K09AM14	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF INC 825 INC 625 TRIM METAL LO FULL SIZE TYPE 2BV		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.				
110								VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055				
120	C		2BVR	M15AM10	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	2 - 2	VAV	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.				
130								VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.				
140	C		2BVR	M15AM14	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	3 - 3	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.				
150								VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.				
160	C	VAW	R2BVR	M15EM10	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.				
170													
180	C	VAW	R2BVR	M09EM10	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# RF INC 825 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR							
190													
200													
210		VB2				***** BALL VALVES *****							
220													
230			Ball V	A15AM14	2 - 2	BALL 1500# RF 825 INC 625 TRIM TRUNNION METAL RB LO							
240	C		Ball V	A09AM14	3 - 3	BALL 900# RF 825 INC 625 TRIM TRUNNION METAL RB LO							
250	C		Ball V	A09AM16	4 - 10	BALL 900# RF 825 INC 625 TRIM TRUNNION METAL RB GO							
260	C		Ball V	A09CM16	12 - 30	BALL 900# CF 825 INC 625 TRIM TRUNNION METAL RB GO							
270													
280	C		Ball V	B15AM10	3/4 - 1 1/2	BALL 1500# RF 825 INC 625 INC TRIM FLOATING METAL FB LO							
290		VA5	Ball V	B15AM14	2 - 2	BALL 1500# RF 825 INC 625 INC TRIM TRUNNION METAL FB LO							
300	C												
310	C	VA5	Ball V	B09AM16	4 - 10	BALL 900# RF 825 INC 625 TRIM TRUNNION METAL FB GO							
320	B	VA5	Ball V	B09CM16	12 - 24	BALL 900# CF 825 INC 625 TRIM TRUNNION METAL FB GO							
330													
340													
350													
360						***** GLOBE VALVES *****							
370	B		Globe	L15AM43	3/4 - 2	GLOBE 1500# RF INC 825 INC TRIM 625 OS&Y HW							
380			Globe	L09AM43	3 - 3	GLOBE 900# RF INC 825 INC TRIM 625 OS&Y HW							
390	B		Globe	L09AM44	4 - 8	GLOBE 900# RF INC 825 INC 625 TRIM OS&Y GO							
400													
410													
420						***** CHECK VALVES *****							
430													
440	B	VC1	Check	C15AM51	3/4 - 1 1/2	CHECK PISTON 1500# RF INC 825 TRIM 625							
450		VC2	Check	C15AM46	2 - 2	CHECK DUAL WAFER LUG 1500# RF INC 825 TRIM 625 THROU BOLTED							
460	B	VC2	Check	C09AM46	3 - 8	CHECK DUAL WAFER LUG 900# RF INC 825 TRIM 625 THROU BOLTED							
470	B	VC2	Check	C09AM53	10 - 10	CHECK DUAL DBL FLG 900# RF INC 825 TRIM 625							
480	B	VC2	Check	C09CM53	12 - 24	CHECK DUAL 900# CF INC 825 TRIM 625							
490													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP						
ALL SIZES- BUTT WELD						CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
							1 MM	17MAR17	MLA	C	1 OF 5	5	900A80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
500						***** PIPE *****							
510													
520	B		Piping	5732001	3/4 - 1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE							
530	C		Piping	5732002	1 1/2 - 1 1/2	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 PE							
540	C		Piping	5361603	2 - 4	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 BE							
550	C		Piping	5361607	6 - 8	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 BE							
560	B		Piping	5361635	10 - 10	PIPE SCH 80S WLD ALLOY 825 B705 CL.2 N08825 BE							
570			Piping	5361637	12 - 24	PIPE SCH 60 WLD ALLOY 825 B705 CL.2 N08825 BE							
580	C		Piping	5361641	30 - 30	PIPE NOM. WT 30MM WLD ALLOY 825 B705 CL.2 N08825 BE							
590													
600	C	810	Piping	5361608	3/4 - 1 1/2	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 BE							
610													
620													
630						***** FITTINGS *****							
640													
650	B		90 Deg	5425111	3/4 - 1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825							
660	B		90 Deg	5424722	1 1/2 - 4	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPS UNS N08825							
670	B		90 Deg	5425111	6 - 8	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825							
680	B		90 Deg	5425109	10 - 10	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPW UNS N08825							
690			90 Deg	5425130	12 - 24	ELL 90 DEG LR SCH 60 ALLOY 825 B366-WPW UNS N08825							
700	C		90 Deg	5425147	30 - 30	ELL 90 DEG LR WT 30MM ALLOY 825 B366-WPW UNS N08825							
710													
720			45 Deg	5425113	3/4 - 1	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825							
730			45 Deg	5424727	1 1/2 - 4	ELL 45 DEG SCH 40S ALLOY 825 B366-WPS UNS N08825							
740			45 Deg	5425113	6 - 8	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825							
750			45 Deg	5425115	10 - 10	ELL 45 DEG SCH 80S ALLOY 825 B366-WPW UNS N08825							
760			45 Deg	5425131	12 - 24	ELL 45 DEG SCH 60 ALLOY 825 B366-WPW UNS N08825							
770													
780	B		Tee	5425119	3/4 - 1	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825							
790	B		Tee	5424725	1 1/2 - 4	TEE SCH 40S ALLOY 825 B366-WPS UNS N08825							
800	B		Tee	5425119	6 - 8	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825							
810	B		Tee	5425118	10 - 10	TEE SCH 80S ALLOY 825 B366-WPW UNS N08825							
820			Tee	5425132	12 - 24	TEE SCH 60 ALLOY 825 B366-WPW UNS N08825							
830	C		Tee	5425152	30 - 30	TEE WT 30MM ALLOY 825 B366-WPW UNS N08825							
840													
850	B		Reduc	5425126	3/4 - 3/4	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1 - 1						
860	B		Reduc	5425124	3/4 - 1	TEE RED SCH 40S/80S ALLOY 825 B366-WPS UNS N08825	1 1/2 - 2						
870	B		Reduc	5424724	1 1/2 - 3	TEE RED SCH 40S ALLOY 825 B366-WPS UNS N08825	2 - 4						
880	B		Reduc	5425125	3 - 4	TEE RED SCH 80S/40S ALLOY 825 B366-WPS UNS N08825	6 - 8						
890	B		Reduc	5425126	6 - 6	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	8 - 8						
900	B		Reduc	5425134	3 - 4	TEE RED SCH 80S/40S ALLOY 825 B366-WPW UNS N08825	10 - 10						
910	B		Reduci	5425117	6 - 8	TEE RED SCH 80S ALLOY 825 B366-WPW UNS N08825	10 - 10						
920			Reduci	5425135	6 - 10	TEE RED SCH 60/80S ALLOY 825 B366-WPW UNS N08825	12 - 24						
930			Reduci	5425133	12 - 20	TEE RED SCH 60 ALLOY 825 B366-WPW UNS N08825	14 - 24						
940	C	new	Reduci	5425129	16 - 16	TEE RED 30MM/SCH 60 ALLOY 825 B366-WPW UNS N08825	30 - 30						
950													
960	B		Cap	5424270	3/4 - 1	CAP SCH 80S ALLOY 825 B366 UNS N08825							
970	B		Cap	5424219	1 1/2 - 4	CAP SCH 40S ALLOY 825 B366 UNS N08825							
980			Cap	5424270	6 - 10	CAP SCH 80S ALLOY 825 B366 UNS N08825							
990			Cap	5425136	12 - 24	CAP SCH 60 ALLOY 825 B366 UNS N08825							
1000													
1010													
1020	B		Concen	5425122	3/4 - 3/4	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1						
1030	B		Concen	5425143	3/4 - 1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY MLA	REV C	SHEET 2 OF 5	PA 5	CLASS 900A80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040	B		Concen	5424733	1 1/2 - 3	REDUCER CONC SCH 40S ALLOY 825 B366-WPS N08825	2 - 4						
1050	B		Concen	5425142	3 - 4	REDUCER CONC SCH 80S/40S ALLOY 825 B366-WPS N08825	6 - 8						
1060	B		Concen	5425122	6 - 6	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	8 - 8						
1070	B		Concen	5425100	4 - 4	REDUCER CONC SCH 80S/40S ALLOY 825 B366-WPW N08825	10 - 10						
1080	B		Concen	5425120	6 - 8	REDUCER CONC SCH 80S ALLOY 825 B366-WPW N08825	10 - 10						
1090	D		Concen	5425141	6 - 10	REDUCER CONC SCH 60/80S ALLOY 825 B366-WPW N08825	12 - 18						
1100			Concen	5425137	12 - 18	REDUCER CONC SCH 60 ALLOY 825 B366-WPW N08825	14 - 24						
1110	C		Concen	5424150	20 - 24	REDUCER CONC WT 30MM/SCH 60 ALLOY 825 B366-WPW N08825	30 - 30						
1120													
1130													
1140	B		Ecccent	5425123	3/4 - 3/4	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	1 - 1						
1150	B		Ecccent	5425145	3/4 - 1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPS N08825	1 1/2 - 2						
1160	B		Ecccent	5424734	1 1/2 - 3	REDUCER ECC SCH 40S ALLOY 825 B366-WPS N08825	2 - 4						
1170	B		Ecccent	5425101	3 - 4	REDUCER ECC SCH 80S/40S ALLOY 825 B366-WPS N08825	6 - 8						
1180	B		Ecccent	5425121	6 - 6	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	8 - 8						
1190	B		Ecccent	5425139	4 - 4	REDUCER ECC SCH 80S/40S ALLOY 825 B366-WPW N08825	10 - 10						
1200	B		Ecccent	5424732	6 - 8	REDUCER ECC SCH 80S ALLOY 825 B366-WPW N08825	10 - 10						
1210			Ecccent	5425140	6 - 10	REDUCER ECC SCH 60/80S ALLOY 825 B366-WPW N08825	12 - 18						
1220			Ecccent	5425138	12 - 20	REDUCER ECC SCH 60 ALLOY 825 B366-WPW N08825	14 - 24						
1230													
1240													
1250						***** OLETS *****							
1260													
1270	B		Nipofl	5513581	3/4 - 1	NIPOFLANGE 1500# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2 - 4						
1280	B		Nipofl	5514631	3/4 - 1	NIPOFLANGE 1500# RF SCH 80S ALLOY 825 B564 UNS N08825	6 - 10						
1290	B		Nipofl	5514641	3/4 - 1	NIPOFLANGE 1500# RF SCH 60/80S ALLOY 825 B564 UNS N08825	12 - 24						
1300													
1310	B		Nipofl	5513680	1 1/2 - 2	NIPOFLANGE 1500# RF SCH 40S ALLOY 825 B564 UNS N08825	2 - 4						
1320	B		Nipofl	5514630	1 1/2 - 2	NIPOFLANGE 1500# RF SCH 80S/40S ALLOY 825 B564 UNS N08825	6 - 10						
1330	B		Nipofl	5514640	1 1/2 - 2	NIPOFLANGE 1500# RF SCH 60/40S ALLOY 825 B564 UNS N08825	12 - 24						
1340													
1350	B	UJ2	Nipofl	5516680	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 40S/160 ALLOY 825 B564 UNS N08825	2 - 4						
1360		UJ2	Nipofl	5516630	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 80S/160 ALLOY 825 B564 UNS N08825	6 - 10						
1370		UJ2	Nipofl	5516640	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 60/160 ALLOY 825 B564 UNS N08825	12 - 24						
1380													
1390													
1400	B	FF5	Endole	5504591	3/4 - 1	NIPOFLANGE FLAT BOTTOM 1500# RF SCH 80S ALLOY 825 B564 UNS N08825	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN PRESSURE: 103.4 BARG, CODE: ASME B31.3 GS, PHC, WSR						
1410	B	FF5	Endole	5504590	1 1/2 - 1 1/2	NIPOFLANGE FLAT BOTTOM 1500# RF SCH 40S ALLOY 825 B564 UNS N08825							
1420													
1430													
1440	B		Weldol	5513413	3/4 - 1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	3 - 4	MATERIAL: ALLOY 825 (NACE) FLANGE RATING: 900 TEMPERATURE RANGE: -46 / 300 °C					
1450	B		Weldol	5513414	3/4 - 1	WELDOLET SCH 80S ALLOY 825 B564 UNS N08825	6 - 10						
1460	B		Weldol	5513433	3/4 - 1	WELDOLET SCH 60/80S ALLOY 825 B564 UNS N08825	12 - 24						
1470	B		Weldol	5513412	1 1/2 - 4	WELDOLET SCH 80S/40S ALLOY 825 B564 UNS N08825	6 - 10						
1480	B		Weldol	5513432	1 1/2 - 4	WELDOLET SCH 60/40S ALLOY 825 B564 UNS N08825	12 - 24						
1490	B		Weldol	5513433	6 - 8	WELDOLET SCH 60/80S ALLOY 825 B564 UNS N08825	18 - 24						
1500	C		Weldol	5513447	3 - 3	WELDOLET WT 30MM/40S ALLOY 825 B564 UNS N08825	30 - 30						
1510													
1520	B	UJ2	Elbole	5516178	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 40S/160 ALLOY 825 B564 UNS N08825	4 - 4						
1530													
1540	B	UJ2	Elbole	5516179	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 80S/160 ALLOY 825 B564 UNS N08825	6 - 10						
1550													
1560	B	UJ2	Elbole	5516180	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 60/160 ALLOY 825 B564 UNS N08825	12 - 24						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP						
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							1 MM	17MAR17	MLA	C	3 OF 5	5	900A80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1570	B		Reduci	5614400	1/2 - 1/2	***** FLANGES *****	3/4 - 3/4							
1580						FLG SCRDR (FNPT) REDUCING 1500# RF ALLOY 825 B564 UNS N08825								
1590						FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
1600						FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
1610	B		Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
1620	C		Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
1630	B		Flange	5570286	3/4 - 1	FLG WN 1500# RF ALLOY 825 SCH 80S B564 UNS N08825								
1640			Flange	5570276	1 1/2 - 2	FLG WN 1500# RF ALLOY 825 SCH 40S B564 UNS N08825								
1650			Flange	5557400	3 - 4	FLG WN 900# RF ALLOY 825 SCH 40S B564 UNS N08825								
1660			Flange	5557401	6 - 10	FLG WN 900# RF ALLOY 825 SCH 80S B564 UNS N08825								
1670			Flange	5557401	6 - 10	FLG WN 900# RF ALLOY 825 SCH 80S B564 UNS N08825								
1680			Flange	5557402	12 - 24	FLG WN 900# RF ALLOY 825 SCH 60 B564 UNS N08825								
1690			Flange	5557402	12 - 24	FLG WN 900# RF ALLOY 825 SCH 60 B564 UNS N08825								
1700			Flange	5557402	12 - 24	FLG WN 900# RF ALLOY 825 SCH 60 B564 UNS N08825								
1710	C	816	Flange	5619820	12 - 24	FLG COMPACT 900# RJ ALLOY 825 SCH 60 B564 UNS N08825								
1720		816	Flange	5619903	30 - 30	FLG COMPACT 900# RJ ALLOY 825 WT 30MM B564 UNS N08825								
1730		816	Flange	5619903	30 - 30	FLG COMPACT 900# RJ ALLOY 825 WT 30MM B564 UNS N08825								
1740		816	Flange	5619903	30 - 30	FLG COMPACT 900# RJ ALLOY 825 WT 30MM B564 UNS N08825								
1750	B		Blind	5617185	1/2 - 2	FLG BLIND 1500# RF ALLOY 825 B564 UNS N08825								
1760			Blind	5617190	3 - 10	FLG BLIND 900# RF ALLOY 825 B564 UNS N08825								
1770			Blind	5617190	3 - 10	FLG BLIND 900# RF ALLOY 825 B564 UNS N08825								
1780			Blind	5619725	12 - 30	FLG COMPACT BLIND 900# RJ ALLOY 825 B564 UNS N08825								
1790			Blind	5617190	12 - 24	PIPE CLASS 900A80								
1800			Blind	5617190	12 - 24	FLG BLIND 900# RF ALLOY 825 B564 UNS N08825								
1810			Blind	5617190	12 - 24	FLG BLIND 900# RF ALLOY 825 B564 UNS N08825								
1820			Blind	5617190	12 - 24	FLG BLIND 900# RF ALLOY 825 B564 UNS N08825								
1830			Orific	5651000	2 - 2	WN ORIF SET 1500# RF ALLOY 825 B564 UNS N08825 SCH 40S								
1840			Orific	5650940	3 - 4	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 40S								
1850			Orific	5650941	6 - 10	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 80S								
1860			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1870			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1880			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1890			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1900			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1910			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1920			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1930			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1940			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1950			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1960			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1970			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1980			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
1990			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2000			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2010			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2020			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2030			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2040			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2050			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2060			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2070			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2080			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2090			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2100			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2110			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2120			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2130			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2140			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2150			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2160			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2170			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2180			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2190			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2200			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2210			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2220			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2230			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2240			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2250			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2260			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								
2270			Orific	5650942	12 - 24	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 60								

			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION			FGP/WPMP								
						ALL SIZES- BUTT WELD			CORR. ALLOW.		DATE		BY	REV	SHEET		PA
						1 MM			17MAR17		MLA	C	4 OF 5		5	900A80	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
2120	C B B B C C	ZA6	Specta	5750300	3/4 - 2	***** BLINDS & SPACERS ***** SPECTACLE BLIND 1500# RF ALLOY 825 B424 UNS N08825 SPECTACLE BLIND 900# RF ALLOY 825 B424 UNS N08825	3 - 30 3 - 30	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN PRESSURE: 103.4 BARG, CODE: ASME B31.3 GS, PHC, WSR MATERIAL: ALLOY 825 (NACE) FLANGE RATING: 900 TEMPERATURE RANGE: -46 / 300 °C							
2130				5750400	3 - 8										
2140			Paddle	5750540	10 - 10	PADDLE BLANK 900# RF ALLOY 825 B424 UNS N08825 PADDLE SPACER 900# RF ALLOY 825 B424 UNS N08825									
2150				5750640	10 - 10										
2160			Paddle	5750540	12 - 24	PADDLE BLANK 900# RF ALLOY 825 B424 UNS N08825 PADDLE SPACER 900# RF ALLOY 825 B424 UNS N08825									
2170				5750640	12 - 24										
2180			Paddle	5750266	12 - 24	PADDLE BLANK COMPACT 900# RJ ALLOY 825 B424 UNS N08825 PADDLE SPACER COMPACT 900# RJ ALLOY 825 B424 UNS N08825									
2190				5750267	12 - 24										
2200			Paddle	***** BRANCHES ***** USE BRANCH CHART 900A80	RWELD	2 - 20									NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN
2210						2 - 20									
2220			***** CONSTRUCTION DETAILS ***** SEAL WELDING-----FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT-----PTFE PASTE												
2230															
2240															
2250															
2260															
2270															
2280															
2290															
2300															
2310															
2320															
2330															
2340															
2350															
2360															
2370															
2380															
2390															
2400															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP								
						ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY MLA	REV C	SHEET 5 OF 5	PA 5	CLASS 900A80		

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
500	C	VC2	Check	C15AM46	2 - 2	CHECK DUAL WAFER LUG 1500# RF INC 825 TRIM 625 THROU BOLTED		VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
510		VC2	Check	C09AM46	3 - 3	CHECK DUAL WAFER LUG 900# RF INC 825 TRIM 625 THROU BOLTED		VC2						
520		VC2	Check	C09AR46	4 - 8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC + 825 CLAD TRIM		VQ2						
530		VC2	Check	C09AR53	10 - 10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC + 825 CLAD TRIM 625		ZA6						
540		VC2	Check	C09CR53	12 - 24	CHECK DUAL 900# CF LTCS A352 LCC + 825 CLAD INC 625 TRIM								
550														
560														
570														
580														
590						*****PIPES*****								
600														
610														
620			Piping	5732001	3/4 - 1 1/2	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE								
630			Piping	5361607	2 - 3	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 BE								
640			Piping	5358510_C80	4 - 6	PIPE SCH 120 SMLS LTCS A333 GR.6 BE WITH 3MM INC 825 INTERNAL CLADDING								
650			Piping	5358509_C80	8 - 16	PIPE SCH 100 SMLS LTCS A333 GR.6 BE WITH 3MM INC 825 INTERNAL CLADDING								
660			Piping	5358521_C80	18 - 24	PIPE SCH 100 EFW LTCS A671 CC60 CL.22 BE WITH 3MM INC 825 INTERNAL CLADDING								
670			Piping	5358635_C80	30 - 30	PIPE WT 48.0MM EFW LTCS A671 CC60 CL.22 BE WITH 3 MM INC 825 INTERNAL CLADDING								
680			Piping	5358533_C80	36 - 36	PIPE WT 57.0MM EFW LTCS A671 CC60 CL.22 BE WITH 3 MM INC 825 INTERNAL CLADDING								
690														
700		810	Piping	5732003	3/4 - 3/4	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE								
710														
720														
730						***** FITTINGS *****								
740														
750			90 Deg	5425111	3/4 - 3	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825								
760			90 Deg	5392700_C80	4 - 6	ELL 90 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING								
770			90 Deg	5392654_C80	8 - 16	ELL 90 DEG LR SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING								
780			90 Deg	5392605_C80	18 - 24	ELL BW 90 DEG LR SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING								
790			90 Deg	5392563_C80	30 - 30	ELL 90 DEG LR WT 48.0MM LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING								
800			90 Deg	5392564_C80	36 - 36	ELL 90 DEG LR WT 57.0MM LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING								
810														
820			45 Deg	5425113	3/4 - 3	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825								
830			45 Deg	5392701_C80	4 - 6	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING								
840			45 Deg	5392660_C80	8 - 16	ELL 45 DEG LR SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING								
850			45 Deg	5392604_C80	18 - 24	ELL 45 DEG LR SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING								
840			45 Deg	5392638_C80	36 - 36	ELL 45 DEG LR WT 57.0MM LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING								
850														
860			Tee	5425119	3/4 - 3	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELD PWHT FOR SIZE 4" AND LARGER	CORR. ALLOW. 1 MM	DATE 17MAR17						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
870			Tee	5392702_C80	4 - 6	TEE BW SCH 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING							
880			Tee	5392663_C80	8 - 16	TEE SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING							
890			Tee	5392602_C80	18 - 24	TEE SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING							
900			Tee	5392639_C80	36 - 36	TEE BW WT 57.0MM LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING							
910													
920			Reduci	5425126	3/4 - 2	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1 - 3						
930			Reduci	5392620	3 - 3	TEE RED SCH 120/80S ALLOY 825 B366-WPS UNS N08825	4 - 6						
940			Reduci	5392622_C80	4 - 6	TEE RED SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	8 - 16						
950			Reduci	5392662_C80	8 - 14	TEE RED SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	10 - 16						
960			Reduci	5392621_C80	8 - 20	TEE RED SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	18 - 24						
970	B	NT6	Reduci	5392836_C80	14 - 14	TEE RED WT 57.0MM/SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	36 - 36						
980													
990			Cap	5424270	3/4 - 3	CAP SCH 80S ALLOY 825 B366 UNS N08825							
1000			Cap	5392703_C80	4 - 6	CAP SCH 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING							
1010			Cap	5392664_C80	8 - 24	CAP SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING							
1020													
1030			Concen	5425122	3/4 - 2	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1 - 3	DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 DHC, DSL,DL, FM, FGH, GHC, GRC, GS, NH, PHC, SG, SH, TH, TM, WBH MATERIAL: CARBON STEEL / CRA FLANGE RATING: 900 RF TEMPERATURE RANGE: -46 / 200 °C					
1040			Concen	5392720	3 - 3	REDUCER CONC SCH 120/80S ALLOY 825 B366-WPS N08825	4 - 6						
1050			Concen	5392722_C80	4 - 6	REDUCER CONC SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	8 - 14						
1060			Concen	5392665_C80	8 - 14	REDUCER CONC SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	10 - 16						
1070			Concen	5392721_C80	10 - 20	REDUCER CONC SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	18 - 24						
1080													
1090			Eccent	5425123	3/4 - 2	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	1 - 3						
1100			Eccent	5392723	3 - 3	REDUCER ECC SCH 120/80S ALLOY 825 B366-WPS N08825	4 - 6						
1110			Eccent	5392725_C80	4 - 6	REDUCER ECC SCH 100/ 120 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	8 - 14						
1120			Eccent	5392666_C80	8 - 14	REDUCER ECC SCH 100 LTCS A420 WPL6 WITH 3MM INC 825 INTERNAL CLADDING	10 - 16						
1130			Eccent	5392724_C80	10 - 20	REDUCER ECC SCH 100 LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	18 - 24						
1140			Eccent	5392213_C80	30 - 30	REDUCER ECC WT 57.0MM/48.0MM LTCS A420 WPL6-W WITH 3MM INC 825 INTERNAL CLADDING	36 - 36						
1150													
1160			Nipofl	5514631	3/4 - 1	NIPOFLANGE 1500# RF SCH 80S ALLOY 825 B564 UNS N08825	2 - 3						
1170			Nipofl	5513421	3/4 - 2	NIPOFLANGE 1500# RF SCH 120/80S ALLOY 825 B564 UNS N08825	4 - 6						
1180			Nipofl	5513422	3/4 - 2	NIPOFLANGE 1500# RF SCH 100/80S ALLOY 825 B564 UNS N08825	8 - 24						
1190			Nipofl	5516678	3/4 - 2	NIPOFLANGE 1500# RF WT 48.0MM /80S ALLOY 825 B564 UNS N08825	30 - 30						
1200			Nipofl	5516679	2 - 2	NIPOFLANGE 1500# RF WT 57.0MM /80S ALLOY 825 B564 UNS N08825	36 - 36						
1210													
1220		UJ2	Nipofl	5513423	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 120/160 ALLOY 825 B564 UNS N08825	4 - 6	FGP/WPMP CORR. ALLOW. 1 MM DATE 17MAR17 BY ET REV C SHEET 3 OF 5 PA 5 CLASS 900C80					
1230		UJ2	Nipofl	5513424	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 100/160 ALLOY 825 B564 UNS N08825	8 - 24						
1240													
1250													
1260		FF5	Endole	5504591	3/4 - 1 1/2	NIPOFLANGE FLAT BOTTOM 1500# RF SCH 80S ALLOY 825 B564 UNS N08825							
1270													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELD PWHT FOR SIZE 4" AND LARGER	CORR. ALLOW. 1 MM	DATE 17MAR17	BY ET	REV C	SHEET 3 OF 5	PA 5	CLASS 900C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
1280	B		Weldol	5513414	3/4 - 1	WELDOLET SCH 80S ALLOY 825 B564 UNS N08825	2 - 3	DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 DHC, DSL,DL, FM, FGH, GHC, GRC, GS, NH, PHC, SG, SH, TH, TM, WBH											
1290			Weldol	5509620	3/4 - 3	WELDOLET SCH 120 /80S ALLOY 825 B564 UNS N08825	4 - 6												
1300			Weldol	5509621	3/4 - 3	WELDOLET SCH 100 /80S ALLOY 825 B564 UNS N08825	8 - 24												
1310			Weldol	5505645	3/4 - 3	WELDOLET WT 48.0MM/SCH 80S ALLOY 825 B564 UNS N08825	30 - 30												
1320			Weldol	5509622_C80	4 - 6	WELDOLET SCH 100/120 LTCS A350 LF2 WITH 3MM INC 825 INTERNAL CLADDING	18 - 24												
1330			Weldol	5505344_C80	8 - 8	WELDOLET SCH 100 LTCS A350 LF2 WITH 3MM INC 825 INTERNAL CLADDING	18 - 24												
1340			UJ2	Elbole	5516020	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 120/160 ALLOY 825 B564 UNS N08825							4 - 6					
1350																			
1360			UJ2	Elbole	5516022	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 100/160 ALLOY 825 B564 UNS N08825							8 - 24					
1370																			
1380			816				***** PIPE FLANGES *****												
1390																			
1400			FA8	Flange	5614400	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 1500# RF ALLOY 825 B564 UNS N08825							3/4 - 1 1/2					
1410																			
1420			Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	2 - 2												
1430																			
1440			Flange	5570286	3/4 - 2	FLG WN 1500# RF ALLOY 825 SCH 80S B564 UNS N08825													
1450																			
1460			Flange	5557401	3 - 3	FLG WN 900# RF ALLOY 825 SCH 80S B564 UNS N08825													
1470																			
1480			Flange	5557143_C80	4 - 6	FLG WN 900# RF LTCS A350 LF2 SCH 120 WITH 3MM INC 825 CLAD FACE & BORE													
1490																			
1500			Flange	5557142_C80	8 - 10	FLG WN 900# RF LTCS A350 LF2 SCH 100 WITH 3MM INC 825 CLAD FACE & BORE													
1510																			
1510			Flange	5557142_C80	12 - 24	FLG WN 900# RF LTCS A350 LF2 SCH 100 WITH 3MM INC 825 CLAD FACE & BORE													
1520																			
1530			Flange	5619887_C80	12 - 16	FLG COMPACT 900# RJ LTCS A350 LF2 SCH 100 WITH 3MM INC 825 CLAD FACE & BORE													
1540																			
1540			Flange	5619888_C80	30 - 30	FLG COMPACT 900# RJ LTCS A350 LF2 WT 48.0MM WITH 3MM INC 825 CLAD FACE & BORE													
1550																			
1550			Flange	5619889_C80	36 - 36	FLG COMPACT 900# RJ LTCS A350 LF2 WT 57.0MM WITH 3MM INC 825 CLAD FACE & BORE													
1560																			
1570			Blind	5617185	1/2 - 2	FLG BLIND 1500# RF ALLOY 825 B564 UNS N08825													
1580																			
1580			Blind	5617190	3 - 3	FLG BLIND 900# RF ALLOY 825 B564 UNS N08825													
1590																			
1590			Blind	5615605_C80	4 - 10	FLG BLIND 900# RF LTCS A350 LF2 WITH 3 MM INC 825 CLAD FACE													
1600																			
1600			Blind	5615605_C80	12 - 24	FLG BLIND 900# RF LTCS A350 LF2 WITH 3 MM INC 825 CLAD FACE													
1610																			
1610			Blind	5619726	12 - 16	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 WITH 3 MM INC 825 CLAD FACE PIPE CLASS 900C80													
1620																			
1620			Blind	5619726	30 - 36	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 WITH 3 MM INC 825 CLAD FACE PIPE CLASS 900C80													
1630																			
1640						***** ORIFICE FLANGES *****													
1650																			
1660			Orific	5651001	2 - 2	WN ORIF SET 1500# RF ALLOY 825 B564 UNS N08825 SCH 80S													
1670																			
1680	Orific	5650941	3 - 3	WN ORIF SET 900# RF ALLOY 825 B564 UNS N08825 SCH 80S															
1690																			
1690	Orific	5624532_C80	4 - 6	WN ORIF SET 900# RF LTCS A350 SCH 120 WITH 3MM INC 825 CLAD FACE & BORE															
1700																			
1700	Orific	5624531_C80	8 - 24	WN ORIF SET 900# RF LTCS A350 SCH 100 WITH 3MM INC 825 CLAD FACE & BORE															
1710																			
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP											
						ALL SIZES BUTT-WELD PWHT FOR SIZE 4" AND LARGER		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS					
								1 MM	17MAR17	ET	C	4 OF 5	5	900C80					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1720	B	810	GASFRS	5673800	1/2 - 2	***** PIPE NIPPLES *****	3 - 36 3 - 36						
1730						NIPPLE PBE TO BE CUT FROM PIPE							
1740													
1750						***** GASKETS *****							
1760													
1770													
1780		815	GASFRS	5673700	3 - 24	GASKET 1500# ALLOY 825 SPIRAL WOUND GRAPHITE IR							
1790			GASFRS			GASKET 900# ALLOY 825 SPIRAL WOUND GRAPHITE IR							
1800													
1810			GASRJ	5673431	12 - 36	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE							
1820						PIPE CLASS 900C80							
1830		B1											
1840						***** BOLTING *****							
1850													
1860		B	BOLS	5675770	1/4 - 20	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1870		B	BA5	5675770	24 - 30	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1880		B	BA5	5675776	36 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
1890		B	BA1	5675776	24 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
1900													
1910		SB6	CPSBOL	5675675	10 - 24	JACK SCREW HEX HEAD A320 GR L7M							
1920													
1930		SB5				***** BLINDS & SPACERS *****							
1940													
1950													
1960			Specta	5750300	3/4 - 2	SPECTACLE BLIND 1500# RF ALLOY 825 B424 UNS N08825							
1970			Specta	5750400	3 - 3	SPECTACLE BLIND 900# RF ALLOY 825 B424 UNS N08825							
1980			Specta	5749596_C80	4 - 8	SPECTACLE BLIND 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE							
1990													
2000			Paddle	5750102_C80	10 - 10	PADDLE BLANK 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE							
2010			Paddle	5750202_C80	10 - 10	PADDLE SPACER 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE							
2020													
2030			Paddle	5750102_C80	12 - 24	PADDLE BLANK 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE							
2040			Paddle	5750202_C80	12 - 24	PADDLE SPACER 900# RF LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE							
2050													
2060			Paddle	5750264_C80	12 - 24	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE							
2070			Paddle	5750265_C80	12 - 24	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70 WITH 3MM INC 825 CLAD FACE & BORE							
2080													
2090													
2100						***** BRANCHES *****							
2110													
2120						USE BRANCH CHART 900C80							
2130													
2140		B	Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN							
2150			Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN							
2160													
2170													
2180					***** CONSTRUCTION DETAILS *****								
2190													
2200	ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY								
2210					PIPE BENDS ----- NONE								
2220					THREAD SEALANT ----- PTFE PASTE								
2230													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	CORR. ALLOW. 1 MM						DATE 17MAR17

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
10						*****VALVES*****		810	NIPPLE TO BE CUT FROM SCH 160 PIPE COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.										
20						*****MODULAR VALVES*****		816											
30																			
40																			
50																			
60	B	VA5	2BV	K15AX35	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF SUPER DUPLEX F55 METAL LO - FULL SIZE TYPE 2BV		B1											
70								FA8											
80	B		2BV	K15AX33	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF SUPER DUPLEX F55 METAL LO - FULL SIZE TYPE 2BV		FS1											
90								SB6											
100	B	VAV	2BV	K09AX33	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF SUPER DUPLEX F55 METAL LO FULL SIZE TYPE 2BV		VA5											
110								VAV											
120	B		2BVR	M15AX35	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF SUPER DUPLEX F55 METAL LO - CONTROLLED BORE TYPE 2BVR		VAW											
130								VB2											
140	B		2BVR	M15AX33	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF SUPER DUPLEX F55 METAL LO - CONTROLLED BORE TYPE 2BVR		VC1											
150																			
160	B	VAW	R2BVR	M15EX35	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF SUPER DUPLEX F55 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	VC2											
170																			
180	B	VAW	R2BVR	M09EX35	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# RF SUPER DUPLEX F55 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	ZA6											
190																			
200																			
210						*****BALL VALVES*****													
220		VB2																	
230			Ball V	A15AX36	2 - 2	BALL 1500# RF SUPER DUPLEX F55 TRUNNION RB SOFT LO													
240	B		Ball V	A09AX36	3 - 3	BALL 900# RF SUPER DUPLEX F55 TRUNNION RB SOFT LO													
250	B		Ball V	A09AX37	4 - 10	BALL 900# RF SUPER DUPLEX F55 TRUNNION RB SOFT GO													
260																			
270			Ball V	B15AX17	3/4 - 1 1/2	BALL 1500# RF SUPER DUPLEX F55 FLOATING FB SOFT LO													
280	B	VA5	Ball V	B15AX36	2 - 3	BALL 1500# RF SUPER DUPLEX F55 TRUNNION FB SOFT LO													
290																			
300	C	VA5	Ball V	B09AX37	4 - 10	BALL 900# RF SUPER DUPLEX F55 TRUNNION FB SOFT GO													
310																			
320																			
330						*****GLOBE VALVES*****													
340																			
350			Globe	L15AX84	3/4 - 1 1/2	GLOBE 1500# RF SUPER DUPLEX F55 OS&Y HW													
360			Globe	L15AX85	2 - 2	GLOBE 1500# RF SUPER DUPLEX A6 OS&Y HW													
370			Globe	L09AX85	3 - 3	GLOBE 900# RF SUPER DUPLEX A6 OS&Y HW													
380			Globe	L09AX83	4 - 8	GLOBE 900# RF SUPER DUPLEX A6 OS&Y GO													
390																			
400																			
410						*****CHECK VALVES*****													
420																			
430		VC1	Check	C15AX86	3/4 - 1 1/2	CHECK PISTON 1500# RF SUPER DUPLEX F55		MATERIAL:								SUPER DUPLEX FLANGE			
440								FLANGE RATING:								900 RF			
450		VC2	Check	C15AX87	2 - 2	CHECK DUAL WAFER LUG 1500# RF SUPER DUPLEX A6 THROU BOLTED		TEMPERATURE RANGE:								-46 / 100 °C			
460																			
470		VC2	Check	C09AX87	3 - 8	CHECK DUAL WAFER LUG 900# RF SUPER DUPLEX A6 THROU BOLTED													
480																			
490		VC2	Check	C09AX88	10 - 10	CHECK DUAL DBL FLG 900# RF SUPER DUPLEX A6													
500																			
510		VC2	Check	C09CX88	12 - 16	CHECK DUAL DBL FLG 900# CF SUPER DUPLEX A6													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP											
						ALL SIZES BUTT-WELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS					
								1 MM	17MAR17	JL	C	1 OF 4	5	900D01					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
520	B	810	Piping	5366231	3/4 - 1	*****PIPES*****								
530														
540						PIPE SCH 80S SMLS SUPER DUPLEX S32760 PE								
550						PIPE SCH 40S SMLS SUPER DUPLEX S32760 PE								
560						PIPE SCH 40S SMLS SUPER DUPLEX S32760 BE								
570						PIPE WT 9.0MM SMLS SUPER DUPLEX S32760 BE								
580						PIPE WT 11.0MM EFW SUPER DUPLEX S32760-WX BE								
590						PIPE WT 13.0MM EFW SUPER DUPLEX S32760-WX BE								
600						PIPE WT 14.0MM EFW SUPER DUPLEX S32760-WX BE								
610						PIPE WT 16.0MM EFW SUPER DUPLEX S32760-WX BE								
620	810	Piping	5366220	1/2 - 1/2	PIPE SCH 160 SMLS SUPER DUPLEX S32760 BE									
630														
640														
650					***** FITTINGS *****									
660														
670														
680					90 Deg							5461101	3/4 - 1	ELL 90 DEG LR SCH 80S SUPER DUPLEX S32760
690					90 Deg							5461102	1 1/2 - 6	ELL 90 DEG LR SCH 40S SUPER DUPLEX S32760
700					90 Deg							5461105	8 - 8	ELL 90 DEG LR WT 9.0MM SUPER DUPLEX S32760
710					90 Deg							5461106	10 - 10	ELL 90 DEG LR 11.0MM EFW SUPER DUPLEX S32760-W
720	90 Deg	5461107	12 - 12	ELL 90 DEG LR 13.0MM EFW SUPER DUPLEX S32760-W										
730	90 Deg	5461108	14 - 14	ELL 90 DEG LR 14.0MM EFW SUPER DUPLEX S32760-W										
740	90 Deg	5461109	16 - 16	ELL 90 DEG LR 16.0MM EFW SUPER DUPLEX S32760-W										
750	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
760														
770														
780														
790														
800														
810														
820														
830														
840														
850	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
860														
870														
880														
890														
900														
910														
920														
930														
940														
950	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
960														
970														
980														
990														
1000														
1010														
1020														
1030														
1040														
1050	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1070														
1080														
1090														
1100														
1110														
1120														
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1140														
1150	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1170														
1180														
1190														
1200														
1210														
1220														
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1240														
1250	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1350	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1450	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1550	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1650	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1750	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1850	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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1920														
1930														
1940														
1950	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
1960														
1970														
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2050	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2110														
2120														
2130														
2140														
2150	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2170														
2180														
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2240														
2250	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2350	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2450	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2550	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2650	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2750	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2850	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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2950	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3050	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3150	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3250	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3350	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3450	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3550	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3650	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3750	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3850	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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3950	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4050	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4150	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4250	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4350	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4450	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4550	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4650	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4750	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4850	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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4950	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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5050	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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5150	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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5250	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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5280														
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5350	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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5450	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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5550	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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5650	B	810	Piping	5366220	1/2 - 1/2	***** FITTINGS *****								
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SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES															
1060	B	FS1	Eccent	5461163	4 - 4	REDUCER ECC SCH 40S SUPER DUPLEX S32760	6 - 6																
1070			Eccent	5461166	6 - 6	REDUCER ECC WT 9.0MM/SCH 40S SMLS SUPER DUPLEX S32760	8 - 8																
1080																							
1090																							
1100						*****BRANCH FITTINGS*****																	
1110																							
1120			Nipofl	5512151	3/4 - 1	NIPOFLANGE 1500# RF SCH 40S/80S SUPER DUPLEX F55	4 - 6																
1130			Nipofl	5512152	3/4 - 1	NIPOFLANGE 1500# RF WT 9.0MM/SCH 80S SUPER DUPLEX F55	8 - 8																
1140			Nipofl	5512153	3/4 - 1	NIPOFLANGE 1500# RF WT 16.0MM/SCH 80S SUPER DUPLEX F55	16 - 16																
1150																							
1160			Weldol	5512124	3/4 - 1	WELDOLET SCH 40S/80S SUPER DUPLEX F55	3 - 6																
1170			Weldol	5512126	1 1/2 - 2	WELDOLET SCH 40S SUPER DUPLEX F55	4 - 6																
1180			Weldol	5512122	2 - 2	WELDOLET SCH 9MM/40S SUPER DUPLEX F55	8 - 8																
1190																							
1200																							
1210						*****PIPE FLANGES*****																	
1220																							
1230			B		Flange	5851205	3/4 - 3/4								FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	3/4 - 1 1/2	DESIGN: ASME B16.5 MG 2.8 CODE: ASME B31.3 WP						
1240			B		Flange	5851205	2 - 2								FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
1250																							
1260				FA8	Reduci	5600675	1/2 - 1/2								FLG SCRD (FNPT) REDUCING 1500# RF SUPER DUPLEX F55								
1270																							
1280					Flange	5590482	3/4 - 1								FLG WN 1500# RF SUPER DUPLEX 80S F55								
1290					Flange	5590481	1 1/2 - 2								FLG WN 1500# RF SUPER DUPLEX 40S F55								
1300					Flange	5590471	3 - 6								FLG WN 900# RF SUPER DUPLEX 40S F55								
1310					Flange	5590474	8 - 8								FLG WN 900# RF SUPER DUPLEX 9.0MM F55								
1320					Flange	5590475	10 - 10								FLG WN 900# RF SUPER DUPLEX 11.0MM F55								
1330			816		Flange	5619907	12 - 12								FLG COMPACT 900# RJ SUPER DUPLEX 13.0MM F55								
1340			816		Flange	5619908	14 - 14								FLG COMPACT 900# RJ SUPER DUPLEX 14.0MM F55								
1350			816		Flange	5619911	16 - 16								FLG COMPACT 900# RJ SUPER DUPLEX 16.0MM F55								
1360																							
1370					Blind	5616876	3/4 - 2								FLG BLIND 1500# RF SUPER DUPLEX F55								
1380					Blind	5616871	3 - 10								FLG BLIND 900# RF SUPER DUPLEX F55								
1390			816		Blind	5619727	12 - 16								FLG COMPACT BLIND 900# RJ SUPER DUPLEX F55 PIPE CLASS 900D01								
1400																							
1410																							
1420															*****ORIFICE FLANGES*****								
1430																							
1440					Orific	5650950	2 - 2								WN ORIF SET 1500# RF SUPER DUPLEX 40S F55								
1450					Orific	5650951	3 - 6								WN ORIF SET 900# RF SUPER DUPLEX 40S F55								
1460																							
1470															***** PIPE NIPPLES *****								
1480																							
1490															NIPPLE PBE CUT FROM PIPE								
1500																							
1510																							
1520															***** GASKETS *****								
1530				FS1																			
1540																							
1550					GASFRS	5673393	3/4 - 2								GASKET 1500# DUPLEX 2205 SP WND GRAPHITE IR								
1560					GASFRS	5673389	3 - 10								GASKET 900# DUPLEX 2205 SP WND GRAPHITE IR								
1570			B		GASRJ	5673432	12 - 16								GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900D01								
1580																							
1590																							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELD	CORR. ALLOW. 1 MM	FGP/WPMP DATE 17MAR17 BY JL REV C SHEET 3 OF 4 PA 5 CLASS 900D01															

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1600	B	B1				***** BOLTING *****	3 - 16 3 - 16						
1610					1/4 - 16	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1620		BOLS	5675770										
1630					12 - 16	JACK SCREW HEX HEAD A320 GR L7M							
1640		SB6	CPSBOL	5675675									
1650						***** BLINDS & SPACERS *****							
1660													
1670													
1680													
1690			Specta	5749840	3/4 - 2	SPECTACLE BLIND 1500# RF SUPER DUPLEX STL S32760							
1700			Specta	5749839	3 - 6	SPECTACLE BLIND 900# RF SUPER DUPLEX STL S32760							
1710													
1720			Paddle	5750109	8 - 10	PADDLE BLANK 900# RF SUPER DUPLEX STL S32760							
1730			Paddle	5750119	12 - 16	PADDLE BLANK COMPACT 900# RJ SUPER DUPLEX STL S32760							
1740													
1750		Paddle	5750337	8 - 10	PADDLE SPACER 900# RF SUPER DUPLEX STL S32760								
1760		Paddle	5750347	12 - 16	PADDLE SPACER COMPACT 900# RJ SUPER DUPLEX STL S32760								
1770													
1780					***** BRANCHES *****								
1790					USE BRANCH CHART 900D01								
1800													
1810													
1820													
1830	B		Reinfo	RWELD	2 - 12	NO-MTO REINFORCING WELD FOR STUB-IN							
1840	B		Reinfo	RWELD	2 - 12	NO-MTO REINFORCING WELD FOR STUB-IN							
1850													
1860						***** CONSTRUCTION DETAILS *****							
1870						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
1880						PIPE BENDS ----- NONE							
1890		ZA6				THREAD SEALANT ----- PTFE PASTE							
1900						FOR FABRICATION AND WELDING DETAILS							
1910						SEE DOC. N° L-ST-2024							
1920													
1930													
1940													
						CONSTRUCTION		FGP/WPMP					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			ALL SIZES BUTT-WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							1 MM	17MAR17	JL	C	4 OF 4	5	900D01

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 20 30 40	D	PP9 VA5	2BV	K15AC02	1/2 - 1 1/2	*****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		O32 810 815 816	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. NIPPLE TO BE CUT FROM SCH 160 PIPE TO BE USED WITH COMPACT FLANGES ONLY. COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FOR FGP ONLY.					
50 60	D		2BV	K15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		B1 BA1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED WITH RAISED FACE FLANGES ONLY.					
70 80	D	VAV	2BV	K09AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 316 SS TRIM METAL LO FULL SIZE TYPE 2BV		BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. FOR FGP ONLY.					
90 100	D		2BVR	M15AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		FA8 FF5	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES					
110 120	D		2BVR	M15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
130 140	D	VAW	R2BVR	M15EC02	1 1/2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 3	SB5	USE JACK SCREWS ONLY WHERE PIPING CAN NOT BE OPENED WITH FLANGE SPREADER.					
150 160	D	VAW	R2BVR	M09EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	SC1	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE AND ORIFICE SETS					
170 180 190 200 210 220		VB2				*****BALL VALVES*****								
230 240			Ball V	A15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO								
250 260	D		Ball V	A15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO								
270 280	D		Ball V	A09AC05	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO								
290 300	D		Ball V	A09AC06	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO								
310 320	D		Ball V	A09AC07	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
330 340			Ball V	A09AC08	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
350 360			Ball V	A09CC07	12 - 24	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
370 380			Ball V	A09CC08	12 - 24	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
390 400			Ball V	B15AC01	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
410 420			Ball V	B15AC02	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
430 440		VA5	Ball V	B15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO								
450 460		VA5	Ball V	B15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO								
470 480	D	VA5	Ball V	B09AC05	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
490 500	D	VA5	Ball V	B09AC06	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
510	D	VA5	Ball V	B09AC07	4- 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
			TENGIZCHEVROI - TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 1.5 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 MLA D 1 OF 5 5 900H21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
520	D	VA5	Ball V	B09AC08	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.						
530		VA5	Ball V	B09CC07	12 - 24	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO		VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS						
540		VA5	Ball V	B09CC08	12 - 24	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO		VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY						
550						*****GATE VALVES*****		VAD	FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE).						
560								VAG	FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY.						
570	C		Gate V	G15AC48	3/4 - 1 1/2	GATE 1500# RF LTCS A350 LF2 TRIM 16 WEDGE HW		VAK	FOR ALL INLINE APPLICATIONS AND LEVEL TRANSMITTER CONNECTIONS						
580	C		Gate V	V15AC41	2 - 2	GATE PARALLEL SLIDE 1500# RF LTCS A352 LCC TRIM 16 HW		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE						
590	C		Gate V	V09AC41	3 - 4	GATE PARALLEL SLIDE 900# RF LTCS A352 LCC TRIM 16 HW		VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055						
600	C		Gate V	V09AC42	6 - 8	GATE PARALLEL SLIDE 900# RF LTCS A352 LCC TRIM 16 GO		VB2	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.						
610						***** BUTTERFLY VALVES *****		VC1	SPRING PISTON CHECK VALVES ARE INSTALLED IN VERTICAL-UP FLOW ONLY. SPRING MAY BE REMOVED WHEN INSTALLED IN HORIZONTAL POSITION.						
620								VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION OR IN VERTICAL - UP FLOW ONLY.						
630								VU2	USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY.						
640								ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
650								SERVICE							
660								DESIGN: ASME B16.5 MG 1.1							
670								CODE: ASME B31.3							
680								BB, DHC, DSL,DL, FM, FGH, GHC, GHP, GRC, NH, PHC, SG, SH, SL, SM, TH, TM, WBL, WBL							
690															
700	C	ZA8	Butter	T09AC61	10 - 10	BUTTERFLY 900# RF LTCS A352 LCC INC TRIM TOV METAL DBL FLG SHORT PAT GO									
710	C	ZA8	Butter	T09CC61	12 - 30	BUTTERFLY 900# COMPACT FLANGE LTCS A352 LCC INC TRIM TOV METAL DBL FLG SHORT PAT GO									
720		PP9				*****GLOBE VALVES*****									
730															
740															
750			Globe	L15AC48	3/4 - 1 1/2	GLOBE 1500# RF LTCS A350 LF2 TRIM 16 OS&Y HW									
760			Globe	L15AC41	2 - 2	GLOBE 1500# RF LTCS A352 LCC TRIM 16 OS&Y HW									
770			Globe	L09AC41	3 - 3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW									
780			Globe	L09AC42	4 - 8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO									
790															
800						*****CHECK VALVES*****									
810															
820															
830		VC1	Check	C15AC50	3/4 - 1 1/2	CHECK PISTON 1500# RF LTCS A350 LF2 TRIM 16									
840		VC2	Check	C15AC45	2 - 2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC TRIM 16 THROU BOLTED									
850		VC2	Check	C09AC45	3 - 8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED									
860		VC2	Check	C09AC52	10 - 10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16									
870		VC2	Check	C09CC52	12 - 24	CHECK DUAL 900# CF LTCS A352 LCC TRIM 16									
880															
890						*****PIPES*****									
900															
910															
920			Piping	5358508	3/4 - 1 1/2	PIPE SCH XS SMLS LTCS A333 GR.6.PE									
930	C		Piping	5358514	2 - 3	PIPE SCH XS SMLS LTCS A333 GR.6 BE									
940	C		Piping	5358510	4 - 16	PIPE SCH 120 SMLS LTCS A333 GR.6 BE									
950			Piping	5358600	18 - 24	PIPE SCH 120 EFW LTCS A671 CC60 CL.22 BE									
960	C		Piping	5358640	30 - 30	PIPE WT 49.0MM EFW LTCS A671 CC60 CL.22 BE									
970															
980	C	810	Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE									
1000						***** FITTINGS *****									
1010															
1020															
1030	C		90 Deg	5392600	3/4 - 3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6									
						CONSTRUCTION		FGP/WPMP							
						ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS		
							1.5 MM	17MAR17	MLA	D	2 OF 5	5	900H21		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040	C		90 Deg	5392700	4 - 16	ELL 90 DEG LR SCH 120 LTCS A420 WPL6							
1050			90 Deg	5392388	18 - 24	ELL 90 DEG LR SCH 120 LTCS A420 WPL6-W							
1060	C		90 Deg	5392567	30 - 30	ELL 90 DEG LR WT 49.0MM LTCS A420 WPL6-W							
1070													
1080	C		45 Deg	5392606	3/4 - 3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6							
1090	C		45 Deg	5392701	4 - 16	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6							
1100			45 Deg	5392392	18 - 24	ELL 45 DEG LR SCH 120 LTCS A420 WPL6-W							
1110	C		45 Deg	5392568	30 - 30	ELL 45 DEG LR WT 49.0MM LTCS A420 WPL6-W							
1120													
1130	C		Tee	5392609	3/4 - 3	TEE BW SCH XS LTCS A420 WPL6							
1140	C		Tee	5392702	4 - 16	TEE BW SCH 120 LTCS A420 WPL6							
1150			Tee	5392404	18 - 24	TEE BW SCH 120 LTCS A420 WPL6-W							
1160													
1170	C		Reduci	5392661	3/4 - 2	TEE RED BW SCH XS LTCS A420 WPL6	1 - 3						
1180	C		Reduci	5392466	1 1/2 - 3	TEE RED BW SCH 120/XS LTCS A420 WPL6	4 - 6						
1190	C		Reduci	5392603	4 - 14	TEE RED SCH 120 LTCS A420 WPL6	6 - 16						
1200			Reduci	5392878	8 - 20	TEE RED SCH 120 LTCS A420 WPL6-W	18 - 24						
1210	C		Reduci	5392888	20 - 24	TEE RED 49.0MM WT / SCH 120 LTCS A420 WPL6-W	30 - 30						
1220													
1230	C		Cap	5392610	3/4 - 3	CAP BW SCH XS LTCS A420 WPL6							
1240	C		Cap	5392703	4 - 24	CAP SCH 120 LTCS A420 WPL6							
1250													
1260	C		Concen	5392458	3/4 - 2	REDUCER CONC SCH XS LTCS A420 WPL6	1 - 3						
1270	C		Concen	5392706	2 - 3	REDUCER CONC SCH 120/ XS LTCS A420 WPL6	4 - 6						
1280	C		Concen	5392704	4 - 14	REDUCER CONC SCH 120 LTCS A420 WPL6	6 - 16						
1290			Concen	5392707	10 - 20	REDUCER CONC SCH 120 LTCS A420 WPL6-W	18 - 24						
1300	C		Concen	5392736	24 - 24	REDUCER CONC WT 49.0MM/SCH 120 LTCS A420 WPL6-W	30 - 30						
1310													
1320	C		Eccent	5392612	3/4 - 2	REDUCER ECC SCH XS LTCS A420 WPL6	1 - 3						
1330	C		Eccent	5392708	3 - 3	REDUCER ECC SCH 120/ XS LTCS A420 WPL6	4 - 6						
1340	C		Eccent	5392705	4 - 14	REDUCER ECC SCH 120 LTCS A420 WPL6	6 - 16						
1350			Eccent	5392709	10 - 20	REDUCER ECC SCH 120 LTCS A420 WPL6-W	18 - 24						
1360	C		Eccent	5392738	24 - 24	REDUCER ECC WT 49.0MM/SCH 120 LTCS A420 WPL6-W	30 - 30						
1370													
1380	C		Nipofl	5504160	3/4 - 1	NIPOFLANGE 1500# RF SCH XS LTCS A350 LF2	2 - 3						
1390	C		Nipofl	5504161	3/4 - 2	NIPOFLANGE 1500# RF SCH 120/XS LTCS A350 LF2	4 - 24						
1400	C		Nipofl	5504193	1 - 1	NIPOFLANGE 1500# RF WT 49.0MM/XS LTCS A350 LF2	30 - 30						
1410	C	UJ2	Nipofl	5504163	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2	4 - 24						
1420	D	UJ2	Nipofl	5504193	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF WT 49.0MM/XS LTCS A350 LF2	30 - 30						
1430													
1440	C	FF5	Endole	5504000	3/4 - 1 1/2	NIPOFLANGE FLAT BOTTOM 1500# RF SCH XS LTCS A350 LF2							
1450													
1460	C		Weldol	5505335	3/4 - 1	WELDOLET SCH XS LTCS A350 LF2	2 - 3						
1470	C		Weldol	5505194	3/4 - 3	WELDOLET SCH 120/ XS LTCS A350 LF2	4 - 24						
1480	C		Weldol	5505195	4 - 8	WELDOLET SCH 120 LTCS A350 LF2	14 - 24						
1490													
1500	C	UJ2	Elbole	5516082	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2	4 - 24						
1510													
1520													
1530						***** OLETS *****							
1540													
1550			Sockol	5533072	3/4 - 1 1/2	SOCKOLET 3000# LTCS A350 LF2	2 - 24						
1560			Endole	5533232	3/4 - 1 1/2	FLAT SOCKOLET 3000# LTCS A350 LF2							
1570		UD7	Thredo	5514772	3/4 - 3/4	THREDOLET 3000# LTCS A350 LF2	2 - 24						
			TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 17MAR17 BY MLA REV D SHEET 3 OF 5 PA 5 CLASS 900H21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1580						*****COUPLINGS, UNIONS & PLUGS *****		SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BB, DHC, DSL,DL, FM, FGH, GHC, GHP, GRC, NH, PHC, SG, SH, SL, SM, TH, TM, WBH, WBL MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -46 / 410 °C						
1590														
1600			Coupli	5478705	3/4 - 1 1/2	COUPLING 3000# SW LTCS A 350 LF2	1 - 1 1/2							
1610			Reduci	5478709	3/4 - 1	COUPLING RED 3000# SW LTCS A350 LF2								
1620			Union	5478706	3/4 - 1 1/2	UNION 3000# SW LTCS A350 LF2								
1630		UD7	Plug	5491303	3/4 - 3/4	PLUG ROUND HEAD SCR D LTCS A350 LF2								
1640														
1650														
1660		816				***** PIPE FLANGES *****								
1670														
1680		FA8	Reduci	5599704	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 1500# RF LTCS A350 LF2	3/4 - 1 1/2							
1690														
1700	C		Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
1710	C		Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
1720														
1730			Flange	5557393	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH XS								
1740	C		Flange	5557141	3 - 3	FLG WN 900# RF LTCS A350 LF2 SCH XS								
1750	C		Flange	5557143	4 - 10	FLG WN 900# RF LTCS A350 LF2 SCH 120 BORE								
1760			Flange	5557143	12 - 24	FLG WN 900# RF LTCS A350 LF2 SCH 120 BORE								
1770	C		Flange	5556499	30 - 30	FLG WN 900# RF LTCS A350 LF2 WT 49.0MM B16.47 SERIES A								
1780														
1790														
1800			Flange	5619821	12 - 24	FLG COMPACT 900# RJ LTCS A350 LF2 SCH 120 BORE								
1810	C		Flange	5619819	30 - 30	FLG COMPACT 900# RJ LTCS A350 LF2 WT 49.0MM								
1820														
1830	C		Blind	5615616	1/2 - 2	FLG BLIND 1500# RF LTCS A350 LF2								
1840			Blind	5615605	3 - 10	FLG BLIND 900# RF LTCS A350 LF2								
1850			Blind	5615605	12 - 24	FLG BLIND 900# RF LTCS A350 LF2								
1860														
1870			Blind	5619770	12 - 30	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 PIPE CLASS 900H21								
1880														
1890	D		Blind	9619758	2 - 2	FLG COMPACT BLIND 10000# RJ LTCS A694 F60 PIPE CLASS 10000H5A - 2 INCH ONLY								
1900	D		Blind	5619756	6 - 6	FLG COMPACT BLIND 10000# RJ 2-1/4CR F22M PIPE CLASS 10000F22 FOR NPS 4 INCH & ABOVE								
1910														
1920														
1930														
1940														
1950			Orific	5624599	2 - 2	WN ORIF SET 1500# RF LTCS A350 LF2 SCH XS								
1960	C		Orific	5624540	3 - 3	WN ORIF SET 900# RF LTCS A350 LF2 SCH XS								
1970	D		Orific	5624532	4 - 10	WN ORIF SET 900# RF LTCS A350 SCH 120 BORE								
1980	D		Orific	5624538	12 - 24	WN ORIF SET COMPACT 900# RJ LTCS A350 SCH 120 BORE								
1990														
2000						***** PIPE NIPPLES *****								
2010														
2020		810				NIPPLE PBE TO BE CUT FROM PIPE								
2030														
2040														
2050						***** GASKETS *****								
2060														
2070	C		GASFRS	5672251	1/2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR								
2080														
			TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 1.5 MM	DATE	BY	REV	SHEET	PA	CLASS	
								17MAR17	MLA	D	4 OF 5	5	900H21	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
2090 2100	C		GASFRS GASFRS	5673117 5674033	3 - 24 30 - 30	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A						
2110 2120	D	815	GASRJ	5673440	12 - 30	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900H21						
2130	D	815	GASRJ	5673550	6 - 6	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 10000F22 FOR NPS 4 INCH & ABOVE						
2140 2150 2160 2170		B1				***** BOLTING *****						
2180	D		BOLS	5675770	1/2 - 20	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS						
2190	D	BA5	BOLS	5675770	24 - 30	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS						
2200	D	BA5	BOLS	5675776	36 - 99	STUD-BOLT WITH 2 NUTS API-6A718						
2210												
2220	D	BA1	BOLS	5675776	24 - 99	STUD-BOLT WITH 2 NUTS API-6A718						
2230		BA5	BOLS	5675778	6 - 99	STUD-BOLT/COMPACT NUT API-6A718						
2240												
2250		SB6	CPSBOL	5675675	10 - 24	JACK SCREW HEX HEAD A320 GR L7M						
2260												
2270		SB5				***** BLINDS & SPACERS *****						
2280												
2290												
2300			Specta	5749595	3/4 - 2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70						
2310			Specta	5749596	3 - 8	SPECTACLE BLIND 900# RF LTCS A516 GR.70						
2320												
2330	C		Paddle	5750102	10 - 10	PADDLE BLANK 900# RF LTCS A516 GR.70						
2340	C		Paddle	5750202	10 - 10	PADDLE SPACER 900# RF LTCS A516 GR.70						
2350												
2360	C		Paddle	5750264	12 - 30	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70						
2370	C		Paddle	5750265	12 - 30	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70						
2380												
2390	C		Paddle	5750102	12 - 24	PADDLE BLANK 900# RF LTCS A516 GR.70						
2400	C		Paddle	5750202	12 - 24	PADDLE SPACER 900# RF LTCS A516 GR.70						
2410												
2420	D		Paddle	5750104	30 - 30	PADDLE BLANK 900# RF LTCS A516 GR.70 MANUFACTURERS STD						
2430												
2440	D		Paddle	5750204	30 - 30	PADDLE SPACER 900# RF LTCS A516 GR.70 MANUFACTURERS STD						
2450												
2460												
2470						***** SP ITEMS *****						
2480						***** BRANCHES *****						
2490												
2500						USE BRANCH CHART 900H21						
2510												
2520												
2530	D		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 30					
2540	D		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 30					
2550												
2560												
2570						***** CONSTRUCTION DETAILS *****						
2580												
2590		ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY						
2600						PIPE BENDS ----- NONE						
2610						THREAD SEALANT ----- PTFE PASTE						
			TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 17MAR17 BY MLA REV D SHEET 5 OF 5 PA 5 CLASS 900H21			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10	B	PP9				*****MODULAR VALVES*****		812	NIPPLE TO BE CUT FROM SCH XXS PIPE					
20								815	TO BE USED WITH COMPACT FLANGES ONLY.					
30								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME'S					
40		VA5	2BV	K15AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.				
50									BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.				
60		B		2BV	K15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.				
70									FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.				
80		B	VAV	2BV	K09AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 316 SS TRIM METAL LO FULL SIZE TYPE 2BV		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156				
90									SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.				
100		B		2BVR	M15AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.				
110														
120		B		2BVR	M15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR							
130														
140		B	VAW	R2BVR	M15EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2						
150														
160		B	VAW	R2BVR	M09EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3						
170														
180														
190							*****BALL VALVES*****							
200														
210						*****GATE VALVES*****								
220														
230														
240			Gate V	G15AC48	3/4 - 1 1/2	GATE 1500# RF LTCS A350 LF2 TRIM 16 WEDGE HW								
250			Gate V	V15AC41	2 - 2	GATE PARALLEL SLIDE 1500# RF LTCS A352 LCC TRIM 16 HW								
260			Gate V	V09AC41	3 - 4	GATE PARALLEL SLIDE 900# RF LTCS A352 LCC TRIM 16 HW								
270			Gate V	V09AC42	6 - 8	GATE PARALLEL SLIDE 900# RF LTCS A352 LCC TRIM 16 GO								
280														
290						*****BUTTERFLY VALVES*****								
300														
310		ZA8	Butter	T09AC61	10 - 10	BUTTERFLY 900# RF LTCS A352 LCC INC TRIM TOV METAL DBL FLG SHORT PAT GO			DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 DL, SH, TH, TM, WU					
320	B	ZA8	Butter	T09CC61	12 - 24	BUTTERFLY 900# COMPACT FLANGE LTCS A352 LCC INC TRIM TOV METAL DBL FLG SHORT PAT GO								
330														
340						*****GLOBE VALVES*****								
350														
360			Globe	L15AC48	3/4 - 1 1/2	GLOBE 1500# RF LTCS A350 LF2 TRIM 16 OS&Y HW								
370			Globe	L15AC41	2 - 2	GLOBE 1500# RF LTCS A352 LCC TRIM 16 OS&Y HW								
380			Globe	L09AC41	3 - 3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW								
390			Globe	L09AC42	4 - 8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO								
400														
410														
420						*****CHECK VALVES*****								
430														
440		VC1	Check	C15AC50	3/4 - 1 1/2	CHECK PISTON 1500# RF LTCS A350 LF2 TRIM 16								
450		VC2	Check	C15AC45	2 - 2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC TRIM 16 THROU BOLTED								
460		VC2	Check	C09AC45	3 - 8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED								
470		VC2	Check	C09AC52	10 - 10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16								
480	B	VC2	Check	C09CC52	12 - 24	CHECK DUAL FLG 900# RF LTCS A352 LCC TRIM 16								
						CONSTRUCTION								
						ALL SIZES- BUTT WELD PWHT FOR WT >19 MM								
							CORR. ALLOW.	DATE	BY	REV	FGP/WPMP			
							3 MM	17MAR17	KD	B	1 OF 4	5	CLASS 900H25	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
490	B		Piping	5358512	3/4 - 1 1/2	***** PIPE ***** PIPE SCH 160 SMLS LTCS A333 GR.6 PE		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. *****TO BE USED IN STEAM AND CONDENSATE SERVICE ONLY*****				
500				5358515	2 - 3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE		VAV					
510				5358510	4 - 16	PIPE SCH 120 SMLS LTCS A333 GR.6 BE		VAV					
520				5358600	18 - 24	PIPE SCH 120 EFW LTCS A671 CC60 CL.22 BE		VC1					
530	812		Piping	5358511	3/4 - 3/4	PIPE SCH XXS SMLS LTCS A333 GR.6 PE		VC2					
540						***** FITTINGS *****		ZA8					
550													
560													
570			90 Deg	5392816	3/4 - 3	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6							
580				5392700	4 - 16	ELL 90 DEG LR SCH 120 LTCS A420 WPL6							
590				5392388	18 - 24	ELL 90 DEG LR SCH 120 LTCS A420 WPL6-W							
600													
610			45 Deg	5392822	3/4 - 3	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6							
620				5392701	4 - 16	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6							
630				5392392	18 - 24	ELL 45 DEG LR SCH 120 LTCS A420 WPL6-W							
640													
650			Tee	5392825	3/4 - 3	TEE BW SCH 160 LTCS A420 WPL6							
660				5392702	4 - 16	TEE BW SCH 120 LTCS A420 WPL6							
670				5392404	18 - 24	TEE BW SCH 120 LTCS A420 WPL6-W							
680													
690			Reduci	5392821	3/4 - 2	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 3						
700				5392470	1 1/2 - 3	TEE RED SCH 120/160 LTCS A420 WPL6							4 - 6
710				5392603	4 - 14	TEE RED SCH 120 LTCS A420 WPL6							6 - 16
720				5392878	8 - 20	TEE RED SCH 120 LTCS A420 WPL6-W							18 - 24
730			Cap	5392826	3/4 - 3	CAP BW SCH 160 LTCS A420 WPL6							
740				5392703	4 - 24	CAP SCH 120 LTCS A420 WPL6							
750													
760													
770			Concen	5392827	3/4 - 2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 3						
780				5392467	1 1/2 - 3	REDUCER CONC SCH 120/160 LTCS A420 WPL6							4 - 6
790				5392704	4 - 14	REDUCER CONC SCH 120 LTCS A420 WPL6							6 - 16
800				5392707	10 - 20	REDUCER CONC SCH 120 LTCS A420 WPL6-W							18 - 24
810			Eccent	5392828	3/4 - 2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 3						
820				5392468	1 1/2 - 3	REDUCER ECC SCH 120/160 LTCS A420 WPL6							4 - 6
830				5392705	4 - 14	REDUCER ECC SCH 120 LTCS A420 WPL6							6 - 16
840				5392709	10 - 20	REDUCER ECC SCH 120 LTCS A420 WPL6-W							18 - 24
850						***** OLETS *****							
860													
870													
880													
890			Nipofl	5504170	3/4 - 2	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2	2 - 3						
900				5504171	3/4 - 2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2							4 - 24
910				5504163	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2							4 - 24
920													
930			Weldol	5505392	3/4 - 1	WELDOLET SCH 160 LTCS A350 LF2	2 - 3						
940				5505394	3/4 - 3	WELDOLET SCH 120/160 LTCS A350 LF2							4 - 24
950				5505195	4 - 8	WELDOLET SCH 120 LTCS A350 LF2							12 - 24
960													
970			Elbole	5516082	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2	4 - 24						
980													
990													
1000													
1010													
1020													
1030													
1040													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY KD REV B SHEET 2 OF 4 PA 5 CLASS 900H25					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1050		816				***** PIPE FLANGES *****							
1060													
1070		FA8	Reduci	5599704	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 1500# RF LTCS A350 LF2	3/4 - 1 1/2						
1080													
1090			Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1100			Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1110													
1120			Flange	5557397	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH 160 BORE							
1130			Flange	5557145	3 - 3	FLG WN 900# RF LTCS A350 LF2 SCH 160							
1140			Flange	5557143	4 - 10	FLG WN 900# RF LTCS A350 LF2 SCH 120 BORE							
1150			Flange	5557143	12 - 24	FLG WN 900# RF LTCS A350 LF2 SCH 120 BORE							
1160													
1170			Flange	5619821	12 - 24	FLG COMPACT 900# RJ LTCS A350 LF2 SCH 120 BORE							
1180													
1190			Blind	5615616	1/2 - 2	FLG BLIND 1500# RF LTCS A350 LF2							
1200			Blind	5615605	3 - 10	FLG BLIND 900# RF LTCS A350 LF2							
1210			Blind	5615605	12 - 24	FLG BLIND 900# RF LTCS A350 LF2							
1220													
1230			Blind	5619771	12 - 24	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 PIPE CLASS 900H25							
1240													
1250													
1260						***** ORIFICE FLANGES *****							
1270													
1280			Orific	5624598	2 - 2	WN ORIF SET 1500# RF LTCS A350 LF2 SCH 160 BORE							
1290			Orific	5624537	3 - 3	WN ORIF SET 900# RF LTCS A350 SCH 160 BORE							
1300			Orific	5624532	4 - 10	WN ORIF SET 900# RF LTCS A350 SCH 120 BORE							
1310													
1320						***** PIPE NIPPLES *****							
1330													
1340													
1350		812				NIPPLE BBE - CUT FROM PIPE							
1360													
1370						***** GASKETS *****							
1380													
1390													
1400			GASFRR	5672251	1/2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
1410			GASFRR	5673117	3 - 24	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
1420													
1430	B	815	GASRJ	5673441	12 - 24	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900H25							
1440													
1450													
1460		B1				***** BOLTING *****							
1470													
1480	B		BOLS	5675770	1/4 - 20	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1490	B	BA5	BOLS	5675770	24 - 30	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1500	B	BA5	BOLS	5675776	36 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
1510													
1520	B	BA1	BOLS	5675776	24 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
1530													
1540		SB6	CPSBOL	5675675	10 - 24	JACK SCREW HEX HEAD A320 GR L7M							
1550													
1560													
1570													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM	CORR. ALLOW. 3 MM	DATE 17MAR17	BY KD	REV B	SHEET 3 OF 4	PA 5	CLASS 900H25

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1580	B	ZA6	Specta Specta Paddle Paddle Paddle Paddle Paddle Paddle	5749595 5749596 5750102 5750202 5750102 5750202 5750215 5750223	3/4 - 2 3 - 8 10 - 10 10 - 10 12 - 24 12 - 24 12 - 24 12 - 24	***** BLINDS AND SPACERS *****	3 - 24 3 - 24							
1590						SPECTACLE BLIND 1500# RF LTCS A516 GR.70								
1600						SPECTACLE BLIND 900# RF LTCS A516 GR.70								
1610														
1620														
1630						PADDLE BLANK 900# RF LTCS A516 GR.70								
1640						PADDLE SPACER 900# RF LTCS A516 GR.70								
1650														
1660						PADDLE BLANK 900# RF LTCS A516 GR.70								
1670						PADDLE SPACER 900# RF LTCS A516 GR.70								
1680														
1690			PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70											
1700			PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70											
1710														
1720			***** BRANCHES *****											
1730														
1740			USE BRANCH CHART 900H25											
1750														
1760			NO-MTO REINFORCING WELD FOR STUB-IN											
1770			NO-MTO REINFORCING WELD FOR STUB-IN											
1780														
1790	***** CONSTRUCTION DETAILS *****													
1800														
1810														
1820	SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY													
1830	PIPE BENDS ----- NONE													
1840	THREAD SEALANT ----- PTFE PASTE													
								SERVICE						
								DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 DL, SH, TH, TM, WU						
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -46 / 410 °C						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT FOR WT >19 MM		CORR. ALLOW. 3 MM	DATE 17MAR17	BY KD	REV B	SHEET 4 OF 4	PA 5	CLASS 900H25

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10 20 30 40 50 60	C	VA5	2BV	K15AC02	1/2 - 1 1/2	***** VALVES ***** ***** MODULAR VALVES ***** MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		817 B1 FF5	USE ONLY FOR TIE IN ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY.					
70 80	C		2BV	K15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		NT5 SB6	TO BE USED IN NON-PIGGED LINES ONLY USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
90 100	C	VAV	2BV	K09AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 316 SS TRIM METAL LO FULL SIZE TYPE 2BV		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
110 120	C		2BVR	M15AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VA5 VAV VAW	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.					
130 140	C		2BVR	M15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
150 160	C	VAW	R2BVR	M15EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
170 180	C	VAW	R2BVR	M09EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
190 200 210 220 230 240 250 260 270 280 290 300 310 320 330 340 350 360 370 380 390 400 410 420 430 440 450 460 470 480 490 500 510	C	VB2	Ball V	A15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO		SERVICE						
	C		Ball V	A15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO		DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 80 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - FG, FGH - FUEL GAS						
	C		Ball V	A09AC05	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO								
	C		Ball V	A09AC06	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO								
	C		Ball V	A09AC07	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
	C		Ball V	A09AC08	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
		VA5	Ball V	B15AC01	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
	C	VA5	Ball V	B15AC02	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
	C	VA5	Ball V	B15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO								
	C	VA5	Ball V	B15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO								
	C	VA5	Ball V	B09AC07	3 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
	C	VA5	Ball V	B09AC08	3 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
						***** ORBIT VALVES *****								
	C	VA5	Ball V	R15AC06	2 - 2	BALL RISING STEM 1500# RF LTCS A352 LCC 316SS TRIM METAL RB HO								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD		CORR. ALLOW. 1.5 MM	DATE 19JAN18	BY DN	REV C	SHEET 1 OF 5	PA 5	CLASS 900H5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
520	C	VA5	Ball V	R09AC06	3 - 4	BALL RISING STEM 900# RF LTCS A352 LCC 316SS TRIM METAL RB HO								
530														
540	C	VA5	Ball V	R09AC08	6 - 6	BALL RISING STEM 900# RF LTCS A352 LCC 316SS TRIM METAL RB GO								
550														
560														
570						***** GATE VALVES *****								
580														
590	C		Gate V	G15AC48	3/4 - 1 1/2	GATE 1500# RF LTCS A350 LF2 TRIM 16 WEDGE HW								
600														
610			Condui	W09AC70	3 - 4	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW								
620														
630	C		Condui	W09AC71	6 - 10	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO								
640														
650														
660						***** BUTTERFLY VALVES *****								
670														
680			Butter	T09AC61	10 - 10	BUTTERFLY 900# RF LTCS A352 LCC INC TRIM TOV METAL DBL FLG SHORT PAT GO								
690														
700						***** CHECK VALVES *****								
710														
720		VC1	Check	C15AC50	3/4 - 1 1/2	CHECK PISTON 1500# RF LTCS A350 LF2 TRIM 16								
730	C	VC2	Check	C15AC45	2 - 2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC TRIM 16 THROU BOLTED								
740	C	VC2	Check	C09AC45	3 - 8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED								
750		VC2	Check	C09AC52	10 - 10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16								
760														
770														
780						***** GLOBE VALVES *****								
790														
800	C		Globe	L15AC48	3/4 - 1 1/2	GLOBE 1500# RF LTCS A350 LF2 TRIM 16 OS&Y HW								
810	C		Globe	L15AC41	2 - 2	GLOBE 1500# RF LTCS A352 LCC TRIM 16 OS&Y HW								
820	C		Globe	L09AC41	3 - 3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW								
830	C		Globe	L09AC42	4 - 8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO								
840														
850														
860						*****PIPE ABOVE GROUND*****								
870														
880			Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE								
890			Piping	5358515	2 - 3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE								
900			Piping	9001257	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE								
910			Piping	9001255	6 - 6	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE								
920														
930						***** PIPE BELOW GROUND*****								
940														
950	C		Piping	5358515CTD	2 - 3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE COATED								
960			Piping	9001257CTD	4 - 4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED								
970			Piping	9001254CTD	6 - 6	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED								
980														
990						***** FITTINGS ABOVE & BELOW GROUND*****								
1000														
1010			90 Deg	5392816	3/4 - 3	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6								
1020			45 Deg	5392822	3/4 - 3	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6								
1030			90 Deg	9006566	4 - 4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60								
1040			45 Deg	9006573	4 - 4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD		CORR. ALLOW. 1.5 MM	DATE 19JAN18	BY DN	REV C	SHEET 2 OF 5	PA 5	CLASS 900H5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1050			90 Deg	9006470	6 - 6	ELL BW 90 DEG LR WT 7.11 MM A860 WPHY 60								
1060			45 Deg	9006574	6 - 6	ELL BW 45 DEG LR WT 7.11MM A860 WPHY 60								
1070														
1080			Tee	5392825	3/4 - 3	TEE BW SCH 160 LTCS A420 WPL6								
1090		NT5	Tee	9301610	4 - 4	TEE BW WT 6.02 MM LTCS A860 WPHY-60								
1100		NT5	Tee	9006600	6 - 6	TEE BW WT 7.11MM A860 WPHY 60								
1110			Reduci	5392821	3/4 - 2	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 3							
1120	C	NT5	Reduci	9006663	2 - 3	TEE RED BW WT 6.02 MM / SCH 160 LTCS A860 WPHY 60	4 - 4							
1130	C	NT5	Reduci	9006661	3 - 3	TEE RED BW WT 7.11 MM/SCH 160 LTCS A860 WPHY 60	6 - 6							
1140		NT5	Reduci	9006610	4 - 4	TEE RED BW WT 7.11 MM/6.02 MM LTCS A860 WPHY 60	6 - 6							
1150														
1160			Cap	5392826	3/4 - 3	CAP BW SCH 160 LTCS A420 WPL6								
1170			Cap	9002480	4 - 4	CAP BW WT 6.02 MM LTCS A860 WPHY-60								
1180			Cap	9005056	6 - 6	CAP BW WT 7.11MM LTCS A860 WPHY-60								
1190														
1200			Concen	5392827	3/4 - 2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 3							
1210			Eccent	5392828	3/4 - 2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 3							
1220	C		Concen	9006917	1 1/2 - 3	REDUCER CONC WT 6.02MM / SCH 160 A860 WPHY 60	4 - 4							
1230	C		Eccent	9002475	1 1/2 - 3	REDUCER ECC WT 6.02MM / SCH 160 LTCS A860 WPHY-60	4 - 4							
1240	C		Concen	9006916	3 - 3	REDUCER CONC WT 7.11MM / SCH 160 A860 WPHY 60	6 - 6							
1250	C		Eccent	9002433	3 - 3	REDUCER ECC WT 7.11MM / SCH 160 LTCS A860 WPHY-60	6 - 6							
1260			Concen	9006907	4 - 4	REDUCER CONC WT 7.11 MM / 6.02 MM LTCS A860 WPHY-60	6 - 6							
1270			Eccent	9001373	4 - 4	REDUCER ECC WT 7.11 MM / 6.02 MM LTCS A860 WPHY-60	6 - 6							
1280														
1290						***** INDUCTION BENDS ABOVE GROUND *****								
1300														
1310			90 Deg	9005380	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60								
1320			90 Deg	9001396	6 - 6	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1330														
1340			45 Deg	9005381	4 - 4	INDUCTION BEND 45DEG X 5D RADIUS WT 6.02 MM API 5L X60								
1350			45 Deg	9001399	6 - 6	C/W 1.5D TANGENT ENDS BW INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1360														
1370			22.5 D	9005382	4 - 4	INDUCTION BEND 22.5DEG X 5D RADIUS WT 6.02 MM								
1380			22.5 D	9002000	6 - 6	API 5L X60 C/W 1.5D TANGENT ENDS BW INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW								
1390														
1400						***** INDUCTION BENDS BELOW GROUND *****								
1450														
1460														
1470			90 Deg	9005380CTD	4 - 4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60								
1480	C		90 Deg	9001396CTD	6 - 6	C/W 1.5D TANGENT ENDS BW COATED INDUCTION BEND 90 DEG X 5D RADIUS WT 7.11MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1490														
1500			45 Deg	9005381CTD	4 - 4	INDUCTION BEND 45DEG X 5D RADIUS WT 6.02 MM API 5L X60								
1510	C		45 Deg	9001399CTD	6 - 6	C/W 1.5D TANGENT ENDS BW COATED INDUCTION BEND 45 DEG X 5D RADIUS WT 7.11 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1520														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD		FGP/WPMP						
								CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								1.5 MM	19JAN18	DN	C	3 OF 5	5	900H5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1530			22.5 D	9005382CTD	4 - 4	INDUCTION BEND 22.5DEG X 5D RADIUS WT 6.02 MM API 5L		SERVICE DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 80 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - FG, FGH - FUEL GAS MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 80 °C COMMENT: LTCS - NON-SOUR SERVICE					
1540	C		22.5 D	9002000CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L							
1550						X60 C/W 1.5D TANGENT ENDS BW COATED							
1560						INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.11 MM API 5L							
1570						X60 C/W 1.5D TANGENT ENDS BW COATED							
1580						***** OLETS ABOVE & BELOW GROUND *****							
1590	C		Weldol	9006400	3/4 - 1 1/2	WELDOLET WT 6.02 MM / SCH 160 LTCS A694 F60	4 - 4						
1600	C		Weldol	9002072	3/4 - 2	WELDOLET WT 7.11MM/ SCH 160 LTCS A694 F60	6 - 6						
1610													
1620	C		Nipofl	5504170	3/4 - 2	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2	2 - 3						
1630			Nipofl	9005388	3/4 - 2	NIPOFLANGE 1500# RF WT 6.02 MM/ SCH 160 LTCS A694 F60	4 - 4						
1640			Nipofl	9002835	3/4 - 2	NIPOFLANGE 1500# RF WT 7.11 MM/ SCH 160 LTCS A694 F60	6 - 6						
1650													
1660	C	FF5	Endole	9504523	3/4 - 3/4	NIPOFLANGE FLAT BOTTOM 1500# RF SCH 160 LTCS A694 F60							
1670													
1680	C	UJ2	Elbole	9002713	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF WT 6.02MM / SCH 160 LTCS A694 F60	4 - 4						
1690	C	UJ2	Elbole	9002714	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF WT 7.11MM / SCH 160 LTCSA694 F60	6 - 6						
1700													
1710						*****PIPE FLANGES ABOVE & BELOW GROUND*****							
1720													
1730													
1740			Flange	5557397	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH 160 BORE							
1750			Flange	5557145	3 - 3	FLG WN 900# RF LTCS A350 LF2 SCH 160							
1760			Flange	9005389	4 - 4	FLG WN 900# RF LTCS A694 F60 WT 6.02MM							
1770			Flange	9002876	6 - 6	FLG WN 900# RF LTCS A694 F60 WT 7.11MM							
1780													
1790	C	817	Flange	5557281	3 - 3	FLG WN 900# RTJ LTCS A350 LF2 SCH 160 BORE		3/4 - 1 1/2					
1800	C	817	Flange	9006715	4 - 4	FLG WN 900# RTJ LTCS A694 F60 WT 6.02MM FLG							
1810		817	Flange	9006700	6 - 6	WN 900# RTJ LTCS A694 F60 WT 7.11MM							
1820													
1830			Reduci	5599704	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 1500# RF LTCS A350 LF2							
1840													
1850	C		Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1860	C		Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1870													
1880			Blind	5615616	1/2 - 2	FLG BLIND 1500# RF LTCS A350 LF2							
1890			Blind	9002863	3 - 12	FLG BLIND 900# RF LTCS A694 F60							
1900	c												
1910			Blind	9001470	6 - 6	FLG BLIND 900# RTJ A694 F60							
1920													
1930													
1940						***** GASKETS *****							
1950													
1960			GASFRS	5672251	3/4 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
1970			GASFRS	5673117	3 - 6	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
1980	C	817	GASRJ	5674491	4 - 6	GASKET 900# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING							
1990													
2000						***** BOLTING *****							
2010		B1											
2020													
2030	C		BOLS	5675770	1/4 - 12	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
2040													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD	CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 19JAN18 BY DN REV C SHEET 4 OF 5 PA 5 CLASS 900HSH					

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10		800				*****MODULAR VALVES*****		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED. THE TEMPERATURE IS LIMITED TO 260°C IN CRUDE OIL SERVICES CONTAINING H2S /AND /OR ORGANIC SULPHUR COMPOUNDS, FOR EXAMPLE, IN THE CRUDE STABILIZATION UNIT. IN HYDROCARBON GAS SERVICE CONTAINING H2S, FOR EXAMPLE, SOUR GAS MOLECULAR SIEVE DEHYDRATION UNITS, USE IS LIMITED TO 230 C. EXCEPTIONS TO THIS ARE THE SULPHUR RECOVERY AND TAIL GAS UNITS WHERE CARBON STEEL SOUR SERVICE LINE CLASSES CAN BE USED UP TO 315°C BECAUSE OF LOWER H2S PARTIAL PRESSURES.				
20		PP9						800					
30													
40	D	VA5	2BV	K15AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2							
50						316 SS TRIM METAL LO - FULL SIZE TYPE 2BV							
60	D		2BV	K15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2							
70						316 SS TRIM METAL LO - FULL SIZE TYPE 2BV							
80	D	VAV	2BV	K09AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2							
90						316 SS TRIM METAL LO FULL SIZE TYPE 2BV							
100	D		2BVR	M15AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2							
110						316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR			812	NIPPLE TO BE CUT FROM SCH XXS PIPE TO BE USED WITH COMPACT FLANGES ONLY. COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's			
120	D		2BVR	M15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2			815				
130						316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR			816				
140	D	VAW	R2BVR	M15EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2	2 - 2		B1				
150						316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR			BA1				
160	D	VAW	R2BVR	M09EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF LTCS A350 LF2	3 - 3		BA5	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED WITH RAISED FACE FLANGES ONLY. TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES. PP9 ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 SB6 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS. SC1 USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.			
170						316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR			FA8				
180	D		2BV	K09AC07	8 - 8	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2			FS1				
190						316SS TRIM TRUNNION SOFT GO FULL BORE TYPE 2BV							
200	D		2BV	K09CC07	14 - 14	MODULAR VALVE DB&B BALL TYPE 900# CF LTCS A350 LF2							
210						316SS TRIM TRUNNION SOFT GO FULL BORE TYPE 2BV							
220		VB2				*****BALL VALVES*****				SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BFH, CI, DHC, FGH, GHC, GRC, GS, PHC, WP MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -46 / 260 °C COMMENT: REFER NOTE 800			
230													
240													
250													
260			Ball V	A15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO							
270													
280			Ball V	A15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO							
290													
300	D		Ball V	A09AC05	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO							
310													
320	D		Ball V	A09AC06	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO							
330													
340	D		Ball V	A09AC07	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO							
350													
360	D		Ball V	A09AC08	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO							
370													
380	D		Ball V	A09CC07-K21	12 - 36	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO							
390													
400	D		Ball V	A09CC08-K21	12 - 36	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO							
410													
420													
430			Ball V	B15AC01	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
440													
450			Ball V	B15AC02	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
460													
470		VA5	Ball V	B15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO							
480													
490		VA5	Ball V	B15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO							
			FGP / WPMP TENGIZCHEVOIL - TENGIZCHEVOIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 17MAR17	BY MC	REV D	SHEET 1 OF 6	PA 5	CLASS 900K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
500	E	VA5	Ball V	B09AC05	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO		UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.						
510								UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS						
520	E	VA5	Ball V	B09AC06	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO			AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST						
530								UDA	FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.						
540	B	VA5	Ball V	B09AC07	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO			UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
550								VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.						
560	B	VA5	Ball V	B09AC08	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO			VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY					
570								VAD	FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE).						
580	D	VA5	Ball V	B09CC07-K21	12 - 36	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO			VAG	FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY.					
590								VAK	FOR ALL INLINE APPLICATIONS AND LEVEL TRANSMITTER CONNECTIONS						
600	D	VA5	Ball V	B09CC08-K21	12 - 36	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO			VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.					
610		VQ2				*****THROUGH CONDUIT GATE VALVES*****			VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055					
620										- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.					
630									VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
640			Condui	W09AC70	3 - 4	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW			VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
650									VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
660			Condui	W09AC71	6 - 10	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO			VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.					
670															
680			Condui	W09CC71	12 - 36	GATE 900# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO									
690						*****GLOBE VALVES*****									
700															
710															
720			Globe	L15AC48	3/4 - 1 1/2	GLOBE 1500# RF LTCS A350 LF2 TRIM 16 OS&Y HW									
730			Globe	L15AC41	2 - 2	GLOBE 1500# RF LTCS A352 LCC TRIM 16 OS&Y HW									
740			Globe	L09AC41	3 - 3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW									
750			Globe	L09AC42	4 - 8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO									
760															
770						*****CHECK VALVES*****									
780															
790															
800		VC1	Check	C15AC50	3/4 - 1 1/2	CHECK PISTON 1500# RF LTCS A350 LF2 TRIM 16				SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BFH, CI, DHC, FGH, GHC, GRC, GS, PHC, WP					
810		VC2	Check	C15AC45	2 - 2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC TRIM 16 THROU BOLTED									
820		VC2	Check	C09AC45	3 - 8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED									
830		VC2	Check	C09AC52	10 - 10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16									
840	D	VC2	Check	C09CC52-K21	12 - 36	CHECK DUAL 900# CF LTCS A352 LCC TRIM 16									
850						*****PIPE*****									
860															
870															
880															
890			Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE				MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 600 RF TEMPERATURE RANGE: -46 / 260 °C COMMENT: REFER NOTE 800					
900	D		Piping	5358515	2 - 3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE									
910			Piping	5358510	4 - 16	PIPE SCH 120 SMLS LTCS A333 GR.6 BE									
920			Piping	5358600	18 - 24	PIPE SCH 120 EFW LTCS A671 CC60 CL.22 BE									
930			Piping	5358627	30 - 30	PIPE WT 50.0MM EFW LTCS A671 CC60 CL.22 BE									
940			Piping	5358625	36 - 36	PIPE WT 59.0MM EFW LTCS A671 CC60 CL.22 BE									
950															
960		812	Piping	5358511	3/4 - 3/4	PIPE SCH XXS SMLS LTCS A333 GR.6 PE									
970															
980						***** FITTINGS *****									
990															
1000															
1010			90 Deg	5392816	3/4 - 3	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6									
1020			90 Deg	5392700	4 - 16	ELL 90 DEG LR SCH 120 LTCS A420 WPL6									
1030			90 Deg	5392388	18 - 24	ELL 90 DEG LR SCH 120 LTCS A420 WPL6-W									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 3 MM	DATE 17MAR17	BY MC	REV D	SHEET 2 OF 6	PA 5	CLASS 900K21		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1040	D		90 Deg	5392547	30 - 30	ELL 90 DEG LR WT 50.0MM LTCS A420 WPL6-W		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
1050			90 Deg	5392538	36 - 36	ELL 90 DEG LR WT 59.0MM LTCS A420 WPL6-W								
1060														
1070			45 Deg	5392822	3/4 - 3	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6								
1080			45 Deg	5392701	4 - 16	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6								
1090			45 Deg	5392392	18 - 24	ELL 45 DEG LR SCH 120 LTCS A420 WPL6-W								
1100			45 Deg	5392549	30 - 30	ELL 45 DEG LR WT 50.0MM LTCS A420 WPL6- W								
1110			45 Deg	5392548	36 - 36	ELL 45 DEG LR WT 59.0MM LTCS A420 WPL6-W								
1120														
1130			Tee	5392825	3/4 - 3	TEE BW SCH 160 LTCS A420 WPL6								
1140			Tee	5392702	4 - 16	TEE BW SCH 120 LTCS A420 WPL6								
1150			Tee	5392404	18 - 24	TEE BW SCH 120 LTCS A420 WPL6-W								
1160			Tee	5392417	30 - 30	TEE BW WT 50.0MM LTCS A420 WPL6- W								
1170			Tee	5392416	36 - 36	TEE BW WT 59.0MM LTCS A420 WPL6-W								
1180														
1190			Reduci	5392821	3/4 - 2	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 3							
1200			Reduci	5392470	1 1/2 - 3	TEE RED SCH 120/160 LTCS A420 WPL6	4 - 6							
1210			Reduci	5392603	4 - 14	TEE RED SCH 120 LTCS A420 WPL6	6 - 16							
1220			Reduci	5392878	8 - 20	TEE RED SCH 120 LTCS A420 WPL6-W	18 - 24							
1230			Reduci	5392872	12 - 24	TEE RED WT 50.0MM/SCH 120 LTCS A420 WPL6-W	30 - 30							
1240			Reduci	5392835	20 - 24	TEE RED WT 59.0MM/SCH 120 LTCS A420 WPL6-W	36 - 36							
1250														
1260			Cap	5392826	3/4 - 3	CAP BW SCH 160 LTCS A420 WPL6								
1270			Cap	5392703	4 - 24	CAP SCH 120 LTCS A420 WPL6								
1280			Cap	5392627	30 - 30	CAP WT 50.0MM LTCS A420 WPL6								
1290			Cap	5392628	36 - 36	CAP SCH 59.0MM LTCS A420 WPL6								
1300														
1310			Concen	5392827	3/4 - 2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 3							
1320			Concen	5392467	1 1/2 - 3	REDUCER CONC SCH 120/160 LTCS A420 WPL6	4 - 6							
1330			Concen	5392704	4 - 14	REDUCER CONC SCH 120 LTCS A420 WPL6	6 - 16							
1340			Concen	5392707	10 - 20	REDUCER CONC SCH 120 LTCS A420 WPL6-W	18 - 24							
1350			Concen	5392314	24 - 24	REDUCER CONC WT 59.0MM/SCH 120 LTCS A420 WPL6-W	36 - 36							
1360			Concen	5392332	20 - 24	REDUCER CONC WT 50.0MM/SCH 120 LTCS A420 WPL6-W	30 - 30							
1370			Concen	5392334	30 - 30	REDUCER CONC WT 59.0MM/50.0MM LTCS A420 WPL6-W	36 - 36							
1380														
1390			Eccent	5392828	3/4 - 2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 3							
1400			Eccent	5392468	1 1/2 - 3	REDUCER ECC SCH 120/160 LTCS A420 WPL6	4 - 6							
1410			Eccent	5392705	4 - 14	REDUCER ECC SCH 120 LTCS A420 WPL6	6 - 16							
1420			Eccent	5392709	10 - 20	REDUCER ECC SCH 120 LTCS A420 WPL6-W	18 - 24							
1430			Eccent	5392317	24 - 24	REDUCER ECC WT 59.0MM/SCH 120 LTCS A420 WPL6-W	36 - 36							
1440			Eccent	5392333	20 - 24	REDUCER ECC WT 50.0MM/SCH 120 LTCS A420 WPL6-W	30 - 30							
1450			Eccent	5392335	30 - 30	REDUCER ECC WT 59.0MM/50.0MM LTCS A420 WPL6-W	36 - 36							
1460														
1470														
1480						***** OLETS *****								
1490														
1500			Nipofl	5504170	3/4 - 2	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2	2 - 3							
1510			Nipofl	5504171	3/4 - 2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2	4 - 24							
1520			Nipofl	5504172	3/4 - 2	NIPOFLANGE 1500# RF 50.0MM/SCH 160 LTCS A350 LF2	30 - 30							
1530			Nipofl	5504173	3/4 - 2	NIPOFLANGE 1500# RF 59.0MM/SCH 160 LTCS A350 LF2	36 - 36							
1540	UJ2	Nipofl	5504163	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2	4 - 24								
1550	UJ2	Nipofl	5504174	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF 50.0MM/160 LTCS A350 LF2	30 - 30								
1560	UJ2	Nipofl	5504175	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF 59.0MM/160 LTCS A350 LF2	36 - 36								
1570														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES		CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY MC REV D SHEET 3 OF 6 PA 5 CLASS 900K21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1580			Weldol	5505392	3/4 - 1	WELDOLET SCH 160 LTCS A350 LF2	2 - 3						
1590			Weldol	5505394	3/4 - 3	WELDOLET SCH 120/160 LTCS A350 LF2	4 - 24						
1600			Weldol	5505625	3/4 - 3	WELDOLET WT 50.0MM/SCH 160 LTCS A350 LF2	30 - 30						
1610			Weldol	5505613	3/4 - 3	WELDOLET WT 59.0MM/SCH 160 LTCS A350 LF2	36 - 36						
1620			Weldol	5505195	4 - 8	WELDOLET SCH 120 LTCS A350 LF2	12 - 24						
1630			Weldol	5505193	4 - 10	WELDOLET WT 50.0MM/SCH 120 LTCS A350 LF2	30 - 30						
1640			Weldol	5505612	4 - 12	WELDOLET WT 59.0MM/SCH 120 LTCS A350 LF2	36 - 36						
1650													
1660			UJ2	Elbole	5516082	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2						4 - 24
1670			UJ2	Elbole	5516083	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF 50.0MM/160 LTCS A350 LF2						30 - 30
1680			UJ2	Elbole	5516084	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF 59.0MM/160 LTCS A350 LF2						36 - 36
1690													
1700	D	UD7	Plug	5491303	3/4 - 1	PLUG ROUND HEAD SCRDR LTCS A350 LF2	2 - 36						
1710	D	UD7	Thredo	5514772	3/4 - 1	THREDOLET 3000# LTCS A350 LF2							
1720													
1730													
1740		816			***** PIPE FLANGES *****								
1750													
1760	C	FA8	Reduci	5599704	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 1500# RF LTCS A350 LF2							3/4 - 1 1/2
1770													
1780	C		Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							3/4 - 1 1/2
1790	C		Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1800													
1810			Flange	5557397	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH 160 BORE							
1820			Flange	5557145	3 - 3	FLG WN 900# RF LTCS A350 LF2 SCH 160							
1830			Flange	5557143	4 - 10	FLG WN 900# RF LTCS A350 LF2 SCH 120 BORE							
1840			Flange	5557143	12 - 24	FLG WN 900# RF LTCS A350 LF2 SCH 120 BORE							
1850			Flange	5556494	30 - 30	FLG WN 900# RF LTCS A350 LF2 WT 50.0MM B16.47 SERIES A							
1860			Flange	5556495	36 - 36	FLG WN 900# RF LTCS A350 LF2 WT 59.0MM B16.47 SERIES A							
1870													
1880	D		Flange	5557395	18 - 18	FLG WN 1500# RF LTCS A350 LF2 SCH 120 BORE	3/4 - 1 1/2						
1890													
1900			Flange	5619821	12 - 24	FLG COMPACT 900# RJ LTCS A350 LF2 SCH 120 BORE							
1910			Flange	5619822	30 - 30	FLG COMPACT 900# RJ LTCS A350 LF2 WT 50.0MM							
1920			Flange	5619823	36 - 36	FLG COMPACT 900# RJ LTCS A350 LF2 WT 59.0MM							
1930													
1940	C		Blind	5615616	1/2 - 2	FLG BLIND 1500# RF LTCS A350 LF2		3/4 - 1 1/2					
1950			Blind	5615605	3 - 10	FLG BLIND 900# RF LTCS A350 LF2							
1960			Blind	5615605	12 - 24	FLG BLIND 900# RF LTCS A350 LF2							
1970			Blind	5621082	30 - 36	FLG BLIND 900# RF LTCS A350 LF2 B16.47 SERIES A							
1980													
1990			Blind	5619773	12 - 36	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 PIPE CLASS 900K21							
2000													
2010													
2020													
2030													
2040			Orific	5624598	2 - 2	WN ORIF SET 1500# RF LTCS A350 LF2 SCH 160 BORE	3/4 - 1 1/2						
2050			Orific	5624537	3 - 3	WN ORIF SET 900# RF LTCS A350 SCH 160 BORE							
2060			Orific	5624532	4 - 10	WN ORIF SET 900# RF LTCS A350 SCH 120 BORE							
2070													
2080	D		Orific	5624538	12 - 24	WN ORIF SET COMPACT 900# RJ LTCS A350 SCH 120 BORE							
2090													
2100													
2110													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES		CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY MC REV D SHEET 4 OF 6 PA 5 CLASS 900K21				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2120 2130 2140 2150 2160 2170 2180 2190 2200 2210 2220 2230 2240 2250 2260 2270 2280 2290 2300 2310 2320 2330 2340 2350 2360 2370 2380 2390 2400 2410 2420 2430 2440 2450 2460 2470 2480 2490 2500 2510 2520 2530 2540 2550 2560 2570 2580 2590 2600 2610 2620 2630 2640	 												

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2650 2660 2670 2680 2690		ZA6				***** CONSTRUCTION DETAILS ***** SEAL WELDING -----FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE							
								SERVICE					
								SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 BFH, CI, DHC, FGH, GHC, GRC, GS, PHC, WP					
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -46 / 260°C COMMENT: REFER NOTE 800					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
						ALL SIZES- BUTT WELD PWHT - ALL SIZES	3 MM	17MAR17	MC	D	6 OF 6	5	900K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
10	A	PP9				*****MODULAR VALVES*****		812	NIPPLE TO BE CUT FROM SCH XXS PIPE								
20								815	TO BE USED WITH COMPACT FLANGES ONLY.								
30								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME'S								
40		VA5	2BV	K15AM10	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	2 - 2	B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.								
60								FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.								
70								PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156								
80		VAV	2BV	K09AM14	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF INC 825 INC 625 TRIM METAL LO FULL SIZE TYPE 2BV		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.								
90								UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.								
100									USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.								
110		2BVR	M15AM10	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	VA5											
120						2BVR		M15AM14	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR							
130											VAW	R2BVR	M15EM10	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 825 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		
140		VAW	R2BVR	M09EM10	1 - 1											MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF INC 825 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	
150						VB2											
160																	
170																	
180																	
190																	
200																	
210																	
220		Ball V	A15AM13	2 - 2	BALL 1500# RF 825 INCONEL 625 TRIM TRUNNION SOFT RB LO												
230						A09AM13		3 - 3	BALL 900# RF 825 INCONEL 625 TRIM TRUNNION SOFT RB LO								
240										A09AR15	4 - 10	BALL 900# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB GO					
250		Ball V	A09CR15-K23	16 - 16	BALL 900# CF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT FB GO												
260																	
270																	
280		Ball V	B15AM09	1 - 1 1/2	BALL 1500# RF 825 INCONEL 625 TRIM FLOATING SOFT FB LO												
290						Ball V		B15AM13	2 - 2	BALL 1500# RF 825 INCONEL 625 TRIM TRUNNION SOFT FB LO							
300											VA5	Ball V	B09AM15	3 - 3	BALL 900# RF 825 INC 625 TRIM TRUNNION SOFT FB LO		
310																	
320	VA5	Ball V	B09AR15	4 - 10	BALL 900# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT FB GO												
330						VA5		Ball V	B09CR15-K23	16 - 16	BALL 900# CF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT FB GO						
340																	
350																	
360	VQ2																
370																	
380																	
390	Condui	W09AM72	3 - 3	GATE 900# RF INC 825 TRIM 625 THROUGH COND SOLID SLAB HW													
400					Condui	W09AR72	4 - 4	GATE 900# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM THROUGH COND SOLID SLAB HW									
									W09AR73	6 - 10	GATE 900# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM THROUGH COND SOLID SLAB GO						
Condui	W09CR73	16 - 16	GATE 900# CF LTCS A352 LCC + 825 CLAD INC 625 TRIM THROUGH COND SOLID SLAB GO														
410																	
420																	
430																	
440																	
450																	
460	Globe	L15AM43	1 - 2	GLOBE 1500# RF INC 825 INC TRIM 625 OS&Y HW													
470					L09AM43	3 - 3	GLOBE 900# RF INC 825 INC TRIM 625 OS&Y HW										
480								L09AR44	4 - 8	GLOBE 900# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM OS&Y GO							
CONSTRUCTION						CORR. ALLOW.	DATE	BY	REV	FGP/WPMP		CLASS					
ALL SIZES- BUTT WELD PWHT - ALL SIZES						6 MM	17NOV16	NPJ	B	1 OF 4	5	900K23					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
490						***** CHECK VALVES *****		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. VB2 SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. VC1 SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. VC2 DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. VQ2 QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
500						VAW																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
510						VC1		Check						C15AM51	1 - 1 1/2	CHECK PISTON 1500# RF INC 825 TRIM 625																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
520						VC2		Check						C15AM46	2 - 2	CHECK DUAL WAFER LUG 1500# RF INC 825 TRIM 625 THROU BOLTED																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
530								VC2						Check	C09AM46	3 - 3	CHECK DUAL WAFER LUG 900# RF INC 825 TRIM 625 THROU BOLTED																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
540														VC2	Check	C09AR46	4 - 8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC + 825 CLAD TRIM INC 625 THROU BOLTED																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
550															VC2	Check	C09AR53	10 - 10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC + 825 CLAD TRIM 625																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
560						VC2										Check	C09CR53-K23	16 - 16	CHECK DUAL 900# CF LTCS A352 LCC + 825 CLAD INC 625 TRIM																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
570								VC2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040			Eccent	5395287	4 - 4	REDUCER ECC SCH 140/XXS LTCS A420 WPL6	8 - 10	VU2 ZA6	USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.				
1050			Eccent	5395288	6 - 6	REDUCER ECC SCH 140/160 LTCS A420 WPL6	8 - 14						
1060			Eccent	5392774	8 - 14	REDUCER ECC SCH 140 LTCS A420 WPL6	10 - 16						
1070													
1080													
1090						***** OLETS *****							
1100													
1110			Nipofl	5504164	1 - 2	NIPOFLANGE 1500# RF XXS LTCS A350 LF2	2 - 4						
1120			Nipofl	5504165	3/4 - 2	NIPOFLANGE 1500# RF SCH 160/XXS LTCS A350 LF2	6 - 6						
1130			Nipofl	5504217	3/4 - 2	NIPOFLANGE 1500# RF SCH 140/XXS LTCS A350 LF2	8 - 16						
1140		UJ2	Nipofl	5504169	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF XXS LTCS A350 LF2	4 - 4						
1150		UJ2	Nipofl	5504167	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 160/XXS LTCS A350 LF2	6 - 6						
1160		UJ2	Nipofl	5504218	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 140/XXS LTCS A350 LF2	8 - 16						
1170													
1180			Weldol	5505197	1 - 1	WELDOLET SCH XXS LTCS A350 LF2	3 - 4						
1190			Weldol	5505396	1 - 2	WELDOLET SCH 160/SCH XXS LTCS A350 LF2	6 - 6						
1200			Weldol	5505181	1 - 4	WELDOLET SCH 140/XXS LTCS A350 LF2	8 - 16						
1210													
1220		UJ2	Elbole	5516077	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF XXS LTCS A350 LF2	4 - 4						
1230		UJ2	Elbole	5516078	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 160/XXS LTCS A350 LF2	6 - 6						
1240		UJ2	Elbole	5516763	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 140/XXS LTCS A350 LF2	8 - 16						
1250													
1260													
1270		816				***** PIPE FLANGES *****							
1280													
1290		FA8	Reduci	5599704	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 1500# RF LTCS A350 LF2	3/4 - 1 1/2						
1300													
1310			Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1320			Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1330													
1340			Flange	5557505	1 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH XXS BORE							
1350													
1360			Flange	5557146	3 - 4	FLG WN 900# RF LTCS A350 LF2 SCH XXS BORE							
1370			Flange	5557145	6 - 6	FLG WN 900# RF LTCS A350 LF2 SCH 160							
1380			Flange	5557717	8 - 10	FLG WN 900# RF LTCS A350 LF2 SCH 140 BORE							
1390				5557717	12 - 16	FLG WN 900# RF LTCS A350 LF2 SCH 140 BORE							
1400													
1410			Flange	5619952	16 - 16	FLG COMPACT 900# RJ LTCS A350 LF2 SCH 140 BORE							
1420													
1430			Blind	5615616	1/2 - 2	FLG BLIND 1500# RF LTCS A350 LF2							
1440			Blind	5615605	3 - 10	FLG BLIND 900# RF LTCS A350 LF2							
1450			Blind	5615605	12 - 16	FLG BLIND 900# RF LTCS A350 LF2							
1460													
1470			Blind	5619774	16 - 16	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 PIPE CLASS 900K23							
1480													
1490													
1500						***** ORIFICE FLANGES *****							
1510													
1520			Orific	5624364	2 - 2	WN ORIF SET 1500# RF LTCS A350 LF2 SCH XXS BORE							
1530			Orific	5624365	3 - 4	WN ORIF SET 900# RF LTCS A350 LF2 SCH XXS BORE							
1540			Orific	5624537	6 - 6	WN ORIF SET 900# RF LTCS A350 SCH 160 BORE							
1550			Orific	5624366	8 - 16	WN ORIF SET 900# RF LTCS A350 LF2 SCH 140 BORE							
1560													
1570													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 6 MM	DATE 17NOV16	BY NPJ	REV A	SHEET 3 OF 4	PA 5	CLASS 900K23

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
1580		812				***** PIPE NIPPLES *****										
1590						NIPPLE BBE - CUT FROM PIPE										
1600																
1610																
1620																
1630						***** GASKETS *****										
1640																
1650						GASFRS								5672251	1/2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR
1660						GASFRS								5673117	3 - 16	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR
1670																
1680		815	GASRJ	5673444	12 - 16	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900K23										
1690																
1700		B1												***** BOLTING *****		
1710																
1720																
1730			BOLS	5675770	1/4 - 16	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS										
1740																
1750		SB6	CPSBOL	5675675	10 - 36	JACK SCREW HEX HEAD A320 GR L7M										
1760																
1770														***** BLINDS AND SPACERS *****		
1780																
1790																
1800			Specta	5749595	3/4 - 2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70										
1810			Specta	5749596	3 - 8	SPECTACLE BLIND 900# RF LTCS A516 GR.70										
1820																
1830			Paddle	5750102	10 - 10	PADDLE BLANK 900# RF LTCS A516 GR.70										
1840			Paddle	5750202	10 - 10	PADDLE SPACER 900# RF LTCS A516 GR.70										
1850																
1860			Paddle	5750102	12 - 16	PADDLE BLANK 900# RF LTCS A516 GR.70										
1870			Paddle	5750202	12 - 16	PADDLE SPACER 900# RF LTCS A516 GR.70										
1880																
1890			Paddle			PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70										
1900		Paddle	5750221	12 - 16	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70											
1910			5750225	12 - 16												
1920						***** BRANCHES *****										
1930																
1940																
1950																
1960						USE BRANCH CHART 900K23										
1970		Reinfo	RWELD	2 - 12	NO-MTO REINFORCING WELD FOR STUB-IN											
1980		Reinfo	RWELD	2 - 12	NO-MTO REINFORCING WELD FOR STUB-IN											
1990																
2000																
2010																
2020						***** CONSTRUCTION DETAILS *****										
2030																
2040						SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY										
2050						PIPE BENDS ----- NONE										
2060						THREAD SEALANT ----- PTFE PASTE										
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD PWHT - ALL SIZES	CORR. ALLOW. 6 MM	DATE 17NOV16	BY NPJ	REV A	FGP/WPMP SHEET 4 OF 4	PA 5	CLASS 900K23			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10						*****VALVES*****		815	TO BE USED WITH COMPACT FLANGES ONLY.				
20								817	USE ONLY FOR TIE IN				
30								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.				
40								FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.				
50								SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE				
60	C	VA5	2BV	K15AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			SPREADERS FOR SPECTACLE BLINDS.				
70								VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW				
80	C		2BV	K15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			DIAGRAMS AND/OR DRAWINGS.				
90								VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.				
100	C	VAV	2BV	K09AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 316 SS TRIM METAL LO FULL SIZE TYPE 2BV		VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055				
110								VB2	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.				
120	C		2BVR	M15AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VC1	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.				
130									SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL				
140	C		2BVR	M15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VC2	POSITION WITH ENGINEERING APPROVAL.				
150									DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.				
160	C	VAW	R2BVR	M15EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2						
170													
180	C	VAW	R2BVR	M09EC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3						
190													
200													
210		VB2				***** BALL VALVES *****							
220													
230	C		Ball V	A15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO							
240													
250	C		Ball V	A15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO							
260													
270	C		Ball V	A09AC05	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO							
280													
290	C		Ball V	A09AC06	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO							
300													
310	C		Ball V	A09AC07	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO							
320													
330	C		Ball V	A09AC08	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO							
340													
350	C		Ball V	A09CC07-K5B	12 - 12	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO							
360													
370	C	VA5	Ball V	A09CC08-K5B	12 - 12	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO							
380													
390													
400	C		Ball V	B15AC01	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO							
410													
420	C		Ball V	B15AC02	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO							
430													
440	C		Ball V	B15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO							
450													
460	C	VA5	Ball V	B15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO							
470													
480													
490	C	VA5	Ball V	B09AC07	3 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO							
500													
510	C	VA5	Ball V	B09AC08	3 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17NOV16	BY JCS	REV C	SHEET 1 OF 5	PA 5	CLASS 900K5B

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
520	C	VA5	Ball V	B09CC07-K5B	12 - 12	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - PHC - MULTIPHASE - RE-ROUTES BD FGH DHC MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS					
530													
540	C	VA5	Ball V	B09CC08-K5B	12 - 12	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO							
550													
560													
570						*****THROUGH CONDUIT GATE VALVES*****							
580													
590	C		Condui	W09AC70	3 - 4	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
600													
610	C		Condui	W09AC71	6 - 10	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
620													
630	C		Condui	W09CC71-K5B	12 - 12	GATE 900# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
640													
650						*****GLOBE VALVES*****							
660													
670													
680	C		Globe	L15AC48	3/4 - 1 1/2	GLOBE 1500# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
690	C		Globe	L15AC41	2 - 2	GLOBE 1500# RF LTCS A352 LCC TRIM 16 OS&Y HW							
700	C		Globe	L09AC41	3 - 3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW							
710	C		Globe	L09AC42	4 - 8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO							
720													
730						*****CHECK VALVES*****							
740													
750													
760		VC1	Check	C15AC50	3/4 - 1 1/2	CHECK PISTON 1500# RF LTCS A350 LF2 TRIM 16							
770		VC2	Check	C15AC45	2 - 2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
780		VC2	Check	C09AC45	3 - 8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
790		VC2	Check	C09AC52	10 - 10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16							
800	C	VC2	Check	C09CC52-K5B	12 - 12	CHECK DUAL 900# CF LTCS A352 LCC TRIM 16							
810													
820						***** PIPE ABOVE GROUND *****							
830													
840													
850			Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE							
860	C		Piping	5358515	2 - 3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE							
870	C		Piping	9005240	4 - 4	LINEPIPE SMLS WT= 8.56 MM API-5L X60 BE							
880			Piping	9001493	6 - 6	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE							
890			Piping	9001495	8 - 8	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE							
900			Piping	9001491	10 - 10	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE							
910			Piping	9005244	12 - 12	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE							
920			Piping	9005243	16 - 16	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE							
930						***** PIPE BELOW GROUND *****							
940													
950													
960			Piping	9001311CTD	6 - 6	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED							
970													
980						*****FITTINGS ABOVE & BELOW GROUND*****							
990													
1000													
1010			45 Deg	5392822	3/4 - 3	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6							
1020			45 Deg	9006567	16 - 16	ELL BW 45 DEG LR WT 19.05MM A860 WPHY 60							
1030													
1040			90 Deg	5392816	3/4 - 3	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6							
1050	C		90 Deg	9006594	4 - 4	ELL BW 90 DEG LR WT 8.56 MM A860 WPHY 60							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17NOV16	BY JCS	REV C	SHEET 2 OF 5	PA 5	CLASS 900K5B

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1060			90 Deg	9006469	12 - 12	ELL BW 90 DEG LR WT 15.88 MM A860 WPHY 60		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - PHC - MULTIPHASE - RE-ROUTES BD FGH DHC MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS						
1070			90 Deg	9006568	16 - 16	ELL BW 90 DEG LR WT 19.05 MM A860 WPHY 60								
1080														
1090			Concen	5392827	3/4 - 2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 3							
1100	C		Concen	9006924	1 1/2 - 3	REDUCER CONC WT 8.56MM / SCH 160 A860 WPHY 60	4 - 4							
1110	C		Concen	9006925	3 - 3	REDUCER CONC WT 9.53MM / SCH 160 A860 WPHY 60	6 - 6							
1120			Concen	9006900	8 - 8	REDUCER CONC WT 14.27MM / 12.7MM LTCS A860 WPHY 60	10 - 10							
1130														
1140			Eccent	5392828	3/4 - 2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 3							
1150	C		Eccent	9004273	1 1/2 - 3	REDUCER ECC WT 8.56MM / SCH 160 LTCS A860 WPHY-60	4 - 4							
1160	C		Eccent	9004274	3 - 3	REDUCER ECC WT 9.53 MM / SCH 160 LTCS A860 WPHY-60	6 - 6							
1170			Eccent	9002410	8 - 8	REDUCER ECC WT 14.27 MM / 12.70 MM LTCS A860 WPHY-60	10 - 10							
1180														
1190			Tee	5392825	3/4 - 3	TEE BW SCH 160 LTCS A420 WPL6								
1200	C		Tee	9004200	12 - 12	TEE BW WT 15.88MM A860 WPHY 60								
1210														
1220			Reduci	5392821	3/4 - 2	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 3							
1230	C		Reduci	9006662	2 - 3	TEE RED BW WT 8.56 MM/SCH 160 LTCS A860 WPHY 60	4 - 4							
1240	C		Reduci	9006665	3 - 3	TEE RED BW WT 9.53 MM/SCH 160 MM LTCS A860 WPHY 60	6 - 6							
1250	C		Reduci	9004207	6 - 6	TEE RED BW WT 15.88 MM/9.53 MM LTCS A860 WPHY 60	12 - 12							
1260			Reduci	9006634	6 - 6	TEE RED BW WT 19.05 MM/9.53 MM LTCS A860 WPHY 60	16 - 16							
1270														
1280	C		Cap	5392826	3/4 - 3	CAP BW SCH 160 LTCS A420 WPL6								
1290	C		Cap	9005200	4 - 4	CAP BW WT 8.56 MM LTCS A860 WPHY-60								
1300	C		Cap	9005060	6 - 6	CAP BW WT 9.53 MM LTCS A860 WPHY-60								
1310	C		Cap	9005067	8 - 8	CAP BW WT 12.70 MM LTCS A860 WPHY-60								
1320	C		Cap	9005068	10 - 10	CAP BW WT 14.27 MM LTCS A860 WPHY-60								
1330	C		Cap	9005069	12 - 12	CAP BW WT 15.88 MM LTCS A860 WPHY-60								
1340	C		Cap	9005072	16 - 16	CAP BW WT 19.05 MM LTCS A860 WPHY-60								
1350														
1360														
1370						***** INDUCTION BENDS BELOW GROUND*****								
1380														
1390			90 Deg	9001815CTD	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1400			45 Deg	9001825CTD	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1410			22.5 D	9001835CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1420														
1430														
1440						***** OLETS ABOVE GROUND *****								
1450														
1460	C		Weldol	9006819	3/4 - 1 1/2	WELDOLET WT 8.56 MM / SCH 160 LTCS A694 F60	4 - 4							
1470	C		Weldol	9002457	3/4 - 2	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	6 - 6							
1480	C		Weldol	9002466	3/4 - 3	WELDOLET WT 12.70 MM/ SCH 160 WT LTCS A694 F60	8 - 8							
1490	C		Weldol	9002470	3/4 - 3	WELDOLET WT 14.27 MM/ SCH 160 WT LTCS A694 F60	10 - 10							
1500	C		Weldol	9002085	3/4 - 3	WELDOLET WT 15.88 MM/ SCH 160 LTCS A694 F60	12 - 12							
1510	C		Weldol	9004335	4 - 4	WELDOLET WT 15.88 MM/ 8.56 MM LTCS A694 F60	12 - 12							
1520	C		Weldol	9004336	3/4 - 3	WELDOLET WT 17.48 MM/SCH 160 LTCS A694 F60	14 - 14							
1530	C		Weldol	9004337	4 - 4	WELDOLET WT 17.48 MM/8.56 MM LTCS A694 F60	14 - 14							
1540	C		Weldol	9002089	3/4 - 3	WELDOLET WT 19.05 MM/ SCH 160 LTCS A694 F60	16 - 16							
1550	C		Weldol	9004338	4 - 4	WELDOLET WT 19.05 MM/ 8.56 MM LTCS A694 F60	16 - 16							
1560	C		Weldol	9004339	6 - 6	WELDOLET WT 19.05 MM/ 9.53 MM LTCS A694 F60	16 - 16							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								3 MM	17NOV16	JCS	C	3 OF 5	5	900K5B

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1570			Nipofl	5504170	1 - 1	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2	2 - 2						
1580													
1590			Nipofl	9002838	3/4 - 2	NIPOFLANGE 1500# RF WT 12.70 MM/ SCH 160 LTCS A694 F60	8 - 8						
1600			Nipofl	9002839	3/4 - 2	NIPOFLANGE 1500# RF WT 14.27 MM/ SCH 160 LTCS A694 F60	10 - 10						
1610			Nipofl	9004322	3/4 - 2	NIPOFLANGE 1500# RF WT 15.88 MM/ SCH 160 LTCS A694 F60	12 - 12						
1620													
1630			Nipofl	9002842	3/4 - 2	NIPOFLANGE 1500# RF WT 19.05 MM/ SCH 160 LTCS A694 F60	16 - 16						
1640													
1650													
1660						*****PIPE FLANGES ABOVE GROUND*****							
1670													
1680			Flange	5557397	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH 160 BORE							
1690			Flange	5557145	3 - 3	FLG WN 900# RF LTCS A350 LF2 SCH 160							
1700			Flange	9004327	4 - 4	FLG WN 900# RF LTCS A694 F60 WT 8.56 MM							
1710			Flange	9004328	6 - 6	FLG WN 900# RF LTCS A694 F60 WT 9.53 MM							
1720			Flange	9002849	8 - 8	FLG WN 900# RF LTCS A694 F60 WT 12.70MM							
1730			Flange	9002850	10 - 10	FLG WN 900# RF LTCS A694 F60 WT 14.27MM							
1740			Flange	9006150	12 - 12	FLG COMPACT 900# RJ A694 F60 WT 15.88 MM							
1750													
1760			Flange	9006154	16 - 16	FLG COMPACT 900# RJ A694 F60 WT 19.05 MM							
1770													
1780			Blind	5615616	3/4 - 2	FLG BLIND 1500# RF LTCS A350 LF2							
1790			Blind	5615605	3 - 3	FLG BLIND 900# RF LTCS A350 LF2							
1800			Blind	9002863	4 - 10	FLG BLIND 900# RF LTCS A694 F60							
1810	C		Blind	9619775	12 - 16	FLG COMPACT BLIND 900# RJ LTCS A694 F60 PIPE CLASS 900K5B							
1820													
1830													
1840		FA8	Reduci	5599704	1/2 - 1	FLG SCRD (FNPT) REDUCING 1500# RF LTCS A350 LF2	3/4 - 1 1/2	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - PHC - MULTIPHASE - RE-ROUTES BD FGH DHC					
1850													
1860	C	817	Flange	5557511	2 - 2	FLG WN 1500# RTJ LTCS A350 LF2 SCH 160 BORE							
1870	C	817	Flange	5557281	3 - 3	FLG WN 900# RTJ LTCS A350 LF2 SCH 160 BORE							
1880	C	817	Flange	9006719	8 - 8	FLG WN 900# RTJ LTCS A694 F60 WT 12.70MM							
1890	C	817	Flange	9006716	10 - 10	FLG WN 900# RTJ LTCS A694 F60 WT 14.27MM							
1900	C	817	Flange	9006718	12 - 12	FLG WN 900# RTJ LTCS A694 F60 WT 15.88MM							
1910													
1920			Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1930			Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1940						***** GASKETS *****							
1950													
1960													
1970	C		GASFRS	5672251	1/2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
1980			GASFRS	5673117	3 - 10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
1990	C	815	GASRJ	5673445	12 - 16	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900K5B							
2000								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS					
2010	C		GASRJ	5674491	3 - 12	GASKET 900# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING							
2020													
2030	C		GASRJ	5674492	2 - 2	GASKET 1500# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING							
2040													
2050													
2060						***** BOLTING *****							
2070		B1											
2080			BOLS	5675770	1/4 - 16	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
2090	D	SB6	CPSBOL	5675675	12 - 16	JACK SCREW HEX HEAD A320 GR L7M							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 17NOV16	BY JCS	REV C	SHEET 4 OF 5	PA 5	CLASS 900K5B

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2100 2110 2120 2130 2140 2150 2160 2170 2180 2190 2200 2210 2220 2230 2240 2250 2260 2270 2280 2290 2300 2310 2320 2330 2340						***** BLINDS & SPACERS		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - PHC - MULTIPHASE - RE-ROUTES BD FGH DHC MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS					
			Specta	5749595	3/4 - 2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70							
			Specta	5749596	3 - 8	SPECTACLE BLIND 900# RF LTCS A516 GR.70							
			Paddle	5750102	10 - 10	PADDLE BLANK 900# RF LTCS A516 GR.70							
			Paddle	5750202	10 - 10	PADDLE SPACER 900# RF LTCS A516 GR.70							
	C		Paddle	5750222	12 - 16	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70	3 - 16						
	C		Paddle	5750228	12 - 16	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70	3 - 16						
						***** BRANCHES *****							
						USE BRANCH CHART 900K5B							
	C		Reinfo	RWELD	2 - 12	NO-MTO REINFORCING WELD FOR STUB-IN							
	C		Reinfo	RWELD	2 - 12	NO-MTO REINFORCING WELD FOR STUB-IN							
						***** CONSTRUCTION DETAILS *****							
						SEAL WELDING ----- REQUIRED PIPE BENDS ----- INDUCTION BENDS							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
						ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	3 MM	17NOV16	JCS	C	5 OF 5	5	900K5B

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
10			Ball V	A15AM14	2 - 2	*****VALVES*****	8 - 8	B1 SB6	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.							
20						*****BALL VALVES*****										
30																
40																
50						BALL 1500# RF 825 INC 625 TRIM TRUNNION METAL RB LO										
60																
70						***** PIPE ABOVE GROUND *****										
80																
90						Piping								9006211	6 - 6	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE EPOXY INTERNALLY COATED
100						Piping								9006226	6 - 6	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE EPOXY INTERNALLY COATED - BELLED ENDS
110						Piping								9001493	6 - 6	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE
120						Piping								9001494	8 - 8	LINEPIPE SMLS WT = 10.31 MM API-5L X60 BE
130																
140																
150																
160																
170			Piping	9006211CTD	6 - 6	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED - EPOXY INTERNALLY COATED										
180			Piping	9006226CTD	6 - 6	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED - EPOXY INTERNALLY COATED - BELLED ENDS										
190			Piping	9001487CTD	6 - 6	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED										
200			Piping	9001493CTD	8 - 8	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED										
210																
220																
230																
240			Cap	9005060	6 - 6	CAP BW WT 9.53 MM LTCS A860 WPHY-60										
250			Cap	9005061	8 - 8	CAP BW WT 10.31 MM LTCS A860 WPHY-60										
260			Eccent	9004137	6 - 6	REDUCER ECC WT 10.31 MM / 9.53 MM LTCS A860 WPHY-60-W										
270			Coupli	9005900	6 - 6	COUPLING INSERT SLEEVE TYPE FOR EPOXY INTERNALLY COATED CARBON STEEL PIPE										
280			Coupli	9005901	6 - 6	COUPLING INSERT SLEEVE TYPE FOR EPOXY INTERNALLY COATED CARBON STEEL PIPE - BELLED END PIPE ONLY										
290																
300																
310																
320	90BND	9001817	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
330	90BND	9001818	8 - 8	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
340																
350	45BND	9001827	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
360	45BND	9001828	8 - 8	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
370																
380	22BND	9006591	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW - EPOXY INTERNALLY COATED												
390	22BND	9006592	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW - EPOXY INTERNALLY COATED - BELLED ENDS												
400																
410	22BND	9001837	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
420																
430	22BND	9001838	8 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW												
440																
450																
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17MAR17 BY JCS REV C SHEET 1 OF 4 PA 5 CLASS 900K5C								

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
460						***** INDUCTION BENDS BELOW GROUND*****		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 140 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - SOUR WATER RE-INJECTION TO BE PE INTERNALLY LINED WITH PULL THROUGH LINER BY THIRD PARTY MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS					
470													
480	C		90BND	9001817CTD	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60							
490	C		90BND	9001818CTD	8 - 8	C/W 1.5D TANGENT ENDS BW COATED INDUCTION BEND 90 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
500													
510	C		45BND	9001827CTD	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60							
520	C		45BND	9001828CTD	8 - 8	C/W 1.5D TANGENT ENDS BW COATED INDUCTION BEND 45 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
530													
540	C		22BND	9006591CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS 9.53 MM API 5L X60							
550						C/W 1.5D TANGENT ENDS BW COATED - EPOXY INTERNALLY COATED							
560	C		22BND	9006592CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS 9.53 MM API 5L X60							
570						C/W 1.5D TGNT ENDS BW COATED - EPOXY INTERNALLY COATED - BELLED							
580	C		22BND	9001837CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L							
590	C		22BND	9001838CTD	8 - 8	X60 C/W 1.5D TANGENT ENDS BW COATED INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.31 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED							
600													
610													
620						***** PIPE FLANGES ABOVE & BELOW GROUND*****							
630													
640			Flange	9004328	6 - 6	FLG WN 900# RF LTCS A694 F60 WT 9.53 MM							
650			Flange	9002848	8 - 8	FLG WN 900# RF LTCS A694 F60 WT 10.31MM							
660			Blind	9002891	2 - 2	FLG BLIND 1500# RF LTCS A694 F60 PE COATED							
670			Blind	9002890	6 - 8	FLG BLIND 900# RF LTCS A694 F60 PE COATED							
680													
690													
700						***** GASKETS *****							
710													
720			GASFRS	5672251	3/4 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
730			GASFRS	5673117	3 - 8	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
740													
750		B1				***** BOLTING *****							
760													
770	C		BOLS	5675770	1/4 - 8	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
780													
790		SB6	BOLS	5675675	10 - 28	JACK SCREW HEX HEAD A320 GR L7M							
800													
810													
820						***** BLINDS & SPACERS							
830													
840			Specta	5749596	6 - 8	SPECTACLE BLIND 900# RF LTCS A516 GR.70							
850													
860													
870						***** BRANCHES *****							
880													
890						USE BRANCH CHART 900K5C							
900													
910	C		Reinfo	RWELD	4 - 6	NO-MTO REINFORCING WELD FOR STUB-IN	6 - 8						
920	C		Reinfo	RWELD	4 - 6	NO-MTO REINFORCING WELD FOR STUB-IN	6 - 8						
930													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 2 OF 3	PA 5	CLASS 900K5C

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
940 950 960 970 980 990 1000 1010 1020	C		BOLS	5675669	1/4 - 99	***** CONSTRUCTION DETAILS ***** EACH COMPLETED LINE TO BE INTERNALLY LINED WITH PULL THROUGH LINER BY THIRD PARTY PIPE BENDS ----- INDUCTION BENDS *****DO NOT USE***** STUD-BOLT A320 GR L7M/A194 GR 7M NUTS								
								SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 140 BARG. @ 90 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE - SOUR WATER RE-INJECTION TO BE PE INTERNALLY LINED WITH PULL THROUGH LINER BY THIRD PARTY						
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 90 °C COMMENT: LTCS						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 3 MM	DATE 17MAR17	BY JCS	REV C	SHEET 3 OF 3	PA 5	CLASS 900K5C

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10						*****VALVES*****		817	USE TO TIE-IN TO EXISTING LINES/EQUIPMENT ONLY						
20								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
30						*****MODULAR VALVES*****		G1	NOT TO BE USED FOR FGP						
40								PL1	PIPE TO BE USED AT A TRANSITION FROM ABOVEGROOUND TO BELOWGROUND						
50		VAV	2BV	K15AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2		SB6	USE FLANGE SPREADERS FOR PADDLE BLANKS/SPACERS AND						
60						316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			FOR SPECTACLE BLINDS.						
70		VA5	2BV	K15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW						
80						316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			DIAGRAMS AND/OR DRAWINGS.						
90		VAV	2BV	K09AC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.						
100						316 SS TRIM METAL LO FULL SIZE TYPE 2BV		VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055						
110			2BVR	M15AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2		VB2	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.						
120						316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VC1	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM						
130			2BVR	M15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2			TEMPERATURE OF 200°C.						
140						316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VC2	SPRING PISTON CHECK VALVES ARE PREFERED IN THE						
150		VAW	R2BVR	M15EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2	2 - 2		HORIZONTAL POSITION						
160						316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR			POSITION WITH ENGINEERING APPROVAL.						
170									DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL						
180									POSITION ONLY.						
190															
200															
210		VB2				***** BALL VALVES *****									
220															
230			Ball V	A15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO									
240															
250			Ball V	A15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO									
260															
270			Ball V	A09AC05	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO									
280															
290			Ball V	A09AC06	3 - 3	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO									
300															
310			Ball V	A09AC07	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO									
320															
330			Ball V	A09AC08	4 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO									
340															
350			Ball V	A09CC07-K5B	12 - 12	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO									
360															
370			Ball V	A09CC08-K5B	12 - 12	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO									
380															
390															
400			Ball V	B15AC01	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
410															
420			Ball V	B15AC02	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO									
430															
440		VA5	Ball V	B15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO									
450															
460		VA5	Ball V	B15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO									
470															
480															
490		VA5	Ball V	B09AC07	3 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
500															
510		VA5	Ball V	B09AC08	3 - 10	BALL 900# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO									
			FE TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FE							
ALL SIZES: BUTTWELD						CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS			
PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND						3 MM	23Jan18	DN	A	1 OF 5	5	900K5D			

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
520		VA5	Ball V	B09CC07-K5B	12 - 12	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO							
530		VA5	Ball V	B09CC08-K5B	12 - 12	BALL 900# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO							
540						*****THROUGH CONDUIT GATE VALVES*****							
550													
560													
570			Condui	W09AC70	3 - 4	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW							
580													
590			Condui	W09AC71	6 - 10	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
600													
610			Condui	W09CC71-K5B	12 - 12	GATE 900# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO							
620													
630						*****GLOBE VALVES*****							
640													
650													
660			Globe	L15AC48	3/4 - 1 1/2	GLOBE 1500# RF LTCS A350 LF2 TRIM 16 OS&Y HW							
670			Globe	L15AC41	2 - 2	GLOBE 1500# RF LTCS A352 LCC TRIM 16 OS&Y HW							
680			Globe	L09AC41	3 - 3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW							
690			Globe	L09AC42	4 - 8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO							
700													
710						*****CHECK VALVES*****							
720													
730													
740		VC1	Check	C15AC50	3/4 - 1 1/2	CHECK PISTON 1500# RF LTCS A350 LF2 TRIM 16							
750		VC2	Check	C15AC45	2 - 2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
760		VC2	Check	C09AC45	3 - 8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED							
770		VC2	Check	C09AC52	10 - 10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16							
780		VC2	Check	C09CC52-K5B	12 - 12	CHECK DUAL DBL FLG 900# CF LTCS A352 LCC TRIM 16							
790													
800						***** PIPE ABOVE GROUND *****							
810													
820													
830			Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE							
840			Piping	5358515	2 - 3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE							
850			Piping	9005240	4 - 4	LINEPIPE SMLS WT= 8.56 MM API-5L X60 BE							
860			Piping	9001493	6 - 6	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE							
870			Piping	9001338	8 - 8	LINEPIPE SMLS WT= 11.1 MM API-5L X60 BE							
880			Piping	9001491	10 - 10	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE							
890			Piping	9005244	12 - 12	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE							
900			Piping	9005243	16 - 16	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE							
910						***** PIPE BELOW GROUND *****							
920													
930													
940			Piping	9001311CTD	6 - 6	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED							
950		PL1	Piping	9001338CTD	8 - 8	LINEPIPE WT 11.1 MM SMLS API 5L X60 BE DUAL COATED							
960			Piping	9001348C1	8 - 8	LINEPIPE WT 9.5 MM SMLS API 5L X60 BE DUAL COATED							
970						*****FITTINGS ABOVE & BELOW GROUND*****							
980													
990													
1000			45 Deg	5392822	3/4 - 3	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6							
1010			45 Deg	9006570	6 - 6	ELL BW 45 DEG LR WT 9.53 MM A860 WPHY 60 - W							
1020			45 Deg	9002174	8 - 8	ELL BW 45 DEG LR WT 11.1MM A860 WPHY 60							
1030			45 Deg	9006567	16 - 16	ELL BW 45 DEG LR WT 19.05MM A860 WPHY 60							
1040													
1050			90 Deg	5392816	3/4 - 3	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6							
1060			90 Deg	9006594	4 - 4	ELL BW 90 DEG LR WT 8.56 MM A860 WPHY 60							
1070			90 Deg	9006471	6 - 6	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60							
1080			90 Deg	9002158	8 - 8	ELL BW 90 DEG LR WT 11.1MM A860 WPHY 60							
			FE TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 23Jan18	BY DN	REV A	SHEET 2 OF 5	PA 5	CLASS 900K5D

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1090			90 Deg	9006469	12 - 12	ELL BW 90 DEG LR WT 15.88 MM A860 WPHY 60		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 120 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - PHC - MULTIPHASE - RE-ROUTES BD FGH DHC MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 120 °C COMMENT: LTCS						
1100			90 Deg	9006568	16 - 16	ELL BW 90 DEG LR WT 19.05 MM A860 WPHY 60								
1110														
1120			Concen	5392827	3/4 - 2	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 3							
1130			Concen	9006924	1 1/2 - 3	REDUCER CONC WT 8.56 MM/ SCH 160 A860 WPHY 60	4 - 4							
1140			Concen	9006925	3 - 3	REDUCER CONC WT 9.53 MM/ SCH 160 A860 WPHY 60	6 - 6							
1150			Concen	9006905	8 - 8	REDUCER CONC WT 14.27 MM/ 11.13MM LTCS A860 WPHY 60	10 - 10							
1160														
1170			Eccent	5392828	3/4 - 2	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 3							
1180			Eccent	9004273	1 1/2 - 3	REDUCER ECC WT 8.56 MM / SCH 160 LTCS A860 WPHY-60	4 - 4							
1190			Eccent	9004274	3 - 3	REDUCER ECC WT 9.53 MM / SCH 160 LTCS A860 WPHY-60	6 - 6							
1200			Eccent	9001394	8 - 8	REDUCER ECC WT 15.88 MM/11.13 MM LTCS A860 WPHY-60	12 - 12							
1210			Eccent	9002409	8 - 8	REDUCER ECC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60	10 - 10							
1220														
1230			Tee	5392825	3/4 - 3	TEE BW SCH 160 LTCS A420 WPL6								
1240			Tee	9004210	6 - 6	TEE BW WT 9.53 MM A860 WPHY 60-W								
1250			Tee	9002114	8 - 8	TEE BW WT 11.1 MM A860 WPHY 60								
1260			Tee	9004200	12 - 12	TEE BW WT 15.88 MM A860 WPHY 60								
1270														
1280			Reduci	5392821	3/4 - 1 1/2	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 2							
1290			Reduci	5392821	1 1/2 - 2	TEE RED BW SCH 160 LTCS A420 WPL6	3 - 3							
1300			Reduci	9006662	2 - 3	TEE RED BW WT 8.56 MM/SCH 160 LTCS A860 WPHY 60	4 - 4							
1310			Reduci	9006665	3 - 3	TEE RED BW WT 9.53 MM/SCH 160 MM LTCS A860 WPHY 60	6 - 6							
1320			Reduci	9006021	4 - 4	TEE RED BW WT 9.53 MM/8.56 MM LTCS A860 WPHY 60	6 - 6							
1330			Reduci	9002134	4 - 4	TEE RED BW WT 11.1 MM/8.56 MM LTCS A860 WPHY 60	8 - 8							
1340			Reduci	9004206	6 - 6	TEE RED BW WT 11.1 MM/9.53 MM LTCS A860 WPHY 60	8 - 8							
1350			Reduci	9004207	6 - 6	TEE RED BW WT 15.88 MM/9.53 MM LTCS A860 WPHY 60	12 - 12							
1360			Reduci	9006634	6 - 6	TEE RED BW WT 19.05 MM/9.53 MM LTCS A860 WPHY 60	16 - 16							
1370			Reduci	9006620	8 - 8	TEE RED BW WT 14.27MM/ WT 11.13MM A860 WPHY 60	10 - 10							
1380			Reduci	9006640	8 - 8	TEE RED BW WT 15.88MM/ WT 11.13MM A860 WPHY 60	12 - 12							
1390														
1400			Cap	5392826	3/4 - 3	CAP BW SCH 160 LTCS A420 WPL6								
1410			Cap	9005200	4 - 4	CAP BW WT 8.56 MM LTCS A860 WPHY-60								
1420			Cap	9005060	6 - 6	CAP BW WT 9.53 MM LTCS A860 WPHY-60								
1430			Cap	9002306	8 - 8	CAP BW WT 11.1MM LTCS A860 WPHY 60								
1440			Cap	9005067	8 - 8	CAP BW WT 12.70 MM LTCS A860 WPHY-60								
1450			Cap	9005068	10 - 10	CAP BW WT 14.27 MM LTCS A860 WPHY-60								
1460			Cap	9005069	12 - 12	CAP BW WT 15.88 MM LTCS A860 WPHY-60								
1470			Cap	9005072	16 - 16	CAP BW WT 19.05 MM LTCS A860 WPHY-60								
1480														
1490						***** INDUCTION BENDS BELOW GROUND*****								
1500														
1510			90 Deg	9001815CTD	6 - 6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1520			45 Deg	9001825CTD	6 - 6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1530			22.5 D	9001835CTD	6 - 6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED								
1540			22.5 D	9001938CTD	8 - 8	INDUCTION BEND 22.5 DEG X 5D RADIUS C/W 1.5D TANGENTS WT 11.1MM SMLS API 5L X60 BE DUAL COATED								
1550														
1560						***** OLETS ABOVE GROUND *****								
1570														
1580			Weldol	5505392	3/4 - 1	WELDOLET SCH 160 LTCS A350 LF2	3 - 3							
1590			Weldol	9006819	3/4 - 1	WELDOLET WT 8.56 MM / SCH 160 LTCS A694 F60	4 - 4							
1600			Weldol	9002457	3/4 - 2	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	6 - 6							
1610			Weldol	9002076	3/4 - 3	WELDOLET WT 11.1 MM/ SCH 160 WT LTCS A694 F60	8 - 8							
			FE TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FE						
						ALL SIZES: BUTTWELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
						PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	3 MM	23Jan18	DN	A	3 OF 5	5	900K5D	

SEQ.NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1620			Weldol	9002470	3/4 - 3	WELDOLET WT 14.27 MM/ SCH 160 LTCS A694 F60	10 - 10	SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 120 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - PHC - MULTIPHASE - RE-ROUTES BD FGH DHC MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 120 °C COMMENT: LTCS					
1630			Weldol	9002085	3/4 - 3	WELDOLET WT 15.88 MM/ SCH 160 LTCS A694 F60	12 - 12						
1640			Weldol	9004335	4 - 4	WELDOLET WT 15.88 MM/ 8.56 MM LTCS A694 F60	12 - 12						
1650			Weldol	9004336	3/4 - 3	WELDOLET WT 17.48 MM/SCH 160 LTCS A694 F60	14 - 14						
1660			Weldol	9004337	4 - 4	WELDOLET WT 17.48 MM/8.56 MM LTCS A694 F60	14 - 14						
1670			Weldol	9002089	3/4 - 3	WELDOLET WT 19.05 MM/ SCH 160 LTCS A694 F60	16 - 16						
1680			Weldol	9004338	4 - 4	WELDOLET WT 19.05 MM/ 8.56 MM LTCS A694 F60	16 - 16						
1690			Weldol	9004339	6 - 6	WELDOLET WT 19.05 MM/ 9.53 MM LTCS A694 F60	16 - 16						
1700													
1710			Nipofl	5504170	3/4 - 1 1/2	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2	2 - 2						
1720													
1730			Nipofl	9004319	3/4 - 2	NIPOFLANGE 1500# RF WT 8.56 MM/ SCH 160 LTCS A694 F60	4 - 4						
1740			Nipofl	9004321	3/4 - 2	NIPOFLANGE 1500# RF WT 11.1 MM/ SCH 160 LTCS A694 F60	8 - 8						
1750			Nipofl	9002839	3/4 - 2	NIPOFLANGE 1500# RF WT 14.27 MM/ SCH 160 LTCS A694 F60	10 - 10						
1760			Nipofl	9004322	3/4 - 2	NIPOFLANGE 1500# RF WT 15.88 MM/ SCH 160 LTCS A694 F60	12 - 12						
1770													
1780			Nipofl	9002842	3/4 - 2	NIPOFLANGE 1500# RF WT 19.05 MM/ SCH 160 LTCS A694 F60	16 - 16						
1790													
1800						*****PIPE FLANGES ABOVE GROUND*****							
1810													
1820													
1830			Flange	5557397	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH 160 BORE							
1840			Flange	5557145	3 - 3	FLG WN 900# RF LTCS A350 LF2 SCH 160							
1850			Flange	9004327	4 - 4	FLG WN 900# RF LTCS A694 F60 WT 8.56 MM							
1860			Flange	9004328	6 - 6	FLG WN 900# RF LTCS A694 F60 WT 9.53 MM							
1870			Flange	9002857	8 - 8	FLG WN 900# RF LTCS A694 F60 WT 11.13MM							
1880													
1890			Flange	9002850	10 - 10	FLG WN 900# RF LTCS A694 F60 WT 14.27MM							
1900			Flange	9006150	12 - 12	FLG COMPACT 900# RJ A694 F60 WT 15.88 MM							
1910													
1920			Flange	9006154	16 - 16	FLG COMPACT 900# RJ A694 F60 WT 19.05 MM							
1930													
1940			Blind	5615616	3/4 - 2	FLG BLIND 1500# RF LTCS A350 LF2							
1950			Blind	5615605	3 - 3	FLG BLIND 900# RF LTCS A350 LF2							
1960			Blind	9002863	4 - 10	FLG BLIND 900# RF LTCS A694 F60							
1970			Blind	9619775	12 - 16	FLG COMPACT BLIND 900# RJ LTCS A694 F60 PIPE CLASS 900K5B							
1980													
1990													
2000		FA8	Reduci	5599704	1/2 - 1	FLG SCR D (FNPT) REDUCING 1500# RF LTCS A350 LF2	3/4 - 1 1/2						
2010													
2020		817	Flange	5557511	3/4 - 2	FLG WN 1500# RTJ LTCS A350 LF2 SCH 160 BORE							
2030		817	Flange	5557281	3 - 3	FLG WN 900# RTJ LTCS A350 LF2 SCH 160 BORE							
2040		817	Flange	9001538	8 - 8	FLG WN 900# RTJ LTCS A694 F60 WT 11.1MM							
2050													
2060		817	Flange	9006716	10 - 10	FLG WN 900# RTJ LTCS A694 F60 WT 14.27MM							
2070		817	Flange	9006718	12 - 12	FLG WN 900# RTJ LTCS A694 F60 WT 15.88MM							
2080													
2090			Flange	5851205	1/2 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
2100			Flange	5851205	2 - 3	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
2110													
2120						***** GASKETS *****							
2130													
2140			GASFRS	5672251	1/2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
2150			GASFRS	5673117	3 - 10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
2160		815	GASRJ	5673445	12 - 16	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900K5B							
2170													
2180													
			FE TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES: BUTTWELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND	CORR. ALLOW. 3 MM	DATE 23Jan18	BY DN	REV A	SHEET 4 OF 5	PA 5	CLASS 900K5D

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2190		B1	GASRJ	5674491	3 - 12	GASKET 900# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING		SERVICE SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 120 DEG.C. (LIMITED BY PIPE WALL THICKNESS) SERVICE PIPELINES/PIPING - PHC - MULTIPHASE - RE-ROUTES BD FGH DHC MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 900 RF TEMPERATURE RANGE: -40 / 120 °C COMMENT: LTCS						
2200			GASRJ	5674492	3/4 - 1 1/2	GASKET 1500# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING								
2210			GASRJ	5674492	2 - 2	GASKET 1500# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING								
2220						***** BOLTING *****								
2230														
2240														
2250			BOLS	5675770	1/2 - 16	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
2260														
2270														
2280														
2290														
2300			Specta	5749595	3/4 - 2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70								
2310			Specta	5749596	3 - 8	SPECTACLE BLIND 900# RF LTCS A516 GR.70								
2320														
2330			Paddle	5750102	10 - 10	PADDLE BLANK 900# RF LTCS A516 GR.70								
2340			Paddle	5750202	10 - 10	PADDLE SPACER 900# RF LTCS A516 GR.70								
2350														
2360			Paddle	5750222	12 - 16	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70								
2370			Paddle	5750228	12 - 16	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70								
2380														
2390														
2400														
2410														
2420														
2430														
2440														
2450														
			FE TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION								
						ALL SIZES: BUTTWELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
						PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND		3 MM	23Jan18	DN	A	5 OF 5	5	900K5D

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10		PP9				*****MODULAR VALVES*****		O32	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.						
20								810	NIPPLE TO BE CUT FROM SCH 160 PIPE.						
30								814	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.						
40	D	VA5	2BV	K15AS22	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		815	TO BE USED WITH COMPACT FLANGES ONLY.						
50								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME'S						
60	D		2BV	K15AS26	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - FULL SIZE TYPE 2BV		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
70								BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.						
80	D	VAV	2BV	K09AS26	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF SS A182 F316 316SS TRIM METAL LO 718 BOLT FULL SIZE TYPE 2BV		BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.						
90								FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.						
100	D		2BVR	M15AS22	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
110								SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
120	D		2BVR	M15AS26	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO 718 BOLT - CONTROLLED BORE TYPE 2BVR		SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTRATRACE.						
130								SP1	8 INCH NB PADDLE BLANKS & SPACERS SHALL ONLY BE USED AROUND THE COOLERS.						
140	D	VAW	R2BVR	M15ES22	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING -CONTROLLED BORE TYPE 2BVR	2 - 2	UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.						
150								UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST						
160	D	VAW	R2BVR	M09ES22	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# RF SS A182 F316SS TRIM METAL LO 718 BOLT REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	UDA	FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.						
170															
180															
190															
200		VB2				*****BALL VALVES*****									
210	D		Ball V	A15AS25	2 - 2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB LO 718 BOLT									
220															
230			Ball V	A15AS26	2 - 2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION METAL RB LO 718 BOLT									
240															
250	D		Ball V	A09AS25	3 - 3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB LO 718 BOLT									
260															
270	D		Ball V	A09AS26	3 - 3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL RB LO 718 BOLT									
280															
290	D		Ball V	A09AS27	4 - 10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT									
300															
310	D		Ball V	A09AS28	4 - 10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT									
320															
330	C		Ball V	A09CS27	12 - 30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO 718 BOLT									
340															
350	C		Ball V	A09CS28	12 - 30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION METAL RB GO 718 BOLT									
360															
370															
380			Ball V	B15AS21	3/4 - 1 1/2	BALL 1500# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO 718 BOLT									
390															
400			Ball V	B15AS22	3/4 - 1 1/2	BALL 1500# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO 718 BOLT									
410															
420		VA5	Ball V	B15AS25	2 - 2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB LO 718 BOLT									
430															
440		VA5	Ball V	B15AS26	2 - 2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION METAL FB LO 718 BOLT									
450															
460															
470	D		Ball V	B09AS25	3 - 3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB LO 718 BOLT									
480															
490	D		Ball V	B09AS26	3 - 3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL FB LO 718 BOLT									
500															
510	D	VA5	Ball V	B09AS27	4 - 10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 1 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17MAR17 MC D 1 OF 6 5 900P21						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
520	D	VA5	Ball V	B09AS28	4 - 10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
530														
540	C	VA5	Ball V	B09CS27	12 - 30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO 718 BOLT		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
550														
560	C	VA5	Ball V	B09CS28	12 - 30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION METAL FB GO 718 BOLT		VAB	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY					
570														
580						*****GLOBE VALVES*****		VAD	FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE).					
590														
600			Globe	L15AS55	3/4 - 1 1/2	GLOBE 1500# RF A182 F316 SS TRIM 16 OS&Y HW 718 BOLT		VAG	FOR INSTRUMENT CONNECTIONS TO VESSELS ONLY.					
610			Globe	L15AS56	2 - 2	GLOBE 1500# RF SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT		VAK	FOR ALL INLINE APPLICATIONS AND LEVEL TRANSMITTER CONNECTIONS					
620			Globe	L09AS56	3 - 3	GLOBE 900# RF SS A351 CF8M TRIM 16 OS&Y HW 718 BOLT 718 BOLT		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.					
630			Globe	L09AS57	4 - 8	GLOBE 900# RF SS A351 CF8M TRIM 16 OS&Y GO 718 BOLT		VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055					
640						*****CHECK VALVES*****								
650														
660														
670		VC1	Check	C15AS54	3/4 - 1 1/2	CHECK PISTON 1500# RF SS A182 F316 TRIM 16 718 BOLT		VB2	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
680		VC2	Check	C15AS45	2 - 2	CHECK DUAL WAFER LUG 1500# RF SS A351 CF8M TRIM 16 THROU BOLTED		VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
690		VC2	Check	C09AS45	3 - 8	CHECK DUAL WAFER LUG 900# RF SS A351 CF8M TRIM 16 THROU BOLTED		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
700		VC2	Check	C09AS52	10 - 10	CHECK DUAL DBL FLG 900# RF SS A351 CF8M TRIM 16		VU2	USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY.					
710	C	VC2	Check	C09CS52	12 - 30	CHECK DUAL 900# CF SS A351 CF8M TRIM 16		ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
720						*****PIPE*****								
730														
740														
750														
760	C		Piping	5365140	3/4 - 1 1/2	PIPE SCH 80S SMLS SS A312 TP316/316L PE								
770	D		Piping	5365138	2 - 3	PIPE SCH 80S SMLS SS A312 TP316/316L BE								
780	C		Piping	5365128	4 - 24	PIPE SCH 120 SMLS SS A312 TP316/316L BE								
790			Piping	5365217	26 - 26	PIPE WT 42.0MM EFW SS A358 316/316L BE								
800			Piping	5365218	28 - 28	PIPE WT 44.0MM EFW SS A358 316/316L BE								
810			Piping	5365216	30 - 30	PIPE WT 47.0MM EFW SS A358 316/316L BE								
820			Piping	5365219	36 - 36	PIPE WT 56.0MM EFW SS A358 316/316L BE								
830														
840		814	Piping	5365207	3/4 - 3/4	PIPE SCH 160 SMLS SS A312 TP316/316L PE								
850														
860						***** FITTINGS *****								
870														
880			90 Deg	5449033	3/4 - 1 1/2	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S								
890	C		90 Deg	5449033	2 - 3	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S								
900	C		90 Deg	5449570	4 - 24	ELL 90 DEG LR SCH 120 SS A403 WP316/316L-S								
910			90 Deg	5451124	26 - 26	ELL 90 DEG LR WT 42.0MM SS A403 WP316/316L-W			AA, BD, BF, BFH, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM					
920			90 Deg	5451125	28 - 28	ELL 90 LR WT 44.0MM 316/316L SS A403 WP316/316L-W								
930			90 Deg	5451136	30 - 30	ELL 90 LR WT 47.0MM SS A403 WP316/316L-W								
940			90 Deg	5451127	36 - 36	ELL 90 LR WT 56.0MM SS A403 WP316/316L-W			VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS					
950														
960			45 Deg	5449146	3/4 - 3	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S			MATERIAL: 316/316L SS DUAL GRADE					
970	C		45 Deg	5449576	4 - 24	ELL 45 DEG LR SCH 120 SS A403 WP316/316L-S			FLANGE RATING: 900 RF					
980			45 Deg	5449084	26 - 26	ELL 45 DEG LR WT 42.0MM SS A403 WP316/316L-W			TEMPERATURE RANGE: -140 / 350 °C					
990			45 Deg	5449085	28 - 28	ELL 45 DEG LR WT 44.0MM SS A403 WP316/316L-W			COMMENT: LOW TEMP BLOW DOWN					
1000			45 Deg	5451126	30 - 30	ELL 45 LR WT 47.0MM SS A403 WP316/316L-W								
1010			45 Deg											
1020														
1030	D		Tee	5449156	3/4 - 3	TEE SCH 80S SS A403 WP316/316L-S								
1040	C		Tee	5449579	4 - 24	TEE SCH 120 SS A403 WP316/316L-S								
1050			Tee	5449087	26 - 26	TEE 42.0MM SS A403 WP316/316L-W								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 1 MM	FGP/WPMP DATE 17MAR17 BY MC REV D SHEET 2 OF 6 PA 5 CLASS 900P21					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1060			Tee	5449088	28 - 28	TEE 44.0MM SS A403 WP316/316L-W							
1070			Tee	5451116	30 - 30	TEE 47.0MM SS A403 WP316/316L-W							
1080			Tee	5449089	36 - 36	TEE 56.0MM SS A403 WP316/316L-W							
1090													
1100	D		Reduci	5448905	3/4 - 2	TEE RED SCH 80S SS A403 WP316/316L-S	1 - 3						
1110	C		Reduci	5449577	2 - 3	TEE RED SCH 120/80S SS A403 WP316/316L-S	4 - 6						
1120	C		Reduci	5449578	4 - 20	TEE RED SCH 120 SS A403 WP316/316L-S	6 - 24						
1130			Reduci	5449094	12 - 24	TEE RED WT 42.0MM X SCH 120 SS A403 WP316/316L-W	26 - 26						
1140			Reduci	5449054	14 - 24	TEE RED WT 47.0MM X SCH 120 SS A403 WP316/316L-W	30 - 30						
1150													
1160	D		Cap	5449049	3/4 - 3	CAP SCH 80S SS A403 WP316/316L-S							
1170	C		Cap	5449052	4 - 24	CAP SCH 120 SS A403 WP316/316L-S							
1180			Cap	5449090	26 - 26	CAP WT 42.0MM SS A403 WP316/316L							
1190			Cap	5449091	28 - 28	CAP WT 44.0MM SS A403 WP316/316L							
1200			Cap	5449097	30 - 30	CAP WT 47.0MM SS A403 WP316/316L							
1210			Cap	5449093	36 - 36	CAP WT 56.0MM SS A403 WP316/316L							
1220													
1230	D		Concen	5449158	3/4 - 2	REDUCER CONC SCH 80S SS A403 WP316/316L-S	1 - 3						
1240	D		Concen	5448916	1 1/2 - 3	REDUCER CONC SCH 120/80S SS A403 WP316/316	4 - 6						
1250	D		Concen	5448915	4 - 20	REDUCER CONC SCH 120 SS A403 WP316/316L-SL-S	6 - 24						
1260			Concen	5449095	18 - 24	REDUCER CONC 42.0MM WT/SCH 120 SS A403 WP316/316L-W	26 - 26						
1270			Concen	5450766	24 - 24	REDUCER CONC 47.0MM / SCH 120 SS A403 WP316/316L-W	30 - 30						
1280			Concen	5450753	24 - 24	REDUCER CONC 56.0MM / SCH 120 SS A403 WP316/316L-W	36 - 36						
1290			Concen	5450757	30 - 30	REDUCER CONC 56.0MM/ 47.0MM SS A403 WP316/316L-W	36 - 36						
1300													
1310	D		Eccent	5449159	3/4 - 2	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1 - 3						
1320	D		Eccent	5449179	1 1/2 - 3	REDUCER ECC SCH 120/80S SS A403 WP316/316L-S	4 - 6						
1330	C		Eccent	5449178	4 - 20	REDUCER ECC SCH 120 SS A403 WP316/316L-S	6 - 24						
1340			Eccent	5449096	18 - 24	REDUCER ECC 42.0MM WT/SCH 120 SS A403 WP316/316L-W	26 - 26						
1350			Eccent	5450716	24 - 24	REDUCER ECC 47.0MM / SCH 120 SS A403 WP316/316L-W	30 - 30						
1360			Eccent	5450755	24 - 24	REDUCER ECC 56.0MM / SCH 120 SS A403 WP316/316L-W	36 - 36						
1370			Eccent	5450734	30 - 30	REDUCER ECC 56.0MM / 47.0MM SS A403 WP316/316L-W	36 - 36						
1380													
1390	D		Nipofl	5504176	3/4 - 1 1/2	NIPOFLANGE 1500# RF SCH 80S SS A182 F316/316L	2 - 3						
1400	D		Nipofl	5504176	2 - 2	NIPOFLANGE 1500# RF SCH 80S SS A182 F316/316L	3 - 3						
1410	D		Nipofl	5504177	3/4 - 2	NIPOFLANGE 1500# RF SCH 120/80S SS A182 F316/316L	4 - 24	SERVICE					
1420	D		Nipofl	5504178	3/4 - 2	NIPOFLANGE 1500# RF 42.0MM WT/SCH 80S SS A182 F316/316L	26 - 26	NACE STD. MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3					
1430	D		Nipofl	5504179	3/4 - 2	NIPOFLANGE 1500# RF 47.0MM WT/SCH 80S SS A182 F316/316L	30 - 30						
1440	D		Nipofl	5504189	3/4 - 2	NIPOFLANGE 1500# RF 56.0MM WT/SCH 80S SS A182 F316/316L	36 - 36						
1450								VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS					
1460		UJ2	Nipofl	5504180	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 120/160 SS A182 F316/316L	4 - 24						
1470		UJ2	Nipofl	5504181	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF 42.0MM WT/SCH 160 SS A182 F316/316L	26 - 26						
1480		UJ2	Nipofl	5504182	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF 47.0MM WT/SCH 160 SS A182 F316/316L	30 - 30						
1490	C	UJ2	Nipofl	5504188	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF 56.0MM WT/SCH 160 SS A182 F316/316L	36 - 36						
1500													
1510	D		Weldol	5509637	3/4 - 1	WELDOLET SCH 80S SS A182 F316/316L	2 - 3						
1520	D		Weldol	5510374	3/4 - 3	WELDOLET SCH 120 /80S SS A182 316/316L	4 - 24						
1530			Weldol	5509649	2 - 3	WELDOLET 42.0MM WT/SCH 80S SS A182 F316/316L	26 - 26						
1540			Weldol	5509647	2 - 3	WELDOLET 47.0MM WT/SCH 80S SS A182 F316/316L	30 - 30						
1550			Weldol	5509640	4 - 8	WELDOLET SCH 120 SS A182 F316/316L	12 - 24	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 900 RF TEMPERATURE RANGE: -140 / 350 °C COMMENT: LOW TEMP BLOW DOWN					
1560			Weldol	5509646	4 - 10	WELDOLET 42.0MM WT/SCH 120 SS A182 F316/316L	26 - 26						
1570			Weldol	5509648	4 - 12	WELDOLET 47.0MM WT/SCH 120 SS A182 F316/316L	30 - 30						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	FGP/WPMP					
								DATE	BY	REV	SHEET	PA	CLASS
								17MAR17	MC	D	3 OF 6	5	900P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1580	C	UJ2	Elbole	5516085	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 120/160 SS A182 F316/316L	4 - 24							
1590		UJ2	Elbole	5516086	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF 42.0MM WT/SCH 160 SS A182 F316/316L	26 - 26							
1600		UJ2	Elbole	5516087	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF 47.0MM WT/SCH 160 SS A182 F316/316L	30 - 30							
1610		UJ2	Elbole	5516088	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF 56.0MM WT/SCH 160 SS A182 F316/316L	36 - 36							
1620														
1630														
1640						*****COUPLINGS,PLUG,UNION*****								
1650														
1660			Coupli	5480142	1/2 - 1 1/2	COUPLING 3000# SW SS A182 F316/316L	1 - 1 1/2							
1670			Reduci	5480134	3/4 - 1	COUPLING RED 3000# SW SS A182 F316/316L								
1680	UD6	Union	5480146	3/4 - 1 1/2	UNION 3000# SW SS A182 F316/316L									
1690		Cap	5498274	3/4 - 3/4	CAP 3000# SCRD A182 F316/316L									
1700	UD7	Plug	5498383	3/4 - 3/4	PLUG ROUND HEAD SCRD SS A182 F316/316L									
1710						*****OLETS*****								
1720														
1730														
1740	UD7	Sockol	5537071	3/4 - 1 1/2	SOCKOLET 3000# SS A182 F316/316L	2 - 26								
1750		Endole	5533235	3/4 - 1 1/2	FLAT SOCKOLET 3000# SS A182 F316/316L	2 - 30								
1760		Thredo	5515771	3/4 - 3/4	THREDOLET 3000# SS A182 F316/316L									
1770														
1780	816					***** PIPE FLANGES *****								
1790														
1800														
1810	FA8	Reduci	5600664	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 1500# RF SS A182 F316/316L	3/4 - 1 1/2								
1820														
1830	C	Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE									
1840	C	Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE									
1850														
1860	D	Flange	5583979	3/4 - 2	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE									
1870		Flange	5583006	3 - 3	FLG WN 900# RF SS A182 F316/316L SCH 80S BORE									
1880		Flange	5583813	4 - 10	FLG WN 900# RF SS A182 F316/316L SCH 120 BORE									
1890		Flange	5583813	12 - 24	FLG WN 900# RF SS A182 F316/316L SCH 120 BORE									
1900	C	Flange	5583018	26 - 26	FLG WN 900# RF SS A182 F316/316L 42.0MM WT B16.47 SERIES A									
1910		Flange	5583019	30 - 30	FLG WN 900# RF SS A182 F316/316L 47.0MM WT B16.47 SERIES A									
1920		Flange	5583020	36 - 36	FLG WN 900# RF SS A182 F316/316L 56.0MM WT B16.47 SERIES A									
1930														
1940	C	Flange	5619824	12 - 24	FLG COMPACT 900# RJ SS A182 F316/316L SCH 120 BORE	SERVICE								
1950		Flange	5619825	26 - 26	FLG COMPACT 900# RJ SS A182 F316/316L WT 42.0MM									
1960		Flange	5619826	28 - 28	FLG COMPACT 900# RJ SS A182 F316/316L WT 44.0MM									
1970		Flange	5619827	30 - 30	FLG COMPACT 900# RJ SS A182 F316/316L WT 47.0MM									
1980		Flange	5619828	36 - 36	FLG COMPACT 900# RJ SS A182 F316/316L WT 56.0MM	AA, BD, BF, BFH, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM								
1990														
2000	C	Blind	5616564	1/2 - 2	FLG BLIND 1500# RF SS A182 F316/316L		VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS							
2010		Blind	5616556	3 - 10	FLG BLIND 900# RF SS A182 F316/316L									
2020		Blind	5616556	12 - 24	FLG BLIND 900# RF SS A182 F316/316L									
2030		Blind	5616533	26 - 30	FLG BLIND 900# RF SS A182 F316/316L B16.47 SERIES A									
2040						MATERIAL: 316/316L SS DUAL GRADE								
2050		Blind	5619752	12 - 36	FLG COMPACT BLIND 900# RJ SS A182 F316/316L PIPE CLASS 900P21	FLANGE RATING: 900 RF								
2060						TEMPERATURE RANGE: -140 / 350 °C								
2070						COMMENT: LOW TEMP BLOW DOWN								
2080														
2090						***** ORIFICE FLANGES *****								
2100	D	Orific	5641519	2 - 2	WN ORIF SET 1500# RF SS A182 F316/316L SCH 80S BORE									
2110	D	Orific	5641287	3 - 3	WN ORIF SET 900# RF SS A182 F316/316L SCH 80S BORE									
2120		Orific	5641293	4 - 24	WN ORIF SET 900# RF SS A182 F316/316L SCH 120 BORE									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP							
						ALL SIZES- BUTT WELD	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS	
							1 MM	17MAR17	MC	D	4 OF 6	5	900P21	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2130			Orific	5640609	26 - 26	WN ORIF SET 900# RF A182 F316/316L 42.0MM WT B16.47 SERIES A		SERVICE NACE STD. MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 AA, BD, BF, BFH, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 900 RF TEMPERATURE RANGE: -140 / 350 °C COMMENT: LOW TEMP BLOW DOWN					
2140			Orific	5640614	30 - 30	WN ORIF SET 900# RF A182 F316/316L 47.0MM WT B16.47 SERIES A							
2150						***** PIPE NIPPLES *****							
2160						NIPPLE PBE CUT FROM PIPE							
2170													
2180													
2190		810											
2200													
2210													
2220						***** GASKETS *****							
2230													
2240	C		GASFRS	5672251	1/2 - 2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
2250			GASFRS	5673117	3 - 24	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR							
2260	C		GASFRS	5674033	26 - 36	GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2270													
2280	D	815	GASRJ	5673450	12 - 36	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900P21							
2290													
2300													
2310		B1				***** BOLTING *****							
2320													
2330	D	BA1	BOLS	5675776	1/4 - 10	STUD-BOLT WITH 2 NUTS API-6A718							
2340	D	BA5	BOLS	5675778	12 - 99	STUD-BOLT/COMPACT NUT API-6A718							
2350	D	BA1	BOLS	5675776	12 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
2360													
2370	D	SB6	CPSBOL	5675676	10 - 30	JACK SCREW HEX HEAD A320 B8M - CLASS 2							
2380		B1											
2390						***** BLINDS AND SPACERS *****							
2400													
2410													
2420			Specta	5749757	1 - 2	SPECTACLE BLIND 1500# RF SS A240 GR 316							
2430			Specta	5749755	3 - 6	SPECTACLE BLIND 900# RF SS A240 GR 316							
2440													
2450	C		Paddle	5750048	8 - 10	PADDLE BLANK 900# RF SS A240 GR 316							
2460	C		Paddle	5750282	8 - 10	PADDLE SPACER 900# RF SS A240 GR 316							
2470													
2480	C		Paddle	5750048	12 - 24	PADDLE BLANK 900# RF SS A240 GR 316							
2490	C		Paddle	5750282	12 - 24	PADDLE SPACER 900# RF SS A240 GR 316							
2500													
2510	C		Paddle	5750532	26 - 36	PADDLE BLANK 900# RF SS A240 GR 316 B16.47 SERIES A							
2520	C		Paddle	5750632	26 - 36	PADDLE SPACER 900# RF SS A240 GR 316 B16.47 SERIES A							
2530													
2540	C		Paddle	5750274	12 - 36	PADDLE BLANK COMPACT 900# RJ SS A240 GR 316							
2550	C		Paddle	5750275	12 - 36	PADDLE SPACER COMPACT 900# RJ SS A240 GR 316							
2560													
2570						***** BRANCHES *****							
2580													
2590						USE BRANCH CHART 900P21							
2600													
2610													
2620	D		Reinfo	RWELD	2 - 24	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 36						
2630	D		Reinfo				3 - 36						
2640													
2650													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY MC	REV D	SHEET 5 OF 6	PA 5	CLASS 900P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2660 2670 2680 2690		ZA6				***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE							
								SERVICE					
								NACE STD. MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 AA, BD, BF, BFH, BFL, BV, CF, CH, CI, DHC, DL, FG, FGH, GS, NL, PHC, PL, SG, VB, VI, WDM VALVES TO BE OPERATED BELOW -100°C WILL BE SP ITEMS					
								MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 900 RF TEMPERATURE RANGE: -140 / 350 °C COMMENT: LOW TEMP BLOW DOWN					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY MC	REV D	FGP/WPMP SHEET 6 OF 6	PA 5	CLASS 900P21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10		PP9				*****MODULAR VALVES*****		814	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.				
20								815	TO BE USED WITH COMPACT FLANGES ONLY.				
30								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's				
40	D	VA5	2BV	K15AS02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			USE ONLY FOR TIE IN				
50								817	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.				
60	D		2BV	K15AS06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		B1	TO BE USED WITH RAISED FACE FLANGES ONLY.				
70								BA1	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.				
80	D	VAV	2BV	K09AS06	3 - 3	MODULAR VALVE DB&B BALL TYPE 900# RF SS A182 F316 316SS TRIM METAL LO FULL SIZE TYPE 2BV		BA5	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.				
90								FA8	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.				
100	D		2BVR	M15AS02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		SB6	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.				
110								UJ2	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.				
120	D		2BVR	M15AS06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VA5	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.				
130								VAV	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055				
140	D	VAW	R2BVR	M15ES02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	VAV	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.				
150								VAW					
160	D	VAW	R2BVR	M09ES02	1 - 1	MODULAR VALVE DB&B BALL TYPE 900# RF SS A182 F316 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3						
170													
180						*****BALL VALVES*****							
190													
200													
210			Ball V	A15AS05	2 - 2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB LO							
220													
230			Ball V	A15AS06	2 - 2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION METAL RB LO							
240													
250	D		Ball V	A09AS05	3 - 3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB LO							
260													
270	D		Ball V	A09AS06	3 - 3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL RB LO							
280													
290	D		Ball V	A09AS07	4 - 10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO							
300													
310	D		Ball V	A09AS08	4 - 10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO							
320													
330	C		Ball V	A09CS07	12 - 30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO							
340													
350	C		Ball V	A09CS08	12 - 30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION METAL RB GO							
360													
370													
380			Ball V	B15AS01	3/4 - 1 1/2	BALL 1500# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO							
390													
400			Ball V	B15AS02	3/4 - 1 1/2	BALL 1500# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO							
410													
420		VA5	Ball V	B15AS05	2 - 2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB LO							
430													
440		VA5	Ball V	B15AS06	2 - 2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION METAL FB LO							
450													
460													
470	D	VA5	Ball V	B09AS07	3 - 3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO							
480													
490	D	VA5	Ball V	B09AS07	3 - 3	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO							
500													
510	D	VA5	Ball V	B09AS07	4 - 10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY KD	REV D	SHEET 1 OF 5	PA 5	CLASS 900P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
520	D	VA5	Ball V	B09AS08	4 - 10	BALL 900# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO		VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.				
530													
540	C	VA5	Ball V	B09CS07	12 - 30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO							
550													
560	C	VA5	Ball V	B09CS08	12 - 30	BALL 900# CF SS A182 F316 316SS TRIM TRUNNION METAL FB GO		VC2					
570									ZA6				
580													
590						*****GLOBE VALVES*****							
600													
610			Globe	L15AS48	3/4 - 1 1/2	GLOBE 1500# RF A182 F316 SS TRIM 16 OS&Y HW							
620			Globe	L15AS41	2 - 2	GLOBE 1500# RF SS A351 CF8M TRIM 16 OS&Y HW							
630			Globe	L09AS41	3 - 3	GLOBE 900# RF SS A351 CF8M TRIM 16 OS&Y HW							
640			Globe	L09AS42	4 - 8	GLOBE 900# RF SS A351 CF8M TRIM 16 OS&Y GO							
650													
660													
670						*****CHECK VALVES*****							
680													
690		VC1	Check	C15AS50	3/4 - 1 1/2	CHECK PISTON 1500# RF SS A182 F316 TRIM 16							
700		VC2	Check	C15AS45	2 - 2	CHECK DUAL WAFER LUG 1500# RF SS A351 CF8M TRIM 16 THROU BOLTED							
710		VC2	Check	C09AS45	3 - 8	CHECK DUAL WAFER LUG 900# RF SS A351 CF8M TRIM 16 THROU BOLTED							
720		VC2	Check	C09AS52	10 - 10	CHECK DUAL DBL FLG 900# RF SS A351 CF8M TRIM 16							
730	C	VC2	Check	C09CS52	12 - 30	CHECK DUAL 900# CF SS A351 CF8M TRIM 16							
740													
750													
760						*****PIPE*****							
770													
780	C		Piping	5365140	3/4 - 1 1/2	PIPE SCH 80S SMLS SS A312 TP316/316L PE							
790	D		Piping	5365138	2 - 3	PIPE SCH 80S SMLS SS A312 TP316/316L BE							
800	C		Piping	5365128	4 - 24	PIPE SCH 120 SMLS SS A312 TP316/316L BE							
810			Piping	5365217	26 - 26	PIPE WT 42.0MM EFW SS A358 316/316L BE							
820			Piping	5365218	28 - 28	PIPE WT 44.0MM EFW SS A358 316/316L BE							
830			Piping	5365216	30 - 30	PIPE WT 47.0MM EFW SS A358 316/316L BE							
840			Piping	5365219	36 - 36	PIPE WT 56.0MM EFW SS A358 316/316L BE							
850													
860	C	814	Piping	5365207	3/4 - 1 1/2	PIPE SCH 160 SMLS SS A312 TP316/316L PE							
870													
880						***** FITTINGS *****							
890													
900			90 Deg	5449033	3/4 - 1 1/2	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S							
910	D		90 Deg	5449033	2 - 3	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S							
920	C		90 Deg	5449570	4 - 24	ELL 90 DEG LR SCH 120 SS A403 WP316/316L-S							
930			90 Deg	5451124	26 - 26	ELL 90 DEG LR WT 42.0MM SS A403 WP316/316L-W							
940			90 Deg	5451125	28 - 28	ELL 90 LR WT 44.0MM 316/316L SS A403 WP316/316L-W							
950			90 Deg	5451136	30 - 30	ELL 90 LR WT 47.0MM SS A403 WP316/316L-W							
960			90 Deg	5451127	36 - 36	ELL 90 LR WT 56.0MM SS A403 WP316/316L-W							
970													
980	D		45 Deg	5449146	3/4 - 3	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S							
990	C		45 Deg	5449576	4 - 24	ELL 45 DEG LR SCH 120 SS A403 WP316/316L-S							
1000			45 Deg	5449084	26 - 26	ELL 45 DEG LR WT 42.0MM SS A403 WP316/316L-W							
1010			45 Deg	5449085	28 - 28	ELL 45 DEG LR WT 44.0MM SS A403 WP316/316L-W							
1020			45 Deg	5451126	30 - 30	ELL 45 LR WT 47.0MM SS A403 WP316/316L-W							
1030			45 Deg	5449086	36 - 36	ELL 45 LR WT 56.0MM SS A403 WP316/316L-W							
1040													
1050			Tee	5449156	3/4 - 1 1/2	TEE SCH 80S SS A403 WP316/316L-S							
1060	D		Tee	5449156	2 - 3	TEE SCH 80S SS A403 WP316/316L-S							
1070	C		Tee	5449579	4 - 24	TEE SCH 120 SS A403 WP316/316L-S							
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY KD	REV D	FGP/WPMP SHEET 2 OF 5	PA 5	CLASS 900P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1080			Tee	5449087	26 - 26	TEE 42.0MM SS A403 WP316/316L-W							
1090			Tee	5449088	28 - 28	TEE 44.0MM SS A403 WP316/316L-W							
1100			Tee	5451116	30 - 30	TEE 47.0MM SS A403 WP316/316L-W							
1110			Tee	5449089	36 - 36	TEE 56.0MM SS A403 WP316/316L-W							
1120													
1130	D		Reduci	5448905	3/4 - 2	TEE RED SCH 80S SS A403 WP316/316L-S	1 - 3						
1140	C		Reduci	5449577	2 - 3	TEE RED SCH 120/80S SS A403 WP316/316L-S	4 - 6						
1150	C		Reduci	5449578	4 - 20	TEE RED SCH 120 SS A403 WP316/316L-S	6 - 24						
1160			Reduci	5449094	12 - 24	TEE RED WT 42.0MM X SCH 120 SS A403 WP316/316L-W	26 - 26						
1170			Reduci	5449054	14 - 24	TEE RED WT 47.0MM X SCH 120 SS A403 WP316/316L-W	30 - 30						
1180													
1190	D		Cap	5449049	3/4 - 3	CAP SCH 80S SS A403 WP316/316L-S							
1200	C		Cap	5449052	4 - 24	CAP SCH 120 SS A403 WP316/316L-S							
1210			Cap	5449090	26 - 26	CAP WT 42.0MM SS A403 WP316/316L							
1220			Cap	5449091	28 - 28	CAP WT 44.0MM SS A403 WP316/316L							
1230			Cap	5449097	30 - 30	CAP WT 47.0MM SS A403 WP316/316L							
1240			Cap	5449093	36 - 36	CAP WT 56.0MM SS A403 WP316/316L							
1250													
1260	D		Concen	5449158	3/4 - 2	REDUCER CONC SCH 80S SS A403 WP316/316L-S	1 - 3						
1270	D		Concen	5448916	1 1/2 - 3	REDUCER CONC SCH 120/80S SS A403 WP316/316 L-S	4 - 6						
1280	D		Concen	5448915	4 - 20	REDUCER CONC SCH 120 SS A403 WP316/316L-S	6 - 24						
1290			Concen	5449095	18 - 24	REDUCER CONC 42.0MM WT/SCH 120 SS A403 WP316/316L-W	26 - 26						
1300			Concen	5450766	24 - 24	REDUCER CONC 47.0MM / SCH 120 SS A403 WP316/316L-W	30 - 30						
1310			Concen	5450753	24 - 24	REDUCER CONC 56.0MM / SCH 120 SS A403 WP316/316L-W	36 - 36						
1320			Concen	5450757	30 - 30	REDUCER CONC 56.0MM/ 47.0MM SS A403 WP316/316L-W	36 - 36						
1330													
1340	D		Eccent	5449159	3/4 - 2	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1 - 3						
1350	D		Eccent	5449179	1 1/2 - 3	REDUCER ECC SCH 120/80S SS A403 WP316/316 L-S	4 - 6						
1360	C		Eccent	5449178	4 - 20	REDUCER ECC SCH 120 SS A403 WP316/316L-S	6 - 24						
1370			Eccent	5449096	18 - 24	REDUCER ECC 42.0MM WT/SCH 120 SS A403 WP316/316L-W	26 - 26						
1380			Eccent	5450716	24 - 24	REDUCER ECC 47.0MM / SCH 120 SS A403 WP316/316L-W	30 - 30						
1390			Eccent	5450755	24 - 24	REDUCER ECC 56.0MM / SCH 120 SS A403 WP316/316L-W	36 - 36						
1400			Eccent	5450734	30 - 30	REDUCER ECC 56.0MM / 47.0MM SS A403 WP316/316L-W	36 - 36						
1410													
1420			Nipofl	5504176	3/4 - 1 1/2	NIPOFLANGE 1500# RF SCH 80S SS A182 F316/316L	2 - 3						
1430			Nipofl	5504177	3/4 - 1 1/2	NIPOFLANGE 1500# RF SCH 120/80S SS A182 F316/316L	4 - 24						
1440			Nipofl	5504178	3/4 - 1 1/2	NIPOFLANGE 1500# RF 42.0MM WT/SCH 80S SS A182 F316/316L	26 - 26						
1450			Nipofl	5504179	3/4 - 1 1/2	NIPOFLANGE 1500# RF 47.0MM WT/SCH 80S SS A182 F316/316L	30 - 30						
1460	C		Nipofl	5504189	3/4 - 1 1/2	NIPOFLANGE 1500# RF 56.0MM WT/SCH 80S SS A182 F316/316L	36 - 36						
1470													
1480		UJ2	Nipofl	5504180	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 120/160 SS A182 F316/316L	4 - 24						
1490		UJ2	Nipofl	5504181	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF 42.0MM WT/SCH 160 SS A182 F316/316L	26 - 26						
1500		UJ2	Nipofl	5504182	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF 47.0MM WT/SCH 160 SS A182 F316/316L	30 - 30						
1510	C	UJ2	Nipofl	5504188	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF 56.0MM WT/SCH 160 SS A182 F316/316L	36 - 36						
1520													
1530			Nipofl	5504183	2 - 2	NIPOFLANGE 1500# RF SCH 80S SS A182 F316/316L	3 - 3						
1540			Nipofl	5504184	2 - 2	NIPOFLANGE 1500# RF SCH 120/80S SS A182 F316/316L	4 - 24						
1550			Nipofl	5504185	2 - 2	NIPOFLANGE 1500# RF 42.0MM WT/SCH 80S SS A182 F316/316L	26 - 26						
1560			Nipofl	5504186	2 - 2	NIPOFLANGE 1500# RF 47.0MM WT/SCH 80 SS A182 F316/316L	30 - 30						
1570			Nipofl	5504187	2 - 2	NIPOFLANGE 1500# RF 56.0MM WT/SCH 80 SS A182 F316/316L	36 - 36						
1580													
1590	D		Weldol	5509637	3/4 - 1	WELDOLET SCH 80S SS A182 F316/316L	2 - 3						
1600	D		Weldol	5510374	3/4 - 3	WELDOLET SCH 120 /80S SS A182 316/316L	4 - 24						
1610	D		Weldol	5509649	2 - 3	WELDOLET 42.0MM WT/SCH 80S SS A182 F316/316L	26 - 26						
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	FGP/WPMP DATE 17MAR17					
								BY KD	REV D	SHEET 3 OF 5	PA 5	CLASS 900P22	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1620	D		Weldol	5509647	2 - 3	WELDOLET 47.0MM WT/SCH 80S SS A182 F316/316L	30 - 30							
1630			Weldol	5509640	4 - 8	WELDOLET SCH 120 SS A182 F316/316L	12 - 24							
1640			Weldol	5509646	4 - 10	WELDOLET 42.0MM WT/SCH 120 SS A182 F316/316L	26 - 26							
1650			Weldol	5509667	1 - 1	WELDOLET 47.0MM WT/SCH 80S SS A182 F316/316L	30 - 30							
1660			Weldol	5509648	4 - 12	WELDOLET 47.0MM WT/SCH 120 SS A182 F316/316L	30 - 30							
1670			Weldol	5509659	4 - 4	WELDOLET 56.0MM WT/ SCH 120 SS A182 F316/316L	36 - 36							
1680														
1690			UJ2	5516085	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 120/160 SS A182 F316/316L	4 - 24							
1700			UJ2	5516086	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF 42.0MM WT/SCH 160 SS A182 F316/316L	26 - 26							
1710			UJ2	5516087	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF 47.0MM WT/SCH 160 SS A182 F316/316L	30 - 30							
1720	C	UJ2	Elbole	5516088	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF 56.0MM WT/SCH 160 SS A182 F316/316L	36 - 36							
1730														
1740														
1750			816			***** PIPE FLANGES *****								
1760														
1770			FA8	Reduci	5600664	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 1500# RF SS A182 F316/316L							3/4 - 1 1/2
1780														
1790			C	Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1800			C	Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1810														
1820	D		Flange	5583979	3/4 - 2	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE								
1830			Flange	5583006	3 - 3	FLG WN 900# RF SS A182 F316/316L SCH 80S BORE								
1840			Flange	5583813	4 - 10	FLG WN 900# RF SS A182 F316/316L SCH 120 BORE								
1850			Flange	5583813	12 - 24	FLG WN 900# RF SS A182 F316/316L SCH 120 BORE								
1860			Flange	5583018	26 - 26	FLG WN 900# RF SS A182 F316/316L 42.0MM WT B16.47 SERIES A								
1870			Flange	5583019	30 - 30	FLG WN 900# RF SS A182 F316/316L 47.0MM WT B16.47 SERIES A								
1880														
1890			Flange	5619824	12 - 24	FLG COMPACT 900# RJ SS A182 F316/316L SCH 120 BORE								
1900			Flange	5619825	26 - 26	FLG COMPACT 900# RJ SS A182 F316/316L WT 42.0MM FLG								
1910			Flange	5619826	28 - 28	COMPACT 900# RJ SS A182 F316/316L WT 44.0MM FLG								
1920	C	817	Flange	5619827	30 - 30	COMPACT 900# RJ SS A182 F316/316L WT 47.0MM FLG								
1930			Flange	5619828	36 - 36	COMPACT 900# RJ SS A182 F316/316L WT 56.0MM								
1940														
1950			D	Flange	5583022	3 - 3	FLG WN 900# RTJ SS A182 F316/316L SCH XS BORE	SERVICE NACE STD. MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 BF, CCD, CI, CT, DHC, FGH, FM, GE, GHC, GS, PHC, SG, WSR						
1960														
1970				Blind	5616564	1/2 - 2	FLG BLIND 1500# RF SS A182 F316/316L							
1980				Blind	5616556	3 - 10	FLG BLIND 900# RF SS A182 F316/316L							
1990				Blind	5616556	12 - 24	FLG BLIND 900# RF SS A182 F316/316L							
2000				Blind	5616533	26 - 30	FLG BLIND 900# RF SS A182 F316/316L B16.47 SERIES A							
2010				Blind	5619753	12 - 36	FLG COMPACT BLIND 900# RJ SS A182 F316/316L PIPE CLASS 900P22							
2020														
2030														
2040														
2050														
2060			Orific	5641519	2 - 2	WN ORIF SET 1500# RF SS A182 F316/316L SCH 80S BORE	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 900 RF TEMPERATURE RANGE: -46 / 350 °C COMMENT:							
2070			Orific	5641287	3 - 3	WN ORIF SET 900# RF SS A182 F316/316L SCH 80S BORE								
2080			Orific	5641293	4 - 24	WN ORIF SET 900# RF SS A182 F316/316L SCH 120 BORE								
2090			Orific	5640609	26 - 26	WN ORIF SET 900# RF A182 F316/316L 42.0MM WT B16.47 SERIES A								
2100														
2110			Orific	5640614	30 - 30	WN ORIF SET 900# RF A182 F316/316L 47.0MM WT B16.47 SERIES A								
2120														
2130														
2140														
2150														
2160														
			814											***** PIPE NIPPLES ***** NIPPLE PBE CUT FROM PIPE
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17MAR17	BY KD	REV D	SHEET 4 OF 5	PA 5	CLASS 900P22	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
2170	C		GASFRS	5672251	1/2 - 2	***** GASKETS ***** GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A										
2180				5673117	3 - 24											
2190				5674033	26 - 30											
2200	D	815	GASRJ	5673451	12 - 36	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 900P22 GASKET 900# RTJ SS OCTAGONAL RING										
2210																
2220																
2230	D	815	GASRJ	5674481	3 - 3	***** BOLTING *****										
2240																
2250																
2260	D	BA1	BOLS	5675770	1/4 - 10	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS STUD-BOLT/COMPACT NUT API-6A718 STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS STUD-BOLT WITH 2 NUTS API-6A718										
2270				5675778	12 - 99											
2280				5675770	12 - 20											
2290	D	BA5	BOLS	5675778	12 - 99											
2300	D	BA1	BOLS	5675770	12 - 20											
2310	D	BA1	BOLS	5675776	24 - 99											
2320		SB6	CPSBOL	5675675	10 - 30	JACK SCREW HEX HEAD A320 GR L7M										
2330																
2340																
2350	C		Specta	5749757	3/4 - 2	***** BLINDS AND SPACERS ***** SPECTACLE BLIND 1500# RF SS A240 GR 316 SPECTACLE BLIND 900# RF SS A240 GR 316										
2360				5749755	3 - 6											
2370																
2380	C		Paddle	5750048	8 - 10	PADDLE BLANK 900# RF SS A240 GR 316 PADDLE SPACER 900# RF SS A240 GR 316										
2390				5750282	8 - 10											
2400																
2410	C		Paddle	5750048	12 - 24	PADDLE BLANK 900# RF SS A240 GR 316 PADDLE SPACER 900# RF SS A240 GR 316										
2420				5750282	12 - 24											
2430																
2440	C		Paddle	5750532	26 - 30	PADDLE BLANK 900# RF SS A240 GR 316 B16.47 SERIES A PADDLE SPACER 900# RF SS A240 GR 316 B16.47 SERIES A										
2450				5750632	26 - 30											
2460																
2470	D		Paddle	5750274	12 - 36	PADDLE BLANK COMPACT 900# RJ SS A240 GR 316 PADDLE SPACER COMPACT 900# RJ SS A240 GR 316										
2480				5750275	12 - 36											
2490																
2500	D		Paddle	5750274	12 - 36	PADDLE BLANK COMPACT 900# RJ SS A240 GR 316 PADDLE SPACER COMPACT 900# RJ SS A240 GR 316										
2510				5750275	12 - 36											
2520																
2530	D		Reinfo	RWELD	2 - 24	USE BRANCH CHART 900P22 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN										
2540				RWELD	2 - 24											
2550																
2560	D		Reinfo	RWELD	2 - 24	***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE										
2570				RWELD	2 - 24											
2580																
2590	D		Reinfo	RWELD	2 - 24	***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE										
2600				RWELD	2 - 24											
2610																
2620	D		Reinfo	RWELD	2 - 24	***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE										
2630				RWELD	2 - 24											
2640																
2650	D		Reinfo	RWELD	2 - 24	***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE										
2660				RWELD	2 - 24											
2670																
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 1 MM		DATE 17MAR17		BY KD	REV D	SHEET 5 OF 5	PA 5	CLASS 900P22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10 20 30 40 50 60	B	PP9 VA5	2BV	K15AL10	1/2 - 1 1/2	*****VALVES***** *****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV		814 815 816	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE. TO BE USED WITH COMPACT FLANGES ONLY. COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's				
70 80	B		2BV	K15AL14	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV		B1 BA1 BA4	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED WITH RAISED FACE FLANGES ONLY. TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.				
90 100	B	VAV	2BV	K15CL14	3 - 3	MODULAR VALVE DB&B BALL TYPE 1500# CF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV		FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.				
110 120	B		2BVR	M15AL10	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		PP9 UJ2	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.				
130 140	B		2BVR	M15AL14	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.				
150 160	B	VAW	R2BVR	M15EL10	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	VAV VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.				
170 180	B	VAW	R2BVR	M15ZL10	1 - 1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	VB2 VC1	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE REFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.				
190 200 210 220 230 240 250 260 270 280 290 300 310		VB2	Ball V	A15AL13	2 - 2	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT RB LO		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.				
320 330 340 350			Ball V	A15AL14	2 - 2	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION METAL RB LO							
360 370 380 390			Ball V	A15CL13	3 - 3	BALL 1500# CF INCONEL 625 TRIM AND BODY TRUNNION SOFT RB LO							
400 410 420 430 440 450 460 470 480 490	B	VA5	Ball V	A15CL14	3 - 3	BALL 1500# CF INCONEL 625 TRIM AND BODY TRUNNION METAL RB LO							
								SERVICE					
								DESIGN: 184 BARG AT 110°C CODE: ASME B31.3 PHC					
								MATERIAL: INCONEL 625 (NACE) FLANGE RATING: 1500 RF TEMPERATURE RANGE: -100 / 110 °C					
								*****GLOBE VALVES***** GLOBE 1500# RF INC 625 B564 INC 625 TRIM OS&Y HW GLOBE 1500# CF INC 625 B564 INC 625 TRIM OS&Y HW					
			FGP / WPMP TENGIZCHEVOIL - TENGIZCHEVOIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP					
						CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS
						1 MM		17NOV16	MLA	B	1 OF 3	5	1500A60

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
500	B B B	814	VC1 VC2 VC2	Check Check Check	C15AL51 C15AL46 C15CL53	3/4 - 1 1/2 2 - 2 3 - 3	*****CHECK VALVES***** CHECK PISTON 1500# RF INC 625 TRIM 625 CHECK DUAL WAFER LUG 1500# RF INC 625 TRIM 625 THROU BOLTED CHECK DUAL 1500# CF INC 625 TRIM 625 ***** PIPE ***** PIPE SCH 80S SMLS ALLOY 625 B444 UNS N06625 PE PIPE SCH 40S SMLS ALLOY 625 B444 UNS N06625 PE PIPE SCH 40S SMLS ALLOY 625 B444 UNS N06625 BE PIPE SCH 160 SMLS ALLOY 625 B444 UNS N06625 BE ***** FITTINGS ***** ELL BW 90 DEG LR SCH 80S ALLOY 625 B366-WPS UNS N06625 ELL BW 90 DEG LR SCH 40S ALLOY 625 B366-WPS UNS N06625 ELL BW 45 DEG LR SCH 80S ALLOY 625 B366-WPS UNS N06625 ELL BW 45 DEG LR SCH 40S ALLOY 625 B366-WPS UNS N06625 TEE BW SCH 80S ALLOY 625 B366-WPS UNS N06625 TEE BW SCH 40S ALLOY 625 B366-WPS UNS N06625 TEE RED SCH 80S ALLOY 625 B366-WPS UNS N06625 TEE RED SCH 40S/80S ALLOY 625 B366-WPS UNS N06625 TEE RED SCH 40S ALLOY 625 B366-WPS UNS N06625 CAP SCH 80S ALLOY 625 B366 UNS N06625 CAP SCH 40S ALLOY 625 B366 UNS N06625 REDUCER CONC SCH 80S ALLOY 625 B366-WPS N06625 REDUCER CONC SCH 40S/80S ALLOY 625 B366-WPS N06625 REDUCER CONC SCH 40S ALLOY 625 B366-WPS N06625 REDUCER ECC SCH 80S ALLOY 625 B366-WPS N06625 REDUCER ECC SCH 40S/80S ALLOY 625 B366-WPS N06625 REDUCER ECC SCH 40S ALLOY 625 B366-WPS N06625 ***** OLETS***** NIPOFLANGE 1500# RF SCH 40S/80S ALLOY 625 B564 UNS N06625 NIPOFLANGE 1500# RF SCH 40S ALLOY 625 B564 UNS N06625 NIPOFLANGE 1500# RF SCH 40S/160 ALLOY 625 B564 UNS N06625 WELDOLET SCH 40S/80S ALLOY 625 B564 UNS N06625 ELBOFLANGE 1500# RF SCH 40S/160 ALLOY 625 B564 UNS N06625	1 - 1 1 1/2 - 2 2 - 3	SERVICE DESIGN: 184 BARG AT 110°C CODE: ASME B31.3 PHC MATERIAL: INCONEL 625 (NACE) FLANGE RATING: 1500 RF TEMPERATURE RANGE: -100 / 110 °C				
510													
520													
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1000													
1010													
1020													
1030													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
						ALL SIZES- BUTT WELD	1 MM	17NOV16	MLA	B	2 OF 3	5	1500A60

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES				
1040		816				***** PIPE FLANGES *****		DESIGN: 184 BARG AT 110°C CODE: ASME B31.3 PHC MATERIAL: INCONEL 625 (NACE) FLANGE RATING: 1500 RF TEMPERATURE RANGE: -100 / 110 °C				
1050		FA8	Reduci	5614410	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 1500# RF ALLOY 625 B564 UNS N06625	3/4 - 1 1/2					
1060												
1070												
1080	B		Flange	5851215	3/4 - 3/4	FLANGE ADAPTER 1500# RF ALLOY 625 B564 UNS N06625 X 12 MM TUBE						
1090	B		Flange	5851215	2 - 2	FLANGE ADAPTER 1500# RF ALLOY 625 B564 UNS N06625 X 12 MM TUBE						
1100												
1110			Flange	5571900	3/4 - 1	FLG WN 1500# RF ALLOY 625 SCH 80S B564 UNS N06625						
1120			Flange	5571901	1 1/2 - 2	FLG WN 1500# RF ALLOY 625 SCH 40S B564 UNS N06625						
1130			Flange	5571901	3 - 3	FLG WN 1500# RF ALLOY 625 SCH 40S B564 UNS N06625						
1140												
1150			Blind	5617186	1/2 - 2	FLG BLIND 1500# RF ALLOY 625 B564 UNS N06625						
1160			Blind	5617186	3 - 3	FLG BLIND 1500# RF ALLOY 625 B564 UNS N06625						
1170												
1180		816	Flange	5619849	3 - 3	FLG COMPACT 1500# RJ ALLOY 625 SCH 40S B564 UNS N06625						
1190												
1200		816	Blind	5619780	3 - 3	FLG COMPACT BLIND 1500# RJ ALLOY 625 B564 UNS N06625 PIPE CLASS 1500A60						
1210												
1220												
1230						*****ORIFICE FLANGES*****						
1240												
1250			Orific	5651201	2 - 3	WN ORIF SET 1500# RF ALLOY 825 B564 UNS N06625 SCH 40S						
1260												
1270												
1280						***** PIPE NIPPLES *****						
1290												
1300		814				NIPPLES BBE TO BE CUT FROM PIPE						
1310												
1320												
1330						***** GASKETS *****						
1340												
1350			GASFRS	5673801	1/2 - 3	GASKET 1500# ALLOY 625 SP WND GRAPHITE IR						
1360	B	815	GASRJ	5673460	3 - 3	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 1500A60						
1370												
1380												
1390		B1				***** BOLTING *****						
1400												
1410	B	BA1	BOLS	5675770	1/4 - 2	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS						
1420	B	BA4	BOLS	5675776	3 - 3	STUD-BOLT WITH 2 NUTS API-6A718						
1430	B	BA1	BOLS	5675770	3 - 3	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS						
1440												
1450												
1460						***** LINE BLANKS *****						
1470												
1480			Specta	5750310	3/4 - 2	SPECTACLE BLIND 1500# RF ALLOY 625 B443 UNS N06625						
1490			Specta	5750310	3 - 3	SPECTACLE BLIND 1500# RF ALLOY 625 B443 UNS N06625						
1500												
1510			Specta	5750411	3 - 3	SPECTACLE BLIND COMPACT 1500# RJ ALLOY 625 B443 UNS N06625						
1520												
1530						***** BRANCHES *****						
1540												
1550						USE BRANCH CHART 1500A60						
1560												
1570	B		Reinfo	RWELD	2 - 2	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 3					
1580	B		Reinfo	RWELD	2 - 2	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 3					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	FGP/WPMP					
						ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM	DATE 17NOV16	BY MLA	REV B	SHEET 3 OF 3	PA 5

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10		PP9				*****VALVES*****		812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.					
20								815	TO BE USED WITH COMPACT FLANGES ONLY.					
30								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's					
40									ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.					
50									TO BE USED WITH RAISED FACE FLANGES ONLY.					
60	D	VA5	2BV	K15AL10	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV		B1	TO BE USED WITH RING TYPE JOINT FLANGES ONLY.					
70								BA1	TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.					
80	D		2BV	K15AL14	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV		BA2	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.					
90								BA4	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.					
100	D	VAV	2BV	K15CL14	3 - 3	MODULAR VALVE DB&B BALL TYPE 1500# CF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV		BA5	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
110								FA8	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
120	D		2BVR	M15AL10	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		PP9	FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.					
130								SB6	USE IN LTCS SMALL BORE LINES ONLY					
140	D		2BVR	M15AL14	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		UDA	USE FOR CONECTION TO EXTRUDED BRANCH FITTINGS ONLY					
150								UDD	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
160	D	VAW	R2BVR	M15EL10	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	UDE						
170								VA5						
180	D	VAW	R2BVR	M15ZL10	1 - 1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3							
190														
200	D	VAW	R2BV	K15ZL14	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# CF X RF INC 625 INC 625 TRIM METAL LO REDUCING - TYPE 2BV	3 - 3							
210														
220														
230		VB2				*****BALL VALVES*****								
240														
250			Ball V	A15AL13	2 - 2	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT RB LO								
260														
270			Ball V	A15CL13	3 - 3	BALL 1500# CF INCONEL 625 TRIM AND BODY TRUNNION SOFT RB LO								
280														
290			Ball V	A15CH15	4 - 24	BALL 1500# CF F22 INCONEL 625 TRIM TRUNNION SOFT RB GO								
300														
310														
320			Ball V	B15AL09	3/4 - 1 1/2	BALL 1500# RF INCONEL 625 BODY AND TRIM FLOATING SOFT FB LO								
330														
340	D	VA5	Ball V	B15AL13	2 - 3	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB LO								
350														
360	D	VA5	Ball V	B15CL15	4 - 4	BALL 1500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO								
370														
380		VA5	Ball V	B15CH15	4 - 24	BALL 1500# CF F22 INCONEL 625 TRIM TRUNNION SOFT FB GO								
390														
400														
410						*****THROUGH CONDUIT GATE VALVES*****								
420														
430	D		Condui	W15CH70	3 - 3	GATE 1500# CF F22 TRIM INC 625 TRIM THROUGH COND SOLID SLAB HW								
440														
450	D		Condui	W15CH71	4 - 24	GATE 1500# CF F22 TRIM INC 625 TRIM THROUGH COND SOLID SLAB GO								
460														
470														
480						*****GLOBE VALVES*****								
490														
500	C		Globe	L15AL43	3/4 - 2	GLOBE 1500# RF INC 625 B564 INC 625 TRIM OS&Y HW								
			FGP / WPMP TENGINZCHEVROIL - TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELDED PWHT - ALL SIZES.		CORR. ALLOW. 3 MM	FGP/WPMP DATE BY REV SHEET PA CLASS 17NOV16 KD D 1 OF 7 5 1500F22					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
510			Globe	L15CL43	3 - 3	GLOBE 1500# CF INC 625 B564 INC 625 TRIM OS&Y HW		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.								
520			Globe	L15CH44	4 - 6	GLOBE 1500# CF F22M INC 625 TRIM OS&Y GO		VAW									
530								VB2									
540						*****CHECK VALVES*****											
550								VC1									
560																	
570	C	VC1	Check	C15AL51	3/4 - 1 1/2	CHECK PISTON 1500# RF INC 625 TRIM 625											
580		VC2	Check	C15AL46	2 - 2	CHECK DUAL WAFER LUG 1500# RF INC 625 TRIM 625 THROU BOLTED											
590																	
600		VC2	Check	C15CL53	3 - 3	CHECK DUAL 1500# CF INC 625 TRIM 625		VC2									
610		VC2	Check	C15CH53	4 - 24	CHECK DUAL 1500# CF F22 TRIM 625		ZA6									
620																	
630						***** PIPE *****											
640																	
650																	
660	D	812	Piping	5358511	3/4 - 1 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE											
670	C		Piping	5360053	2 - 2	PIPE WT 7.0MM SMLS 2-1/4CR F22M BE											
680	C		Piping	5360054	3 - 3	PIPE WT 8.5MM SMLS 2-1/4CR F22M BE											
690	C		Piping	5360055	4 - 4	PIPE WT 10.0MM SMLS 2-1/4CR F22M BE											
700	C		Piping	5360056	6 - 6	PIPE WT 13.0MM SMLS 2-1/4CR F22M BE											
710	C		Piping	5360057	8 - 8	PIPE WT 16.0MM SMLS 2-1/4CR F22M BE											
720	C		Piping	5360059	10 - 10	PIPE WT 19.0MM SMLS 2-1/4CR F22M BE											
730	C		Piping	5360060	12 - 12	PIPE WT 22.0MM SMLS 2-1/4CR F22M BE											
740	C		Piping	5360072	14 - 14	PIPE WT 24.0MM SMLS 2-1/4CR F22M BE											
750	C		Piping	5360073	16 - 16	PIPE WT 27.0MM SMLS 2-1/4CR F22M BE											
760	C		Piping	5360076	18 - 18	PIPE WT 30.0MM SMLS 2-1/4CR F22M BE											
770	C		Piping	5360077	20 - 20	PIPE WT 33.0MM SMLS 2-1/4CR F22M BE											
780	C		Piping	5360052	24 - 24	PIPE WT 38.5MM SMLS 2-1/4CR F22M BE											
790	C		Piping	5360093	30 - 30	PIPE WT 47.5MM SMLS 2-1/4CR F22M BE											
800																	
810																	
820						***** FITTINGS *****											
830																	
840			90 Deg	5392870	3/4 - 1 1/2	ELL 90 DEG LR XXS LTCS A420 WPL6											
850	C		90 Deg	5406638	2 - 2	ELL 90 DEG LR WT 7.0MM 2-1/4CR F22M											
860	C		90 Deg	5406639	3 - 3	ELL 90 DEG LR WT 8.5MM 2-1/4CR F22M											
870	C		90 Deg	5406640	4 - 4	ELL 90 DEG LR WT 10.0MM 2-1/4CR F22M											
880	C		90 Deg	5406641	6 - 6	ELL 90 DEG LR WT 13.0MM 2-1/4CR F22M											
890	C		90 Deg	5406642	8 - 8	ELL 90 DEG LR WT 16.0MM 2-1/4CR F22M											
900	C		90 Deg	5406643	10 - 10	ELL 90 DEG LR WT 19.0MM 2-1/4CR F22M											
910	C		90 Deg	5406644	12 - 12	ELL 90 DEG LR WT 22.0MM 2-1/4CR F22M											
920	C		90 Deg	5406602	14 - 14	ELL 90 DEG LR WT 24.0MM 2-1/4CR F22M											
930	C		90 Deg	5406603	16 - 16	ELL 90 DEG LR WT 27.0MM 2-1/4CR F22M											
940	C		90 Deg	5406606	18 - 18	ELL 90 DEG LR WT 30.0MM 2-1/4CR F22M											
950	C		90 Deg	5406607	20 - 20	ELL 90 DEG LR WT 33.0MM 2-1/4CR F22M											
960	C		90 Deg	5406610	24 - 24	ELL 90 DEG LR WT 38.5MM 2-1/4CR F22M											
970	C		90 Deg	5406633	30 - 30	ELL 90 DEG LR WT 47.5MM 2-1/4CR F22M											
980																	
990			45 Deg	5392876	3/4 - 1 1/2	ELL 45 DEG LR XXS LTCS A420 WPL6											
1000	C		45 Deg	5407031	2 - 2	ELL 45 DEG LR WT 7.0MM 2-1/4CR F22M											
1010	C		45 Deg	5407032	3 - 3	ELL 45 DEG LR WT 8.5MM 2-1/4CR F22M											
1020	C		45 Deg	5407033	4 - 4	ELL 45 DEG LR WT 10.0MM 2-1/4CR F22M											
1030	C		45 Deg	5407034	6 - 6	ELL 45 DEG LR WT 13.0MM 2-1/4CR F22M											
1040	C		45 Deg	5407035	8 - 8	ELL 45 DEG LR WT 16.0MM 2-1/4CR F22M											
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM		DATE 17NOV16		BY KD	REV D	FGP/WPMP SHEET 2 OF 7		PA 5	CLASS 1500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES													
1050	C		45 Deg	5407036	10 - 10	ELL 45 DEG LR WT 19.0MM 2-1/4CR F22M															
1060	C		45 Deg	5407037	12 - 12	ELL 45 DEG LR WT 22.0MM 2-1/4CR F22M															
1070	C		45 Deg	5407062	14 - 14	ELL 45 DEG LR WT 24.0MM 2-1/4CRF22M															
1080	C		45 Deg	5407063	16 - 16	ELL 45 DEG LR WT 27.0MM 2-1/4CR F22M															
1090	C		45 Deg	5407066	18 - 18	ELL 45 DEG LR WT 30.0MM 2-1/4CR F22M															
1100	C		45 Deg	5407067	20 - 20	ELL 45 DEG LR WT 33.0MM 2-1/4CR F22M															
1110	C		45 Deg	5407042	24 - 24	ELL 45 DEG LR WT 38.5MM 2-1/4CR F22M															
1120	C		45 Deg	5407096	30 - 30	ELL 45 DEG LR WT 47.5MM 2-1/4CR F22M															
1130																					
1140			Tee	5392879	3/4 - 1 1/2	TEE SCH XXS LTCS A420 WPL6															
1150	C		Tee	5407214	2 - 2	TEE WT 7.0MM 2-1/4CR F22M		1 - 1 1/2 2 - 2 3 - 3 3 - 3 4 - 4 4 - 4 12 - 12 14 - 14 14 - 14 16 - 16 16 - 16 18 - 18 20 - 20 20 - 20 24 - 24 30 - 30													
1160	C		Tee	5407215	3 - 3	TEE WT 8.5MM 2-1/4CR F22M															
1170	C		Tee	5407216	4 - 4	TEE WT 10.0MM 2-1/4CR F22M															
1180	C		Tee	5407217	6 - 6	TEE WT 13.0MM 2-1/4CR F22M															
1190	C		Tee	5407218	8 - 8	TEE WT 16.0MM 2-1/4CR F22M															
1200	C		Tee	5407219	10 - 10	TEE WT 19.0MM 2-1/4CR F22M															
1210	C		Tee	5407220	12 - 12	TEE WT 22.0MM 2-1/4CR F22M															
1220	C		Tee	5407162	14 - 14	TEE WT 24.0MM 2-1/4CR F22M															
1230	C		Tee	5407163	16 - 16	TEE WT 27.0MM 2-1/4CR F22M															
1240	C		Tee	5407166	18 - 18	TEE WT 30.0MM 2-1/4CR F22M															
1250	C		Tee	5407167	20 - 20	TEE WT 33.0MM 2-1/4CR F22M		SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 GS- SOUR GAS													
1260	C		Tee	5407159	24 - 24	TEE WT 38.5MM 2-1/4CR F22M															
1270	C		Tee	5407211	30 - 30	TEE WT 47.5MM 2-1/4CR F22M															
1280																					
1290			Reduci	5392877	3/4 - 1	TEE RED SCH XXS LTCS A420 WPL6															
1300	C		Reduci	5407451	3/4 - 1 1/2	TEE RED BW WT 7.0MM / XXS 2-1/4CR F22M															
1310	C		Reduci	5407452	3/4 - 1 1/2	TEE RED BW WT 8.5MM / XXS 2-1/4CR F22M															
1320	C		Reduci	5407453	2 - 2	TEE RED BW WT 8.5MM / 7.0MM 2-1/4CR F22M															
1330	D		Reduci	5407460	2 - 2	TEE RED BW WT 10.0MM / 7.0MM 2-1/4CR F22M															
1340	C		Reduci	5407454	3 - 3	TEE RED BW WT 10.0MM / 8.5MM 2-1/4CR F22M															
1350	C		Reduci	5407455	10 - 10	TEE RED BW WT 22.0MM / 19.0MM 2-1/4CR F22M		MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 1500 RF TEMPERATURE RANGE: -40 / 155 °C													
1360	C		Reduci	5407456	10 - 10	TEE RED BW WT 24.0MM / 19.0MM 2-1/4CR F22M															
1370	C		Reduci	5407457	12 - 12	TEE RED BW WT 24.0MM / 22.0MM 2-1/4CR F22M															
1380	C		Reduci	5407458	10 - 10	TEE RED BW WT 27.0MM / 19.0MM 2-1/4CR F22M															
1390	C		Reduci	5407459	12 - 12	TEE RED BW WT 27.0MM / 22.0MM 2-1/4CR F22M															
1400	C		Reduci	5407386	16 - 16	TEE RED BW WT 30.0MM / 27.0MM 2-1/4CR F22M															
1410	C		Reduci	5407412	10 - 10	TEE RED BW WT 33.0MM / 19.0MM 2-1/4CR F22M															
1420	C		Reduci	5407413	12 - 12	TEE RED BW WT 33.0MM / 22.0MM 2-1/4CR F22M															
1430	C		Reduci	5407384	14 - 14	TEE RED BW WT 33.0MM / 24.0MM 2-1/4CR F22															
1440	C		Reduci	5407383	16 - 16	TEE RED BW WT 33.0MM / 27.0MM 2-1/4CR F22M															
1450	C		Reduci	5407387	18 - 18	TEE RED BW WT 33.0MM / 30.0MM 2-1/4CR F22M															
1460	C		Reduci	5407393	16 - 16	TEE RED BW WT 38.5MM / 27.0MM 2-1/4CR F22M		CONSTRUCTION	CORR. ALLOW.	DATE	BY	REV	FGP/WPMP								
1470	C		Reduci	5407396	18 - 18	TEE RED BW WT 38.5MM / 30.0MM 2-1/4CR F22M							SHEET	PA	CLASS						
1480	C		Reduci	5407391	20 - 20	TEE RED BW WT 38.5MM / 33.0MM 2-1/4CR F22M															
1490	C		Reduci	5407380	24 - 24	TEE RED BW WT 47.5MM / 38.5MM 2-1/4CR F22M															
1500																					
1510			Cap	5392880	3/4 - 1 1/2	CAP SCH XXS LTCS A420 WPL6															
1520	C		Cap	5407154	2 - 2	CAP WT 7.0MM 2-1/4CR F22M															
1530	C		Cap	5407155	3 - 3	CAP WT 8.5MM 2-1/4CR F22M															
1540	C		Cap	5407156	4 - 4	CAP WT 10.0MM 2-1/4CR F22M															
1550	C		Cap	5407157	6 - 6	CAP WT 13.0MM 2-1/4CR F22M															
1560	C		Cap	5407158	8 - 8	CAP WT 16.0MM 2-1/4CR F22M															
1570	C		Cap	5407185	10 - 10	CAP WT 19.0MM 2-1/4CR F22M															
1580	C		Cap	5407186	12 - 12	CAP WT 22.0MM 2-1/4CR F22M															
1590	C		Cap	5407102	14 - 14	CAP WT 24.0MM SMLS 2-1/4CR F22M															
1600	C		Cap	5407103	16 - 16	CAP WT 27.0MM 2-1/4CR F22M															
1610	C		Cap	5407106	18 - 18	CAP WT 30.0MM SMLS 2-1/4CR F22M															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			ALL SIZES BUTTWELD PWHT - ALL SIZES.		3 MM		17NOV16		KD		D		3 OF 7		5		1500F22	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1620	C		Cap	5407107	20 - 20	CAP WT 33.0MM SMLS 2-1/4CR F22M							
1630	C		Cap	5407129	24 - 24	CAP WT 38.5MM SMLS 2-1/4CR F22M							
1640	C		Cap	5407151	30 - 30	CAP WT 47.5MM 2-1/4CR F22M							
1650													
1660			Concen	5392881	3/4 - 1	REDUCER CONC SCH XXS LTCS A420 WPL6	1 - 1 1/2						
1670	C		Concen	5407203	3/4 - 1 1/2	REDUCER CONC WT 7.0MM/SCH XXS 2-1/4CR F22M	2 - 2						
1680	C		Concen	5407204	1 1/2 - 1 1/2	REDUCER CONC WT 8.5MM/SCH XXS 2-1/4CR F22M	3 - 3						
1690	C		Concen	5407205	1 1/2 - 1 1/2	REDUCER CONC WT 10.0MM/SCH XXS 2-1/4CR F22M	4 - 4						
1700	C		Concen	5407206	2 - 2	REDUCER CONC WT 8.5MM/7.0MM 2-1/4CR F22M R	3 - 3						
1710	C		Concen	5407237	2 - 2	REDUCER CONC WT 10.0MM/7.0MM 2-1/4CR F22M	4 - 4						
1720	C		Concen	5407238	3 - 3	REDUCER CONC WT 10.0MM/8.5MM 2-1/4CR F22M	4 - 4						
1730	C		Concen	5407207	10 - 10	REDUCER CONC WT 22.0MM/19.0MM 2-1/4CR F22M	12 - 12						
1740	C		Concen	5407531	10 - 10	REDUCER CONC WT 24.0MM/19.0MM 2-1/4CR F22M	14 - 14						
1750			Concen	5407532	12 - 12	REDUCER CONC WT 24.0MM/22.0MM 2-1/4CR F22M	14 - 14						
1760	C		Concen	5407533	10 - 10	REDUCER CONC WT 27.0MM/19.0MM 2-1/4CR F22M	16 - 16						
1770			Concen	5407534	12 - 12	REDUCER CONC WT 27.0MM/22.0MM 2-1/4CR F22M	16 - 16						
1780	C		Concen	5407266	16 - 16	REDUCER CONC WT 30.0MM/ 27.0MM 2-1/4CR F22M	18 - 18						
1790	C		Concen	5407264	14 - 14	REDUCER CONC WT 33.0MM/ 24.0MM 2-1/4CR F22M	20 - 20						
1800	C		Concen	5407263	16 - 16	REDUCER CONC WT 33.0MM/ 27.0MM 2-1/4CR F22M	20 - 20						
1810	C		Concen	5407267	18 - 18	REDUCER CONC WT 33.0MM/ 30.0MM 2-1/4CR F22M	20 - 20						
1820	C		Concen	5407273	16 - 16	REDUCER CONC WT 38.5MM/ 27.0MM 2-1/4CR F22M	24 - 24						
1830	C		Concen	5407276	18 - 18	REDUCER CONC WT 38.5MM/ 30.0MM 2-1/4CR F22M	24 - 24						
1840	C		Concen	5407271	20 - 20	REDUCER CONC WT 38.5MM/ 33.0MM 2-1/4CR F22M	24 - 24						
1850	C		Concen	5407260	24 - 24	REDUCER CONC WT 47.5MM/ 38.5MM 2-1/4CR F22M	30 - 30						
1860													
1870			Eccent	5392882	3/4 - 1	REDUCER ECC SCH XXS LTCS A420 WPL6	1 - 1 1/2						
1880	C		Eccent	5407508	3/4 - 1 1/2	REDUCER ECC WT 7.0MM / XXS 2-1/4CR F22M	2 - 2						
1890	C		Eccent	5407509	3/4 - 1 1/2	REDUCER ECC WT 8.5MM / XXS 2-1/4CR F22M	3 - 3						
1900	C		Eccent	5407510	1 1/2 - 1 1/2	REDUCER ECC WT 10.0MM / XXS 2-1/4CR F22M	4 - 4						
1910	C		Eccent	5407511	2 - 2	REDUCER ECC WT 8.5MM / 7.0MM 2-1/4CR F22M	3 - 3						
1920	C		Eccent	5407350	2 - 2	REDUCER ECC 10.0MM / 7.0MM 2-1/4CR F22M	4 - 4						
1930	C		Eccent	5407351	3 - 3	REDUCER ECC 10.0MM / 8.5MM 2-1/4CR F22M	4 - 4						
1940	C		Eccent	5407512	10 - 10	REDUCER ECC WT 22.0MM / 19.0MM 2-1/4CR F22M	12 - 12						
1950	C		Eccent	5407513	10 - 10	REDUCER ECC WT 24.0MM / 19.0MM 2-1/4CR F22M	14 - 14						
1960	C		Eccent	5407514	12 - 12	REDUCER ECC WT 24.0MM / 22.0MM 2-1/4CR F22M	14 - 14						
1970	C		Eccent	5407340	10 - 10	REDUCER ECC WT 27.0MM/ 19.0MM 2-1/4CR F22M	16 - 16						
1980	C		Eccent	5407535	12 - 12	REDUCER ECC WT 27.0MM / 22.0MM 2-1/4CR F22M	16 - 16						
1990	C		Eccent	5407326	16 - 16	REDUCER ECC WT 30.0MM/ 27.0MM 2-1/4CR F22M	18 - 18	SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 GS- SOUR GAS					
2000	C		Eccent	5407324	14 - 14	REDUCER ECC WT 33.0MM/ 24.0MM 2-1/4CR F22M	20 - 20						
2010	C		Eccent	5407323	16 - 16	REDUCER ECC WT 33.0MM/ 27.0MM 2-1/4CR F22M	20 - 20						
2020	C		Eccent	5407327	18 - 18	REDUCER ECC WT 33.0MM/ 30.0MM 2-1/4CR F22M	20 - 20						
2030	C		Eccent	5407333	16 - 16	REDUCER ECC WT 38.5MM/ 27.0MM 2-1/4CR F22M	24 - 24						
2040	C		Eccent	5407336	18 - 18	REDUCER ECC WT 38.5MM/ 30.0MM 2-1/4CR F22M	24 - 24						
2050	C		Eccent	5407331	20 - 20	REDUCER ECC WT 38.5MM/ 33.0MM 2-1/4CR F22M	24 - 24						
2060	C		Eccent	5407320	24 - 24	REDUCER ECC WT 47.5MM/ 38.5MM 2-1/4CR F22M	30 - 30						
2070													
2080													
2090						*****EXTRUDED BRANCHES*****		MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 1500 RF TEMPERATURE RANGE: -40 / 155 °C					
2110	D		Extrud	5506805E	3/4 - 1 1/2	NO MTO-EXTRUDED BRANCH - 8.5MM x XXS - 2-1/4CR F22M NO	3 - 3						
2120	D		Extrud	5506806E	3/4 - 1 1/2	MTO-EXTRUDED BRANCH - 10.0MM x XXS - 2-1/4CR F22M NO	4 - 4						
2130	D		Extrud	5506811E	3/4 - 1 1/2	MTO-EXTRUDED BRANCH - 19.0MM x XXS - 2-1/4CR F22M NO	10 - 10						
2140	D		Extrud	5506812E	3/4 - 1 1/2	MTO-EXTRUDED BRANCH - 24.0MM x XXS - 2-1/4CR F22M NO	14 - 14						
2150	D		Extrud	5506814E	3/4 - 1 1/2	MTO-EXTRUDED BRANCH - 33.0MM x XXS - 2-1/4CR F22M NO	20 - 20						
2160	D		Extrud	5506821E	3/4 - 1 1/2	MTO-EXTRUDED BRANCH - 38.5MM x XXS - 2-1/4CR F22M NO	24 - 24						
2170	D		Extrud	5506826E	3/4 - 1 1/2	MTO-EXTRUDED BRANCH - 47.5MM x XXS - 2-1/4CR F22M NO	30 - 30						
2180	D		Extrud	5506875E	2 - 2	MTO-EXTRUDED BRANCH - 13.0MM x 7.0MM - 2-1/4CR F22M NO	6 - 6						
						CONSTRUCTION	FGP/WPMP						
						ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							3 MM	17NOV16	KD	D	4 OF 7	5	1500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2190	D		Extrud	5506824E	2 - 2	MTO-EXTRUDED BRANCH - 38.5MM x 7.0MM - 2-1/4CR F22M NO	24 - 24							
2200	D		Extrud	5506825E	2 - 2	MTO-EXTRUDED BRANCH - 47.5MM x 7.0MM - 2-1/4CR F22M NO	30 - 30							
2210	D		Extrud	5506831E	3 - 3	MTO-EXTRUDED BRANCH - 38.5MM x 8.5MM - 2-1/4CR F22M NO	24 - 24							
2220	D		Extrud	5506832E	3 - 3	MTO-EXTRUDED BRANCH - 47.5MM x 8.5MM - 2-1/4CR F22M	30 - 30							
2230														
2240														
2250		816				***** PIPE FLANGES *****								
2260														
2270	C	FA8	Reduci	5599704	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 1500# RF LTCS A350 LF2	3/4 - 3/4							
2280														
2290	C		Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
2300	C		Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE								
2310														
2320		UDA	Flange	5557505	3/4 - 1 1/2	FLG WN 1500# RF LTCS A350 LF2 SCH XXS BORE								
2330		UDD												
2340														
2350	D	UDE	Flange	5563338	3/4 - 1 1/2	FLG WN 1500# RF 2-1/4CR F22M SCH XXS								
2360														
2370														
2380	C		Flange	5563350	2 - 2	FLG WN 1500# RF 2-1/4CR F22M WT 7.0MM								
2390	C		Flange	5563351	3 - 3	FLG WN 1500# RF 2-1/4CR F22M WT 8.5MM								
2400	C		Flange	5563352	4 - 4	FLG WN 1500# RF 2-1/4CR F22M WT 10.0MM								
2410	C		Flange	5563353	6 - 6	FLG WN 1500# RF 2-1/4CR F22M WT 13.0MM								
2420	C		Flange	5563354	8 - 8	FLG WN 1500# RF 2-1/4CR F22M WT 16.0MM								
2430	C		Flange	5563355	10 - 10	FLG WN 1500# RF 2-1/4CR F22M WT 19.0MM								
2440	C		Flange	5563356	12 - 12	FLG WN 1500# RF 2-1/4CR F22M WT 22.0MM								
2450	C		Flange	5563380	14 - 14	FLG WN 1500# RF 2-1/4CR F22M WT 24.0MM								
2460	C		Flange	5563381	16 - 16	FLG WN 1500# RF 2-1/4CR F22M WT 27.0MM								
2470	C		Flange	5563382	18 - 18	FLG WN 1500# RF 2-1/4CR F22M WT 30.0MM								
2480	C		Flange	5563383	20 - 20	FLG WN 1500# RF 2-1/4CR F22M WT 33.0MM								
2490	C		Flange	5563385	24 - 24	FLG WN 1500# RF 2-1/4CR F22M WT 38.5MM								
2500	C		Flange	5563429	30 - 30	FLG WN 1500# RF 2-1/4CR F22M WT 47.5MM MANUFACTURERS STANDARD								
2510														
2520														
2530	C		Flange	5563363	3 - 3	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 8.5MM								
2540	C		Flange	5563364	4 - 4	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 10.0MM								
2550	C		Flange	5563365	6 - 6	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 13.0MM								
2560	C		Flange	5563366	8 - 8	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 16.0MM								
2570	C		Flange	5563367	10 - 10	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 19.0MM								
2580	C		Flange	5563368	12 - 12	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 22.0MM								
2590	C		Flange	5619832	14 - 14	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 24.0MM								
2600	C		Flange	5619833	16 - 16	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 27.0MM								
2610	C		Flange	5619834	18 - 18	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 30.0MM								
2620	C		Flange	5619835	20 - 20	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 33.0MM								
2630	C		Flange	5619836	24 - 24	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 38.5MM								
2640	C		Flange	5619837	30 - 30	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 47.5MM								
2650														
2660			Flange	5563311	10 - 10	FLG WN 2500# RTJ 2-1/4CR F22M WT 19.0MM								
2670														
2680														
2690	C		Blind	5615616	1/2 - 1 1/2	FLG BLIND 1500# RF LTCS A350 LF2								
2700			Blind	5407447	2 - 2	FLG BLIND 1500# RF 2-1/4CR F22M								
2710			Blind	5407447	3 - 24	FLG BLIND 1500# RF 2-1/4CR F22M								
2720			Blind	5407496	30 - 30	FLG BLIND 1500# RF 2-1/4CR F22M MANUFACTURERS STANDARD								
2730														
2740	D		Blind	5619782	3 - 3	FLG COMPACT BLIND 1500# RJ 2-1/4CR F22M PIPE CLASS 1500F22 FOR NPS 3 INCH ONLY								
2750														
				FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY KD	REV D	FGP/WPMP SHEET 5 OF 7		PA 5	CLASS 1500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2760	D		Blind	5619781	4 - 30	FLG COMPACT BLIND 1500# RJ 2-1/4CR F22M PIPE CLASS 1500F22 FOR NPS 4 INCH & ABOVE							
2770						***** ORIFICE FLANGES *****							
2780													
2790													
2800	C		Orific	5407585	24 - 24	WN ORIF SET 1500# RF 2-1/4CR F22M WT 38.5MM							
2810													
2820		812				***** PIPE NIPPLES *****							
2830													
2840						BBE NIPPLES - USE PIPE CUT TO REQUIRED LENGTH							
2850													
2860						***** GASKETS *****							
2870													
2880													
2890													
2900	C		GASFRS	5672251	1/2 - 24	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
2910			GASFRS	5674013	30 - 30	GASKET 1500# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A							
2920													
2930			GASRJ	5674480	10 - 10	GASKET 2500# RTJ SS OCTAGONAL RING							
2940													
2950	D	815	GASRJ	5673471	3 - 3	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 1500F22 FOR NPS 3 INCH ONLY							
2960	D	815	GASRJ	5673470	4 - 30	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 1500F22 FOR NPS 4 INCH & ABOVE							
2970													
2980		B1				***** BOLTING *****							
2990													
3000	D		BOLS	5675770	3/4 - 4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3010	D	BA5	BOLS	5675778	6 - 99	STUD-BOLT/COMPACT NUT API-6A718							
3020													
3030	D	BA1	BOLS	5675770	6 - 14	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3040	D	BA1	BOLS	5675776	16 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
3050													
3060		SB6	CPSBOL	5675675	12 - 30	JACK SCREW HEX HEAD A320 GR L7M							
3070													
3080						***** BLINDS & SPACERS *****							
3090													
3100													
3110	D		Specta	5749570	3 - 3	SPECTACLE BLIND COMPACT 1500# RJ 2-1/4 CR F22M PIPE CLASS 1500F22 FOR NPS 3 INCH & BELOW		SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 GS- SOUR GAS					
3120			Specta	5749577	4 - 4	SPECTACLE BLIND COMPACT 1500# RJ 2-1/4 CR F22M PIPE CLASS 1500F22 FOR NPS 4 INCH & ABOVE							
3130													
3140	C		Paddle	5750286	10 - 30	PADDLE SPACER COMPACT 1500# RJ 2-1/4CR F22M PIPE CLASS 1500F22 FOR NPS 4 INCH & ABOVE							
3150	C		Paddle	5750285	10 - 30	PADDLE BLANK COMPACT 1500# RJ 2-1/4CR F22M PIPE CLASS 1500F22 FOR NPS 4 INCH & ABOVE		MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 1500 RF TEMPERATURE RANGE: -40 / 155 °C					
3160													
3170	C		Specta	5407647	2 - 2	SPECTACLE BLIND 1500# RF 2-1/4 CR F22M							
3180	C		Specta	5407647	3 - 4	SPECTACLE BLIND 1500# RF 2-1/4 CR F22M							
3190													
3200			Paddle	5407781	10 - 24	PADDLE SPACER 1500# RF 2-1/4CR F22M							
3210			Paddle	5407881	10 - 24	PADDLE BLANK 1500# RF 2-1/4CR F22M							
3220													
3230			Paddle	5407782	30 - 30	PADDLE SPACER 1500# RF 2-1/4CR F22M MANUFACTURERS STANDARD							
3240													
3250			Paddle	5407882	30 - 30	PADDLE BLANK 1500# RF 2-1/4CR F22M MANUFACTURERS STANDARD							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY KD	REV D	FGP/WPMP SHEET 6 OF 7	PA 5	CLASS 1500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3260						***** BRANCHES *****		SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 GS- SOUR GAS MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 1500 RF TEMPERATURE RANGE: -40 / 155 °C					
3270						USE BRANCH CHART 1500F22							
3280													
3290													
3300													
3310	C		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 30						
3320	C		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 30						
3330													
3340						***** CONSTRUCTION DETAILS *****							
3350													
3360		ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY							
3370						PIPE BENDS ----- NONE							
3380						THREAD SEALANT ----- NONE							
3390													
3400						*****DO NOT USE*****							
3410													
3420													
3430													
3440	C		Nipofl	5514113	3/4 - 1 1/2	NIPOFLANGE 1500# RF 13.0MM/XXS 2-1/4CR F22M	6 - 6						
3450	C		Nipofl	5514114	3/4 - 1 1/2	NIPOFLANGE 1500# RF 16.0MM/XXS 2-1/4CR F22M	8 - 8						
3460	C		Nipofl	5514136	3/4 - 1 1/2	NIPOFLANGE 1500# RF 33.0MM/XXS 2-1/4CR F22M	20 - 20						
3470	C		Nipofl	5514137	3/4 - 1 1/2	NIPOFLANGE 1500# RF 38.5MM/XXS 2-1/4CR F22M	24 - 24						
3480	C		Nipofl	5514138	3/4 - 1 1/2	NIPOFLANGE 1500# RF 47.5MM/XXS 2-1/4CR F22M	30 - 30						
3490													
3500	C		Weldol	5506908	3/4 - 1 1/2	WELDOLET WT 7.0MM / SCH XXS 2-1/4CR F22M	2 - 2						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY KD	REV D	FGP/WPMP SHEET 7 OF 7	PA 5	CLASS 1500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10		PP9				*****VALVES*****		032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.					
20								812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.					
30								815	TO BE USED WITH COMPACT FLANGES ONLY.					
40						*****MODULAR VALVES*****		816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's					
50														
60	D	VA5	2BV	K15AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			B1 ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.					
70								BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.					
80	D		2BV	K15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			FA8 USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.					
90								PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
100	D	VAV	2BV	K15CC06	3 - 3	MODULAR VALVE DB&B BALL TYPE 1500# CF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV			SC1 USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.					
110														
120	D		2BVR	M15AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR			UD6 USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.					
130														
140	D		2BVR	M15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR			UD7 USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST					
150								2 - 2	FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.					
160	D	VAW	R2BVR	M15EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR			UJ2 USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
170								3 - 3	VA5 USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
180	D	VAW	R2BVR	M15ZC02	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF LTCS A350LF2 316SSTRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR			VAB TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY					
190														
200														
210		VB2				*****BALL VALVES*****								
220														
230			Ball V	A15AC05	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO								
240														
250			Ball V	A15AC06	2 - 2	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO								
260														
270			Ball V	A15CC05	3 - 3	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO								
280														
290			Ball V	A15CC06	3 - 3	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB LO								
300														
310	D		Ball V	A15CC07-H1	4 - 4	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
320														
330			Ball V	A15CC08	4 - 4	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL RB GO								
340														
350														
360			Ball V	B15AC01	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO								
370														
380			Ball V	B15AC02	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING METAL FB LO								
390														
400	D	VA5	Ball V	B15AC05	2 - 3	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO								
410														
420	D	VA5	Ball V	B15AC06	2 - 3	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB LO								
430														
440	D	VA5	Ball V	B15CC07-H1	4 - 4	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
450														
460	D	VA5	Ball V	B15CC08	4 - 4	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION METAL FB GO								
470														
480														
490						*****GLOBE VALVES*****								
500														
510			Globe	L15AC48	3/4 - 1 1/2	GLOBE 1500# RF LTCS A350 LF2 TRIM 16 OS&Y HW								
520			Globe	L15AC41	2 - 2	GLOBE 1500# RF LTCS A352 LCC TRIM 16 OS&Y HW								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED PWHT FOR WT > 19MM	CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY MLA	REV D	FGP/WPMP SHEET 1 OF 4	PA 5	CLASS 1500H1	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
530	D		Globe	L15CC41	3 - 3	GLOBE 1500# CF LTCS A352 LCC TRIM 16 OS&Y HW		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.						
540		Globe	L15CC42-H1	4 - 4	GLOBE 1500# CF LTCS A352 LCC TRIM 16 OS&Y GO	VAW									
550								VB2							
560						*****CHECK VALVES*****		VC1							
570															
580		VC1	Check	C15AC50	3/4 - 1 1/2	CHECK PISTON 1500# RF LTCS A350 LF2 TRIM 16									
590		VC2	Check	C15AC45	2 - 2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC TRIM 16 THROU BOLTED									
600						CHECK DUAL 1500# CF LTCS A352 LCC TRIM 16									
610	D	VC2	Check	C15CC52-H1	3 - 4	CHECK DUAL 1500# CF LTCS A352 LCC TRIM 16									
620															
630															
640						***** PIPE *****									
650															
660			Piping	5358512	3/4 - 1 1/2	PIPE SCH 160 SMLS LTCS A333 GR.6 PE									
670			Piping	5358515	2 - 4	PIPE SCH 160 SMLS LTCS A333 GR.6 BE									
680	D		Piping	5358513	6 - 6	PIPE SCH XXS SMLS LTCS A333 GR.6 BE									
690	D		Piping	5358528	12 - 12	PIPE WT 34.0MM SMLS LTCS A333 GR.6 BE									
700	D		Piping	5358531	18 - 18	PIPE WT 47.0MM EFW LTCS A671 CC60 CL.22 BE									
710															
720		812	Piping	5358513	3/4 - 3/4	PIPE SCH XXS SMLS LTCS A333 GR.6 BE									
730															
740															
750						***** FITTINGS *****									
760															
770			90 Deg	5392816	3/4 - 4	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6									
780			45 Deg	5392822	3/4 - 4	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6									
790			Tee	5392825	3/4 - 4	TEE BW SCH 160 LTCS A420 WPL6									
800			Reduci	5392821	3/4 - 3	TEE RED BW SCH 160 LTCS A420 WPL6	1 - 4								
810			Cap	5392826	3/4 - 4	CAP BW SCH 160 LTCS A420 WPL6									
820															
830			Concen	5392827	3/4 - 3	REDUCER CONC SCH 160 LTCS A420 WPL6	1 - 4								
840			Eccent	5392828	3/4 - 3	REDUCER ECC SCH 160 LTCS A420 WPL6	1 - 4								
850	D		Eccent	5392950	3 - 3	REDUCER ECC SCH XXS/SCH 160 LTCS A420 WPL6	6 - 6								
860	D		Eccent	5392951	4 - 4	REDUCER ECC WT 24.0MM/SCH 160 LTCS A420 WPL6	8 - 8								
870	D		Eccent	5392952	6 - 6	REDUCER ECC WT 34.0MM/SCH XXS LTCS A420 WPL6	12 - 12								
880	D		Eccent	5392953	8 - 8	REDUCER ECC WT 34.0MM/24.0MM LTCS A420 WPL6	12 - 12								
890	D		Eccent	5392954	12 - 12	REDUCER ECC WT 47.0MM/34.0MM LTCS A420 WPL6-W	18 - 18								
900	D		Eccent	5392954	12 - 12	REDUCER ECC WT 47.0MM/34.0MM LTCS A420 WPL6-W	18 - 18								
910															
920						***** OLETS*****									
930															
940															
950			Nipofl	5504159	3/4 - 1	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2	2 - 4								
960		UJ2	Nipofl	5504170	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2	4 - 4								
970															
980			Weldol	5505392	3/4 - 1 1/2	WELDOLET SCH 160 LTCS A350 LF2	2 - 4								
990															
1000		UJ2	Elbole	5516080	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF SCH 160 LTCS A350 LF2	4 - 4								
1010															
1020															
1030		UD7	Thredo	5514772	3/4 - 1	THREDOLET 3000# LTCS A350 LF2	2 - 4								
1040			Sockol	5533071	3/4 - 1 1/2	SOCKOLET 6000# LTCS A350 LF2	2 - 4								
1050															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED PWHT FOR WT > 19MM		CORR. ALLOW. 1.5 MM	DATE 17MAR17	BY MLA	REV D	FGP/WPMP SHEET 2 OF 4		PA 5	CLASS 1500H1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1060	C	UD7	Coupli Reduci Union Plug	5478807	3/4 - 1 1/2	***** COUPLINGS UNION & PLUGS ***** COUPLING 6000# SW LTCS A350 LF2 COUPLING RED 6000# SW LTCS A350 LF2 UNION 6000# SW LTCS STL A350 LF2 PLUG ROUND HEAD SCRD LTCS A350 LF2	1 - 1 1/2								
1070				5478812	3/4 - 1										
1080				5478809	3/4 - 1 1/2										
1090				5491303	3/4 - 1										
1100			816	Reduci	5599704	1/2 - 1/2	***** PIPE FLANGES ***** FLG SCRD (FNPT) REDUCING 1500# RF LTCS A350 LF2						3/4 - 1 1/2		
1110					5851205	3/4 - 3/4									
1120					5851205	2 - 2									
1130					5557397	3/4 - 2									
1140			C	Flange	5557397	3 - 4	FLG WN 1500# RF LTCS A350 LF2 SCH 160 BORE FLG WN 1500# RF LTCS A350 LF2 SCH 160 BORE								
1150					5615616	1/2 - 2							FLG BLIND 1500# RF LTCS A350 LF2		
1160					5619838	3 - 4								FLG COMPACT 1500# RJ LTCS A350 LF2 SCH 160 BORE FLG COMPACT 1500# RJ LTCS A350 LF2 SCH XXS	
1170					5619841	6 - 6									
1180			D	Blind	5619785	3 - 6	FLG COMPACT BLIND 1500# RJ LTCS A350 LF2 PIPE CLASS 1500H1								
1190					812	Orific							5624598		2 - 4
1200													***** PIPE NIPPLES ***** NIPPLES BBE TO BE CUT FROM PIPE		
1210														***** GASKETS *****	
1220	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR														
1230		GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 1500H1													
1240			***** BOLTING *****												
1250				STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS											
1260	***** LINE BLANKS *****														
1270		SPECTACLE BLIND 1500# RF LTCS A516 GR.70													
1280			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
1290				SPECTACLE BLIND 1500# RF LTCS A516 GR.70											
1300	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
1310		SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70													
1320			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
1330				SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70											
1340	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
1350		SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70													
1360			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
1370				SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70											
1380	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
1390		SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70													
1400			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
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1420	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
1430		SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70													
1440			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
1450				SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70											
1460	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
1470		SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70													
1480			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
1490				SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70											
1500	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
1510		SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70													
1520			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
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1540	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
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1560			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
1570				SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70											
1580	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
1590		SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70													
1600			SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70												
1610				SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70											
	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70														
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				SPECTACLE BLIND COMP											

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1620 1630 1640 1650 1660 1670 1680 1690 1700 1710 1720 1730 1740			Reinfo Reinfo	RWELD RWELD	2 - 12 2 - 12	***** BRANCHES ***** USE BRANCH CHART 1500H1 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE	3 - 18 3 - 18						
								DESIGN: ASME B16.5 CODE: ASME B31.3 MG 1.1 DHC, PHC, SG					
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 1500 RF TEMPERATURE RANGE: -46 / 343°C					
		FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN				CONSTRUCTION ALL SIZES BUTT-WELDED PWHT FOR WT > 19MM	CORR. ALLOW. 1.5MM	DATE 17MAR17	BY MLA	REV D	FGP/WPMP SHEET 4 OF 4	PA 5	CLASS 1500H1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10		PP9				*****VALVES*****		032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.						
20						*****MODULAR VALVES*****		812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.						
30								815	TO BE USED WITH COMPACT FLANGES ONLY.						
40								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's						
50	D	VA5	2BV	K15AC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
60								BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.						
70	D		2BV	K15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV		BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.						
80								FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.						
90	D		2BVR	M15AC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
100								SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
110	D		2BVR	M15AC06	2 - 2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.						
120								UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.						
130	D	VAW	R2BVR	M15EC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.						
140								UDA	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.						
150		VB2				***** BALL VALVES *****		UI2	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.						
160	D		Ball V	A15AC05	2 - 3	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB LO		VA5	TO BE USED FOR PRESSURE CONNECTION WITH CHEMICAL SEAL ONLY						
170								VAB							
180			Ball V	A15CC07	4 - 4	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO									
190															
200			Ball V	A15CC07	10 - 12	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO									
210															
220			Ball V	A15CC07	20 - 20	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO									
230															
240			Ball V	B15AC01	3/4 - 1 1/2	BALL 1500# RF LTCS A350 LF2 316SS TRIM FLOATING SOFT FB LO									
250	D	VA5	Ball V	B15AC05	2 - 3	BALL 1500# RF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB LO									
260															
270	D		Ball V	B15CC07	4 - 4	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
280															
290	D	VA5	Ball V	B15CC07	10 - 12	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
300															
310	D		Ball V	B15CC07	20 - 20	BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
320															
330		VA5	Ball V	B15CC07		BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
340															
350		VA5	Ball V	B15CC07		BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
360															
370		VA5	Ball V	B15CC07		BALL 1500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO									
380															
390		VQ2				*****THROUGH CONDUIT GATE VALVES*****									
400															
410	D		Condui	W15CC70	3 - 3	GATE 1500# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW									
420															
430	D		Condui	W15CC71	4 - 12	GATE 1500# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO									
440															
450	D		Condui	W15CC71	20 - 20	GATE 1500# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO									
460															
470															
480															
490						***** GLOBE VALVES *****									
500															
510			Globe	L15AC48	3/4 - 1 1/2	GLOBE 1500# RF LTCS A350 LF2 TRIM 16 OS&Y HW									
520	D		Globe	L15AC41	2 - 3	GLOBE 1500# RF LTCS A352 LCC TRIM 16 OS&Y HW									
530			Globe	L15CC42	4 - 4	GLOBE 1500# CF LTCS A352 LCC TRIM 16 OS&Y GO									
540															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP							
						ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS
								3 MM		17NOV16	MLA	D	1 OF 4	5	1500K1

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1100	C		Nipofl	5504149	3/4 - 2	NIPOFLANGE 1500# RF WT 33.0MM/XXS LTCS A350 LF2	10 - 10						
1110	C		Nipofl	5504153	3/4 - 2	NIPOFLANGE 1500# RF WT 37.0MM/XXS LTCS A350 LF2	12 - 12						
1120	C		Nipofl	5504135	3/4 - 2	NIPOFLANGE 1500# RF WT 54.0MM/XXS LTCS A350 LF2	20 - 20						
1130													
1140	C	UJ2	Nipofl	5504169	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF XXS LTCS A350 LF2	4 - 4						
1150	C	UJ2	Nipofl	5504190	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF WT 33.0MM/XXS LTCS A350 LF2	10 - 10						
1160	C	UJ2	Nipofl	5504191	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF WT 37.0MM/XXS LTCS A350 LF2	12 - 12						
1170	C	UJ2	Nipofl	5504192	1 1/2 - 1 1/2	NIPOFLANGE 1500# RF WT 54.0MM/XXS LTCS A350 LF2	20 - 20						
1180													
1190	C		Weldol	5505197	3/4 - 3/4	WELDOLET SCH XXS LTCS A350 LF2	2 - 2						
1200	C		Weldol	5505380	3/4 - 1	WELDOLET WT 37.0MM/SCH XXS LTCS A350 LF2	12 - 12						
1210	C		Weldol	5505381	2 - 3	WELDOLET WT 54.0MM / SCH XXS LTCS A350 LF2	20 - 20						
1220													
1230	C	UJ2	Elbole	5516077	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF XXS LTCS A350 LF2	4 - 4						
1240	C	UJ2	Elbole	5516069	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF WT 33.0MM/XXS LTCS A350 LF2	10 - 10						
1250	C	UJ2	Elbole	5516070	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF WT 37.0MM/XXS LTCS A350 LF2	12 - 12						
1260	C	UJ2	Elbole	5516071	1 1/2 - 1 1/2	ELBOFLANGE 1500# RF WT 54.0MM/SCH XXS LTCS A350 LF2	20 - 20						
1270													
1280													
1290		816				***** PIPE FLANGES *****							
1300													
1310		FA8	Reduci	5599704	1/2 - 1/2	FLG SCR D (FNPT) REDUCING 1500# RF LTCS A350 LF2	3/4 - 1 1/2						
1320													
1330	C		Flange	5851205	3/4 - 3/4	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1340	C		Flange	5851205	2 - 2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE							
1350													
1360	C		Flange	5557505	3/4 - 2	FLG WN 1500# RF LTCS A350 LF2 SCH XXS BORE							
1370	C		Flange	5557505	3 - 4	FLG WN 1500# RF LTCS A350 LF2 SCH XXS BORE							
1380	C		Flange	5557514	10 - 10	FLG WN 1500# RF LTCS A350 LF2 WT 33.0MM BORE							
1390	C		Flange	5557515	12 - 12	FLG WN 1500# RF LTCS A350 LF2 WT 37.0MM BORE							
1400	C		Flange	5557516	20 - 20	FLG WN 1500# RF LTCS A350 LF2 WT 54.0MM BORE							
1410													
1420	D		Flange	5557706	2 - 2	FLG WN 2500# RTJ LTCS A350 LF2 SCH XXS							
1430													
1440	C		Flange	5619984	4 - 4	FLG COMPACT 1500# RJ LTCS A350 LF2 SCH XXS							
1450	C		Flange	5619846	10 - 10	FLG COMPACT 1500# RJ LTCS A350 LF2 WT 33.0MM							
1460	C		Flange	5619847	12 - 12	FLG COMPACT 1500# RJ LTCS A350 LF2 WT 37.0MM							
1470	C		Flange	5619848	20 - 20	FLG COMPACT 1500# RJ LTCS A350 LF2 WT 54.0MM							
1480													
1490	C		Blind	5615616	1/2 - 2	FLG BLIND 1500# RF LTCS A350 LF2							
1500			Blind	5615616	3 - 20	FLG BLIND 1500# RF LTCS A350 LF2							
1510			Blind	5615614	3/4 - 20	FLG BLIND 1500# RTJ LTCS A350 LF2							
1520													
1530			Blind	5619786	4 - 4	FLG COMPACT BLIND 1500# RJ LTCS A350 LF2 PIPE CLASS 1500K1							
1540			Blind	5619786	10 - 12	FLG COMPACT BLIND 1500# RJ LTCS A350 LF2 PIPE CLASS 1500K1							
1550			Blind	5619786	20 - 20	FLG COMPACT BLIND 1500# RJ LTCS A350 LF2 PIPE CLASS 1500K1							
1560													
1570						***** PIPE NIPPLES *****							
1580													
1590		812				BBE NIPPLES - USE PIPE CUT TO REQUIRED LENGTH							
1600													
1610													
1620						***** GASKETS *****							
1630													
1640	C		GASFRS	5672251	1/2 - 20	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR							
1650													
1660	D	815	GASRJ	5673481	3 - 20	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 1500K1							
			FGP / WPMP TENGINZCHEVROIL - TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17NOV16 BY MLA REV D SHEET 3 OF 4 PA 5 CLASS 1500K1					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1670	D		GASRJ	5674460	2 - 2	GASKET 2500# ALLOY INC 625 OCTAGONAL RING								
1680						***** BOLTING *****								
1690		B1												
1700														
1710	D	BA5	BOLS	5675770	1/4 - 14	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
1720	D	BA5	BOLS	5675770	16 - 18	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
1730	D	BA5	BOLS	5675776	20 - 99	STUD-BOLT WITH 2 NUTS API-6A718								
1740														
1750	D	BA1	BOLS	5675776	16 - 99	STUD-BOLT WITH 2 NUTS API-6A718								
1760		SB6	CPSBOL	5675675	10 - 20	JACK SCREW HEX HEAD A320 GR L7M								
1770														
1780														
1790		B1				***** BLINDS & SPACERS *****								
1800														
1810														
1820			Specta	5749595	1 - 2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70								
1830														
1840			Paddle	5750049	10 - 20	PADDLE BLANK 1500# RF LTCS A516-GR70								
1850			Paddle	5750249	10 - 20	PADDLE SPACER 1500# RF LTCS A516-GR70								
1860														
1870			Paddle	5750283	10 - 20	PADDLE BLANK COMPACT 1500# RJ LTCS A516-GR70								
1880	C		Paddle	5750284	10 - 20	PADDLE SPACER COMPACT 1500# RJ LTCS A516-GR70								
1890														
1900														
1910														
1920						***** BRANCHES *****								
1930														
1940						USE BRANCH CHART 1500K1								
1950														
1960	D		Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 20							
1970	D		Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 20							
1980														
1990						***** CONSTRUCTION DETAILS *****								
2000														
2010		ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY								
2020						PIPE BENDS ----- NONE								
2030						THREAD SEALANT ----- PTFE PASTE								
								SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 DHC,GS,HOS,PHC						
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 1500 RF TEMPERATURE RANGE: -46 / 126°C						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM	DATE 17NOV16	BY MLA	REV D	SHEET 4 OF 4	PA 5	CLASS 1500K1



SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
530	C	812	Piping	5365206	3/4 - 3/4	PIPE SCH XXS SMLS SS A312 316/316L SS BE	1 - 2	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. USE THIS VALVE FOR ORIFICE TAP BLOCK VALVE ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
540			VAW											
550			VB2											
560														
570														
580			90 Deg	5449140	3/4 - 2	ELL BW 90 DEG LR SCH 160 SS A403 WP316/316L-S		VC1						
590			45 Deg	5449142	3/4 - 2	ELL BW 45 DEG LR SCH 160 SS A403 WP316/316L-S		VC2						
600			Tee	5449190	3/4 - 2	TEE SCH 160 SS A403 WP316/316L-S								
610			Reduci	5449686	3/4 - 1 1/2	TEE RED SCH 160 316/316L SS A403		VU2						
620			Cap	5449055	3/4 - 2	CAP SCH 160 SS A403 WP316/316L-S		ZA6						
630														
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1060														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED		FGP/WPMP						
						CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS	
						1 MM		17MAR17	MLA	D	2 OF 3	5	1500P1	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1060 1070 1080 1090 1100 1110 1120 1130 1140 1150 1160 1170 1180 1190 1200 1210 1220 1230 1240 1250 1260 1270 1280 1290 1300 1310 1320 1330 1340 1350 1360 1370 1380 1390 1400 1410 1420 1430 1440 1450 1460 1470						***** ORIFICE FLANGES ***** WN ORIF SET 1500# RF SS A182 F316/316L SCH 160 BORE WN ORIF SET 1500# RTJ SS A182 F316/316L SCH 160 ***** PIPE NIPPLES ***** NIPPLE BBE CUT FROM PIPE ***** GASKETS ***** GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 1500P1 ***** BOLTING ***** STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS ***** BLINDS & SPACERS ***** SPECTACLE BLIND 1500# RF SS A240 GR 316 ***** BRANCHES ***** USE BRANCH CHART 1500P1 NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN ***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE								
	C		Orific	5641520	2 - 2									
		812	Orific	5640613	2 - 2									
	D	815	GASFRS	5672251	1/2 - 2									
	D		GASRJ	5673490	4 - 4									
		B1												
	D	B1	BOLS	5675770	1/4 - 4									
			Specta	5749757	1 - 2									
	D		Reinfo	RWELD	2 - 3		3 - 4							
	D		Reinfo	RWELD	2 - 3		3 - 4							
								SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 FGH, HOS, PHC						
								MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 1500 RF TEMPERATURE RANGE: -50 / 100°C						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED		CORR. ALLOW. 1 MM	DATE 17MAR17	BY MLA	REV D	SHEET 3 OF 3	PA 5	CLASS 1500P1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10	B	PP9	2BV	K25RL10	1/2 - 2	*****VALVES*****	2 - 2	814 815 816 B1 BA2 BA4 FA8 PP9 UJ2 VA5 VAU VAV VAV VB2 VC1 VC2	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE. TO BE USED WITH COMPACT FLANGES ONLY. COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED WITH RING TYPE JOINT FLANGES ONLY. TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS. USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS. COMPACT FLANGE AT LARGE END RING TYPE JOINT AT SMALL END USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.						
20		*****MODULAR VALVES*****													
30		MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL LO - FULL SIZE TYPE 2BV													
40		MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV													
50		MODULAR VALVE DB&B BALL TYPE 2500# CF INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV													
60		MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV													
70		MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV													
80		MODULAR VALVE DB&B BALL TYPE 2500# RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR													
90		MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR													
100		*****BALL VALVES*****													
110		BALL 2500# RTJ INCONEL 625 BODY AND TRIM TRUNNION SOFT RB GO													
120		BALL 2500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT RB GO													
130		BALL 2500# RTJ INCONEL 625 BODY AND TRIM FLOATING SOFT FB LO													
140		BALL 2500# RTJ INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO													
150		BALL 2500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO													
160		*****GLOBE VALVES*****													
170		GLOBE 2500# RTJ INC 625 B564 INC 625 TRIM OS&Y HW													
180		GLOBE 2500# CF INC 625 B564 INC 625 TRIM OS&Y GO													
190		*****CHECK VALVES*****													
200		CHECK PISTON 2500# RTJ INC 625 TRIM 625													
210		CHECK DUAL 2500# RTJ INC 625 TRIM 625													
220		CHECK DUAL 2500# CF INC 625 TRIM 625													
230		*****PIPE *****													
240		PIPE SCH 80S SMLS ALLOY 625 B444 UNS N06625 PE													
250															
260															
270															
280															
290															
300															
310															
320															
330															
340															
350															
360															
370															
380															
390															
400															
410															
420															
430															
440															
450															
460															
470															
480															
490															
500															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES- BUTT WELD		CORR. ALLOW. 1 MM		FGP/WPMP DATE 17NOV16 BY MLA REV B SHEET 1 OF 3 PA 5 CLASS 2500A60					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
510	B	814	Piping	5362284	2 - 4	PIPE SCH 80S SMLS ALLOY 625 B444 UNS N06625 BE	1 - 3						
520			Piping	5362289	3/4 - 3/4	PIPE SCH 160 SMLS ALLOY 625 B444 UNS N06625 BE							
530						***** FITTINGS *****							
540													
550			90 Deg	5426552	3/4 - 3	ELL BW 90 DEG LR SCH 80S ALLOY 625 B366-WPS UNS N06625							
560													
570			45 Deg	5426557	3/4 - 3	ELL BW 45 DEG LR SCH 80S ALLOY 625 B366-WPS UNS N06625							
580													
590			Tee	5426560	3/4 - 3	TEE BW SCH 80S ALLOY 625 B366-WPS UNS N06625							
600			Reduci	5426559	3/4 - 2	TEE RED SCH 80S ALLOY 625 B366-WPS UNS N06625							
610	UJ2	816	Cap	5426561	3/4 - 3	CAP SCH 80S ALLOY 625 B366 UNS N06625	3/4 - 1 1/2	SERVICE DESIGN: 184 BARG AT 110°C CODE: ASME B31.3 PHC MATERIAL: INCONEL 625 (NACE) FLANGE RATING: 2500 RF TEMPERATURE RANGE: -100 / 110 °C					
620			Concen	5426562	3/4 - 3	REDUCER CONC SCH 80S ALLOY 625 B366-WPS N06625							
630			Eccent	5426563	3/4 - 3	REDUCER ECC SCH 80S ALLOY 625 B366-WPS N06625							
640						***** OLETS*****							
650													
660			Nipofl	5513887	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT SCH 80S ALLOY 625 B564 UNS N06625							
670													
680			Nipofl	5513888	1 1/2 - 1 1/2	NIPOFLANGE 2500# RTJ WT SCH 80S/160 ALLOY 625 B564 UNS N06625							
690													
700			Weldol	5513557	3/4 - 1 1/2	WELDOLET SCH 80S ALLOY 625 B564 UNS N06625							
710	FA8	814				***** PIPE FLANGES *****	3/4 - 1 1/2						
720			Reduci	5599725	1/2 - 3/4	FLG SCRD (FNPT) REDUCING 2500# RTJ ALLOY 625 B564 UNS N06625							
730			Flange	5851212	3/4 - 3/4	FLANGE ADAPTER RTJ 2500# ALLOY 625 B564 UNS N06625 X 12 MM TUBE							
740			Flange	5851212	2 - 2	FLANGE ADAPTER RTJ 2500# ALLOY 625 B564 UNS N06625 X 12 MM TUBE							
750													
760			Flange	5619951	3/4 - 2	FLG WN 2500# RTJ ALLOY 625 SCH 80S B564 UNS N06625							
770			Flange	5619950	3 - 3	FLG COMPACT 2500# RJ ALLOY 625 SCH 80S B564 UNS N06625							
780													
790			Blind	5619735	3/4 - 2	FLG BLIND 2500# RTJ ALLOY 625 B564 UNS N06625							
800			Blind	5619790	3 - 3	FLG COMPACT BLIND 2500# RJ ALLOY 625 B564 UNS N06625 PIPE CLASS 2500A60							
810	B					***** PIPE NIPPLES *****							
820						NIPPLES BBE TO BE CUT FROM PIPE							
830						***** GASKETS *****							
840													
850													
860													
870													
880													
890													
900													
910			GASRJ	5674460	1/2 - 2	GASKET 2500# ALLOY INC 625 OCTAGONAL RING							
920													
930													
940													
950													
960													
970													
980													
990													
1000													
1010			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION	CORR. ALLOW. 1 MM	FGP/WPMP					
1020						ALL SIZES- BUTT WELD		DATE	BY	REV	SHEET	PA	CLASS
1030								17NOV16	MLA	B	2 OF 3	5	
1040													2500A60

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10		PP9				*****VALVES*****		812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.						
20								815	TO BE USED WITH COMPACT FLANGES ONLY.						
30								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's						
40		VAT				*****MODULAR VALVES*****									
50															
60	D	VA5	2BV	K25RL10	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL LO - FULL SIZE TYPE 2BV		B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.						
70								BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.						
80	D		2BV	K25RL16	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV		BA2	TO BE USED WITH RING TYPE JOINT FLANGES ONLY.						
90								BA4	TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.						
100	D	VAV	2BV	K25CL16	3 - 3	MODULAR VALVE DB&B BALL TYPE 2500# CF INCONEL 625 BODY AND TRIM METAL GO - FULL SIZE TYPE 2BV		BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.						
110								FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.						
120	D		2BVR	M25RL10	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
130								SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
140	D		2BVR	M25RL16	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INCONEL 625 BODY AND TRIM METAL GO - CONTROLLED BORE TYPE 2BVR		UDD	USE IN LTCS SMALL BORE LINES ONLY						
150								UDE	USE FOR CONECTION TO EXTRUDED BRANCH FITTINGS ONLY						
160	D	VAW	R2BVR	M25HL10	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.						
170															
180	D	VAU	R2BVR	M25YL10	1 - 1	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ INC 625 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3 - 3	VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.						
190		VAW						VAT	CONTROLLED BORE (2BVR) TYPE VALVE TO BE USED FOR INSTRUMENT ISOLATION ONLY						
200	D	VAW	R2BV	K25YL16	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ INC 625 INC TRIM 625 METAL GO REDUCING - TYPE 2BV	3 - 3	VAU	COMPACT FLANGE AT LARGE END RING TYPE JOINT AT SMALL END						
210															
220															
230		VB2				*****BALL VALVES*****									
240															
250	D		Ball V	A25RL15	2 - 2	BALL 2500# RTJ INCONEL 625 BODY AND TRIM TRUNNION SOFT RB GO									
260															
270	D		Ball V	A25CL15	3 - 3	BALL 2500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT RB GO									
280															
290	D		Ball V	A25CH15	6 - 14	BALL 2500# CF F22 INCONEL 625 TRUNNION SOFT RB GO									
300															
310	D		Ball V	B25RL09	3/4 - 1 1/2	BALL 2500# RTJ INCONEL 625 BODY AND TRIM FLOATING SOFT FB LO									
320															
330	D	VA5	Ball V	B25RL15	2 - 2	BALL 2500# RTJ INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO									
340															
350	D	VA5	Ball V	B25CL15	3 - 3	BALL 2500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO									
360															
370	C	VA5	Ball V	B25CH15	6 - 14	BALL 2500# CF F22 INCONEL 625 TRIM TRUNNION SOFT FB GO									
380															
390															
400						*****THROUGH CONDUIT GATE VALVES*****									
410															
420			Condui	W25CH70	2 - 2	GATE 2500# CF F22 TRIM INC 625 TRIM THROUGH COND SOLID SLAB HW									
430															
440			Condui	W25CH71	3 - 12	GATE 2500# CF F22 TRIM INC 625 TRIM THROUGH COND SOLID SLAB GO									
450															
460															
470						*****GLOBE VALVES*****									
480															
490	D		Globe	L25RL43	3/4 - 2	GLOBE 2500# RTJ INC 625 B564 INC 625 TRIM OS&Y HW									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM	FGP/WPMP DATE 17NOV16 BY KD REV D SHEET 1 OF 7 PA 5 CLASS 2500F22						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
500	D		Globe	L25CL44	3 - 3	GLOBE 2500# CF INC 625 B564 INC 625 TRIM OS&Y GO		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.				
510								VAW					
520								VB2	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.				
530						*****CHECK VALVES*****							
540													
550	D	VC1	Check	C25RL51	3/4 - 1 1/2	CHECK PISTON 2500# RTJ INC 625 TRIM 625		VC1					
560	D	VC2	Check	C25RL53	2 - 2	CHECK DUAL 2500# RTJ INC 625 TRIM 625							
570	D	VC2	Check	C25CL53	3 - 3	CHECK DUAL 2500# CF INC 625 TRIM 625							
580	D	VC2	Check	C25CH53	6 - 24	CHECK DUAL 2500# CF F22 TRIM 625		VC2					
590													
600													
610						***** PIPE *****							
620													
630	D	812	Piping	5358511	3/4 - 1	PIPE SCH XXS SMLS LTCS A333 GR.6 PE							
640	D		Piping	5358523	1 1/2 - 1 1/2	PIPE WT 11.0MM SMLS LTCS A333 GR.6 PE							
650	C		Piping	5360061	2 - 2	PIPE WT 9.0MM SMLS 2-1/4CR F22M BE							
660	C		Piping	5360064	3 - 3	PIPE WT 11.5MM SMLS 2-1/4CR F22M BE							
670	C		Piping	5360067	4 - 4	PIPE WT 14.0MM SMLS 2-1/4CR F22M BE							
680	C		Piping	5360068	6 - 6	PIPE WT 19.5MM SMLS 2-1/4CR F22M BE							
690	C		Piping	5360072	8 - 8	PIPE WT 24.0MM SMLS 2-1/4CR F22M BE							
700	C		Piping	5360074	10 - 10	PIPE WT 29.0MM SMLS 2-1/4CR F22M BE							
710	C		Piping	5360078	12 - 12	PIPE WT 34.0MM SMLS 2-1/4CR F22M BE							
720	C		Piping	5360081	14 - 14	PIPE WT 37.0MM SMLS 2-1/4CR F22M BE							
730	C		Piping	5360084	16 - 16	PIPE WT 42.0MM SMLS 2-1/4CR F22M BE							
740	C		Piping	5360086	18 - 18	PIPE WT 46.5MM SMLS 2-1/4CR F22M BE							
750	C		Piping	5360087	20 - 20	PIPE WT 51.5MM SMLS 2-1/4CR F22M BE							
760	C		Piping	5360094	26 - 26	PIPE WT 66.0MM SMLS 2-1/4CR F22M BE							
770													
780													
790						***** FITTINGS *****							
800													
810			90 Deg	5392870	3/4 - 1	ELL 90 DEG LR XXS LTCS A420 WPL6							
820			90 Deg	5392890	1 1/2 - 1 1/2	ELL 90 DEG LR WT 11.0MM LTCS A420 WPL6							
830	C		90 Deg	5406623	2 - 2	ELL 90 DEG LR WT 9.0MM 2-1/4CR F22M							
840	C		90 Deg	5406624	3 - 3	ELL 90 DEG LR WT 11.5MM 2-1/4CR F22M							
850	C		90 Deg	5406637	4 - 4	ELL 90 DEG LR WT 14.0MM 2-1/4CR F22M							
860	C		90 Deg	5406625	6 - 6	ELL 90 DEG LR WT 19.5MM 2-1/4CR F22M							
870	C		90 Deg	5406602	8 - 8	ELL 90 DEG LR WT 24.0MM 2-1/4CR F22M							
880	C		90 Deg	5406604	10 - 10	ELL 90 DEG LR WT 29.0MM 2-1/4CR F22M							
890	C		90 Deg	5406608	12 - 12	ELL 90 DEG LR WT 34.0MM 2-1/4CR F22M							
900	C		90 Deg	5406611	14 - 14	ELL 90 DEG LR WT 37.0MM 2-1/4CR F22M							
910	C		90 Deg	5406614	16 - 16	ELL 90 DEG LR WT 42.0MM 2-1/4CR F22M							
920	C		90 Deg	5406616	18 - 18	ELL 90 DEG LR WT 46.5MM 2-1/4CR F22M							
930	C		90 Deg	5406617	20 - 20	ELL 90 DEG LR WT 51.5MM 2-1/4CR F22M							
940	C		90 Deg	5406634	26 - 26	ELL 90 DEG LR WT 66.0MM 2-1/4CR F22M							
950													
960			45 Deg	5392876	3/4 - 1	ELL 45 DEG LR XXS LTCS A420 WPL6							
970			45 Deg	5392891	1 1/2 - 1 1/2	ELL 45 DEG LR WT 11.0MM LTCS A420 WPL6							
980	C		45 Deg	5407085	2 - 2	ELL 45 DEG LR WT 9.0MM 2-1/4CR F22M							
990	C		45 Deg	5407086	3 - 3	ELL 45 DEG LR WT 11.5MM 2-1/4CR F22M							
1000	C		45 Deg	5407084	4 - 4	ELL 45 DEG LR WT 14.0MM 2-1/4CR F22M							
1010	C		45 Deg	5407087	6 - 6	ELL 45 DEG LR WT 19.5MM 2-1/4CR F22M							
1020	C		45 Deg	5407062	8 - 8	ELL 45 DEG LR WT 24.0MM 2-1/4CR F22M							
1030	C		45 Deg	5407068	12 - 12	ELL 45 DEG LR WT 34.0MM 2-1/4CR F22M							
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17NOV16 BY KD REV D SHEET 2 OF 7 PA 5 CLASS 2500F22					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1040	C		45 Deg	5407071	14 - 14	ELL 45 DEG LR WT 37.0MM 2-1/4CR F22M							
1050	C		45 Deg	5407074	16 - 16	ELL 45 DEG LR WT 42.0MM 2-1/4CR F22M							
1060	C		45 Deg	5407076	18 - 18	ELL 45 DEG LR WT 46.5MM 2-1/4CR F22M							
1070	C		45 Deg	5407077	20 - 20	ELL 45 DEG LR WT 51.5MM 2-1/4CR F22M							
1080	C		45 Deg	5407094	26 - 26	ELL 45 DEG LR WT 66.0MM 2-1/4CR F22M							
1090													
1100			Tee	5392879	3/4 - 1	TEE SCH XXS LTCS A420 WPL6							
1110			Tee	5392892	1 1/2 - 1 1/2	TEE WT 11.0MM LTCS A420 WPL6							
1120	C		Tee	5407182	2 - 2	TEE WT 9.0MM 2-1/4CR F22M							
1130	C		Tee	5407183	3 - 3	TEE WT 11.5MM 2-1/4CR F22M							
1140	C		Tee	5407181	4 - 4	TEE WT 14.0MM 2-1/4CR F22M							
1150	C		Tee	5407184	6 - 6	TEE WT 19.5MM 2-1/4CR F22M							
1160	C		Tee	5407162	8 - 8	TEE WT 24.0MM 2-1/4CR F22M							
1170	C		Tee	5407168	12 - 12	TEE WT 34.0MM 2-1/4CR F22M							
1180	C		Tee	5407170	14 - 14	TEE WT 37.0MM 2-1/4CR F22M							
1190	C		Tee	5407173	16 - 16	TEE WT 42.0MM 2-1/4CR F22M							
1200	C		Tee	5407175	18 - 18	TEE WT 46.5MM 2-1/4CR F22M							
1210	C		Tee	5407176	20 - 20	TEE WT 51.5MM 2-1/4CR F22M							
1220	C		Tee	5407212	26 - 26	TEE WT 66.0MM 2-1/4CR F22M							
1230													
1240			Reduci	5392877	3/4 - 3/4	TEE RED SCH XXS LTCS A420 WPL6	1 - 1						
1250			Reduci	5392894	3/4 - 1	TEE RED WT 11.0MM / SCH XXS LTCS A420 WPL6	1 1/2 - 1 1/2						
1260	C		Reduci	5407397	3/4 - 1	TEE RED BW WT 9.0MM / XXS 2-1/4CR F22M	2 - 2						
1270	C		Reduci	5407398	1 1/2 - 1 1/2	TEE RED BW WT 9.0MM / 11.0MM 2-1/4CR F22M	2 - 2						
1280	C		Reduci	5407399	1 1/2 - 1 1/2	TEE RED BW WT 11.5MM / 11.0MM 2-1/4CR F22M	3 - 3						
1290	C		Reduci	5407400	2 - 2	TEE RED BW WT 11.5MM / 9.0MM 2-1/4CR F22M	3 - 3						
1300	C		Reduci	5407409	8 - 8	TEE RED BW WT 51.5MM / 24.0MM 2-1/4CR F22M	20 - 20						
1310	C		Reduci	5407411	10 - 10	TEE RED BW WT 37.0MM / 29.0MM 2-1/4CR F22M	14 - 14						
1320	C		Reduci	5407404	10 - 10	TEE RED BW WT 46.5MM / 29.0MM 2-1/4CR F22M	18 - 18						
1330	C		Reduci	5407410	12 - 12	TEE RED BW WT 46.5MM / 34.0MM 2-1/4CR F22M	18 - 18						
1340	C		Reduci	5407405	16 - 16	TEE RED BW WT 46.5MM / 42.0MM 2-1/4CR F22M	18 - 18						
1350	C		Reduci	5407406	16 - 16	TEE RED BW WT 51.5MM / 42.0MM 2-1/4CR F22M	20 - 20						
1360	C		Reduci	5407407	18 - 18	TEE RED BW WT 51.5MM / 46.5MM 2-1/4CR F22M	20 - 20						
1370	C		Reduci	5407408	20 - 20	TEE RED BW WT 66.0MM / 51.5MM 2-1/4CR F22M	26 - 26						
1380													
1390			Cap	5392880	3/4 - 1	CAP SCH XXS LTCS A420 WPL6		SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 GS,PHC					
1400			Cap	5392893	1 1/2 - 1 1/2	CAP WT 11.0MM LTCS A420 WPL6							
1410	C		Cap	5407126	2 - 2	CAP WT 9.0MM 2-1/4CR F22M							
1420	C		Cap	5407127	3 - 3	CAP WT 11.5MM 2-1/4CR F22M							
1430	C		Cap	5407125	4 - 4	CAP WT 14.0MM 2-1/4CR F22M							
1440	C		Cap	5407128	6 - 6	CAP WT 19.5MM 2-1/4CR F22M							
1450	C		Cap	5407102	8 - 8	CAP WT 24.0MM SMLS 2-1/4CR F22M							
1460	C		Cap	5407108	12 - 12	CAP WT 34.0MM SMLS 2-1/4CR F22M							
1470	C		Cap	5407111	14 - 14	CAP WT 37.0MM SMLS 2-1/4CR F22M							
1480	C		Cap	5407114	16 - 16	CAP WT 42.0MM SMLS 2-1/4CR F22M							
1490	C		Cap	5407116	18 - 18	CAP WT 46.5MM 2-1/4CR F22M							
1500	C		Cap	5407117	20 - 20	CAP WT 51.5MM 2-1/4CR F22M							
1510	C		Cap	5407123	26 - 26	CAP WT 66.0MM 2-1/4CR F22M							
1520								MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 2500 RF TEMPERATURE RANGE: -40 / 180 °C					
1530			Concen	5392881	3/4 - 3/4	REDUCER CONC SCH XXS LTCS A420 WPL6	1 - 1						
1540			Concen	5392895	3/4 - 1	REDUCER CONC WT 11.0MM / SCH XXS LTCS A420 WPL6	1 1/2 - 1 1/2						
1550			Concen	5407246	3/4 - 1	REDUCER CONC WT 9.0MM / XXS 2-1/4CR F22M	2 - 2						
1560	C		Concen	5407247	1 1/2 - 1 1/2	REDUCER CONC WT 9.0MM / 11.0MM 2-1/4CR F22M	2 - 2						
1570	C		Concen	5407248	1 1/2 - 1 1/2	REDUCER CONC WT 11.5MM / 11.0MM 2-1/4CR F22M	3 - 3						
1580	C		Concen	5407249	1 1/2 - 1 1/2	REDUCER CONC WT 14.0MM / 11.0MM 2-1/4CR F22M	4 - 4						
1590	C		Concen	5407250	2 - 2	REDUCER CONC WT 11.5MM / 9.0MM 2-1/4CR F22M	3 - 3						
1600	C		Concen	5407251	2 - 2	REDUCER CONC WT 14.0MM / 9.0MM 2-1/4CR F22M	4 - 4						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM						DATE 17NOV16

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1610	C		Concen	5407252	3 - 3	REDUCER CONC WT 14.0MM / 11.5MM 2-1/4CR F22M	4 - 4								
1620	C		Concen	5407293	8 - 8	REDUCER CONC WT 29.0MM/24.0MM 2-1/4CR F22M	10 - 10								
1630	C		Concen	5407295	10 - 10	REDUCER CONC WT 37.0MM/ 29.0MM 2-1/4CR F22M	14 - 14								
1640	C		Concen	5407294	12 - 12	REDUCER CONC WT 51.5MM/34.0MM 2-1/4CR F22M	20 - 20								
1650	C		Concen	5407296	16 - 16	REDUCER CONC WT 46.5MM/ 42.0MM 2-1/4CR F22M	18 - 18								
1660	C		Concen	5407297	16 - 16	REDUCER CONC WT 51.5MM/42.0MM 2-1/4CR F22M	20 - 20								
1670	C		Concen	5407299	18 - 18	REDUCER CONC WT 51.5MM/46.5MM 2-1/4CR F22M	20 - 20								
1680	C		Concen	5407277	20 - 20	REDUCER CONC WT 66.0MM/51.5MM 2-1/4CR F22M	26 - 26								
1690															
1700			Eccent	5392882	3/4 - 3/4	REDUCER ECC SCH XXS LTCS A420 WPL6	1 - 1								
1710			Eccent	5392896	3/4 - 1	REDUCER ECC WT 11.0MM / SCH XXS LTCS A420 WPL6	1 1/2 - 1 1/2								
1720	C		Eccent	5407305	3/4 - 1	REDUCER ECC WT 9.0MM / XXS 2-1/4CR F22M	2 - 2								
1730	C		Eccent	5407306	1 1/2 - 1 1/2	REDUCER ECC WT 9.0MM / 11.0MM 2-1/4CR F22M	2 - 2								
1740	C		Eccent	5407307	1 1/2 - 1 1/2	REDUCER ECC WT 11.5MM / 11.0MM 2-1/4CR F22M	3 - 3								
1750	C		Eccent	5407308	1 1/2 - 1 1/2	REDUCER ECC WT 14.0MM / 11.0MM 2-1/4CR F22M	4 - 4								
1760	C		Eccent	5407309	2 - 2	REDUCER ECC WT 11.5MM / 9.0MM 2-1/4CR F22M	3 - 3								
1770	C		Eccent	5407341	2 - 2	REDUCER ECC WT 14.0MM / 9.0MM 2-1/4CR F22M	4 - 4								
1780	C		Eccent	5407342	3 - 3	REDUCER ECC WT 14.0MM / 11.5MM 2-1/4CR F22M	4 - 4								
1790	C		Eccent	5407322	6 - 6	REDUCER ECC WT 34.0MM/ 19.5MM 2-1/4CR F22M	12 - 12								
1800	C		Eccent	5407329	6 - 6	REDUCER ECC WT 24.0MM/ 19.5MM 2-1/4CR F22M	8 - 8								
1810	C		Eccent	5407339	8 - 8	REDUCER ECC WT 29.0MM/ 24.0MM 2-1/4CR F22M	10 - 10								
1820	C		Eccent	5407334	8 - 8	REDUCER ECC WT 34.0MM/ 24.0MM 2-1/4CR F22M	12 - 12								
1830	C		Eccent	5407353	8 - 8	REDUCER ECC WT 37.0MM/ 24.0MM 2-1/4CR F22M	14 - 14								
1840	C		Eccent	5407354	10 - 10	REDUCER ECC WT 37.0MM/ 29.0MM 2-1/4CR F22M	14 - 14								
1850	C		Eccent	5407358	8 - 8	REDUCER ECC WT 42.0MM/ 24.0MM 2-1/4CR F22M	16 - 16								
1860	C		Eccent	5407332	10 - 10	REDUCER ECC WT 34.0MM/ 29.0MM 2-1/4CR F22M	12 - 12								
1870	C		Eccent	5407355	10 - 10	REDUCER ECC WT 42.0MM/ 29.0MM 2-1/4CR F22M	16 - 16								
1880	C		Eccent	5407356	16 - 16	REDUCER ECC WT 46.5MM/ 42.0MM 2-1/4CR F22M	18 - 18								
1890	C		Eccent	5407357	16 - 16	REDUCER ECC WT 51.5MM/ 42.0MM 2-1/4CR F22M	20 - 20								
1900	C		Eccent	5407359	18 - 18	REDUCER ECC WT 51.5MM/ 46.5MM 2-1/4CR F22M	20 - 20								
1910	C		Eccent	5407337	20 - 20	REDUCER ECC WT 66.0MM/ 51.5MM 2-1/4CR F22M	26 - 26								
1920															
1930															
1940						*****EXTRUDED BRANCHES*****									
1950															
1960	D		Extrud	5506807E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 11.5MM x XXS - 2-1/4CR F22M	3 - 3								
1970	D		Extrud	5506809E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 14.0MM x XXS - 2-1/4CR F22M	4 - 4								
1980	D		Extrud	5506812E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 24.0MM x XXS - 2-1/4CR F22M	8 - 8								
1990	D		Extrud	5506815E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 34.0MM x XXS - 2-1/4CR F22M	12 - 12								
2000	D		Extrud	5506816E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 37.0MM x XXS - 2-1/4CR F22M	14 - 14								
2010	D		Extrud	5506818E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 42.0MM x XXS - 2-1/4CR F22M	16 - 16								
2020	D		Extrud	5506819E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 46.5MM x XXS - 2-1/4CR F22M	18 - 18								
2030	D		Extrud	5506822E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 51.5MM x XXS - 2-1/4CR F22M	20 - 20								
2040															
2050	D		Extrud	5506857E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 11.5MM x 11.0MM - 2-1/4CR F22M	3 - 3								
2060															
2070	D		Extrud	5506859E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 14.0MM x 11.0MM - 2-1/4CR F22M	4 - 4								
2080															
2090	D		Extrud	5506862E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 24.0MM x 11.0MM - 2-1/4CR F22M	8 - 8								
2100															
2110	D		Extrud	5506865E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 34.0MM x 11.0MM - 2-1/4CR F22M	12 - 12								
2120															
2130	D		Extrud	5506866E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 37.0MM x 11.0MM - 2-1/4CR F22M	14 - 14								
2140															
2150	D		Extrud	5506868E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 42.0MM x 11.0MM - 2-1/4CR F22M	16 - 16								
2160															
2170	D		Extrud	5506869E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 46.5MM x 11.0MM - 2-1/4CR F22M	18 - 18								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY KD	REV D	FGP/WPMP SHEET 4 OF 7			PA 5	CLASS 2500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2180	D	816	Extrud	5506872E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 51.5MM x 11.0MM - 2-1/4CR F22M	20 - 20						
2190	D		Extrud	5506873E	1 1/2 - 1 1/2	NO MTO-EXTRUDED BRANCH - 66.0MM x 11.0MM - 2-1/4CR F22M	26 - 26						
2200													
2210													
2220	D		Extrud	5506827E	2 - 2	NO MTO-EXTRUDED BRANCH - 51.5MM x 9.0MM - 2-1/4CR F22M	20 - 20						
2230	D		Extrud	5506830E	3 - 3	NO MTO-EXTRUDED BRANCH - 51.5MM x 11.5MM - 2-1/4CR F22M	20 - 20						
2240	D												
2250			Extrud	5506829E	4 - 4	NO MTO-EXTRUDED BRANCH - 51.5MM x 14.0MM - 2-1/4CR F22M	20 - 20						
2260													
2270	D		Extrud	5506828E	2 - 2	NO MTO-EXTRUDED BRANCH - 66.0MM x 9.0MM - 2-1/4CR F22M	26 - 26						
2280	D		Extrud	5506823E	3/4 - 1	NO MTO-EXTRUDED BRANCH - 66.0MM x XXS - 2-1/4CR F22M	26 - 26						
2290	C					***** PIPE FLANGES *****							
2300													
2310													
2320													
2330			Flange	5563396	3 - 3	FLG WN 2500# RF 2-1/4CR F22M WT 11.5MM							
2340			Flange	5563397	4 - 4	FLG WN 2500# RF 2-1/4CR F22M WT 14.0MM							
2350			Flange	5563398	6 - 6	FLG WN 2500# RF 2-1/4CR F22M WT 19.5MM							
2360			Flange	5563400	8 - 8	FLG WN 2500# RF 2-1/4CR F22M WT 24.0MM							
2370			Flange	5563401	10 - 10	FLG WN 2500# RF 2-1/4CR F22M WT 29.0MM							
2380			Flange	5563402	12 - 12	FLG WN 2500# RF 2-1/4CR F22M WT 34.0MM							
2390													
2400	C		Flange	5563403	14 - 14	FLG WN 2500# RF 2-1/4CR F22M WT 37.0MM MANUFACTURERS STANDARD							
2410	C		Flange	5563404	16 - 16	FLG WN 2500# RF 2-1/4CR F22M WT 42.0MM MANUFACTURERS STANDARD							
2420	C		Flange	5563405	18 - 18	FLG WN 2500# RF 2-1/4CR F22M WT 46.5MM MANUFACTURERS STANDARD							
2430	C		Flange	5563406	20 - 20	FLG WN 2500# RF 2-1/4CR F22M WT 51.5MM MANUFACTURERS STANDARD							
2440	C		Flange	5563430	26 - 26	FLG WN 2500# RF 2-1/4CR F22M WT 66.0MM MANUFACTURERS STANDARD							
2450	D	FA8	Reduci	5599713	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 2500# RTJ LTCS A350 LF2	3/4 - 3/4						
2460													
2470													
2480	D	UDE	Flange	5851202	3/4 - 3/4	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE							
2490	D		Flange	5851202	2 - 2	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE							
2500													
2510	D	UDE	Flange	5563339	3/4 - 1	FLG WN 2500# RTJ 2-1/4CR F22M SCH XXS							
2520	D		Flange	5563340	1 1/2 - 1 1/2	FLG WN 2500# RTJ 2-1/4CR F22M WT 11.0MM							
2530	D	UDD	Flange	5557706	3/4 - 1	FLG WN 2500# RTJ LTCS A350 LF2 SCH XXS	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 GS,PHC						
2540			Flange	5619935	1 1/2 - 1 1/2	FLG WN 2500# RTJ LTCS A350 LF2 WT 11.0MM							
2550													
2560	D		Flange	5619932	2 - 2	FLG WN 2500# RTJ 2-1/4CR F22M WT 9MM	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 2500 RF TEMPERATURE RANGE: -40 / 180 °C						
2570	C												
2580	C		Flange	5619921	3 - 3	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 11.5MM							
2590	C		Flange	5619922	4 - 4	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 14.0MM							
2600	C		Flange	5619923	6 - 6	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 19.5MM							
2610	C		Flange	5619853	8 - 8	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 24.0MM							
2620	C		Flange	5619854	10 - 10	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 29.0MM							
2630	C		Flange	5619855	12 - 12	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 34.0MM							
2640	C		Flange	5619869	14 - 14	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 37.0MM							
2650	C		Flange	5619856	16 - 16	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 42.0MM							
2660	C		Flange	5619857	18 - 18	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 46.5MM							
2670	C		Flange	5619858	20 - 20	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 51.5MM							
2680	C		Flange	5619859	26 - 26	FLG COMPACT 2500# RJ 2-1/4CR F22M WT 66.0MM							
2690	D												
2700				Flange	5563312	10 - 10		FLG WN 2500# RTJ 2-1/4CR F22M WT 29.0MM					
2710													
2720	D			Blind	5619736	3/4 - 1		FLG BLIND 2500# RTJ LTCS A350 LF2					
2730	D	Blind		5619737	1 1/2 - 2	FLG BLIND 2500# RTJ 2-1/4CR F22M							
2740													
2750													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY KD	REV D	FGP/WPMP SHEET 5 OF 7	PA 5	CLASS 2500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2760	D		Blind	5619792	3 - 3	FLG COMPACT BLIND 2500# RJ 2-1/4CR F22M PIPE CLASS 2500F22 FOR NPS 3 INCH ONLY							
2770													
2780	D		Blind	5619791	4 - 26	FLG COMPACT BLIND 2500# RJ 2-1/4CR F22M PIPE CLASS 2500F22 FOR NPS 4 INCH & ABOVE							
2790													
2800													
2810	C		Blind	5407493	3 - 12	FLG BLIND 2500# RF 2-1/4CR F22M							
2820			Blind	5407494	14 - 26	FLG BLIND 2500# RF 2-1/4CR F22M MANUFACTURERS STANDARD							
2830													
2840													
2850						***** GASKETS *****							
2860													
2870	C		GASFRR	5672271	3 - 12	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR							
2880	C		GASFRR	5672272	14 - 26	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR MANUFACTURERS STANDARD							
2890													
2900	D		GASRJ	5674480	1/2 - 2	GASKET 2500# RTJ SS OCTAGONAL RING							
2910	D		GASRJ	5674480	10 - 10	GASKET 2500# RTJ SS OCTAGONAL RING							
2920													
2930			GASRJ	5674486	8 - 8	GASKET 10000 API 6A BX 156 7 1/16" NS 316SS RING							
2940													
2950	D	815	GASRJ	5673511	3 - 3	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 2500F22 FOR NPS 3 INCH ONLY							
2960	D	815	GASRJ	5673510	4 - 26	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 2500F22 FOR NPS 4 INCH & ABOVE							
2970													
2980													
2990		B1				***** BOLTING *****							
3000													
3010	D	BA2	BOLS	5675770	1/4 - 2	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3020	D	BA4	BOLS	5675770	3 - 4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3030	D	BA5	BOLS	5675778	6 - 99	STUD-BOLT/COMPACT NUT API-6A718							
3040													
3050	D	BA1	BOLS	5675770	3 - 8	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
3060	D	BA1	BOLS	5675776	10 - 99	STUD-BOLT WITH 2 NUTS API-6A718							
3070													
3080		SB6	CPSBOL	5675675	12 - 26	JACK SCREW HEX HEAD A320 GR L7M							
3090		B1											
3100													
3110						***** BLINDS & SPACERS *****							
3120													
3130	D		Specta	5407692	2 - 2	SPECTACLE BLIND RTJ 2500# 2-1/4 CR F22M							
3140	D		Specta	5407694	3 - 3	SPECTACLE BLIND COMPACT 2500# RJ 2-1/4 CR F22M PIPE CLASS 2500F22 FOR NPS 3 INCH & BELOW							
3150													
3160	C		Paddle	5407805	8 - 26	PADDLE BLANK COMPACT 2500# RJ 2-1/4CR F22M FOR PIPE CLASS 2500F22 NPS 4 INCH & ABOVE							
3170	C		Paddle	5407705	8 - 26	PADDLE SPACER COMPACT 2500# RJ 2-1/4CR F22M FOR PIPE CLASS 2500F22 NPS 4 INCH & ABOVE							
3180													
3190	C		Specta	5407693	3 - 3	SPECTACLE BLIND 2500# RF 2-1/4 CR F22M PADDLE BLANK 2500# RF 2-1/4CR F22M							
3200			Paddle	5407803	8 - 8	PADDLE BLANK 2500# RF 2-1/4CR F22M							
3210			Paddle	5407703	8 - 8	PADDLE SPACER 2500# RF 2-1/4CR F22M							
3220													
3230			Paddle	5407803	12 - 12	PADDLE BLANK 2500# RF 2-1/4CR F22M							
3240			Paddle	5407703	12 - 12	PADDLE SPACER 2500# RF 2-1/4CR F22M							
3250													
3260	C		Paddle	5407804	14 - 26	PADDLE BLANK 2500# RF 2-1/4CR F22M MANUFACTURERS STANDARD							
						CONSTRUCTION		FGP/WPMP					
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
							3 MM	17NOV16	KD	D	6 OF 7	5	2500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3270	C		Paddle	5407704	14 - 26	PADDLE SPACER 2500# RF 2-1/4CR F22M MANUFACTURERS STANDARD		SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 GS,PHC MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 2500 RF TEMPERATURE RANGE: -40 / 180 °C					
3280						***** BRANCHES *****							
3290						USE BRANCH CHART 2500F22							
3300													
3310													
3320													
3330													
3340	D		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 26						
3350	D		Reinfo	RWELD	2 - 20	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 26						
3360													
3370						***** CONSTRUCTION DETAILS *****							
3380													
3390		ZA6				SEAL WELDING REQUIRED----- FOR THREADED CONNECTIONS ONLY							
3400						PIPE BENDS ----- NONE							
3410						THREAD SEALANT -----NONE							
3420													
3430													
3440						*****DO NOT USE*****							
3450													
3460	D		Nipofl	5504551	3/4 - 1	NIPOFLANGE 2500# RTJ WT 11.5MM / SCH XXS 2-1/4CR F22M	3 - 3						
3470	D		Nipofl	5504554	3/4 - 1	NIPOFLANGE 2500# RTJ WT 24.0MM/ SCH XXS 2-1/4CR F22M	8 - 8						
3480	D		Nipofl	5504560	3/4 - 1	NIPOFLANGE 2500# RTJ WT 51.5MM/ SCH XXS 2-1/4CR F22M	20 - 20						
3490	D		Nipofl	5504572	1 1/2 - 1 1/2	NIPOFLANGE 2500# RTJ WT 51.5MM/ 11.0MM 2-1/4CR F22M	20 - 20						
3500	D		Nipofl	5504573	1 1/2 - 1 1/2	NIPOFLANGE 2500# RTJ WT 66.0MM/ 11.0MM 2-1/4CR F22M	26 - 26						
3510	D	UJ2	Nipofl	5504617	1 1/2 - 1 1/2	NIPOFLANGE 2500# RTJ 14.0MM / 11.0MM 2-1/4CR F22M	4 - 4						
3520													
3530	C		Weldol	5514281	2 - 2	WELDOLET WT 51.5MM / 9.0MM 2-1/4CR F22M	20 - 20						
3540	C		Weldol	5514282	3 - 3	WELDOLET WT 51.5MM / 11.5MM 2-1/4CR F22M	20 - 20						
3550	C		Weldol	5514283	4 - 4	WELDOLET WT 51.5MM / 14.0MM 2-1/4CR F22M	20 - 20						
3560	C		Weldol	5506987	3/4 - 1	WELDOLET WT 51.5MM / SCH XXS 2-1/4CR F22M	20 - 20						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY KD	REV D	FGP/WPMP SHEET 7 OF 7	PA 5	CLASS 2500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10		PP9				*****VALVES*****		032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.					
20						*****MODULAR VALVES*****		812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.					
30								815	TO BE USED WITH COMPACT FLANGES ONLY.					
40								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's					
50														
60	D	VA5	2BV	K25RC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM METAL LO - FULL SIZE TYPE 2BV			ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.					
70								B1	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.					
80	D		2BV	K25RC08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM METAL GO - FULL SIZE TYPE 2BV		BA5	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.					
90								FA8	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
100	D	VAV	2BV	K25CC08	3 - 3	MODULAR VALVE DB&B BALL TYPE 2500# CF LTCS ASTM A350 LF2 316SS TRIM METAL GO - FULL SIZE TYPE 2BV		PP9	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
110								SB6	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.					
120	D	VAV	R2BV	M25HC08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ REDUCING LTCS A350LF2 316SS TRIM METAL GO - FULL SIZE TYPE R2BV	3 - 3	SC1						
130														
140	D		2BVR	M25RC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS.					
150								UD7	SEAL WELD AFTER HYDROSTATIC TEST.					
160	D		2BVR	M25RC08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM METAL GO - CONTROLLED BORE TYPE 2BVR			USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST					
170								UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
180	D	VAW	R2BVR	M25HC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
190								VAD	FOR INSTRUMENT CONNECTIONS ONLY AND STEAM BOOTS (WHERE APPLICABLE).					
200	D	VAU	R2BVR	M25YC02	1 - 1	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ LTCS A350LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE 2BVR	3 - 3	VAF	FOR METHANOL AND CHEMICAL INJECTION POINTS ONLY.					
210								VAU	COMPACT FLANGE AT LARGE END RING TYPE JOINT AT SMALL END					
220														
230		VB2				*****BALL VALVES*****								
240														
250	D		Ball V	A25RC07	2 - 2	BALL 2500# RTJ LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
260														
270	D		Ball V	A25CC07	3 - 3	BALL 2500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
280														
290			Ball V	A25CC07	12 - 12	BALL 2500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO								
300														
310														
320			Ball V	B25RC01	3/4 - 1 1/2	BALL 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM FLOATING SOFT FB LO								
330														
340		VA5	Ball V	B25RC07	2 - 2	BALL 2500# RTJ LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
350														
360		VA5	Ball V	B25CC07	3 - 3	BALL 2500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
370														
380		VA5	Ball V	B25CC07	12 - 12	BALL 2500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO								
390														
400														
410						*****GLOBE VALVES*****								
420														
430			Globe	L25RC48	3/4 - 1 1/2	GLOBE 2500# RTJ LTCS A350 LF2 TRIM 16 OS&Y HW								
440			Globe	L25RC41	2 - 2	GLOBE 2500# RTJ LTCS A352 LCC TRIM 16 OS&Y HW								
450	D		Globe	L25CC42	3 - 3	GLOBE 2500# CF LTCS A352 LCC TRIM 16 OS&Y GO								
460														
470														
480						*****CHECK VALVES*****								
490														
500	D	VC1	Check	C25RC50	3/4 - 1 1/2	CHECK PISTON 2500# RTJ LTCS A350 LF2 TRIM 16								
510	D	VC2	Check	C25RC52	2 - 2	CHECK DUAL 2500# RTJ LTCS A352 LCC TRIM 16								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED PWHT FOR WT > 19MM	CORR. ALLOW. 1.5 MM	FGP/WPMP DATE 17NOV16 BY MLA REV D SHEET 1 OF 4 PA 5 CLASS 2500H1						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
520	D	VC2	Check	C25CC52	3 - 3	CHECK DUAL 2500# CF LTCS A352 LCC TRIM 16		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.										
530		VC2	Check	C25CC52	12 - 12	CHECK DUAL 2500# CF LTCS A352 LCC TRIM 16		VAW											
540								VB2											
550						*****PIPES*****													
560																			
570								VC1											
580	D	812	Piping	5358511	3/4 - 1 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE													
590	C		Piping	5358516	2 - 2	PIPE WT 12.0MM SMLS LTCS A333 GR.6 BE													
600	C		Piping	5358536	3 - 3	PIPE WT 16.0MM SMLS LTCS A333 GR.6 BE													
610	C		Piping	5358537	4 - 4	PIPE WT 20.0MM SMLS LTCS A333 GR.6 BE		VC2											
620	C		Piping	5358539	6 - 6	PIPE WT 29.0MM SMLS LTCS A333 GR.6 BE													
630	C		Piping	5358538	12 - 12	PIPE WT 53.0MM SMLS LTCS A333 GR.6 BE		ZA6											
640	D		Piping	5358532	16 - 16	PIPE WT 66.0MM SMLS LTCS A333 GR.6 BE													
650						***** FITTINGS *****													
660																			
670																			
680	C		90 Deg	5392870	3/4 - 1 1/2	ELL 90 DEG LR XXS LTCS A420 WPL6													
690	C		90 Deg	5392871	2 - 2	ELL 90 DEG LR WT 12.0MM LTCS A420 WPL6													
700	C		90 Deg	5392818	3 - 3	ELL 90 DEG LR WT 16.0MM LTCS A420 WPL6													
710	C		90 Deg	5392819	12 - 12	ELL 90 DEG LR WT 53.0MM LTCS A420 WPL6													
720																			
730	C		45 Deg	5392876	3/4 - 1 1/2	ELL 45 DEG LR XXS LTCS A420 WPL6													
740	C		45 Deg	5392875	2 - 2	ELL 45 DEG LR WT 12.0MM LTCS A420 WPL6													
750	C		45 Deg	5392832	3 - 3	ELL 45 DEG LR WT 16.0MM LTCS A420 WPL6													
760	C		45 Deg	5392833	12 - 12	ELL 45 DEG LR WT 53.0MM LTCS A420 WPL6													
770																			
780	C		Tee	5392879	3/4 - 1 1/2	TEE SCH XXS LTCS A420 WPL6													
790	C		Tee	5392889	2 - 2	TEE WT 12.0MM LTCS A420 WPL6													
800	C		Tee	5392837	3 - 3	TEE WT 16.0MM LTCS A420 WPL6													
810	C		Tee	5392838	12 - 12	TEE WT 53.0MM LTCS A420 WPL6													
820																			
830	C		Reduci	5392877	3/4 - 1	TEE RED SCH XXS LTCS A420 WPL6		1 - 1 1/2											
840	C		Reduci	5392887	3/4 - 1 1/2	TEE RED WT 12.0MM / SCH XXS LTCS A420 WPL6		2 - 2											
850	C		Reduci	5392842	1 1/2 - 1 1/2	TEE RED WT 16.0MM/SCH XXS LTCS A420 WPL6		3 - 3											
860	C		Reduci	5392843	2 - 2	TEE RED WT 16.0MM/12.0MM LTCS A420 WPL6		3 - 3											
870	C		Reduci	5392844	2 - 2	TEE RED WT 20.0MM/12.0MM LTCS A420 WPL6		4 - 4											
880																			
890	C		Cap	5392880	3/4 - 1 1/2	CAP SCH XXS LTCS A420 WPL6													
900	C		Cap	5392883	2 - 2	CAP WT 12.0MM LTCS A420 WPL6													
910	C		Cap	5392846	3 - 3	CAP WT 16.0MM LTCS A420 WPL6													
920	D		Cap	5392624	6 - 6	CAP WT 29.0MM LTCS A420 WPL6													
930	C		Cap	5392847	12 - 12	CAP WT 53.0MM LTCS A420 WPL6													
940																			
950	C		Concen	5392881	3/4 - 1	REDUCER CONC SCH XXS LTCS A420 WPL6		1 - 1 1/2											
960	C		Concen	5392884	3/4 - 1 1/2	REDUCER CONC WT 12.0MM / SCH XXS LTCS A420 WPL6		2 - 2											
970	C		Concen	5392859	1 1/2 - 1 1/2	REDUCER CONC WT 16.0MM/XXS LTCS A420 WPL6		3 - 3											
980	C		Concen	5392858	2 - 2	REDUCER CONC WT 16.0MM/12.0MM LTCS A420 WPL6		3 - 3											
990	C		Concen	5392852	2 - 2	REDUCER CONC WT 20.0MM/12.0MM LTCS A420 WPL6		4 - 4											
1000	C		Concen	5392853	3 - 3	REDUCER CONC WT 20.0MM/16.0MM LTCS A420 WPL6		4 - 4											
1010																			
1020	C		Eccent	5392882	3/4 - 1	REDUCER ECC SCH XXS LTCS A420 WPL6		1 - 1 1/2											
1030	C		Eccent	5392885	3/4 - 1 1/2	REDUCER ECC WT 12.0MM / SCH XXS LTCS A420 WPL6		2 - 2											
1040	C		Eccent	5392862	1 1/2 - 1 1/2	REDUCER ECC WT 16.0MM/XXS LTCS A420 WPL6		3 - 3											
1050	C		Eccent	5392861	2 - 2	REDUCER ECC WT 16.0MM/12.0MM LTCS A420 WPL6		3 - 3											
						CONSTRUCTION								FGP/WPMP					
						ALL SIZES BUTT-WELDED PWHT FOR WT > 19MM		CORR. ALLOW.						DATE	BY	REV	SHEET	PA	CLASS
								1.5 MM						17NOV16	MLA	D	2 OF 4	5	2500H1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1060	C		Eccent	5392863	1 1/2 - 1 1/2	REDUCER ECC WT 20.0MM / SCH XXS LTCS A420 WPL6	4 - 4						
1070	C		Eccent	5392867	2 - 2	REDUCER ECC WT 20.0MM / 12.0MM LTCS A420 WPL6	4 - 4						
1080	C		Eccent	5392866	3 - 3	REDUCER ECC WT 20.0MM / 16.0MM LTCS A420 WPL6	4 - 4						
1090	C		Eccent	5392864	3 - 3	REDUCER ECC WT 29.0MM / 16.0MM LTCS A420 WPL6	6 - 6						
1100	D		Eccent	5392869	4 - 4	REDUCER ECC WT 37.0MM / 20.0MM LTCS A420 WPL6	8 - 8						
1110	C		Eccent	5392865	6 - 6	REDUCER ECC WT 53.0MM / 29MM LTCS A420 WPL6	12 - 12						
1120	D		Eccent	5392874	8 - 8	REDUCER ECC WT 53.0MM / 37MM LTCS A420 WPL6	12 - 12						
1130	D		Eccent	5392873	8 - 8	REDUCER ECC WT 66.0MM / 37.0MM LTCS A420 WPL6	16 - 16						
1140													
1150													
1160						***** OLETS *****							
1170													
1180	D		Nipofl	5504285	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2	2 - 2						
1190	D		Nipofl	5504286	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2	3 - 3						
1200	D		Nipofl	5504287	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 20.0MM / SCH XXS LTCS A350 LF2	4 - 4						
1210	D		Nipofl	5504288	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 53.0MM / SCH XXS LTCS A350 LF2	12 - 12						
1220													
1230	D	UJ2	Nipofl	5504290	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2	2 - 2						
1240	D	UJ2	Nipofl	5504291	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2	3 - 3						
1250	D	UJ2	Nipofl	5504292	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 20.0MM / SCH XXS LTCS A350 LF2	4 - 4						
1260	D	UJ2	Nipofl	5504293	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 53.0MM / SCH XXS LTCS A350 LF2	12 - 12						
1270													
1280	C		Weldol	5505158	3/4 - 1	WELDOLET WT 12.0MM/SCH XXS LTCS A350 LF2	2 - 2						
1290	C		Weldol	5505159	3/4 - 1	WELDOLET WT 16.0MM/SCH XXS LTCS A350 LF2	3 - 3						
1300	C		Weldol	5505160	3/4 - 1	WELDOLET WT 20.0 MM / SCH XXS LTCS A350 LF2	4 - 4						
1310	D		Weldol	5505156	2 - 2	WELDOLET WT 29.0MM/WT 12.0MM LTCS A350 LF2	6 - 6						
1320	C		Weldol	5505161	3/4 - 1	WELDOLET WT 53.0 MM / SCH XXS LTCS A350 LF2	12 - 12						
1330	C		Weldol	5505162	2 - 2	WELDOLET WT 53.0 MM / 12.0MM LTCS A350 LF2	12 - 12						
1340	C		Weldol	5505163	3 - 3	WELDOLET WT 53.0 MM / 16.0MM LTCS A350 LF2	12 - 12						
1350													
1360	D	UJ2	Elbole	5516025	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2	2 - 2						
1370	D	UJ2	Elbole	5516026	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2	3 - 3						
1380	D	UJ2	Elbole	5516027	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 20.0MM /SCH XXS LTCS A350 LF2	4 - 4						
1390	D	UJ2	Elbole	5516028	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 53.0MM / SCH XXS LTCS A350 LF2	12 - 12						
1400													
1410		816				***** PIPE FLANGES *****							
1420													
1430	D		Flange	5557706	3/4 - 1 1/2	FLG WN 2500# RTJ LTCS A350 LF2 SCH XXS	DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 GHC, GHP, GS, PHC, PL, SG NOT TO BE USED IN SOUR SERVICES						
1440	D		Flange	5619890	2 - 2	FLG WN 2500# RTJ LTCS A350 LF2 WT 12.0MM							
1450	D		Flange	5619936	3 - 3	FLG WN 2500# RTJ LTCS A350 LF2 WT 16.0MM							
1460													
1470	D		Flange	5619893	3 - 3	FLG COMPACT 2500# RJ LTCS A350 LF2 WT 16.0MM							
1480	D		Flange	5619894	12 - 12	FLG COMPACT 2500# RJ LTCS A350 LF2 WT 53.0MM							
1490													
1500	C		Flange	5557714	3 - 3	FLG WN 2500# RF LTCS A350 LF2 WT 16.0MM							
1510	C		Flange	5557715	4 - 4	FLG WN 2500# RF LTCS A350 LF2 WT 20.0MM							
1520	C		Flange	5557716	12 - 12	FLG WN 2500# RF LTCS A350 LF2 WT 53.0MM							
1530													
1540	D	FA8	Reduci	5599713	1/2 - 3/4	FLG SCRDR (FNPT) REDUCING 2500# RTJ LTCS A350 LF2	3/4 - 1 1/2	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 2500 RF TEMPERATURE RANGE: -46 / 200°C					
1550													
1560	D		Flange	5851202	3/4 - 3/4	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE							
1570	D		Flange	5851202	2 - 2	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE							
1580													
1590	D		Blind	5615615	3/4 - 2	FLG BLIND 2500# RTJ LTCS A350 LF2							
1600	D		Blind	5615617	3 - 12	FLG BLIND 2500# RF LTCS A350 LF2							
1610	D		Blind	5619795	3 - 3	FLG COMPACT BLIND 2500# RJ LTCS A350 LF2 PIPE CLASS 2500H1							
1620	D		Blind	5619795	12 - 12	FLG COMPACT BLIND 2500# RJ LTCS A350 LF2 PIPE CLASS 2500H1							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED PWHT FOR WT > 19MM	CORR. ALLOW. 1.5 MM						DATE 17NOV16

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1630						***** PIPE NIPPLES *****								
1640		812				NIPPLE BBE CUT FROM PIPE								
1650														
1660														
1670														
1680						***** GASKETS *****								
1690														
1700	C		GASFRS	5672271	1/2 - 12	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR								
1710	D		GASRJ	5674480	1/2 - 3	GASKET 2500# RTJ SS OCTAGONAL RING								
1720	D	815	GASRJ	5673520	3 - 12	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 2500H1								
1730														
1740														
1750		B1				***** BOLTING *****								
1760														
1770	D		BOLS	5675770	1/4 - 8	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
1780	D	BA5	BOLS	5675770	10 - 12	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
1790														
1800	D	BA1	BOLS	5675776	10 - 99	STUD-BOLT WITH 2 NUTS API-6A718								
1810														
1820		SB6	CPSBOL	5675675	12 - 12	JACK SCREW HEX HEAD A320 GR L7M								
1830														
1840														
1850						***** BLINDS & SPACERS *****								
1860														
1870			Specta	5749590	3/4 - 2	SPECTACLE BLIND 2500# RTJ LTCS A516-GR70								
1880	C		Specta	5749597	3 - 3	SPECTACLE BLIND 2500# RF LTCS A516-GR70								
1890														
1900	D		Specta	5749598	3 - 3	SPECTACLE BLIND COMPACT 2500# RJ LTCS A516-GR70								
1910														
1920														
1930						***** BRANCHES *****								
1940						USE BRANCH CHART2500H1								
1950														
1960														
1970	D		Reinfo	RWELD	2 - 12	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 16							
1980	D		Reinfo	RWELD	2 - 12	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 16							
1990														
2000						***** CONSTRUCTION DETAILS *****								
2010														
2020		ZA6				SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY								
2030						PIPE BENDS ----- NONE								
2040						THREAD SEALANT ----- PTFE PASTE								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED PWHT FOR WT > 19MM	CORR. ALLOW. 1.5MM						DATE 17NOV16	BY MLA

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
10		PP9				*****VALVES*****		812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.				
20								815	TO BE USED WITH COMPACT FLANGES ONLY.				
30								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING				
40						*****MODULAR VALVES*****			SME's				
50								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.				
60	B	VA5	2BV	K25RC02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM METAL LO - FULL SIZE TYPE 2BV		FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.				
70								PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156				
80	B		2BV	K25RC08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM METAL GO - FULL SIZE TYPE 2BV		SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.				
90								UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.				
100	B		2BVR	M25RC02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.				
110								VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055				
120	B		2BVR	M25RC08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM METAL GO - CONTROLLED BORE TYPE 2BVR		VB2	- VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.				
130							2 - 2	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.				
140	B	VAW	R2BVR	M25HC02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS A350 LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR		VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.				
150								ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.				
160						*****BALL VALVES*****							
170		VB2							SERVICE				
180									DESIGN: 365 AT -46/200 DEG C				
190	B		Ball V	A25RC07	2 - 2	BALL 2500# RTJ LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO			CODE: ASME B31.3				
200									DHC, PHC				
210	B		Ball V	A25CC07-K1	4 - 4	BALL 2500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT RB GO							
220													
230													
240	B		Ball V	B25RC01	3/4 - 1 1/2	BALL 2500# RTJ LTCS ASTM A350 LF2 316SS TRIM FLOATING SOFT FB LO							
250													
260	B	VA5	Ball V	B25RC07	2 - 2	BALL 2500# RTJ LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO							
270													
280	B	VA5	Ball V	B25CC07-K1	4 - 4	BALL 2500# CF LTCS A350 LF2 316SS TRIM TRUNNION SOFT FB GO							
290													
300						*****GLOBE VALVES*****							
310													
320													
330	B		Globe	L25RC48	3/4 - 1 1/2	GLOBE 2500# RTJ LTCS A350 LF2 TRIM 16 OS&Y HW							
340	B		Globe	L25RC41	2 - 2	GLOBE 2500# RTJ LTCS A352 LCC TRIM 16 OS&Y HW							
350	B		Globe	L25CC42-K1	4 - 4	GLOBE 2500# CF LTCS A352 LCC TRIM 16 OS&Y GO							
360													
370						****CHECK VALVES****							
380													
390													
400	B	VC1	Check	C25RC50	3/4 - 1 1/2	CHECK PISTON 2500# RTJ LTCS A350 LF2 TRIM 16							
410	B	VC2	Check	C25RC52	2 - 2	CHECK DUAL 2500# RTJ LTCS A352 LCC TRIM 16							
420	B	VC2	Check	C25CC52-K1	4 - 4	CHECK DUAL 2500# CF LTCS A352 LCC TRIM 16							
430													
440						*****PIPES*****							
450													
460													
470	B	812	Piping	5358511	3/4 - 1 1/2	PIPE SCH XXS SMLS LTCS A333 GR.6 PE							
480			Piping	5358516	2 - 2	PIPE WT 12.0MM SMLS LTCS A333 GR.6 BE							
490			Piping	5358536	3 - 3	PIPE WT 16.0MM SMLS LTCS A333 GR.6 BE							
500			Piping	5358537	4 - 4	PIPE WT 20.0MM SMLS LTCS A333 GR.6 BE							
510													
520						***** FITTINGS *****							
530													
540													
550			90 Deg	5392870	3/4 - 1 1/2	ELL 90 DEG LR XXS LTCS A420 WPL6							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY ET	REV B	SHEET 1 OF 4	PA 5	CLASS 2500K1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
560	B		90 Deg	5392871	2 - 2	ELL 90 DEG LR WT 12.0MM LTCS A420 WPL6	1 - 1 1/2							
570			90 Deg	5392818	3 - 3	ELL 90 DEG LR WT 16.0MM LTCS A420 WPL6								
580														
590			45 Deg	5392876	3/4 - 1 1/2	ELL 45 DEG LR XXS LTCS A420 WPL6								
600			45 Deg	5392875	2 - 2	ELL 45 DEG LR WT 12.0MM LTCS A420 WPL6								
610			45 Deg	5392832	3 - 3	ELL 45 DEG LR WT 16.0MM LTCS A420 WPL6								
620														
630			Tee	5392879	3/4 - 1 1/2	TEE SCH XXS LTCS A420 WPL6								
640			Tee	5392889	2 - 2	TEE WT 12.0MM LTCS A420 WPL6								
650			Tee	5392837	3 - 3	TEE WT 16.0MM LTCS A420 WPL6								
660														
670			Reduci	5392877	3/4 - 1	TEE RED SCH XXS LTCS A420 WPL6								
680			Reduci	5392887	1 - 1	TEE RED WT 12.0MM / SCH XXS LTCS A420 WPL6								
690			Reduci	5392842	1 1/2 - 1 1/2	TEE RED WT 16.0MM/SCH XXS LTCS A420 WPL6								
700			Reduci	5392843	2 - 2	TEE RED WT 16.0MM/12.0MM LTCS A420 WPL6								
710			Reduci	5392844	2 - 2	TEE RED WT 20.0MM/12.0MM LTCS A420 WPL6								
720														
730			Cap	5392880	3/4 - 1 1/2	CAP SCH XXS LTCS A420 WPL6								
740			Cap	5392883	2 - 2	CAP WT 12.0MM LTCS A420 WPL6								
750			Cap	5392846	3 - 3	CAP WT 16.0MM LTCS A420 WPL6								
760														
770			Concen	5392881	3/4 - 1	REDUCER CONC SCH XXS LTCS A420 WPL6								
780			Concen	5392884	3/4 - 1 1/2	REDUCER CONC WT 12.0MM / SCH XXS LTCS A420 WPL6								
790			Concen	5392859	1 1/2 - 1 1/2	REDUCER CONC WT 16.0MM/XXS LTCS A420 WPL6								
800			Concen	5392852	2 - 2	REDUCER CONC WT 20.0MM/12.0MM LTCS A420 WPL6								
810			Concen	5392853	3 - 3	REDUCER CONC WT 20.0MM/16.0MM LTCS A420 WPL6								
820														
830			Eccent	5392882	3/4 - 1	REDUCER ECC SCH XXS LTCS A420 WPL6								
840			Eccent	5392885	3/4 - 1 1/2	REDUCER ECC WT 12.0MM / SCH XXS LTCS A420 WPL6								
850			Eccent	5392862	1 1/2 - 1 1/2	REDUCER ECC WT 16.0MM/XXS LTCS A420 WPL6								
860			Eccent	5392861	2 - 2	REDUCER ECC WT 16.0MM/12.0MM LTCS A420 WPL6								
870			Eccent	5392863	1 1/2 - 1 1/2	REDUCER ECC WT 20.0MM / SCH XXS LTCS A420 WPL6								
880			Eccent	5392867	2 - 2	REDUCER ECC WT 20.0MM / 12.0MM LTCS A420 WPL6								
890			Eccent	5392866	3 - 3	REDUCER ECC WT 20.0MM / 16.0MM LTCS A420 WPL6								
900														
910														
920						***** OLETS *****								
930														
940	B		Nipofl	5504285	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2	2 - 2	DESIGN: 365 AT -46/200 DEG C CODE: ASME B31.3 DHC, PHC						
950	B		Nipofl	5504286	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2	3 - 3							
960	B		Nipofl	5504287	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 20.0MM / SCH XXS LTCS A350 LF2	4 - 4							
970														
980	B	UJ2	Nipofl	5504290	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2	2 - 2							
990	B	UJ2	Nipofl	5504291	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2	3 - 3							
1000	B	UJ2	Nipofl	5504292	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 20.0MM / SCH XXS LTCS A350 LF2	4 - 4							
1010														
1020			Weldol	5505158	3/4 - 1	WELDOLET WT 12.0MM/SCH XXS LTCS A350 LF2	2 - 2	MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 2500 RF TEMPERATURE RANGE: -46 / 200°C COMMENT: LIMITED TO 365 barg						
1030			Weldol	5505159	3/4 - 1	WELDOLET WT 16.0MM/SCH XXS LTCS A350 LF2	3 - 3							
1040			Weldol	5505160	3/4 - 1	WELDOLET WT 20.0 MM / SCH XXS LTCS A350 LF2	4 - 4							
1050														
1060	B	UJ2	Elbole	5516025	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2	2 - 2							
1070	B	UJ2	Elbole	5516026	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2	3 - 3							
1080	B	UJ2	Elbole	5516027	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 20.0MM /SCH XXS LTCS A350 LF2	4 - 4							
1090														
1100														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELDED PWHT - ALL SIZES.		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								3 MM	17NOV16	ET	B	2 OF 4	5	2500K1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
1110		816				***** PIPE FLANGES *****	3/4 - 1 1/2								
1120															
1130	B		Flange	5557706	3/4 - 1 1/2	FLG WN 2500# RTJ LTCS A350 LF2 SCH XXS									
1140	B		Flange	5619890	2 - 2	FLG WN 2500# RTJ LTCS A350 LF2 WT 12.0MM									
1150															
1160			Flange	5557714	3 - 3	FLG WN 2500# RF LTCS A350 LF2 WT 16.0MM									
1170			Flange	5557715	4 - 4	FLG WN 2500# RF LTCS A350 LF2 WT 20.0MM									
1180															
1190	B		FA8	Reduci	5599713	1/2 - 3/4								FLG SCRD (FNPT) REDUCING 2500# RTJ LTCS A350 LF2	
1200															
1210	B		Flange	5851202	3/4 - 3/4	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE		DESIGN: 365 AT -46/200 DEG C CODE: ASME B31.3 DHC, PHC							
1220	B		Flange	5851202	2 - 2	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE									
1230															
1240	B		Flange	5619902	4 - 4	FLG COMPACT 2500# RJ LTCS A350 LF2 WT 20.0MM									
1250															
1260			Blind	5615617	3 - 4	FLG BLIND 2500# RF LTCS A350 LF2									
1270	B		Blind	5615615	3/4 - 2	FLG BLIND 2500# RTJ LTCS A350 LF2									
1280	B		Blind	5619796	4 - 4	FLG COMPACT BLIND 2500# RJ LTCS A350 LF2 PIPE CLASS 2500K1									
1290															
1300															
1310		812				***** PIPE NIPPLES *****									
1320						NIPPLE BBE CUT FROM PIPE									
1330															
1340															
1350															
1360						***** GASKETS *****									
1370															
1380				GASFRS	5672271	1/2 - 4								GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR	
1390	B			GASRJ	5674480	3/4 - 2								GASKET 2500# RTJ SS OCTAGONAL RING	
1400	B		815	GASRJ	5673521	3 - 4								GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 2500K1	
1410								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 2500 RF TEMPERATURE RANGE: -46 / 200°C COMMENT: LIMITED TO 365 barg							
1420															
1430		B1				***** BOLTING *****									
1440															
1450															
1460	B		BOLS	5675770	1/4 - 4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS									
1470		SB6													
1480			CPSBOL	5675675	12 - 12	JACK SCREW HEX HEAD A320 GR L7M									
1490															
1500						***** BLINDS & SPACERS *****									
1510															
1520															
1530			Specta	5749597	3 - 3	SPECTACLE BLIND 2500# RF LTCS A516-GR70									
1540															
1550			Specta	5749590	3/4 - 2	SPECTACLE BLIND 2500# RTJ LTCS A516-GR70									
1560	B		Specta	5749575	3 - 4	SPECTACLE BLIND COMPACT 2500# RJ LTCS A516-GR70									
1570															
1580						***** BRANCHES *****									
1590															
1600						USE BRANCH CHART2500K1									
1610							3 - 4								
1620															
1630															
1640	B		Reinfo	RWELD	2 - 3	NO-MTO REINFORCING WELD FOR STUB-IN									
1650	B		Reinfo	RWELD	2 - 3	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 4								
1660															
1670															
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM	DATE 17NOV16	BY ET	REV B	FGP/WPMP SHEET 3 OF 4		PA 5	CLASS 2500K1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1680 1690 1700 1710 1720 1730 1740 1750 1760 1770 1780 1790 1800		ZA6	Blind Specta	5619730 5749598	3/4 - 2 2 - 2	***** CONSTRUCTION DETAILS ***** SEAL WELDING ----- FOR THREADED CONNECTIONS ONLY PIPE BENDS ----- NONE THREAD SEALANT ----- PTFE PASTE *****DO NOT USE***** FLG COMPACT BLIND 2500# RJ LTCS A350 LF2 SPECTACLE BLIND COMPACT 2500# RJ LTCS A516-GR70							
								DESIGN: 365 AT -46/200 DEG C CODE: ASME B31.3 DHC, PHC					
								MATERIAL: LOW TEMP CARBON STEEL FLANGE RATING: 2500 RF TEMPERATURE RANGE: -46 / 200°C COMMENT: LIMITED TO 365 barg					
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY ET	REV B	FGP/WPMP SHEET 4 OF 4	PA 5	CLASS 2500K1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10 20 30 40 50 60	D	VA5	2BV	K25RS02	1/2 - 1 1/2	*****VALVES***** *****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F316 316SS TRIM METAL LO - FULL SIZE TYPE 2BV		032	ENGINEERING SME'S SHALL BE CONSULTED FOR APPROVAL WHEN LEGACY COMPONENTS ARE TO BE USED.						
70 80	D		2BV	K25RS08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F316 316SS TRIM METAL GO - FULL SIZE TYPE 2BV		812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.						
90 100	D	VAV	2BV	K25CS08	3 - 3	MODULAR VALVE DB&B BALL TYPE 2500# CF SS A182 F316 316SS TRIM METAL GO - FULL SIZE TYPE 2BV		815 816	TO BE USED WITH COMPACT FLANGES ONLY. COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's						
110 120	D		2BVR	M25RS02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F316 316SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR		B1 FA8 PP9	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY. ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
130 140	D		2BVR	M25RS08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F316 316SS TRIM METAL GO - CONTROLLED BORE TYPE 2BVR		SC1	USE 3/4 IN SCREWED FLANGES FOR SAMPLE SUPPLY AND RETURN CONNECTIONS AND 1/2 IN & 3/4 IN SCREWED FLANGES STEAM SUPPLY/CONDENSATE RETURN LINES FOR CONTROTRACE.						
150 160	D	VAW	R2BVR	M25HS02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F316 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2	UD6	USE FOR UNVALVED CAPPED HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST.						
170 180	D	VAU	R2BVR	M25YS02	1 - 1	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ SS A182 F316 316SS TRIM METAL LO REDUCING - CONTROLLED BORE 2BVR	3 - 3	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST						
190 200 210 220		VAW						UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.						
230 240	D		Ball V	A25RS07	2 - 2	*****BALL VALVES***** BALL 2500# RTJ SS A182 F316 316SS TRIM TRUNNION SOFT RB GO BALL 2500# CF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO		VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.						
250 260	D		Ball V	A25CS07	3 - 4			VAU	COMPACT FLANGE AT LARGE END RING TYPE JOINT AT SMALL END						
270 280	D		Ball V	B25RS01	3/4 - 1 1/2										
290 300	D	VA5	Ball V	B25RS07	2 - 2										
310 320 330 340 350	D	VA5	Ball V	B25CS07	3 - 4	BALL 2500# CF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO		SERVICE							
360 370 380 390 400 410 420	D D		Globe Globe Globe	L25RS48 L25RS41 L25CS42	3/4 - 1 1/2 2 - 2 3 - 4	*****GLOBE VALVES***** GLOBE 2500# RTJ SS A182 F316 TRIM 16 OS&Y HW GLOBE 2500# RTJ SS A351 CF8M TRIM 16 OS&Y HW GLOBE 2500# CF SS A351 CF8M TRIM 16 OS&Y GO		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 BF, SG, VB							
430 440 450 460 470 480 490	D D D	VC1 VC2 VC2	Check Check Check	C25RS50 C25RS52 C25CS52	3/4 - 1 1/2 2 - 2 3 - 4	*****CHECK VALVES***** CHECK PISTON 2500# RTJ SS A182 F316 TRIM 16 CHECK DUAL 2500# RTJ SS A351 CF8M TRIM 16 CHECK DUAL 2500# CF SS A351 CF8M TRIM 16		MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 2500 RF TEMPERATURE RANGE: -46 / 120°C COMMENT: SHORT TERM 350 BARG @ 120°C							
500 510 520	D	812	Piping Piping Piping	5365202 5365208 5365209	3/4 - 1 1/2 2 - 2 3 - 3	***** PIPE ***** PIPE SCH XXS SMLS SS A312 316/316L SS PE PIPE WT 12.7MM SMLS SS A312 316/316L BE PIPE WT 19.05MM SMLS SS A312 316/316L BE									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED		FGP/WPMP							
						CORR. ALLOW. 1 MM		DATE 17NOV16	BY MLA	REV D	SHEET 1 OF 4	PA 5	CLASS 2500P1		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
530			Piping	5365210	4 - 4	PIPE WT 22.00MM SMLS SS A312 316/316L BE		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.					
540								VAW						
550						***** FITTINGS *****		VB2	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.					
560								VC1						
570								VC2	DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY.					
580			90 Deg	5449141	3/4 - 1 1/2	ELL BW 90 DEG LR SCH XXS SS A403 WP316/316L-S		ZA6						
590			90 Deg	5449057	2 - 2	ELL BW 90 DEG LR WT 12.7MM SS A403 WP316/316L-S			REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
600			90 Deg	5449058	3 - 3	ELL BW 90 DEG LR WT 19.05MM SS A403 WP316/316L-S								
610			90 Deg	5449082	4 - 4	ELL BW 90 DEG LR WT 22.00MM SS A403 WP316/316L-S								
620														
630			45 Deg	5449143	3/4 - 1 1/2	ELL 45 DEG LR XXS SS A403 WP316/316L-S								
640			45 Deg	5449059	2 - 2	ELL BW 45 DEG LR WT 12.7MM SS A403 WP316/316L-S								
650			45 Deg	5449060	3 - 3	ELL BW 45 DEG LR WT 19.05MM SS A403 WP316/316L-S								
660			45 Deg	5449083	4 - 4	ELL BW 45 DEG LR WT 22.00MM SS A403 WP316/316L-S								
670														
680			Tee	5449191	3/4 - 1 1/2	TEE SCH XXS SS A403 WP316/316L-S								
690			Tee	5449061	2 - 2	TEE WT 12.7MM SS A403 WP316/316L-S								
700			Tee	5449062	3 - 3	TEE WT 19.05MM SS A403 WP316/316L-S								
710			Tee	5449183	4 - 4	TEE WT 22.00MM SS A403 WP316/316L-S								
720														
730			Reduci	5449192	3/4 - 1	TEE RED SCH XXS SS A403 WP316/316L-S	1 - 1 1/2							
740			Reduci	5449063	3/4 - 1 1/2	TEE RED WT 12.7MM X SCH XXS SS A403 WP316/316L-S	2 - 2							
750			Reduci	5449064	1 1/2 - 1 1/2	TEE RED WT 19.05MM X SCH XXS SS A403 WP316/316L-S	3 - 3							
760			Reduci	5449065	2 - 2	TEE RED WT 19.05MM/WT 12.7 MM SS A403 WP316/316L-S	3 - 3							
770			Reduci	5449098	3 - 3	TEE RED WT 22.0MM/WT 19.05 MM SS A403 WP316/316L-S	4 - 4							
780														
790			Cap	5449056	3/4 - 1 1/2	CAP SCH XXS SS A403 WP316/316L-S								
800			Cap	5449066	2 - 2	CAP WT 12.7MM SS A403 WP316/316L-S								
810			Cap	5449067	3 - 3	CAP WT 19.05MM SS A403 WP316/316L-S								
820			Cap	5449099	4 - 4	CAP WT 22.00MM SS A403 WP316/316L-S								
830														
840			Concen	5449181	3/4 - 1	REDUCER CONC SCH XXS SS A403 WP316/316L-S	1 - 1 1/2							
850			Concen	5449068	3/4 - 1 1/2	REDUCER CONC WT 12.7MM/SCH XXS SS A403 WP316/316L-S	2 - 2							
860			Concen	5449069	1 1/2 - 1 1/2	REDUCER CONC WT 19.05MM/SCH XXS SS A403 WP316/316L-S	3 - 3							
870	C		Concen	5392365	1 1/2 - 1 1/2	REDUCER CONC WT 22.0MM/ SCH XXS SS A403 WP316/316L-S	4 - 4							
880			Concen	5449070	2 - 2	REDUCER CONC WT 19.05MM/WT 12.7MM SS A403 WP316/316L-S	3 - 3							
890	C		Concen	5392349	2 - 2	REDUCER CONC WT 22.0MM/ 12.7MM SS A403 WP316/316L-S	4 - 4							
900			Concen	5449074	3 - 3	REDUCER CONC WT 22.0MM/19.05MM SS A403 WP316/316L-S	4 - 4							
910														
920			Eccent	5449186	3/4 - 1	REDUCER ECC SCH XXS SS A403 WP316/316L-S	1 - 1 1/2							
930			Eccent	5449071	3/4 - 1 1/2	REDUCER ECC WT 12.7MM/SCH XXS SS A403 WP316/316L-S	2 - 2							
940			Eccent	5449072	1 1/2 - 1 1/2	REDUCER ECC WT 19.05MM / SCH XXS SS A403 WP316/316L-S	3 - 3							
950	C		Eccent	5449182	1 1/2 - 1 1/2	REDUCER ECC WT 22.0MM / SCH XXS SS A403 WP316/316L-S	4 - 4							
960			Eccent	5449073	2 - 2	REDUCER ECC WT 19.05MM / WT 12.7MM SS A403WP316/316L-S	3 - 3							
970														
980	C		Eccent	5449188	2 - 2	REDUCER ECC WT 22.0MM / 12.7MM SS A403 WP316/316L-S	4 - 4							
990			Eccent	5449075	3 - 3	REDUCER ECC WT 22.0MM / 19.05MM SS A403 WP316/316L-S	4 - 4							
1000														
1010						***** OLETS *****								
1020														
1030	D	UJ2	Nipofl	5504575	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 12.7MM/ SCH XXS SS A182 F316/316L	2 - 2							
1040	D	UJ2	Nipofl	5504446	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 19.05 MM / SCH XXS SS A182 F316/316L	3 - 3							
1050	D	UJ2	Nipofl	5504447	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 22.00MM / SCH XXS SS A182 F316/316L	4 - 4							
1060														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED		CORR. ALLOW. 1 MM		FGP/WPMP DATE 17NOV16 BY MLA REV D SHEET 2 OF 4 PA 5 CLASS 2500P1				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1070			Weldol	5509662	3/4 - 1	WELDOLET19.05MM WT/SCH XXS SS A182 F316/316L	3 - 3						
1080			Weldol	5509663	3/4 - 1 1/2	WELDOLET 22.00MM WT/SCH XXS SS A182 F316/316L	4 - 4						
1090													
1100	D	UJ2	Elbole	5516452	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 12.7MM/ SCH XXS SS A182 F316/316L	2 - 2						
1110	D	UJ2	Elbole	5516446	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 19.05MM / SCH XXS SS A182 F316/316L	3 - 3						
1120	D	UJ2	Elbole	5516447	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 22.00MM / SCH XXS SS A182 F316/316L	4 - 4						
1130													
1140													
1150		816				***** PIPE FLANGES *****							
1160													
1170	D		Flange	5584166	3/4 - 1 1/2	FLG WN 2500# RTJ SS A182 F316/316L SCH XXS		3/4 - 3/4					
1180	D		Flange	5584170	2 - 2	FLG WN 2500# RTJ SS A182 F316/316L WT 12.7MM							
1190	D		Flange	5584171	3 - 3	FLG WN 2500# RTJ SS A182 F316/316L WT 19.05MM							
1200													
1210			Flange	5619863	3 - 3	FLG COMPACT 2500# RJ SS A182 F316/316L WT 19.05MM							
1220			Flange	5619864	4 - 4	FLG COMPACT 2500# RJ SS A182 F316/316L WT 22.0MM							
1230													
1240			Flange	5584174	3 - 3	FLG WN 2500# RF SS A182 F316/316L WT 19.05MM							
1250			Flange	5584175	4 - 4	FLG WN 2500# RF SS A182 F316/316L WT 22.0MM							
1260													
1270	D	FA8	Reduci	5599830	1/2 - 1/2	FLG SCRDR (FNPT) REDUCING 2500# RTJ SS A182 F316/316L							
1280													
1290	D		Flange	5851202	3/4 - 3/4	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE							
1300	D		Flange	5851202	2 - 2	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE							
1310													
1320	C		Blind	5616574	1/2 - 4	FLG BLIND 2500# RF SS A182 316/316L							
1330	D		Blind	5616570	3/4 - 2	FLG BLIND 2500# RTJ SS A182 316/316L							
1340	D		Blind	5619797	3 - 4	FLG COMPACT BLIND 2500# RJ SS A182 316/316L PIPE CLASS 2500P1							
1350													
1360						***** GASKETS *****							
1370													
1380													
1390	C		GASFRS	5672272	1/2 - 4	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR MANUFACTURERS STANDARD		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 BF, SG, VB					
1400	D		GASRJ	5674480	1/2 - 2	GASKET 2500# RTJ SS OCTAGONAL RING							
1410	D	815	GASRJ	5673530	3 - 4	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 2500P1							
1420													
1430		B1				***** BOLTING *****							
1440													
1450													
1460	D		BOLS	5675770	1/4 - 4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS							
1470													
1480						***** BLINDS & SPACERS *****							
1490													
1500													
1510			Specta	5749514	3 - 4	SPECTACLE BLIND 2500# RF SS A240 GR 316		MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 2500 RF TEMPERATURE RANGE: -46 / 120°C COMMENT: SHORT TERM 350 BARG @ 120°C					
1520	D		Specta	5749592	1 - 2	SPECTACLE BLIND 2500# RTJ SS A240 GR 316							
1530	D		Specta	5749592	3 - 3	SPECTACLE BLIND 2500# RTJ SS A240 GR 316							
1540													
1550	D		Specta	5749573	3 - 3	SPECTACLE BLIND COMPACT 2500# RJ SS A240 GR 316							
1560													
1570						***** BRANCHES *****							
1580													
1590						USE BRANCH CHART2500P1							
1600													
1610													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED	CORR. ALLOW. 1 MM	DATE 17NOV16	BY MLA	REV D	SHEET 3 OF 4	PA 5	CLASS 2500P1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES													
1620 1630 1640 1650 1660 1670 1680 1690 1700	D D	ZA6	Reinfo Reinfo	RWELD RWELD	2 - 3 2 - 3	NO-MTO REINFORCING WELD FOR STUB-IN NO-MTO REINFORCING WELD FOR STUB-IN *****CONSTRUCTION DETAILS***** SEAL WELDING-----FOR THREADED CONNECTIONS ONLY PIPE BENDS-----NONE THREAD SEALANT-----PTFE PASTE	3 - 4 3 - 4														
								SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 BF, SG, VB													
								MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 2500 RF TEMPERATURE RANGE: -46 / 120°C COMMENT: SHORT TERM 350 BARG @ 120°C													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION															
						ALL SIZES BUTT-WELDED															
						CORR. ALLOW. 1 MM															
												DATE	BY	REV	SHEET	PA	CLASS				
												17NOV16	MLA	D	4 OF 4	5	2500P1				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
10		PP9				*****VALVES*****		812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.					
20								815	TO BE USED WITH COMPACT FLANGES ONLY.					
30								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's					
40								B1	ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA.					
50								FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.					
60	B	VA5	2BV	K25RT02	1/2 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO - FULL SIZE TYPE 2BV		FF5	FLAT BOTTOM NIPOFLANGES FOR USE IN BLEED VALVE ASSEMBLIES ONLY.					
70								PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156					
80	B		2BV	K25RT08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL GO - FULL SIZE TYPE 2BV		UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.					
90								VA5	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.					
100	B	VAV	2BV	K25CT08	3 - 3	MODULAR VALVE DB&B BALL TYPE 2500# CF SS A182 F321 316SS TRIM METAL GO - FULL SIZE TYPE 2BV		VAU	COMPACT FLANGE AT LARGE END RING TYPE JOINT AT SMALL END					
110														
120	B		2BVR	M25RT02	3/4 - 1 1/2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR								
130														
140	B		2BVR	M25RT08	2 - 2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL GO - CONTROLLED BORE TYPE 2BVR								
150														
160	B	VAW	R2BVR	M25HT02	3/4 - 1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ SS A182 F321 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2 - 2							
170														
180	B	VAU	R2BVR	M25YT02	1 - 1	MODULAR VALVE DB&B BALL TYPE 2500# CF x RTJ SS A182 F321 316SS TRIM METAL LO REDUCING - CONTROLLED BORE 2BVR	3 - 3							
190														
200		VAW												
210		VB2												
220						*****BALL VALVES*****								
230	B		Ball V	A25RT07	2 - 2	BALL 2500# RTJ SS A182 F321 316SS TRIM TRUNNION SOFT RB GO								
240														
250	B		Ball V	A25CT07	3 - 3	BALL 2500# CF SS A182 F321 316SS TRIM TRUNNION SOFT RB GO								
260														
270														
280	B		Ball V	B25RT01	3/4 - 1 1/2	BALL 2500# RTJ SS A182 F321 316SS TRIM FLOATING SOFT FB LO								
290														
300	B	VA5	Ball V	B25RT07	2 - 2	BALL 2500# RTJ SS A182 F321 316SS TRIM TRUNNION SOFT FB GO								
310														
320	B	VA5	Ball V	B25CT07	3 - 3	BALL 2500# CF SS A182 F321 316SS TRIM TRUNNION SOFT FB GO								
330														
340														
350						*****GLOBE VALVES*****								
360														
370	B		Globe	L25RT48	3/4 - 1 1/2	GLOBE 2500# RTJ SS A182 F321 TRIM 16 OS&Y HW								
380														
390						*****CHECK VALVES*****								
400														
410														
420	B	VC1	Check	C25RT50	3/4 - 1 1/2	CHECK PISTON 2500# RTJ SS A182 F321 TRIM 16								
430	B	VC2	Check	C25RT52	2 - 2	CHECK DUAL 2500# RTJ SS A182 F321 TRIM 16								
440	B	VC2	Check	C25CT52	3 - 3	CHECK DUAL 2500# CF SS A182 F321 TRIM 16								
450														
460						***** PIPE *****								
470														
480														
490	B		Piping	5365202	3/4 - 1 1/2	PIPE SCH XXS SMLS SS A312 316/316L SS PE								
500			Piping	5365208	2 - 2	PIPE WT 12.7MM SMLS SS A312 316/316L BE								
510			Piping	5365209	3 - 3	PIPE WT 19.05MM SMLS SS A312 316/316L BE								
520			Piping	5365210	4 - 4	PIPE WT 22.00MM SMLS SS A312 316/316L BE								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED	CORR. ALLOW. 1 MM	DATE 17NOV16	BY MLA	REV B	FGP/WPMP SHEET 1 OF 4	PA 5	CLASS 2500P2	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES									
530	B	812	Piping	5365211	6 - 6	PIPE WT 24.00MM SMLS SS A312 316/316L BE	1 - 1 1/2 2 - 2 3 - 3 4 - 4	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C. SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. DUAL PLATE CHECK VALVES TO BE INSTALLED IN THE HORIZONTAL POSITION ONLY. REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.								
540			Piping	5365206	3/4 - 3/4	PIPE SCH XXS SMLS SS A312 316/316L SS BE		VAW									
550								VB2									
560								VC1									
570								VC2									
580								ZA6									
590						***** FITTINGS *****											
600			90 Deg	5449141	3/4 - 1 1/2	ELL BW 90 DEG LR SCH XXS SS A403 WP316/316L-S											
610			90 Deg	5449057	2 - 2	ELL BW 90 DEG LR WT 12.7MM SS A403 WP316/316L-S											
620			90 Deg	5449058	3 - 3	ELL BW 90 DEG LR WT 19.05MM SS A403 WP316/316L-S											
630			90 Deg	5449082	4 - 4	ELL BW 90 DEG LR WT 22.00MM SS A403 WP316/316L-S											
640																	
650			45 Deg	5449143	3/4 - 1 1/2	ELL 45 DEG LR XXS SS A403 WP316/316L-S											
660			45 Deg	5449059	2 - 2	ELL BW 45 DEG LR WT 12.7MM SS A403 WP316/316L-S											
670			45 Deg	5449060	3 - 3	ELL BW 45 DEG LR WT 19.05MM SS A403 WP316/316L-S											
680			45 Deg	5449083	4 - 4	ELL BW 45 DEG LR WT 22.00MM SS A403 WP316/316L-S											
690																	
700			Tee	5449191	3/4 - 1 1/2	TEE SCH XXS SS A403 WP316/316L-S											
710			Tee	5449061	2 - 2	TEE WT 12.7MM SS A403 WP316/316L-S											
720			Tee	5449062	3 - 3	TEE WT 19.05MM SS A403 WP316/316L-S											
730			Tee	5449183	4 - 4	TEE WT 22.00MM SS A403 WP316/316L-S											
740																	
750			Reduci	5449192	3/4 - 1	TEE RED SCH XXS SS A403 WP316/316L-S											
760			Reduci	5449063	3/4 - 1 1/2	TEE RED WT 12.7MM X SCH XXS SS A403 WP316/316L-S											
770			Reduci	5449064	1 1/2 - 1 1/2	TEE RED WT 19.05MM X SCH XXS SS A403 WP316/316L-S											
780			Reduci	5449065	2 - 2	TEE RED WT 19.05MM/WT 12.7 MM SS A403 WP316/316L-S											
790			Reduci	5449029	2 - 2	TEE RED WT 22.0MM/WT 12.7 MM SS A403 WP316/316L-S											
800			Reduci	5449098	3 - 3	TEE RED WT 22.0MM/WT 19.05 MM SS A403 WP316/316L-S											
810																	
820			Cap	5449056	3/4 - 1 1/2	CAP SCH XXS SS A403 WP316/316L-S											
830			Cap	5449066	2 - 2	CAP WT 12.7MM SS A403 WP316/316L-S											
840			Cap	5449067	3 - 3	CAP WT 19.05MM SS A403 WP316/316L-S											
850			Cap	5449099	4 - 4	CAP WT 22.00MM SS A403 WP316/316L-S											
860			Cap	5449100	6 - 6	CAP WT 24.00MM SS A403 WP316/316L-S											
870																	
880			Concen	5449181	3/4 - 1	REDUCER CONC SCH XXS SS A403 WP316/316L-S											
890			Concen	5449068	3/4 - 1 1/2	REDUCER CONC WT 12.7MM/SCH XXS SS A403 WP316/316L-S											
900			Concen	5449069	1 1/2 - 1 1/2	REDUCER CONC WT 19.05MM/SCH XXS SS A403 WP316/316L-S											
910			Concen	5392365	1 1/2 - 1 1/2	REDUCER CONC WT 22.0MM/ SCH XXS SS A403 WP316/316L-S											
920			Concen	5449070	2 - 2	REDUCER CONC WT 19.05MM/WT 12.7MM SS A403WP316/316L-S											
930																	
940			Concen	5392349	2 - 2	REDUCER CONC WT 22.0MM/ 12.7MM SS A403 WP316/316L-S											
950			Concen	5449074	3 - 3	REDUCER CONC WT 22.0MM/19.05MM SS A403 WP316/316L-S											
960																	
970			Ecccent	5449186	3/4 - 1	REDUCER ECC SCH XXS SS A403 WP316/316L-S											
980			Ecccent	5449071	3/4 - 1 1/2	REDUCER ECC WT 12.7MM/SCH XXS SS A403 WP316/316L-S											
990			Ecccent	5449072	1 1/2 - 1 1/2	REDUCER ECC WT 19.05MM / SCH XXS SS A403 WP316/316L-S											
1000			Ecccent	5449182	1 1/2 - 1 1/2	REDUCER ECC WT 22.0MM / SCH XXS SS A403 WP316/316L-S											
1010			Ecccent	5449073	2 - 2	REDUCER ECC WT 19.05MM / WT 12.7MM SS A403WP316/316L-S											
1020																	
1030			Ecccent	5449188	2 - 2	REDUCER ECC WT 22.0MM / 12.7MM SS A403 WP316/316L-S											
1040			Ecccent	5449075	3 - 3	REDUCER ECC WT 22.0MM / 19.05MM SS A403 WP316/316L-S											
1050																	
1060																	
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED		FGP/WPMP									
						CORR. ALLOW.		DATE	BY	REV	SHEET	PA	CLASS				
						1 MM		17NOV16	MLA	B	2 OF 4	5	2500P2				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
1070						***** OLETS *****								
1080														
1090	B	UJ2	Nipofl	5504586	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 12.7MM / SCH XXS SS A182 F321	2 - 2							
1100	B	UJ2	Nipofl	5504587	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 19.05 MM / SCH XXS SS A182 F321	3 - 3							
1110	B	UJ2	Nipofl	5504588	3/4 - 1 1/2	NIPOFLANGE 2500# RTJ WT 22.00MM/XXS SS A182 F321	4 - 4							
1120														
1130	B	FF5	Endole	5504610	3/4 - 3/4	NIPOFLANGE FLAT BOTTOM 2500# RTJ SCH XXS SS A182 F321								
1140														
1150			Weldol	5509662	3/4 - 1	WELDOLET19.05MM WT/SCH XXS SS A182 F316/316L	3 - 3							
1160			Weldol	5509663	3/4 - 1 1/2	WELDOLET 22.00MM WT/SCH XXS SS A182 F316/316L	4 - 4							
1170			Weldol	5509664	2 - 2	WELDOLET 24.00MM WT/ 12.7MM SS A182 F316/316L	6 - 6							
1180														
1190		UJ2	Elbole	5561561	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 12.7MM / SCH XXS SS A182 F321	2 - 2							
1200		UJ2	Elbole	5561562	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 19.05MM / SCH XXS SS A182 F321	3 - 3							
1210		UJ2	Elbole	5561563	1 1/2 - 1 1/2	ELBOFLANGE 2500# RTJ WT 22.0MM / SCH XXS SS A182 F321	4 - 4							
1220														
1230		816				***** PIPE FLANGES *****								
1240														
1250	B		Flange	5584161	1/2 - 1 1/2	FLG WN 2500# RTJ SS A182 F321 SCH XXS								
1260	B		Flange	5884162	2 - 2	FLG WN 2500# RTJ SS A182 F321 WT 12.7MM								
1270	B		Flange	5584163	3 - 3	FLG WN 2500# RTJ SS A182 F321 WT 19.05MM								
1280														
1290			Flange	5619867	3 - 3	FLG COMPACT 2500# RJ SS A182 F321 WT 19.05MM								
1300														
1310			Flange	5584274	3 - 3	FLG WN 2500# RF SS A182 F321 WT 19.05MM								
1320			Flange	5584275	4 - 4	FLG WN 2500# RF SS A182 F321 WT 22.0MM								
1330														
1340	B	FA8	Flange	5599831	1/2 - 1/2	FLG SCRD (FNPT) REDUCING 2500# RTJ SS A182 F321	3/4 - 3/4							
1350														
1360	B		Flange	5851213	3/4 - 3/4	FLANGE ADAPTER 2500# RTJ 321 SS X 12 MM TUBE								
1370	B		Flange	5851213	2 - 2	FLANGE ADAPTER 2500# RTJ 321 SS X 12 MM TUBE								
1380														
1390														
1400			Blind	5616584	3 - 4	FLG BLIND 2500# RF SS A182 F321								
1410	B		Blind	5619738	3/4 - 2	FLG BLIND 2500# RTJ SS A182 F321								
1420	B		Blind	5619798	3 - 3	FLG COMPACT BLIND 2500# RJ SS A182 F321 PIPE CLASS 2500P2								
1430														
1440														
1450						***** GASKETS *****								
1460														
1470			GASFRS	5672272	1/2 - 4	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR MANUFACTURERS STANDARD								
1480	B		GASRJ	5674480	1/2 - 3	GASKET 2500# RTJ SS OCTAGONAL RING								
1490	B	815	GASRJ	5673531	3 - 4	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 2500P2								
1500														
1510														
1520		B1				***** BOLTING *****								
1530														
1540	B		BOLS	5675770	1/4 - 4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
1550														
1560														
1570						***** BLINDS & SPACERS *****								
1580														
1590	B		Specta	5749592	1 - 2	SPECTACLE BLIND 2500# RTJ SS A240 GR 316								
1600	B		Specta	5749573	3 - 4	SPECTACLE BLIND COMPACT 2500# RJ SS A240 GR 316								
1610														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTT-WELDED	CORR. ALLOW. 1 MM	DATE 17NOV16	BY MLA	REV B	FGP/WPMP SHEET 3 OF 4		PA 5	CLASS 2500P2

[illegible]

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES								
10 20 30 40 50 60	B	PP9	2BV	K10CL16	2 - 2	*****VALVES***** *****MODULAR VALVES***** MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV	3 - 3	812 815 816	NIPPLE TO BE CUT FROM SS SCH XXS PIPE. TO BE USED WITH COMPACT FLANGES ONLY. COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED WITH API 6A FLANGES ONLY. TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.							
70 80	B	VAV	2BV	K10CL16	3 - 3	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV		B1 BA3 BA4	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY.							
90 100	B	VAW	R2BV	K10PL16	2 - 2	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV REDUCING FLANGES		FAX	API 6BX FLANGE TO NB PIPE DESIGNATION							
110 120 130 140	B		Ball V	B10CL16-A60	3/4 - 6	BALL 10000 CF INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO			NB PIPE 6BX FLANGE 2" 1 13/16" 3" 3 1/16" 4" 4 1/16" 6" 5 1/8" 8" 7 1/16"							
150 160 170									B	Ball V	B10LL16	3/4 - 6	BALL 10000 API 6A RJ INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO	PP9 VAV VAW	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.	
180 190 200 210 220	B	VQ2	Condui	W10CL72-A60	2 - 2	GATE 10000 CF INCONEL 625 BODY AND TRIM THROUGH CONDUIT SOLID SLAB HW		3 - 3	VC7	10000PSI RATED CHECK VALVES SHALL BE SP ITEMS. REFER TO SP LIST FOR DETAILS						
230			Condui	W10CL73	3 - 6	GATE 10000 CF INCONEL 625 BODY AND TRIM THROUGH CONDUIT SOLID SLAB GO			VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.						
240 250 260 270 280 290 300 310	B		Globe	L10CL43-A60	2 - 2	GLOBE 10000# CF INC 625 B564 INC 625 TRIM OS&Y HW				ZA6	REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
320 330 340 350										VC7	*****CHECK VALVES*****					
360 370 380 390 400 410	B		Piping	5362279	3/4 - 1 1/2	PIPE SCH 160 SMLS ALLOY 625 B444 UNS N06625 PE			3 - 3	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG GS - SOUR GAS, PHC, GHP						
420 430 440 450 460	B		Piping	5362289	2 - 3	PIPE SCH 160 SMLS ALLOY 625 B444 UNS N06625 BE				MATERIAL: INCONEL 625 (NACE) FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -100 / 110 °C						
470 480 490 500 510	B		Piping	5362281	4 - 4	PIPE WT 14MM SMLS ALLOY 625 B444 UNS N06625 BE										
			Piping	5362282	6 - 6	PIPE WT 20MM SMLS ALLOY 625 B444 UNS N06625 BE										
			Piping	5362283	8 - 8	PIPE WT 26MM SMLS ALLOY 625 B444 UNS N06625 BE										
		812	Piping	5362285	3/4 - 3/4	PIPE SCH XXS SMLS ALLOY 625 B444 UNS N06625 BE										
						***** FITTINGS *****										
			90 Deg	5426512	3/4 - 3	ELL BW 90 DEG LR SCH 160 ALLOY 625 B366-WPS UNS N06625										
			90 Deg	5426513	4 - 4	ELL BW 90 DEG LR WT 14.0MM ALLOY 625 B366-WPS UNS N06625										
			90 Deg	5426514	6 - 6	ELL BW 90 DEG LR WT 20.00MM ALLOY 625 B366-WPS UNS N06625										
FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN						CONSTRUCTION ALL SIZES- BUTT WELD	CORR. ALLOW. 1 MM			DATE 17NOV16	BY KD	REV B	SHEET 1 OF 4	PA 5	CLASS 10000A60	

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
520			45 Deg	5426517	3/4 - 3	ELL BW 45 DEG LR SCH 160 ALLOY 625 B366-WPS UNS N06625								
530			45 Deg	5426523	4 - 4	ELL BW 45 DEG LR WT 14.0MM ALLOY 625 B366-WPS UNS N06625								
540														
550			45 Deg	5426524	6 - 6	ELL BW 45 DEG LR WT 20.0MM ALLOY 625 B366-WPS UNS N06625								
560														
570														
580			Tee	5426564	3/4 - 3	TEE BW SCH 160 ALLOY 625 B366-WPS UNS N06625								
590			Tee	5426565	4 - 4	TEE BW WT 14.0MM ALLOY 625 B366-WPS UNS N06625								
600			Tee	5426566	6 - 6	TEE BW WT 20.0MM ALLOY 625 B366-WPS UNS N06625								
610														
620			Reduci	5426551	3/4 - 2	TEE RED SCH 160 ALLOY 625 B366-WPS UNS N06625	1 - 3							
630			Reduci	5426553	1 1/2 - 3	TEE RED WT 14.0MM/ SCH 160 ALLOY 625 B366-WPS UNS N06625	4 - 4							
640			Reduci	5426554	3 - 3	TEE RED WT 20.0MM/ SCH 160 ALLOY 625 B366-WPS UNS N06625	6 - 6							
650			Reduci	5426555	4 - 4	TEE RED WT 20.0MM/ 14MM ALLOY 625 B366-WPS UNS N06625	6 - 6							
660														
670			Cap	5426526	3/4 - 3	CAP SCH 160 ALLOY 625 B366 UNS N06625								
680			Cap	5426527	4 - 4	CAP WT 14.0MM ALLOY 625 B366 UNS N06625								
690			Cap	5426528	6 - 6	CAP WT 20.0MM ALLOY 625 B366 UNS N06625								
700														
710			Concen	5426570	3/4 - 2	REDUCER CONC SCH 160 ALLOY 625 B366-WPS N06625	1 - 3							
720			Concen	5426571	1 1/2 - 3	REDUCER CONC WT 14.0MM/ SCH 160 ALLOY 625 B366-WPS N06625	4 - 4							
730			Concen	5426572	3 - 3	REDUCER CONC WT 20.0MM/ SCH 160 ALLOY 625 B366-WPS N06625	6 - 6							
740			Concen	5426573	4 - 4	REDUCER CONC WT 20.0MM/ 14.0MM ALLOY 625 B366-WPS N06625	6 - 6							
750														
760														
770			Eccent	5426576	3/4 - 2	REDUCER ECC SCH 160 ALLOY 625 B366-WPS N06625	1 - 3							
780			Eccent	5426577	1 1/2 - 3	REDUCER ECC WT 14.0MM/SCH 160 ALLOY 625 B366-WPS N06625	4 - 4							
790														
800			Eccent	5426578	3 - 3	REDUCER ECC WT 20.0MM/SCH 160 ALLOY 625 B366-WPS N06625	6 - 6							
810														
820			Eccent	5426579	4 - 4	REDUCER ECC WT 20.0MM/14.0MM ALLOY 625 B366-WPS N06625	6 - 6							
830														
840														
850						***** OLETS *****								
860														
870			Sweepo	5506928	2 - 2	REINFORCED BRANCH INSERT CONNECTION 20.0MM X SCH 160, BW OUTLET API ALLOY 625 B564 UNS N06625	6 - 6							
880														
890														
900														
910		816 FAX				***** FLANGES *****								
920														
930			Reduci	5851251	2 - 2	FLANGE COMPACT 10000# RJ WITH 9/16" FEMALE AUTOCLAVE CONNECTION ALLOY 625 B564 UNS N06625								
940														
950		816	Flange	5619910	3/4 - 3/4	FLG COMPACT 10000# RJ BORE 15.6MM ALLOY 625 B564 UNS N06625								
960														
970		816	Flange	5619909	1 - 1	FLG COMPACT 10000# RJ BORE 20.7MM ALLOY 625 B564 UNS N06625								
980														
990		816	Flange	5619912	1 1/2 - 1 1/2	FLG COMPACT 10000# RJ BORE 34.0MM ALLOY 625 B564 UNS N06625								
1000														
1010		816	Flange	5619913	2 - 2	FLG COMPACT 10000# RJ BORE 42.8MM ALLOY 625 B564 UNS N06625								
1020														
1030		816	Flange	5619914	3 - 3	FLG COMPACT 10000# RJ BORE 66.6MM ALLOY 625 B564 UNS N06625								
1040														
1050		816	Flange	5619915	4 - 4	FLG COMPACT 10000# RJ BORE 86.3MM ALLOY 625 B564 UNS N06625								
						CONSTRUCTION								FGP/WPMP
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			ALL SIZES- BUTT WELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								1 MM	17NOV16	KD	B	2 OF 4	5	10000A60

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES										
1060	B	816	Flange	5619916	6 - 6	FLG COMPACT 10000# RJ BORE 128.3MM ALLOY 625 B564 UNS N06625												
1070		816	Flange	5619917	8 - 8	FLG COMPACT 10000# RJ BORE 167.1MM ALLOY 625 B564 UNS N06625												
1080			Flange	5563432	2 - 2	FLG WN 10000 API 6A TYPE 6BX 1 13/16" NS (2" NB) RTJ BORE 42.8MM ALLOY 625 B564 UNS N06625												
1090																		
1100																		
1110																		
1120																		
1130																		
1140																		
1150																		
1160	Blind	5407440	2 - 2	FLG BLIND 10000 API 6A TYPE 6BX 1 13/16" NS RTJ ALLOY 625 B564 UNS N06625														
1170	Blind	5407441	3 - 3	FLG BLIND 10000 API 6A TYPE 6BX 3 1/16" NS RTJ ALLOY 625 B564 UNS N06625														
1180	Blind	5407442	4 - 4	FLG BLIND 10000 API 6A TYPE 6BX 4 1/6" NS RTJ ALLOY 625 B564 UNS N06625														
1190	Blind	5407443	6 - 6	FLG BLIND 10000 API 6A TYPE 6BX 5 1/8" NS RTJ ALLOY 625 B564 UNS N06625														
1200	B		Blind	5619751	1/2 - 8	FLG COMPACT BLIND 10000# RJ ALLOY 625 B564 UNS N06625 PIPE CLASS 10000A60												
1210																		
1220																		
1230																		
1240																		
1250																		
1260																		
1270																		
1280																		
1290																		
1300		812				SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG GS - SOUR GAS, PHC, GHP												
1310																		
1320																		
1330																		
1340																		
1350																		
1360																		
1370																		
1380																		
1390													GASRJ		5674476	2 - 2	MATERIAL: INCONEL 625 (NACE) FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -100 / 110 °C	
1400																		
1410																		
1420																		
1430																		
1440	B	815	GASRJ	5673540	1/2 - 8	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 10000A60												
1450	B	B1				***** BOLTING *****												
1460																		
1470																		
1480																		
1490																		
	B	BA4	BOLS	5675776	3/4 - 4	STUD-BOLT WITH 2 NUTS API-6A718												
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP										
						ALL SIZES- BUTT WELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS				
								1 MM	17NOV16	KD	B	3 OF 4	5	10000A60				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES											
1500	B	BA5	BOLS	5675778	6 - 99	STUD-BOLT/COMPACT NUT API-6A718	3 - 6 3 - 6	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG GS - SOUR GAS, PHC, GHP											
1510	B	BA3	BOLS	5675770	2 - 6	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS													
1520																			
1530																			
1540																			
1550	FAX		Specta	5749572	2 - 4	SPECTACLE BLIND COMPACT 10000# RJ FEMALE ALLOY 625 B443 UNS N06625 PIPE CLASS 10000A60													
1560																			
1570																			
1580																			
1590																			
1600																			
1610																			
1620																			
1630																			
1640																			
1650																			
1660																			
1670																			
1680																			
1690																			
1700	B	ZA6	Reinfo	RWELD	2 - 4	NO-MTO REINFORCING WELD FOR STUB-IN													
1710	B		Reinfo	RWELD	2 - 4	NO-MTO REINFORCING WELD FOR STUB-IN													
1720						*****CONSTRUCTION DETAILS*****													
1730																			
1740																			
1750																			
1760																			
1770																			
1780																			
1790																			
1800																			
1810																			
1820																			
1830																			
1840																			
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP											
						ALL SIZES- BUTT WELD		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS					
								1 MM	17NOV16	KD	B	4 OF 4	5	10000A60					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10		PP9				*****VALVES*****		812	NIPPLE TO BE CUT FROM SS SCH 160 XXS PIPE.						
20								815	TO BE USED WITH COMPACT FLANGES ONLY.						
30								816	COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME's						
40		VAT				*****MODULAR VALVES*****		BA3	TO BE USED WITH API 6A FLANGES ONLY.						
50	D		2BV	K10CL16	2 - 2	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV		BA4	TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.						
60								BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY						
70															
80	D	VAV	2BV	K10CL16	3 - 3	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV									
90															
100	D	VAW	R2BV	K10PL16	2 - 2	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO FULL SIZE TYPE 2BV REDUCING FLANGES	3 - 3	FA1	API 6BX FLANGE TO NB PIPE DESIGNATION						
110									NB PIPE		6BX FLANGE				
120									2"		1 13/16"				
130									3"		2 9/16"				
140									4"		4 1/16"				
150									6"		5 1/8"				
160									8"		7 1/16"				
170			Ball V	B10CL16	3/4 - 1 1/2	BALL 10000 CF INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO			10"		9"				
180			Ball V	B10CL16	2 - 3	BALL 10000 CF INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO			12"		11"				
190			Ball V	B10CH16	4 - 22	BALL 10000 CF F22 INCONEL 625 TRIM TRUNNION METAL FB GO			14"		11"				
200									16"		13 5/8"				
210									18"		13 5/8"				
220			Ball V	B10LL16	3/4 - 1 1/2	BALL 10000 API 6A RJ INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO			20"		16 3/4 "				
230									22"		16 3/4 "				
240			Ball V	B10LL16	2 - 3	BALL 10000 API 6A RJ INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO		NA	NOT TO BE USED						
250			Ball V	B10LH16	4 - 22	BALL 10000 API 6A RJ F22 INCONEL 625 TRIM TRUNNION METAL FB GO		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
260								SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.						
270															
280															
290						*****THROUGH CONDUIT GATE VALVES*****									
300															
310	C		Condui	W10CL72	2 - 2	GATE 10000 CF INCONEL 625 BODY AND TRIM THROUGH CONDUIT SOLID SLAB HW									
320	D		Condui	W10CH75	3 - 3	GATE 10000 CF F22 TRIM INC 625 THROUGH CONDUIT SOLID SLAB GO									
330															
340	D		Condui	W10CH73	4 - 12	GATE 10000 CF F22 TRIM INC 625 THROUGH CONDUIT SOLID SLAB GO									
350															
360															
370						*****GLOBE VALVES*****									
380															
390	C		Globe	L10CL43	2 - 2	GLOBE 10000# CF INC 625 B564 INC 625 TRIM OS&Y HW									
400															
410						*****CHECK VALVES*****									
420		VC7													
430															
440						*****PIPE*****									
450															
460															
470	C	812	Piping	5358511	3/4 - 1	PIPE SCH XXS SMLS LTCS A333 GR.6 PE									
480	C		Piping	5358541	1 1/2 - 1 1/2	PIPE WT 12.0MM SMLS LTCS A333 GR.6 PE									
490	C		Piping	5360062	2 - 2	PIPE WT 10.5MM SMLS 2-1/4CR F22M BE									
500	C		Piping	5360067	3 - 3	PIPE WT 14.0MM SMLS 2-1/4CR F22M BE									
510	C		Piping	5360070	4 - 4	PIPE WT 17.0MM SMLS 2-1/4CR F22M BE									
520	C		Piping	5360071	6 - 6	PIPE WT 23.5MM SMLS 2-1/4CR F22M BE									
530	C		Piping	5360075	8 - 8	PIPE WT 29.5MM SMLS 2-1/4CR F22M BE									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM		FGP/WPMP DATE 17NOV16 BY MLA REV D SHEET 1 OF 8 PA 5 CLASS 10000F22					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
540	C		Piping	5360079	10 - 10	PIPE WT 36.0MM SMLS 2-1/4CR F22M BE		VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE. VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS. 10000PSI RATED CHECK VALVES SHALL BE SP ITEMS. REFER TO SP LIST FOR DETAILS REFER TO PIM-SU-5112-TCO, SECTION 6.1,19-20 FOR SEAL WELDING.					
550	C		Piping	5360084	12 - 12	PIPE WT 42.0MM SMLS 2-1/4CR F22M BE		VAW						
560	C		Piping	5360085	14 - 14	PIPE WT 46.0MM SMLS 2-1/4CR F22M BE								
570	C		Piping	5360088	16 - 16	PIPE WT 52.0MM SMLS 2-1/4CR F22M BE		VC7						
580	C		Piping	5360090	18 - 18	PIPE WT 58.0MM SMLS 2-1/4CR F22M BE								
590	C		Piping	5360092	20 - 20	PIPE WT 64.5MM SMLS 2-1/4CR F22M BE		ZA6						
600	C		Piping	5360095	22 - 22	PIPE WT 70.5MM SMLS 2-1/4CR F22M BE								
610														
620														
630														
640						***** FITTINGS *****								
650			90 Deg	5392870	3/4 - 1	ELL 90 DEG LR XXS LTCS A420 WPL6								
660			90 Deg	5392871	1 1/2 - 1 1/2	ELL 90 DEG LR WT 12.0MM LTCS A420 WPL6								
670	C		90 Deg	5406636	2 - 2	ELL 90 DEG LR WT 10.5MM 2-1/4CR F22M								
680	C		90 Deg	5406637	3 - 3	ELL 90 DEG LR WT 14.0MM 2-1/4CR F22M								
690	C		90 Deg	5406600	4 - 4	ELL 90 DEG LR WT 17.0MM 2-1/4CR F22M								
700	C		90 Deg	5406601	6 - 6	ELL 90 DEG LR WT 23.5MM 2-1/4CR F22M								
710	C		90 Deg	5406605	8 - 8	ELL 90 DEG LR WT 29.5MM 2-1/4CR F22M								
720	C		90 Deg	5406609	10 - 10	ELL 90 DEG LR WT 36.0MM 2-1/4CR F22M								
730	C		90 Deg	5406614	12 - 12	ELL 90 DEG LR WT 42.0MM 2-1/4CR F22M								
740	C		90 Deg	5406615	14 - 14	ELL 90 DEG LR WT 46.0MM 2-1/4CR F22M								
750	C		90 Deg	5406618	16 - 16	ELL 90 DEG LR WT 52.0MM 2-1/4CR F22M			SERVICE SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG DHC, GS, GHP, KM MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -40 / 130 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF					
760	C		90 Deg	5406620	18 - 18	ELL 90 DEG LR WT 58.0MM 2-1/4CR F22M								
770	C		90 Deg	5406622	20 - 20	ELL 90 DEG LR WT 64.5MM 2-1/4CR F22M								
780	C		90 Deg	5406635	22 - 22	ELL 90 DEG LR WT 70.5MM 2-1/4CR F22M								
790														
800			45 Deg	5392876	3/4 - 1	ELL 45 DEG LR XXS LTCS A420 WPL6								
810			45 Deg	5392875	1 1/2 - 1 1/2	ELL 45 DEG LR WT 12.0MM LTCS A420 WPL6								
820	C		45 Deg	5407083	2 - 2	ELL 45 DEG LR WT 10.5MM 2-1/4CR F22M								
830	C		45 Deg	5407084	3 - 3	ELL 45 DEG LR WT 14.0MM 2-1/4CR F22M								
840	C		45 Deg	5407060	4 - 4	ELL 45 DEG LR WT 17.0MM 2-1/4CR F22M								
850	C		45 Deg	5407061	6 - 6	ELL 45 DEG LR WT 23.5MM 2-1/4CR F22M								
860	C		45 Deg	5407065	8 - 8	ELL 45 DEG LR WT 29.5MM 2-1/4CR F22M								
870	C		45 Deg	5407069	10 - 10	ELL 45 DEG LR WT 36.0MM 2-1/4CR F22M								
880	C		45 Deg	5407074	12 - 12	ELL 45 DEG LR WT 42.0MM 2-1/4CR F22M								
890	C		45 Deg	5407075	14 - 14	ELL 45 DEG LR WT 46.0MM 2-1/4CR F22M								
900	C		45 Deg	5407078	16 - 16	ELL 45 DEG LR WT 52.0MM 2-1/4CR F22M								
910	C		45 Deg	5407080	18 - 18	ELL 45 DEG LR WT 58.0MM 2-1/4CR F22M								
920	C		45 Deg	5407082	20 - 20	ELL 45 DEG LR WT 64.5MM 2-1/4CR F22M								
930	C		45 Deg	5407095	22 - 22	ELL 45 DEG LR WT 70.5MM 2-1/4CR F22M								
940	D		22.5 D	5407045	8 - 8	ELL 22.5 DEG LR WT 29.5 MM 2-1/4CR F22M								
950														
960			Tee	5392879	3/4 - 1	TEE SCH XXS LTCS A420 WPL6								
970			Tee	5392889	1 1/2 - 1 1/2	TEE WT 12.0MM LTCS A420 WPL6								
980	C		Tee	5407180	2 - 2	TEE WT 10.5MM 2-1/4CR F22M								
990	C		Tee	5407181	3 - 3	TEE WT 14.0MM 2-1/4CR F22M								
1000	C		Tee	5407160	4 - 4	TEE WT 17.0MM 2-1/4CR F22M								
1010	C		Tee	5407161	6 - 6	TEE WT 23.5MM 2-1/4CR F22M								
1020	C		Tee	5407165	8 - 8	TEE WT 29.5MM 2-1/4CR F22M								
1030	C		Tee	5407169	10 - 10	TEE WT 36.0MM 2-1/4CR F22M								
1040	C		Tee	5407173	12 - 12	TEE WT 42.0MM 2-1/4CR F22M								
1050	C		Tee	5407174	14 - 14	TEE WT 46.0MM 2-1/4CR F22M								
1060	C		Tee	5407177	16 - 16	TEE WT 52.0MM 2-1/4CR F22M								
1070	C		Tee	5407178	18 - 18	TEE WT 58.0MM 2-1/4CR F22M								
1080	C		Tee	5407213	22 - 22	TEE WT 70.5MM 2-1/4CR F22M								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM		FGP/WPMP DATE 17NOV16 BY MLA REV D SHEET 2 OF 8 PA 5 CLASS 10000F22				

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES										
1090	C		Reduci	5392877	3/4 - 3/4	TEE RED SCH XXS LTCS A420 WPL6	1 - 1	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG DHC, GS, GHP, KM										
1100			Reduci	5392887	3/4 - 1	TEE RED WT 12.0MM / SCH XXS LTCS A420 WPL6	1 1/2 - 1 1/2											
1110			Reduci	5407366	3/4 - 1	TEE RED SCH 10.5MM / XXS 2-1/4CR F22M	2 - 2											
1120			Reduci	5407367	1 1/2 - 1 1/2	TEE RED BW WT 10.5MM / 12.0MM 2-1/4CR F22M	2 - 2											
1130			Reduci	5407368	1 1/2 - 1 1/2	TEE RED BW 14.0MM / 12.0MM 2-1/4CR F22M	3 - 3											
1140			Reduci	5407369	2 - 2	TEE RED BW WT 14.0MM / 10.5MM 2-1/4CR F22M	3 - 3											
1150			Reduci	5407370	2 - 2	TEE RED BW SCH 17.0MM / 10.5MM 2-1/4CR F22M	4 - 4											
1160			Reduci	5407371	3 - 3	TEE RED BW WT 17.0MM / 14.0MM 2-1/4CR F22M	4 - 4											
1170			Reduci	5407372	3 - 3	TEE RED BW WT 23.5MM / 14.0MM 2-1/4CR F22M	6 - 6											
1180			Reduci	5407381	6 - 6	TEE RED BW WT 29.5MM / 23.5MM 2-1/4CR F22M	8 - 8											
1190			Reduci	5407403	6 - 6	TEE RED BW WT 46.0MM / 23.5MM 2-1/4CR F22M	14 - 14											
1200			Reduci	5407365	8 - 8	TEE RED BW WT 46.0MM / 29.5MM 2-1/4CR F22M	14 - 14											
1210			Reduci	5407395	10 - 10	TEE RED BW WT 46.0MM / 36.0MM 2-1/4CR F22M	14 - 14											
1220			Reduci	5407373	6 - 6	TEE RED BW WT 52.0MM / 23.5MM 2-1/4CR F22M	16 - 16											
1230			Reduci	5407376	8 - 8	TEE RED BW WT 52.0MM / 29.5MM 2-1/4CR F22M	16 - 16											
1240			Reduci	5407392	10 - 10	TEE RED BW WT 52.0MM / 36.0MM 2-1/4CR F22M	16 - 16											
1250			Reduci	5407374	14 - 14	TEE RED BW WT 52.0MM / 46.0MM 2-1/4CR F22M	16 - 16											
1260			Reduci	5407375	8 - 8	TEE RED BW WT 58.0MM / 29.5MM 2-1/4CR F22M	18 - 18											
1270			Reduci	5407378	14 - 14	TEE RED BW WT 58.0MM / 46.0MM 2-1/4CR F22M	18 - 18											
1280			Reduci	5407377	16 - 16	TEE RED BW WT 58.0MM / 52.0MM 2-1/4CR F22M	18 - 18											
1290			Reduci	5407379	18 - 18	TEE RED BW WT 70.5MM / 58.0MM 2-1/4CR F22M	22 - 22											
1300																		
1310					Cap	5392880	3/4 - 1						CAP SCH XXS LTCS A420 WPL6					
1320					Cap	5392883	1 1/2 - 1 1/2						CAP WT 12.0MM LTCS A420 WPL6					
1330			C		Cap	5407124	2 - 2						CAP WT 10.5MM 2-1/4CR F22M					
1340			C		Cap	5407125	3 - 3						CAP WT 14.0MM 2-1/4CR F22M					
1350			C		Cap	5407100	4 - 4						CAP WT 17.0MM SMLS 2-1/4CR F22M					
1360			C		Cap	5407101	6 - 6						CAP WT 23.5MM SMLS 2-1/4CR F22M					
1370			C		Cap	5407105	8 - 8						CAP WT 29.5MM SMLS 2-1/4CR F22M					
1380			C		Cap	5407109	10 - 10						CAP WT 36.0MM SMLS 2-1/4CR F22M					
1390			C		Cap	5407114	12 - 12						CAP WT 42.0MM SMLS 2-1/4CR F22M					
1400			C		Cap	5407115	14 - 14						CAP WT 46.0MM SMLS 2-1/4CR F22M					
1410			C		Cap	5407118	16 - 16						CAP WT 52MM SMLS 2-1/4CR F22M					
1420			C		Cap	5407120	18 - 18						CAP WT 58.0MM SMLS 2-1/4CR F22M					
1430			C		Cap	5407122	20 - 20						CAP WT 64.5MM SMLS 2-1/4CR F22M					
1440			C		Cap	5407153	22 - 22						CAP WT 70.5MM 2-1/4CR F22M					
1450																		
1460	C		Concen	5392881	3/4 - 3/4	REDUCER CONC SCH XXS LTCS A420 WPL6	1 - 1	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -40 / 130 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF										
1470			Concen	5392884	3/4 - 1	REDUCER CONC WT 12.0MM / SCH XXS LTCS A420 WPL6	1 1/2 - 1 1/2											
1480			Concen	5407231	3/4 - 1	REDUCER CONC WT 10.5MM/SCH XXS 2-1/4CR F22M	2 - 2											
1490			Concen	5407232	1 1/2 - 1 1/2	REDUCER CONC WT 10.5MM/12.0MM 2-1/4CR F22M	2 - 2											
1500			Concen	5407233	1 1/2 - 1 1/2	REDUCER CONC WT 14.0MM/12.0MM 2-1/4CR F22M	3 - 3											
1510			Concen	5407234	2 - 2	REDUCER CONC WT 14.0MM/10.5MM 2-1/4CR F22M	3 - 3											
1520			Concen	5407236	2 - 2	REDUCER CONC WT 17.0MM/10.5MM 2-1/4CR F22M	4 - 4											
1530			Concen	5407235	4 - 4	REDUCER CONC WT 29.5MM/17.0MM 2-1/4CR F22M	8 - 8											
1540			Concen	5407255	6 - 6	REDUCER CONC WT 29.5MM/23.5MM 2-1/4CR F22M	8 - 8											
1550			Concen	5407278	6 - 6	REDUCER CONC WT 36.0MM/ 23.5MM 2-1/4CR F22M	10 - 10											
1560			Concen	5407253	8 - 8	REDUCER CONC WT 46.0MM/29.5MM 2-1/4CR F22M	14 - 14											
1570			Concen	5407256	8 - 8	REDUCER CONC WT 52.0MM/29.5MM 2-1/4CR F22M	16 - 16											
1580			Concen	5407254	14 - 14	REDUCER CONC WT 52.0MM/46.0MM 2-1/4CR F22M	16 - 16											
1590			Concen	5407258	14 - 14	REDUCER CONC WT 58.0MM/46.0MM 2-1/4CR F22M	18 - 18											
1600			Concen	5407298	12 - 12	REDUCER CONC WT 52.0MM/42.0MM 2-1/4CR F22M	16 - 16											
1610			Concen	5407257	16 - 16	REDUCER CONC WT 58.0MM/52.0MM 2-1/4CR F22M	18 - 18											
1620			Concen	5407259	18 - 18	REDUCER CONC WT 70.5MM/58.0MM 2-1/4CR F22M	22 - 22											
1630																		
1640	C		Eccent	5392882	3/4 - 3/4	REDUCER ECC SCH XXS LTCS A420 WPL6	1 - 1											
1650			Eccent	5392885	3/4 - 1	REDUCER ECC WT 12.0MM / SCH XXS LTCS A420 WPL6	1 1/2 - 1 1/2											
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY MLA	REV D	FGP/WPMP SHEET 3 OF 8	PA 5	CLASS 10000F22					

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
1660	C		Eccent	5407345	3/4 - 1	REDUCER ECC 10.5MM / SCH XXS 2-1/4CR F22M	2 - 2						
1670	C		Eccent	5407346	1 1/2 - 1 1/2	REDUCER ECC 10.5MM / 12.0MM 2-1/4CR F22M	2 - 2						
1680	C		Eccent	5407347	1 1/2 - 1 1/2	REDUCER ECC 14.0MM / 12.0MM 2-1/4CR F22M	3 - 3						
1690	C		Eccent	5407348	2 - 2	REDUCER ECC 14.0MM / 10.5MM 2-1/4CR F22M	3 - 3						
1700	C		Eccent	5407349	2 - 2	REDUCER ECC 17.0MM / 10.5MM 2-1/4CR F22M	4 - 4						
1710	C		Eccent	5407312	6 - 6	REDUCER ECC WT 29.5MM/23.5MM 2-1/4CR F22M	8 - 8						
1720	C		Eccent	5407313	6 - 6	REDUCER ECC WT 36.0MM/23.5MM 2-1/4CR F22M	10 - 10						
1730	C		Eccent	5407338	8 - 8	REDUCER ECC WT 36.0MM/ 29.5MM 2-1/4CR F22M	10 - 10						
1740	C		Eccent	5407321	8 - 8	REDUCER ECC WT 42.0MM/ 29.5MM 2-1/4CR F22M	12 - 12						
1750	C		Eccent	5407311	8 - 8	REDUCER ECC WT 46.0MM/29.5MM 2-1/4CR F22M	14 - 14						
1760	C		Eccent	5407315	8 - 8	REDUCER ECC WT 52.0MM/29.5MM 2-1/4CR F22M	16 - 16						
1770	C		Eccent	5407310	10 - 10	REDUCER ECC WT 52.0MM/36.0MM 2-1/4CR F22M	16 - 16						
1780	C		Eccent	5407314	14 - 14	REDUCER ECC WT 52.0MM/46.0MM 2-1/4CR F22M	16 - 16						
1790	C		Eccent	5407318	14 - 14	REDUCER ECC WT 58.0MM/46.0MM 2-1/4CR F22M	18 - 18						
1800	C		Eccent	5407316	12 - 12	REDUCER ECC WT 58.0MM/42.0MM 2-1/4CR F22M	18 - 18						
1810	C		Eccent	5407317	16 - 16	REDUCER ECC WT 58.0MM/52.0MM 2-1/4CR F22M	18 - 18						
1820	C		Eccent	5407319	18 - 18	REDUCER ECC WT 70.5MM/58.0MM 2-1/4CR F22M	22 - 22						
1830													
1840						***** EXTRUDED BRANCHES *****							
1850													
1860													
1870	D		Extrud	5506927E	2 - 2	NO MTO-EXTRUDED BRANCH - 17.0MM X 10.5MM-2-1/4CR F22M	4 - 4	SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG DHC, GS, GHP, KM					
1880													
1890	D		Extrud	5506940E	2 - 2	NO MTO-EXTRUDED BRANCH - 23.5MM X 10.5MM-2-1/4CR F22M	6 - 6						
1900													
1910	D		Extrud	5506941E	2 - 2	NO MTO-EXTRUDED BRANCH - 29.5MM X 10.5MM-2-1/4CR F22M	8 - 8						
1920													
1930	D		Extrud	5506952E	3 - 3	NO MTO-EXTRUDED BRANCH - 29.5MM X 14.0MM-2-1/4CR F22M	8 - 8						
1940													
1950	D		Extrud	5506942E	2 - 2	NO MTO-EXTRUDED BRANCH - 36.0MM X 10.5MM-2-1/4CR F22M	10 - 10						
1960													
1970	D		Extrud	5506943E	2 - 2	NO MTO-EXTRUDED BRANCH - 42.0MMX 10.5MM-2-1/4CR F22M	12 - 12	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -40 / 130 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF					
1980													
1990	D		Extrud	5506944E	2 - 2	NO MTO-EXTRUDED BRANCH - 46.0MM X 10.5MM- 2-1/4CR F22M	14 - 14						
2000													
2010	D		Extrud	5506945E	3 - 3	NO MTO-EXTRUDED BRANCH - 46.0MM X 14.0MM-2-1/4CR F22M	14 - 14						
2020													
2030	D		Extrud	5506936E	4 - 4	NO MTO-EXTRUDED BRANCH - 46.0MM X 17.0MM-2-1/4CR F22M	14 - 14						
2040													
2050	D		Extrud	5506946E	2 - 2	NO MTO-EXTRUDED BRANCH - 52.0MM X 10.5MM-2-1/4CR F22M	16 - 16						
2060													
2070	D		Extrud	5506947E	3 - 3	NO MTO-EXTRUDED BRANCH - 52.0MM X 14.0MM-2-1/4CR F22M	16 - 16						
2080													
2090	D		Extrud	5506939E	4 - 4	NO MTO-EXTRUDED BRANCH - 52.0MM X 17.0MM-2-1/4CR F22M	16 - 16						
2100													
2110	D		Extrud	5506948E	2 - 2	NO MTO-EXTRUDED BRANCH - 58.0MM X 10.5MM-2-1/4CR F22M	18 - 18						
2120													
2130	D		Extrud	5506949E	3 - 3	NO MTO-EXTRUDED BRANCH - 58.0MM X 14.0MM-2-1/4CR F22M	18 - 18						
2140													
2150	D		Extrud	5506950E	2 - 2	NO MTO-EXTRUDED BRANCH - 70.5MM X 10.5MM-2-1/4CR F22M	22 - 22						
2160													
2170	D		Extrud	5506951E	3 - 3	NO MTO-EXTRUDED BRANCH - 70.5MM X 14.0MM-2-1/4CR F22M	22 - 22						
2180													
2190													
2200	D		Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 22						
2210	D		Reinfo	RWELD	2 - 16	NO-MTO REINFORCING WELD FOR STUB-IN	3 - 22						
2220													
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY MLA	REV D	FGP/WPMP SHEET 4 OF 8	PA 5	CLASS 10000F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
2230		816				***** FLANGES *****		SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG DHC, GS, GHP, KM MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -40 / 130 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF					
2240		FA1											
2250													
2260	C		Flange	5851252	2 - 2	FLANGE COMPACT 10000# RJ WITH 9/16" FEMALE AUTOCLAVE CONNECTION 2-1/4CR F22M							
2270													
2280													
2290	C	816	Flange	5619871	3/4 - 3/4	FLG COMPACT 10000# RJ BORE 11.0MM LTCS A350 LF2							
2300	C	816	Flange	5619872	1 - 1	FLG COMPACT 10000# RJ BORE 15.2MM LTCS A350 LF2							
2310	C	816	Flange	5619873	1 1/2 - 1 1/2	FLG COMPACT 10000# RJ BORE 24.3MM LTCS A350 LF2							
2320	C	816	Flange	5619874	2 - 2	FLG COMPACT 10000# RJ BORE 39.3MM 2-1/4CR F22M							
2330	C	816	Flange	5619875	3 - 3	FLG COMPACT 10000# RJ BORE 60.9MM 2-1/4CR F22M							
2340	C	816	Flange	5619896	4 - 4	FLG COMPACT 10000# RJ BORE 80.3MM 2-1/4CR F22M							
2350	C	816	Flange	5619897	6 - 6	FLG COMPACT 10000# RJ BORE 121.3MM 2-1/4CR F22M							
2360	C	816	Flange	5619876	8 - 8	FLG COMPACT 10000# RJ BORE 160.1MM 2-1/4CR F22M							
2370	C	816	Flange	5619898	10 - 10	FLG COMPACT 10000# RJ BORE 201.0MM 2-1/4CR F22M							
2380	C	816	Flange	5619899	12 - 12	FLG COMPACT 10000# RJ BORE 239.8MM 2-1/4CR F22M							
2390	C	816	Flange	5619877	14 - 14	FLG COMPACT 10000# RJ BORE 263.6MM 2-1/4CR F22M							
2400	C	816	Flange	5619878	16 - 16	FLG COMPACT 10000# RJ BORE 302.4MM 2-1/4CR F22M							
2410	D	816	Flange	5619879	18 - 18	FLG COMPACT 10000# RJ BORE 341.2MM 2-1/4CR F22M							
2420	C	816	Flange	5619880	20 - 20	FLG COMPACT 10000# RJ BORE 379.0MM 2-1/4CR F22M							
2430	D	816	Flange	5619881	22 - 22	FLG COMPACT 10000# RJ BORE 417.8MM 2-1/4CR F22M							
2440													
2450	C		Flange	5563412	2 - 2	FLG WN 10000 API 6A TYPE 6BX 1 13/16" NS (2" NB) RTJ BORE 39.3MM 2-1/4CR F22M							
2460													
2470	C		Flange	5563413	3 - 3	FLG WN 10000 API 6A TYPE 6BX 2 9/16" NS (3" NB) RTJ BORE 60.9MM 2-1/4CR F22M							
2480													
2490	C		Flange	5563420	4 - 4	FLG WN 10000 API 6A TYPE 6BX 4 1/16" NS (4" NB) RTJ BORE 80.3MM 2-1/4CR F22M							
2500													
2510	C		Flange	5563421	6 - 6	FLG WN 10000 API 6A TYPE 6BX 5 1/8" NS (6" NB) RTJ BORE 121.3MM 2-1/4CR F22M							
2520													
2530	C		Flange	5563422	8 - 8	FLG WN 10000 API 6A TYPE 6BX 7 1/16" NS (8" NB) RTJ BORE 160.1MM 2-1/4CR F22M							
2540													
2550	C		Flange	5563423	10 - 10	FLG WN 10000 API 6A TYPE 6BX 9" NS (10" NB) RTJ BORE 201.0MM 2-1/4CR F22M							
2560													
2570	C		Flange	5563424	12 - 12	FLG WN 10000 API 6A TYPE 6BX 11" NS (12" NB) RTJ BORE 239.8MM 2-1/4CR F22M							
2580													
2590	C		Flange	5563425	14 - 14	FLG WN 10000 API 6A TYPE 6BX 11" (14" NB) NS RTJ BORE 263.6MM 2-1/4CR F22M							
2600													
2610	C		Flange	5563426	16 - 16	FLG WN 10000 API 6A TYPE 6BX 13 5/8" NS (16" NB) RTJ BORE 302.4MM 2-1/4CR F22M							
2620													
2630	C		Flange	5563427	18 - 18	FLG WN 10000 API 6A TYPE 6BX 13 5/8" NS (18" NB) RTJ BORE 341.0MM 2-1/4CR F22M							
2640													
2650	C		Flange	5563428	20 - 20	FLG WN 10000 API 6A TYPE 6BX 16 3/4" NS (20" NB) RTJ BORE 379.0MM 2-1/4CR F22M							
2660													
2670			Flange	5563431	22 - 22	FLG WN 10000 API 6A TYPE 6BX 16 3/4" NS (22" NB) RTJ BORE 418.0MM 2-1/4CR F22M							
2680													
2690													
2700			Blind	5407422	2 - 2	FLG BLIND 10000 API 6A TYPE 6BX 1 13/16" NS RTJ 2-1/4CR F22M							
2710													
2720			Blind	5407423	3 - 3	FLG BLIND 10000 API 6A TYPE 6BX 2 9/16" NS RTJ 2-1/4CR F22M							
2730													
2740			Blind	5407430	4 - 4	FLG BLIND 10000 API 6A TYPE 6BX 4 1/6" NS RTJ 2-1/4CR F22M							
2750			Blind	5407431	6 - 6	FLG BLIND 10000 API 6A TYPE 6BX 5 1/8" NS RTJ 2-1/4CR F22M							
2760			Blind	5407432	8 - 8	FLG BLIND 10000 API 6A TYPE 6BX 7 1/16" NS RTJ 2-1/4CR F22M							
2770													
2780			Blind	5407433	10 - 10	FLG BLIND 10000 API 6A TYPE 6BX 9" NS RTJ 2-1/4CR F22M							
2790			Blind	5407435	12 - 14	FLG BLIND 10000 API 6A TYPE 6BX 11" NS RTJ 2-1/4CR F22M							
2800			Blind	5407436	16 - 18	FLG BLIND 10000 API 6A TYPE 6BX 13 5/8" NS RTJ 2-1/4CR F22M							
			FGP / WPMP TENGINZCHEVROIL - TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY MLA	REV D	SHEET 5 OF 8	PA 5	CLASS 10000F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
2810			Blind	5407438	20 - 22	FLG BLIND 10000 API 6A TYPE 6BX 16 3/4" NS RTJ 2-1/4CR F22M								
2820														
2830														
2840	C		Blind	5619755	3/4 - 1 1/2	FLG COMPACT BLIND 10000# RJ LTCS A350 LF2 PIPE CLASS 10000F22 FOR NPS 1 1/2 INCH & BELOW	new							
2850														
2850	D		Blind	5619754	2 - 3	FLG COMPACT BLIND 10000# RJ 2-1/4CR F22M PIPE CLASS 10000F22 FOR NPS 2 TO 3 INCH	new							
2870														
2880	D		Blind	5619756	4 - 22	FLG COMPACT BLIND 10000# RJ 2-1/4CR F22M PIPE CLASS 10000F22 FOR NPS 4 INCH & ABOVE	new							
2890														
2900														
2910														
2920		FA1				***** ORIFICE FLANGES *****								
2930														
2940	C		Orific	5407522	2 - 2	WN ORIF SET 10000 API 6A TYPE 6BX 1 13/16" NS (2" NB) RTJ BORE 39.3MM 2-1/4CR F22M								
2950	C		Orific	5407545	14 - 14	WN ORIF SET 10000 API 6A TYPE 6BX 11" NS (14" NB) RTJ BORE 263.6MM 2-1/4CR F22M								
2960	C		Orific	5407546	16 - 16	WN ORIF SET 10000 API 6A TYPE 6BX 13 5/8" NS (16" NB) RTJ BORE 302.4MM 2-1/4CR F22M								
2970	C		Orific	5407547	18 - 18	WN ORIF SET 10000 API 6A TYPE 6BX 13 5/8" NS (18" NB) RTJ BORE 341.0MM 2-1/4CR F22M								
2980	C		Orific	5407711	22 - 22	WN ORIF SET 10000 API 6A TYPE 6BX 16 3/4" NS (22" NB) RTJ BORE 418.0MM 2-1/4CR F22M								
2990														
3000														
3010						*****PIPE NIPPLES*****								
3020														
3030		812				NIPPLE PBE CUT FROM PIPE								
3040														
3050														
3060		FA1				***** GASKETS *****								
3070														
3080			GASRJ	5674471	2 - 2	GASKET 10000 API 6A BX 151 1 13/16" NS 316 SS RING								
3090			GASRJ	5674475	3 - 3	GASKET 10000 API 6A BX 153 2 9/16" NS 316SS RING								
3100			GASRJ	5674498	4 - 4	GASKET 10000 API 6A BX 155 4 1/16" NS 316 SS RING								
3110			GASRJ	5674497	6 - 6	GASKET 10000 API 6A BX 169 5 1/8" NS 316SS RING								
3120			GASRJ	5674486	8 - 8	GASKET 10000 API 6A BX 156 7 1/16" NS 316SS RING								
3130			GASRJ	5674487	10 - 10	GASKET 10000 API 6A BX 157 9" NS 316SS RING								
3140			GASRJ	5674483	12 - 14	GASKET 10000 API 6A BX 158 11" NS 316SS RING								
3150			GASRJ	5674484	16 - 18	GASKET 10000 API 6A BX 159 13 5/8" NS 316SS RING6A BX								
3160			GASRJ	5674488	20 - 22	GASKET 10000 API 162 16 3/4" NS 316SS RING								
3170														
3180	D	815	GASRJ	5673554	3/4 - 3	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 10000F22 FOR NPS 3 INCH & BELOW								
3190	D	815	GASRJ	5673550	4 - 22	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 10000F22 FOR NPS 4 INCH & ABOVE								
3200														
3210														
3220						***** BOLTING *****								
3230														
3240	D	BA4	BOLS	5675770	3/4 - 4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
3250	D	BA5	BOLS	5675778	6 - 99	STUD-BOLT/COMPACT NUT API-6A718								
3260														
3270	D	BA3	BOLS	5675770	2 - 18	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
3280	D	BA3	BOLS	5675776	20 - 99	STUD-BOLT WITH 2 NUTS API-6A718								
3290														
3300		SB6	CPSBOL	5675675	8 - 22	JACK SCREW HEX HEAD A320 GR L7M								
3310														
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17NOV16 BY MLA REV D SHEET 6 OF 8 PA 5 CLASS 10000F22						

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3320 3330 3340	C	FA1	Specta	5407622	2 - 2	SPECTACLE BLIND 10000 API 6A 1 13/16" NS FOR 2" PIPE RTJ FEMALE 2-1/4CR F22M							
3350			Specta	5407623	3 - 3	SPECTACLE BLIND 10000 API 6A 2 9/16" NS FOR 3" PIPE RTJ FEMALE 2-1/4CR F22M							
3360 3370 3380 3390			Specta	5407624	4 - 4	SPECTACLE BLIND 10000 API 6A 4 1/16" NS FOR 4" PIPE RTJ FEMALE 2-1/4CR F22M							
3400			Paddle	5407742	8 - 8	PADDLE BLANK 10000 API 6A 7 1/16" NS FOR 8" PIPE RTJ FEMALE 2-1/4CR F22M							
3410			Paddle	5407842	8 - 8	PADDLE SPACER 10000 API 6A 7 1/16" NS FOR 8" PIPE RTJ FEMALE 2-1/4CR F22M							
3420			Paddle	5407745	10 - 10	PADDLE BLANK 10000 API 6A 9" NS FOR 10" PIPE RTJ FEMALE 2-1/4CR F22M							
3430 3440			Paddle	5407845	10 - 10	PADDLE SPACER 10000 API 6A 9" NS FOR 10" PIPE RTJ FEMALE 2-1/4CR F22M							
3450			Paddle	5407747	14 - 14	PADDLE BLANK 10000 API 6A 11" NS FOR 14" PIPE RTJ FEMALE 2-1/4CR F22M							
3460 3470			Paddle	5407847	14 - 14	PADDLE SPACER 10000 API 6A 11" NS FOR 14" PIPE RTJ FEMALE 2-1/4CR F22M							
3480			Paddle	5407746	16 - 16	PADDLE BLANK 10000 API 6A 13 5/8" NS FOR 16" PIPE RTJ FEMALE 2-1/4CR F22M							
3490 3500	D	ZA6	Paddle	5407846	16 - 16	PADDLE SPACER 10000 API 6A 13 5/8" NS FOR 16" PIPE RTJ FEMALE 2-1/4CR F22M		SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG DHC, GS, GHP, KM					
3510			Paddle	5407744	18 - 18	PADDLE BLANK 10000 API 6A 13 5/8" NS FOR 18" PIPE RTJ FEMALE 2-1/4CR F22M							
3520 3530			Paddle	5407844	18 - 18	PADDLE SPACER 10000 API 6A 13 5/8" NS FOR 18" PIPE RTJ FEMALE 2-1/4CR F22M							
3540			Paddle	5407784	22 - 22	PADDLE BLANK 10000 API 6A 16 3/4" NS FOR 22" PIPE RTJ FEMALE 2-1/4CR F22M							
3550 3560 3570			Paddle	5407884	22 - 22	PADDLE SPACER 10000 API 6A 16 3/4" NS FOR 22" PIPE RTJ FEMALE 2-1/4CR F22M							
3580			Specta	5749571	2 - 3	SPECTACLE BLIND COMPACT 10000# RJ FEMALE 2-1/4CR F22M PIPE CLASS 10000F22 FOR NPS 3 INCH & BELOW							
3590			Specta	5749576	4 - 4	SPECTACLE BLIND COMPACT 10000# RJ FEMALE 2-1/4CR F22M PIPE CLASS 10000F22 FOR NPS 4 INCH							
3600 3610			Paddle	5750276	8 - 22	PADDLE BLANK COMPACT 10000# RJ 2-1/4CR F22M PIPE CLASS 10000F22							
3620 3630 3640 3650 3660 3670 3680 3690 3700 3710 3720 3730			Paddle	5750288	8 - 22	PADDLE SPACER COMPACT 10000# RJ 2-1/4CR F22M PIPE CLASS 10000F22							
						*****CONSTRUCTION DETAILS*****							
						SEAL WELDING-----FOR THREADED CONNECTIONS ONLY PIPE BENDS-----NONE THREAD SEALANT-----NONE							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.	CORR. ALLOW. 3 MM	DATE 17NOV16	BY MLA	REV D	FGP/WPMP SHEET 7 OF 8	PA 5	CLASS 10000F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
3740 3750 3760	D	NA	Ball V	M10CL10	3/4 - 2	*****DO NOT USE VALVES***** MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL LO		SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.3 CHAPTER IX / API 6A DESIGN PRESSURE - 600 BARG DHC, GS, GHP, KM MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -40 / 130 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF					
3770	C		Sweepo	5506940	2 - 2	REINFORCED BRANCH INSERT CONNECTION 23.5MM X 10.5MM, BW OUTLET API 2-1/4CR F22M	6 - 6						
3780	C		Sweepo	5506941	2 - 2	REINFORCED BRANCH INSERT CONNECTION 29.5MM X 10.5MM, BW OUTLET API 2-1/4CR F22M	8 - 8						
3790	C		Sweepo	5506946	2 - 2	REINFORCED BRANCH INSERT CONNECTION 52.0MM X 10.5MM, BW OUTLET API 2-1/4CR F22M	16 - 16						
3800	C		Flange	5851207	3/4 - 2	FLANGE ADAPTER COMPACT 10000# RJ 316 SS X 12 MM TUBE							
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM	DATE 17NOV16	BY MLA	REV D	SHEET 8 OF 8	PA 5	CLASS 10000F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES														
10						*****VALVES*****		815	TO BE USED WITH COMPACT FLANGES ONLY.													
20						BA4		TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.														
30						FA1		API 6BX FLANGE TO NB PIPE DESIGNATION														
40								NB	6BX													
50								2"	1 13/16"													
60								3"	2 9/16"													
70								4"	4 1/16"													
80								6"	5 1/8"													
90								8"	7 1/16"													
100								10"	9"													
110								12"	11"													
120								14"	11"													
130								16"	13 5/8"													
140								18"	13 5/8"													
150								20"	16 3/4 "													
160								22"	16 3/4 "													
170						VU1		USE THIS VALVE FOR PRESSURE INSTRUMENT CONNECTIONS, VENTS AND DRAINS.														
180								ALL BOLTS ARE METRIC DIA.														
190								COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME'S														
200																						
210																						
220																						
230																						
240																						
250																						
260																						
270																						
280																						
290																						
300																						
310																						
320																						
330						90 Deg		9006586	2 - 2	ELL BW 90 DEG LR WT 11.07MM A860 WPHY 60				SERVICE								
340									2 - 2	ELL BW 45 DEG LR WT 11.07MM A860 WPHY 60												
350									4 - 4	ELL BW 90 DEG LR WT 19.51MM A860 WPHY 60												
360									4 - 4	ELL BW 45 DEG LR WT 19.51MM A860 WPHY 60												
370						45 Deg		9006585	4 - 4	*****FITTINGS ABOVE GROUND*****				NON SOUR SERVICE DESIGN CODE ASME B31.8 / API 6A DESIGN PRESSURE - 600 BARG PIPING: KM, FGH - ABOVE GROUND								
380																	Tee	9006653	2 - 2	TEE BW WT 11.07 MM LTCS A860 WPHY 60		
390																			4 - 4	TEE BW WT 19.51 MM LTCS A860 WPHY 60		
400																			2 - 2	TEE RED BW WT 19.51/11.07 MM LTCS A860 WPHY 60		
410						Eccent		9006913	2 - 2	REDUCER ECC WT 19.51 MM / 11.07 MM LTCS A860 WPHY-60				MATERIAL: API-5L X60 LTCS FLANGE FLANGE RATING: 10000 CF TEMPERATURE RANGE: -40 / 110 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF								
420									Concen	9006914	2 - 2	REDUCER CONC WT 19.51 MM / 11.07 MM LTCS A860 WPHY-60										
430											FA1	***** PIPE FLANGES ABOVE GROUND *****										
440																						
450						816		Reduci	9006725	2 - 2	FLANGE COMPACT 10000# RJ WITH 9/16" FEMALE AUTOCLAVE CONNECTION LTCS A694 F60											
460																						
470																						
480						816		Flange	9006721	2 - 2	FLG COMPACT 10000# RJ A694 F60 WT 11.07 MM											
490																						
500																						
510						816		Flange	9006722	4 - 4	FLG COMPACT 10000# RJ A694 F60 WT 19.51 MM											
520																						
530																						
540																						
550																						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP														
						ALL SIZES BUTTWELD PWHT - 4" ONLY		CORR. ALLOW. 1.5 MM		DATE 17NOV16	BY JCS	REV A	SHEET 1 OF 2	PA 5	CLASS 10000H5A							

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
560			Blind	9619758	2 - 2	FLG COMPACT BLIND 10000# RJ LTCS A694 F60 PIPE CLASS 10000H5A - 2 INCH ONLY								
570			Blind	9619759	4 - 4	FLG COMPACT BLIND 10000# RJ LTCS A694 F60 PIPE CLASS 10000H5A - 4 INCH ONLY								
580														
590														
600						***** GASKETS *****								
610														
620		815	GASRJ	5673562	2 - 2	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 10000H5A FOR NPS 3 INCH & BELOW								
630		815	GASRJ	5673560	4 - 4	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 10000H5A NPS 4 INCH & ABOVE								
640														
650			GASRJ	5674472	2 - 2	GASKET 10000 API 6A BX 152 2 1/16" NS 316 SS RING								
660														
670														
680		B1				***** BOLTING *****								
690														
700		BA4	BOLS	5675770	2 - 4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
710														
720														
730						*****BRANCHES*****								
740														
750						USE BRANCH CHART 10000H5A								
760														
770			Reinfo	RWELD	3 - 3	NO-MTO REINFORCING WELD FOR STUB-IN	4 - 4							
780			Reinfo	RWELD	3 - 3	NO-MTO REINFORCING WELD FOR STUB-IN	4 - 4							
790														
800						*****SP ITEMS*****								
810														
820														
830			Flange	SP-9521	2 - 2	FLG WN 10000 API 6A TYPE 6BX 2 1/16" NS (2" NB) RTJ 38.16MM BORE ASTM A694 F60								
840														
850														
860						*****DO NOT USE*****								
870														
880			Ball V	B10CL16	3/4 - 2	BALL 10000 CF INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO								
								SERVICE						
								NON SOUR SERVICE DESIGN CODE ASME B31.8 / API 6A DESIGN PRESSURE - 600 BARG PIPING: KM, FGH - ABOVE GROUND						
								MATERIAL: API-5L X60 LTCS FLANGE FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -40 / 110 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF						
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION		FGP/WPMP						
						ALL SIZES BUTTWELD PWHT - 4" ONLY		CORR. ALLOW.	DATE	BY	REV	SHEET	PA	CLASS
								1.5 MM	17NOV16	JCS	A	2 OF 2	5	10000H5A

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10			Condui	W10CL72-K5A	3 - 3	*****VALVES*****		G1	FA	NOT TO BE USED ON FGP API 6BX FLANGE TO NB PIPE DESIGNATION NB PIPE 6BX FLANGE 3" 2 1/16" REPLACE 2 9/16" AND 3 1/16" FLANGES AND VALVES MOUNTED ON EXISTING TUBING & CASING KILL LINES WITH 2 1/16" COUNTERPARTS WHEN PERFORMING REPAIRS/REPLACEMENTS AND/OR MODIFICATIONS AT LEGACY INSTALLATIONS.					
20						***** THROUGH CONDUIT GATE VALVES*****									
30						GATE 10000 CF INCONEL 625 BODY AND TRIM THROUGH CONDUIT SOLID SLAB HW									
40															
50															
60															
70			CHECK V	GH3C2 - K5A	3 - 3	*****CHECK VALVES*****									
80						CHECK API 10000 TYPE 6BX RTJ BOLT THROUGH LUG DUAL PLATE 60K MATL/INCONEL TRIM									
90															
100															
110			Piping	9005242	3 - 3	*****PIPE *****									
120						LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE									
130															
140						*****FITTINGS *****									
150															
160						ELL BW 90 DEG LR WT 17.48 MM A860 WPHY 60-W									
170			90 Deg	9006474	3 - 3	ELL BW 45 DEG LR WT 17.48MM A860 WPHY 60									
180															
190			45 Deg	9006582	3 - 3										
200															
210			TEE	9006489	3 - 3	TEE BW WT 17.48MM A860 WPHY 60-W									
220															
230			FLANGE	5301668	3 - 3	*****PIPE FLANGES *****									
240						FLG WN 10000 API 6A TYPE 6BX 2 1/16" NS (3" NB) RTJ 1.688" BORE 60K MATL									
250			BLIND	5407423	3 - 3										
260						FLG BLIND 10000 API 6A TYPE 6BX 2 9/16" NS RTJ 2-1/4CR F22M									
270			GASRJ	5674472	3 - 3	***** GASKETS *****									
280						GASKET 10000 API 6A BX 152 2 1/16" NS 316 SS RING									
290			BOLTS	5675770		***** BOLTING*****									
300						STUD BOLT A 320 GR L7M/ A 194 GR 7M NUTS									
310															
320						***** BRANCHES *****									
330						USE BRANCH CHART 10000K5A IN PIM-SU-5112-TCO									
340															
350															
360															
370															
380															
390															
400															
410															
420															
430															
440															
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500															
510															
520															
530															
540															
550															
560															
570															
580															
			ISSUED FOR BASE BUSINESS USE ONLY TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - FOR WT >19MM NO FIELD WELDS AT COMISSIONED WELLS - SEE NOTE PP	CORR. ALLOW. 3 MM	TCO BASE BUSINESS							
								DATE	BY	REV	SHEET	PA	CLASS		
								10JAN18	KM	A	1 OF 1	5	10000K5A		

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES							
10		PP9	Piping	9005278CTD	14 - 14	*****PIPE BELOW GROUND*****		NA P** PP9	NOT TO BE USED *** PIPE NOTES *** ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156						
20															
30															
40															
50															
60															
70															
80															
90															
100															
110															
120															
130															
140															
150															
160															
170															
180															
190															
200															
210															
220															
230															
240															
250															
260															
270		NA	Ball V	M10CL16	2 - 2	MODULAR VALVE DB&B BALL TYPE 10000 CF INCONEL 625 BODY AND TRIM METAL GO									
280		NA	Ball V	B10CL16	3/4 - 2	BALL 10000 CF INCONEL 625 BODY AND TRIM TRUNNION METAL FB GO									
290		NA	Gate V	W10CL72	2 - 2	GATE 10000 CF INCONEL 625 BODY AND TRIM THROUGH CONDUIT SOLID SLAB HW									
300															
310		NA	Globe	L10CL43	2 - 2	GLOBE 10000# CF INC 625 B564 INC 625 TRIM OS&Y HW									
320		NA	Piping	9001492	2 - 2	LINEPIPE SMLS WT= 14 MM API-5L X60 BE									
330		NA	90 Deg	9006466	2 - 2	ELL BW 90 DEG LR WT 14MM A860 WPHY 60									
340		NA	45 Deg	9006467	2 - 2	ELL BW 45 DEG LR WT 14MM A860 WPHY 60									
350		NA	Tee	9006206	2 - 2	TEE BW WT 14MM LTCS A860 WPHY 60									
360		NA	Flange	5851207	2 - 2	FLANGE ADAPTER COMPACT 10000# RJ 316 SS X 12 MM TUBE									
370		NA	Blind	9006224	2 - 14	FLG COMPACT BLIND 10000# RJ LTCS A694 F60									
380		NA	Flange	9006223	2 - 2	FLG COMPACT 10000# RJ A694 F60 WT 14.00 MM SPECTACLE									
390		NA	Specta	9002768	2 - 2	BLIND COMPACT API 10000 RJ LTCS A516-GR70									
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL WELDS	CORR. ALLOW. 3 MM	FGP/WPMP DATE 17NOV16 BY JCS REV B SHEET 1 OF 1 PA 5 CLASS 10000X1							

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES						
20	B	PP9	Piping	9005277	14 - 14	*****PIPE ABOVE GROUND*****		815	TO BE USED WITH COMPACT FLANGES ONLY. COMPACT FLANGES WHEREVER AVAILABLE SHOULD BE THE PREFERRED CONNECTION. FLANGE DESIGN SHALL BE CONFIRMED BY ENGINEERING SME'S ALL BOLTS USED ON THIS PROJECT ARE METRIC DIA. TO BE USED WITH API 6A FLANGES ONLY. TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. API 6BX FLANGE TO NB PIPE DESIGNATION NB PIPE 6BX FLANGE 2" 1 13/16" 3" 2 9/16" 4" 4 1/16" 6" 5 1/8" 8" 7 1/16" 10" 9" 12" 11" 14" 11" 16" 13 5/8" 18" 13 5/8" 20" 16 3/4 " 22" 16 3/4 "					
30														
40														
50														
60														
70														
80														
90														
100														
110														
120														
130														
140														
150														
160														
170														
180														
190														
200														
210														
220														
230														
240														
250														
260														
270														
280														
290														
300														
310														
320														
330														
340														
350														
360														
370														
380														
390														
400														
410														
420														
430														
440														
450														
460														
470														
480														
490														
500														
510														
520														
530														
		816	Flange	9005279	14 - 14	FLG COMPACT 10000# RJ BORE 245.6MM 2-1/4CR F22M		PP9	ALL PIPING COMPONENTS SHALL MEET THE REQUIREMENTS OF NACE MR0175/ISO 15156 USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.					
			Flange	9006192	14 - 14	FLG WN 10000 API 6A TYPE 6BX 11" (14" NB) NS RTJ BORE 249.0MM 2-1/4CR F22M		SB6						
			Blind	5407435	14 - 14	FLG BLIND 10000 API 6A TYPE 6BX 11" NS RTJ 2-1/4CR F22M								
			Blind	9619757	14 - 14	FLG COMPACT BLIND 10000# RJ 2-1/4CR F22M PIPE CLASS 10000X22								
		FA1				***** GASKETS *****			SERVICE SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.8 / API 6A DESIGN PRESSURE - 600 BARG GHP - GAS INJECTION HIGH PRESSURE - ABOVE GROUND					
			GASRJ	5674483	14 - 14	GASKET 10000 API 6A BX 158 11" NS 316SS RING								
		815	GASRJ	5673551	14 - 14	GASKET 10000# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE PIPE CLASS 10000X22								
						***** BOLTING *****								
	B	BA5	BOLS	5675778	14 - 14	STUD-BOLT/COMPACT NUT API-6A718		MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -40 / 110 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF						
	B	BA3	BOLS	5675770	14 - 14	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS								
		SB6	Y	5675675	14 - 14	JACK SCREW HEX HEAD A320 GR L7M								
		FA1				***** BLINDS AND SPACERS *****								
			Paddle	5407747	14 - 14	PADDLE BLANK 10000 API 6A 11" NS FOR 14" PIPE RTJ FEMALE 2-1/4CR F22M								
			FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM	DATE 17NOV16	BY JCS	REV B	SHEET 1 OF 2	PA 5	CLASS 10000X22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES	DESCRIPTION	REFERENCE / LARGE END SIZES	NOTES					
540 550 560 570 580 590 600 610 620 630 640			Paddle	5407845	14 - 14	PADDLE SPACER 10000 API 6A 9" NS FOR 10" PIPE RTJ FEMALE 2-1/4CR F22M *****BRANCHES***** USE BRANCH CHART 10000X22 *****CONSTRUCTION DETAILS***** PIPE BENDS----- F22 = FORGED							
								SERVICE					
								SOUR SERVICE NACE STD. MR0175 / ISO 15156 DESIGN CODE ASME B31.8 / API 6A DESIGN PRESSURE - 600 BARG GHP - GAS INJECTION HIGH PRESSURE - ABOVE GROUND					
								MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 10000 RJ TEMPERATURE RANGE: -40 / 110 °C COMMENT: FLG FACE RJ REFERS COMPACT FLANGE - CF					
		FGP / WPMP TENGIZCHEVROIL - TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN			CONSTRUCTION ALL SIZES BUTTWELD PWHT - ALL SIZES.		CORR. ALLOW. 3 MM	DATE 17NOV16	BY JCS	REV B	SHEET 2 OF 2	PA 5	CLASS 10000X22